Symbolic gate atlas

Exact identities. Explicit phase policies. Native entangler counts.

11 targets × 32 basis panels

Each target has one page with 32 basis panels in alphabetical order. Blue labels denote literal matrix equality. Brown labels denote a projective circuit with its exact scalar factor. Both appear when the projective construction is cheaper or their secondary-score comparison remains unresolved. Physical GP operations are forbidden.
E counts native two-qubit primitives, L counts one-qubit operations. Evaluated scores are displayed rounded; the archive stores exact expressions. Shared-expression scores that exceeded their evaluation budget are labelled unresolved. Objective version 2 clamps negative denominator penalties; an exact compatibility audit preserves the earlier retained scores. Only an explicit matching certificate establishes a minimum entangler count. A verified upper bound does not establish minimum local cost. A minimum uniform E budget concerns a circuit for the whole parameterized family; a certified hard member forces that budget, while special parameter values can require fewer pulses. Published-cap notes identify constructions above known universal two-qubit bounds; these remain optimization gaps. Proof conditions, source hashes and attempt histories remain in the archive.
Open cases are shown as open. Proved obstructions preserve the supplied basis and may be conditional for parameterized families. Register constructors use the recorded integer-width or matrix fixtures. These pages do not claim a solution for every register size.
Circuits are drawn by Qiskit's LaTeX drawer. Long circuits continue on successive rows in time order. An explicit $[G]^{\times n}$ box denotes n consecutive identical primitive operations; their full physical count remains in E.
Long exact expressions use numbered η_j labels defined in the expression glossary. No parameter is rounded or dropped.
The separate Clifford volume lists every exact phase in each distinct one-qubit alphabet, with 24 circuits per page. Identical alphabets shared by multiple presets are printed once.

Archive SHA256: 47d7e2ef23978fcf1544be38878e3df41f0c0784e5465a3cc6f847f258cc99bb

Alphabetical target index

Links lead to the 32-basis page for each target.

C3SX . . . 4
C4X . . . 5
Canonical . . . 6
CZS . . . 7
fFredkin . . . 8
fSim . . . 9
Peres . . . 10
PhasedFSim . . . 11
SWAPPow . . . 12
SYC . . . 13
TK2 . . . 14

Native basis key

Every panel preserves its supplied alphabet. Native units and macro counts matter.

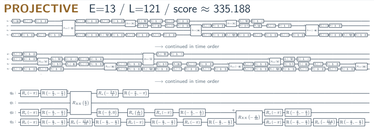
Preset	Entangler	Local alphabet	Scope / units
aqt_rxx	Rxx	R, Rz	exact continuous
berkeley_b	B	U3	exact continuous
clifford_only	CX	H, S	Clifford analysis/control; not universal.
clifford_t_cx	CX	H, S, T, Tdg	Dense universal; exact only for representable targets, not arbitrary symbolic angles.
clifford_t_cz	CZ	H, S, T, Tdg	approximate discrete
cx_h_phase	CX	H, P	exact continuous
cx_rz_ry	CX	Rz, Ry	exact continuous
cx_u	CX	U3	exact continuous
cz_rz_rx	CZ	Rz, Rx	exact continuous
cz_u	CZ	U3	exact continuous
google_cz	CZ	PhasedXZ	exact continuous
google_fsim	fSim	PhasedXZ	exact continuous
google_sqrt_iswap	SiSWAP	PhasedXZ	exact continuous
google_syc	SYC	PhasedXZ	exact continuous
ibm_cx	CX	Rz, Sx, X	exact continuous
ibm_cz	CZ	Rz, Sx, X	exact continuous
ibm_ecr	ECR	Rz, Sx, X	exact continuous
ionq_aria	IonQMS	GPi, GPi2	Native turns; interaction theta restricted to [0,1/4] on hardware.
ionq_aria_full	IonQMSMax	GPi, GPi2	exact continuous
ionq_forte	IonQZZ	GPi, GPi2	exact continuous
iqm_cz	CZ	R	R(theta,phi) represents PRX in radians; IQM API units require conversion.
iswap_u	iSWAP	U3	exact continuous
quantinuum_zzmax	ZZMax	PhasedX, Rz	exact continuous
quantinuum_zzphase	ZZPhase	PhasedX, Rz	exact continuous
rigetti_cz	CZ	Rz, Sx, X	Sx differs from native Rx(pi/2) by a tracked global phase.
sqrt_iswap_u	SiSWAP	U3	exact continuous
sqrt_swap_u	SSWAP	U3	exact continuous
tket_tk2	TK2	PhasedX, Rz	Compiler macro count; expand TK2 before comparing physical pulse counts.
xx90_u	XX90	U3	exact continuous
xx_rz_ry	Rxx	Rz, Ry	exact continuous
zx_u	Rzx	U3	exact continuous
zz_u	Rzz	U3	exact continuous

C3SX

4 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=13 / L=121 / score ≈ 335.188 — E upper bound



$U = -(-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=40 / L=115 / score ≈ 442.290 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

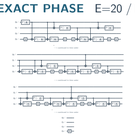
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

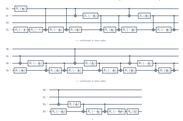
cx_h_phase

EXACT PHASE E=20 / L=35 / score ≈ 166.842 — E upper bound



cx_rz_ry

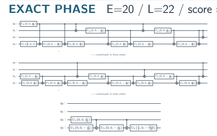
PROJECTIVE E=20 / L=24 / score ≈ 145.248 — E upper bound



$U = e^{-\frac{31i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=20 / L=22 / score ≈ 141.332 — E upper bound



cz_rz_rx

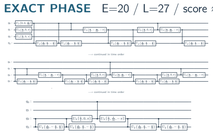
PROJECTIVE E=20 / L=40 / score ≈ 179.680 — E upper bound



$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=20 / L=27 / score ≈ 158.962 — E upper bound



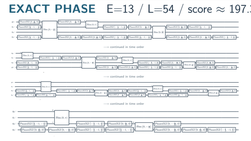
google_cz

EXACT PHASE E=20 / L=75 / score ≈ 268.850 — E upper bound



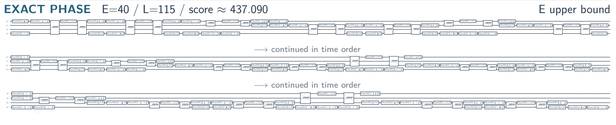
google_fsim

EXACT PHASE E=13 / L=54 / score ≈ 197.376 — E upper bound



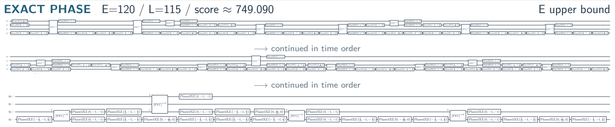
google_sqrt_iswap

EXACT PHASE E=40 / L=115 / score ≈ 437.090 — E upper bound



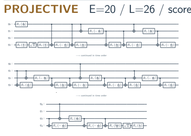
google_syc

EXACT PHASE E=120 / L=115 / score ≈ 749.090 — E upper bound



ibm_cx

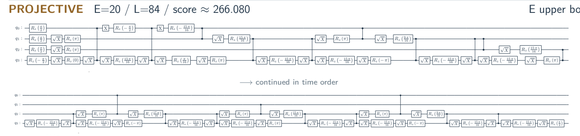
PROJECTIVE E=20 / L=26 / score ≈ 149.448 — E upper bound



$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

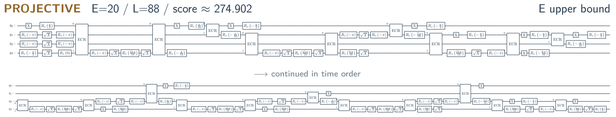
PROJECTIVE E=20 / L=84 / score ≈ 266.080 — E upper bound



$U = e^{-\frac{31i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

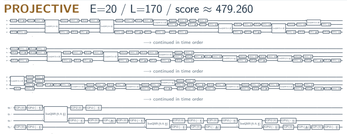
PROJECTIVE E=20 / L=88 / score ≈ 274.902 — E upper bound



$U = e^{-\frac{15i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

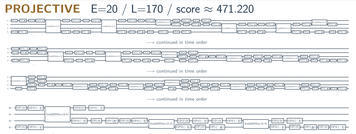
PROJECTIVE E=20 / L=170 / score ≈ 479.260 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

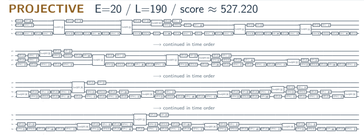
PROJECTIVE E=20 / L=170 / score ≈ 471.220 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

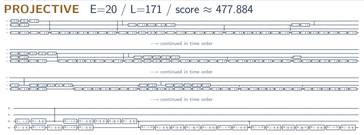
PROJECTIVE E=20 / L=190 / score ≈ 527.220 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=20 / L=171 / score ≈ 477.884 — E upper bound



$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=26 / L=41 / score ≈ 200.446 — E upper bound



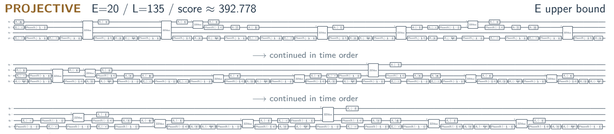
PROJECTIVE E=26 / L=40 / score ≈ 196.440 — E upper bound



$U = \eta_{15}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

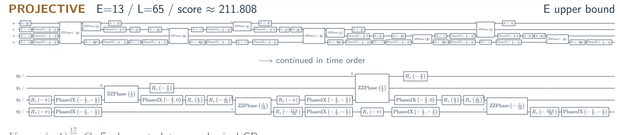
PROJECTIVE E=20 / L=135 / score ≈ 392.778 — E upper bound



$U = \sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

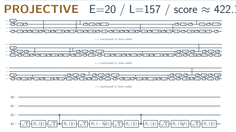
PROJECTIVE E=13 / L=65 / score ≈ 211.808 — E upper bound



$U = -(-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

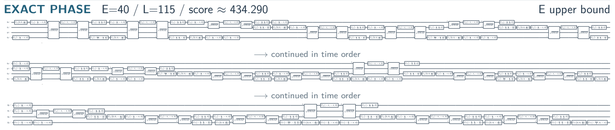
PROJECTIVE E=20 / L=157 / score ≈ 422.154 — E upper bound



$U = e^{\frac{1i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

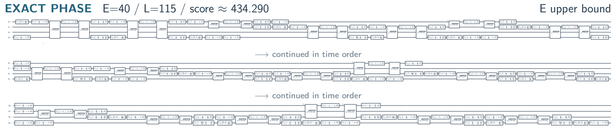
sqrt_iswap_u

EXACT PHASE E=40 / L=115 / score ≈ 434.290 — E upper bound



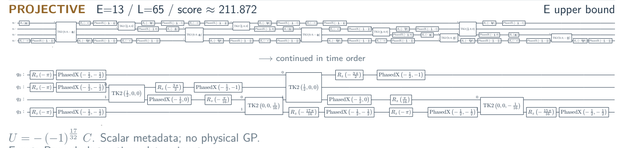
sqrt_swap_u

EXACT PHASE E=40 / L=115 / score ≈ 434.290 — E upper bound



tket_tk2

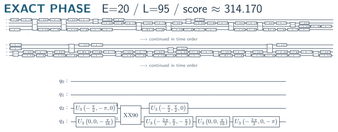
PROJECTIVE E=13 / L=65 / score ≈ 211.872 — E upper bound



$U = -(-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

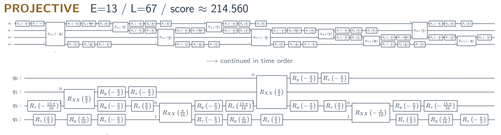
xx90_u

EXACT PHASE E=20 / L=95 / score ≈ 314.170 — E upper bound



xx_rz_ry

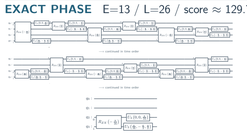
PROJECTIVE E=13 / L=67 / score ≈ 214.560 — E upper bound



$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=13 / L=26 / score ≈ 129.782 — E upper bound

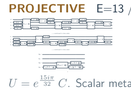


zz_u

EXACT PHASE E=13 / L=29 / score ≈ 135.000 — E upper bound



PROJECTIVE E=13 / L=27 / score ≈ 128.988 — E upper bound



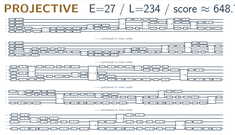
$U = e^{\frac{15i\pi}{32}}\, C$. Scalar metadata; no physical GP.

C4X

5 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

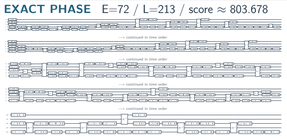
PROJECTIVE E=27 / L=234 / score ≈ 648.754 — E upper bound



$U = (-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=72 / L=213 / score ≈ 803.678 — E upper bound



clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

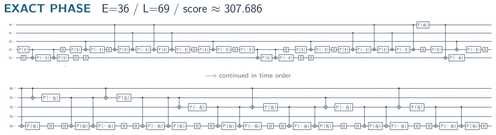
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: determinant and clifford t field

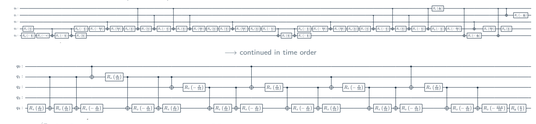
cx_h_phase

EXACT PHASE E=36 / L=69 / score ≈ 307.686 — E upper bound



cx_rz_ry

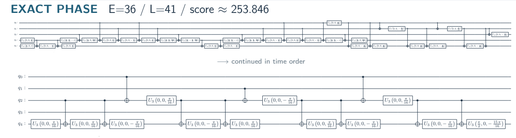
PROJECTIVE E=36 / L=54 / score ≈ 279.708 — E upper bound



$U = e^{\frac{i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

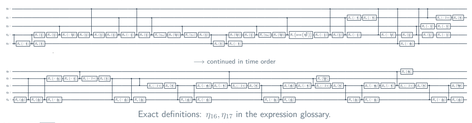
cx_u

EXACT PHASE E=36 / L=41 / score ≈ 253.846 — E upper bound



cz_rz_rx

PROJECTIVE E=36 / L=69 / score ≈ 310.372 — E upper bound

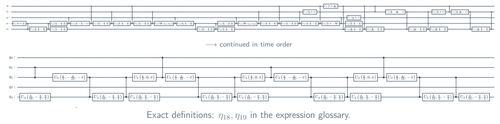


Exact definitions: η_{16}, η_{17} in the expression glossary.

$U = \sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

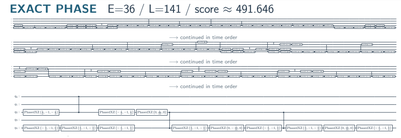
EXACT PHASE E=36 / L=46 / score ≈ 275.510 — E upper bound



Exact definitions: η_{18}, η_{19} in the expression glossary.

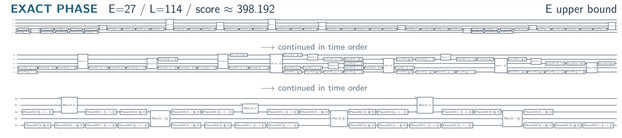
google_cz

EXACT PHASE E=36 / L=141 / score ≈ 491.646 — E upper bound



google_fsim

EXACT PHASE E=27 / L=114 / score ≈ 398.192 — E upper bound



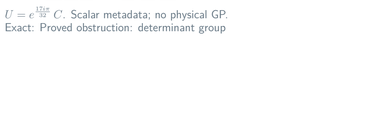
google_sqrt_iswap

EXACT PHASE E=72 / L=213 / score ≈ 794.478 — E upper bound



google_syc

EXACT PHASE E=216 / L=213 / score ≈ 1356.078 — E upper bound



ibm_cx

PROJECTIVE E=36 / L=59 / score ≈ 288.898 — E upper bound



$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=36 / L=130 / score ≈ 429.746 — E upper bound



$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=36 / L=169 / score ≈ 513.392 — E upper bound

$U = e^{-\frac{7i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=36 / L=318 / score ≈ 884.452 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=36 / L=318 / score ≈ 869.980 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=36 / L=354 / score ≈ 970.780 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=36 / L=325 / score ≈ 894.500 — E upper bound

$U = -\sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=54 / L=79 / score ≈ 404.674 — E upper bound

Exact definitions: η_{28}, η_{29} in the expression glossary.

PROJECTIVE E=54 / L=78 / score ≈ 400.668 — E upper bound

$U = \eta_{29}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=36 / L=241 / score ≈ 698.678 — E upper bound

$U = \sqrt[32]{-1}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=27 / L=151 / score ≈ 468.080 — E upper bound

$U = (-1)^{\frac{15}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=36 / L=130 / score ≈ 429.746 — E upper bound

$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=72 / L=213 / score ≈ 789.278 — E upper bound

sqrt_swap_u

EXACT PHASE E=72 / L=213 / score ≈ 789.278 — E upper bound

tket_tk2

PROJECTIVE E=27 / L=162 / score ≈ 490.246 — E upper bound

$U = (-1)^{\frac{17}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=36 / L=177 / score ≈ 573.062 — E upper bound

xx_rz_ry

PROJECTIVE E=27 / L=147 / score ≈ 453.948 — E upper bound

$U = e^{\frac{17i\pi}{32}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=27 / L=56 / score ≈ 265.190 — E upper bound

PROJECTIVE E=27 / L=54 / score ≈ 259.978 — E upper bound

$U = e^{\frac{7i\pi}{8}}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=27 / L=55 / score ≈ 243.618 — E upper bound

Exact definitions: $\eta_{14}, \eta_{15}, \eta_{16}, \eta_{17}, \eta_{18}$ in the expression glossary.

PROJECTIVE E=27 / L=53 / score ≈ 237.606 — E upper bound

Exact definitions: $\eta_{14}, \eta_{15}, \eta_{16}, \eta_{17}, \eta_{18}, \eta_{19}$ in the expression glossary.

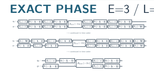
$U = (-1)^{\frac{15}{32}}\, C$. Scalar metadata; no physical GP.

Canonical (a, b, c)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=41 / score ≈ 102.928 — minimum uniform E budget



PROJECTIVE E=3 / L=20 / score ≈ 56.872 — minimum uniform E budget



$U = \eta_{36}\, C$. Scalar metadata; no physical GP.

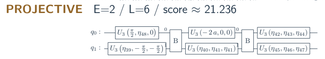
berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget



Exact definitions: $\eta_{27}, \eta_{28}, \eta_{29}, \eta_{30}, \eta_{31}, \eta_{32}, \eta_{33}, \eta_{34}, \eta_{35}, \eta_{36}, \eta_{37}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget



Exact definitions: $\eta_{39}, \eta_{40}, \eta_{41}, \eta_{42}, \eta_{43}, \eta_{44}, \eta_{45}, \eta_{46}, \eta_{47}, \eta_{48}$ in the expression glossary.

$U = \eta_{49}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 52.620 — minimum uniform E budget



PROJECTIVE E=3 / L=12 / score ≈ 36.612 — minimum uniform E budget



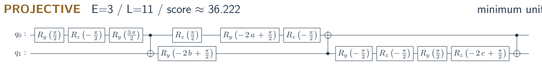
$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=4 / L=17 / score ≈ 52.834 — minimum uniform E budget



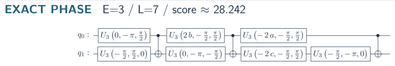
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget



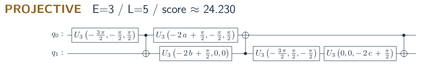
$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — minimum uniform E budget



PROJECTIVE E=3 / L=5 / score ≈ 24.230 — minimum uniform E budget



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=28 / score ≈ 77.256 — minimum uniform E budget



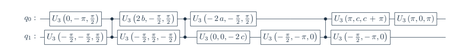
PROJECTIVE E=3 / L=11 / score ≈ 35.822 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 34.660 — minimum uniform E budget



PROJECTIVE E=3 / L=8 / score ≈ 30.648 — minimum uniform E budget



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 35.111 — minimum uniform E budget



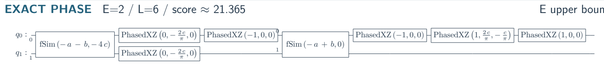
PROJECTIVE E=3 / L=8 / score ≈ 30.439 — minimum uniform E budget



$U = e^{i(a-b+c)}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 21.365 — E upper bound



PROJECTIVE E=2 / L=4 / score ≈ 16.693 — E upper bound



$U = e^{ic}\, C$. Scalar metadata; no physical GP.

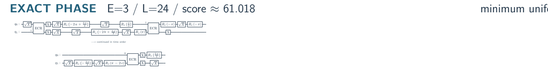
google_sqrt_iswap

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 1.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 1.

$U = e^{i\xi_{1,585}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=19 / score ≈ 115.365 — E upper bound



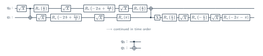
PROJECTIVE E=18 / L=17 / score ≈ 110.693 — E upper bound



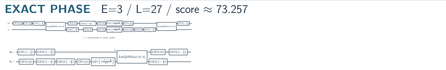
$U = \eta_{50}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.216 — minimum uniform E budget



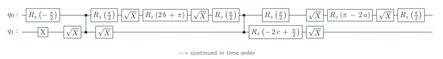
PROJECTIVE E=3 / L=13 / score ≈ 39.012 — minimum uniform E budget



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget



ibm_ecr

EXACT PHASE E=3 / L=24 / score ≈ 61.018 — minimum uniform E budget

PROJECTIVE E=3 / L=22 / score ≈ 57.214 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=3 / L=27 / score ≈ 74.463 — minimum uniform E budget

PROJECTIVE E=3 / L=27 / score ≈ 74.263 — minimum uniform E budget



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=3 / L=27 / score ≈ 73.257 — minimum uniform E budget

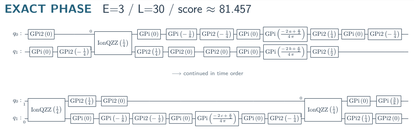
PROJECTIVE E=3 / L=27 / score ≈ 73.057 — minimum uniform E budget



$U = \frac{\sqrt[3]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=3 / L=30 / score ≈ 81.457 — minimum uniform E budget



iqm_cz

EXACT PHASE E=4 / L=27 / score ≈ 75.708 — minimum uniform E budget



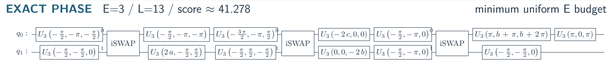
PROJECTIVE E=3 / L=13 / score ≈ 40.452 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=3 / L=13 / score ≈ 41.278 — minimum uniform E budget



PROJECTIVE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget



$U = \frac{\sqrt{2}i(1+i)e^{ih}}{2}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=25 / score ≈ 71.206 — E upper bound



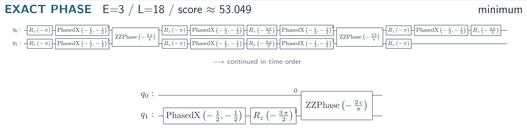
PROJECTIVE E=3 / L=10 / score ≈ 34.293 — minimum uniform E budget



$U = -i\, C$. Scalar metadata; no physical GP.

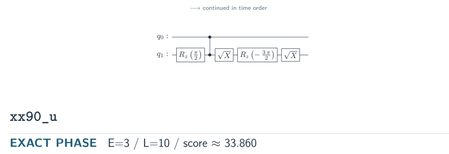
quantinuum_zzphase

EXACT PHASE E=3 / L=18 / score ≈ 53.049 — minimum uniform E budget



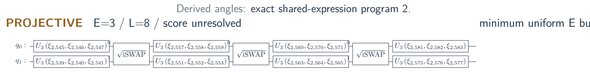
rigetti_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget



sqrt_iswap_u

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 2.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 2.

$U = e^{i\xi_{2,584}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.714 — E upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound



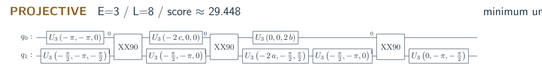
$U = -\sqrt[4]{-1}e^{ic}\, C$. Scalar metadata; no physical GP.

tket_tk2

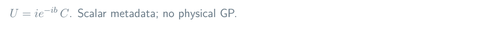
EXACT PHASE E=1 / L=0 / score ≈ 4.997 — minimum uniform E budget

xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 33.860 — minimum uniform E budget



PROJECTIVE E=3 / L=8 / score ≈ 29.448 — minimum uniform E budget



$U = ie^{-ih}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=41 / score ≈ 98.888 — minimum uniform E budget



PROJECTIVE E=3 / L=31 / score ≈ 80.068 — minimum uniform E budget



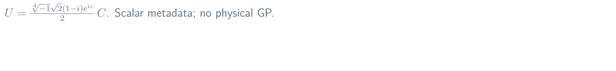
$U = \eta_{36}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.284 — minimum uniform E budget



PROJECTIVE E=3 / L=11 / score ≈ 39.272 — minimum uniform E budget



$U = \frac{\sqrt[3]{-1}\sqrt{2}(1-i)e^{ic}}{2}\, C$. Scalar metadata; no physical GP.

zz_u

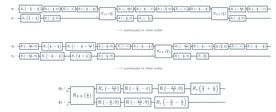
EXACT PHASE E=3 / L=12 / score ≈ 38.478 — minimum uniform E budget

CZS (θ, ϕ, γ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

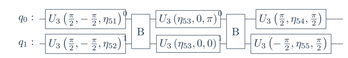
PROJECTIVE E=4 / L=40 / score ≈ 106.532 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=6 / score ≈ 21.636 — minimum uniform E budget



Exact definitions: $\eta_{51}, \eta_{52}, \eta_{53}, \eta_{54}, \eta_{55}$ in the expression glossary.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

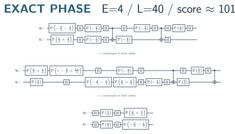
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

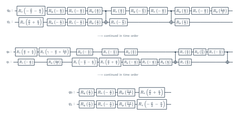
cx_h_phase

EXACT PHASE E=4 / L=40 / score ≈ 101.640 — E upper bound



cx_rz_ry

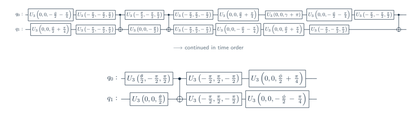
PROJECTIVE E=4 / L=40 / score ≈ 105.680 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

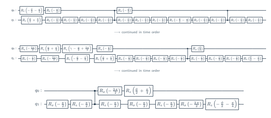
cx_u

EXACT PHASE E=4 / L=21 / score ≈ 67.726 — E upper bound



cz_rz_rx

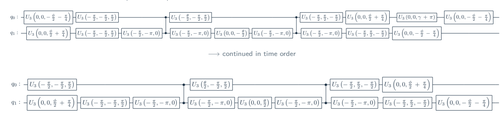
PROJECTIVE E=4 / L=42 / score ≈ 110.084 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

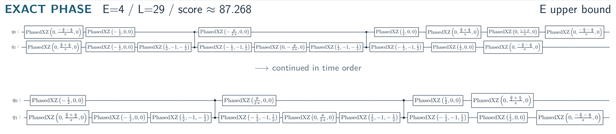
cz_u

EXACT PHASE E=4 / L=29 / score ≈ 85.374 — E upper bound



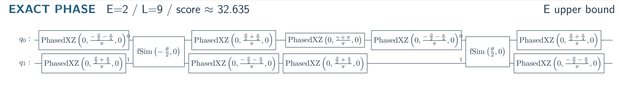
google_cz

EXACT PHASE E=4 / L=29 / score ≈ 87.268 — E upper bound



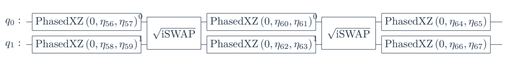
google_fsim

EXACT PHASE E=2 / L=9 / score ≈ 32.635 — E upper bound



google_sqrt_iswap

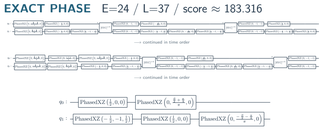
EXACT PHASE E=2 / L=6 / score ≈ 21.757 — minimum uniform E budget



Exact definitions: $\eta_{56}, \eta_{57}, \eta_{58}, \eta_{59}, \eta_{60}, \eta_{61}, \eta_{62}, \eta_{63}, \eta_{64}, \eta_{65}, \eta_{66}, \eta_{67}$ in the expression glossary.

google_syc

EXACT PHASE E=24 / L=37 / score ≈ 183.316 — E upper bound



ibm_cx

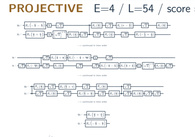
PROJECTIVE E=4 / L=32 / score ≈ 84.832 — E upper bound



$U = -e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

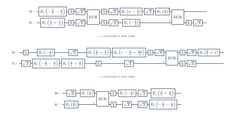
PROJECTIVE E=4 / L=54 / score ≈ 130.448 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

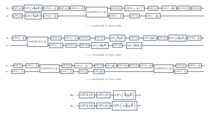
PROJECTIVE E=4 / L=44 / score ≈ 108.832 — E upper bound



$U = -e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

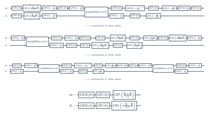
PROJECTIVE E=4 / L=58 / score ≈ 153.634 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

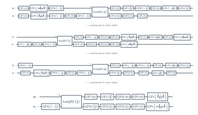
PROJECTIVE E=4 / L=58 / score ≈ 152.026 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

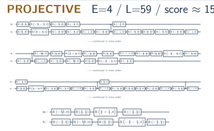
PROJECTIVE E=4 / L=62 / score ≈ 163.226 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

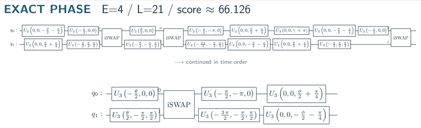
PROJECTIVE E=4 / L=59 / score ≈ 151.836 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

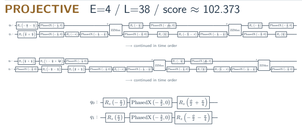
iswap_u

EXACT PHASE E=4 / L=21 / score ≈ 66.126 — E upper bound



quantinuum_zzmax

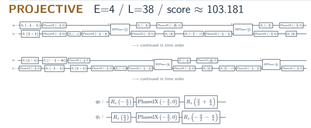
PROJECTIVE E=4 / L=38 / score ≈ 102.373 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

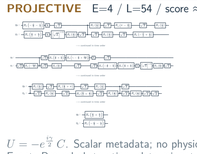
PROJECTIVE E=4 / L=38 / score ≈ 103.181 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

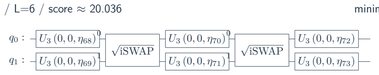
PROJECTIVE E=4 / L=54 / score ≈ 130.448 — E upper bound



$U = -e^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

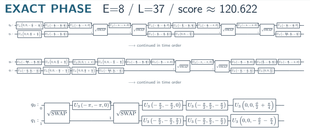
EXACT PHASE E=2 / L=6 / score ≈ 20.036 — minimum uniform E budget



Exact definitions: $\eta_{68}, \eta_{69}, \eta_{70}, \eta_{71}, \eta_{72}, \eta_{73}$ in the expression glossary.

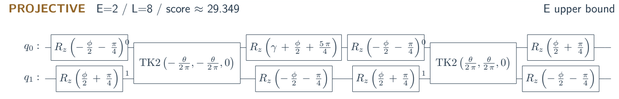
sqrt_swap_u

EXACT PHASE E=8 / L=37 / score ≈ 120.622 — E upper bound



tket_tk2

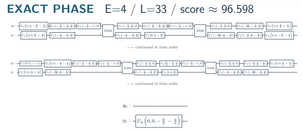
PROJECTIVE E=2 / L=8 / score ≈ 29.349 — E upper bound



$U = e^{\frac{i(\gamma+\pi)}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

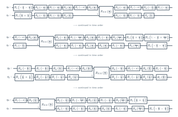
xx90_u

EXACT PHASE E=4 / L=33 / score ≈ 96.598 — E upper bound



xx_rz_ry

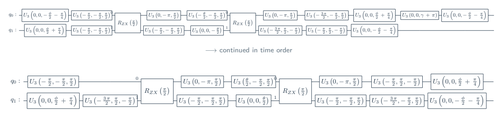
PROJECTIVE E=4 / L=66 / score ≈ 162.540 — E upper bound



$U = -ie^{\frac{5\pi}{2}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

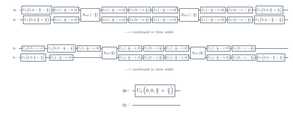
zx_u

EXACT PHASE E=4 / L=29 / score ≈ 87.782 — E upper bound



zz_u

EXACT PHASE E=4 / L=33 / score ≈ 91.006 — E upper bound

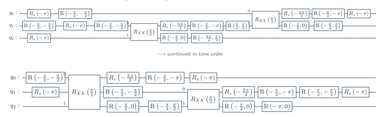


fFredkin

3 qubits / 32 preserved native bases / alphabetical panel order

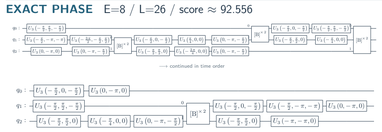
aqt_rxx

EXACT PHASE E=4 / L=30 / score ≈ 84.106 — proved E minimum



berkeley_b

EXACT PHASE E=8 / L=26 / score ≈ 92.556 — E upper bound



clifford_only

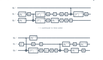
Exact: Proved obstruction: not a clifford
Projective: Proved obstruction: not a clifford

clifford_t_cx

EXACT PHASE E=4 / L=46 / score ≈ 108.000 — proved E minimum



PROJECTIVE E=4 / L=42 / score ≈ 100.000 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

clifford_t_cz

EXACT PHASE E=4 / L=54 / score ≈ 124.000 — proved E minimum



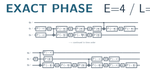
PROJECTIVE E=4 / L=50 / score ≈ 116.000 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

cx_h_phase

EXACT PHASE E=4 / L=32 / score ≈ 83.640 — proved E minimum



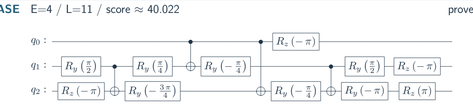
PROJECTIVE E=4 / L=30 / score ≈ 79.236 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

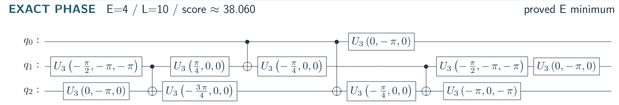
cx_rz_ry

EXACT PHASE E=4 / L=11 / score ≈ 40.022 — proved E minimum



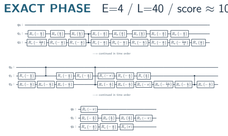
cx_u

EXACT PHASE E=4 / L=10 / score ≈ 38.060 — proved E minimum



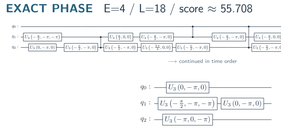
cz_rz_rx

EXACT PHASE E=4 / L=40 / score ≈ 103.880 — proved E minimum



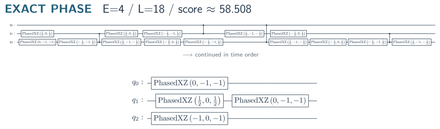
cz_u

EXACT PHASE E=4 / L=18 / score ≈ 55.708 — proved E minimum



google_cz

EXACT PHASE E=4 / L=18 / score ≈ 58.508 — proved E minimum



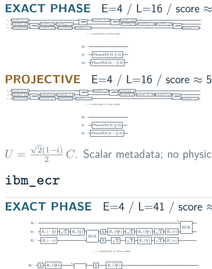
google_fsim

EXACT PHASE E=3 / L=12 / score ≈ 41.684 — E upper bound

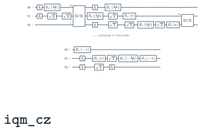


google_sqrt_iswap

EXACT PHASE E=4 / L=16 / score ≈ 53.896 — E upper bound



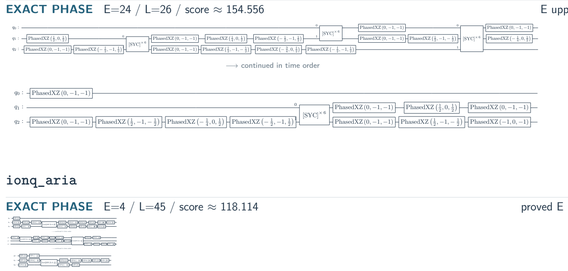
PROJECTIVE E=4 / L=16 / score ≈ 53.296 — E upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.

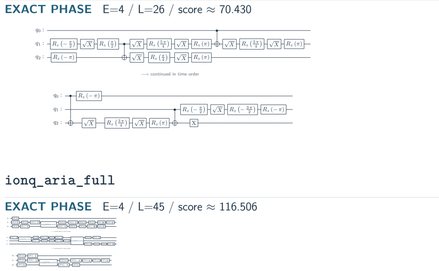
google_syc

EXACT PHASE E=24 / L=26 / score ≈ 154.556 — E upper bound



ibm_cx

EXACT PHASE E=4 / L=26 / score ≈ 70.430 — proved E minimum



ibm_cz

EXACT PHASE E=4 / L=48 / score ≈ 116.850 — proved E minimum



PROJECTIVE E=4 / L=48 / score ≈ 116.446 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

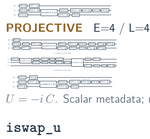
ibm_ecr

EXACT PHASE E=4 / L=41 / score ≈ 100.836 — proved E minimum

ionq_aria

EXACT PHASE E=4 / L=45 / score ≈ 118.114 — proved E minimum

PROJECTIVE E=4 / L=45 / score ≈ 117.714 — proved E minimum

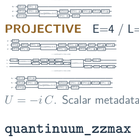


$U = -iC$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=4 / L=45 / score ≈ 116.506 — proved E minimum

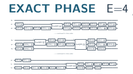
PROJECTIVE E=4 / L=45 / score ≈ 116.106 — proved E minimum



$U = -iC$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=4 / L=49 / score ≈ 127.706 — proved E minimum

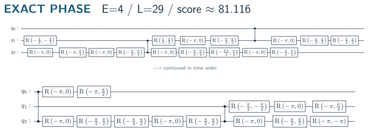


PROJECTIVE E=4 / L=49 / score ≈ 127.306 — proved E minimum

$U = -iC$. Scalar metadata; no physical GP.

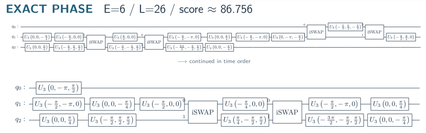
iqm_cz

EXACT PHASE E=4 / L=29 / score ≈ 81.116 — proved E minimum



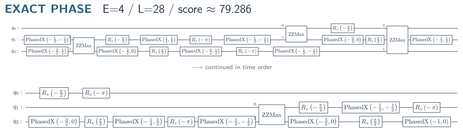
iswap_u

EXACT PHASE E=6 / L=26 / score ≈ 86.756 — E upper bound



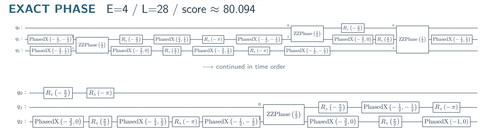
quantinuum_zzmax

EXACT PHASE E=4 / L=28 / score ≈ 79.286 — proved E minimum



quantinuum_zzphase

EXACT PHASE E=4 / L=28 / score ≈ 80.094 — proved E minimum



rigetti_cz

EXACT PHASE E=4 / L=48 / score ≈ 116.850 — proved E minimum



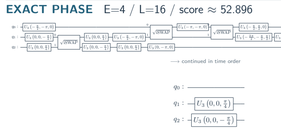
PROJECTIVE E=4 / L=48 / score ≈ 116.446 — proved E minimum



$U = iC$. Scalar metadata; no physical GP.

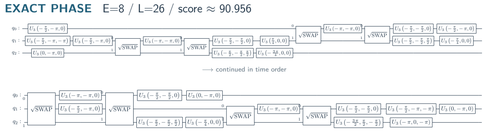
sqrt_iswap_u

EXACT PHASE E=4 / L=16 / score ≈ 52.896 — E upper bound



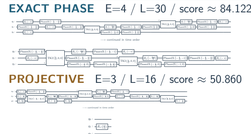
sqrt_swap_u

EXACT PHASE E=8 / L=26 / score ≈ 90.956 — E upper bound

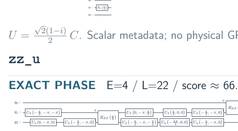


tket_tk2

EXACT PHASE E=4 / L=30 / score ≈ 84.122 — E upper bound



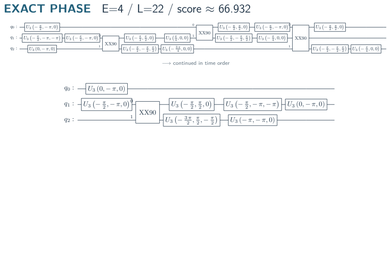
PROJECTIVE E=3 / L=16 / score ≈ 50.860 — E upper bound



$U = \frac{\sqrt{2}(1-i)}{2} C$. Scalar metadata; no physical GP.

xx90_u

EXACT PHASE E=4 / L=22 / score ≈ 66.932 — proved E minimum



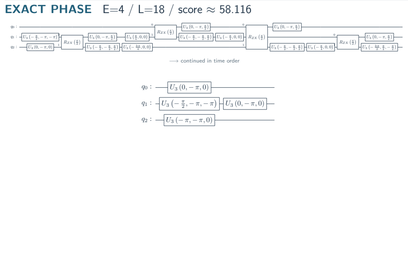
xx_rz_ry

EXACT PHASE E=4 / L=39 / score ≈ 101.686 — proved E minimum



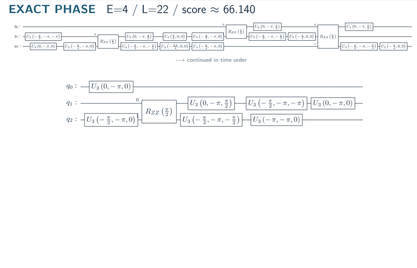
zx_u

EXACT PHASE E=4 / L=18 / score ≈ 58.116 — proved E minimum



zz_u

EXACT PHASE E=4 / L=22 / score ≈ 66.140 — proved E minimum

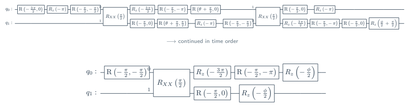


fSim (θ, ϕ)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

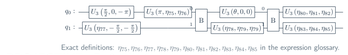
PROJECTIVE E=3 / L=22 / score ≈ 61.276 — minimum uniform E budget



$U = \eta_{74}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget



Exact definitions: $\eta_{75}, \eta_{76}, \eta_{77}, \eta_{78}, \eta_{79}, \eta_{80}, \eta_{81}, \eta_{82}, \eta_{83}, \eta_{84}, \eta_{85}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.436 — minimum uniform E budget



Exact definitions: $\eta_{77}, \eta_{78}, \eta_{79}, \eta_{80}, \eta_{81}, \eta_{82}, \eta_{83}, \eta_{84}, \eta_{85}, \eta_{86}$ in the expression glossary.

$U = \eta_{87}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

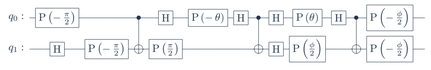
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

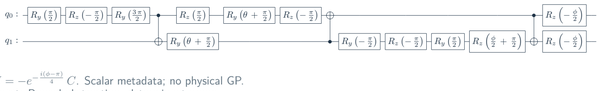
cx_h_phase

EXACT PHASE E=3 / L=14 / score ≈ 41.216 — minimum uniform E budget



cx_rz_ry

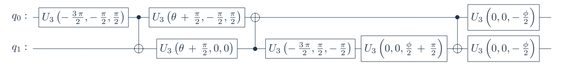
PROJECTIVE E=3 / L=13 / score ≈ 40.626 — minimum uniform E budget



$U = -e^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

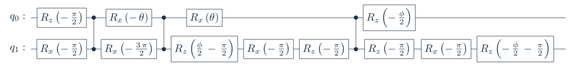
cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.642 — minimum uniform E budget



cz_rz_rx

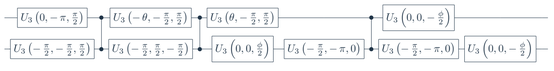
PROJECTIVE E=3 / L=12 / score ≈ 38.024 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

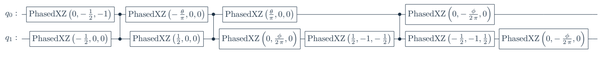
cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget



google_cz

EXACT PHASE E=3 / L=10 / score ≈ 35.711 — minimum uniform E budget



google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.004 — minimum uniform E budget



google_sqrt_iswap

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 3.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget

Derived angles: exact shared-expression program 3.

$U = e^{i\xi_{3,589}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=21 / score ≈ 120.438 — E upper bound



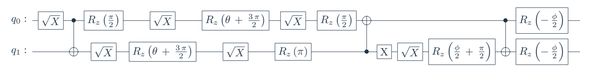
PROJECTIVE E=18 / L=19 / score ≈ 115.765 — E upper bound



$U = -ie^{-i\theta}\, C$. Scalar metadata; no physical GP.

ibm_cx

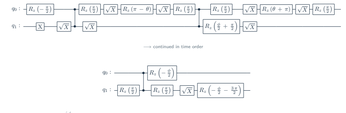
PROJECTIVE E=3 / L=15 / score ≈ 43.416 — minimum uniform E budget



$U = -e^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

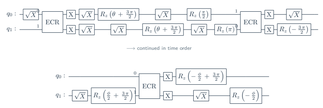
PROJECTIVE E=3 / L=21 / score ≈ 56.024 — minimum uniform E budget



$U = \sqrt[4]{-1}\, e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

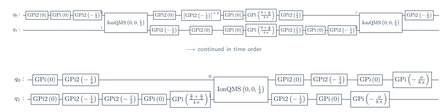
PROJECTIVE E=3 / L=23 / score ≈ 59.416 — minimum uniform E budget



$U = -ie^{-\frac{i(\phi-\pi)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

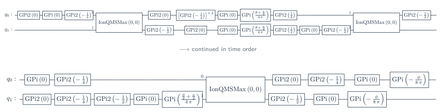
PROJECTIVE E=3 / L=31 / score ≈ 84.131 — minimum uniform E budget



$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

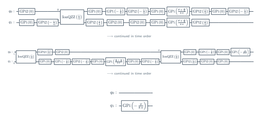
PROJECTIVE E=3 / L=31 / score ≈ 82.925 — minimum uniform E budget



$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

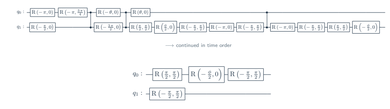
PROJECTIVE E=3 / L=34 / score ≈ 91.325 — minimum uniform E budget



$U = \eta_{88}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

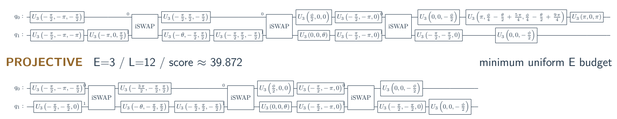
PROJECTIVE E=3 / L=19 / score ≈ 54.676 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=15 / score ≈ 46.690 — minimum uniform E budget



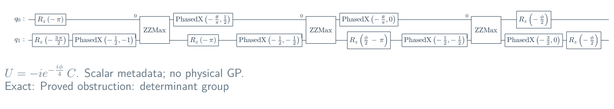
PROJECTIVE E=3 / L=12 / score ≈ 39.872 — minimum uniform E budget



$U = \eta_{89}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

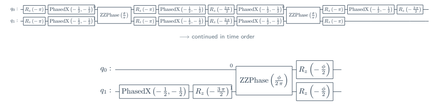
PROJECTIVE E=3 / L=12 / score ≈ 38.897 — minimum uniform E budget



$U = -ie^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

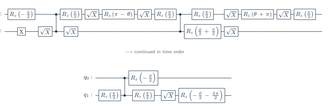
PROJECTIVE E=3 / L=20 / score ≈ 57.653 — minimum uniform E budget



$U = e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

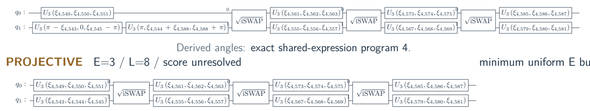
PROJECTIVE E=3 / L=21 / score ≈ 56.024 — minimum uniform E budget



$U = \sqrt[4]{-1}\, e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

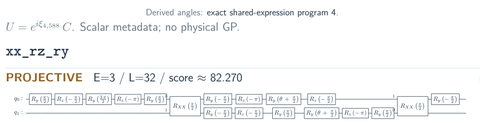
sqrt_iswap_u

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 4.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget

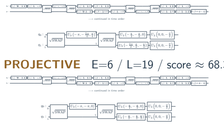


Derived angles: exact shared-expression program 4.

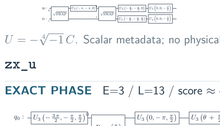
$U = e^{i\xi_{4,588}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 69.114 — E upper bound



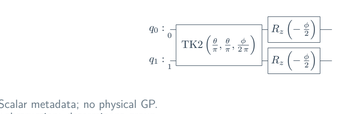
PROJECTIVE E=6 / L=19 / score ≈ 68.314 — E upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

tket_tk2

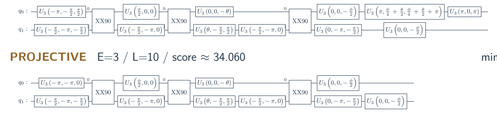
PROJECTIVE E=1 / L=2 / score ≈ 9.601 — minimum uniform E budget



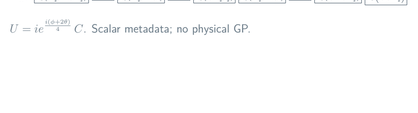
$U = e^{-\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=12 / score ≈ 39.272 — minimum uniform E budget



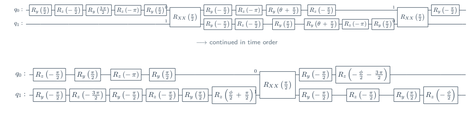
PROJECTIVE E=3 / L=10 / score ≈ 34.060 — minimum uniform E budget



$U = ie^{\frac{i(\phi+2\theta)}{4}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

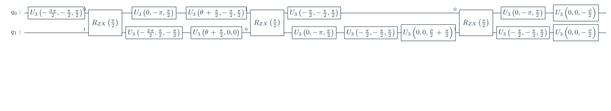
PROJECTIVE E=3 / L=32 / score ≈ 82.270 — minimum uniform E budget



$U = \eta_{74}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

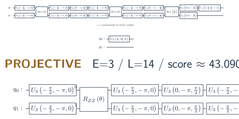
zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.684 — minimum uniform E budget

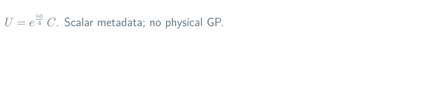


zz_u

EXACT PHASE E=3 / L=16 / score ≈ 47.902 — minimum uniform E budget



PROJECTIVE E=3 / L=14 / score ≈ 43.090 — minimum uniform E budget



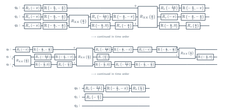
$U = e^{\frac{i\phi}{4}}\, C$. Scalar metadata; no physical GP.

Peres

3 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

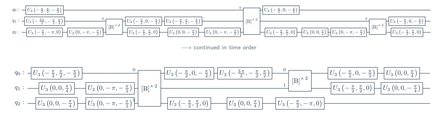
PROJECTIVE E=5 / L=35 / score ≈ 99.714 — E upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=10 / L=29 / score ≈ 109.374 — E upper bound

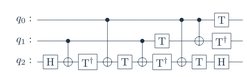


clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

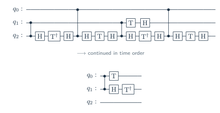
clifford_t_cx

EXACT PHASE E=5 / L=9 / score ≈ 38.000 — proved E minimum



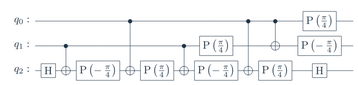
clifford_t_cz

EXACT PHASE E=5 / L=17 / score ≈ 54.000 — proved E minimum



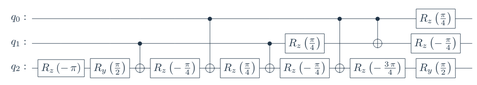
cx_h_phase

EXACT PHASE E=5 / L=9 / score ≈ 40.814 — proved E minimum



cx_rz_ry

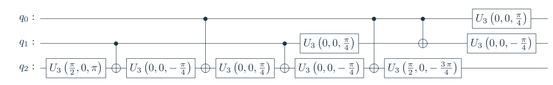
PROJECTIVE E=5 / L=10 / score ≈ 43.220 — proved E minimum



$U = -\sqrt[8]{-1}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=5 / L=8 / score ≈ 39.248 — proved E minimum



cz_rz_rx

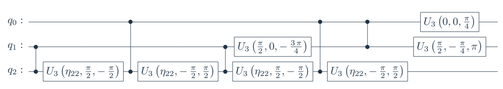
PROJECTIVE E=5 / L=12 / score ≈ 47.224 — proved E minimum



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

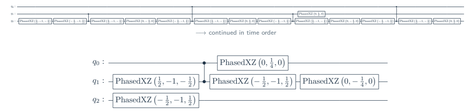
EXACT PHASE E=5 / L=7 / score ≈ 37.242 — proved E minimum



Exact definitions: η_{22} in the expression glossary.

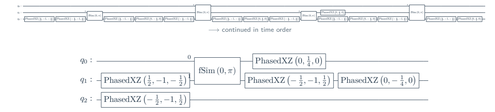
google_cz

EXACT PHASE E=5 / L=19 / score ≈ 65.714 — proved E minimum



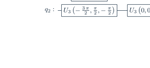
google_fsim

EXACT PHASE E=5 / L=19 / score ≈ 65.734 — E upper bound



google_sqrt_iswap

EXACT PHASE E=10 / L=29 / score ≈ 108.374 — E upper bound



PROJECTIVE E=10 / L=29 / score ≈ 107.774 — E upper bound

$U = \frac{\sqrt{2}(1-i)^5}{8}\,C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=30 / L=29 / score ≈ 185.774 — E upper bound



ibm_cx

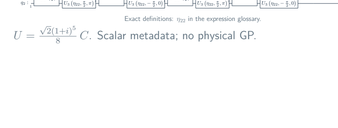
PROJECTIVE E=5 / L=12 / score ≈ 47.218 — proved E minimum



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

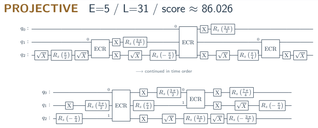
PROJECTIVE E=5 / L=26 / score ≈ 75.226 — proved E minimum



$U = e^{-\frac{3i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

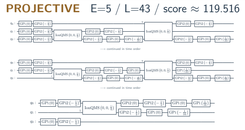
PROJECTIVE E=5 / L=31 / score ≈ 86.026 — proved E minimum



$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

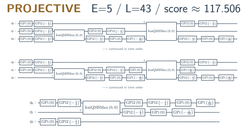
PROJECTIVE E=5 / L=43 / score ≈ 119.516 — E upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=5 / L=43 / score ≈ 117.506 — E upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

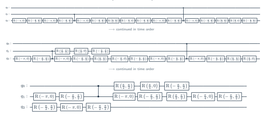
PROJECTIVE E=5 / L=48 / score ≈ 131.506 — E upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=5 / L=45 / score ≈ 123.380 — proved E minimum



$U = e^{\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=10 / L=18 / score ≈ 84.908 — E upper bound



Exact definitions: η_{22} in the expression glossary.

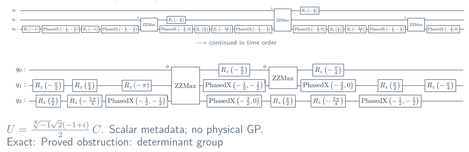
PROJECTIVE E=10 / L=16 / score ≈ 79.696 — E upper bound



$U = e^{-\frac{i\pi}{8}}\,C$. Scalar metadata; no physical GP.

quantinuum_zzmax

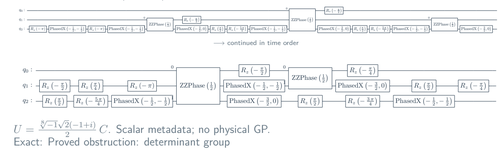
PROJECTIVE E=5 / L=31 / score ≈ 90.486 — proved E minimum



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

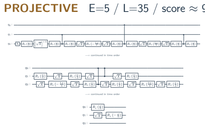
PROJECTIVE E=5 / L=31 / score ≈ 91.496 — E upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=5 / L=35 / score ≈ 95.238 — proved E minimum



$U = -(-1)^{\frac{7}{8}}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

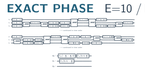
sqrt_iswap_u

EXACT PHASE E=10 / L=29 / score ≈ 107.374 — E upper bound



sqrt_swap_u

EXACT PHASE E=10 / L=29 / score ≈ 108.174 — E upper bound



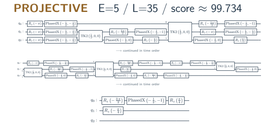
PROJECTIVE E=10 / L=29 / score ≈ 107.374 — E upper bound



$U = e^{-\frac{i\pi}{4}}\,C$. Scalar metadata; no physical GP.

tket_tk2

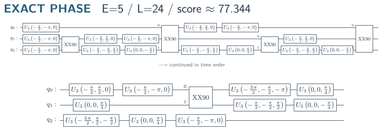
PROJECTIVE E=5 / L=35 / score ≈ 99.734 — E upper bound



$U = \frac{\sqrt[8]{-1}\sqrt{2}(-1+i)}{2}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=5 / L=24 / score ≈ 77.344 — proved E minimum



xx_rz_ry

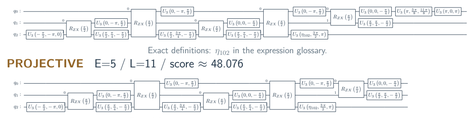
PROJECTIVE E=5 / L=35 / score ≈ 94.080 — E upper bound



$U = \eta_{101}\,C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=5 / L=13 / score ≈ 53.288 — E upper bound



Exact definitions: η_{102} in the expression glossary.

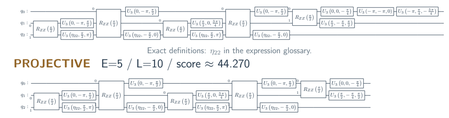
PROJECTIVE E=5 / L=11 / score ≈ 48.076 — E upper bound

Exact definitions: η_{102} in the expression glossary.

$U = (-1)^{\frac{3}{8}}\,C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=5 / L=12 / score ≈ 49.082 — E upper bound



Exact definitions: η_{22} in the expression glossary.

PROJECTIVE E=5 / L=10 / score ≈ 44.270 — E upper bound

Exact definitions: η_{22} in the expression glossary.

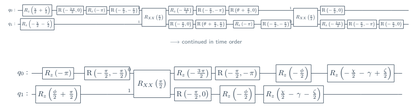
$U = \frac{\sqrt{2}(1+i)^5}{8}\,C$. Scalar metadata; no physical GP.

PhasedFSim $(\theta, \zeta, \chi, \gamma, \phi)$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

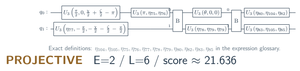
PROJECTIVE E=3 / L=26 / score ≈ 70.084 — minimum uniform E budget



$U = \eta_{108} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

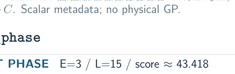
berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.642 — minimum uniform E budget



Exact definitions: $\eta_{104}, \eta_{105}, \eta_{71}, \eta_{76}, \eta_{77}, \eta_{78}, \eta_{79}, \eta_{80}, \eta_{92}, \eta_{93}, \eta_{95}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.636 — minimum uniform E budget



Exact definitions: $\eta_{104}, \eta_{105}, \eta_{77}, \eta_{78}, \eta_{79}, \eta_{80}, \eta_{92}, \eta_{93}, \eta_{95}, \eta_{96}$ in the expression glossary.

$U = \eta_{87} C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

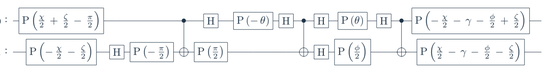
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

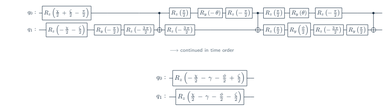
cx_h_phase

EXACT PHASE E=3 / L=15 / score ≈ 43.418 — minimum uniform E budget



cx_rz_ry

PROJECTIVE E=3 / L=17 / score ≈ 49.034 — minimum uniform E budget



$U = \eta_{106} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=11 / score ≈ 37.466 — minimum uniform E budget



PROJECTIVE E=3 / L=11 / score ≈ 37.266 — minimum uniform E budget



$U = e^{\frac{i\phi}{2}} C$. Scalar metadata; no physical GP.

cz_rz_rx

PROJECTIVE E=3 / L=13 / score ≈ 40.226 — minimum uniform E budget

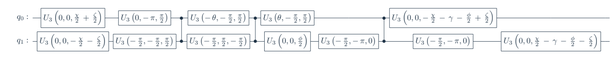


Exact definitions: η_{107} in the expression glossary.

$U = \eta_{106} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

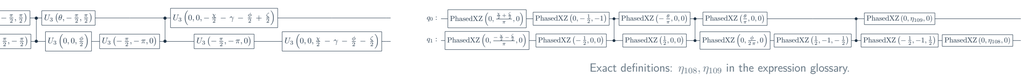
cz_u

EXACT PHASE E=3 / L=12 / score ≈ 39.672 — minimum uniform E budget



google_cz

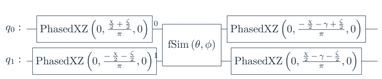
EXACT PHASE E=3 / L=12 / score ≈ 40.784 — minimum uniform E budget



Exact definitions: η_{108}, η_{109} in the expression glossary.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 14.149 — minimum uniform E budget



google_sqrt_iswap

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 5.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget

Derived angles: exact shared-expression program 5.

$U = e^{i\xi_{5,597}} C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=25 / score ≈ 130.583 — E upper bound



PROJECTIVE E=18 / L=23 / score ≈ 125.911 — E upper bound

$U = -ie^{-i\theta} C$. Scalar metadata; no physical GP.

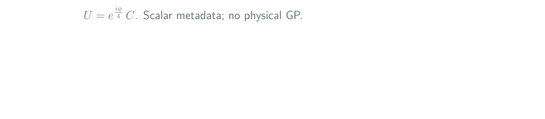
ibm_cx

PROJECTIVE E=3 / L=19 / score ≈ 52.224 — minimum uniform E budget

$U = -e^{-\frac{i(\phi_3+\phi-\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — minimum uniform E budget



$U = \sqrt[4]{-1} e^{-\frac{i(\phi_3+\phi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=27 / score ≈ 68.224 — minimum uniform E budget

$U = -ie^{-\frac{i(4\gamma+\phi-\pi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=39 / score ≈ 103.869 — minimum uniform E budget

$U = \eta_{111} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=39 / score ≈ 102.663 — minimum uniform E budget

$U = \eta_{111} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=42 / score ≈ 111.063 — minimum uniform E budget

$U = \eta_{111} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=25 / score ≈ 68.700 — minimum uniform E budget

$U = \eta_{106} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=17 / score ≈ 51.102 — minimum uniform E budget

PROJECTIVE E=3 / L=14 / score ≈ 44.284 — minimum uniform E budget

$U = \eta_{89} C$. Scalar metadata; no physical GP.

quantinuum_zzmax

PROJECTIVE E=3 / L=13 / score ≈ 41.299 — minimum uniform E budget

$U = -ie^{-\frac{i(4\gamma+\phi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

PROJECTIVE E=3 / L=22 / score ≈ 62.057 — minimum uniform E budget

$U = e^{-\frac{i(4\gamma+\phi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

PROJECTIVE E=3 / L=22 / score ≈ 58.226 — minimum uniform E budget

Exact definitions: η_{110} in the expression glossary.

$U = \sqrt[4]{-1} e^{-\frac{i(4\gamma+\phi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget

Derived angles: exact shared-expression program 6.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget

Derived angles: exact shared-expression program 6.

$U = e^{i\xi_{6,596}} C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=23 / score ≈ 77.938 — E upper bound

PROJECTIVE E=6 / L=23 / score ≈ 77.138 — E upper bound

$U = -\sqrt[4]{-1} C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=4 / score ≈ 14.005 — minimum uniform E budget

$U = e^{-\frac{i(4\gamma+\phi)}{4}} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=14 / score ≈ 43.684 — minimum uniform E budget

PROJECTIVE E=3 / L=12 / score ≈ 38.472 — minimum uniform E budget

$U = ie^{\frac{i(\phi+2\theta)}{4}} C$. Scalar metadata; no physical GP.

xx_rz_ry

PROJECTIVE E=3 / L=36 / score ≈ 91.078 — minimum uniform E budget

$U = \eta_{103} C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=17 / score ≈ 52.508 — minimum uniform E budget

zz_u

EXACT PHASE E=3 / L=18 / score ≈ 52.314 — minimum uniform E budget

PROJECTIVE E=3 / L=16 / score ≈ 47.502 — minimum uniform E budget

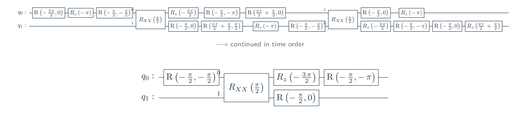
$U = e^{\frac{i\phi}{4}} C$. Scalar metadata; no physical GP.

SWAPPow $(t, global_{shift})$

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

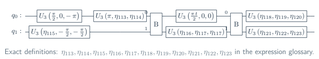
PROJECTIVE E=3 / L=20 / score ≈ 56.872 — minimum uniform E budget



$U = \eta_{112}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.442 — minimum uniform E budget



Exact definitions: $\eta_{113}, \eta_{114}, \eta_{115}, \eta_{116}, \eta_{117}, \eta_{118}, \eta_{119}, \eta_{120}, \eta_{121}, \eta_{122}, \eta_{123}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.436 — minimum uniform E budget



Exact definitions: $\eta_{115}, \eta_{116}, \eta_{117}, \eta_{118}, \eta_{119}, \eta_{120}, \eta_{121}, \eta_{122}, \eta_{123}, \eta_{124}$ in the expression glossary.

$U = \eta_{125}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 53.220 — minimum uniform E budget

PROJECTIVE E=3 / L=12 / score ≈ 37.212 — minimum uniform E budget



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget



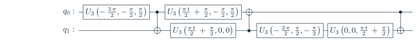
$U = \eta_{126}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — minimum uniform E budget



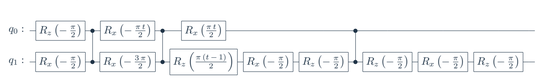
PROJECTIVE E=3 / L=5 / score ≈ 24.230 — minimum uniform E budget



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

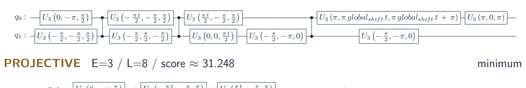
PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget



$U = \eta_{127}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget



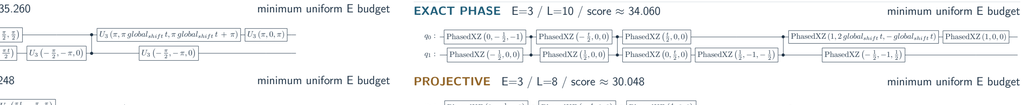
PROJECTIVE E=3 / L=8 / score ≈ 31.248 — minimum uniform E budget



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 34.060 — minimum uniform E budget



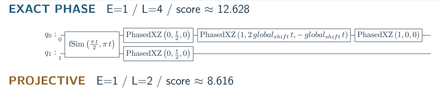
PROJECTIVE E=3 / L=8 / score ≈ 30.048 — minimum uniform E budget



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=1 / L=4 / score ≈ 12.628 — minimum uniform E budget



PROJECTIVE E=1 / L=2 / score ≈ 8.616 — minimum uniform E budget



$U = e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

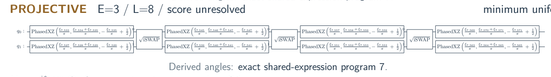
google_sqrt_iswap

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 7.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 7.

$U = e^{i\xi_{7,573}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=19 / score ≈ 113.914 — E upper bound



PROJECTIVE E=18 / L=17 / score ≈ 109.702 — E upper bound



$U = \eta_{129}\, C$. Scalar metadata; no physical GP.

ibm_cx

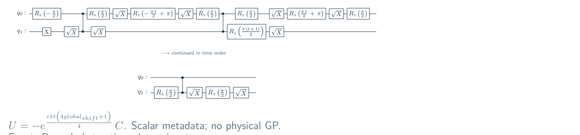
PROJECTIVE E=3 / L=13 / score ≈ 39.012 — minimum uniform E budget



$U = \eta_{126}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

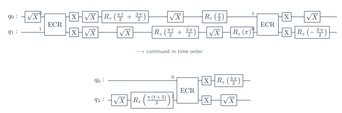
PROJECTIVE E=3 / L=20 / score ≈ 54.020 — minimum uniform E budget



$U = -e^{\frac{i\pi\left(4 g l o b a l_{shift}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

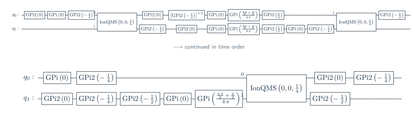
PROJECTIVE E=3 / L=22 / score ≈ 57.214 — minimum uniform E budget



$U = \eta_{130}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

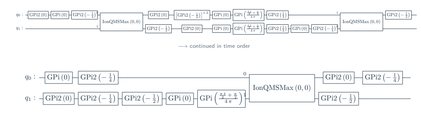
PROJECTIVE E=3 / L=27 / score ≈ 73.272 — minimum uniform E budget



$U = \eta_{131}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

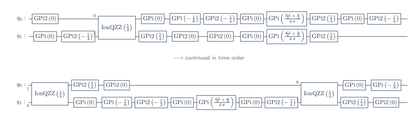
PROJECTIVE E=3 / L=27 / score ≈ 72.066 — minimum uniform E budget



$U = \eta_{131}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

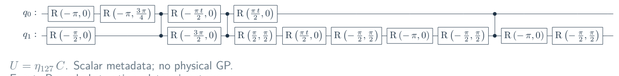
PROJECTIVE E=3 / L=30 / score ≈ 80.466 — minimum uniform E budget



$U = \eta_{131}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

PROJECTIVE E=3 / L=13 / score ≈ 41.052 — minimum uniform E budget



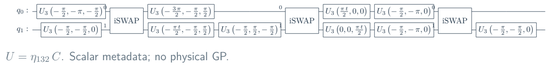
$U = \eta_{127}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=13 / score ≈ 41.878 — minimum uniform E budget



PROJECTIVE E=3 / L=10 / score ≈ 35.860 — minimum uniform E budget



$U = \eta_{132}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

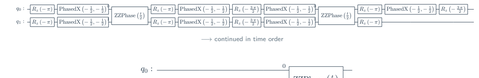
PROJECTIVE E=3 / L=10 / score ≈ 34.232 — minimum uniform E budget



$U = -ie^{\frac{i\pi\left(4 g l o b a l_{shift}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

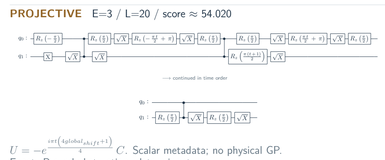
PROJECTIVE E=3 / L=18 / score ≈ 52.658 — minimum uniform E budget



$U = e^{\frac{i\pi\left(4 g l o b a l_{shift}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

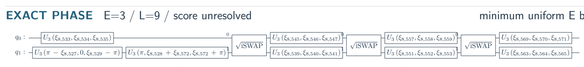
PROJECTIVE E=3 / L=20 / score ≈ 54.020 — minimum uniform E budget



$U = -e^{\frac{i\pi\left(4 g l o b a l_{shift}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 8.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget



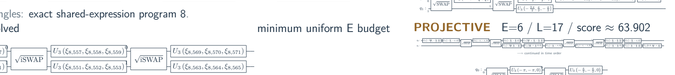
Derived angles: exact shared-expression program 8.

$U = e^{i\xi_{8,572}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 68.714 — E upper bound

PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound



$U = -\sqrt[4]{-1}e^{i\pi g l o b a l_{shift}\, t}\, C$. Scalar metadata; no physical GP.

tket_tk2

PROJECTIVE E=1 / L=0 / score ≈ 4.606 — minimum uniform E budget



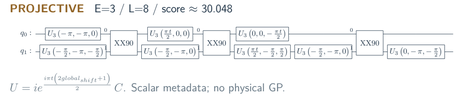
$U = e^{\frac{i\pi\left(4 g l o b a l_{shift}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 34.860 — minimum uniform E budget



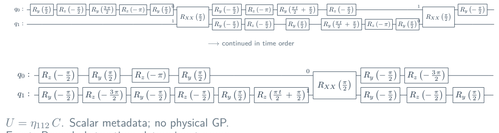
PROJECTIVE E=3 / L=8 / score ≈ 30.048 — minimum uniform E budget



$U = ie^{\frac{i\pi\left(2 g l o b a l_{shift}+1\right)}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

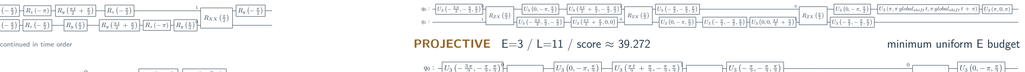
PROJECTIVE E=3 / L=31 / score ≈ 80.068 — minimum uniform E budget



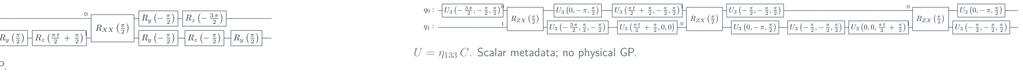
$U = \eta_{112}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.284 — minimum uniform E budget



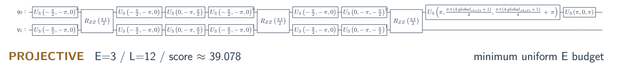
PROJECTIVE E=3 / L=11 / score ≈ 39.272 — minimum uniform E budget



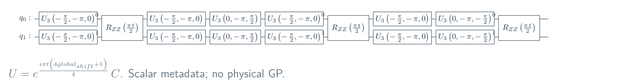
$U = \eta_{133}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=14 / score ≈ 43.890 — minimum uniform E budget



PROJECTIVE E=3 / L=12 / score ≈ 39.078 — minimum uniform E budget



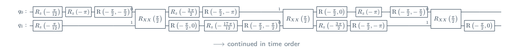
$U = e^{\frac{i\pi\left(4 g l o b a l_{shift}+1\right)}{4}}\, C$. Scalar metadata; no physical GP.

SYC

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

PROJECTIVE E=3 / L=17 / score ≈ 51.809 — proved E minimum



$U = \eta_{134}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

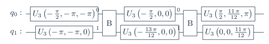
berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 24.076 — proved E minimum



Exact definitions: $\eta_{130}, \eta_{136}, \eta_{137}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 22.787 — proved E minimum



$U = e^{\frac{i\pi}{24}}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: not a clifford

clifford_t_cx

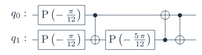
Exact: Proved obstruction: determinant group
Projective: Proved obstruction: outside clifford t number field

clifford_t_cz

Exact: Proved obstruction: determinant group
Projective: Proved obstruction: outside clifford t number field

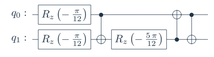
cx_h_phase

EXACT PHASE E=3 / L=3 / score ≈ 20.157 — proved E minimum



cx_rz_ry

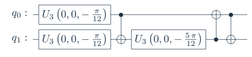
PROJECTIVE E=3 / L=3 / score ≈ 20.157 — proved E minimum



$U = e^{-\frac{7i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

cx_u

EXACT PHASE E=3 / L=3 / score ≈ 20.169 — proved E minimum



cz_rz_rx

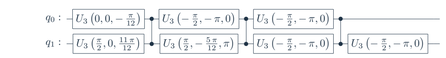
PROJECTIVE E=3 / L=12 / score ≈ 39.975 — proved E minimum



$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

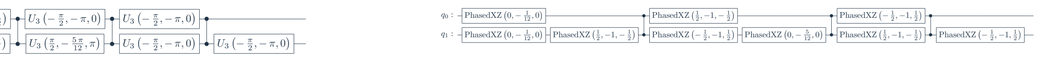
cz_u

EXACT PHASE E=3 / L=7 / score ≈ 29.393 — proved E minimum



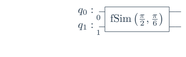
google_cz

EXACT PHASE E=3 / L=9 / score ≈ 34.605 — proved E minimum



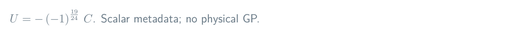
google_fsim

EXACT PHASE E=1 / L=0 / score ≈ 4.721 — proved E minimum



google_sqrt_iswap

EXACT PHASE E=3 / L=9 / score ≈ 39.025 — proved E minimum



Exact definitions: $\eta_{138}, \eta_{139}, \eta_{140}, \eta_{141}, \eta_{142}, \eta_{143}, \eta_{144}, \eta_{145}, \eta_{146}, \eta_{147}$ in the expression glossary.

PROJECTIVE E=3 / L=8 / score ≈ 34.924 — proved E minimum

Exact definitions: $\eta_{140}, \eta_{141}, \eta_{142}, \eta_{143}, \eta_{144}, \eta_{145}, \eta_{146}, \eta_{147}$ in the expression glossary.
$U = \eta_{148}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

google_syc

EXACT PHASE E=1 / L=0 / score ≈ 4.000 — proved E minimum

ibm_cx

PROJECTIVE E=3 / L=3 / score ≈ 20.157 — proved E minimum

$U = e^{-\frac{7i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_cz

PROJECTIVE E=3 / L=14 / score ≈ 42.763 — proved E minimum



$U = e^{-\frac{19i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ibm_ecr

PROJECTIVE E=3 / L=15 / score ≈ 44.763 — proved E minimum



$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria

PROJECTIVE E=3 / L=25 / score ≈ 70.219 — proved E minimum



$U = \eta_{134}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_aria_full

PROJECTIVE E=3 / L=21 / score ≈ 60.405 — proved E minimum



$U = \eta_{149}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

ionq_forte

PROJECTIVE E=3 / L=28 / score ≈ 77.413 — proved E minimum



$U = \eta_{134}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iqm_cz

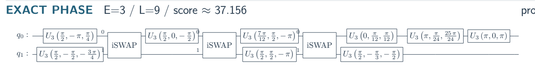
PROJECTIVE E=3 / L=19 / score ≈ 56.627 — proved E minimum



$U = (-1)^{\frac{17}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

iswap_u

EXACT PHASE E=3 / L=9 / score ≈ 37.156 — proved E minimum



PROJECTIVE E=3 / L=7 / score ≈ 31.310 — proved E minimum



$U = e^{\frac{i\pi}{24}}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

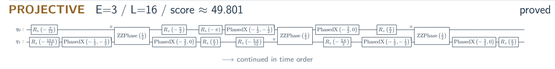
PROJECTIVE E=3 / L=12 / score ≈ 40.187 — proved E minimum



$U = -(-1)^{\frac{13}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

quantinuum_zzphase

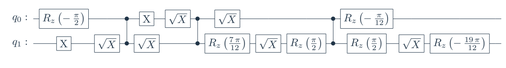
PROJECTIVE E=3 / L=16 / score ≈ 49.801 — proved E minimum



$U = \eta_{134}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

rigetti_cz

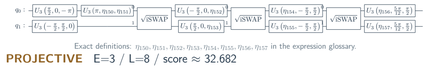
PROJECTIVE E=3 / L=14 / score ≈ 42.763 — proved E minimum



$U = e^{-\frac{19i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

sqrt_iswap_u

EXACT PHASE E=3 / L=9 / score ≈ 36.322 — proved E minimum



Exact definitions: $\eta_{150}, \eta_{151}, \eta_{152}, \eta_{153}, \eta_{154}, \eta_{155}, \eta_{156}, \eta_{157}$ in the expression glossary.

PROJECTIVE E=3 / L=8 / score ≈ 32.682 — proved E minimum



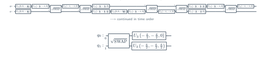
Exact definitions: $\eta_{150}, \eta_{151}, \eta_{154}, \eta_{155}, \eta_{156}, \eta_{157}$ in the expression glossary.
$U = \eta_{158}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=15 / score ≈ 60.641 — E upper bound



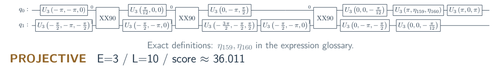
PROJECTIVE E=6 / L=15 / score ≈ 59.841 — E upper bound



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

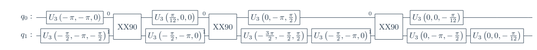
xx90_u

EXACT PHASE E=3 / L=12 / score ≈ 40.023 — proved E minimum



Exact definitions: η_{159}, η_{160} in the expression glossary.

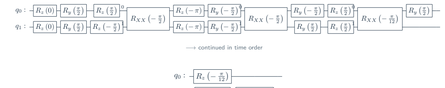
PROJECTIVE E=3 / L=10 / score ≈ 36.011 — proved E minimum



$U = -(-1)^{\frac{19}{24}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

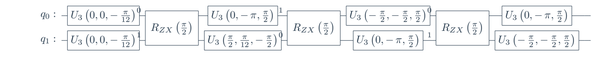
PROJECTIVE E=3 / L=18 / score ≈ 52.993 — proved E minimum



$U = e^{\frac{11i\pi}{24}}\, C$. Scalar metadata; no physical GP.
Exact: Proved obstruction: determinant group

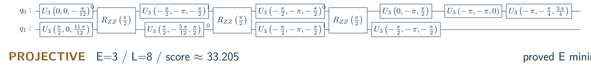
zx_u

EXACT PHASE E=3 / L=8 / score ≈ 33.005 — proved E minimum



zz_u

EXACT PHASE E=3 / L=10 / score ≈ 38.017 — proved E minimum



PROJECTIVE E=3 / L=8 / score ≈ 33.205 — proved E minimum



$U = \frac{\sqrt{2}(1+i)^3}{4}\, C$. Scalar metadata; no physical GP.

TK2 (a, b, c)

2 qubits / 32 preserved native bases / alphabetical panel order

aqt_rxx

EXACT PHASE E=3 / L=41 / score ≈ 102.928 — minimum uniform E budget



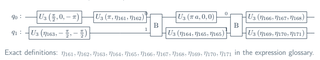
PROJECTIVE E=3 / L=20 / score ≈ 56.872 — minimum uniform E budget



$U = \eta_{36}\, C$. Scalar metadata; no physical GP.

berkeley_b

EXACT PHASE E=2 / L=7 / score ≈ 23.242 — minimum uniform E budget



Exact definitions: $\eta_{161}, \eta_{162}, \eta_{163}, \eta_{164}, \eta_{165}, \eta_{166}, \eta_{167}, \eta_{168}, \eta_{169}, \eta_{170}, \eta_{171}$ in the expression glossary.

PROJECTIVE E=2 / L=6 / score ≈ 21.236 — minimum uniform E budget



Exact definitions: $\eta_{161}, \eta_{164}, \eta_{165}, \eta_{166}, \eta_{167}, \eta_{168}, \eta_{169}, \eta_{170}, \eta_{171}, \eta_{172}$ in the expression glossary.

$U = \eta_{173}\, C$. Scalar metadata; no physical GP.

clifford_only

Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cx

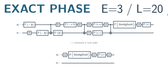
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

clifford_t_cz

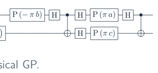
Exact: Proved obstruction: discrete alphabet nonconstant family
Projective: Proved obstruction: discrete alphabet nonconstant family

cx_h_phase

EXACT PHASE E=3 / L=20 / score ≈ 53.020 — minimum uniform E budget



PROJECTIVE E=3 / L=12 / score ≈ 36.612 — minimum uniform E budget



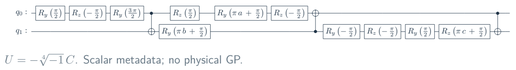
$U = e^{-\frac{i\pi(a-b+c)}{2}}\, C$. Scalar metadata; no physical GP.

cx_rz_ry

EXACT PHASE E=4 / L=17 / score ≈ 52.834 — minimum uniform E budget



PROJECTIVE E=3 / L=11 / score ≈ 36.222 — minimum uniform E budget



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

cx_u

EXACT PHASE E=3 / L=7 / score ≈ 28.242 — minimum uniform E budget



PROJECTIVE E=3 / L=5 / score ≈ 24.230 — minimum uniform E budget



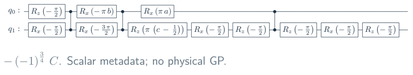
$U = e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

cz_rz_rx

EXACT PHASE E=4 / L=28 / score ≈ 77.256 — minimum uniform E budget



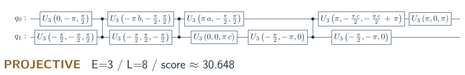
PROJECTIVE E=3 / L=11 / score ≈ 35.822 — minimum uniform E budget



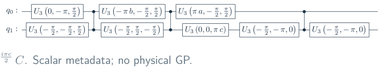
$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

cz_u

EXACT PHASE E=3 / L=10 / score ≈ 35.060 — minimum uniform E budget



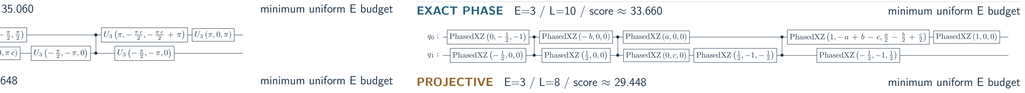
PROJECTIVE E=3 / L=8 / score ≈ 30.648 — minimum uniform E budget



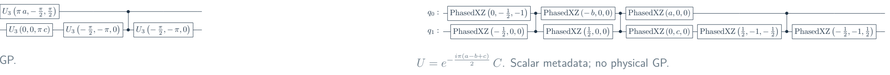
$U = e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

google_cz

EXACT PHASE E=3 / L=10 / score ≈ 33.660 — minimum uniform E budget



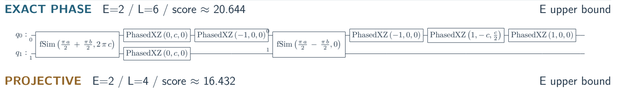
PROJECTIVE E=3 / L=8 / score ≈ 29.448 — minimum uniform E budget



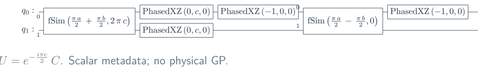
$U = e^{\frac{i\pi(a-b+c)}{2}}\, C$. Scalar metadata; no physical GP.

google_fsim

EXACT PHASE E=2 / L=6 / score ≈ 20.644 — E upper bound



PROJECTIVE E=2 / L=4 / score ≈ 16.432 — E upper bound



$U = e^{-\frac{i\phi_{ZZ}}{2}}\, C$. Scalar metadata; no physical GP.

google_sqrt_iswap

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 9.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 9.

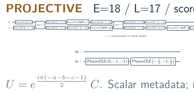
$U = e^{i\xi_{0,585}}\, C$. Scalar metadata; no physical GP.

google_syc

EXACT PHASE E=18 / L=19 / score ≈ 113.914 — E upper bound



PROJECTIVE E=18 / L=17 / score ≈ 109.702 — E upper bound



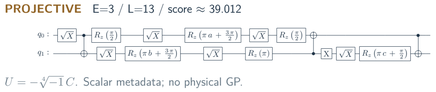
$U = e^{\frac{i\pi(-a-b-c-1)}{2}}\, C$. Scalar metadata; no physical GP.

ibm_cx

EXACT PHASE E=3 / L=16 / score ≈ 45.216 — minimum uniform E budget



PROJECTIVE E=3 / L=13 / score ≈ 39.012 — minimum uniform E budget



$U = -\sqrt[4]{-1}\, C$. Scalar metadata; no physical GP.

ibm_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget



ibm_ecr

EXACT PHASE E=3 / L=24 / score ≈ 61.018 — minimum uniform E budget



PROJECTIVE E=3 / L=22 / score ≈ 57.214 — minimum uniform E budget



$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

ionq_aria

EXACT PHASE E=3 / L=27 / score ≈ 73.472 — minimum uniform E budget



PROJECTIVE E=3 / L=27 / score ≈ 73.272 — minimum uniform E budget



$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_aria_full

EXACT PHASE E=3 / L=27 / score ≈ 72.266 — minimum uniform E budget



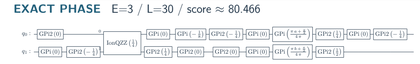
PROJECTIVE E=3 / L=27 / score ≈ 72.066 — minimum uniform E budget



$U = \frac{\sqrt[4]{-1}\sqrt{2}(-1+i)}{2}\, C$. Scalar metadata; no physical GP.

ionq_forte

EXACT PHASE E=3 / L=30 / score ≈ 80.466 — minimum uniform E budget

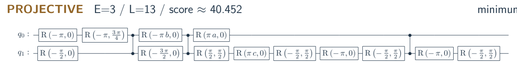


iqm_cz

EXACT PHASE E=4 / L=27 / score ≈ 75.708 — minimum uniform E budget



PROJECTIVE E=3 / L=13 / score ≈ 40.452 — minimum uniform E budget



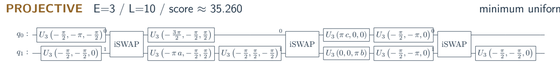
$U = -(-1)^{\frac{3}{4}}\, C$. Scalar metadata; no physical GP.

iswap_u

EXACT PHASE E=3 / L=12 / score ≈ 40.072 — minimum uniform E budget



PROJECTIVE E=3 / L=10 / score ≈ 35.260 — minimum uniform E budget



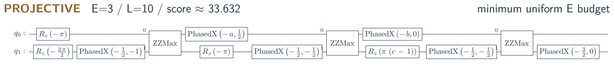
$U = \eta_{174}\, C$. Scalar metadata; no physical GP.

quantinuum_zzmax

EXACT PHASE E=4 / L=25 / score ≈ 70.876 — E upper bound



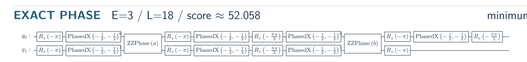
PROJECTIVE E=3 / L=10 / score ≈ 33.632 — minimum uniform E budget



$U = -i\, C$. Scalar metadata; no physical GP.

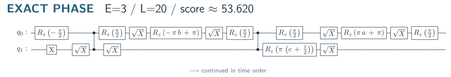
quantinuum_zzphase

EXACT PHASE E=3 / L=18 / score ≈ 52.058 — minimum uniform E budget



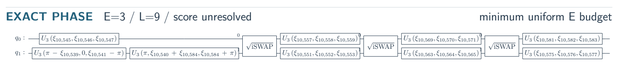
rigetti_cz

EXACT PHASE E=3 / L=20 / score ≈ 53.620 — minimum uniform E budget



sqrt_iswap_u

EXACT PHASE E=3 / L=9 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 10.

PROJECTIVE E=3 / L=8 / score unresolved — minimum uniform E budget



Derived angles: exact shared-expression program 10.

$U = e^{i\xi_{10,584}}\, C$. Scalar metadata; no physical GP.

sqrt_swap_u

EXACT PHASE E=6 / L=19 / score ≈ 69.114 — E upper bound



PROJECTIVE E=6 / L=17 / score ≈ 63.902 — E upper bound



$U = -\sqrt[4]{-1}e^{-\frac{i\pi c}{2}}\, C$. Scalar metadata; no physical GP.

tket_tk2

EXACT PHASE E=1 / L=0 / score ≈ 4.006 — minimum uniform E budget



xx90_u

EXACT PHASE E=3 / L=10 / score ≈ 34.260 — minimum uniform E budget



PROJECTIVE E=3 / L=8 / score ≈ 29.448 — minimum uniform E budget



$U = ie^{-\frac{i\pi b}{2}}\, C$. Scalar metadata; no physical GP.

xx_rz_ry

EXACT PHASE E=3 / L=41 / score ≈ 98.888 — minimum uniform E budget



PROJECTIVE E=3 / L=31 / score ≈ 80.068 — minimum uniform E budget



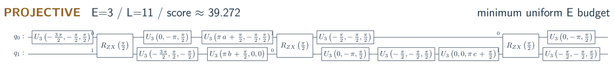
$U = \eta_{36}\, C$. Scalar metadata; no physical GP.

zx_u

EXACT PHASE E=3 / L=13 / score ≈ 43.684 — minimum uniform E budget



PROJECTIVE E=3 / L=11 / score ≈ 39.272 — minimum uniform E budget



$U = \eta_{175}\, C$. Scalar metadata; no physical GP.

zz_u

EXACT PHASE E=3 / L=12 / score ≈ 38.478 — minimum uniform E budget



Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{1} = -2\pi + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{2} = -\frac{\pi}{32} + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{3} = \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right) + \frac{31\pi}{32}$$

$$\eta_{4} = -2i\log\left(\frac{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}i\sqrt{\frac{(-1)^{\frac{1}{16}}\left(\sin\left(\frac{\pi}{16}\right) - i\cos\left(\frac{\pi}{16}\right)\right)}{2}}}{2}}{\sqrt{\frac{1}{2} + \frac{(-1)^{\frac{1}{16}}\left(\sin\left(\frac{\pi}{16}\right) - i\cos\left(\frac{\pi}{16}\right)\right)}{2}}}\right)$$

$$\eta_{5} = \operatorname{atan}\left(\frac{\sin\left(\frac{7\pi}{16}\right)}{\cos\left(\frac{7\pi}{16}\right)}\right)$$

$$\eta_{6} = -2i\log\left(\frac{\frac{\sqrt{\frac{-(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^{2}}{2}} + \frac{i\sqrt{\frac{(-1)^{\frac{1}{16}}\left(1-1^{\frac{1}{8}}(-1)\right)\left(1-e^{\frac{i\pi}{16}}\right)}{2}}}{2}}{\sqrt{\frac{(-1)^{\frac{1}{16}}\left(1-1^{\frac{1}{8}}(-1)\right)\left(1-e^{\frac{i\pi}{16}}\right)}{4} - \frac{(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^{2}}{4}}}\right)$$

$$\eta_{7} = -\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{1-\cos\left(\frac{\pi}{16}\right)}\right) - \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right) + \pi$$

$$\eta_{8} = -2i\log\left(\frac{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}i\sqrt{\frac{1^{\frac{1}{8}}(-1_{e}-\frac{i\pi}{16}}}{2}}}{2}}{\sqrt{\frac{1}{2} + \frac{1^{\frac{1}{8}}(-1_{e}-\frac{i\pi}{16}}{2}}}\right)$$

$$\eta_{9} = \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}}{-2+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{10} = -2i\log\left(\frac{\frac{\sqrt{\left(1-(-1)^{\frac{1}{16}}\right)\left(1+e^{\frac{i\pi}{16}}\right)}}{2} + \frac{i\sqrt{\left(1+(-1)^{\frac{1}{16}}\right)\left(1-e^{\frac{i\pi}{16}}\right)}}{2}}{\sqrt{\frac{\left(1+(-1)^{\frac{1}{16}}\right)\left(1-e^{\frac{i\pi}{16}}\right)}{4} + \frac{\left(1-(-1)^{\frac{1}{16}}\right)\left(1+e^{\frac{i\pi}{16}}\right)}{4}}}\right)$$

$$\eta_{11} = -\pi + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right) + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{1-\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{12} = -2i\log\left(\frac{\frac{\sqrt{\frac{-(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^{2}}{2}} + \frac{i\sqrt{\frac{\left(1-1^{\frac{1}{8}}(-1)\right)\left(1-e^{-\frac{i\pi}{16}}\right)}{2}}}{2}}{\sqrt{\frac{\left(1-1^{\frac{1}{8}}(-1)\right)\left(1-e^{-\frac{i\pi}{16}}\right)}{4} - \frac{(-1)^{\frac{1}{16}}\left(1+e^{\frac{i\pi}{16}}\right)^{2}}{4}}}\right)$$

$$\eta_{13} = -\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{1-\cos\left(\frac{\pi}{16}\right)}\right) - \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{14} = -\pi + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{15} = e^{i\left(-\frac{\pi}{32}+\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)\right)}$$

$$\eta_{16} = -\pi - \operatorname{acos}\left(\frac{\sqrt{3}}{3}\right)$$

$$\eta_{17} = -\pi + \operatorname{acos}\left(\frac{\sqrt{3}}{3}\right)$$

$$\eta_{18} = -\frac{\pi}{2} + \operatorname{acos}\left(\frac{\sqrt{3}}{3}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{19} = -\frac{\pi}{2} - \operatorname{acos}\left(\frac{\sqrt{3}}{3}\right)$$

$$\eta_{20} = -\frac{37\pi}{32} + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{21} = -\frac{5\pi}{32} + \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)$$

$$\eta_{22} = 2\operatorname{atan}\left(\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{23} = -2i\log\left(\frac{\frac{\sqrt{2}}{2}+\frac{\sqrt[8]{2}i\sqrt{\sqrt[4]{-1}(1-i)}}{2}}{\sqrt{\frac{1}{2}+\frac{\sqrt[8]{-1}\sqrt{2}(1-i)}{4}}}\right)$$

$$\eta_{24} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{\left(1-\sqrt[4]{-1}\right)\left(-\sqrt{2}+2+\sqrt{2}i\right)}}{4}+\frac{\sqrt{2}i\sqrt{\left(-i-(-1)^{\frac{3}{4}}\right)\left(\sqrt{2}+\sqrt{2}i+2i\right)}}{4}}{\sqrt{\frac{\left(1-\sqrt[4]{-1}\right)\left(-\sqrt{2}+2+\sqrt{2}i\right)}{8}-\frac{\left(-1)^{\frac{3}{4}}+i\right)\left(\sqrt{2}+\sqrt{2}i+2i\right)}{8}}}\right)$$

$$\eta_{25} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{\left(1-\sqrt[4]{-1}\right)\left(-\sqrt{2}+2+\sqrt{2}i\right)}}{4}+\frac{\sqrt{2}i\sqrt{\left(1-(-1)^{\frac{3}{4}}\right)\left(\sqrt{2}+2+\sqrt{2}i\right)}}{4}}{\sqrt{\frac{\left(1-\sqrt[4]{-1}\right)\left(-\sqrt{2}+2+\sqrt{2}i\right)}{8}+\frac{\left(1-(-1)^{\frac{3}{4}}\right)\left(\sqrt{2}+2+\sqrt{2}i\right)}{8}}}\right)$$

$$\eta_{26} = -2i\log\left(\frac{\frac{\sqrt{-(-1)^{\frac{3}{4}}\left(1+e^{\frac{i\pi}{4}}\right)^2}}{2}+\frac{i\sqrt{(-1)^{\frac{3}{4}}\left(1-\sqrt[4]{-1}\right)\left(1-e^{\frac{i\pi}{4}}\right)}}{2}}{\sqrt{\frac{(-1)^{\frac{3}{4}}\left(1-\sqrt[4]{-1}\right)\left(1-e^{\frac{i\pi}{4}}\right)}{4}-\frac{(-1)^{\frac{3}{4}}\left(1+e^{\frac{i\pi}{4}}\right)^2}{4}}}\right)$$

$$\eta_{27} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{\left(1-(-1)^{\frac{3}{4}}\right)\left(\sqrt{2}+2+\sqrt{2}i\right)}}{4}+\frac{\sqrt{2}i\sqrt{\left(-1-(-1)^{\frac{3}{4}}\right)\left(-2+\sqrt{2}+\sqrt{2}i\right)}}{4}}{\sqrt{\frac{\left(-1-(-1)^{\frac{3}{4}}\right)\left(-2+\sqrt{2}+\sqrt{2}i\right)}{8}+\frac{\left(1-(-1)^{\frac{3}{4}}\right)\left(\sqrt{2}+2+\sqrt{2}i\right)}{8}}}\right)$$

$$\eta_{28} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{\left(1-\sqrt[4]{-1}\right)\left(-\sqrt{2}+2+\sqrt{2}i\right)}}{4}+\frac{\sqrt{2}i\sqrt{\left(1+\sqrt[4]{-1}\right)\left(\sqrt{2}+2-\sqrt{2}i\right)}}{4}}{\sqrt{\frac{\left(1-\sqrt[4]{-1}\right)\left(-\sqrt{2}+2+\sqrt{2}i\right)}{8}+\frac{\left(1+\sqrt[4]{-1}\right)\left(\sqrt{2}+2-\sqrt{2}i\right)}{8}}}\right)$$

$$\eta_{29} = e^{i\left(-\frac{37\pi}{32}+\operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)+1}\right)\right)}$$

$$\eta_{30} = -\operatorname{atan}\left(\frac{\cos\left(\frac{\pi}{16}\right)}{\sin\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{31} = \operatorname{atan}\left(\frac{-2+\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{32} = \operatorname{atan}\left(\frac{\sin\left(\frac{\pi}{16}\right)}{\cos\left(\frac{\pi}{16}\right)}\right)$$

$$\eta_{33} = \operatorname{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{34} = -\operatorname{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{35} = -\pi + \operatorname{atan}\left(\frac{2-\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{36} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1+i)^3}{4}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{37} = \frac{\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor+\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}=0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sqrt{2}\sin\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{38} = \frac{\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor+\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}=0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sqrt{2}\sin\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{39} = -\frac{\begin{cases} 0 & \text{for } \cos\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi$$

$$\eta_{40} = -2\operatorname{atan_2}\left(\sqrt{2}\sin\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right),\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}\right)$$

$$\eta_{41} = -\begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}=0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sqrt{2}\sin\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{42} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{43} = \left(\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{44} = -\frac{\begin{cases} 0 & \text{for } \cos\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \left(\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) - 2\pi$$

$$\eta_{45} = \left((1-\left(2\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)\left(\frac{\begin{cases} 0 & \text{for } \cos\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi\right)\right) + \left(\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{46} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{47} = \left(\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \frac{\pi}{2}$$

$$\eta_{48} = -\frac{\begin{cases} 0 & \text{for } \cos\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \pi$$

$$\eta_{49} = e^{i\left(\frac{\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor+\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)+2\sin^2\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)}=0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sqrt{2}\sin\left(b-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(c-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(b+c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}-\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)=0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}\cos(2b)\cos(2c)},\sin\left(b-c-\frac{\pi\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor}{2}+\frac{\pi\left\lfloor \frac{2c}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\left\lfloor \frac{2b}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right) + 3\pi\right)}$$

$$\eta_{50} = \sin(a+b+c) - i\cos(a+b+c)$$

$$\eta_{51} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}=0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi),-\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{5\pi}{4}$$

$$\eta_{52} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}=0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi),-\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4}$$

$$\eta_{53} = -\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)},\sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}\right)$$

$$\eta_{54} = \frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)+\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}=0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi),-\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2+\sin^2\left(\frac{\gamma}{2}\right)}=0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2}+\frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2},-\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{55} = \frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4}$$

$$\eta_{56} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{57} = -\frac{\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{3\pi}{4}}{2} + \frac{1}{2}$$

$$\eta_{58} = -\frac{\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\gamma}{4}$$

$$\eta_{59} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\pi}{4}}{\pi} + \frac{1}{2}$$

$$\eta_{60} = \frac{\pi - \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right)}{\pi}$$

$$\eta_{61} = \frac{\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right) - \pi}{\pi} + \frac{1}{2}$$

$$\eta_{62} = \frac{\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right)}{\pi}$$

$$\eta_{63} = -\frac{\operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right)}{\pi} + \frac{1}{2}$$

$$\eta_{64} = \frac{\frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{65} = \frac{-\frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{3\pi}{4}}{2} + \frac{1}{2}$$

$$\eta_{66} = \frac{\frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{67} = \frac{-\frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2}}{\pi} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{3\pi}{4}}{2} + \frac{1}{2}$$

$$\eta_{68} = \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{69} = -\frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\pi}{4}$$

$$\eta_{70} = \pi - \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right)$$

$$\eta_{71} = \operatorname{atan_2}\left(\sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)}, \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)}\right)$$

$$\eta_{72} = \frac{\gamma}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{73} = \frac{\gamma}{2} + \frac{\begin{cases} 0 & \text{for } \sqrt{\sin^2(\phi)\sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right) + \sin^2(\theta)\cos^2\left(\frac{\gamma}{2}\right)\cos^2(\phi)} = 0 \\ \operatorname{atan_2}\left(-\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\cos(\phi), -\sin(\phi)\sin(\theta)\cos\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \sqrt{\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}\right)^2 + \sin^2\left(\frac{\gamma}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\frac{\cos\left(\frac{\gamma}{2}-\theta\right)}{2} + \frac{\cos\left(\frac{\gamma}{2}+\theta\right)}{2}, -\sin\left(\frac{\gamma}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{3\pi}{4}$$

$$\eta_{74} = \frac{\sqrt{2}i(1+i)^3 e^{\frac{i(\pi-\phi)}{4}}}{4}$$

$$\eta_{75} = \frac{\phi}{4} + \frac{\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor + \left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sqrt{2}\sin\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} - \left(\frac{3\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right) + \pi$$

$$\eta_{76} = \frac{\phi}{4} + \frac{\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor + \left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sqrt{2}\sin\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right) + 3\pi$$

$$\eta_{77} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi$$

$$\eta_{78} = -2\operatorname{atan_2}\left(-\sqrt{2}\sin\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right), \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)}\right)$$

$$\eta_{79} = -\begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sqrt{2}\sin\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{80} = \left(-\frac{\pi(1-(2(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2)))}{2}\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{81} = -\frac{\phi}{2} + \left(\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{82} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + (\pi(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2)) - 2\pi$$

$$\eta_{83} = \left((1 - (2(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2))) \left(\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi\right)\right) + (\pi(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2))$$

$$\eta_{84} = -\frac{\phi}{2} - \left(\frac{\pi(1-(2(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2)))}{2}\right) + \left(\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{85} = \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \frac{\pi}{2}$$

$$\eta_{86} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \pi$$

$$\eta_{87} = e^{i\left(\frac{\phi}{4} + \frac{\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor + \left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta) + 2\sin^2\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sqrt{2}\sin\left(\frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\cos\left(\frac{\phi}{4} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, -\sin\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\phi}{4} + \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} + \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}(-1)^{\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}\cos\left(\frac{\phi}{2}\right)\cos(\theta)}, \sin\left(\frac{\phi}{4} - \frac{\theta}{2} + \frac{\pi\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor}{2} - \frac{\pi\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)}{2}\right) + 2\pi\right)}$$

$$\eta_{88} = \frac{\sqrt{2}(1+i)^3 e^{\frac{i(\pi-\phi)}{4}}}{4}$$

$$\eta_{89} = \frac{\sqrt{2}i(1+i)e^{\frac{i(\phi-2\theta)}{4}}}{2}$$

$$\eta_{90} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) + \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{91} = -2i\log\left(\frac{\frac{\sqrt{(-1)^{\frac{3}{4}}\left(\sqrt{2}-\sqrt{2}i+\sqrt{\sqrt{2}+2}\right)}}{2}+\frac{i\sqrt{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2}-\sqrt{2}\right)}}{2}}{\sqrt{\frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2}-\sqrt{2}\right)}{4}}+\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2}-\sqrt{2}i-\sqrt{\sqrt{2}+2}\right)}{4}}\right)$$

$$\eta_{92} = \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{93} = -\pi - \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{94} = -2i\log\left(\frac{\frac{\sqrt{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}}{2}+\frac{i\sqrt{(-1)^{\frac{3}{4}}\left(\sqrt{2}-\sqrt{2}-i\sqrt{\sqrt{2}+2}\right)}}{2}}{\sqrt{\frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}{4}}+\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2}-\sqrt{2}-i\sqrt{\sqrt{2}+2}\right)}{4}}\right)$$

$$\eta_{95} = \operatorname{atan}\left(\frac{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}\right) - \operatorname{atan}\left(\frac{-\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{96} = -2i\log\left(\frac{\frac{\sqrt{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}}{2}+\frac{i\sqrt{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}}{2}}{\sqrt{\frac{\sqrt[4]{-1}\left(\sqrt{\sqrt{2}+2}-i\sqrt{2-\sqrt{2}}\right)}{4}}+\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}{4}}\right)$$

$$\eta_{97} = -\pi + 2\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}}{\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{98} = -\pi - \operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}\right) - \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1-\sqrt{2})\sqrt{\sqrt{2}+2}}{(1-\sqrt{2})\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{99} = -2i\log\left(\frac{\frac{\sqrt{2}\sqrt{\sqrt[4]{-1}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)}\left((-1)^{\frac{3}{4}}(1+i)+1-i\right)\left(-\sqrt{2}(1-i)(1+i)+2+2i\right)}}{4}+\frac{\sqrt{2}i\sqrt{-(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)(2+\sqrt{2}(1-i)(1+i)+2i)\left((-1)^{\frac{3}{4}}(1+i)-1+i\right)}}{4}}{\sqrt{\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-\sqrt{-2-\sqrt{2}}\right)\left((-1)^{\frac{3}{4}}(1+i)+1-i\right)\left(-\sqrt{2}(1-i)(1+i)+2+2i\right)}{32}}-\frac{(-1)^{\frac{3}{4}}\left(\sqrt{2-\sqrt{2}}-i\sqrt{\sqrt{2}+2}\right)(2+\sqrt{2}(1-i)(1+i)+2i)\left((-1)^{\frac{3}{4}}(1+i)-1+i\right)}{32}}\right)$$

$$\eta_{100} = -\pi - \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1-\sqrt{2})\sqrt{\sqrt{2}+2}}{(1-\sqrt{2})\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right) + \operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}\right)$$

$$\eta_{101} = e^{i\left(-\frac{\operatorname{atan}\left(\frac{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{-\sqrt{2-\sqrt{2}}+(1+\sqrt{2})\sqrt{\sqrt{2}+2}}{\sqrt{\sqrt{2}+2}+(1+\sqrt{2})\sqrt{2-\sqrt{2}}}\right)}{2}+\frac{7\pi}{8}\right)}$$

$$\eta_{102} = 2\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}\right)$$

$$\eta_{103} = \frac{\sqrt{2}i(1+i)^{\frac{3}{4}}e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{104} = -\frac{\chi}{2} - \gamma - \frac{\phi}{2} + \frac{\zeta}{2} + \left(\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{105} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \left(\frac{\pi\left(1-\left(2\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor -\frac{\phi}{2\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor -\frac{\theta}{\pi}+\frac{1}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{106} = -(-1)^{\frac{3}{4}}e^{-\frac{i(4\gamma+\phi)}{4}}$$

$$\eta_{107} = \frac{\chi}{2} - \gamma - \frac{\phi}{2} - \frac{\zeta}{2} - \frac{\pi}{2}$$

$$\eta_{108} = \frac{\frac{\chi}{2}-\gamma-\frac{\phi}{2}-\frac{\zeta}{2}}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{109} = \frac{-\frac{3\pi}{2} - \gamma - \frac{\phi}{2} + \frac{\lambda}{2}}{\pi}$$

$$\eta_{110} = \frac{3\pi}{2} - \gamma - \frac{\phi}{2} - \frac{\lambda}{2} - \frac{3\pi}{2}$$

$$\eta_{111} = \frac{\sqrt{2}(1+i)^3 e^{\frac{i(-4\gamma-\phi+\pi)}{4}}}{4}$$

$$\eta_{112} = \frac{\sqrt{2}i(1+i)^3 e^{\frac{i\pi\left(t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)+1\right)}{4}}}{4}$$

$$\eta_{113} = \frac{\pi t(4\mathrm{global}_{\mathrm{shift}}+1)}{4} + \begin{cases} 0 & \text{for } \sqrt{2\sin^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sqrt{2}\sin\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} - \left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor \bmod 2\right)}{2}\right) + \pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor + \pi$$

$$\eta_{114} = \frac{\pi t(4\mathrm{global}_{\mathrm{shift}}+1)}{4} + \frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{2} + \begin{cases} 0 & \text{for } \sqrt{2\sin^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sqrt{2}\sin\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} - \left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor \bmod 2\right)}{2}\right) + \frac{\operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, -(-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} + \pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor + 3\pi$$

$$\eta_{115} = \frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{2} - \frac{\operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, -(-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} + \pi$$

$$\eta_{116} = -2\operatorname{atan_2}\left(-\sqrt{2}\sin\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right), \sqrt{2\sin^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)}\right)$$

$$\eta_{117} = -\begin{cases} 0 & \text{for } \sqrt{2\sin^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sqrt{2}\sin\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{118} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{119} = 2\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)$$

$$\eta_{120} = -\frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{2} + \left(\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right) - \frac{\operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, -(-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} - 2\pi$$

$$\eta_{121} = \left(\left(1 - \left(2\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right)\right)\left(-\frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{2} + \frac{\operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, -(-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} + \pi\right)\right) + \left(\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{122} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(2\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right)$$

$$\eta_{123} = \left(\pi\left(\left\lfloor\frac{1}{2}-\frac{t}{2}\right\rfloor \bmod 2\right)\right) + \frac{\pi}{2}$$

$$\eta_{124} = -\frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{2} - \frac{\operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, -(-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} - \pi$$

$$\eta_{125} = e^{i\left(\frac{\pi t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)}{4} + \frac{\begin{cases} 0 & \text{for } (-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\cos\left(\frac{\pi t}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, 0\right) & \text{otherwise} \end{cases}}{2} + i\begin{cases} 0 & \text{for } \sqrt{2\sin^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos^2\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right) + \cos^2\left(\frac{\pi t}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\left|\cos\left(\frac{\pi t}{2}\right)\right|, -\sqrt{2}\sin\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\cos\left(\frac{\pi t}{4} + \frac{\pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} - \left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor \bmod 2\right)}{2}\right) + \frac{\operatorname{atan_2}\left(-\left|\cos\left(\frac{\pi t}{2}\right)\right|, -(-1)^{\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor}\sin\left(\frac{\pi t}{2}\right)\right)}{2} + \pi\left\lfloor\frac{1}{2}-\frac{\lambda}{2}\right\rfloor + 2\pi\right)}$$

$$\eta_{126} = -e^{\frac{i\pi\left(t\left(4\mathrm{global}_{\mathrm{shift}}+1\right)+1\right)}{4}}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{127} = -(-1)^{\frac{3}{4}}\, e^{\frac{i\pi\left(4g_{global_{shift}}+1\right)}{4}}$$

$$\eta_{128} = -\frac{\frac{\pi\left(2g_{global_{shift}}-1\right)}{2}}{\pi} - \pi$$

$$\eta_{129} = e^{\frac{i\pi\left(t\left(4g_{global_{shift}}+1\right)-3t-2\right)}{4}}$$

$$\eta_{130} = -ie^{\frac{i\pi\left(t\left(4g_{global_{shift}}+1\right)+1\right)}{4}}$$

$$\eta_{131} = \frac{\sqrt{2}(1+i)^2 e^{\frac{i\pi\left(t\left(4g_{global_{shift}}+1\right)+1\right)}{4}}}{4}$$

$$\eta_{132} = \frac{\sqrt{2}i(1+i)e^{i\pi g_{global_{shift}}t}}{2}$$

$$\eta_{133} = \frac{\sqrt[4]{-1}\sqrt{2}(1-i)e^{i\pi g_{global_{shift}}t}}{2}$$

$$\eta_{134} = \frac{(-1)^{\frac{3}{4}}\sqrt{2}(1-i)}{2}$$

$$\eta_{135} = -\pi + \operatorname{atan}\left(\frac{-\sqrt{3\sqrt{2}+6}+\sqrt{2-\sqrt{2}}+\sqrt{6-3\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{136} = \operatorname{atan}\left(\frac{-\sqrt{3\sqrt{2}+6}+\sqrt{2-\sqrt{2}}+\sqrt{6-3\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right) + \pi$$

$$\eta_{137} = \pi - 2\operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{138} = \frac{\operatorname{atan}\left(\frac{2\sqrt{\sqrt{2}+\sqrt{6}}}{-\sqrt{\sqrt{3}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)+\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)+\frac{7\pi}{12}-2\operatorname{atan}\left(\frac{2\sqrt[4]{3}\sqrt{1+\sqrt{3}}}{-\sqrt{\sqrt{3}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)}{\pi}$$

$$\eta_{139} = \frac{-\frac{13\pi}{24}+\operatorname{atan}\left(\frac{2\sqrt[4]{3}\sqrt{1+\sqrt{3}}}{-\sqrt{\sqrt{3}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)-\frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2-\sqrt{2}}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}-\frac{\operatorname{atan}\left(\frac{2\sqrt{\sqrt{2}+\sqrt{6}}}{-\sqrt{\sqrt{3}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)}{2}}{\pi}+\frac{1}{2}$$

$$\eta_{140} = \frac{-\frac{5}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}$$

$$\eta_{141} = \frac{1}{2}-\frac{-\frac{5}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}$$

$$\eta_{142} = \frac{-\frac{5}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}$$

$$\eta_{143} = \frac{-\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\frac{\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{\frac{\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}}{2}+\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}{2}}\right)}{2}}{\pi}+\frac{\pi}{2}+\frac{1}{2}$$

$$\eta_{144} = \frac{-\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)-\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{145} = \frac{-\operatorname{atan}\left(\frac{\sqrt{3}\sqrt{\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)+\operatorname{atan}\left(\frac{\sqrt{3}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{146} = \frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{\pi}$$

$$\eta_{147} = \frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}$$

$$\eta_{148} = e^{i\left(\frac{\operatorname{atan}\left(\frac{2\sqrt{\sqrt{2}+\sqrt{6}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)}{2}+\frac{\pi}{24}+\frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2}-\sqrt{3}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}-\operatorname{atan}\left(\frac{2\sqrt[4]{3}\sqrt{1+\sqrt{3}}}{-\sqrt{\sqrt{2}+2}+\sqrt{3}\sqrt{2-\sqrt{2}}}\right)\right)}$$

$$\eta_{149} = \frac{(-1)^{\frac{3}{24}}\sqrt{2}(1+i)}{2}$$

$$\eta_{150} = \frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2}-\sqrt{2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}+\frac{13\pi}{24}$$

$$\eta_{151} = \frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2}-\sqrt{2}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}+\frac{25\pi}{24}$$

$$\eta_{152} = -\frac{\pi}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}$$

$$\eta_{153} = -\frac{\pi}{2}-\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}$$

$$\eta_{154} = -\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)-\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{155} = -\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)+\operatorname{atan}\left(\frac{\sqrt{2}\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)$$

$$\eta_{156} = \frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{2}+\frac{\sqrt{6}}{2}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}$$

$$\eta_{157} = -\frac{\operatorname{atan}\left(\frac{\sqrt{-\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{3}\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}+\frac{\operatorname{atan}\left(\frac{\sqrt{\frac{\sqrt{2}}{4}+\frac{\sqrt{6}}{4}}}{-\sqrt{3}\sqrt{\frac{1}{2}-\frac{\sqrt{2}}{4}}+\sqrt{\frac{\sqrt{2}}{4}+\frac{1}{2}}}\right)}{2}$$

$$\eta_{158} = e^{i\left(\frac{\pi}{24}+\frac{\operatorname{atan}\left(\frac{2\sqrt{-\sqrt{2}+\sqrt{6}}}{-\sqrt{2}-\sqrt{3}+\sqrt{3}\sqrt{\sqrt{2}+2}}\right)}{2}\right)}$$

$$\eta_{159} = \operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)$$

$$\eta_{160} = \operatorname{atan}\left(\frac{-\sqrt{3}\sqrt{\sqrt{2}+2}+\sqrt{2-\sqrt{2}}}{\sqrt{3}\sqrt{2-\sqrt{2}}+\sqrt{\sqrt{2}+2}}\right)+\pi$$

$$\eta_{161} = \frac{\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor+\left\lfloor\frac{1}{2}-c\right\rfloor\right)}{2}+\begin{cases}0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)+2\sin^2\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)}=0\\ \operatorname{atan}_2\left(\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)},-\sqrt{2}\sin\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}-\left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)}{2}\right)+\pi$$

$$\eta_{162} = \frac{\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor+\left\lfloor\frac{1}{2}-c\right\rfloor\right)}{2}+\begin{cases}0 & \text{for } \sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)+2\sin^2\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)}=0\\ \operatorname{atan}_2\left(\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)},-\sqrt{2}\sin\left(\frac{\pi b}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}+\frac{\begin{cases}0 & \text{for } \cos\left(\frac{\pi b}{2}-\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)=0\\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)},-\sin\left(\frac{\pi b}{2}+\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}}{2}+\frac{\begin{cases}0 & \text{for } \cos\left(\frac{\pi b}{2}+\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}+\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)=0\\ \operatorname{atan}_2\left(-\sqrt{(-1)^{\left\lfloor\frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor\frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)},-\sin\left(\frac{\pi b}{2}-\frac{\pi c}{2}+\frac{\pi\left\lfloor\frac{1}{2}-b\right\rfloor}{2}-\frac{\pi\left\lfloor\frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise}\end{cases}}{2}-\left(\frac{3\pi\left(\left\lfloor\frac{1}{2}-b\right\rfloor \bmod 2\right)}{2}\right)+3\pi$$

Exact expression glossary

Definitions shared by the numbered circuit parameters and scalar factors.

$$\eta_{163} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi$$

$$\eta_{164} = -2\operatorname{atan_2}\left(-\sqrt{2}\sin\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\cos\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2}\right), \sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)}\right)$$

$$\eta_{165} = -\begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sqrt{2}\sin\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}$$

$$\eta_{166} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right)$$

$$\eta_{167} = \left(\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor \frac{1}{2}-c\right\rfloor \bmod 2\right)\right)$$

$$\eta_{168} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \left(\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right) - 2\pi$$

$$\eta_{169} = \left(\left(1-\left(2\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right)\right)\left(-\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \pi\right)\right) + \left(\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right)$$

$$\eta_{170} = \left(-\frac{\pi\left(1-\left(2\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right)\right)}{2}\right) + \left(\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right) + \left(\pi\left(\left\lfloor \frac{1}{2}-c\right\rfloor \bmod 2\right)\right)$$

$$\eta_{171} = \left(\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)\right) + \frac{\pi}{2}$$

$$\eta_{172} = -\frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \pi$$

$$\eta_{173} = e^{i\left(\frac{\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor + \left\lfloor \frac{1}{2}-c\right\rfloor\right)}{2} + \begin{cases} 0 & \text{for } \sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c) + 2\sin^2\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2}\right)\cos^2\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)} = 0 \\ \operatorname{atan_2}\left(\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sqrt{2}\sin\left(\frac{\pi b}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2}\right)\cos\left(\frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} + \frac{\begin{cases} 0 & \text{for } \cos\left(\frac{\pi b}{2} + \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} + \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right) = 0 \\ \operatorname{atan_2}\left(-\sqrt{(-1)^{\left\lfloor \frac{1}{2}-b\right\rfloor}(-1)^{\left\lfloor \frac{1}{2}-c\right\rfloor}\cos(\pi b)\cos(\pi c)}, -\sin\left(\frac{\pi b}{2} - \frac{\pi c}{2} + \frac{\pi\left\lfloor \frac{1}{2}-b\right\rfloor}{2} - \frac{\pi\left\lfloor \frac{1}{2}-c\right\rfloor}{2}\right)\right) & \text{otherwise} \end{cases}}{2} - \left(\frac{3\pi\left(\left\lfloor \frac{1}{2}-b\right\rfloor \bmod 2\right)}{2}\right) + 2\pi\right)}$$

$$\eta_{174} = \frac{\sqrt{2}i(1+i)e^{-\frac{i\pi b}{2}}}{2}$$

$$\eta_{175} = \frac{\sqrt[4]{-1}\sqrt{2}(1-i)e^{-\frac{i\pi c}{2}}}{2}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,0} = \left\lfloor \frac{2a}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{1,1} = \left\lfloor \frac{2b}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{1,2} = \left\lfloor \frac{2c}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{1,3} = -\frac{\pi \xi_{1,0}}{2} + a$$

$$\xi_{1,4} = -\frac{\pi \xi_{1,1}}{2} + b$$

$$\xi_{1,5} = -\frac{\pi \xi_{1,2}}{2} + c$$

$$\xi_{1,6} = \begin{cases} \xi_{1,3} & \text{for } (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,3}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \\ \xi_{1,4} & \text{for } (|\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,4}|) \wedge (|\xi_{1,3}| \leq |\xi_{1,4}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \\ \xi_{1,5} & \text{otherwise} \end{cases}$$

$$\xi_{1,7} = \begin{cases} \xi_{1,4} & \text{for } |\xi_{1,3}| \geq |\xi_{1,4}| \wedge |\xi_{1,4}| \geq |\xi_{1,5}| \\ \xi_{1,5} & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge |\xi_{1,4}| \leq |\xi_{1,5}| \\ \xi_{1,3} & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge |\xi_{1,3}| \leq |\xi_{1,4}| \\ \xi_{1,5} & \text{for } |\xi_{1,4}| \geq |\xi_{1,5}| \wedge |\xi_{1,3}| \leq |\xi_{1,5}| \\ \xi_{1,3} & \text{for } |\xi_{1,3}| \geq |\xi_{1,4}| \wedge |\xi_{1,3}| \leq |\xi_{1,5}| \\ \xi_{1,4} & \text{otherwise} \end{cases}$$

$$\xi_{1,8} = \begin{cases} \xi_{1,5} & \text{for } |\xi_{1,3}| \geq |\xi_{1,4}| \wedge |\xi_{1,4}| \geq |\xi_{1,5}| \\ \xi_{1,4} & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge |\xi_{1,4}| \leq |\xi_{1,5}| \\ \xi_{1,5} & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge |\xi_{1,3}| \leq |\xi_{1,4}| \\ \xi_{1,3} & \text{for } |\xi_{1,4}| \geq |\xi_{1,5}| \wedge |\xi_{1,3}| \leq |\xi_{1,5}| \\ \xi_{1,4} & \text{for } |\xi_{1,3}| \geq |\xi_{1,4}| \wedge |\xi_{1,3}| \leq |\xi_{1,5}| \\ \xi_{1,3} & \text{otherwise} \end{cases}$$

$$\xi_{1,9} = \begin{cases} 1 & \text{for } |\xi_{1,3}| \geq |\xi_{1,4}| \wedge |\xi_{1,4}| \geq |\xi_{1,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge |\xi_{1,4}| \leq |\xi_{1,5}| \\ 0 & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge |\xi_{1,3}| \leq |\xi_{1,4}| \\ \frac{1}{2} & \text{for } |\xi_{1,3}| \leq |\xi_{1,5}| \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,4}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{1,10} = \begin{cases} 0 & \text{for } |\xi_{1,3}| \geq |\xi_{1,4}| \wedge |\xi_{1,4}| \geq |\xi_{1,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge (|\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,4}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \leq |\xi_{1,4}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \\ \frac{1}{2} & \text{for } |\xi_{1,3}| \leq |\xi_{1,5}| \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,4}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,11} = \begin{cases} 0 & \text{for } (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,3}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{1,3}| \geq |\xi_{1,5}| \wedge |\xi_{1,3}| \leq |\xi_{1,4}| \\ \frac{1}{2} & \text{for } |\xi_{1,3}| \leq |\xi_{1,5}| \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,4}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{1,12} = \begin{cases} 0 & \text{for } (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,3}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \geq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,4}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,3}| \geq |\xi_{1,4}| \vee |\xi_{1,3}| \geq |\xi_{1,5}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \wedge (|\xi_{1,4}| \geq |\xi_{1,5}| \vee |\xi_{1,3}| \leq |\xi_{1,4}| \vee |\xi_{1,4}| \leq |\xi_{1,5}|) \\ \frac{1}{2} & \text{for } |\xi_{1,4}| \geq |\xi_{1,3}| \wedge |\xi_{1,3}| \leq |\xi_{1,5}| \\ -\frac{1}{2} & \text{for } |\xi_{1,3}| \geq |\xi_{1,4}| \wedge |\xi_{1,3}| \leq |\xi_{1,5}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,13} = \begin{cases} -1 & \text{for } \xi_{1,6} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{1,14} = \begin{cases} -1 & \text{for } \xi_{1,7} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{1,15} = \begin{cases} 1 & \text{for } \xi_{1,13} = 1 \wedge \xi_{1,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,16} = \begin{cases} 0 & \text{for } \xi_{1,13} = 1 \wedge \xi_{1,14} = 1 \\ 1 & \text{for } \xi_{1,13} = 1 \wedge \xi_{1,14} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,17} = \begin{cases} 0 & \text{for } \xi_{1,13} = 1 \wedge (\xi_{1,13} = 1 \vee \xi_{1,14} = -1) \wedge (\xi_{1,13} = 1 \vee \xi_{1,14} = 1) \wedge (\xi_{1,14} = -1 \vee \xi_{1,14} = 1) \\ 1 & \text{for } \xi_{1,13} = -1 \wedge \xi_{1,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,18} = \begin{cases} 0 & \text{for } (\xi_{1,13} = -1 \vee \xi_{1,13} = 1) \wedge (\xi_{1,13} = 1 \vee \xi_{1,14} = 1) \wedge (\xi_{1,14} = -1 \vee \xi_{1,14} = 1) \wedge (\xi_{1,13} = -1 \vee \xi_{1,13} = 1 \vee \xi_{1,14} = -1) \wedge (\xi_{1,13} = -1 \vee \xi_{1,13} = 1 \vee \xi_{1,14} = 1) \wedge (\xi_{1,13} = -1 \vee \xi_{1,14} = -1 \vee \xi_{1,14} = 1) \wedge (\xi_{1,13} = 1 \vee \xi_{1,14} = -1 \vee \xi_{1,14} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{1,19} = \cos\left(\frac{\pi \xi_{1,0}}{2}\right)$$

$$\xi_{1,20} = -\sin\left(\frac{\pi \xi_{1,0}}{2}\right)$$

$$\xi_{1,21} = \cos\left(\frac{\pi \xi_{1,1}}{2}\right)$$

$$\xi_{1,22} = -\sin\left(\frac{\pi \xi_{1,1}}{2}\right)$$

$$\xi_{1,23} = \xi_{1,19}\xi_{1,21}$$

$$\xi_{1,24} = \xi_{1,20}\xi_{1,21}$$

$$\xi_{1,25} = \xi_{1,19}\xi_{1,22}$$

$$\xi_{1,26} = -\xi_{1,20}\xi_{1,22}$$

$$\xi_{1,27} = \cos\left(\frac{\pi \xi_{1,2}}{2}\right)$$

$$\xi_{1,28} = -\sin\left(\frac{\pi \xi_{1,2}}{2}\right)$$

$$\xi_{1,29} = \xi_{1,23}\xi_{1,27} - \xi_{1,26}\xi_{1,28}$$

$$\xi_{1,30} = \xi_{1,24}\xi_{1,27} - \xi_{1,25}\xi_{1,28}$$

$$\xi_{1,31} = \xi_{1,24}\xi_{1,28} + \xi_{1,25}\xi_{1,27}$$

$$\xi_{1,32} = \xi_{1,23}\xi_{1,28} + \xi_{1,26}\xi_{1,27}$$

$$\xi_{1,33} = \xi_{1,15}\left(-\xi_{1,10}\xi_{1,30} - \xi_{1,11}\xi_{1,31} - \xi_{1,12}\xi_{1,32} + \xi_{1,29}\xi_{1,9}\right) - \xi_{1,16}\left(\xi_{1,10}\xi_{1,29} - \xi_{1,11}\xi_{1,32} + \xi_{1,12}\xi_{1,31} + \xi_{1,30}\xi_{1,9}\right) - \xi_{1,17}\left(\xi_{1,10}\xi_{1,32} + \xi_{1,11}\xi_{1,29} - \xi_{1,12}\xi_{1,30} + \xi_{1,31}\xi_{1,9}\right) - \xi_{1,18}\left(-\xi_{1,10}\xi_{1,31} + \xi_{1,11}\xi_{1,30} + \xi_{1,12}\xi_{1,29} + \xi_{1,32}\xi_{1,9}\right)$$

$$\xi_{1,34} = \xi_{1,15}\left(\xi_{1,10}\xi_{1,29} - \xi_{1,11}\xi_{1,32} + \xi_{1,12}\xi_{1,31} + \xi_{1,30}\xi_{1,9}\right) + \xi_{1,16}\left(-\xi_{1,10}\xi_{1,30} - \xi_{1,11}\xi_{1,31} - \xi_{1,12}\xi_{1,32} + \xi_{1,29}\xi_{1,9}\right) - \xi_{1,17}\left(-\xi_{1,10}\xi_{1,31} + \xi_{1,11}\xi_{1,30} + \xi_{1,12}\xi_{1,29} + \xi_{1,32}\xi_{1,9}\right) + \xi_{1,18}\left(\xi_{1,10}\xi_{1,32} + \xi_{1,11}\xi_{1,29} - \xi_{1,12}\xi_{1,30} + \xi_{1,31}\xi_{1,9}\right)$$

$$\xi_{1,35} = \xi_{1,15}\left(\xi_{1,10}\xi_{1,32} + \xi_{1,11}\xi_{1,29} - \xi_{1,12}\xi_{1,30} + \xi_{1,31}\xi_{1,9}\right) + \xi_{1,16}\left(-\xi_{1,10}\xi_{1,31} + \xi_{1,11}\xi_{1,30} + \xi_{1,12}\xi_{1,29} + \xi_{1,32}\xi_{1,9}\right) + \xi_{1,17}\left(-\xi_{1,10}\xi_{1,30} - \xi_{1,11}\xi_{1,31} - \xi_{1,12}\xi_{1,32} + \xi_{1,29}\xi_{1,9}\right) - \xi_{1,18}\left(\xi_{1,10}\xi_{1,29} - \xi_{1,11}\xi_{1,32} + \xi_{1,12}\xi_{1,31} + \xi_{1,30}\xi_{1,9}\right)$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,36} = \xi_{1,15}\left(-\xi_{1,10}\xi_{1,31} + \xi_{1,11}\xi_{1,30} + \xi_{1,12}\xi_{1,29} + \xi_{1,32}\xi_{1,9}\right) - \xi_{1,16}\left(\xi_{1,10}\xi_{1,32} + \xi_{1,11}\xi_{1,29} - \xi_{1,12}\xi_{1,30} + \xi_{1,31}\xi_{1,9}\right) + \xi_{1,17}\left(\xi_{1,10}\xi_{1,29} - \xi_{1,11}\xi_{1,32} + \xi_{1,12}\xi_{1,31} + \xi_{1,30}\xi_{1,9}\right) + \xi_{1,18}\left(-\xi_{1,10}\xi_{1,30} - \xi_{1,11}\xi_{1,31} - \xi_{1,12}\xi_{1,32} + \xi_{1,29}\xi_{1,9}\right)$$

$$\xi_{1,37} = -\xi_{1,10}\xi_{1,30} - \xi_{1,11}\xi_{1,31} - \xi_{1,12}\xi_{1,32} + \xi_{1,29}\xi_{1,9}$$

$$\xi_{1,38} = \xi_{1,10}\xi_{1,29} - \xi_{1,11}\xi_{1,32} + \xi_{1,12}\xi_{1,31} + \xi_{1,30}\xi_{1,9}$$

$$\xi_{1,39} = \xi_{1,10}\xi_{1,32} + \xi_{1,11}\xi_{1,29} - \xi_{1,12}\xi_{1,30} + \xi_{1,31}\xi_{1,9}$$

$$\xi_{1,40} = -\xi_{1,10}\xi_{1,31} + \xi_{1,11}\xi_{1,30} + \xi_{1,12}\xi_{1,29} + \xi_{1,32}\xi_{1,9}$$

$$\xi_{1,41} = -\xi_{1,10}\xi_{1,16} - \xi_{1,11}\xi_{1,17} - \xi_{1,12}\xi_{1,18} + \xi_{1,15}\xi_{1,9}$$

$$\xi_{1,42} = -\xi_{1,10}\xi_{1,15} - \xi_{1,11}\xi_{1,18} + \xi_{1,12}\xi_{1,17} - \xi_{1,16}\xi_{1,9}$$

$$\xi_{1,43} = \xi_{1,10}\xi_{1,18} - \xi_{1,11}\xi_{1,15} - \xi_{1,12}\xi_{1,16} - \xi_{1,17}\xi_{1,9}$$

$$\xi_{1,44} = -\xi_{1,10}\xi_{1,17} + \xi_{1,11}\xi_{1,16} - \xi_{1,12}\xi_{1,15} - \xi_{1,18}\xi_{1,9}$$

$$\xi_{1,45} = \xi_{1,13}\xi_{1,6}$$

$$\xi_{1,46} = \xi_{1,14}\xi_{1,7}$$

$$\xi_{1,47} = \xi_{1,13}\xi_{1,14}\xi_{1,8}$$

$$\xi_{1,48} = -\frac{\pi(\xi_{1,0}+\xi_{1,1}+\xi_{1,2})}{2}$$

$$\xi_{1,49} = \begin{cases} 0 & \text{for } \left(\xi_{1,46} > \frac{\pi}{8} \vee \xi_{1,47} < 0\right) \wedge \xi_{1,46} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{1,50} = \frac{\pi}{8}$$

$$\xi_{1,51} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{1,46} > \frac{\pi}{8} \wedge \xi_{1,47} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{1,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,52} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{1,46} > \frac{\pi}{8} \wedge \xi_{1,47} < 0 \\ \frac{1}{2} & \text{for } \xi_{1,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,53} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{1,46} > \frac{\pi}{8} \vee \xi_{1,47} < 0\right) \wedge \xi_{1,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,54} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{1,46} > \frac{\pi}{8} \vee \xi_{1,47} < 0\right) \wedge \xi_{1,46} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{1,55} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{1,46} > \frac{\pi}{8} \vee \xi_{1,47} < 0\right) \wedge \xi_{1,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,56} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{1,46} > \frac{\pi}{8} \vee \xi_{1,47} < 0\right) \wedge \xi_{1,46} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{1,57} = \begin{cases} \frac{1}{2} & \text{for } \xi_{1,46} > \frac{\pi}{8} \wedge \xi_{1,47} < 0 \\ -\frac{1}{2} & \text{for } \xi_{1,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,58} = \left\lfloor \frac{2\xi_{1,45} - 2\xi_{1,49}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{1,59} = \left\lfloor \frac{2\xi_{1,46} - 2\xi_{1,50}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{1,60} = \left\lfloor \frac{2\xi_{1,47} - 2\xi_{1,51}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{1,61} = \xi_{1,45} - \xi_{1,49} - \frac{\pi\xi_{1,58}}{2}$$

$$\xi_{1,62} = \xi_{1,46} - \xi_{1,50} - \frac{\pi\xi_{1,59}}{2}$$

$$\xi_{1,63} = \xi_{1,47} - \xi_{1,51} - \frac{\pi\xi_{1,60}}{2}$$

$$\xi_{1,64} = \begin{cases} \xi_{1,61} & \text{for } \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,61}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,62}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,63}| \vee |\xi_{1,62}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,62}| \geq |\xi_{1,63}| \vee |\xi_{1,62}| \leq |\xi_{1,63}|\right) \\ \xi_{1,62} & \text{for } \left(|\xi_{1,61}| \geq |\xi_{1,63}| \vee |\xi_{1,62}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,63}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,62}| \geq |\xi_{1,63}| \vee |\xi_{1,61}| \leq |\xi_{1,62}|\right) \wedge \left(|\xi_{1,61}| \leq |\xi_{1,62}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \\ \xi_{1,63} & \text{otherwise} \end{cases}$$

$$\xi_{1,65} = \begin{cases} \xi_{1,62} & \text{for } |\xi_{1,61}| \geq |\xi_{1,62}| \wedge |\xi_{1,62}| \geq |\xi_{1,63}| \\ \xi_{1,63} & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge |\xi_{1,62}| \leq |\xi_{1,63}| \\ \xi_{1,61} & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge |\xi_{1,61}| \leq |\xi_{1,62}| \\ \xi_{1,63} & \text{for } |\xi_{1,62}| \geq |\xi_{1,63}| \wedge |\xi_{1,61}| \leq |\xi_{1,63}| \\ \xi_{1,61} & \text{for } |\xi_{1,61}| \geq |\xi_{1,62}| \wedge |\xi_{1,61}| \leq |\xi_{1,63}| \\ \xi_{1,62} & \text{otherwise} \end{cases}$$

$$\xi_{1,66} = \begin{cases} \xi_{1,63} & \text{for } |\xi_{1,61}| \geq |\xi_{1,62}| \wedge |\xi_{1,62}| \geq |\xi_{1,63}| \\ \xi_{1,62} & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge |\xi_{1,62}| \leq |\xi_{1,63}| \\ \xi_{1,63} & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge |\xi_{1,61}| \leq |\xi_{1,62}| \\ \xi_{1,61} & \text{for } |\xi_{1,62}| \geq |\xi_{1,63}| \wedge |\xi_{1,61}| \leq |\xi_{1,63}| \\ \xi_{1,62} & \text{for } |\xi_{1,61}| \geq |\xi_{1,62}| \wedge |\xi_{1,61}| \leq |\xi_{1,63}| \\ \xi_{1,61} & \text{otherwise} \end{cases}$$

$$\xi_{1,67} = \begin{cases} 1 & \text{for } |\xi_{1,61}| \geq |\xi_{1,62}| \wedge |\xi_{1,62}| \geq |\xi_{1,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge |\xi_{1,62}| \leq |\xi_{1,63}| \\ 0 & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge |\xi_{1,61}| \leq |\xi_{1,62}| \\ \frac{1}{2} & \text{for } |\xi_{1,61}| \leq |\xi_{1,63}| \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,62}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,62}| \geq |\xi_{1,63}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{1,68} = \begin{cases} 0 & \text{for } |\xi_{1,61}| \geq |\xi_{1,62}| \wedge |\xi_{1,62}| \geq |\xi_{1,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge \left(|\xi_{1,61}| \geq |\xi_{1,63}| \vee |\xi_{1,61}| \leq |\xi_{1,62}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,63}| \vee |\xi_{1,62}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \leq |\xi_{1,62}| \vee |\xi_{1,62}| \leq |\xi_{1,63}|\right) \\ \frac{1}{2} & \text{for } |\xi_{1,61}| \leq |\xi_{1,63}| \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,62}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,62}| \geq |\xi_{1,63}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,69} = \begin{cases} 0 & \text{for } \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,61}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,62}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,63}| \vee |\xi_{1,62}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,62}| \geq |\xi_{1,63}| \vee |\xi_{1,62}| \leq |\xi_{1,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{1,61}| \geq |\xi_{1,63}| \wedge |\xi_{1,61}| \leq |\xi_{1,62}| \\ \frac{1}{2} & \text{for } |\xi_{1,61}| \leq |\xi_{1,63}| \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,62}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,61}| \geq |\xi_{1,62}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,62}| \geq |\xi_{1,63}| \vee |\xi_{1,61}| \leq |\xi_{1,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{1,70} = \begin{cases} 0 & \text{for } \left(|\xi_{1,64}| \geq |\xi_{1,62}| \vee |\xi_{1,64}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,64}| \geq |\xi_{1,64}| \vee |\xi_{1,62}| \geq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,64}| \geq |\xi_{1,64}| \vee |\xi_{1,62}| \geq |\xi_{1,64}| \vee |\xi_{1,63}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,64}| \geq |\xi_{1,64}| \vee |\xi_{1,62}| \geq |\xi_{1,64}| \vee |\xi_{1,63}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,64}| \geq |\xi_{1,64}| \vee |\xi_{1,62}| \geq |\xi_{1,64}| \vee |\xi_{1,63}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,64}| \geq |\xi_{1,64}| \vee |\xi_{1,62}| \geq |\xi_{1,64}| \vee |\xi_{1,63}| \leq |\xi_{1,63}|\right) \wedge \left(|\xi_{1,64}| \geq |\xi_{1,64}| \vee |\xi_{1,62}| \leq |\xi_{1,64}| \vee |\xi_{1,63}| \leq |\xi_{1,63}|\right) \\ \frac{1}{2} & \text{for } |\xi_{1,64}| \geq |\xi_{1,64}| \wedge |\xi_{1,64}| \leq |\xi_{1,64}| \\ -\frac{1}{2} & \text{for } |\xi_{1,64}| \geq |\xi_{1,64}| \wedge |\xi_{1,64}| \leq |\xi_{1,64}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,71} = \begin{cases} -1 & \text{for } \xi_{1,64} < 0 \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 1

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$$\xi_{1,72} = \begin{cases} -1 & \text{for } \xi_{1,65} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{1,73} = \begin{cases} 1 & \text{for } \xi_{1,71} = 1 \wedge \xi_{1,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,74} = \begin{cases} 0 & \text{for } \xi_{1,71} = 1 \wedge \xi_{1,72} = 1 \\ 1 & \text{for } \xi_{1,71} = 1 \wedge \xi_{1,72} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,75} = \begin{cases} 0 & \text{for } \xi_{1,71} = 1 \wedge (\xi_{1,71} = 1 \vee \xi_{1,72} = -1) \wedge (\xi_{1,71} = 1 \vee \xi_{1,72} = 1) \wedge (\xi_{1,72} = -1 \vee \xi_{1,72} = 1) \\ 1 & \text{for } \xi_{1,71} = -1 \wedge \xi_{1,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,76} = \begin{cases} 0 & \text{for } (\xi_{1,71} = -1 \vee \xi_{1,71} = 1) \wedge (\xi_{1,71} = 1 \vee \xi_{1,72} = 1) \wedge (\xi_{1,72} = -1 \vee \xi_{1,72} = 1) \wedge (\xi_{1,71} = -1 \vee \xi_{1,71} = 1 \vee \xi_{1,72} = -1) \wedge (\xi_{1,71} = -1 \vee \xi_{1,71} = 1 \vee \xi_{1,72} = 1) \wedge (\xi_{1,71} = -1 \vee \xi_{1,72} = -1 \vee \xi_{1,72} = 1) \wedge (\xi_{1,71} = 1 \vee \xi_{1,72} = -1 \vee \xi_{1,72} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{1,77} = \cos\left(\frac{\pi\xi_{1,58}}{2}\right)$$

$$\xi_{1,78} = -\sin\left(\frac{\pi\xi_{1,58}}{2}\right)$$

$$\xi_{1,79} = \cos\left(\frac{\pi\xi_{1,59}}{2}\right)$$

$$\xi_{1,80} = -\sin\left(\frac{\pi\xi_{1,59}}{2}\right)$$

$$\xi_{1,81} = \xi_{1,77}\xi_{1,79}$$

$$\xi_{1,82} = \xi_{1,78}\xi_{1,79}$$

$$\xi_{1,83} = \xi_{1,77}\xi_{1,80}$$

$$\xi_{1,84} = -\xi_{1,78}\xi_{1,80}$$

$$\xi_{1,85} = \cos\left(\frac{\pi\xi_{1,60}}{2}\right)$$

$$\xi_{1,86} = -\sin\left(\frac{\pi\xi_{1,60}}{2}\right)$$

$$\xi_{1,87} = \xi_{1,81}\xi_{1,85} - \xi_{1,84}\xi_{1,86}$$

$$\xi_{1,88} = \xi_{1,82}\xi_{1,85} - \xi_{1,83}\xi_{1,86}$$

$$\xi_{1,89} = \xi_{1,82}\xi_{1,86} + \xi_{1,83}\xi_{1,85}$$

$$\xi_{1,90} = \xi_{1,81}\xi_{1,86} + \xi_{1,84}\xi_{1,85}$$

$$\xi_{1,91} = \xi_{1,73}\left(\xi_{1,67}\xi_{1,87} - \xi_{1,68}\xi_{1,88} - \xi_{1,69}\xi_{1,89} - \xi_{1,70}\xi_{1,90}\right) - \xi_{1,74}\left(\xi_{1,67}\xi_{1,88} + \xi_{1,68}\xi_{1,87} - \xi_{1,69}\xi_{1,90} + \xi_{1,70}\xi_{1,89}\right) - \xi_{1,75}\left(\xi_{1,67}\xi_{1,89} + \xi_{1,68}\xi_{1,90} + \xi_{1,69}\xi_{1,87} - \xi_{1,70}\xi_{1,88}\right) - \xi_{1,76}\left(\xi_{1,67}\xi_{1,90} - \xi_{1,68}\xi_{1,89} + \xi_{1,69}\xi_{1,88} + \xi_{1,70}\xi_{1,87}\right)$$

$$\xi_{1,92} = \xi_{1,73}\left(\xi_{1,67}\xi_{1,88} + \xi_{1,68}\xi_{1,87} - \xi_{1,69}\xi_{1,90} + \xi_{1,70}\xi_{1,89}\right) + \xi_{1,74}\left(\xi_{1,67}\xi_{1,87} - \xi_{1,68}\xi_{1,88} - \xi_{1,69}\xi_{1,89} - \xi_{1,70}\xi_{1,90}\right) - \xi_{1,75}\left(\xi_{1,67}\xi_{1,90} - \xi_{1,68}\xi_{1,89} + \xi_{1,69}\xi_{1,88} + \xi_{1,70}\xi_{1,87}\right) + \xi_{1,76}\left(\xi_{1,67}\xi_{1,89} + \xi_{1,68}\xi_{1,90} + \xi_{1,69}\xi_{1,87} - \xi_{1,70}\xi_{1,88}\right)$$

$$\xi_{1,93} = \xi_{1,73}\left(\xi_{1,67}\xi_{1,89} + \xi_{1,68}\xi_{1,90} + \xi_{1,69}\xi_{1,87} - \xi_{1,70}\xi_{1,88}\right) + \xi_{1,74}\left(\xi_{1,67}\xi_{1,90} - \xi_{1,68}\xi_{1,89} + \xi_{1,69}\xi_{1,88} + \xi_{1,70}\xi_{1,87}\right) + \xi_{1,75}\left(\xi_{1,67}\xi_{1,87} - \xi_{1,68}\xi_{1,88} - \xi_{1,69}\xi_{1,89} - \xi_{1,70}\xi_{1,90}\right) - \xi_{1,76}\left(\xi_{1,67}\xi_{1,88} + \xi_{1,68}\xi_{1,87} - \xi_{1,69}\xi_{1,90} + \xi_{1,70}\xi_{1,89}\right)$$

$$\xi_{1,94} = \xi_{1,73}\left(\xi_{1,67}\xi_{1,90} - \xi_{1,68}\xi_{1,89} + \xi_{1,69}\xi_{1,88} + \xi_{1,70}\xi_{1,87}\right) - \xi_{1,74}\left(\xi_{1,67}\xi_{1,89} + \xi_{1,68}\xi_{1,90} + \xi_{1,69}\xi_{1,87} - \xi_{1,70}\xi_{1,88}\right) + \xi_{1,75}\left(\xi_{1,67}\xi_{1,88} + \xi_{1,68}\xi_{1,87} - \xi_{1,69}\xi_{1,90} + \xi_{1,70}\xi_{1,89}\right) + \xi_{1,76}\left(\xi_{1,67}\xi_{1,87} - \xi_{1,68}\xi_{1,88} - \xi_{1,69}\xi_{1,89} - \xi_{1,70}\xi_{1,90}\right)$$

$$\xi_{1,95} = \xi_{1,67}\xi_{1,87} - \xi_{1,68}\xi_{1,88} - \xi_{1,69}\xi_{1,89} - \xi_{1,70}\xi_{1,90}$$

$$\xi_{1,96} = \xi_{1,67}\xi_{1,88} + \xi_{1,68}\xi_{1,87} - \xi_{1,69}\xi_{1,90} + \xi_{1,70}\xi_{1,89}$$

$$\xi_{1,97} = \xi_{1,67}\xi_{1,89} + \xi_{1,68}\xi_{1,90} + \xi_{1,69}\xi_{1,87} - \xi_{1,70}\xi_{1,88}$$

$$\xi_{1,98} = \xi_{1,67}\xi_{1,90} - \xi_{1,68}\xi_{1,89} + \xi_{1,69}\xi_{1,88} + \xi_{1,70}\xi_{1,87}$$

$$\xi_{1,99} = \xi_{1,67}\xi_{1,73} - \xi_{1,68}\xi_{1,74} - \xi_{1,69}\xi_{1,75} - \xi_{1,70}\xi_{1,76}$$

$$\xi_{1,100} = -\xi_{1,67}\xi_{1,74} - \xi_{1,68}\xi_{1,73} - \xi_{1,69}\xi_{1,76} + \xi_{1,70}\xi_{1,75}$$

$$\xi_{1,101} = -\xi_{1,67}\xi_{1,75} + \xi_{1,68}\xi_{1,76} - \xi_{1,69}\xi_{1,73} - \xi_{1,70}\xi_{1,74}$$

$$\xi_{1,102} = -\xi_{1,67}\xi_{1,76} - \xi_{1,68}\xi_{1,75} + \xi_{1,69}\xi_{1,74} - \xi_{1,70}\xi_{1,73}$$

$$\xi_{1,103} = \xi_{1,64}\xi_{1,71}$$

$$\xi_{1,104} = \xi_{1,65}\xi_{1,72}$$

$$\xi_{1,105} = \xi_{1,66}\xi_{1,71}\xi_{1,72}$$

$$\xi_{1,106} = -\frac{\pi(\xi_{1,58}+\xi_{1,59}+\xi_{1,60})}{2}$$

$$\xi_{1,107} = \cos\left(2\xi_{1,103}\right)$$

Exact shared-expression program 1

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$$\xi_{1,108} = \cos\left(2\xi_{1,104}\right)$$

$$\xi_{1,109} = \cos\left(2\xi_{1,105}\right)$$

$$\xi_{1,110} = \sqrt{\xi_{1,107}^2 - 2\xi_{1,107}\xi_{1,108}\xi_{1,109} + \xi_{1,108}^2 + \xi_{1,109}^2 - 1}$$

$$\xi_{1,111} = \xi_{1,107} - \xi_{1,108} + \xi_{1,109}$$

$$\xi_{1,112} = \sqrt{-\frac{\xi_{1,110}}{2} + \frac{\xi_{1,111}}{2} + \frac{1}{2}}$$

$$\xi_{1,113} = \sqrt{\frac{\xi_{1,110}}{2} - \frac{\xi_{1,111}}{2} + \frac{1}{2}}$$

$$\xi_{1,114} = \sqrt{-\frac{\xi_{1,110}}{2} - \frac{\xi_{1,111}}{2} + \frac{1}{2}}$$

$$\xi_{1,115} = \frac{(1-\xi_{1,108})(\xi_{1,107}+1)(\xi_{1,109}+1)}{2}$$

$$\xi_{1,116} = \begin{cases} -1 & \text{for } \xi_{1,105} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{1,117} = \begin{cases} 0 & \text{for } \xi_{1,112} = 0 \\ \frac{\sqrt{\xi_{1,115}\xi_{1,116}}}{\xi_{1,112}} & \text{otherwise} \end{cases}$$

$$\xi_{1,118} = \begin{cases} 1 & \text{for } \xi_{1,112} = 0 \\ \frac{\sqrt{\xi_{1,107}\xi_{1,108}\xi_{1,109}}}{\xi_{1,112}} & \text{otherwise} \end{cases}$$

$$\xi_{1,119} = 2\xi_{1,112}\xi_{1,113}\xi_{1,114}\xi_{1,118}$$

$$\xi_{1,120} = \sqrt{2}\xi_{1,118}\left(\xi_{1,112}^2\xi_{1,114} - \xi_{1,114}\right)$$

$$\xi_{1,121} = 2\xi_{1,112}\xi_{1,113}\xi_{1,114}\xi_{1,117}$$

$$\xi_{1,122} = \sqrt{2}\left(2\xi_{1,112}^2\xi_{1,114}\xi_{1,117} - \xi_{1,114}\xi_{1,117}\right)$$

$$\xi_{1,123} = \sqrt{2}\xi_{1,112}\xi_{1,113}\xi_{1,117}\xi_{1,118}$$

$$\xi_{1,124} = \sqrt{2}\xi_{1,112}\xi_{1,113}\left(-\xi_{1,114}^2 + \xi_{1,117}^2\right)$$

$$\xi_{1,125} = -\xi_{1,112}^2\xi_{1,114}^2 + \xi_{1,112}^2\xi_{1,117}^2 + \xi_{1,114}^2$$

$$\xi_{1,126} = \xi_{1,112}^2\xi_{1,114}^2 - \xi_{1,112}^2\xi_{1,117}^2 + \xi_{1,117}^2$$

$$\xi_{1,127} = 2\xi_{1,118}\left(\xi_{1,112}^2\xi_{1,117} - \frac{\xi_{1,117}}{2}\right)$$

$$\xi_{1,128} = -\sqrt{2}\xi_{1,112}\xi_{1,113}\xi_{1,117}\xi_{1,118}$$

$$\xi_{1,129} = -\xi_{1,117}\xi_{1,118}$$

$$\xi_{1,130} = \xi_{1,112}^2\xi_{1,114}^2 + \xi_{1,112}^2\xi_{1,117}^2 - \xi_{1,112}^2 - \xi_{1,114}^2 - \xi_{1,117}^2 + 1$$

$$\xi_{1,131} = -\sqrt{2}\left(\xi_{1,112}\xi_{1,113}\xi_{1,114}^2 + \xi_{1,112}\xi_{1,113}\xi_{1,117}^2 - \xi_{1,112}\xi_{1,113}\right)$$

$$\xi_{1,132} = \sqrt{2}\xi_{1,112}^2\xi_{1,114}\xi_{1,118}$$

$$\xi_{1,133} = \xi_{1,123}^2 - \xi_{1,126}^2 - \xi_{1,129}^2 - \xi_{1,132}^2$$

$$\xi_{1,134} = \xi_{1,123}\xi_{1,124} - \xi_{1,126}\xi_{1,127} - \xi_{1,129}\xi_{1,130}$$

$$\xi_{1,135} = \xi_{1,123}\xi_{1,125} - \xi_{1,126}\xi_{1,128} - \xi_{1,129}\xi_{1,131}$$

$$\xi_{1,136} = \xi_{1,119}^2 + \xi_{1,121}^2 + \xi_{1,124}^2 - \xi_{1,127}^2 - \xi_{1,130}^2$$

$$\xi_{1,137} = \xi_{1,119}\xi_{1,120} + \xi_{1,121}\xi_{1,122} + \xi_{1,124}\xi_{1,125} - \xi_{1,127}\xi_{1,128} - \xi_{1,130}\xi_{1,131}$$

$$\xi_{1,138} = \xi_{1,120}^2 + \xi_{1,122}^2 + \xi_{1,125}^2 - \xi_{1,128}^2 - \xi_{1,131}^2$$

$$\xi_{1,139} = \sin\left(2\xi_{1,104}\right)\sin\left(2\xi_{1,105}\right)\cos\left(2\xi_{1,103}\right)$$

$$\xi_{1,140} = \sin\left(2\xi_{1,103}\right)\sin\left(2\xi_{1,105}\right)\cos\left(2\xi_{1,104}\right)$$

$$\xi_{1,141} = \sin\left(2\xi_{1,103}\right)\sin\left(2\xi_{1,104}\right)\cos\left(2\xi_{1,105}\right)$$

$$\xi_{1,142} = \sin\left(2\xi_{1,103}\right)\cos\left(2\xi_{1,104}\right)\cos\left(2\xi_{1,105}\right)$$

$$\xi_{1,143} = \sin\left(2\xi_{1,104}\right)\cos\left(2\xi_{1,103}\right)\cos\left(2\xi_{1,105}\right)$$

Exact shared-expression program 1

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$$\xi_{1,144} = \sin\left(2\xi_{1,105}\right)\cos\left(2\xi_{1,103}\right)\cos\left(2\xi_{1,104}\right)$$

$$\xi_{1,145} = \xi_{1,139}^2 - \xi_{1,142}^2$$

$$\xi_{1,146} = \xi_{1,140}^2 - \xi_{1,143}^2$$

$$\xi_{1,147} = \xi_{1,141}^2 - \xi_{1,144}^2$$

$$\xi_{1,148} = \frac{\xi_{1,133} - \xi_{1,146}}{\xi_{1,145} - \xi_{1,146}}$$

$$\xi_{1,149} = \frac{\xi_{1,134}}{\xi_{1,145} - \xi_{1,146}}$$

$$\xi_{1,150} = \frac{\xi_{1,135}}{\xi_{1,145} - \xi_{1,146}}$$

$$\xi_{1,151} = \frac{\xi_{1,136} - \xi_{1,146}}{\xi_{1,145} - \xi_{1,146}}$$

$$\xi_{1,152} = \frac{\xi_{1,137}}{\xi_{1,145} - \xi_{1,146}}$$

$$\xi_{1,153} = \frac{\xi_{1,138} - \xi_{1,146}}{\xi_{1,145} - \xi_{1,146}}$$

$$\xi_{1,154} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,148} & \text{otherwise} \end{cases}$$

$$\xi_{1,155} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,149} & \text{otherwise} \end{cases}$$

$$\xi_{1,156} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,150} & \text{otherwise} \end{cases}$$

$$\xi_{1,157} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,151} & \text{otherwise} \end{cases}$$

$$\xi_{1,158} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,152} & \text{otherwise} \end{cases}$$

$$\xi_{1,159} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,153} & \text{otherwise} \end{cases}$$

$$\xi_{1,160} = \frac{\xi_{1,133} - \xi_{1,147}}{\xi_{1,145} - \xi_{1,147}}$$

$$\xi_{1,161} = \frac{\xi_{1,134}}{\xi_{1,145} - \xi_{1,147}}$$

$$\xi_{1,162} = \frac{\xi_{1,135}}{\xi_{1,145} - \xi_{1,147}}$$

$$\xi_{1,163} = \frac{\xi_{1,136} - \xi_{1,147}}{\xi_{1,145} - \xi_{1,147}}$$

$$\xi_{1,164} = \frac{\xi_{1,137}}{\xi_{1,145} - \xi_{1,147}}$$

$$\xi_{1,165} = \frac{\xi_{1,138} - \xi_{1,147}}{\xi_{1,145} - \xi_{1,147}}$$

$$\xi_{1,166} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,147} \\ \xi_{1,160} & \text{otherwise} \end{cases}$$

$$\xi_{1,167} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,147} \\ \xi_{1,161} & \text{otherwise} \end{cases}$$

$$\xi_{1,168} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,147} \\ \xi_{1,162} & \text{otherwise} \end{cases}$$

$$\xi_{1,169} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,147} \\ \xi_{1,163} & \text{otherwise} \end{cases}$$

$$\xi_{1,170} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,147} \\ \xi_{1,164} & \text{otherwise} \end{cases}$$

$$\xi_{1,171} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,147} \\ \xi_{1,165} & \text{otherwise} \end{cases}$$

$$\xi_{1,172} = \xi_{1,154}\xi_{1,166} + \xi_{1,155}\xi_{1,167} + \xi_{1,156}\xi_{1,168}$$

$$\xi_{1,173} = \xi_{1,154}\xi_{1,167} + \xi_{1,155}\xi_{1,169} + \xi_{1,156}\xi_{1,170}$$

$$\xi_{1,174} = \xi_{1,154}\xi_{1,168} + \xi_{1,155}\xi_{1,170} + \xi_{1,156}\xi_{1,171}$$

$$\xi_{1,175} = \xi_{1,155}\xi_{1,166} + \xi_{1,157}\xi_{1,167} + \xi_{1,158}\xi_{1,168}$$

$$\xi_{1,176} = \xi_{1,155}\xi_{1,167} + \xi_{1,157}\xi_{1,169} + \xi_{1,158}\xi_{1,170}$$

$$\xi_{1,177} = \xi_{1,155}\xi_{1,168} + \xi_{1,157}\xi_{1,170} + \xi_{1,158}\xi_{1,171}$$

$$\xi_{1,178} = \xi_{1,156}\xi_{1,166} + \xi_{1,158}\xi_{1,167} + \xi_{1,159}\xi_{1,168}$$

$$\xi_{1,179} = \xi_{1,156}\xi_{1,167} + \xi_{1,158}\xi_{1,169} + \xi_{1,159}\xi_{1,170}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,180} = \xi_{1,156}\xi_{1,168} + \xi_{1,158}\xi_{1,170} + \xi_{1,159}\xi_{1,171}$$

$$\xi_{1,181} = \sqrt{\xi_{1,172}}$$

$$\xi_{1,182} = \frac{\xi_{1,172}}{\xi_{1,181}}$$

$$\xi_{1,183} = \frac{\xi_{1,175}}{\xi_{1,181}}$$

$$\xi_{1,184} = \frac{\xi_{1,178}}{\xi_{1,181}}$$

$$\xi_{1,185} = \sqrt{\xi_{1,176}}$$

$$\xi_{1,186} = \frac{\xi_{1,173}}{\xi_{1,185}}$$

$$\xi_{1,187} = \frac{\xi_{1,176}}{\xi_{1,185}}$$

$$\xi_{1,188} = \frac{\xi_{1,179}}{\xi_{1,185}}$$

$$\xi_{1,189} = \sqrt{\xi_{1,180}}$$

$$\xi_{1,190} = \frac{\xi_{1,174}}{\xi_{1,189}}$$

$$\xi_{1,191} = \frac{\xi_{1,177}}{\xi_{1,189}}$$

$$\xi_{1,192} = \frac{\xi_{1,180}}{\xi_{1,189}}$$

$$\xi_{1,193} = \begin{cases} \xi_{1,182} & \text{for } \xi_{1,172} \geq \xi_{1,176} \wedge \xi_{1,172} \geq \xi_{1,180} \\ \xi_{1,186} & \text{for } \xi_{1,176} \geq \xi_{1,180} \wedge \xi_{1,172} \leq \xi_{1,176} \\ \xi_{1,190} & \text{otherwise} \end{cases}$$

$$\xi_{1,194} = \begin{cases} \xi_{1,183} & \text{for } \xi_{1,172} \geq \xi_{1,176} \wedge \xi_{1,172} \geq \xi_{1,180} \\ \xi_{1,187} & \text{for } \xi_{1,176} \geq \xi_{1,180} \wedge \xi_{1,172} \leq \xi_{1,176} \\ \xi_{1,191} & \text{otherwise} \end{cases}$$

$$\xi_{1,195} = \begin{cases} \xi_{1,184} & \text{for } \xi_{1,172} \geq \xi_{1,176} \wedge \xi_{1,172} \geq \xi_{1,180} \\ \xi_{1,188} & \text{for } \xi_{1,176} \geq \xi_{1,180} \wedge \xi_{1,172} \leq \xi_{1,176} \\ \xi_{1,192} & \text{otherwise} \end{cases}$$

$$\xi_{1,196} = \frac{\xi_{1,133} - \xi_{1,145}}{-\xi_{1,145} + \xi_{1,146}}$$

$$\xi_{1,197} = \frac{\xi_{1,134}}{-\xi_{1,145} + \xi_{1,146}}$$

$$\xi_{1,198} = \frac{\xi_{1,135}}{-\xi_{1,145} + \xi_{1,146}}$$

$$\xi_{1,199} = \frac{\xi_{1,136} - \xi_{1,145}}{-\xi_{1,145} + \xi_{1,146}}$$

$$\xi_{1,200} = \frac{\xi_{1,137}}{-\xi_{1,145} + \xi_{1,146}}$$

$$\xi_{1,201} = \frac{\xi_{1,138} - \xi_{1,145}}{-\xi_{1,145} + \xi_{1,146}}$$

$$\xi_{1,202} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,196} & \text{otherwise} \end{cases}$$

$$\xi_{1,203} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,197} & \text{otherwise} \end{cases}$$

$$\xi_{1,204} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,198} & \text{otherwise} \end{cases}$$

$$\xi_{1,205} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,199} & \text{otherwise} \end{cases}$$

$$\xi_{1,206} = \begin{cases} 0 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,200} & \text{otherwise} \end{cases}$$

$$\xi_{1,207} = \begin{cases} 1 & \text{for } \xi_{1,145} = \xi_{1,146} \\ \xi_{1,201} & \text{otherwise} \end{cases}$$

$$\xi_{1,208} = \frac{\xi_{1,133} - \xi_{1,147}}{\xi_{1,146} - \xi_{1,147}}$$

$$\xi_{1,209} = \frac{\xi_{1,134}}{\xi_{1,146} - \xi_{1,147}}$$

$$\xi_{1,210} = \frac{\xi_{1,135}}{\xi_{1,146} - \xi_{1,147}}$$

$$\xi_{1,211} = \frac{\xi_{1,136} - \xi_{1,147}}{\xi_{1,146} - \xi_{1,147}}$$

$$\xi_{1,212} = \frac{\xi_{1,137}}{\xi_{1,146} - \xi_{1,147}}$$

$$\xi_{1,213} = \frac{\xi_{1,138} - \xi_{1,147}}{\xi_{1,146} - \xi_{1,147}}$$

$$\xi_{1,214} = \begin{cases} 1 & \text{for } \xi_{1,146} = \xi_{1,147} \\ \xi_{1,208} & \text{otherwise} \end{cases}$$

$$\xi_{1,215} = \begin{cases} 0 & \text{for } \xi_{1,146} = \xi_{1,147} \\ \xi_{1,209} & \text{otherwise} \end{cases}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,216} = \begin{cases} 0 & \text{for } \xi_{1,146} = \xi_{1,147} \\ \xi_{1,210} & \text{otherwise} \end{cases}$$

$$\xi_{1,217} = \begin{cases} 1 & \text{for } \xi_{1,146} = \xi_{1,147} \\ \xi_{1,211} & \text{otherwise} \end{cases}$$

$$\xi_{1,218} = \begin{cases} 0 & \text{for } \xi_{1,146} = \xi_{1,147} \\ \xi_{1,212} & \text{otherwise} \end{cases}$$

$$\xi_{1,219} = \begin{cases} 1 & \text{for } \xi_{1,146} = \xi_{1,147} \\ \xi_{1,213} & \text{otherwise} \end{cases}$$

$$\xi_{1,220} = \xi_{1,202}\xi_{1,214} + \xi_{1,203}\xi_{1,215} + \xi_{1,204}\xi_{1,216}$$

$$\xi_{1,221} = \xi_{1,202}\xi_{1,215} + \xi_{1,203}\xi_{1,217} + \xi_{1,204}\xi_{1,218}$$

$$\xi_{1,222} = \xi_{1,202}\xi_{1,216} + \xi_{1,203}\xi_{1,218} + \xi_{1,204}\xi_{1,219}$$

$$\xi_{1,223} = \xi_{1,203}\xi_{1,214} + \xi_{1,205}\xi_{1,215} + \xi_{1,206}\xi_{1,216}$$

$$\xi_{1,224} = \xi_{1,203}\xi_{1,215} + \xi_{1,205}\xi_{1,217} + \xi_{1,206}\xi_{1,218}$$

$$\xi_{1,225} = \xi_{1,203}\xi_{1,216} + \xi_{1,205}\xi_{1,218} + \xi_{1,206}\xi_{1,219}$$

$$\xi_{1,226} = \xi_{1,204}\xi_{1,214} + \xi_{1,206}\xi_{1,215} + \xi_{1,207}\xi_{1,216}$$

$$\xi_{1,227} = \xi_{1,204}\xi_{1,215} + \xi_{1,206}\xi_{1,217} + \xi_{1,207}\xi_{1,218}$$

$$\xi_{1,228} = \xi_{1,204}\xi_{1,216} + \xi_{1,206}\xi_{1,218} + \xi_{1,207}\xi_{1,219}$$

$$\xi_{1,229} = \begin{cases} \xi_{1,193}^2 & \text{for } \xi_{1,145} = \xi_{1,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,230} = \begin{cases} \xi_{1,193}\xi_{1,194} & \text{for } \xi_{1,145} = \xi_{1,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,231} = \begin{cases} \xi_{1,193}\xi_{1,195} & \text{for } \xi_{1,145} = \xi_{1,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,232} = \begin{cases} \xi_{1,194}^2 & \text{for } \xi_{1,145} = \xi_{1,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,233} = \begin{cases} \xi_{1,194}\xi_{1,195} & \text{for } \xi_{1,145} = \xi_{1,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,234} = \begin{cases} \xi_{1,195}^2 & \text{for } \xi_{1,145} = \xi_{1,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{1,235} = \xi_{1,220} - \xi_{1,229}$$

$$\xi_{1,236} = \xi_{1,221} - \xi_{1,230}$$

$$\xi_{1,237} = \xi_{1,222} - \xi_{1,231}$$

$$\xi_{1,238} = \xi_{1,223} - \xi_{1,230}$$

$$\xi_{1,239} = \xi_{1,224} - \xi_{1,232}$$

$$\xi_{1,240} = \xi_{1,225} - \xi_{1,233}$$

$$\xi_{1,241} = \xi_{1,226} - \xi_{1,231}$$

$$\xi_{1,242} = \xi_{1,227} - \xi_{1,233}$$

$$\xi_{1,243} = \xi_{1,228} - \xi_{1,234}$$

$$\xi_{1,244} = \sqrt{\xi_{1,235}}$$

$$\xi_{1,245} = \frac{\xi_{1,235}}{\xi_{1,244}}$$

$$\xi_{1,246} = \frac{\xi_{1,238}}{\xi_{1,244}}$$

$$\xi_{1,247} = \frac{\xi_{1,241}}{\xi_{1,244}}$$

$$\xi_{1,248} = \sqrt{\xi_{1,239}}$$

$$\xi_{1,249} = \frac{\xi_{1,236}}{\xi_{1,248}}$$

$$\xi_{1,250} = \frac{\xi_{1,239}}{\xi_{1,248}}$$

$$\xi_{1,251} = \frac{\xi_{1,242}}{\xi_{1,248}}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,252} = \sqrt{\xi_{1,243}}$$

$$\xi_{1,253} = \frac{\xi_{1,237}}{\xi_{1,252}}$$

$$\xi_{1,254} = \frac{\xi_{1,240}}{\xi_{1,252}}$$

$$\xi_{1,255} = \frac{\xi_{1,243}}{\xi_{1,252}}$$

$$\xi_{1,256} = \begin{cases} \xi_{1,245} & \text{for } \xi_{1,235} \geq \xi_{1,239} \wedge \xi_{1,235} \geq \xi_{1,243} \\ \xi_{1,249} & \text{for } \xi_{1,239} \geq \xi_{1,243} \wedge \xi_{1,235} \leq \xi_{1,239} \\ \xi_{1,253} & \text{otherwise} \end{cases}$$

$$\xi_{1,257} = \begin{cases} \xi_{1,246} & \text{for } \xi_{1,235} \geq \xi_{1,239} \wedge \xi_{1,235} \geq \xi_{1,243} \\ \xi_{1,250} & \text{for } \xi_{1,239} \geq \xi_{1,243} \wedge \xi_{1,235} \leq \xi_{1,239} \\ \xi_{1,254} & \text{otherwise} \end{cases}$$

$$\xi_{1,258} = \begin{cases} \xi_{1,247} & \text{for } \xi_{1,235} \geq \xi_{1,239} \wedge \xi_{1,235} \geq \xi_{1,243} \\ \xi_{1,251} & \text{for } \xi_{1,239} \geq \xi_{1,243} \wedge \xi_{1,235} \leq \xi_{1,239} \\ \xi_{1,255} & \text{otherwise} \end{cases}$$

$$\xi_{1,259} = \xi_{1,194}\xi_{1,258} - \xi_{1,195}\xi_{1,257}$$

$$\xi_{1,260} = -\xi_{1,193}\xi_{1,258} + \xi_{1,195}\xi_{1,256}$$

$$\xi_{1,261} = \xi_{1,193}\xi_{1,257} - \xi_{1,194}\xi_{1,256}$$

$$\xi_{1,262} = \xi_{1,139}^2 + \xi_{1,142}^2$$

$$\xi_{1,263} = \xi_{1,140}^2 + \xi_{1,143}^2$$

$$\xi_{1,264} = \xi_{1,141}^2 + \xi_{1,144}^2$$

$$\xi_{1,265} = \frac{\xi_{1,126}\xi_{1,142}\xi_{1,193} + \xi_{1,194}(\xi_{1,119}\xi_{1,139} + \xi_{1,127}\xi_{1,142}) + \xi_{1,195}(\xi_{1,120}\xi_{1,139} + \xi_{1,128}\xi_{1,142})}{\xi_{1,262}}$$

$$\xi_{1,266} = \frac{\xi_{1,129}\xi_{1,142}\xi_{1,193} + \xi_{1,194}(\xi_{1,121}\xi_{1,139} + \xi_{1,130}\xi_{1,142}) + \xi_{1,195}(\xi_{1,122}\xi_{1,139} + \xi_{1,131}\xi_{1,142})}{\xi_{1,262}}$$

$$\xi_{1,267} = \frac{\xi_{1,124}\xi_{1,139}\xi_{1,194} + \xi_{1,125}\xi_{1,139}\xi_{1,195} + \xi_{1,193}(\xi_{1,123}\xi_{1,139} + \xi_{1,132}\xi_{1,142})}{\xi_{1,262}}$$

$$\xi_{1,268} = \frac{\xi_{1,126}\xi_{1,143}\xi_{1,256} + \xi_{1,257}(\xi_{1,119}\xi_{1,140} + \xi_{1,127}\xi_{1,143}) + \xi_{1,258}(\xi_{1,120}\xi_{1,140} + \xi_{1,128}\xi_{1,143})}{\xi_{1,263}}$$

$$\xi_{1,269} = \frac{\xi_{1,129}\xi_{1,143}\xi_{1,256} + \xi_{1,257}(\xi_{1,121}\xi_{1,140} + \xi_{1,130}\xi_{1,143}) + \xi_{1,258}(\xi_{1,122}\xi_{1,140} + \xi_{1,131}\xi_{1,143})}{\xi_{1,263}}$$

$$\xi_{1,270} = \frac{\xi_{1,124}\xi_{1,140}\xi_{1,257} + \xi_{1,125}\xi_{1,140}\xi_{1,258} + \xi_{1,256}(\xi_{1,123}\xi_{1,140} + \xi_{1,132}\xi_{1,143})}{\xi_{1,263}}$$

$$\xi_{1,271} = \frac{\xi_{1,126}\xi_{1,144}\xi_{1,259} + \xi_{1,260}(\xi_{1,119}\xi_{1,141} + \xi_{1,127}\xi_{1,144}) + \xi_{1,261}(\xi_{1,120}\xi_{1,141} + \xi_{1,128}\xi_{1,144})}{\xi_{1,264}}$$

$$\xi_{1,272} = \frac{\xi_{1,129}\xi_{1,144}\xi_{1,259} + \xi_{1,260}(\xi_{1,121}\xi_{1,141} + \xi_{1,130}\xi_{1,144}) + \xi_{1,261}(\xi_{1,122}\xi_{1,141} + \xi_{1,131}\xi_{1,144})}{\xi_{1,264}}$$

$$\xi_{1,273} = \frac{\xi_{1,124}\xi_{1,141}\xi_{1,260} + \xi_{1,125}\xi_{1,141}\xi_{1,261} + \xi_{1,259}(\xi_{1,123}\xi_{1,141} + \xi_{1,132}\xi_{1,144})}{\xi_{1,264}}$$

$$\xi_{1,274} = 1 - \xi_{1,265}^2$$

$$\xi_{1,275} = -\xi_{1,265}\xi_{1,266}$$

$$\xi_{1,276} = -\xi_{1,265}\xi_{1,267}$$

$$\xi_{1,277} = 1 - \xi_{1,266}^2$$

$$\xi_{1,278} = -\xi_{1,266}\xi_{1,267}$$

$$\xi_{1,279} = 1 - \xi_{1,267}^2$$

$$\xi_{1,280} = \xi_{1,274}^2 + \xi_{1,275}^2 + \xi_{1,276}^2$$

$$\xi_{1,281} = \xi_{1,275}^2 + \xi_{1,277}^2 + \xi_{1,278}^2$$

$$\xi_{1,282} = \xi_{1,276}^2 + \xi_{1,278}^2 + \xi_{1,279}^2$$

$$\xi_{1,283} = \sqrt{\xi_{1,280}}$$

$$\xi_{1,284} = \frac{\xi_{1,274}}{\xi_{1,283}}$$

$$\xi_{1,285} = \frac{\xi_{1,275}}{\xi_{1,283}}$$

$$\xi_{1,286} = \frac{\xi_{1,276}}{\xi_{1,283}}$$

$$\xi_{1,287} = \sqrt{\xi_{1,281}}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,288} = \frac{\xi_{1,277}}{\xi_{1,287}}$$

$$\xi_{1,289} = \frac{\xi_{1,275}}{\xi_{1,287}}$$

$$\xi_{1,290} = \frac{\xi_{1,278}}{\xi_{1,287}}$$

$$\xi_{1,291} = \sqrt{\xi_{1,282}}$$

$$\xi_{1,292} = \frac{\xi_{1,279}}{\xi_{1,291}}$$

$$\xi_{1,293} = \frac{\xi_{1,276}}{\xi_{1,291}}$$

$$\xi_{1,294} = \frac{\xi_{1,278}}{\xi_{1,291}}$$

$$\xi_{1,295} = \begin{cases} \xi_{1,284} & \text{for } \xi_{1,280} \geq \xi_{1,281} \wedge \xi_{1,280} \geq \xi_{1,282} \\ \xi_{1,289} & \text{for } \xi_{1,281} \geq \xi_{1,282} \wedge \xi_{1,280} \leq \xi_{1,281} \\ \xi_{1,293} & \text{otherwise} \end{cases}$$

$$\xi_{1,296} = \begin{cases} \xi_{1,285} & \text{for } \xi_{1,280} \geq \xi_{1,281} \wedge \xi_{1,280} \geq \xi_{1,282} \\ \xi_{1,288} & \text{for } \xi_{1,281} \geq \xi_{1,282} \wedge \xi_{1,280} \leq \xi_{1,281} \\ \xi_{1,294} & \text{otherwise} \end{cases}$$

$$\xi_{1,297} = \begin{cases} \xi_{1,286} & \text{for } \xi_{1,280} \geq \xi_{1,281} \wedge \xi_{1,280} \geq \xi_{1,282} \\ \xi_{1,290} & \text{for } \xi_{1,281} \geq \xi_{1,282} \wedge \xi_{1,280} \leq \xi_{1,281} \\ \xi_{1,292} & \text{otherwise} \end{cases}$$

$$\xi_{1,298} = \xi_{1,266}\xi_{1,297} - \xi_{1,267}\xi_{1,296}$$

$$\xi_{1,299} = -\xi_{1,265}\xi_{1,297} + \xi_{1,267}\xi_{1,295}$$

$$\xi_{1,300} = \xi_{1,265}\xi_{1,296} - \xi_{1,266}\xi_{1,295}$$

$$\xi_{1,301} = 1 - \xi_{1,268}^2$$

$$\xi_{1,302} = -\xi_{1,268}\xi_{1,269}$$

$$\xi_{1,303} = -\xi_{1,268}\xi_{1,270}$$

$$\xi_{1,304} = 1 - \xi_{1,269}^2$$

$$\xi_{1,305} = -\xi_{1,269}\xi_{1,270}$$

$$\xi_{1,306} = 1 - \xi_{1,270}^2$$

$$\xi_{1,307} = \xi_{1,301}^2 + \xi_{1,302}^2 + \xi_{1,303}^2$$

$$\xi_{1,308} = \xi_{1,302}^2 + \xi_{1,304}^2 + \xi_{1,305}^2$$

$$\xi_{1,309} = \xi_{1,303}^2 + \xi_{1,305}^2 + \xi_{1,306}^2$$

$$\xi_{1,310} = \sqrt{\xi_{1,307}}$$

$$\xi_{1,311} = \frac{\xi_{1,301}}{\xi_{1,310}}$$

$$\xi_{1,312} = \frac{\xi_{1,302}}{\xi_{1,310}}$$

$$\xi_{1,313} = \frac{\xi_{1,303}}{\xi_{1,310}}$$

$$\xi_{1,314} = \sqrt{\xi_{1,308}}$$

$$\xi_{1,315} = \frac{\xi_{1,304}}{\xi_{1,314}}$$

$$\xi_{1,316} = \frac{\xi_{1,302}}{\xi_{1,314}}$$

$$\xi_{1,317} = \frac{\xi_{1,305}}{\xi_{1,314}}$$

$$\xi_{1,318} = \sqrt{\xi_{1,309}}$$

$$\xi_{1,319} = \frac{\xi_{1,306}}{\xi_{1,318}}$$

$$\xi_{1,320} = \frac{\xi_{1,303}}{\xi_{1,318}}$$

$$\xi_{1,321} = \frac{\xi_{1,305}}{\xi_{1,318}}$$

$$\xi_{1,322} = \begin{cases} \xi_{1,311} & \text{for } \xi_{1,307} \geq \xi_{1,308} \wedge \xi_{1,307} \geq \xi_{1,309} \\ \xi_{1,316} & \text{for } \xi_{1,308} \geq \xi_{1,309} \wedge \xi_{1,307} \leq \xi_{1,308} \\ \xi_{1,320} & \text{otherwise} \end{cases}$$

$$\xi_{1,323} = \begin{cases} \xi_{1,312} & \text{for } \xi_{1,307} \geq \xi_{1,308} \wedge \xi_{1,307} \geq \xi_{1,309} \\ \xi_{1,315} & \text{for } \xi_{1,308} \geq \xi_{1,309} \wedge \xi_{1,307} \leq \xi_{1,308} \\ \xi_{1,321} & \text{otherwise} \end{cases}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{1,324} = \begin{cases} \xi_{1,313} & \text{for } \xi_{1,307} \geq \xi_{1,308} \wedge \xi_{1,307} \geq \xi_{1,309} \\ \xi_{1,317} & \text{for } \xi_{1,308} \geq \xi_{1,309} \wedge \xi_{1,307} \leq \xi_{1,308} \\ \xi_{1,319} & \text{otherwise} \end{cases}$$

$$\xi_{1,325} = \xi_{1,269}\xi_{1,324} - \xi_{1,270}\xi_{1,323}$$

$$\xi_{1,326} = -\xi_{1,268}\xi_{1,324} + \xi_{1,270}\xi_{1,322}$$

$$\xi_{1,327} = \xi_{1,268}\xi_{1,323} - \xi_{1,269}\xi_{1,322}$$

$$\xi_{1,328} = \xi_{1,266}\xi_{1,270} - \xi_{1,267}\xi_{1,269}$$

$$\xi_{1,329} = -\xi_{1,265}\xi_{1,270} + \xi_{1,267}\xi_{1,268}$$

$$\xi_{1,330} = \xi_{1,265}\xi_{1,269} - \xi_{1,266}\xi_{1,268}$$

$$\xi_{1,331} = 1 - \xi_{1,271}^2$$

$$\xi_{1,332} = -\xi_{1,271}\xi_{1,272}$$

$$\xi_{1,333} = -\xi_{1,271}\xi_{1,273}$$

$$\xi_{1,334} = 1 - \xi_{1,272}^2$$

$$\xi_{1,335} = -\xi_{1,272}\xi_{1,273}$$

$$\xi_{1,336} = 1 - \xi_{1,273}^2$$

$$\xi_{1,337} = \xi_{1,331}^2 + \xi_{1,332}^2 + \xi_{1,333}^2$$

$$\xi_{1,338} = \xi_{1,332}^2 + \xi_{1,334}^2 + \xi_{1,335}^2$$

$$\xi_{1,339} = \xi_{1,333}^2 + \xi_{1,335}^2 + \xi_{1,336}^2$$

$$\xi_{1,340} = \sqrt{\xi_{1,337}}$$

$$\xi_{1,341} = \frac{\xi_{1,331}}{\xi_{1,340}}$$

$$\xi_{1,342} = \frac{\xi_{1,332}}{\xi_{1,340}}$$

$$\xi_{1,343} = \frac{\xi_{1,333}}{\xi_{1,340}}$$

$$\xi_{1,344} = \sqrt{\xi_{1,338}}$$

$$\xi_{1,345} = \frac{\xi_{1,334}}{\xi_{1,344}}$$

$$\xi_{1,346} = \frac{\xi_{1,332}}{\xi_{1,344}}$$

$$\xi_{1,347} = \frac{\xi_{1,335}}{\xi_{1,344}}$$

$$\xi_{1,348} = \sqrt{\xi_{1,339}}$$

$$\xi_{1,349} = \frac{\xi_{1,336}}{\xi_{1,348}}$$

$$\xi_{1,350} = \frac{\xi_{1,333}}{\xi_{1,348}}$$

$$\xi_{1,351} = \frac{\xi_{1,335}}{\xi_{1,348}}$$

$$\xi_{1,352} = \begin{cases} \xi_{1,341} & \text{for } \xi_{1,337} \geq \xi_{1,338} \wedge \xi_{1,337} \geq \xi_{1,339} \\ \xi_{1,346} & \text{for } \xi_{1,338} \geq \xi_{1,339} \wedge \xi_{1,337} \leq \xi_{1,338} \\ \xi_{1,350} & \text{otherwise} \end{cases}$$

$$\xi_{1,353} = \begin{cases} \xi_{1,342} & \text{for } \xi_{1,337} \geq \xi_{1,338} \wedge \xi_{1,337} \geq \xi_{1,339} \\ \xi_{1,345} & \text{for } \xi_{1,338} \geq \xi_{1,339} \wedge \xi_{1,337} \leq \xi_{1,338} \\ \xi_{1,351} & \text{otherwise} \end{cases}$$

$$\xi_{1,354} = \begin{cases} \xi_{1,343} & \text{for } \xi_{1,337} \geq \xi_{1,338} \wedge \xi_{1,337} \geq \xi_{1,339} \\ \xi_{1,347} & \text{for } \xi_{1,338} \geq \xi_{1,339} \wedge \xi_{1,337} \leq \xi_{1,338} \\ \xi_{1,349} & \text{otherwise} \end{cases}$$

$$\xi_{1,355} = \xi_{1,272}\xi_{1,354} - \xi_{1,273}\xi_{1,353}$$

$$\xi_{1,356} = -\xi_{1,271}\xi_{1,354} + \xi_{1,273}\xi_{1,352}$$

$$\xi_{1,357} = \xi_{1,271}\xi_{1,353} - \xi_{1,272}\xi_{1,352}$$

$$\xi_{1,358} = -\xi_{1,266}\xi_{1,273} + \xi_{1,267}\xi_{1,272}$$

$$\xi_{1,359} = \xi_{1,265}\xi_{1,273} - \xi_{1,267}\xi_{1,271}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,360} = -\xi_{1,265}\xi_{1,272} + \xi_{1,266}\xi_{1,271}$$

$$\xi_{1,361} = \xi_{1,269}\xi_{1,273} - \xi_{1,270}\xi_{1,272}$$

$$\xi_{1,362} = -\xi_{1,268}\xi_{1,273} + \xi_{1,270}\xi_{1,271}$$

$$\xi_{1,363} = \xi_{1,268}\xi_{1,272} - \xi_{1,269}\xi_{1,271}$$

$$\xi_{1,364} = \begin{cases} 1 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,265} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,325} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,265} & \text{for } \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,352} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,265} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,361} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,265} & \text{for } \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,365} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,295} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,268} & \text{for } \xi_{1,264} = 0 \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} = 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,262} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge \xi_{1,263} \neq 0 \\ \xi_{1,355} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,358} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,268} & \text{for } (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,263} \neq 0 \vee \xi_{1,264} \neq 0) \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,366} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,266} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,326} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,266} & \text{for } \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,353} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,266} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,362} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,266} & \text{for } \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,367} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,296} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,269} & \text{for } \xi_{1,264} = 0 \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} = 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,262} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge \xi_{1,263} \neq 0 \\ \xi_{1,356} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,359} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,269} & \text{for } (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,263} \neq 0 \vee \xi_{1,264} \neq 0) \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,368} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,267} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,327} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,354} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,363} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,369} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,297} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,270} & \text{for } \xi_{1,264} = 0 \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} = 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,262} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge \xi_{1,263} \neq 0 \\ \xi_{1,357} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,360} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,270} & \text{for } (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,263} \neq 0 \vee \xi_{1,264} \neq 0) \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,370} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,267} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,327} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,354} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,363} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,371} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,297} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,270} & \text{for } \xi_{1,264} = 0 \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} = 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,262} \neq 0) \wedge (\xi_{1,264} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge \xi_{1,263} \neq 0 \\ \xi_{1,357} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,360} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,270} & \text{for } (\xi_{1,262} = 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} = 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,263} \neq 0) \wedge (\xi_{1,262} \neq 0 \vee \xi_{1,264} \neq 0) \wedge (\xi_{1,263} \neq 0 \vee \xi_{1,264} \neq 0) \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,372} = \begin{cases} 0 & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \\ \xi_{1,267} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \\ \xi_{1,327} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,264} = 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,264} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \\ \xi_{1,354} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} = 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,263} = 0 \wedge \xi_{1,262} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,363} & \text{for } \xi_{1,262} = 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \\ \xi_{1,267} & \text{for } \xi_{1,262} \neq 0 \wedge \xi_{1,263} \neq 0 \wedge \xi_{1,264} \neq 0 \end{cases}$$

$$\xi_{1,373} = \frac{\xi_{1,364}}{4} + \frac{\xi_{1,368}}{4} + \frac{\xi_{1,372}}{4} + \frac{1}{4}$$

$$\xi_{1,374} = \frac{\xi_{1,364}}{4} - \frac{\xi_{1,368}}{4} - \frac{\xi_{1,372}}{4} + \frac{1}{4}$$

$$\xi_{1,375} = -\frac{\xi_{1,364}}{4} + \frac{\xi_{1,368}}{4} - \frac{\xi_{1,372}}{4} + \frac{1}{4}$$

$$\xi_{1,376} = -\frac{\xi_{1,364}}{4} - \frac{\xi_{1,368}}{4} + \frac{\xi_{1,372}}{4} + \frac{1}{4}$$

$$\xi_{1,377} = \sqrt{\xi_{1,373}}$$

$$\xi_{1,378} = \frac{-\xi_{1,369}+\xi_{1,371}}{4\xi_{1,377}}$$

$$\xi_{1,379} = \frac{\xi_{1,366}-\xi_{1,370}}{4\xi_{1,377}}$$

$$\xi_{1,380} = \frac{-\xi_{1,365}+\xi_{1,367}}{4\xi_{1,377}}$$

$$\xi_{1,381} = \sqrt{\xi_{1,374}}$$

$$\xi_{1,382} = \frac{-\xi_{1,369}+\xi_{1,371}}{4\xi_{1,381}}$$

$$\xi_{1,383} = \frac{\xi_{1,365}+\xi_{1,367}}{4\xi_{1,381}}$$

$$\xi_{1,384} = \frac{\xi_{1,366}+\xi_{1,370}}{4\xi_{1,381}}$$

$$\xi_{1,385} = \sqrt{\xi_{1,375}}$$

$$\xi_{1,386} = \frac{\xi_{1,366}-\xi_{1,370}}{4\xi_{1,385}}$$

$$\xi_{1,387} = \frac{\xi_{1,365}+\xi_{1,367}}{4\xi_{1,385}}$$

$$\xi_{1,388} = \frac{\xi_{1,369}+\xi_{1,371}}{4\xi_{1,385}}$$

$$\xi_{1,389} = \sqrt{\xi_{1,376}}$$

$$\xi_{1,390} = \frac{-\xi_{1,365}+\xi_{1,367}}{4\xi_{1,389}}$$

$$\xi_{1,391} = \frac{\xi_{1,366}+\xi_{1,370}}{4\xi_{1,389}}$$

$$\xi_{1,392} = \frac{\xi_{1,369}+\xi_{1,371}}{4\xi_{1,389}}$$

$$\xi_{1,393} = \begin{cases} \xi_{1,377} & \text{for } \xi_{1,373} \geq \xi_{1,374} \wedge \xi_{1,373} \geq \xi_{1,375} \wedge \xi_{1,373} \geq \xi_{1,376} \\ \xi_{1,382} & \text{for } \xi_{1,374} \geq \xi_{1,375} \wedge \xi_{1,374} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,374} \\ \xi_{1,386} & \text{for } \xi_{1,375} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,375} \wedge \xi_{1,374} \leq \xi_{1,375} \\ \xi_{1,390} & \text{otherwise} \end{cases}$$

$$\xi_{1,394} = \begin{cases} \xi_{1,378} & \text{for } \xi_{1,373} \geq \xi_{1,374} \wedge \xi_{1,373} \geq \xi_{1,375} \wedge \xi_{1,373} \geq \xi_{1,376} \\ \xi_{1,381} & \text{for } \xi_{1,374} \geq \xi_{1,375} \wedge \xi_{1,374} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,374} \\ \xi_{1,387} & \text{for } \xi_{1,375} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,375} \wedge \xi_{1,374} \leq \xi_{1,375} \\ \xi_{1,391} & \text{otherwise} \end{cases}$$

$$\xi_{1,395} = \begin{cases} \xi_{1,379} & \text{for } \xi_{1,373} \geq \xi_{1,374} \wedge \xi_{1,373} \geq \xi_{1,375} \wedge \xi_{1,373} \geq \xi_{1,376} \\ \xi_{1,383} & \text{for } \xi_{1,374} \geq \xi_{1,375} \wedge \xi_{1,374} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,374} \\ \xi_{1,385} & \text{for } \xi_{1,375} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,375} \wedge \xi_{1,374} \leq \xi_{1,375} \\ \xi_{1,392} & \text{otherwise} \end{cases}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,396} = \begin{cases} \xi_{1,380} & \text{for } \xi_{1,373} \geq \xi_{1,374} \wedge \xi_{1,373} \geq \xi_{1,375} \wedge \xi_{1,373} \geq \xi_{1,376} \\ \xi_{1,384} & \text{for } \xi_{1,374} \geq \xi_{1,375} \wedge \xi_{1,374} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,374} \\ \xi_{1,388} & \text{for } \xi_{1,375} \geq \xi_{1,376} \wedge \xi_{1,373} \leq \xi_{1,375} \wedge \xi_{1,374} \leq \xi_{1,375} \\ \xi_{1,389} & \text{otherwise} \end{cases}$$

$$\xi_{1,397} = \frac{\xi_{1,193}}{4} + \frac{\xi_{1,257}}{4} + \frac{\xi_{1,261}}{4} + \frac{1}{4}$$

$$\xi_{1,398} = \frac{\xi_{1,193}}{4} - \frac{\xi_{1,257}}{4} - \frac{\xi_{1,261}}{4} + \frac{1}{4}$$

$$\xi_{1,399} = -\frac{\xi_{1,193}}{4} + \frac{\xi_{1,257}}{4} - \frac{\xi_{1,261}}{4} + \frac{1}{4}$$

$$\xi_{1,400} = -\frac{\xi_{1,193}}{4} - \frac{\xi_{1,257}}{4} + \frac{\xi_{1,261}}{4} + \frac{1}{4}$$

$$\xi_{1,401} = \sqrt{\xi_{1,397}}$$

$$\xi_{1,402} = \frac{\xi_{1,258} - \xi_{1,260}}{4\xi_{1,401}}$$

$$\xi_{1,403} = \frac{-\xi_{1,195} + \xi_{1,259}}{4\xi_{1,401}}$$

$$\xi_{1,404} = \frac{\xi_{1,194} - \xi_{1,256}}{4\xi_{1,401}}$$

$$\xi_{1,405} = \sqrt{\xi_{1,398}}$$

$$\xi_{1,406} = \frac{\xi_{1,258} - \xi_{1,260}}{4\xi_{1,405}}$$

$$\xi_{1,407} = \frac{\xi_{1,194} + \xi_{1,256}}{4\xi_{1,405}}$$

$$\xi_{1,408} = \frac{\xi_{1,195} + \xi_{1,259}}{4\xi_{1,405}}$$

$$\xi_{1,409} = \sqrt{\xi_{1,399}}$$

$$\xi_{1,410} = \frac{-\xi_{1,195} + \xi_{1,259}}{4\xi_{1,409}}$$

$$\xi_{1,411} = \frac{\xi_{1,194} + \xi_{1,256}}{4\xi_{1,409}}$$

$$\xi_{1,412} = \frac{\xi_{1,258} + \xi_{1,260}}{4\xi_{1,409}}$$

$$\xi_{1,413} = \sqrt{\xi_{1,400}}$$

$$\xi_{1,414} = \frac{\xi_{1,194} - \xi_{1,256}}{4\xi_{1,413}}$$

$$\xi_{1,415} = \frac{\xi_{1,195} + \xi_{1,259}}{4\xi_{1,413}}$$

$$\xi_{1,416} = \frac{\xi_{1,258} + \xi_{1,260}}{4\xi_{1,413}}$$

$$\xi_{1,417} = \begin{cases} \xi_{1,401} & \text{for } \xi_{1,397} \geq \xi_{1,398} \wedge \xi_{1,397} \geq \xi_{1,399} \wedge \xi_{1,397} \geq \xi_{1,400} \\ \xi_{1,406} & \text{for } \xi_{1,398} \geq \xi_{1,399} \wedge \xi_{1,398} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,398} \\ \xi_{1,410} & \text{for } \xi_{1,399} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,399} \wedge \xi_{1,398} \leq \xi_{1,399} \\ \xi_{1,414} & \text{otherwise} \end{cases}$$

$$\xi_{1,418} = \begin{cases} \xi_{1,402} & \text{for } \xi_{1,397} \geq \xi_{1,398} \wedge \xi_{1,397} \geq \xi_{1,399} \wedge \xi_{1,397} \geq \xi_{1,400} \\ \xi_{1,405} & \text{for } \xi_{1,398} \geq \xi_{1,399} \wedge \xi_{1,398} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,398} \\ \xi_{1,411} & \text{for } \xi_{1,399} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,399} \wedge \xi_{1,398} \leq \xi_{1,399} \\ \xi_{1,415} & \text{otherwise} \end{cases}$$

$$\xi_{1,419} = \begin{cases} \xi_{1,403} & \text{for } \xi_{1,397} \geq \xi_{1,398} \wedge \xi_{1,397} \geq \xi_{1,399} \wedge \xi_{1,397} \geq \xi_{1,400} \\ \xi_{1,407} & \text{for } \xi_{1,398} \geq \xi_{1,399} \wedge \xi_{1,398} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,398} \\ \xi_{1,409} & \text{for } \xi_{1,399} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,399} \wedge \xi_{1,398} \leq \xi_{1,399} \\ \xi_{1,416} & \text{otherwise} \end{cases}$$

$$\xi_{1,420} = \begin{cases} \xi_{1,404} & \text{for } \xi_{1,397} \geq \xi_{1,398} \wedge \xi_{1,397} \geq \xi_{1,399} \wedge \xi_{1,397} \geq \xi_{1,400} \\ \xi_{1,408} & \text{for } \xi_{1,398} \geq \xi_{1,399} \wedge \xi_{1,398} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,398} \\ \xi_{1,412} & \text{for } \xi_{1,399} \geq \xi_{1,400} \wedge \xi_{1,397} \leq \xi_{1,399} \wedge \xi_{1,398} \leq \xi_{1,399} \\ \xi_{1,413} & \text{otherwise} \end{cases}$$

$$\xi_{1,421} = \xi_{1,112}\left(\xi_{1,117} + i\xi_{1,118}\right)$$

$$\xi_{1,422} = -\frac{\sqrt{2}\xi_{1,112}\xi_{1,114}}{2} + \frac{\sqrt{2}i\xi_{1,113}(\xi_{1,117}+i\xi_{1,118})}{2}$$

$$\xi_{1,423} = \frac{\sqrt{2}i\xi_{1,112}\xi_{1,114}}{2} - \frac{\sqrt{2}\xi_{1,113}(\xi_{1,117}+i\xi_{1,118})}{2}$$

$$\xi_{1,424} = -\xi_{1,113}\xi_{1,114}$$

$$\xi_{1,425} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{1,112}(\xi_{1,117}+i\xi_{1,118})}{2} - \frac{\sqrt{2}i\xi_{1,113}\xi_{1,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{1,112}(\xi_{1,117}-i\xi_{1,118})}{2} - \frac{\sqrt{2}\xi_{1,113}\xi_{1,114}}{2}\right)}{2}$$

$$\xi_{1,426} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{1,112}(\xi_{1,117}+i\xi_{1,118})}{2} - \frac{\sqrt{2}i\xi_{1,113}\xi_{1,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{1,112}(\xi_{1,117}-i\xi_{1,118})}{2} - \frac{\sqrt{2}\xi_{1,113}\xi_{1,114}}{2}\right)}{2}$$

$$\xi_{1,427} = \frac{\sqrt{2}i\xi_{1,112}\xi_{1,114}}{2} - \frac{\sqrt{2}\xi_{1,113}(\xi_{1,117}-i\xi_{1,118})}{2}$$

$$\xi_{1,428} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{1,112}(\xi_{1,117}-i\xi_{1,118})}{2} - \frac{\sqrt{2}i\xi_{1,113}\xi_{1,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{1,112}(\xi_{1,117}+i\xi_{1,118})}{2} - \frac{\sqrt{2}\xi_{1,113}\xi_{1,114}}{2}\right)}{2}$$

$$\xi_{1,429} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{1,112}(\xi_{1,117}-i\xi_{1,118})}{2} - \frac{\sqrt{2}i\xi_{1,113}\xi_{1,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{1,112}(\xi_{1,117}+i\xi_{1,118})}{2} - \frac{\sqrt{2}\xi_{1,113}\xi_{1,114}}{2}\right)}{2}$$

$$\xi_{1,430} = -\frac{\sqrt{2}\xi_{1,112}\xi_{1,114}}{2} + \frac{\sqrt{2}i\xi_{1,113}(\xi_{1,117}-i\xi_{1,118})}{2}$$

$$\xi_{1,431} = \xi_{1,112}\left(\xi_{1,117} - i\xi_{1,118}\right)$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,432} = (\xi_{1,393} - i\xi_{1,396})\,(\xi_{1,417} - i\xi_{1,420})$$

$$\xi_{1,433} = (\xi_{1,393} - i\xi_{1,396})\,(-i\xi_{1,418} - \xi_{1,419})$$

$$\xi_{1,434} = (\xi_{1,417} - i\xi_{1,420})\,(-i\xi_{1,394} - \xi_{1,395})$$

$$\xi_{1,435} = (-i\xi_{1,394} - \xi_{1,395})\,(-i\xi_{1,418} - \xi_{1,419})$$

$$\xi_{1,436} = (\xi_{1,393} - i\xi_{1,396})\,(-i\xi_{1,418} + \xi_{1,419})$$

$$\xi_{1,437} = (\xi_{1,393} - i\xi_{1,396})\,(\xi_{1,417} + i\xi_{1,420})$$

$$\xi_{1,438} = (-i\xi_{1,394} - \xi_{1,395})\,(-i\xi_{1,418} + \xi_{1,419})$$

$$\xi_{1,439} = (\xi_{1,417} + i\xi_{1,420})\,(-i\xi_{1,394} - \xi_{1,395})$$

$$\xi_{1,440} = (\xi_{1,417} - i\xi_{1,420})\,(-i\xi_{1,394} + \xi_{1,395})$$

$$\xi_{1,441} = (-i\xi_{1,394} + \xi_{1,395})\,(-i\xi_{1,418} - \xi_{1,419})$$

$$\xi_{1,442} = (\xi_{1,393} + i\xi_{1,396})\,(\xi_{1,417} - i\xi_{1,420})$$

$$\xi_{1,443} = (\xi_{1,393} + i\xi_{1,396})\,(-i\xi_{1,418} - \xi_{1,419})$$

$$\xi_{1,444} = (-i\xi_{1,394} + \xi_{1,395})\,(-i\xi_{1,418} + \xi_{1,419})$$

$$\xi_{1,445} = (\xi_{1,417} + i\xi_{1,420})\,(-i\xi_{1,394} + \xi_{1,395})$$

$$\xi_{1,446} = (\xi_{1,393} + i\xi_{1,396})\,(-i\xi_{1,418} + \xi_{1,419})$$

$$\xi_{1,447} = (\xi_{1,393} + i\xi_{1,396})\,(\xi_{1,417} + i\xi_{1,420})$$

$$\xi_{1,448} = \xi_{1,421}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}}\right) + \xi_{1,422}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}}\right) + \xi_{1,423}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}}\right) + \xi_{1,424}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,444}}\right)$$

$$\xi_{1,449} = \xi_{1,422}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}}\right) + \xi_{1,425}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}}\right) + \xi_{1,427}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,444}}\right) + \xi_{1,428}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}}\right)$$

$$\xi_{1,450} = \xi_{1,423}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}}\right) + \xi_{1,426}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}}\right) + \xi_{1,429}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}}\right) + \xi_{1,430}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,444}}\right)$$

$$\xi_{1,451} = \xi_{1,424}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}}\right) + \xi_{1,427}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}}\right) + \xi_{1,430}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}}\right) + \xi_{1,431}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,444}}\right)$$

$$\xi_{1,452} = \xi_{1,421}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}}\right) + \xi_{1,422}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}}\right) + \xi_{1,423}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}}\right) + \xi_{1,424}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,441}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}}\right)$$

$$\xi_{1,453} = \xi_{1,422}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}}\right) + \xi_{1,425}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}}\right) + \xi_{1,427}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,444}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}}\right) + \xi_{1,428}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}}\right)$$

$$\xi_{1,454} = \xi_{1,423}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}}\right) + \xi_{1,426}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}}\right) + \xi_{1,429}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}}\right) + \xi_{1,430}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,441}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}}\right)$$

$$\xi_{1,455} = \xi_{1,424}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,432}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,435}}\right) + \xi_{1,427}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,436}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,439}}\right) + \xi_{1,430}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,440}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,443}}\right) + \xi_{1,431}\left(-ie^{-i\xi_{1,105}}\sin\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,441}} + e^{-i\xi_{1,105}}\cos\left(\xi_{1,103} - \xi_{1,104}\right)\overline{\xi_{1,447}}\right)$$

$$\xi_{1,456} = \xi_{1,421}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}}\right) + \xi_{1,422}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}}\right) + \xi_{1,423}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}}\right) + \xi_{1,424}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}}\right)$$

$$\xi_{1,457} = \xi_{1,422}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}}\right) + \xi_{1,425}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}}\right) + \xi_{1,427}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}}\right) + \xi_{1,428}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}}\right)$$

$$\xi_{1,458} = \xi_{1,423}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}}\right) + \xi_{1,426}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}}\right) + \xi_{1,429}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}}\right) + \xi_{1,430}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}}\right)$$

$$\xi_{1,459} = \xi_{1,424}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}}\right) + \xi_{1,427}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}}\right) + \xi_{1,430}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}}\right) + \xi_{1,431}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}}\right)$$

$$\xi_{1,460} = \xi_{1,421}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}}\right) + \xi_{1,422}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}}\right) + \xi_{1,423}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}}\right) + \xi_{1,424}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}}\right)$$

$$\xi_{1,461} = \xi_{1,422}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}}\right) + \xi_{1,425}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}}\right) + \xi_{1,427}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}}\right) + \xi_{1,428}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}}\right)$$

$$\xi_{1,462} = \xi_{1,423}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}}\right) + \xi_{1,426}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}}\right) + \xi_{1,429}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}}\right) + \xi_{1,430}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}}\right)$$

$$\xi_{1,463} = \xi_{1,424}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,433}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,434}}\right) + \xi_{1,427}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,437}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,438}}\right) + \xi_{1,430}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,441}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,442}}\right) + \xi_{1,431}\left(-ie^{i\xi_{1,105}}\sin\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,445}} + e^{i\xi_{1,105}}\cos\left(\xi_{1,103} + \xi_{1,104}\right)\overline{\xi_{1,446}}\right)$$

$$\xi_{1,464} = \frac{\operatorname{re}\left(\xi_{1,448}\right)}{4} + \frac{\operatorname{re}\left(\xi_{1,455}\right)}{4} + \frac{\operatorname{re}\left(\xi_{1,457}\right)}{4} + \frac{\operatorname{re}\left(\xi_{1,462}\right)}{4}$$

$$\xi_{1,465} = -\frac{\operatorname{im}\left(\xi_{1,449}\right)}{4} - \frac{\operatorname{im}\left(\xi_{1,454}\right)}{4} - \frac{\operatorname{im}\left(\xi_{1,456}\right)}{4} - \frac{\operatorname{im}\left(\xi_{1,463}\right)}{4}$$

$$\xi_{1,466} = -\frac{\operatorname{re}\left(\xi_{1,449}\right)}{4} + \frac{\operatorname{re}\left(\xi_{1,454}\right)}{4} + \frac{\operatorname{re}\left(\xi_{1,456}\right)}{4} - \frac{\operatorname{re}\left(\xi_{1,463}\right)}{4}$$

$$\xi_{1,467} = -\frac{\operatorname{im}\left(\xi_{1,448}\right)}{4} + \frac{\operatorname{im}\left(\xi_{1,455}\right)}{4} + \frac{\operatorname{im}\left(\xi_{1,457}\right)}{4} - \frac{\operatorname{im}\left(\xi_{1,462}\right)}{4}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,468} = -\frac{\operatorname{im}(\xi_{1,450})}{4} - \frac{\operatorname{im}(\xi_{1,453})}{4} - \frac{\operatorname{im}(\xi_{1,459})}{4} - \frac{\operatorname{im}(\xi_{1,460})}{4}$$

$$\xi_{1,469} = -\frac{\operatorname{re}(\xi_{1,451})}{4} - \frac{\operatorname{re}(\xi_{1,452})}{4} - \frac{\operatorname{re}(\xi_{1,458})}{4} - \frac{\operatorname{re}(\xi_{1,461})}{4}$$

$$\xi_{1,470} = \frac{\operatorname{im}(\xi_{1,451})}{4} - \frac{\operatorname{im}(\xi_{1,452})}{4} - \frac{\operatorname{im}(\xi_{1,458})}{4} + \frac{\operatorname{im}(\xi_{1,461})}{4}$$

$$\xi_{1,471} = -\frac{\operatorname{re}(\xi_{1,450})}{4} + \frac{\operatorname{re}(\xi_{1,453})}{4} + \frac{\operatorname{re}(\xi_{1,459})}{4} - \frac{\operatorname{re}(\xi_{1,460})}{4}$$

$$\xi_{1,472} = -\frac{\operatorname{re}(\xi_{1,450})}{4} + \frac{\operatorname{re}(\xi_{1,453})}{4} - \frac{\operatorname{re}(\xi_{1,459})}{4} + \frac{\operatorname{re}(\xi_{1,460})}{4}$$

$$\xi_{1,473} = \frac{\operatorname{im}(\xi_{1,451})}{4} - \frac{\operatorname{im}(\xi_{1,452})}{4} + \frac{\operatorname{im}(\xi_{1,458})}{4} - \frac{\operatorname{im}(\xi_{1,461})}{4}$$

$$\xi_{1,474} = \frac{\operatorname{re}(\xi_{1,451})}{4} + \frac{\operatorname{re}(\xi_{1,452})}{4} - \frac{\operatorname{re}(\xi_{1,458})}{4} - \frac{\operatorname{re}(\xi_{1,461})}{4}$$

$$\xi_{1,475} = \frac{\operatorname{im}(\xi_{1,450})}{4} + \frac{\operatorname{im}(\xi_{1,453})}{4} - \frac{\operatorname{im}(\xi_{1,459})}{4} - \frac{\operatorname{im}(\xi_{1,460})}{4}$$

$$\xi_{1,476} = -\frac{\operatorname{im}(\xi_{1,448})}{4} + \frac{\operatorname{im}(\xi_{1,455})}{4} - \frac{\operatorname{im}(\xi_{1,457})}{4} + \frac{\operatorname{im}(\xi_{1,462})}{4}$$

$$\xi_{1,477} = -\frac{\operatorname{re}(\xi_{1,449})}{4} + \frac{\operatorname{re}(\xi_{1,454})}{4} - \frac{\operatorname{re}(\xi_{1,456})}{4} + \frac{\operatorname{re}(\xi_{1,463})}{4}$$

$$\xi_{1,478} = \frac{\operatorname{im}(\xi_{1,449})}{4} + \frac{\operatorname{im}(\xi_{1,454})}{4} - \frac{\operatorname{im}(\xi_{1,456})}{4} - \frac{\operatorname{im}(\xi_{1,463})}{4}$$

$$\xi_{1,479} = -\frac{\operatorname{re}(\xi_{1,448})}{4} - \frac{\operatorname{re}(\xi_{1,455})}{4} + \frac{\operatorname{re}(\xi_{1,457})}{4} + \frac{\operatorname{re}(\xi_{1,462})}{4}$$

$$\xi_{1,480} = \xi_{1,464}^2 + \xi_{1,468}^2 + \xi_{1,472}^2 + \xi_{1,476}^2$$

$$\xi_{1,481} = \xi_{1,465}^2 + \xi_{1,469}^2 + \xi_{1,473}^2 + \xi_{1,477}^2$$

$$\xi_{1,482} = \xi_{1,466}^2 + \xi_{1,470}^2 + \xi_{1,474}^2 + \xi_{1,478}^2$$

$$\xi_{1,483} = \xi_{1,467}^2 + \xi_{1,471}^2 + \xi_{1,475}^2 + \xi_{1,479}^2$$

$$\xi_{1,484} = \sqrt{\xi_{1,480}}$$

$$\xi_{1,485} = \frac{\xi_{1,464}}{\xi_{1,484}}$$

$$\xi_{1,486} = \frac{\xi_{1,468}}{\xi_{1,484}}$$

$$\xi_{1,487} = \frac{\xi_{1,472}}{\xi_{1,484}}$$

$$\xi_{1,488} = \frac{\xi_{1,476}}{\xi_{1,484}}$$

$$\xi_{1,489} = \sqrt{\xi_{1,481}}$$

$$\xi_{1,490} = \frac{\xi_{1,465}}{\xi_{1,489}}$$

$$\xi_{1,491} = \frac{\xi_{1,469}}{\xi_{1,489}}$$

$$\xi_{1,492} = \frac{\xi_{1,473}}{\xi_{1,489}}$$

$$\xi_{1,493} = \frac{\xi_{1,477}}{\xi_{1,489}}$$

$$\xi_{1,494} = \sqrt{\xi_{1,482}}$$

$$\xi_{1,495} = \frac{\xi_{1,466}}{\xi_{1,494}}$$

$$\xi_{1,496} = \frac{\xi_{1,470}}{\xi_{1,494}}$$

$$\xi_{1,497} = \frac{\xi_{1,474}}{\xi_{1,494}}$$

$$\xi_{1,498} = \frac{\xi_{1,478}}{\xi_{1,494}}$$

$$\xi_{1,499} = \sqrt{\xi_{1,483}}$$

$$\xi_{1,500} = \frac{\xi_{1,467}}{\xi_{1,499}}$$

$$\xi_{1,501} = \frac{\xi_{1,471}}{\xi_{1,499}}$$

$$\xi_{1,502} = \frac{\xi_{1,475}}{\xi_{1,499}}$$

$$\xi_{1,503} = \frac{\xi_{1,479}}{\xi_{1,499}}$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,504} = \begin{cases} \xi_{1,485} & \text{for } \xi_{1,480} \geq \xi_{1,481} \wedge \xi_{1,480} \geq \xi_{1,482} \wedge \xi_{1,480} \geq \xi_{1,483} \\ \xi_{1,490} & \text{for } \xi_{1,481} \geq \xi_{1,482} \wedge \xi_{1,481} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,481} \\ \xi_{1,495} & \text{for } \xi_{1,482} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,482} \wedge \xi_{1,481} \leq \xi_{1,482} \\ \xi_{1,500} & \text{otherwise} \end{cases}$$

$$\xi_{1,505} = \begin{cases} \xi_{1,486} & \text{for } \xi_{1,480} \geq \xi_{1,481} \wedge \xi_{1,480} \geq \xi_{1,482} \wedge \xi_{1,480} \geq \xi_{1,483} \\ \xi_{1,491} & \text{for } \xi_{1,481} \geq \xi_{1,482} \wedge \xi_{1,481} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,481} \\ \xi_{1,496} & \text{for } \xi_{1,482} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,482} \wedge \xi_{1,481} \leq \xi_{1,482} \\ \xi_{1,501} & \text{otherwise} \end{cases}$$

$$\xi_{1,506} = \begin{cases} \xi_{1,487} & \text{for } \xi_{1,480} \geq \xi_{1,481} \wedge \xi_{1,480} \geq \xi_{1,482} \wedge \xi_{1,480} \geq \xi_{1,483} \\ \xi_{1,492} & \text{for } \xi_{1,481} \geq \xi_{1,482} \wedge \xi_{1,481} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,481} \\ \xi_{1,497} & \text{for } \xi_{1,482} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,482} \wedge \xi_{1,481} \leq \xi_{1,482} \\ \xi_{1,502} & \text{otherwise} \end{cases}$$

$$\xi_{1,507} = \begin{cases} \xi_{1,488} & \text{for } \xi_{1,480} \geq \xi_{1,481} \wedge \xi_{1,480} \geq \xi_{1,482} \wedge \xi_{1,480} \geq \xi_{1,483} \\ \xi_{1,493} & \text{for } \xi_{1,481} \geq \xi_{1,482} \wedge \xi_{1,481} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,481} \\ \xi_{1,498} & \text{for } \xi_{1,482} \geq \xi_{1,483} \wedge \xi_{1,480} \leq \xi_{1,482} \wedge \xi_{1,481} \leq \xi_{1,482} \\ \xi_{1,503} & \text{otherwise} \end{cases}$$

$$\xi_{1,508} = \xi_{1,464}\xi_{1,504} + \xi_{1,468}\xi_{1,505} + \xi_{1,472}\xi_{1,506} + \xi_{1,476}\xi_{1,507}$$

$$\xi_{1,509} = \xi_{1,465}\xi_{1,504} + \xi_{1,469}\xi_{1,505} + \xi_{1,473}\xi_{1,506} + \xi_{1,477}\xi_{1,507}$$

$$\xi_{1,510} = \xi_{1,466}\xi_{1,504} + \xi_{1,470}\xi_{1,505} + \xi_{1,474}\xi_{1,506} + \xi_{1,478}\xi_{1,507}$$

$$\xi_{1,511} = \xi_{1,467}\xi_{1,504} + \xi_{1,471}\xi_{1,505} + \xi_{1,475}\xi_{1,506} + \xi_{1,479}\xi_{1,507}$$

$$\xi_{1,512} = \xi_{1,41}\xi_{1,52} - \xi_{1,42}\xi_{1,55} - \xi_{1,43}\xi_{1,56} - \xi_{1,44}\xi_{1,57}$$

$$\xi_{1,513} = \xi_{1,41}\xi_{1,55} + \xi_{1,42}\xi_{1,52} - \xi_{1,43}\xi_{1,57} + \xi_{1,44}\xi_{1,56}$$

$$\xi_{1,514} = \xi_{1,41}\xi_{1,56} + \xi_{1,42}\xi_{1,57} + \xi_{1,43}\xi_{1,52} - \xi_{1,44}\xi_{1,55}$$

$$\xi_{1,515} = \xi_{1,41}\xi_{1,57} - \xi_{1,42}\xi_{1,56} + \xi_{1,43}\xi_{1,55} + \xi_{1,44}\xi_{1,52}$$

$$\xi_{1,516} = \xi_{1,10}\xi_{1,55} + \xi_{1,11}\xi_{1,55} + \xi_{1,12}\xi_{1,55} + \xi_{1,54}\xi_{1,9}$$

$$\xi_{1,517} = -\xi_{1,10}\xi_{1,54} + \xi_{1,11}\xi_{1,55} - \xi_{1,12}\xi_{1,55} + \xi_{1,55}\xi_{1,9}$$

$$\xi_{1,518} = -\xi_{1,10}\xi_{1,55} - \xi_{1,11}\xi_{1,54} + \xi_{1,12}\xi_{1,55} + \xi_{1,55}\xi_{1,9}$$

$$\xi_{1,519} = \xi_{1,10}\xi_{1,55} - \xi_{1,11}\xi_{1,55} - \xi_{1,12}\xi_{1,54} + \xi_{1,55}\xi_{1,9}$$

$$\xi_{1,520} = \xi_{1,52}\left(\xi_{1,100}\xi_{1,505} + \xi_{1,101}\xi_{1,506} + \xi_{1,102}\xi_{1,507} + \xi_{1,504}\xi_{1,99}\right) - \xi_{1,52}\left(\xi_{1,100}\xi_{1,506} - \xi_{1,101}\xi_{1,505} + \xi_{1,102}\xi_{1,504} - \xi_{1,507}\xi_{1,99}\right) - \xi_{1,53}\left(\xi_{1,100}\xi_{1,504} + \xi_{1,101}\xi_{1,507} - \xi_{1,102}\xi_{1,506} - \xi_{1,505}\xi_{1,99}\right) - \xi_{1,54}\left(-\xi_{1,100}\xi_{1,507} + \xi_{1,101}\xi_{1,504} + \xi_{1,102}\xi_{1,505} - \xi_{1,506}\xi_{1,99}\right)$$

$$\xi_{1,521} = \xi_{1,52}\left(\xi_{1,100}\xi_{1,504} + \xi_{1,101}\xi_{1,507} - \xi_{1,102}\xi_{1,506} - \xi_{1,505}\xi_{1,99}\right) + \xi_{1,52}\left(-\xi_{1,100}\xi_{1,507} + \xi_{1,101}\xi_{1,504} + \xi_{1,102}\xi_{1,505} - \xi_{1,506}\xi_{1,99}\right) + \xi_{1,53}\left(\xi_{1,100}\xi_{1,505} + \xi_{1,101}\xi_{1,506} + \xi_{1,102}\xi_{1,507} + \xi_{1,504}\xi_{1,99}\right) - \xi_{1,54}\left(\xi_{1,100}\xi_{1,506} - \xi_{1,101}\xi_{1,505} + \xi_{1,102}\xi_{1,504} - \xi_{1,507}\xi_{1,99}\right)$$

$$\xi_{1,522} = -\xi_{1,52}\left(\xi_{1,100}\xi_{1,504} + \xi_{1,101}\xi_{1,507} - \xi_{1,102}\xi_{1,506} - \xi_{1,505}\xi_{1,99}\right) + \xi_{1,52}\left(-\xi_{1,100}\xi_{1,507} + \xi_{1,101}\xi_{1,504} + \xi_{1,102}\xi_{1,505} - \xi_{1,506}\xi_{1,99}\right) + \xi_{1,53}\left(\xi_{1,100}\xi_{1,506} - \xi_{1,101}\xi_{1,505} + \xi_{1,102}\xi_{1,504} - \xi_{1,507}\xi_{1,99}\right) + \xi_{1,54}\left(\xi_{1,100}\xi_{1,505} + \xi_{1,101}\xi_{1,506} + \xi_{1,102}\xi_{1,507} + \xi_{1,504}\xi_{1,99}\right)$$

$$\xi_{1,523} = \xi_{1,52}\left(\xi_{1,100}\xi_{1,505} + \xi_{1,101}\xi_{1,506} + \xi_{1,102}\xi_{1,507} + \xi_{1,504}\xi_{1,99}\right) + \xi_{1,52}\left(\xi_{1,100}\xi_{1,506} - \xi_{1,101}\xi_{1,505} + \xi_{1,102}\xi_{1,504} - \xi_{1,507}\xi_{1,99}\right) - \xi_{1,53}\left(-\xi_{1,100}\xi_{1,507} + \xi_{1,101}\xi_{1,504} + \xi_{1,102}\xi_{1,505} - \xi_{1,506}\xi_{1,99}\right) + \xi_{1,54}\left(\xi_{1,100}\xi_{1,504} + \xi_{1,101}\xi_{1,507} - \xi_{1,102}\xi_{1,506} - \xi_{1,505}\xi_{1,99}\right)$$

$$\xi_{1,524} = -\xi_{1,53}\left(-\xi_{1,508}\xi_{1,68} - \xi_{1,509}\xi_{1,67} + \xi_{1,510}\xi_{1,70} - \xi_{1,511}\xi_{1,69}\right) - \xi_{1,53}\left(-\xi_{1,508}\xi_{1,69} - \xi_{1,509}\xi_{1,70} - \xi_{1,510}\xi_{1,67} + \xi_{1,511}\xi_{1,68}\right) - \xi_{1,53}\left(-\xi_{1,508}\xi_{1,70} + \xi_{1,509}\xi_{1,69} - \xi_{1,510}\xi_{1,68} - \xi_{1,511}\xi_{1,67}\right) + \xi_{1,54}\left(\xi_{1,508}\xi_{1,67} - \xi_{1,509}\xi_{1,68} - \xi_{1,510}\xi_{1,69} - \xi_{1,511}\xi_{1,70}\right)$$

$$\xi_{1,525} = \xi_{1,53}\left(\xi_{1,508}\xi_{1,67} - \xi_{1,509}\xi_{1,68} - \xi_{1,510}\xi_{1,69} - \xi_{1,511}\xi_{1,70}\right) + \xi_{1,53}\left(-\xi_{1,508}\xi_{1,69} - \xi_{1,509}\xi_{1,70} - \xi_{1,510}\xi_{1,67} + \xi_{1,511}\xi_{1,68}\right) - \xi_{1,53}\left(-\xi_{1,508}\xi_{1,70} + \xi_{1,509}\xi_{1,69} - \xi_{1,510}\xi_{1,68} - \xi_{1,511}\xi_{1,67}\right) + \xi_{1,54}\left(-\xi_{1,508}\xi_{1,68} - \xi_{1,509}\xi_{1,67} + \xi_{1,510}\xi_{1,70} - \xi_{1,511}\xi_{1,69}\right)$$

$$\xi_{1,526} = \xi_{1,53}\left(\xi_{1,508}\xi_{1,67} - \xi_{1,509}\xi_{1,68} - \xi_{1,510}\xi_{1,69} - \xi_{1,511}\xi_{1,70}\right) - \xi_{1,53}\left(-\xi_{1,508}\xi_{1,68} - \xi_{1,509}\xi_{1,67} + \xi_{1,510}\xi_{1,70} - \xi_{1,511}\xi_{1,69}\right) + \xi_{1,53}\left(-\xi_{1,508}\xi_{1,70} + \xi_{1,509}\xi_{1,69} - \xi_{1,510}\xi_{1,68} - \xi_{1,511}\xi_{1,67}\right) + \xi_{1,54}\left(-\xi_{1,508}\xi_{1,69} - \xi_{1,509}\xi_{1,70} - \xi_{1,510}\xi_{1,67} + \xi_{1,511}\xi_{1,68}\right)$$

$$\xi_{1,527} = \xi_{1,53}\left(\xi_{1,508}\xi_{1,67} - \xi_{1,509}\xi_{1,68} - \xi_{1,510}\xi_{1,69} - \xi_{1,511}\xi_{1,70}\right) + \xi_{1,53}\left(-\xi_{1,508}\xi_{1,68} - \xi_{1,509}\xi_{1,67} + \xi_{1,510}\xi_{1,70} - \xi_{1,511}\xi_{1,69}\right) - \xi_{1,53}\left(-\xi_{1,508}\xi_{1,69} - \xi_{1,509}\xi_{1,70} - \xi_{1,510}\xi_{1,67} + \xi_{1,511}\xi_{1,68}\right) + \xi_{1,54}\left(-\xi_{1,508}\xi_{1,70} + \xi_{1,509}\xi_{1,69} - \xi_{1,510}\xi_{1,68} - \xi_{1,511}\xi_{1,67}\right)$$

$$\xi_{1,528} = \xi_{1,393}\left(\xi_{1,33}\xi_{1,91} - \xi_{1,34}\xi_{1,92} - \xi_{1,35}\xi_{1,93} - \xi_{1,36}\xi_{1,94}\right) + \xi_{1,394}\left(\xi_{1,33}\xi_{1,92} + \xi_{1,34}\xi_{1,91} + \xi_{1,35}\xi_{1,94} - \xi_{1,36}\xi_{1,93}\right) + \xi_{1,395}\left(\xi_{1,33}\xi_{1,93} - \xi_{1,34}\xi_{1,94} + \xi_{1,35}\xi_{1,91} + \xi_{1,36}\xi_{1,92}\right) + \xi_{1,396}\left(\xi_{1,33}\xi_{1,94} + \xi_{1,34}\xi_{1,93} - \xi_{1,35}\xi_{1,92} + \xi_{1,36}\xi_{1,91}\right)$$

$$\xi_{1,529} = \xi_{1,393}\left(\xi_{1,33}\xi_{1,92} + \xi_{1,34}\xi_{1,91} + \xi_{1,35}\xi_{1,94} - \xi_{1,36}\xi_{1,93}\right) - \xi_{1,394}\left(\xi_{1,33}\xi_{1,91} - \xi_{1,34}\xi_{1,92} - \xi_{1,35}\xi_{1,93} - \xi_{1,36}\xi_{1,94}\right) + \xi_{1,395}\left(\xi_{1,33}\xi_{1,94} + \xi_{1,34}\xi_{1,93} - \xi_{1,35}\xi_{1,92} + \xi_{1,36}\xi_{1,91}\right) - \xi_{1,396}\left(\xi_{1,33}\xi_{1,93} - \xi_{1,34}\xi_{1,94} + \xi_{1,35}\xi_{1,91} + \xi_{1,36}\xi_{1,92}\right)$$

$$\xi_{1,530} = \xi_{1,393}\left(\xi_{1,33}\xi_{1,93} - \xi_{1,34}\xi_{1,94} + \xi_{1,35}\xi_{1,91} + \xi_{1,36}\xi_{1,92}\right) - \xi_{1,394}\left(\xi_{1,33}\xi_{1,94} + \xi_{1,34}\xi_{1,93} - \xi_{1,35}\xi_{1,92} + \xi_{1,36}\xi_{1,91}\right) - \xi_{1,395}\left(\xi_{1,33}\xi_{1,91} - \xi_{1,34}\xi_{1,92} - \xi_{1,35}\xi_{1,93} - \xi_{1,36}\xi_{1,94}\right) + \xi_{1,396}\left(\xi_{1,33}\xi_{1,92} + \xi_{1,34}\xi_{1,91} + \xi_{1,35}\xi_{1,94} - \xi_{1,36}\xi_{1,93}\right)$$

$$\xi_{1,531} = \xi_{1,393}\left(\xi_{1,33}\xi_{1,94} + \xi_{1,34}\xi_{1,93} - \xi_{1,35}\xi_{1,92} + \xi_{1,36}\xi_{1,91}\right) + \xi_{1,394}\left(\xi_{1,33}\xi_{1,93} - \xi_{1,34}\xi_{1,94} + \xi_{1,35}\xi_{1,91} + \xi_{1,36}\xi_{1,92}\right) - \xi_{1,395}\left(\xi_{1,33}\xi_{1,92} + \xi_{1,34}\xi_{1,91} + \xi_{1,35}\xi_{1,94} - \xi_{1,36}\xi_{1,93}\right) - \xi_{1,396}\left(\xi_{1,33}\xi_{1,91} - \xi_{1,34}\xi_{1,92} - \xi_{1,35}\xi_{1,93} - \xi_{1,36}\xi_{1,94}\right)$$

$$\xi_{1,532} = \xi_{1,417}\left(\xi_{1,37}\xi_{1,95} - \xi_{1,38}\xi_{1,96} - \xi_{1,39}\xi_{1,97} - \xi_{1,40}\xi_{1,98}\right) + \xi_{1,418}\left(\xi_{1,37}\xi_{1,96} + \xi_{1,38}\xi_{1,95} + \xi_{1,39}\xi_{1,98} - \xi_{1,40}\xi_{1,97}\right) + \xi_{1,419}\left(\xi_{1,37}\xi_{1,97} - \xi_{1,38}\xi_{1,98} + \xi_{1,39}\xi_{1,95} + \xi_{1,40}\xi_{1,96}\right) - \xi_{1,420}\left(\xi_{1,37}\xi_{1,98} + \xi_{1,38}\xi_{1,97} - \xi_{1,39}\xi_{1,96} + \xi_{1,40}\xi_{1,95}\right)$$

$$\xi_{1,533} = \xi_{1,417}\left(\xi_{1,37}\xi_{1,96} + \xi_{1,38}\xi_{1,95} + \xi_{1,39}\xi_{1,98} - \xi_{1,40}\xi_{1,97}\right) - \xi_{1,418}\left(\xi_{1,37}\xi_{1,95} - \xi_{1,38}\xi_{1,96} - \xi_{1,39}\xi_{1,97} - \xi_{1,40}\xi_{1,98}\right) + \xi_{1,419}\left(\xi_{1,37}\xi_{1,98} + \xi_{1,38}\xi_{1,97} - \xi_{1,39}\xi_{1,96} + \xi_{1,40}\xi_{1,95}\right) - \xi_{1,420}\left(\xi_{1,37}\xi_{1,97} - \xi_{1,38}\xi_{1,98} + \xi_{1,39}\xi_{1,95} + \xi_{1,40}\xi_{1,96}\right)$$

$$\xi_{1,534} = \xi_{1,417}\left(\xi_{1,37}\xi_{1,97} - \xi_{1,38}\xi_{1,98} + \xi_{1,39}\xi_{1,95} + \xi_{1,40}\xi_{1,96}\right) - \xi_{1,418}\left(\xi_{1,37}\xi_{1,98} + \xi_{1,38}\xi_{1,97} - \xi_{1,39}\xi_{1,96} + \xi_{1,40}\xi_{1,95}\right) - \xi_{1,419}\left(\xi_{1,37}\xi_{1,95} - \xi_{1,38}\xi_{1,96} - \xi_{1,39}\xi_{1,97} - \xi_{1,40}\xi_{1,98}\right) + \xi_{1,420}\left(\xi_{1,37}\xi_{1,96} + \xi_{1,38}\xi_{1,95} + \xi_{1,39}\xi_{1,98} - \xi_{1,40}\xi_{1,97}\right)$$

$$\xi_{1,535} = \xi_{1,417}\left(\xi_{1,37}\xi_{1,98} + \xi_{1,38}\xi_{1,97} - \xi_{1,39}\xi_{1,96} + \xi_{1,40}\xi_{1,95}\right) + \xi_{1,418}\left(\xi_{1,37}\xi_{1,97} - \xi_{1,38}\xi_{1,98} + \xi_{1,39}\xi_{1,95} + \xi_{1,40}\xi_{1,96}\right) - \xi_{1,419}\left(\xi_{1,37}\xi_{1,96} + \xi_{1,38}\xi_{1,95} + \xi_{1,39}\xi_{1,98} - \xi_{1,40}\xi_{1,97}\right) - \xi_{1,420}\left(\xi_{1,37}\xi_{1,95} - \xi_{1,38}\xi_{1,96} - \xi_{1,39}\xi_{1,97} - \xi_{1,40}\xi_{1,98}\right)$$

$$\xi_{1,536} = \xi_{1,106} + \xi_{1,48}$$

$$\xi_{1,537} = \begin{cases} 0 & \text{for } \xi_{1,512}^2 + \xi_{1,515}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,515}, \xi_{1,512}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,538} = \begin{cases} 0 & \text{for } \xi_{1,513}^2 + \xi_{1,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,513}, \xi_{1,514}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,539} = 2\operatorname{atan}_2\left(\sqrt{\xi_{1,513}^2 + \xi_{1,514}^2}, \sqrt{\xi_{1,512}^2 + \xi_{1,515}^2}\right)$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,540} = -\xi_{1,537} + \xi_{1,538}$$

$$\xi_{1,541} = -\xi_{1,537} - \xi_{1,538}$$

$$\xi_{1,542} = \xi_{1,536} + \xi_{1,537}$$

$$\xi_{1,543} = \begin{cases} 0 & \text{for } \xi_{1,516}^2 + \xi_{1,519}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,519}, \xi_{1,516}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,544} = \begin{cases} 0 & \text{for } \xi_{1,517}^2 + \xi_{1,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,517}, \xi_{1,518}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,545} = 2\operatorname{atan}_2\left(\sqrt{\xi_{1,517}^2 + \xi_{1,518}^2}, \sqrt{\xi_{1,516}^2 + \xi_{1,519}^2}\right)$$

$$\xi_{1,546} = -\xi_{1,543} + \xi_{1,544}$$

$$\xi_{1,547} = -\xi_{1,543} - \xi_{1,544}$$

$$\xi_{1,548} = \xi_{1,542} + \xi_{1,543}$$

$$\xi_{1,549} = \begin{cases} 0 & \text{for } \xi_{1,520}^2 + \xi_{1,523}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,523}, \xi_{1,520}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,550} = \begin{cases} 0 & \text{for } \xi_{1,521}^2 + \xi_{1,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,521}, \xi_{1,522}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,551} = 2\operatorname{atan}_2\left(\sqrt{\xi_{1,521}^2 + \xi_{1,522}^2}, \sqrt{\xi_{1,520}^2 + \xi_{1,523}^2}\right)$$

$$\xi_{1,552} = -\xi_{1,549} + \xi_{1,550}$$

$$\xi_{1,553} = -\xi_{1,549} - \xi_{1,550}$$

$$\xi_{1,554} = \xi_{1,548} + \xi_{1,549}$$

$$\xi_{1,555} = \begin{cases} 0 & \text{for } \xi_{1,524}^2 + \xi_{1,527}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,527}, \xi_{1,524}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,556} = \begin{cases} 0 & \text{for } \xi_{1,525}^2 + \xi_{1,526}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,525}, \xi_{1,526}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,557} = 2\operatorname{atan}_2\left(\sqrt{\xi_{1,525}^2 + \xi_{1,526}^2}, \sqrt{\xi_{1,524}^2 + \xi_{1,527}^2}\right)$$

$$\xi_{1,558} = -\xi_{1,555} + \xi_{1,556}$$

$$\xi_{1,559} = -\xi_{1,555} - \xi_{1,556}$$

$$\xi_{1,560} = \xi_{1,554} + \xi_{1,555}$$

$$\xi_{1,561} = \begin{cases} 0 & \text{for } \xi_{1,117}^2 + \xi_{1,118}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{1,118}, \xi_{1,117}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,562} = \begin{cases} 0 & \text{for } \xi_{1,114}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{1,114}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,563} = 2\operatorname{atan}_2\left(\left|\xi_{1,114}\right|, \sqrt{\xi_{1,117}^2 + \xi_{1,118}^2}\right)$$

$$\xi_{1,564} = -\xi_{1,561} + \xi_{1,562}$$

$$\xi_{1,565} = -\xi_{1,561} - \xi_{1,562}$$

$$\xi_{1,566} = \xi_{1,560} + \xi_{1,561}$$

$$\xi_{1,567} = \begin{cases} 0 & \text{for } \xi_{1,112}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{1,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,568} = \begin{cases} 0 & \text{for } \xi_{1,113}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{1,113}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,569} = 2\operatorname{atan}_2\left(\left|\xi_{1,113}\right|, \left|\xi_{1,112}\right|\right)$$

$$\xi_{1,570} = -\xi_{1,567} + \xi_{1,568}$$

$$\xi_{1,571} = -\xi_{1,567} - \xi_{1,568}$$

$$\xi_{1,572} = \xi_{1,566} + \xi_{1,567}$$

$$\xi_{1,573} = \begin{cases} 0 & \text{for } \xi_{1,528}^2 + \xi_{1,531}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,531}, \xi_{1,528}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,574} = \begin{cases} 0 & \text{for } \xi_{1,529}^2 + \xi_{1,530}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,529}, \xi_{1,530}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,575} = 2\operatorname{atan}_2\left(\sqrt{\xi_{1,529}^2 + \xi_{1,530}^2}, \sqrt{\xi_{1,528}^2 + \xi_{1,531}^2}\right)$$

Exact shared-expression program 1

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{1,576} = -\xi_{1,573} + \xi_{1,574}$$

$$\xi_{1,577} = -\xi_{1,573} - \xi_{1,574}$$

$$\xi_{1,578} = \xi_{1,572} + \xi_{1,573}$$

$$\xi_{1,579} = \begin{cases} 0 & \text{for } \xi_{1,532}^2 + \xi_{1,535}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,535}, \xi_{1,532}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,580} = \begin{cases} 0 & \text{for } \xi_{1,533}^2 + \xi_{1,534}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{1,533}, \xi_{1,534}\right) & \text{otherwise} \end{cases}$$

$$\xi_{1,581} = 2\operatorname{atan}_2\left(\sqrt{\xi_{1,533}^2 + \xi_{1,534}^2}, \sqrt{\xi_{1,532}^2 + \xi_{1,535}^2}\right)$$

$$\xi_{1,582} = -\xi_{1,579} + \xi_{1,580}$$

$$\xi_{1,583} = -\xi_{1,579} - \xi_{1,580}$$

$$\xi_{1,584} = \xi_{1,578} + \xi_{1,579}$$

$$\xi_{1,585} = -\frac{\xi_{1,539}}{2} - \frac{\xi_{1,545}}{2} - \frac{\xi_{1,551}}{2} - \frac{\xi_{1,557}}{2} - \frac{\xi_{1,563}}{2} - \frac{\xi_{1,569}}{2} - \frac{\xi_{1,575}}{2} - \frac{\xi_{1,581}}{2} + \xi_{1,584}$$

Derived parameters for program 1

Exact output expressions used by the native gate labels.

$$v_{1,0} = \frac{2\xi_{1,539}+2\xi_{1,585}+\pi\left(-\frac{\xi_{1,541}}{w}+\frac{\xi_{1,540}+\xi_{1,541}}{w}+\frac{1}{2}\right)-\frac{3\pi}{2}}{\pi}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,0} = \left\lfloor \frac{2a}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{2,1} = \left\lfloor \frac{2b}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{2,2} = \left\lfloor \frac{2c}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{2,3} = -\frac{\pi \xi_{2,0}}{2} + a$$

$$\xi_{2,4} = -\frac{\pi \xi_{2,1}}{2} + b$$

$$\xi_{2,5} = -\frac{\pi \xi_{2,2}}{2} + c$$

$$\xi_{2,6} = \begin{cases} \xi_{2,3} & \text{for } (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,3}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \\ \xi_{2,4} & \text{for } (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,4}|) \wedge (|\xi_{2,3}| \leq |\xi_{2,4}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \\ \xi_{2,5} & \text{otherwise} \end{cases}$$

$$\xi_{2,7} = \begin{cases} \xi_{2,4} & \text{for } |\xi_{2,3}| \geq |\xi_{2,4}| \wedge |\xi_{2,4}| \geq |\xi_{2,5}| \\ \xi_{2,5} & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge |\xi_{2,4}| \leq |\xi_{2,5}| \\ \xi_{2,3} & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge |\xi_{2,3}| \leq |\xi_{2,4}| \\ \xi_{2,5} & \text{for } |\xi_{2,4}| \geq |\xi_{2,5}| \wedge |\xi_{2,3}| \leq |\xi_{2,5}| \\ \xi_{2,3} & \text{for } |\xi_{2,3}| \geq |\xi_{2,4}| \wedge |\xi_{2,3}| \leq |\xi_{2,5}| \\ \xi_{2,4} & \text{otherwise} \end{cases}$$

$$\xi_{2,8} = \begin{cases} \xi_{2,5} & \text{for } |\xi_{2,3}| \geq |\xi_{2,4}| \wedge |\xi_{2,4}| \geq |\xi_{2,5}| \\ \xi_{2,4} & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge |\xi_{2,4}| \leq |\xi_{2,5}| \\ \xi_{2,5} & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge |\xi_{2,3}| \leq |\xi_{2,4}| \\ \xi_{2,3} & \text{for } |\xi_{2,4}| \geq |\xi_{2,5}| \wedge |\xi_{2,3}| \leq |\xi_{2,5}| \\ \xi_{2,4} & \text{for } |\xi_{2,3}| \geq |\xi_{2,4}| \wedge |\xi_{2,3}| \leq |\xi_{2,5}| \\ \xi_{2,3} & \text{otherwise} \end{cases}$$

$$\xi_{2,9} = \begin{cases} 1 & \text{for } |\xi_{2,3}| \geq |\xi_{2,4}| \wedge |\xi_{2,4}| \geq |\xi_{2,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge |\xi_{2,4}| \leq |\xi_{2,5}| \\ 0 & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge |\xi_{2,3}| \leq |\xi_{2,4}| \\ \frac{1}{2} & \text{for } |\xi_{2,3}| \leq |\xi_{2,5}| \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,4}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{2,10} = \begin{cases} 0 & \text{for } |\xi_{2,3}| \geq |\xi_{2,4}| \wedge |\xi_{2,4}| \geq |\xi_{2,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,4}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \leq |\xi_{2,4}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \\ \frac{1}{2} & \text{for } |\xi_{2,3}| \leq |\xi_{2,5}| \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,4}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,11} = \begin{cases} 0 & \text{for } (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,3}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{2,3}| \geq |\xi_{2,5}| \wedge |\xi_{2,3}| \leq |\xi_{2,4}| \\ \frac{1}{2} & \text{for } |\xi_{2,3}| \leq |\xi_{2,5}| \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,4}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{2,12} = \begin{cases} 0 & \text{for } (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,3}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,5}| \geq |\xi_{2,4}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,5}| \geq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,5}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,4}| \vee |\xi_{2,5}| \leq |\xi_{2,4}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,4}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \geq |\xi_{2,4}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,4}|) \wedge (|\xi_{2,3}| \geq |\xi_{2,5}| \vee |\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,5}|) \wedge (|\xi_{2,4}| \geq |\xi_{2,5}| \vee |\xi_{2,3}| \leq |\xi_{2,4}| \vee |\xi_{2,4}| \leq |\xi_{2,5}|) \\ \frac{1}{2} & \text{for } |\xi_{2,4}| \geq |\xi_{2,5}| \wedge |\xi_{2,3}| \leq |\xi_{2,5}| \\ -\frac{1}{2} & \text{for } |\xi_{2,3}| \geq |\xi_{2,4}| \wedge |\xi_{2,3}| \leq |\xi_{2,5}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,13} = \begin{cases} -1 & \text{for } \xi_{2,6} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{2,14} = \begin{cases} -1 & \text{for } \xi_{2,7} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{2,15} = \begin{cases} 1 & \text{for } \xi_{2,13} = 1 \wedge \xi_{2,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,16} = \begin{cases} 0 & \text{for } \xi_{2,13} = 1 \wedge \xi_{2,14} = 1 \\ 1 & \text{for } \xi_{2,13} = 1 \wedge \xi_{2,14} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,17} = \begin{cases} 0 & \text{for } \xi_{2,13} = 1 \wedge (\xi_{2,13} = 1 \vee \xi_{2,14} = -1) \wedge (\xi_{2,13} = 1 \vee \xi_{2,14} = 1) \wedge (\xi_{2,14} = -1 \vee \xi_{2,14} = 1) \\ 1 & \text{for } \xi_{2,13} = -1 \wedge \xi_{2,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,18} = \begin{cases} 0 & \text{for } (\xi_{2,13} = -1 \vee \xi_{2,13} = 1) \wedge (\xi_{2,13} = 1 \vee \xi_{2,14} = 1) \wedge (\xi_{2,14} = -1 \vee \xi_{2,14} = 1) \wedge (\xi_{2,13} = -1 \vee \xi_{2,13} = 1 \vee \xi_{2,14} = -1) \wedge (\xi_{2,13} = -1 \vee \xi_{2,13} = 1 \vee \xi_{2,14} = 1) \wedge (\xi_{2,13} = -1 \vee \xi_{2,14} = -1 \vee \xi_{2,14} = 1) \wedge (\xi_{2,13} = 1 \vee \xi_{2,14} = -1 \vee \xi_{2,14} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{2,19} = \cos\left(\frac{\pi \xi_{2,0}}{2}\right)$$

$$\xi_{2,20} = -\sin\left(\frac{\pi \xi_{2,0}}{2}\right)$$

$$\xi_{2,21} = \cos\left(\frac{\pi \xi_{2,1}}{2}\right)$$

$$\xi_{2,22} = -\sin\left(\frac{\pi \xi_{2,1}}{2}\right)$$

$$\xi_{2,23} = \xi_{2,19}\xi_{2,21}$$

$$\xi_{2,24} = \xi_{2,20}\xi_{2,21}$$

$$\xi_{2,25} = \xi_{2,19}\xi_{2,22}$$

$$\xi_{2,26} = -\xi_{2,20}\xi_{2,22}$$

$$\xi_{2,27} = \cos\left(\frac{\pi \xi_{2,2}}{2}\right)$$

$$\xi_{2,28} = -\sin\left(\frac{\pi \xi_{2,2}}{2}\right)$$

$$\xi_{2,29} = \xi_{2,23}\xi_{2,27} - \xi_{2,26}\xi_{2,28}$$

$$\xi_{2,30} = \xi_{2,24}\xi_{2,27} - \xi_{2,25}\xi_{2,28}$$

$$\xi_{2,31} = \xi_{2,24}\xi_{2,28} + \xi_{2,25}\xi_{2,27}$$

$$\xi_{2,32} = \xi_{2,23}\xi_{2,28} + \xi_{2,26}\xi_{2,27}$$

$$\xi_{2,33} = \xi_{2,15}\left(-\xi_{2,10}\xi_{2,30} - \xi_{2,11}\xi_{2,31} - \xi_{2,12}\xi_{2,32} + \xi_{2,29}\xi_{2,9}\right) - \xi_{2,16}\left(\xi_{2,10}\xi_{2,29} - \xi_{2,11}\xi_{2,32} + \xi_{2,12}\xi_{2,31} + \xi_{2,30}\xi_{2,9}\right) - \xi_{2,17}\left(\xi_{2,10}\xi_{2,32} + \xi_{2,11}\xi_{2,29} - \xi_{2,12}\xi_{2,30} + \xi_{2,31}\xi_{2,9}\right) - \xi_{2,18}\left(-\xi_{2,10}\xi_{2,31} + \xi_{2,11}\xi_{2,30} + \xi_{2,12}\xi_{2,29} + \xi_{2,32}\xi_{2,9}\right)$$

$$\xi_{2,34} = \xi_{2,15}\left(\xi_{2,10}\xi_{2,29} - \xi_{2,11}\xi_{2,32} + \xi_{2,12}\xi_{2,31} + \xi_{2,30}\xi_{2,9}\right) + \xi_{2,16}\left(-\xi_{2,10}\xi_{2,30} - \xi_{2,11}\xi_{2,31} - \xi_{2,12}\xi_{2,32} + \xi_{2,29}\xi_{2,9}\right) - \xi_{2,17}\left(-\xi_{2,10}\xi_{2,31} + \xi_{2,11}\xi_{2,30} + \xi_{2,12}\xi_{2,29} + \xi_{2,32}\xi_{2,9}\right) + \xi_{2,18}\left(\xi_{2,10}\xi_{2,32} + \xi_{2,11}\xi_{2,29} - \xi_{2,12}\xi_{2,30} + \xi_{2,31}\xi_{2,9}\right)$$

$$\xi_{2,35} = \xi_{2,15}\left(\xi_{2,10}\xi_{2,32} + \xi_{2,11}\xi_{2,29} - \xi_{2,12}\xi_{2,30} + \xi_{2,31}\xi_{2,9}\right) + \xi_{2,16}\left(-\xi_{2,10}\xi_{2,31} + \xi_{2,11}\xi_{2,30} + \xi_{2,12}\xi_{2,29} + \xi_{2,32}\xi_{2,9}\right) + \xi_{2,17}\left(-\xi_{2,10}\xi_{2,30} - \xi_{2,11}\xi_{2,31} - \xi_{2,12}\xi_{2,32} + \xi_{2,29}\xi_{2,9}\right) - \xi_{2,18}\left(\xi_{2,10}\xi_{2,29} - \xi_{2,11}\xi_{2,32} + \xi_{2,12}\xi_{2,31} + \xi_{2,30}\xi_{2,9}\right)$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,36} = \xi_{2,15}\left(-\xi_{2,10}\xi_{2,31} + \xi_{2,11}\xi_{2,30} + \xi_{2,12}\xi_{2,29} + \xi_{2,32}\xi_{2,9}\right) - \xi_{2,16}\left(\xi_{2,10}\xi_{2,32} + \xi_{2,11}\xi_{2,29} - \xi_{2,12}\xi_{2,30} + \xi_{2,31}\xi_{2,9}\right) + \xi_{2,17}\left(\xi_{2,10}\xi_{2,29} - \xi_{2,11}\xi_{2,32} + \xi_{2,12}\xi_{2,31} + \xi_{2,30}\xi_{2,9}\right) + \xi_{2,18}\left(-\xi_{2,10}\xi_{2,30} - \xi_{2,11}\xi_{2,31} - \xi_{2,12}\xi_{2,32} + \xi_{2,29}\xi_{2,9}\right)$$

$$\xi_{2,37} = -\xi_{2,10}\xi_{2,30} - \xi_{2,11}\xi_{2,31} - \xi_{2,12}\xi_{2,32} + \xi_{2,29}\xi_{2,9}$$

$$\xi_{2,38} = \xi_{2,10}\xi_{2,29} - \xi_{2,11}\xi_{2,32} + \xi_{2,12}\xi_{2,31} + \xi_{2,30}\xi_{2,9}$$

$$\xi_{2,39} = \xi_{2,10}\xi_{2,32} + \xi_{2,11}\xi_{2,29} - \xi_{2,12}\xi_{2,30} + \xi_{2,31}\xi_{2,9}$$

$$\xi_{2,40} = -\xi_{2,10}\xi_{2,31} + \xi_{2,11}\xi_{2,30} + \xi_{2,12}\xi_{2,29} + \xi_{2,32}\xi_{2,9}$$

$$\xi_{2,41} = -\xi_{2,10}\xi_{2,16} - \xi_{2,11}\xi_{2,17} - \xi_{2,12}\xi_{2,18} + \xi_{2,15}\xi_{2,9}$$

$$\xi_{2,42} = -\xi_{2,10}\xi_{2,15} - \xi_{2,11}\xi_{2,18} + \xi_{2,12}\xi_{2,17} - \xi_{2,16}\xi_{2,9}$$

$$\xi_{2,43} = \xi_{2,10}\xi_{2,18} - \xi_{2,11}\xi_{2,15} - \xi_{2,12}\xi_{2,16} - \xi_{2,17}\xi_{2,9}$$

$$\xi_{2,44} = -\xi_{2,10}\xi_{2,17} + \xi_{2,11}\xi_{2,16} - \xi_{2,12}\xi_{2,15} - \xi_{2,18}\xi_{2,9}$$

$$\xi_{2,45} = \xi_{2,13}\xi_{2,6}$$

$$\xi_{2,46} = \xi_{2,14}\xi_{2,7}$$

$$\xi_{2,47} = \xi_{2,13}\xi_{2,14}\xi_{2,8}$$

$$\xi_{2,48} = -\frac{\pi(\xi_{2,0}+\xi_{2,1}+\xi_{2,2})}{2}$$

$$\xi_{2,49} = \begin{cases} 0 & \text{for } \left(\xi_{2,46} > \frac{\pi}{8} \vee \xi_{2,47} < 0\right) \wedge \xi_{2,46} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{2,50} = \frac{\pi}{8}$$

$$\xi_{2,51} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{2,46} > \frac{\pi}{8} \wedge \xi_{2,47} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{2,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,52} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{2,46} > \frac{\pi}{8} \wedge \xi_{2,47} < 0 \\ \frac{1}{2} & \text{for } \xi_{2,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,53} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{2,46} > \frac{\pi}{8} \vee \xi_{2,47} < 0\right) \wedge \xi_{2,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,54} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{2,46} > \frac{\pi}{8} \vee \xi_{2,47} < 0\right) \wedge \xi_{2,46} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{2,55} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{2,46} > \frac{\pi}{8} \vee \xi_{2,47} < 0\right) \wedge \xi_{2,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,56} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{2,46} > \frac{\pi}{8} \vee \xi_{2,47} < 0\right) \wedge \xi_{2,46} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{2,57} = \begin{cases} \frac{1}{2} & \text{for } \xi_{2,46} > \frac{\pi}{8} \wedge \xi_{2,47} < 0 \\ -\frac{1}{2} & \text{for } \xi_{2,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,58} = \left\lfloor \frac{2\xi_{2,45} - 2\xi_{2,49}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{2,59} = \left\lfloor \frac{2\xi_{2,46} - 2\xi_{2,50}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{2,60} = \left\lfloor \frac{2\xi_{2,47} - 2\xi_{2,51}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{2,61} = \xi_{2,45} - \xi_{2,49} - \frac{\pi\xi_{2,58}}{2}$$

$$\xi_{2,62} = \xi_{2,46} - \xi_{2,50} - \frac{\pi\xi_{2,59}}{2}$$

$$\xi_{2,63} = \xi_{2,47} - \xi_{2,51} - \frac{\pi\xi_{2,60}}{2}$$

$$\xi_{2,64} = \begin{cases} \xi_{2,61} & \text{for } \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \\ \xi_{2,62} & \text{for } \left(|\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,62}|\right) \wedge \left(|\xi_{2,61}| \leq |\xi_{2,62}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \\ \xi_{2,63} & \text{otherwise} \end{cases}$$

$$\xi_{2,65} = \begin{cases} \xi_{2,62} & \text{for } |\xi_{2,61}| \geq |\xi_{2,62}| \wedge |\xi_{2,62}| \geq |\xi_{2,63}| \\ \xi_{2,63} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,62}| \leq |\xi_{2,63}| \\ \xi_{2,61} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,62}| \\ \xi_{2,63} & \text{for } |\xi_{2,62}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,63}| \\ \xi_{2,61} & \text{for } |\xi_{2,61}| \geq |\xi_{2,62}| \wedge |\xi_{2,61}| \leq |\xi_{2,63}| \\ \xi_{2,62} & \text{otherwise} \end{cases}$$

$$\xi_{2,66} = \begin{cases} \xi_{2,63} & \text{for } |\xi_{2,61}| \geq |\xi_{2,62}| \wedge |\xi_{2,62}| \geq |\xi_{2,63}| \\ \xi_{2,62} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,62}| \leq |\xi_{2,63}| \\ \xi_{2,63} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,62}| \\ \xi_{2,61} & \text{for } |\xi_{2,62}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,63}| \\ \xi_{2,62} & \text{for } |\xi_{2,61}| \geq |\xi_{2,62}| \wedge |\xi_{2,61}| \leq |\xi_{2,63}| \\ \xi_{2,61} & \text{otherwise} \end{cases}$$

$$\xi_{2,67} = \begin{cases} 1 & \text{for } |\xi_{2,61}| \geq |\xi_{2,62}| \wedge |\xi_{2,62}| \geq |\xi_{2,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,62}| \leq |\xi_{2,63}| \\ 0 & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,62}| \\ \frac{1}{2} & \text{for } |\xi_{2,61}| \leq |\xi_{2,63}| \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{2,68} = \begin{cases} 0 & \text{for } |\xi_{2,61}| \geq |\xi_{2,62}| \wedge |\xi_{2,62}| \geq |\xi_{2,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge \left(|\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,62}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \leq |\xi_{2,62}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \\ \frac{1}{2} & \text{for } |\xi_{2,61}| \leq |\xi_{2,63}| \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,69} = \begin{cases} 0 & \text{for } \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,62}| \\ \frac{1}{2} & \text{for } |\xi_{2,61}| \leq |\xi_{2,63}| \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{2,70} = \begin{cases} 0 & \text{for } \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \geq |\xi_{2,63}| \vee |\xi_{2,62}| \geq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \leq |\xi_{2,63}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,62}| \vee |\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,62}| \vee |\xi_{2,61}| \leq |\xi_{2,63}|\right) \wedge \left(|\xi_{2,61}| \geq |\xi_{2,62}| \vee |\xi_{2,62}| \geq |\xi_{2,63}| \vee |\xi_{2,61}| \leq |\xi_{2,62}| \vee |\xi_{2,62}| \leq |\xi_{2,63}|\right) \\ \frac{1}{2} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,62}| \\ -\frac{1}{2} & \text{for } |\xi_{2,61}| \geq |\xi_{2,63}| \wedge |\xi_{2,61}| \leq |\xi_{2,63}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,71} = \begin{cases} -1 & \text{for } \xi_{2,64} < 0 \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,72} = \begin{cases} -1 & \text{for } \xi_{2,65} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{2,73} = \begin{cases} 1 & \text{for } \xi_{2,71} = 1 \wedge \xi_{2,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,74} = \begin{cases} 0 & \text{for } \xi_{2,71} = 1 \wedge \xi_{2,72} = 1 \\ 1 & \text{for } \xi_{2,71} = 1 \wedge \xi_{2,72} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,75} = \begin{cases} 0 & \text{for } \xi_{2,71} = 1 \wedge (\xi_{2,71} = 1 \vee \xi_{2,72} = -1) \wedge (\xi_{2,71} = 1 \vee \xi_{2,72} = 1) \wedge (\xi_{2,72} = -1 \vee \xi_{2,72} = 1) \\ 1 & \text{for } \xi_{2,71} = -1 \wedge \xi_{2,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,76} = \begin{cases} 0 & \text{for } (\xi_{2,71} = -1 \vee \xi_{2,71} = 1) \wedge (\xi_{2,71} = 1 \vee \xi_{2,72} = 1) \wedge (\xi_{2,72} = -1 \vee \xi_{2,72} = 1) \wedge (\xi_{2,71} = -1 \vee \xi_{2,71} = 1 \vee \xi_{2,72} = -1) \wedge (\xi_{2,71} = -1 \vee \xi_{2,71} = 1 \vee \xi_{2,72} = 1) \wedge (\xi_{2,71} = -1 \vee \xi_{2,72} = -1 \vee \xi_{2,72} = 1) \wedge (\xi_{2,71} = 1 \vee \xi_{2,72} = -1 \vee \xi_{2,72} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{2,77} = \cos\left(\frac{\pi\xi_{2,58}}{2}\right)$$

$$\xi_{2,78} = -\sin\left(\frac{\pi\xi_{2,58}}{2}\right)$$

$$\xi_{2,79} = \cos\left(\frac{\pi\xi_{2,59}}{2}\right)$$

$$\xi_{2,80} = -\sin\left(\frac{\pi\xi_{2,59}}{2}\right)$$

$$\xi_{2,81} = \xi_{2,77}\xi_{2,79}$$

$$\xi_{2,82} = \xi_{2,78}\xi_{2,79}$$

$$\xi_{2,83} = \xi_{2,77}\xi_{2,80}$$

$$\xi_{2,84} = -\xi_{2,78}\xi_{2,80}$$

$$\xi_{2,85} = \cos\left(\frac{\pi\xi_{2,60}}{2}\right)$$

$$\xi_{2,86} = -\sin\left(\frac{\pi\xi_{2,60}}{2}\right)$$

$$\xi_{2,87} = \xi_{2,81}\xi_{2,85} - \xi_{2,84}\xi_{2,86}$$

$$\xi_{2,88} = \xi_{2,82}\xi_{2,85} - \xi_{2,83}\xi_{2,86}$$

$$\xi_{2,89} = \xi_{2,82}\xi_{2,86} + \xi_{2,83}\xi_{2,85}$$

$$\xi_{2,90} = \xi_{2,81}\xi_{2,86} + \xi_{2,84}\xi_{2,85}$$

$$\xi_{2,91} = \xi_{2,73}\left(\xi_{2,67}\xi_{2,87} - \xi_{2,68}\xi_{2,88} - \xi_{2,69}\xi_{2,89} - \xi_{2,70}\xi_{2,90}\right) - \xi_{2,74}\left(\xi_{2,67}\xi_{2,88} + \xi_{2,68}\xi_{2,87} - \xi_{2,69}\xi_{2,90} + \xi_{2,70}\xi_{2,89}\right) - \xi_{2,75}\left(\xi_{2,67}\xi_{2,89} + \xi_{2,68}\xi_{2,90} + \xi_{2,69}\xi_{2,87} - \xi_{2,70}\xi_{2,88}\right) - \xi_{2,76}\left(\xi_{2,67}\xi_{2,90} - \xi_{2,68}\xi_{2,89} + \xi_{2,69}\xi_{2,88} + \xi_{2,70}\xi_{2,87}\right)$$

$$\xi_{2,92} = \xi_{2,73}\left(\xi_{2,67}\xi_{2,88} + \xi_{2,68}\xi_{2,87} - \xi_{2,69}\xi_{2,90} + \xi_{2,70}\xi_{2,89}\right) + \xi_{2,74}\left(\xi_{2,67}\xi_{2,87} - \xi_{2,68}\xi_{2,88} - \xi_{2,69}\xi_{2,89} - \xi_{2,70}\xi_{2,90}\right) - \xi_{2,75}\left(\xi_{2,67}\xi_{2,90} - \xi_{2,68}\xi_{2,89} + \xi_{2,69}\xi_{2,88} + \xi_{2,70}\xi_{2,87}\right) + \xi_{2,76}\left(\xi_{2,67}\xi_{2,89} + \xi_{2,68}\xi_{2,90} + \xi_{2,69}\xi_{2,87} - \xi_{2,70}\xi_{2,88}\right)$$

$$\xi_{2,93} = \xi_{2,73}\left(\xi_{2,67}\xi_{2,89} + \xi_{2,68}\xi_{2,90} + \xi_{2,69}\xi_{2,87} - \xi_{2,70}\xi_{2,88}\right) + \xi_{2,74}\left(\xi_{2,67}\xi_{2,90} - \xi_{2,68}\xi_{2,89} + \xi_{2,69}\xi_{2,88} + \xi_{2,70}\xi_{2,87}\right) + \xi_{2,75}\left(\xi_{2,67}\xi_{2,87} - \xi_{2,68}\xi_{2,88} - \xi_{2,69}\xi_{2,89} - \xi_{2,70}\xi_{2,90}\right) - \xi_{2,76}\left(\xi_{2,67}\xi_{2,88} + \xi_{2,68}\xi_{2,87} - \xi_{2,69}\xi_{2,90} + \xi_{2,70}\xi_{2,89}\right)$$

$$\xi_{2,94} = \xi_{2,73}\left(\xi_{2,67}\xi_{2,90} - \xi_{2,68}\xi_{2,89} + \xi_{2,69}\xi_{2,88} + \xi_{2,70}\xi_{2,87}\right) - \xi_{2,74}\left(\xi_{2,67}\xi_{2,89} + \xi_{2,68}\xi_{2,90} + \xi_{2,69}\xi_{2,87} - \xi_{2,70}\xi_{2,88}\right) + \xi_{2,75}\left(\xi_{2,67}\xi_{2,88} + \xi_{2,68}\xi_{2,87} - \xi_{2,69}\xi_{2,90} + \xi_{2,70}\xi_{2,89}\right) + \xi_{2,76}\left(\xi_{2,67}\xi_{2,87} - \xi_{2,68}\xi_{2,88} - \xi_{2,69}\xi_{2,89} - \xi_{2,70}\xi_{2,90}\right)$$

$$\xi_{2,95} = \xi_{2,67}\xi_{2,87} - \xi_{2,68}\xi_{2,88} - \xi_{2,69}\xi_{2,89} - \xi_{2,70}\xi_{2,90}$$

$$\xi_{2,96} = \xi_{2,67}\xi_{2,88} + \xi_{2,68}\xi_{2,87} - \xi_{2,69}\xi_{2,90} + \xi_{2,70}\xi_{2,89}$$

$$\xi_{2,97} = \xi_{2,67}\xi_{2,89} + \xi_{2,68}\xi_{2,90} + \xi_{2,69}\xi_{2,87} - \xi_{2,70}\xi_{2,88}$$

$$\xi_{2,98} = \xi_{2,67}\xi_{2,90} - \xi_{2,68}\xi_{2,89} + \xi_{2,69}\xi_{2,88} + \xi_{2,70}\xi_{2,87}$$

$$\xi_{2,99} = \xi_{2,67}\xi_{2,73} - \xi_{2,68}\xi_{2,74} - \xi_{2,69}\xi_{2,75} - \xi_{2,70}\xi_{2,76}$$

$$\xi_{2,100} = -\xi_{2,67}\xi_{2,74} - \xi_{2,68}\xi_{2,73} - \xi_{2,69}\xi_{2,76} + \xi_{2,70}\xi_{2,75}$$

$$\xi_{2,101} = -\xi_{2,67}\xi_{2,75} + \xi_{2,68}\xi_{2,76} - \xi_{2,69}\xi_{2,73} - \xi_{2,70}\xi_{2,74}$$

$$\xi_{2,102} = -\xi_{2,67}\xi_{2,76} - \xi_{2,68}\xi_{2,75} + \xi_{2,69}\xi_{2,74} - \xi_{2,70}\xi_{2,73}$$

$$\xi_{2,103} = \xi_{2,64}\xi_{2,71}$$

$$\xi_{2,104} = \xi_{2,65}\xi_{2,72}$$

$$\xi_{2,105} = \xi_{2,66}\xi_{2,71}\xi_{2,72}$$

$$\xi_{2,106} = -\frac{\pi(\xi_{2,58}+\xi_{2,59}+\xi_{2,60})}{2}$$

$$\xi_{2,107} = \cos\left(2\xi_{2,103}\right)$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,108} = \cos\left(2\xi_{2,104}\right)$$

$$\xi_{2,109} = \cos\left(2\xi_{2,105}\right)$$

$$\xi_{2,110} = \sqrt{\xi_{2,107}^2 - 2\xi_{2,107}\xi_{2,108}\xi_{2,109} + \xi_{2,108}^2 + \xi_{2,109}^2 - 1}$$

$$\xi_{2,111} = \xi_{2,107} - \xi_{2,108} + \xi_{2,109}$$

$$\xi_{2,112} = \sqrt{-\frac{\xi_{2,110}}{2} + \frac{\xi_{2,111}}{2} + \frac{1}{2}}$$

$$\xi_{2,113} = \sqrt{\frac{\xi_{2,110}}{2} - \frac{\xi_{2,111}}{2} + \frac{1}{2}}$$

$$\xi_{2,114} = \sqrt{-\frac{\xi_{2,110}}{2} - \frac{\xi_{2,111}}{2} + \frac{1}{2}}$$

$$\xi_{2,115} = \frac{(1-\xi_{2,108})(\xi_{2,107}+1)(\xi_{2,109}+1)}{2}$$

$$\xi_{2,116} = \begin{cases} -1 & \text{for } \xi_{2,105} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{2,117} = \begin{cases} 0 & \text{for } \xi_{2,112} = 0 \\ \frac{\sqrt{\xi_{2,115}\xi_{2,116}}}{\xi_{2,112}} & \text{otherwise} \end{cases}$$

$$\xi_{2,118} = \begin{cases} 1 & \text{for } \xi_{2,112} = 0 \\ \frac{\sqrt{\xi_{2,107}\xi_{2,108}\xi_{2,109}}}{\xi_{2,112}} & \text{otherwise} \end{cases}$$

$$\xi_{2,119} = 2\xi_{2,112}\xi_{2,113}\xi_{2,114}\xi_{2,118}$$

$$\xi_{2,120} = \sqrt{2}\xi_{2,118}\left(\xi_{2,112}^2\xi_{2,114} - \xi_{2,114}\right)$$

$$\xi_{2,121} = 2\xi_{2,112}\xi_{2,113}\xi_{2,114}\xi_{2,117}$$

$$\xi_{2,122} = \sqrt{2}\left(2\xi_{2,112}^2\xi_{2,114}\xi_{2,117} - \xi_{2,114}\xi_{2,117}\right)$$

$$\xi_{2,123} = \sqrt{2}\xi_{2,112}\xi_{2,113}\xi_{2,117}\xi_{2,118}$$

$$\xi_{2,124} = \sqrt{2}\xi_{2,112}\xi_{2,113}\left(-\xi_{2,114}^2 + \xi_{2,117}^2\right)$$

$$\xi_{2,125} = -\xi_{2,112}^2\xi_{2,114}^2 + \xi_{2,112}^2\xi_{2,117}^2 + \xi_{2,114}^2$$

$$\xi_{2,126} = \xi_{2,112}^2\xi_{2,114}^2 - \xi_{2,112}^2\xi_{2,117}^2 + \xi_{2,117}^2$$

$$\xi_{2,127} = 2\xi_{2,118}\left(\xi_{2,112}^2\xi_{2,117} - \frac{\xi_{2,117}}{2}\right)$$

$$\xi_{2,128} = -\sqrt{2}\xi_{2,112}\xi_{2,113}\xi_{2,117}\xi_{2,118}$$

$$\xi_{2,129} = -\xi_{2,117}\xi_{2,118}$$

$$\xi_{2,130} = \xi_{2,112}^2\xi_{2,114}^2 + \xi_{2,112}^2\xi_{2,117}^2 - \xi_{2,112}^2 - \xi_{2,114}^2 - \xi_{2,117}^2 + 1$$

$$\xi_{2,131} = -\sqrt{2}\left(\xi_{2,112}\xi_{2,113}\xi_{2,114}^2 + \xi_{2,112}\xi_{2,113}\xi_{2,117}^2 - \xi_{2,112}\xi_{2,113}\right)$$

$$\xi_{2,132} = \sqrt{2}\xi_{2,112}^2\xi_{2,114}\xi_{2,118}$$

$$\xi_{2,133} = \xi_{2,123}^2 - \xi_{2,126}^2 - \xi_{2,129}^2 - \xi_{2,132}^2$$

$$\xi_{2,134} = \xi_{2,123}\xi_{2,124} - \xi_{2,126}\xi_{2,127} - \xi_{2,129}\xi_{2,130}$$

$$\xi_{2,135} = \xi_{2,123}\xi_{2,125} - \xi_{2,126}\xi_{2,128} - \xi_{2,129}\xi_{2,131}$$

$$\xi_{2,136} = \xi_{2,119}^2 + \xi_{2,121}^2 + \xi_{2,124}^2 - \xi_{2,127}^2 - \xi_{2,130}^2$$

$$\xi_{2,137} = \xi_{2,119}\xi_{2,120} + \xi_{2,121}\xi_{2,122} + \xi_{2,124}\xi_{2,125} - \xi_{2,127}\xi_{2,128} - \xi_{2,130}\xi_{2,131}$$

$$\xi_{2,138} = \xi_{2,120}^2 + \xi_{2,122}^2 + \xi_{2,125}^2 - \xi_{2,128}^2 - \xi_{2,131}^2$$

$$\xi_{2,139} = \sin\left(2\xi_{2,104}\right)\sin\left(2\xi_{2,105}\right)\cos\left(2\xi_{2,103}\right)$$

$$\xi_{2,140} = \sin\left(2\xi_{2,103}\right)\sin\left(2\xi_{2,105}\right)\cos\left(2\xi_{2,104}\right)$$

$$\xi_{2,141} = \sin\left(2\xi_{2,103}\right)\sin\left(2\xi_{2,104}\right)\cos\left(2\xi_{2,105}\right)$$

$$\xi_{2,142} = \sin\left(2\xi_{2,103}\right)\cos\left(2\xi_{2,104}\right)\cos\left(2\xi_{2,105}\right)$$

$$\xi_{2,143} = \sin\left(2\xi_{2,104}\right)\cos\left(2\xi_{2,103}\right)\cos\left(2\xi_{2,105}\right)$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{2,144} = \sin\left(2\xi_{2,105}\right)\cos\left(2\xi_{2,103}\right)\cos\left(2\xi_{2,104}\right)$$
$$\xi_{2,145} = \xi_{2,139}^2 - \xi_{2,142}^2$$
$$\xi_{2,146} = \xi_{2,140}^2 - \xi_{2,143}^2$$
$$\xi_{2,147} = \xi_{2,141}^2 - \xi_{2,144}^2$$
$$\xi_{2,148} = \frac{\xi_{2,133}-\xi_{2,146}}{\xi_{2,145}-\xi_{2,146}}$$
$$\xi_{2,149} = \frac{\xi_{2,134}}{\xi_{2,145}-\xi_{2,146}}$$
$$\xi_{2,150} = \frac{\xi_{2,135}}{\xi_{2,145}-\xi_{2,146}}$$
$$\xi_{2,151} = \frac{\xi_{2,136}-\xi_{2,146}}{\xi_{2,145}-\xi_{2,146}}$$
$$\xi_{2,152} = \frac{\xi_{2,137}}{\xi_{2,145}-\xi_{2,146}}$$
$$\xi_{2,153} = \frac{\xi_{2,138}-\xi_{2,146}}{\xi_{2,145}-\xi_{2,146}}$$
$$\xi_{2,154} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,148} & \text{otherwise} \end{cases}$$
$$\xi_{2,155} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,149} & \text{otherwise} \end{cases}$$
$$\xi_{2,156} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,150} & \text{otherwise} \end{cases}$$
$$\xi_{2,157} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,151} & \text{otherwise} \end{cases}$$
$$\xi_{2,158} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,152} & \text{otherwise} \end{cases}$$
$$\xi_{2,159} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,153} & \text{otherwise} \end{cases}$$
$$\xi_{2,160} = \frac{\xi_{2,133}-\xi_{2,147}}{\xi_{2,145}-\xi_{2,147}}$$
$$\xi_{2,161} = \frac{\xi_{2,134}}{\xi_{2,145}-\xi_{2,147}}$$
$$\xi_{2,162} = \frac{\xi_{2,135}}{\xi_{2,145}-\xi_{2,147}}$$
$$\xi_{2,163} = \frac{\xi_{2,136}-\xi_{2,147}}{\xi_{2,145}-\xi_{2,147}}$$
$$\xi_{2,164} = \frac{\xi_{2,137}}{\xi_{2,145}-\xi_{2,147}}$$
$$\xi_{2,165} = \frac{\xi_{2,138}-\xi_{2,147}}{\xi_{2,145}-\xi_{2,147}}$$
$$\xi_{2,166} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,147} \\ \xi_{2,160} & \text{otherwise} \end{cases}$$
$$\xi_{2,167} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,147} \\ \xi_{2,161} & \text{otherwise} \end{cases}$$
$$\xi_{2,168} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,147} \\ \xi_{2,162} & \text{otherwise} \end{cases}$$
$$\xi_{2,169} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,147} \\ \xi_{2,163} & \text{otherwise} \end{cases}$$
$$\xi_{2,170} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,147} \\ \xi_{2,164} & \text{otherwise} \end{cases}$$
$$\xi_{2,171} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,147} \\ \xi_{2,165} & \text{otherwise} \end{cases}$$
$$\xi_{2,172} = \xi_{2,154}\xi_{2,166} + \xi_{2,155}\xi_{2,167} + \xi_{2,156}\xi_{2,168}$$
$$\xi_{2,173} = \xi_{2,154}\xi_{2,167} + \xi_{2,155}\xi_{2,169} + \xi_{2,156}\xi_{2,170}$$
$$\xi_{2,174} = \xi_{2,154}\xi_{2,168} + \xi_{2,155}\xi_{2,170} + \xi_{2,156}\xi_{2,171}$$
$$\xi_{2,175} = \xi_{2,155}\xi_{2,166} + \xi_{2,157}\xi_{2,167} + \xi_{2,158}\xi_{2,168}$$
$$\xi_{2,176} = \xi_{2,155}\xi_{2,167} + \xi_{2,157}\xi_{2,169} + \xi_{2,158}\xi_{2,170}$$
$$\xi_{2,177} = \xi_{2,155}\xi_{2,168} + \xi_{2,157}\xi_{2,170} + \xi_{2,158}\xi_{2,171}$$
$$\xi_{2,178} = \xi_{2,156}\xi_{2,166} + \xi_{2,158}\xi_{2,167} + \xi_{2,159}\xi_{2,168}$$
$$\xi_{2,179} = \xi_{2,156}\xi_{2,167} + \xi_{2,158}\xi_{2,169} + \xi_{2,159}\xi_{2,170}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{2,180} = \xi_{2,156}\xi_{2,168} + \xi_{2,158}\xi_{2,170} + \xi_{2,159}\xi_{2,171}$$

$$\xi_{2,181} = \sqrt{\xi_{2,172}}$$

$$\xi_{2,182} = \frac{\xi_{2,172}}{\xi_{2,181}}$$

$$\xi_{2,183} = \frac{\xi_{2,175}}{\xi_{2,181}}$$

$$\xi_{2,184} = \frac{\xi_{2,178}}{\xi_{2,181}}$$

$$\xi_{2,185} = \sqrt{\xi_{2,176}}$$

$$\xi_{2,186} = \frac{\xi_{2,173}}{\xi_{2,185}}$$

$$\xi_{2,187} = \frac{\xi_{2,176}}{\xi_{2,185}}$$

$$\xi_{2,188} = \frac{\xi_{2,179}}{\xi_{2,185}}$$

$$\xi_{2,189} = \sqrt{\xi_{2,180}}$$

$$\xi_{2,190} = \frac{\xi_{2,174}}{\xi_{2,189}}$$

$$\xi_{2,191} = \frac{\xi_{2,177}}{\xi_{2,189}}$$

$$\xi_{2,192} = \frac{\xi_{2,180}}{\xi_{2,189}}$$

$$\xi_{2,193} = \begin{cases} \xi_{2,182} & \text{for } \xi_{2,172} \geq \xi_{2,176} \wedge \xi_{2,172} \geq \xi_{2,180} \\ \xi_{2,186} & \text{for } \xi_{2,176} \geq \xi_{2,180} \wedge \xi_{2,172} \leq \xi_{2,176} \\ \xi_{2,190} & \text{otherwise} \end{cases}$$

$$\xi_{2,194} = \begin{cases} \xi_{2,183} & \text{for } \xi_{2,172} \geq \xi_{2,176} \wedge \xi_{2,172} \geq \xi_{2,180} \\ \xi_{2,187} & \text{for } \xi_{2,176} \geq \xi_{2,180} \wedge \xi_{2,172} \leq \xi_{2,176} \\ \xi_{2,191} & \text{otherwise} \end{cases}$$

$$\xi_{2,195} = \begin{cases} \xi_{2,184} & \text{for } \xi_{2,172} \geq \xi_{2,176} \wedge \xi_{2,172} \geq \xi_{2,180} \\ \xi_{2,188} & \text{for } \xi_{2,176} \geq \xi_{2,180} \wedge \xi_{2,172} \leq \xi_{2,176} \\ \xi_{2,192} & \text{otherwise} \end{cases}$$

$$\xi_{2,196} = \frac{\xi_{2,133} - \xi_{2,145}}{-\xi_{2,145} + \xi_{2,146}}$$

$$\xi_{2,197} = \frac{\xi_{2,134}}{-\xi_{2,145} + \xi_{2,146}}$$

$$\xi_{2,198} = \frac{\xi_{2,135}}{-\xi_{2,145} + \xi_{2,146}}$$

$$\xi_{2,199} = \frac{\xi_{2,136} - \xi_{2,145}}{-\xi_{2,145} + \xi_{2,146}}$$

$$\xi_{2,200} = \frac{\xi_{2,137}}{-\xi_{2,145} + \xi_{2,146}}$$

$$\xi_{2,201} = \frac{\xi_{2,138} - \xi_{2,145}}{-\xi_{2,145} + \xi_{2,146}}$$

$$\xi_{2,202} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,196} & \text{otherwise} \end{cases}$$

$$\xi_{2,203} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,197} & \text{otherwise} \end{cases}$$

$$\xi_{2,204} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,198} & \text{otherwise} \end{cases}$$

$$\xi_{2,205} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,199} & \text{otherwise} \end{cases}$$

$$\xi_{2,206} = \begin{cases} 0 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,200} & \text{otherwise} \end{cases}$$

$$\xi_{2,207} = \begin{cases} 1 & \text{for } \xi_{2,145} = \xi_{2,146} \\ \xi_{2,201} & \text{otherwise} \end{cases}$$

$$\xi_{2,208} = \frac{\xi_{2,133} - \xi_{2,147}}{\xi_{2,146} - \xi_{2,147}}$$

$$\xi_{2,209} = \frac{\xi_{2,134}}{\xi_{2,146} - \xi_{2,147}}$$

$$\xi_{2,210} = \frac{\xi_{2,135}}{\xi_{2,146} - \xi_{2,147}}$$

$$\xi_{2,211} = \frac{\xi_{2,136} - \xi_{2,147}}{\xi_{2,146} - \xi_{2,147}}$$

$$\xi_{2,212} = \frac{\xi_{2,137}}{\xi_{2,146} - \xi_{2,147}}$$

$$\xi_{2,213} = \frac{\xi_{2,138} - \xi_{2,147}}{\xi_{2,146} - \xi_{2,147}}$$

$$\xi_{2,214} = \begin{cases} 1 & \text{for } \xi_{2,146} = \xi_{2,147} \\ \xi_{2,208} & \text{otherwise} \end{cases}$$

$$\xi_{2,215} = \begin{cases} 0 & \text{for } \xi_{2,146} = \xi_{2,147} \\ \xi_{2,209} & \text{otherwise} \end{cases}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{2,216} = \begin{cases} 0 & \text{for } \xi_{2,146} = \xi_{2,147} \\ \xi_{2,210} & \text{otherwise} \end{cases}$$

$$\xi_{2,217} = \begin{cases} 1 & \text{for } \xi_{2,146} = \xi_{2,147} \\ \xi_{2,211} & \text{otherwise} \end{cases}$$

$$\xi_{2,218} = \begin{cases} 0 & \text{for } \xi_{2,146} = \xi_{2,147} \\ \xi_{2,212} & \text{otherwise} \end{cases}$$

$$\xi_{2,219} = \begin{cases} 1 & \text{for } \xi_{2,146} = \xi_{2,147} \\ \xi_{2,213} & \text{otherwise} \end{cases}$$

$$\xi_{2,220} = \xi_{2,202}\xi_{2,214} + \xi_{2,203}\xi_{2,215} + \xi_{2,204}\xi_{2,216}$$

$$\xi_{2,221} = \xi_{2,202}\xi_{2,215} + \xi_{2,203}\xi_{2,217} + \xi_{2,204}\xi_{2,218}$$

$$\xi_{2,222} = \xi_{2,202}\xi_{2,216} + \xi_{2,203}\xi_{2,218} + \xi_{2,204}\xi_{2,219}$$

$$\xi_{2,223} = \xi_{2,203}\xi_{2,214} + \xi_{2,205}\xi_{2,215} + \xi_{2,206}\xi_{2,216}$$

$$\xi_{2,224} = \xi_{2,203}\xi_{2,215} + \xi_{2,205}\xi_{2,217} + \xi_{2,206}\xi_{2,218}$$

$$\xi_{2,225} = \xi_{2,203}\xi_{2,216} + \xi_{2,205}\xi_{2,218} + \xi_{2,206}\xi_{2,219}$$

$$\xi_{2,226} = \xi_{2,204}\xi_{2,214} + \xi_{2,206}\xi_{2,215} + \xi_{2,207}\xi_{2,216}$$

$$\xi_{2,227} = \xi_{2,204}\xi_{2,215} + \xi_{2,206}\xi_{2,217} + \xi_{2,207}\xi_{2,218}$$

$$\xi_{2,228} = \xi_{2,204}\xi_{2,216} + \xi_{2,206}\xi_{2,218} + \xi_{2,207}\xi_{2,219}$$

$$\xi_{2,229} = \begin{cases} \xi_{2,193}^2 & \text{for } \xi_{2,145} = \xi_{2,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,230} = \begin{cases} \xi_{2,193}\xi_{2,194} & \text{for } \xi_{2,145} = \xi_{2,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,231} = \begin{cases} \xi_{2,193}\xi_{2,195} & \text{for } \xi_{2,145} = \xi_{2,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,232} = \begin{cases} \xi_{2,194}^2 & \text{for } \xi_{2,145} = \xi_{2,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,233} = \begin{cases} \xi_{2,194}\xi_{2,195} & \text{for } \xi_{2,145} = \xi_{2,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,234} = \begin{cases} \xi_{2,195}^2 & \text{for } \xi_{2,145} = \xi_{2,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{2,235} = \xi_{2,220} - \xi_{2,229}$$

$$\xi_{2,236} = \xi_{2,221} - \xi_{2,230}$$

$$\xi_{2,237} = \xi_{2,222} - \xi_{2,231}$$

$$\xi_{2,238} = \xi_{2,223} - \xi_{2,230}$$

$$\xi_{2,239} = \xi_{2,224} - \xi_{2,232}$$

$$\xi_{2,240} = \xi_{2,225} - \xi_{2,233}$$

$$\xi_{2,241} = \xi_{2,226} - \xi_{2,231}$$

$$\xi_{2,242} = \xi_{2,227} - \xi_{2,233}$$

$$\xi_{2,243} = \xi_{2,228} - \xi_{2,234}$$

$$\xi_{2,244} = \sqrt{\xi_{2,235}}$$

$$\xi_{2,245} = \frac{\xi_{2,235}}{\xi_{2,244}}$$

$$\xi_{2,246} = \frac{\xi_{2,238}}{\xi_{2,244}}$$

$$\xi_{2,247} = \frac{\xi_{2,241}}{\xi_{2,244}}$$

$$\xi_{2,248} = \sqrt{\xi_{2,239}}$$

$$\xi_{2,249} = \frac{\xi_{2,236}}{\xi_{2,248}}$$

$$\xi_{2,250} = \frac{\xi_{2,239}}{\xi_{2,248}}$$

$$\xi_{2,251} = \frac{\xi_{2,242}}{\xi_{2,248}}$$

Exact shared-expression program 2

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$$\xi_{2,252} = \sqrt{\xi_{2,243}}$$

$$\xi_{2,253} = \frac{\xi_{2,237}}{\xi_{2,252}}$$

$$\xi_{2,254} = \frac{\xi_{2,240}}{\xi_{2,252}}$$

$$\xi_{2,255} = \frac{\xi_{2,243}}{\xi_{2,252}}$$

$$\xi_{2,256} = \begin{cases} \xi_{2,245} & \text{for } \xi_{2,235} \geq \xi_{2,239} \wedge \xi_{2,235} \geq \xi_{2,243} \\ \xi_{2,249} & \text{for } \xi_{2,239} \geq \xi_{2,243} \wedge \xi_{2,235} \leq \xi_{2,239} \\ \xi_{2,253} & \text{otherwise} \end{cases}$$

$$\xi_{2,257} = \begin{cases} \xi_{2,246} & \text{for } \xi_{2,235} \geq \xi_{2,239} \wedge \xi_{2,235} \geq \xi_{2,243} \\ \xi_{2,250} & \text{for } \xi_{2,239} \geq \xi_{2,243} \wedge \xi_{2,235} \leq \xi_{2,239} \\ \xi_{2,254} & \text{otherwise} \end{cases}$$

$$\xi_{2,258} = \begin{cases} \xi_{2,247} & \text{for } \xi_{2,235} \geq \xi_{2,239} \wedge \xi_{2,235} \geq \xi_{2,243} \\ \xi_{2,251} & \text{for } \xi_{2,239} \geq \xi_{2,243} \wedge \xi_{2,235} \leq \xi_{2,239} \\ \xi_{2,255} & \text{otherwise} \end{cases}$$

$$\xi_{2,259} = \xi_{2,194}\xi_{2,258} - \xi_{2,195}\xi_{2,257}$$

$$\xi_{2,260} = -\xi_{2,193}\xi_{2,258} + \xi_{2,195}\xi_{2,256}$$

$$\xi_{2,261} = \xi_{2,193}\xi_{2,257} - \xi_{2,194}\xi_{2,256}$$

$$\xi_{2,262} = \xi_{2,139}^2 + \xi_{2,142}^2$$

$$\xi_{2,263} = \xi_{2,140}^2 + \xi_{2,143}^2$$

$$\xi_{2,264} = \xi_{2,141}^2 + \xi_{2,144}^2$$

$$\xi_{2,265} = \frac{\xi_{2,126}\xi_{2,142}\xi_{2,193} + \xi_{2,194}(\xi_{2,119}\xi_{2,139} + \xi_{2,127}\xi_{2,142}) + \xi_{2,195}(\xi_{2,120}\xi_{2,139} + \xi_{2,128}\xi_{2,142})}{\xi_{2,262}}$$

$$\xi_{2,266} = \frac{\xi_{2,129}\xi_{2,142}\xi_{2,193} + \xi_{2,194}(\xi_{2,121}\xi_{2,139} + \xi_{2,130}\xi_{2,142}) + \xi_{2,195}(\xi_{2,122}\xi_{2,139} + \xi_{2,131}\xi_{2,142})}{\xi_{2,262}}$$

$$\xi_{2,267} = \frac{\xi_{2,124}\xi_{2,139}\xi_{2,194} + \xi_{2,125}\xi_{2,139}\xi_{2,195} + \xi_{2,193}(\xi_{2,123}\xi_{2,139} + \xi_{2,132}\xi_{2,142})}{\xi_{2,262}}$$

$$\xi_{2,268} = \frac{\xi_{2,126}\xi_{2,143}\xi_{2,256} + \xi_{2,257}(\xi_{2,119}\xi_{2,140} + \xi_{2,127}\xi_{2,143}) + \xi_{2,258}(\xi_{2,120}\xi_{2,140} + \xi_{2,128}\xi_{2,143})}{\xi_{2,263}}$$

$$\xi_{2,269} = \frac{\xi_{2,129}\xi_{2,143}\xi_{2,256} + \xi_{2,257}(\xi_{2,121}\xi_{2,140} + \xi_{2,130}\xi_{2,143}) + \xi_{2,258}(\xi_{2,122}\xi_{2,140} + \xi_{2,131}\xi_{2,143})}{\xi_{2,263}}$$

$$\xi_{2,270} = \frac{\xi_{2,124}\xi_{2,140}\xi_{2,257} + \xi_{2,125}\xi_{2,140}\xi_{2,258} + \xi_{2,256}(\xi_{2,123}\xi_{2,140} + \xi_{2,132}\xi_{2,143})}{\xi_{2,263}}$$

$$\xi_{2,271} = \frac{\xi_{2,126}\xi_{2,144}\xi_{2,259} + \xi_{2,260}(\xi_{2,119}\xi_{2,141} + \xi_{2,127}\xi_{2,144}) + \xi_{2,261}(\xi_{2,120}\xi_{2,141} + \xi_{2,128}\xi_{2,144})}{\xi_{2,264}}$$

$$\xi_{2,272} = \frac{\xi_{2,129}\xi_{2,144}\xi_{2,259} + \xi_{2,260}(\xi_{2,121}\xi_{2,141} + \xi_{2,130}\xi_{2,144}) + \xi_{2,261}(\xi_{2,122}\xi_{2,141} + \xi_{2,131}\xi_{2,144})}{\xi_{2,264}}$$

$$\xi_{2,273} = \frac{\xi_{2,124}\xi_{2,141}\xi_{2,260} + \xi_{2,125}\xi_{2,141}\xi_{2,261} + \xi_{2,259}(\xi_{2,123}\xi_{2,141} + \xi_{2,132}\xi_{2,144})}{\xi_{2,264}}$$

$$\xi_{2,274} = 1 - \xi_{2,265}^2$$

$$\xi_{2,275} = -\xi_{2,265}\xi_{2,266}$$

$$\xi_{2,276} = -\xi_{2,265}\xi_{2,267}$$

$$\xi_{2,277} = 1 - \xi_{2,266}^2$$

$$\xi_{2,278} = -\xi_{2,266}\xi_{2,267}$$

$$\xi_{2,279} = 1 - \xi_{2,267}^2$$

$$\xi_{2,280} = \xi_{2,274}^2 + \xi_{2,275}^2 + \xi_{2,276}^2$$

$$\xi_{2,281} = \xi_{2,275}^2 + \xi_{2,277}^2 + \xi_{2,278}^2$$

$$\xi_{2,282} = \xi_{2,276}^2 + \xi_{2,278}^2 + \xi_{2,279}^2$$

$$\xi_{2,283} = \sqrt{\xi_{2,280}}$$

$$\xi_{2,284} = \frac{\xi_{2,274}}{\xi_{2,283}}$$

$$\xi_{2,285} = \frac{\xi_{2,275}}{\xi_{2,283}}$$

$$\xi_{2,286} = \frac{\xi_{2,276}}{\xi_{2,283}}$$

$$\xi_{2,287} = \sqrt{\xi_{2,281}}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,288} = \frac{\xi_{2,277}}{\xi_{2,287}}$$

$$\xi_{2,289} = \frac{\xi_{2,275}}{\xi_{2,287}}$$

$$\xi_{2,290} = \frac{\xi_{2,278}}{\xi_{2,287}}$$

$$\xi_{2,291} = \sqrt{\xi_{2,282}}$$

$$\xi_{2,292} = \frac{\xi_{2,279}}{\xi_{2,291}}$$

$$\xi_{2,293} = \frac{\xi_{2,276}}{\xi_{2,291}}$$

$$\xi_{2,294} = \frac{\xi_{2,278}}{\xi_{2,291}}$$

$$\xi_{2,295} = \begin{cases} \xi_{2,284} & \text{for } \xi_{2,280} \geq \xi_{2,281} \wedge \xi_{2,280} \geq \xi_{2,282} \\ \xi_{2,289} & \text{for } \xi_{2,281} \geq \xi_{2,282} \wedge \xi_{2,280} \leq \xi_{2,281} \\ \xi_{2,293} & \text{otherwise} \end{cases}$$

$$\xi_{2,296} = \begin{cases} \xi_{2,285} & \text{for } \xi_{2,280} \geq \xi_{2,281} \wedge \xi_{2,280} \geq \xi_{2,282} \\ \xi_{2,288} & \text{for } \xi_{2,281} \geq \xi_{2,282} \wedge \xi_{2,280} \leq \xi_{2,281} \\ \xi_{2,294} & \text{otherwise} \end{cases}$$

$$\xi_{2,297} = \begin{cases} \xi_{2,286} & \text{for } \xi_{2,280} \geq \xi_{2,281} \wedge \xi_{2,280} \geq \xi_{2,282} \\ \xi_{2,290} & \text{for } \xi_{2,281} \geq \xi_{2,282} \wedge \xi_{2,280} \leq \xi_{2,281} \\ \xi_{2,292} & \text{otherwise} \end{cases}$$

$$\xi_{2,298} = \xi_{2,266}\xi_{2,297} - \xi_{2,267}\xi_{2,296}$$

$$\xi_{2,299} = -\xi_{2,265}\xi_{2,297} + \xi_{2,267}\xi_{2,295}$$

$$\xi_{2,300} = \xi_{2,265}\xi_{2,296} - \xi_{2,266}\xi_{2,295}$$

$$\xi_{2,301} = 1 - \xi_{2,268}^2$$

$$\xi_{2,302} = -\xi_{2,268}\xi_{2,269}$$

$$\xi_{2,303} = -\xi_{2,268}\xi_{2,270}$$

$$\xi_{2,304} = 1 - \xi_{2,269}^2$$

$$\xi_{2,305} = -\xi_{2,269}\xi_{2,270}$$

$$\xi_{2,306} = 1 - \xi_{2,270}^2$$

$$\xi_{2,307} = \xi_{2,301}^2 + \xi_{2,302}^2 + \xi_{2,303}^2$$

$$\xi_{2,308} = \xi_{2,302}^2 + \xi_{2,304}^2 + \xi_{2,305}^2$$

$$\xi_{2,309} = \xi_{2,303}^2 + \xi_{2,305}^2 + \xi_{2,306}^2$$

$$\xi_{2,310} = \sqrt{\xi_{2,307}}$$

$$\xi_{2,311} = \frac{\xi_{2,301}}{\xi_{2,310}}$$

$$\xi_{2,312} = \frac{\xi_{2,302}}{\xi_{2,310}}$$

$$\xi_{2,313} = \frac{\xi_{2,303}}{\xi_{2,310}}$$

$$\xi_{2,314} = \sqrt{\xi_{2,308}}$$

$$\xi_{2,315} = \frac{\xi_{2,304}}{\xi_{2,314}}$$

$$\xi_{2,316} = \frac{\xi_{2,302}}{\xi_{2,314}}$$

$$\xi_{2,317} = \frac{\xi_{2,305}}{\xi_{2,314}}$$

$$\xi_{2,318} = \sqrt{\xi_{2,309}}$$

$$\xi_{2,319} = \frac{\xi_{2,306}}{\xi_{2,318}}$$

$$\xi_{2,320} = \frac{\xi_{2,303}}{\xi_{2,318}}$$

$$\xi_{2,321} = \frac{\xi_{2,305}}{\xi_{2,318}}$$

$$\xi_{2,322} = \begin{cases} \xi_{2,311} & \text{for } \xi_{2,307} \geq \xi_{2,308} \wedge \xi_{2,307} \geq \xi_{2,309} \\ \xi_{2,316} & \text{for } \xi_{2,308} \geq \xi_{2,309} \wedge \xi_{2,307} \leq \xi_{2,308} \\ \xi_{2,320} & \text{otherwise} \end{cases}$$

$$\xi_{2,323} = \begin{cases} \xi_{2,312} & \text{for } \xi_{2,307} \geq \xi_{2,308} \wedge \xi_{2,307} \geq \xi_{2,309} \\ \xi_{2,315} & \text{for } \xi_{2,308} \geq \xi_{2,309} \wedge \xi_{2,307} \leq \xi_{2,308} \\ \xi_{2,321} & \text{otherwise} \end{cases}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{2,324} = \begin{cases} \xi_{2,313} & \text{for } \xi_{2,307} \geq \xi_{2,308} \wedge \xi_{2,307} \geq \xi_{2,309} \\ \xi_{2,317} & \text{for } \xi_{2,308} \geq \xi_{2,309} \wedge \xi_{2,307} \leq \xi_{2,308} \\ \xi_{2,319} & \text{otherwise} \end{cases}$$

$$\xi_{2,325} = \xi_{2,269}\xi_{2,324} - \xi_{2,270}\xi_{2,323}$$

$$\xi_{2,326} = -\xi_{2,268}\xi_{2,324} + \xi_{2,270}\xi_{2,322}$$

$$\xi_{2,327} = \xi_{2,268}\xi_{2,323} - \xi_{2,269}\xi_{2,322}$$

$$\xi_{2,328} = \xi_{2,266}\xi_{2,270} - \xi_{2,267}\xi_{2,269}$$

$$\xi_{2,329} = -\xi_{2,265}\xi_{2,270} + \xi_{2,267}\xi_{2,268}$$

$$\xi_{2,330} = \xi_{2,265}\xi_{2,269} - \xi_{2,266}\xi_{2,268}$$

$$\xi_{2,331} = 1 - \xi_{2,271}^2$$

$$\xi_{2,332} = -\xi_{2,271}\xi_{2,272}$$

$$\xi_{2,333} = -\xi_{2,271}\xi_{2,273}$$

$$\xi_{2,334} = 1 - \xi_{2,272}^2$$

$$\xi_{2,335} = -\xi_{2,272}\xi_{2,273}$$

$$\xi_{2,336} = 1 - \xi_{2,273}^2$$

$$\xi_{2,337} = \xi_{2,331}^2 + \xi_{2,332}^2 + \xi_{2,333}^2$$

$$\xi_{2,338} = \xi_{2,332}^2 + \xi_{2,334}^2 + \xi_{2,335}^2$$

$$\xi_{2,339} = \xi_{2,333}^2 + \xi_{2,335}^2 + \xi_{2,336}^2$$

$$\xi_{2,340} = \sqrt{\xi_{2,337}}$$

$$\xi_{2,341} = \frac{\xi_{2,331}}{\xi_{2,340}}$$

$$\xi_{2,342} = \frac{\xi_{2,332}}{\xi_{2,340}}$$

$$\xi_{2,343} = \frac{\xi_{2,333}}{\xi_{2,340}}$$

$$\xi_{2,344} = \sqrt{\xi_{2,338}}$$

$$\xi_{2,345} = \frac{\xi_{2,334}}{\xi_{2,344}}$$

$$\xi_{2,346} = \frac{\xi_{2,332}}{\xi_{2,344}}$$

$$\xi_{2,347} = \frac{\xi_{2,335}}{\xi_{2,344}}$$

$$\xi_{2,348} = \sqrt{\xi_{2,339}}$$

$$\xi_{2,349} = \frac{\xi_{2,336}}{\xi_{2,348}}$$

$$\xi_{2,350} = \frac{\xi_{2,333}}{\xi_{2,348}}$$

$$\xi_{2,351} = \frac{\xi_{2,335}}{\xi_{2,348}}$$

$$\xi_{2,352} = \begin{cases} \xi_{2,341} & \text{for } \xi_{2,337} \geq \xi_{2,338} \wedge \xi_{2,337} \geq \xi_{2,339} \\ \xi_{2,346} & \text{for } \xi_{2,338} \geq \xi_{2,339} \wedge \xi_{2,337} \leq \xi_{2,338} \\ \xi_{2,350} & \text{otherwise} \end{cases}$$

$$\xi_{2,353} = \begin{cases} \xi_{2,342} & \text{for } \xi_{2,337} \geq \xi_{2,338} \wedge \xi_{2,337} \geq \xi_{2,339} \\ \xi_{2,345} & \text{for } \xi_{2,338} \geq \xi_{2,339} \wedge \xi_{2,337} \leq \xi_{2,338} \\ \xi_{2,351} & \text{otherwise} \end{cases}$$

$$\xi_{2,354} = \begin{cases} \xi_{2,343} & \text{for } \xi_{2,337} \geq \xi_{2,338} \wedge \xi_{2,337} \geq \xi_{2,339} \\ \xi_{2,347} & \text{for } \xi_{2,338} \geq \xi_{2,339} \wedge \xi_{2,337} \leq \xi_{2,338} \\ \xi_{2,349} & \text{otherwise} \end{cases}$$

$$\xi_{2,355} = \xi_{2,272}\xi_{2,354} - \xi_{2,273}\xi_{2,353}$$

$$\xi_{2,356} = -\xi_{2,271}\xi_{2,354} + \xi_{2,273}\xi_{2,352}$$

$$\xi_{2,357} = \xi_{2,271}\xi_{2,353} - \xi_{2,272}\xi_{2,352}$$

$$\xi_{2,358} = -\xi_{2,266}\xi_{2,273} + \xi_{2,267}\xi_{2,272}$$

$$\xi_{2,359} = \xi_{2,265}\xi_{2,273} - \xi_{2,267}\xi_{2,271}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,360} = -\xi_{2,265}\xi_{2,272} + \xi_{2,266}\xi_{2,271}$$

$$\xi_{2,361} = \xi_{2,269}\xi_{2,273} - \xi_{2,270}\xi_{2,272}$$

$$\xi_{2,362} = -\xi_{2,268}\xi_{2,273} + \xi_{2,270}\xi_{2,271}$$

$$\xi_{2,363} = \xi_{2,268}\xi_{2,272} - \xi_{2,269}\xi_{2,271}$$

$$\xi_{2,364} = \begin{cases} 1 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,265} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,325} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,263} \neq 0 \\ \xi_{2,265} & \text{for } \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,263} \neq 0 \\ \xi_{2,352} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,265} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,361} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,265} & \text{for } \xi_{2,262} \neq 0 \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,365} = \begin{cases} 0 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,295} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,268} & \text{for } \xi_{2,264} = 0 \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} = 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,262} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge \xi_{2,263} \neq 0 \\ \xi_{2,355} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,358} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,268} & \text{for } (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,263} \neq 0 \vee \xi_{2,264} \neq 0) \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,366} = \begin{cases} 0 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,326} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,266} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,263} \neq 0 \\ \xi_{2,353} & \text{for } \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,263} \neq 0 \\ \xi_{2,266} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,362} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,266} & \text{for } \xi_{2,262} \neq 0 \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,367} = \begin{cases} 0 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,296} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,269} & \text{for } \xi_{2,264} = 0 \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} = 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,262} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge \xi_{2,263} \neq 0 \\ \xi_{2,356} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,359} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,269} & \text{for } (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,263} \neq 0 \vee \xi_{2,264} \neq 0) \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,368} = \begin{cases} 0 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,267} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,327} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,263} \neq 0 \\ \xi_{2,267} & \text{for } \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,263} \neq 0 \\ \xi_{2,354} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,267} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,363} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,267} & \text{for } \xi_{2,262} \neq 0 \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,369} = \begin{cases} 0 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,297} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,270} & \text{for } \xi_{2,264} = 0 \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} = 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,262} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge \xi_{2,263} \neq 0 \\ \xi_{2,357} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,360} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,270} & \text{for } (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,263} \neq 0 \vee \xi_{2,264} \neq 0) \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,371} = \begin{cases} 0 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,297} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,270} & \text{for } \xi_{2,264} = 0 \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} = 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,262} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge \xi_{2,263} \neq 0 \\ \xi_{2,357} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,360} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,270} & \text{for } (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,263} \neq 0 \vee \xi_{2,264} \neq 0) \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,372} = \begin{cases} 0 & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \\ \xi_{2,298} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,264} = 0 \wedge \xi_{2,262} \neq 0 \\ \xi_{2,271} & \text{for } \xi_{2,264} = 0 \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} = 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,262} \neq 0) \wedge (\xi_{2,264} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge \xi_{2,263} \neq 0 \\ \xi_{2,358} & \text{for } \xi_{2,262} = 0 \wedge \xi_{2,263} = 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,361} & \text{for } \xi_{2,263} = 0 \wedge \xi_{2,262} \neq 0 \wedge \xi_{2,264} \neq 0 \\ \xi_{2,271} & \text{for } (\xi_{2,262} = 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} = 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,263} \neq 0) \wedge (\xi_{2,262} \neq 0 \vee \xi_{2,264} \neq 0) \wedge (\xi_{2,263} \neq 0 \vee \xi_{2,264} \neq 0) \wedge \xi_{2,263} \neq 0 \wedge \xi_{2,264} \neq 0 \end{cases}$$

$$\xi_{2,373} = \frac{\xi_{2,364}}{4} + \frac{\xi_{2,368}}{4} + \frac{\xi_{2,372}}{4} + \frac{1}{4}$$

$$\xi_{2,374} = \frac{\xi_{2,364}}{4} - \frac{\xi_{2,368}}{4} - \frac{\xi_{2,372}}{4} + \frac{1}{4}$$

$$\xi_{2,375} = -\frac{\xi_{2,364}}{4} + \frac{\xi_{2,368}}{4} - \frac{\xi_{2,372}}{4} + \frac{1}{4}$$

$$\xi_{2,376} = -\frac{\xi_{2,364}}{4} - \frac{\xi_{2,368}}{4} + \frac{\xi_{2,372}}{4} + \frac{1}{4}$$

$$\xi_{2,377} = \sqrt{\xi_{2,373}}$$

$$\xi_{2,378} = \frac{-\xi_{2,369} + \xi_{2,371}}{4\xi_{2,377}}$$

$$\xi_{2,379} = \frac{\xi_{2,366} - \xi_{2,370}}{4\xi_{2,377}}$$

$$\xi_{2,380} = \frac{-\xi_{2,365} + \xi_{2,367}}{4\xi_{2,377}}$$

$$\xi_{2,381} = \sqrt{\xi_{2,374}}$$

$$\xi_{2,382} = \frac{-\xi_{2,369} + \xi_{2,371}}{4\xi_{2,381}}$$

$$\xi_{2,383} = \frac{\xi_{2,365} + \xi_{2,367}}{4\xi_{2,381}}$$

$$\xi_{2,384} = \frac{\xi_{2,366} + \xi_{2,370}}{4\xi_{2,381}}$$

$$\xi_{2,385} = \sqrt{\xi_{2,375}}$$

$$\xi_{2,386} = \frac{\xi_{2,366} - \xi_{2,370}}{4\xi_{2,385}}$$

$$\xi_{2,387} = \frac{\xi_{2,365} + \xi_{2,367}}{4\xi_{2,385}}$$

$$\xi_{2,388} = \frac{\xi_{2,369} + \xi_{2,371}}{4\xi_{2,385}}$$

$$\xi_{2,389} = \sqrt{\xi_{2,376}}$$

$$\xi_{2,390} = \frac{-\xi_{2,365} + \xi_{2,367}}{4\xi_{2,389}}$$

$$\xi_{2,391} = \frac{\xi_{2,366} + \xi_{2,370}}{4\xi_{2,389}}$$

$$\xi_{2,392} = \frac{\xi_{2,369} + \xi_{2,371}}{4\xi_{2,389}}$$

$$\xi_{2,393} = \begin{cases} \xi_{2,377} & \text{for } \xi_{2,373} \geq \xi_{2,374} \wedge \xi_{2,373} \geq \xi_{2,375} \wedge \xi_{2,373} \geq \xi_{2,376} \\ \xi_{2,382} & \text{for } \xi_{2,374} \geq \xi_{2,375} \wedge \xi_{2,374} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,374} \\ \xi_{2,386} & \text{for } \xi_{2,375} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,375} \wedge \xi_{2,374} \leq \xi_{2,375} \\ \xi_{2,390} & \text{otherwise} \end{cases}$$

$$\xi_{2,394} = \begin{cases} \xi_{2,378} & \text{for } \xi_{2,373} \geq \xi_{2,374} \wedge \xi_{2,373} \geq \xi_{2,375} \wedge \xi_{2,373} \geq \xi_{2,376} \\ \xi_{2,381} & \text{for } \xi_{2,374} \geq \xi_{2,375} \wedge \xi_{2,374} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,374} \\ \xi_{2,387} & \text{for } \xi_{2,375} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,375} \wedge \xi_{2,374} \leq \xi_{2,375} \\ \xi_{2,391} & \text{otherwise} \end{cases}$$

$$\xi_{2,395} = \begin{cases} \xi_{2,379} & \text{for } \xi_{2,373} \geq \xi_{2,374} \wedge \xi_{2,373} \geq \xi_{2,375} \wedge \xi_{2,373} \geq \xi_{2,376} \\ \xi_{2,383} & \text{for } \xi_{2,374} \geq \xi_{2,375} \wedge \xi_{2,374} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,374} \\ \xi_{2,385} & \text{for } \xi_{2,375} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,375} \wedge \xi_{2,374} \leq \xi_{2,375} \\ \xi_{2,392} & \text{otherwise} \end{cases}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,396} = \begin{cases} \xi_{2,380} & \text{for } \xi_{2,373} \geq \xi_{2,374} \wedge \xi_{2,373} \geq \xi_{2,375} \wedge \xi_{2,373} \geq \xi_{2,376} \\ \xi_{2,384} & \text{for } \xi_{2,374} \geq \xi_{2,375} \wedge \xi_{2,374} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,374} \\ \xi_{2,388} & \text{for } \xi_{2,375} \geq \xi_{2,376} \wedge \xi_{2,373} \leq \xi_{2,375} \wedge \xi_{2,374} \leq \xi_{2,375} \\ \xi_{2,389} & \text{otherwise} \end{cases}$$

$$\xi_{2,397} = \frac{\xi_{2,193}}{4} + \frac{\xi_{2,257}}{4} + \frac{\xi_{2,261}}{4} + \frac{1}{4}$$

$$\xi_{2,398} = \frac{\xi_{2,193}}{4} - \frac{\xi_{2,257}}{4} - \frac{\xi_{2,261}}{4} + \frac{1}{4}$$

$$\xi_{2,399} = -\frac{\xi_{2,193}}{4} + \frac{\xi_{2,257}}{4} - \frac{\xi_{2,261}}{4} + \frac{1}{4}$$

$$\xi_{2,400} = -\frac{\xi_{2,193}}{4} - \frac{\xi_{2,257}}{4} + \frac{\xi_{2,261}}{4} + \frac{1}{4}$$

$$\xi_{2,401} = \sqrt{\xi_{2,397}}$$

$$\xi_{2,402} = \frac{\xi_{2,258} - \xi_{2,260}}{4\xi_{2,401}}$$

$$\xi_{2,403} = \frac{-\xi_{2,195} + \xi_{2,259}}{4\xi_{2,401}}$$

$$\xi_{2,404} = \frac{\xi_{2,194} - \xi_{2,256}}{4\xi_{2,401}}$$

$$\xi_{2,405} = \sqrt{\xi_{2,398}}$$

$$\xi_{2,406} = \frac{\xi_{2,258} - \xi_{2,260}}{4\xi_{2,405}}$$

$$\xi_{2,407} = \frac{\xi_{2,194} + \xi_{2,256}}{4\xi_{2,405}}$$

$$\xi_{2,408} = \frac{\xi_{2,195} + \xi_{2,259}}{4\xi_{2,405}}$$

$$\xi_{2,409} = \sqrt{\xi_{2,399}}$$

$$\xi_{2,410} = \frac{-\xi_{2,195} + \xi_{2,259}}{4\xi_{2,409}}$$

$$\xi_{2,411} = \frac{\xi_{2,194} + \xi_{2,256}}{4\xi_{2,409}}$$

$$\xi_{2,412} = \frac{\xi_{2,258} + \xi_{2,260}}{4\xi_{2,409}}$$

$$\xi_{2,413} = \sqrt{\xi_{2,400}}$$

$$\xi_{2,414} = \frac{\xi_{2,194} - \xi_{2,256}}{4\xi_{2,413}}$$

$$\xi_{2,415} = \frac{\xi_{2,195} + \xi_{2,259}}{4\xi_{2,413}}$$

$$\xi_{2,416} = \frac{\xi_{2,258} + \xi_{2,260}}{4\xi_{2,413}}$$

$$\xi_{2,417} = \begin{cases} \xi_{2,401} & \text{for } \xi_{2,397} \geq \xi_{2,398} \wedge \xi_{2,397} \geq \xi_{2,399} \wedge \xi_{2,397} \geq \xi_{2,400} \\ \xi_{2,406} & \text{for } \xi_{2,398} \geq \xi_{2,399} \wedge \xi_{2,398} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,398} \\ \xi_{2,410} & \text{for } \xi_{2,399} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,399} \wedge \xi_{2,398} \leq \xi_{2,399} \\ \xi_{2,414} & \text{otherwise} \end{cases}$$

$$\xi_{2,418} = \begin{cases} \xi_{2,402} & \text{for } \xi_{2,397} \geq \xi_{2,398} \wedge \xi_{2,397} \geq \xi_{2,399} \wedge \xi_{2,397} \geq \xi_{2,400} \\ \xi_{2,405} & \text{for } \xi_{2,398} \geq \xi_{2,399} \wedge \xi_{2,398} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,398} \\ \xi_{2,411} & \text{for } \xi_{2,399} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,399} \wedge \xi_{2,398} \leq \xi_{2,399} \\ \xi_{2,415} & \text{otherwise} \end{cases}$$

$$\xi_{2,419} = \begin{cases} \xi_{2,403} & \text{for } \xi_{2,397} \geq \xi_{2,398} \wedge \xi_{2,397} \geq \xi_{2,399} \wedge \xi_{2,397} \geq \xi_{2,400} \\ \xi_{2,407} & \text{for } \xi_{2,398} \geq \xi_{2,399} \wedge \xi_{2,398} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,398} \\ \xi_{2,409} & \text{for } \xi_{2,399} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,399} \wedge \xi_{2,398} \leq \xi_{2,399} \\ \xi_{2,416} & \text{otherwise} \end{cases}$$

$$\xi_{2,420} = \begin{cases} \xi_{2,404} & \text{for } \xi_{2,397} \geq \xi_{2,398} \wedge \xi_{2,397} \geq \xi_{2,399} \wedge \xi_{2,397} \geq \xi_{2,400} \\ \xi_{2,408} & \text{for } \xi_{2,398} \geq \xi_{2,399} \wedge \xi_{2,398} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,398} \\ \xi_{2,412} & \text{for } \xi_{2,399} \geq \xi_{2,400} \wedge \xi_{2,397} \leq \xi_{2,399} \wedge \xi_{2,398} \leq \xi_{2,399} \\ \xi_{2,413} & \text{otherwise} \end{cases}$$

$$\xi_{2,421} = \xi_{2,112}\left(\xi_{2,117} + i\xi_{2,118}\right)$$

$$\xi_{2,422} = -\frac{\sqrt{2}\xi_{2,112}\xi_{2,114}}{2} + \frac{\sqrt{2}i\xi_{2,113}\left(\xi_{2,117} + i\xi_{2,118}\right)}{2}$$

$$\xi_{2,423} = \frac{\sqrt{2}i\xi_{2,112}\xi_{2,114}}{2} - \frac{\sqrt{2}\xi_{2,113}\left(\xi_{2,117} + i\xi_{2,118}\right)}{2}$$

$$\xi_{2,424} = -\xi_{2,113}\xi_{2,114}$$

$$\xi_{2,425} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{2,112}\left(\xi_{2,117} + i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}i\xi_{2,113}\xi_{2,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{2,112}\left(\xi_{2,117} - i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}\xi_{2,113}\xi_{2,114}}{2}\right)}{2}$$

$$\xi_{2,426} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{2,112}\left(\xi_{2,117} + i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}i\xi_{2,113}\xi_{2,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{2,112}\left(\xi_{2,117} - i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}\xi_{2,113}\xi_{2,114}}{2}\right)}{2}$$

$$\xi_{2,427} = \frac{\sqrt{2}i\xi_{2,112}\xi_{2,114}}{2} - \frac{\sqrt{2}\xi_{2,113}\left(\xi_{2,117} - i\xi_{2,118}\right)}{2}$$

$$\xi_{2,428} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{2,112}\left(\xi_{2,117} - i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}i\xi_{2,113}\xi_{2,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{2,112}\left(\xi_{2,117} + i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}\xi_{2,113}\xi_{2,114}}{2}\right)}{2}$$

$$\xi_{2,429} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{2,112}\left(\xi_{2,117} - i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}i\xi_{2,113}\xi_{2,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{2,112}\left(\xi_{2,117} + i\xi_{2,118}\right)}{2} - \frac{\sqrt{2}\xi_{2,113}\xi_{2,114}}{2}\right)}{2}$$

$$\xi_{2,430} = -\frac{\sqrt{2}\xi_{2,112}\xi_{2,114}}{2} + \frac{\sqrt{2}i\xi_{2,113}\left(\xi_{2,117} - i\xi_{2,118}\right)}{2}$$

$$\xi_{2,431} = \xi_{2,112}\left(\xi_{2,117} - i\xi_{2,118}\right)$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,432} = (\xi_{2,393} - i\xi_{2,396})\,(\xi_{2,417} - i\xi_{2,420})$$

$$\xi_{2,433} = (\xi_{2,393} - i\xi_{2,396})\,(-i\xi_{2,418} - \xi_{2,419})$$

$$\xi_{2,434} = (\xi_{2,417} - i\xi_{2,420})\,(-i\xi_{2,394} - \xi_{2,395})$$

$$\xi_{2,435} = (-i\xi_{2,394} - \xi_{2,395})\,(-i\xi_{2,418} - \xi_{2,419})$$

$$\xi_{2,436} = (\xi_{2,393} - i\xi_{2,396})\,(-i\xi_{2,418} + \xi_{2,419})$$

$$\xi_{2,437} = (\xi_{2,393} - i\xi_{2,396})\,(\xi_{2,417} + i\xi_{2,420})$$

$$\xi_{2,438} = (-i\xi_{2,394} - \xi_{2,395})\,(-i\xi_{2,418} + \xi_{2,419})$$

$$\xi_{2,439} = (\xi_{2,417} + i\xi_{2,420})\,(-i\xi_{2,394} - \xi_{2,395})$$

$$\xi_{2,440} = (\xi_{2,417} - i\xi_{2,420})\,(-i\xi_{2,394} + \xi_{2,395})$$

$$\xi_{2,441} = (-i\xi_{2,394} + \xi_{2,395})\,(-i\xi_{2,418} - \xi_{2,419})$$

$$\xi_{2,442} = (\xi_{2,393} + i\xi_{2,396})\,(\xi_{2,417} - i\xi_{2,420})$$

$$\xi_{2,443} = (\xi_{2,393} + i\xi_{2,396})\,(-i\xi_{2,418} - \xi_{2,419})$$

$$\xi_{2,444} = (-i\xi_{2,394} + \xi_{2,395})\,(-i\xi_{2,418} + \xi_{2,419})$$

$$\xi_{2,445} = (\xi_{2,417} + i\xi_{2,420})\,(-i\xi_{2,394} + \xi_{2,395})$$

$$\xi_{2,446} = (\xi_{2,393} + i\xi_{2,396})\,(-i\xi_{2,418} + \xi_{2,419})$$

$$\xi_{2,447} = (\xi_{2,393} + i\xi_{2,396})\,(\xi_{2,417} + i\xi_{2,420})$$

$$\xi_{2,448} = \xi_{2,421}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}}\right) + \xi_{2,422}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}}\right) + \xi_{2,423}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}}\right) + \xi_{2,424}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,444}}\right)$$

$$\xi_{2,449} = \xi_{2,422}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}}\right) + \xi_{2,425}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}}\right) + \xi_{2,427}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,444}}\right) + \xi_{2,428}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}}\right)$$

$$\xi_{2,450} = \xi_{2,423}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}}\right) + \xi_{2,426}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}}\right) + \xi_{2,429}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}}\right) + \xi_{2,430}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,444}}\right)$$

$$\xi_{2,451} = \xi_{2,424}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}}\right) + \xi_{2,427}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}}\right) + \xi_{2,430}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}}\right) + \xi_{2,431}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,444}}\right)$$

$$\xi_{2,452} = \xi_{2,421}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}}\right) + \xi_{2,422}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}}\right) + \xi_{2,423}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}}\right) + \xi_{2,424}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,441}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}}\right)$$

$$\xi_{2,453} = \xi_{2,422}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}}\right) + \xi_{2,425}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}}\right) + \xi_{2,427}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,444}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}}\right) + \xi_{2,428}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}}\right)$$

$$\xi_{2,454} = \xi_{2,423}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}}\right) + \xi_{2,426}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}}\right) + \xi_{2,429}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}}\right) + \xi_{2,430}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,441}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}}\right)$$

$$\xi_{2,455} = \xi_{2,424}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,432}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,435}}\right) + \xi_{2,427}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,436}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,439}}\right) + \xi_{2,430}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,440}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,443}}\right) + \xi_{2,431}\left(-ie^{-i\xi_{2,105}}\sin\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,441}} + e^{-i\xi_{2,105}}\cos\left(\xi_{2,103} - \xi_{2,104}\right)\overline{\xi_{2,447}}\right)$$

$$\xi_{2,456} = \xi_{2,421}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}}\right) + \xi_{2,422}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}}\right) + \xi_{2,423}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}}\right) + \xi_{2,424}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}}\right)$$

$$\xi_{2,457} = \xi_{2,422}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}}\right) + \xi_{2,425}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}}\right) + \xi_{2,427}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}}\right) + \xi_{2,428}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}}\right)$$

$$\xi_{2,458} = \xi_{2,423}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}}\right) + \xi_{2,426}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}}\right) + \xi_{2,429}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}}\right) + \xi_{2,430}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}}\right)$$

$$\xi_{2,459} = \xi_{2,424}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}}\right) + \xi_{2,427}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}}\right) + \xi_{2,430}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}}\right) + \xi_{2,431}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}}\right)$$

$$\xi_{2,460} = \xi_{2,421}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}}\right) + \xi_{2,422}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}}\right) + \xi_{2,423}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}}\right) + \xi_{2,424}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}}\right)$$

$$\xi_{2,461} = \xi_{2,422}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}}\right) + \xi_{2,425}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}}\right) + \xi_{2,427}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}}\right) + \xi_{2,428}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}}\right)$$

$$\xi_{2,462} = \xi_{2,423}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}}\right) + \xi_{2,426}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}}\right) + \xi_{2,429}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}}\right) + \xi_{2,430}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}}\right)$$

$$\xi_{2,463} = \xi_{2,424}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,433}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,434}}\right) + \xi_{2,427}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,437}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,438}}\right) + \xi_{2,430}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,441}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,442}}\right) + \xi_{2,431}\left(-ie^{i\xi_{2,105}}\sin\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,445}} + e^{i\xi_{2,105}}\cos\left(\xi_{2,103} + \xi_{2,104}\right)\overline{\xi_{2,446}}\right)$$

$$\xi_{2,464} = \frac{\operatorname{re}\left(\xi_{2,448}\right)}{4} + \frac{\operatorname{re}\left(\xi_{2,455}\right)}{4} + \frac{\operatorname{re}\left(\xi_{2,457}\right)}{4} + \frac{\operatorname{re}\left(\xi_{2,462}\right)}{4}$$

$$\xi_{2,465} = -\frac{\operatorname{im}\left(\xi_{2,449}\right)}{4} - \frac{\operatorname{im}\left(\xi_{2,454}\right)}{4} - \frac{\operatorname{im}\left(\xi_{2,456}\right)}{4} - \frac{\operatorname{im}\left(\xi_{2,463}\right)}{4}$$

$$\xi_{2,466} = -\frac{\operatorname{re}\left(\xi_{2,449}\right)}{4} + \frac{\operatorname{re}\left(\xi_{2,454}\right)}{4} + \frac{\operatorname{re}\left(\xi_{2,456}\right)}{4} - \frac{\operatorname{re}\left(\xi_{2,463}\right)}{4}$$

$$\xi_{2,467} = -\frac{\operatorname{im}\left(\xi_{2,448}\right)}{4} + \frac{\operatorname{im}\left(\xi_{2,455}\right)}{4} + \frac{\operatorname{im}\left(\xi_{2,457}\right)}{4} - \frac{\operatorname{im}\left(\xi_{2,462}\right)}{4}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,468} = -\frac{\operatorname{im}(\xi_{2,450})}{4} - \frac{\operatorname{im}(\xi_{2,453})}{4} - \frac{\operatorname{im}(\xi_{2,459})}{4} - \frac{\operatorname{im}(\xi_{2,460})}{4}$$

$$\xi_{2,469} = -\frac{\operatorname{re}(\xi_{2,451})}{4} - \frac{\operatorname{re}(\xi_{2,452})}{4} - \frac{\operatorname{re}(\xi_{2,458})}{4} - \frac{\operatorname{re}(\xi_{2,461})}{4}$$

$$\xi_{2,470} = \frac{\operatorname{im}(\xi_{2,451})}{4} - \frac{\operatorname{im}(\xi_{2,452})}{4} - \frac{\operatorname{im}(\xi_{2,458})}{4} + \frac{\operatorname{im}(\xi_{2,461})}{4}$$

$$\xi_{2,471} = -\frac{\operatorname{re}(\xi_{2,450})}{4} + \frac{\operatorname{re}(\xi_{2,453})}{4} + \frac{\operatorname{re}(\xi_{2,459})}{4} - \frac{\operatorname{re}(\xi_{2,460})}{4}$$

$$\xi_{2,472} = -\frac{\operatorname{re}(\xi_{2,450})}{4} + \frac{\operatorname{re}(\xi_{2,453})}{4} - \frac{\operatorname{re}(\xi_{2,459})}{4} + \frac{\operatorname{re}(\xi_{2,460})}{4}$$

$$\xi_{2,473} = \frac{\operatorname{im}(\xi_{2,451})}{4} - \frac{\operatorname{im}(\xi_{2,452})}{4} + \frac{\operatorname{im}(\xi_{2,458})}{4} - \frac{\operatorname{im}(\xi_{2,461})}{4}$$

$$\xi_{2,474} = \frac{\operatorname{re}(\xi_{2,451})}{4} + \frac{\operatorname{re}(\xi_{2,452})}{4} - \frac{\operatorname{re}(\xi_{2,458})}{4} - \frac{\operatorname{re}(\xi_{2,461})}{4}$$

$$\xi_{2,475} = \frac{\operatorname{im}(\xi_{2,450})}{4} + \frac{\operatorname{im}(\xi_{2,453})}{4} - \frac{\operatorname{im}(\xi_{2,459})}{4} - \frac{\operatorname{im}(\xi_{2,460})}{4}$$

$$\xi_{2,476} = -\frac{\operatorname{im}(\xi_{2,448})}{4} + \frac{\operatorname{im}(\xi_{2,455})}{4} - \frac{\operatorname{im}(\xi_{2,457})}{4} + \frac{\operatorname{im}(\xi_{2,462})}{4}$$

$$\xi_{2,477} = -\frac{\operatorname{re}(\xi_{2,449})}{4} + \frac{\operatorname{re}(\xi_{2,454})}{4} - \frac{\operatorname{re}(\xi_{2,456})}{4} + \frac{\operatorname{re}(\xi_{2,463})}{4}$$

$$\xi_{2,478} = \frac{\operatorname{im}(\xi_{2,449})}{4} + \frac{\operatorname{im}(\xi_{2,454})}{4} - \frac{\operatorname{im}(\xi_{2,456})}{4} - \frac{\operatorname{im}(\xi_{2,463})}{4}$$

$$\xi_{2,479} = -\frac{\operatorname{re}(\xi_{2,448})}{4} - \frac{\operatorname{re}(\xi_{2,455})}{4} + \frac{\operatorname{re}(\xi_{2,457})}{4} + \frac{\operatorname{re}(\xi_{2,462})}{4}$$

$$\xi_{2,480} = \xi_{2,464}^2 + \xi_{2,468}^2 + \xi_{2,472}^2 + \xi_{2,476}^2$$

$$\xi_{2,481} = \xi_{2,465}^2 + \xi_{2,469}^2 + \xi_{2,473}^2 + \xi_{2,477}^2$$

$$\xi_{2,482} = \xi_{2,466}^2 + \xi_{2,470}^2 + \xi_{2,474}^2 + \xi_{2,478}^2$$

$$\xi_{2,483} = \xi_{2,467}^2 + \xi_{2,471}^2 + \xi_{2,475}^2 + \xi_{2,479}^2$$

$$\xi_{2,484} = \sqrt{\xi_{2,480}}$$

$$\xi_{2,485} = \frac{\xi_{2,464}}{\xi_{2,484}}$$

$$\xi_{2,486} = \frac{\xi_{2,468}}{\xi_{2,484}}$$

$$\xi_{2,487} = \frac{\xi_{2,472}}{\xi_{2,484}}$$

$$\xi_{2,488} = \frac{\xi_{2,476}}{\xi_{2,484}}$$

$$\xi_{2,489} = \sqrt{\xi_{2,481}}$$

$$\xi_{2,490} = \frac{\xi_{2,465}}{\xi_{2,489}}$$

$$\xi_{2,491} = \frac{\xi_{2,469}}{\xi_{2,489}}$$

$$\xi_{2,492} = \frac{\xi_{2,473}}{\xi_{2,489}}$$

$$\xi_{2,493} = \frac{\xi_{2,477}}{\xi_{2,489}}$$

$$\xi_{2,494} = \sqrt{\xi_{2,482}}$$

$$\xi_{2,495} = \frac{\xi_{2,466}}{\xi_{2,494}}$$

$$\xi_{2,496} = \frac{\xi_{2,470}}{\xi_{2,494}}$$

$$\xi_{2,497} = \frac{\xi_{2,474}}{\xi_{2,494}}$$

$$\xi_{2,498} = \frac{\xi_{2,478}}{\xi_{2,494}}$$

$$\xi_{2,499} = \sqrt{\xi_{2,483}}$$

$$\xi_{2,500} = \frac{\xi_{2,467}}{\xi_{2,499}}$$

$$\xi_{2,501} = \frac{\xi_{2,471}}{\xi_{2,499}}$$

$$\xi_{2,502} = \frac{\xi_{2,475}}{\xi_{2,499}}$$

$$\xi_{2,503} = \frac{\xi_{2,479}}{\xi_{2,499}}$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,504} = \begin{cases} \xi_{2,485} & \text{for } \xi_{2,480} \geq \xi_{2,481} \wedge \xi_{2,480} \geq \xi_{2,482} \wedge \xi_{2,480} \geq \xi_{2,483} \\ \xi_{2,490} & \text{for } \xi_{2,481} \geq \xi_{2,482} \wedge \xi_{2,481} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,481} \\ \xi_{2,495} & \text{for } \xi_{2,482} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,482} \wedge \xi_{2,481} \leq \xi_{2,482} \\ \xi_{2,500} & \text{otherwise} \end{cases}$$

$$\xi_{2,505} = \begin{cases} \xi_{2,486} & \text{for } \xi_{2,480} \geq \xi_{2,481} \wedge \xi_{2,480} \geq \xi_{2,482} \wedge \xi_{2,480} \geq \xi_{2,483} \\ \xi_{2,491} & \text{for } \xi_{2,481} \geq \xi_{2,482} \wedge \xi_{2,481} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,481} \\ \xi_{2,496} & \text{for } \xi_{2,482} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,482} \wedge \xi_{2,481} \leq \xi_{2,482} \\ \xi_{2,501} & \text{otherwise} \end{cases}$$

$$\xi_{2,506} = \begin{cases} \xi_{2,487} & \text{for } \xi_{2,480} \geq \xi_{2,481} \wedge \xi_{2,480} \geq \xi_{2,482} \wedge \xi_{2,480} \geq \xi_{2,483} \\ \xi_{2,492} & \text{for } \xi_{2,481} \geq \xi_{2,482} \wedge \xi_{2,481} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,481} \\ \xi_{2,497} & \text{for } \xi_{2,482} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,482} \wedge \xi_{2,481} \leq \xi_{2,482} \\ \xi_{2,502} & \text{otherwise} \end{cases}$$

$$\xi_{2,507} = \begin{cases} \xi_{2,488} & \text{for } \xi_{2,480} \geq \xi_{2,481} \wedge \xi_{2,480} \geq \xi_{2,482} \wedge \xi_{2,480} \geq \xi_{2,483} \\ \xi_{2,493} & \text{for } \xi_{2,481} \geq \xi_{2,482} \wedge \xi_{2,481} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,481} \\ \xi_{2,498} & \text{for } \xi_{2,482} \geq \xi_{2,483} \wedge \xi_{2,480} \leq \xi_{2,482} \wedge \xi_{2,481} \leq \xi_{2,482} \\ \xi_{2,503} & \text{otherwise} \end{cases}$$

$$\xi_{2,508} = \xi_{2,464}\xi_{2,504} + \xi_{2,468}\xi_{2,505} + \xi_{2,472}\xi_{2,506} + \xi_{2,476}\xi_{2,507}$$

$$\xi_{2,509} = \xi_{2,465}\xi_{2,504} + \xi_{2,469}\xi_{2,505} + \xi_{2,473}\xi_{2,506} + \xi_{2,477}\xi_{2,507}$$

$$\xi_{2,510} = \xi_{2,466}\xi_{2,504} + \xi_{2,470}\xi_{2,505} + \xi_{2,474}\xi_{2,506} + \xi_{2,478}\xi_{2,507}$$

$$\xi_{2,511} = \xi_{2,467}\xi_{2,504} + \xi_{2,471}\xi_{2,505} + \xi_{2,475}\xi_{2,506} + \xi_{2,479}\xi_{2,507}$$

$$\xi_{2,512} = \xi_{2,41}\xi_{2,52} - \xi_{2,42}\xi_{2,55} - \xi_{2,43}\xi_{2,56} - \xi_{2,44}\xi_{2,57}$$

$$\xi_{2,513} = \xi_{2,41}\xi_{2,55} + \xi_{2,42}\xi_{2,52} - \xi_{2,43}\xi_{2,57} + \xi_{2,44}\xi_{2,56}$$

$$\xi_{2,514} = \xi_{2,41}\xi_{2,56} + \xi_{2,42}\xi_{2,57} + \xi_{2,43}\xi_{2,52} - \xi_{2,44}\xi_{2,55}$$

$$\xi_{2,515} = \xi_{2,41}\xi_{2,57} - \xi_{2,42}\xi_{2,56} + \xi_{2,43}\xi_{2,55} + \xi_{2,44}\xi_{2,52}$$

$$\xi_{2,516} = \xi_{2,10}\xi_{2,55} + \xi_{2,11}\xi_{2,55} + \xi_{2,12}\xi_{2,55} + \xi_{2,54}\xi_{2,9}$$

$$\xi_{2,517} = -\xi_{2,10}\xi_{2,54} + \xi_{2,11}\xi_{2,55} - \xi_{2,12}\xi_{2,55} + \xi_{2,55}\xi_{2,9}$$

$$\xi_{2,518} = -\xi_{2,10}\xi_{2,55} - \xi_{2,11}\xi_{2,54} + \xi_{2,12}\xi_{2,55} + \xi_{2,55}\xi_{2,9}$$

$$\xi_{2,519} = \xi_{2,10}\xi_{2,55} - \xi_{2,11}\xi_{2,55} - \xi_{2,12}\xi_{2,54} + \xi_{2,55}\xi_{2,9}$$

$$\xi_{2,520} = \xi_{2,52}\left(\xi_{2,100}\xi_{2,505} + \xi_{2,101}\xi_{2,506} + \xi_{2,102}\xi_{2,507} + \xi_{2,504}\xi_{2,99}\right) - \xi_{2,52}\left(\xi_{2,100}\xi_{2,506} - \xi_{2,101}\xi_{2,505} + \xi_{2,102}\xi_{2,504} - \xi_{2,507}\xi_{2,99}\right) - \xi_{2,53}\left(\xi_{2,100}\xi_{2,504} + \xi_{2,101}\xi_{2,507} - \xi_{2,102}\xi_{2,506} - \xi_{2,505}\xi_{2,99}\right) - \xi_{2,54}\left(-\xi_{2,100}\xi_{2,507} + \xi_{2,101}\xi_{2,504} + \xi_{2,102}\xi_{2,505} - \xi_{2,506}\xi_{2,99}\right)$$

$$\xi_{2,521} = \xi_{2,52}\left(\xi_{2,100}\xi_{2,504} + \xi_{2,101}\xi_{2,507} - \xi_{2,102}\xi_{2,506} - \xi_{2,505}\xi_{2,99}\right) + \xi_{2,52}\left(-\xi_{2,100}\xi_{2,507} + \xi_{2,101}\xi_{2,504} + \xi_{2,102}\xi_{2,505} - \xi_{2,506}\xi_{2,99}\right) + \xi_{2,53}\left(\xi_{2,100}\xi_{2,505} + \xi_{2,101}\xi_{2,506} + \xi_{2,102}\xi_{2,507} + \xi_{2,504}\xi_{2,99}\right) - \xi_{2,54}\left(\xi_{2,100}\xi_{2,506} - \xi_{2,101}\xi_{2,505} + \xi_{2,102}\xi_{2,504} - \xi_{2,507}\xi_{2,99}\right)$$

$$\xi_{2,522} = -\xi_{2,52}\left(\xi_{2,100}\xi_{2,504} + \xi_{2,101}\xi_{2,507} - \xi_{2,102}\xi_{2,506} - \xi_{2,505}\xi_{2,99}\right) + \xi_{2,52}\left(-\xi_{2,100}\xi_{2,507} + \xi_{2,101}\xi_{2,504} + \xi_{2,102}\xi_{2,505} - \xi_{2,506}\xi_{2,99}\right) + \xi_{2,53}\left(\xi_{2,100}\xi_{2,506} - \xi_{2,101}\xi_{2,505} + \xi_{2,102}\xi_{2,504} - \xi_{2,507}\xi_{2,99}\right) + \xi_{2,54}\left(\xi_{2,100}\xi_{2,505} + \xi_{2,101}\xi_{2,506} + \xi_{2,102}\xi_{2,507} + \xi_{2,504}\xi_{2,99}\right)$$

$$\xi_{2,523} = \xi_{2,52}\left(\xi_{2,100}\xi_{2,505} + \xi_{2,101}\xi_{2,506} + \xi_{2,102}\xi_{2,507} + \xi_{2,504}\xi_{2,99}\right) + \xi_{2,52}\left(\xi_{2,100}\xi_{2,506} - \xi_{2,101}\xi_{2,505} + \xi_{2,102}\xi_{2,504} - \xi_{2,507}\xi_{2,99}\right) - \xi_{2,53}\left(-\xi_{2,100}\xi_{2,507} + \xi_{2,101}\xi_{2,504} + \xi_{2,102}\xi_{2,505} - \xi_{2,506}\xi_{2,99}\right) + \xi_{2,54}\left(\xi_{2,100}\xi_{2,504} + \xi_{2,101}\xi_{2,507} - \xi_{2,102}\xi_{2,506} - \xi_{2,505}\xi_{2,99}\right)$$

$$\xi_{2,524} = -\xi_{2,53}\left(-\xi_{2,508}\xi_{2,68} - \xi_{2,509}\xi_{2,67} + \xi_{2,510}\xi_{2,70} - \xi_{2,511}\xi_{2,69}\right) - \xi_{2,53}\left(-\xi_{2,508}\xi_{2,69} - \xi_{2,509}\xi_{2,70} - \xi_{2,510}\xi_{2,67} + \xi_{2,511}\xi_{2,68}\right) - \xi_{2,53}\left(-\xi_{2,508}\xi_{2,70} + \xi_{2,509}\xi_{2,69} - \xi_{2,510}\xi_{2,68} - \xi_{2,511}\xi_{2,67}\right) + \xi_{2,54}\left(\xi_{2,508}\xi_{2,67} - \xi_{2,509}\xi_{2,68} - \xi_{2,510}\xi_{2,69} - \xi_{2,511}\xi_{2,70}\right)$$

$$\xi_{2,525} = \xi_{2,53}\left(\xi_{2,508}\xi_{2,67} - \xi_{2,509}\xi_{2,68} - \xi_{2,510}\xi_{2,69} - \xi_{2,511}\xi_{2,70}\right) + \xi_{2,53}\left(-\xi_{2,508}\xi_{2,69} - \xi_{2,509}\xi_{2,70} - \xi_{2,510}\xi_{2,67} + \xi_{2,511}\xi_{2,68}\right) - \xi_{2,53}\left(-\xi_{2,508}\xi_{2,70} + \xi_{2,509}\xi_{2,69} - \xi_{2,510}\xi_{2,68} - \xi_{2,511}\xi_{2,67}\right) + \xi_{2,54}\left(-\xi_{2,508}\xi_{2,68} - \xi_{2,509}\xi_{2,67} + \xi_{2,510}\xi_{2,70} - \xi_{2,511}\xi_{2,69}\right)$$

$$\xi_{2,526} = \xi_{2,53}\left(\xi_{2,508}\xi_{2,67} - \xi_{2,509}\xi_{2,68} - \xi_{2,510}\xi_{2,69} - \xi_{2,511}\xi_{2,70}\right) - \xi_{2,53}\left(-\xi_{2,508}\xi_{2,68} - \xi_{2,509}\xi_{2,67} + \xi_{2,510}\xi_{2,70} - \xi_{2,511}\xi_{2,69}\right) + \xi_{2,53}\left(-\xi_{2,508}\xi_{2,70} + \xi_{2,509}\xi_{2,69} - \xi_{2,510}\xi_{2,68} - \xi_{2,511}\xi_{2,67}\right) + \xi_{2,54}\left(-\xi_{2,508}\xi_{2,69} - \xi_{2,509}\xi_{2,70} - \xi_{2,510}\xi_{2,67} + \xi_{2,511}\xi_{2,68}\right)$$

$$\xi_{2,527} = \xi_{2,53}\left(\xi_{2,508}\xi_{2,67} - \xi_{2,509}\xi_{2,68} - \xi_{2,510}\xi_{2,69} - \xi_{2,511}\xi_{2,70}\right) + \xi_{2,53}\left(-\xi_{2,508}\xi_{2,68} - \xi_{2,509}\xi_{2,67} + \xi_{2,510}\xi_{2,70} - \xi_{2,511}\xi_{2,69}\right) - \xi_{2,53}\left(-\xi_{2,508}\xi_{2,69} - \xi_{2,509}\xi_{2,70} - \xi_{2,510}\xi_{2,67} + \xi_{2,511}\xi_{2,68}\right) + \xi_{2,54}\left(-\xi_{2,508}\xi_{2,70} + \xi_{2,509}\xi_{2,69} - \xi_{2,510}\xi_{2,68} - \xi_{2,511}\xi_{2,67}\right)$$

$$\xi_{2,528} = \xi_{2,393}\left(\xi_{2,33}\xi_{2,91} - \xi_{2,34}\xi_{2,92} - \xi_{2,35}\xi_{2,93} - \xi_{2,36}\xi_{2,94}\right) + \xi_{2,394}\left(\xi_{2,33}\xi_{2,92} + \xi_{2,34}\xi_{2,91} + \xi_{2,35}\xi_{2,94} - \xi_{2,36}\xi_{2,93}\right) + \xi_{2,395}\left(\xi_{2,33}\xi_{2,93} - \xi_{2,34}\xi_{2,94} + \xi_{2,35}\xi_{2,91} + \xi_{2,36}\xi_{2,92}\right) + \xi_{2,396}\left(\xi_{2,33}\xi_{2,94} + \xi_{2,34}\xi_{2,93} - \xi_{2,35}\xi_{2,92} + \xi_{2,36}\xi_{2,91}\right)$$

$$\xi_{2,529} = \xi_{2,393}\left(\xi_{2,33}\xi_{2,92} + \xi_{2,34}\xi_{2,91} + \xi_{2,35}\xi_{2,94} - \xi_{2,36}\xi_{2,93}\right) - \xi_{2,394}\left(\xi_{2,33}\xi_{2,91} - \xi_{2,34}\xi_{2,92} - \xi_{2,35}\xi_{2,93} - \xi_{2,36}\xi_{2,94}\right) + \xi_{2,395}\left(\xi_{2,33}\xi_{2,94} + \xi_{2,34}\xi_{2,93} - \xi_{2,35}\xi_{2,92} + \xi_{2,36}\xi_{2,91}\right) - \xi_{2,396}\left(\xi_{2,33}\xi_{2,93} - \xi_{2,34}\xi_{2,94} + \xi_{2,35}\xi_{2,91} + \xi_{2,36}\xi_{2,92}\right)$$

$$\xi_{2,530} = \xi_{2,393}\left(\xi_{2,33}\xi_{2,93} - \xi_{2,34}\xi_{2,94} + \xi_{2,35}\xi_{2,91} + \xi_{2,36}\xi_{2,92}\right) - \xi_{2,394}\left(\xi_{2,33}\xi_{2,94} + \xi_{2,34}\xi_{2,93} - \xi_{2,35}\xi_{2,92} + \xi_{2,36}\xi_{2,91}\right) - \xi_{2,395}\left(\xi_{2,33}\xi_{2,91} - \xi_{2,34}\xi_{2,92} - \xi_{2,35}\xi_{2,93} - \xi_{2,36}\xi_{2,94}\right) + \xi_{2,396}\left(\xi_{2,33}\xi_{2,92} + \xi_{2,34}\xi_{2,91} + \xi_{2,35}\xi_{2,94} - \xi_{2,36}\xi_{2,93}\right)$$

$$\xi_{2,531} = \xi_{2,393}\left(\xi_{2,33}\xi_{2,94} + \xi_{2,34}\xi_{2,93} - \xi_{2,35}\xi_{2,92} + \xi_{2,36}\xi_{2,91}\right) + \xi_{2,394}\left(\xi_{2,33}\xi_{2,93} - \xi_{2,34}\xi_{2,94} + \xi_{2,35}\xi_{2,91} + \xi_{2,36}\xi_{2,92}\right) - \xi_{2,395}\left(\xi_{2,33}\xi_{2,92} + \xi_{2,34}\xi_{2,91} + \xi_{2,35}\xi_{2,94} - \xi_{2,36}\xi_{2,93}\right) - \xi_{2,396}\left(\xi_{2,33}\xi_{2,91} - \xi_{2,34}\xi_{2,92} - \xi_{2,35}\xi_{2,93} - \xi_{2,36}\xi_{2,94}\right)$$

$$\xi_{2,532} = \xi_{2,417}\left(\xi_{2,37}\xi_{2,95} - \xi_{2,38}\xi_{2,96} - \xi_{2,39}\xi_{2,97} - \xi_{2,40}\xi_{2,98}\right) + \xi_{2,418}\left(\xi_{2,37}\xi_{2,96} + \xi_{2,38}\xi_{2,95} + \xi_{2,39}\xi_{2,98} - \xi_{2,40}\xi_{2,97}\right) + \xi_{2,419}\left(\xi_{2,37}\xi_{2,97} - \xi_{2,38}\xi_{2,98} + \xi_{2,39}\xi_{2,95} + \xi_{2,40}\xi_{2,96}\right) + \xi_{2,420}\left(\xi_{2,37}\xi_{2,98} + \xi_{2,38}\xi_{2,97} - \xi_{2,39}\xi_{2,96} + \xi_{2,40}\xi_{2,95}\right)$$

$$\xi_{2,533} = \xi_{2,417}\left(\xi_{2,37}\xi_{2,96} + \xi_{2,38}\xi_{2,95} + \xi_{2,39}\xi_{2,98} - \xi_{2,40}\xi_{2,97}\right) - \xi_{2,418}\left(\xi_{2,37}\xi_{2,95} - \xi_{2,38}\xi_{2,96} - \xi_{2,39}\xi_{2,97} - \xi_{2,40}\xi_{2,98}\right) + \xi_{2,419}\left(\xi_{2,37}\xi_{2,98} + \xi_{2,38}\xi_{2,97} - \xi_{2,39}\xi_{2,96} + \xi_{2,40}\xi_{2,95}\right) - \xi_{2,420}\left(\xi_{2,37}\xi_{2,97} - \xi_{2,38}\xi_{2,98} + \xi_{2,39}\xi_{2,95} + \xi_{2,40}\xi_{2,96}\right)$$

$$\xi_{2,534} = \xi_{2,417}\left(\xi_{2,37}\xi_{2,97} - \xi_{2,38}\xi_{2,98} + \xi_{2,39}\xi_{2,95} + \xi_{2,40}\xi_{2,96}\right) - \xi_{2,418}\left(\xi_{2,37}\xi_{2,98} + \xi_{2,38}\xi_{2,97} - \xi_{2,39}\xi_{2,96} + \xi_{2,40}\xi_{2,95}\right) - \xi_{2,419}\left(\xi_{2,37}\xi_{2,95} - \xi_{2,38}\xi_{2,96} - \xi_{2,39}\xi_{2,97} - \xi_{2,40}\xi_{2,98}\right) + \xi_{2,420}\left(\xi_{2,37}\xi_{2,96} + \xi_{2,38}\xi_{2,95} + \xi_{2,39}\xi_{2,98} - \xi_{2,40}\xi_{2,97}\right)$$

$$\xi_{2,535} = \xi_{2,417}\left(\xi_{2,37}\xi_{2,98} + \xi_{2,38}\xi_{2,97} - \xi_{2,39}\xi_{2,96} + \xi_{2,40}\xi_{2,95}\right) + \xi_{2,418}\left(\xi_{2,37}\xi_{2,97} - \xi_{2,38}\xi_{2,98} + \xi_{2,39}\xi_{2,95} + \xi_{2,40}\xi_{2,96}\right) - \xi_{2,419}\left(\xi_{2,37}\xi_{2,96} + \xi_{2,38}\xi_{2,95} + \xi_{2,39}\xi_{2,98} - \xi_{2,40}\xi_{2,97}\right) - \xi_{2,420}\left(\xi_{2,37}\xi_{2,95} - \xi_{2,38}\xi_{2,96} - \xi_{2,39}\xi_{2,97} - \xi_{2,40}\xi_{2,98}\right)$$

$$\xi_{2,536} = \xi_{2,106} + \xi_{2,48}$$

$$\xi_{2,537} = \begin{cases} 0 & \text{for } \xi_{2,512}^2 + \xi_{2,515}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,515}, \xi_{2,512}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,538} = \begin{cases} 0 & \text{for } \xi_{2,513}^2 + \xi_{2,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,513}, \xi_{2,514}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,539} = 2\operatorname{atan}_2\left(\sqrt{\xi_{2,513}^2 + \xi_{2,514}^2}, \sqrt{\xi_{2,512}^2 + \xi_{2,515}^2}\right)$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,540} = -\xi_{2,537} + \xi_{2,538}$$

$$\xi_{2,541} = -\xi_{2,537} - \xi_{2,538}$$

$$\xi_{2,542} = \xi_{2,536} + \xi_{2,537}$$

$$\xi_{2,543} = \begin{cases} 0 & \text{for } \xi_{2,516}^2 + \xi_{2,519}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,519}, \xi_{2,516}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,544} = \begin{cases} 0 & \text{for } \xi_{2,517}^2 + \xi_{2,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,517}, \xi_{2,518}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,545} = 2\operatorname{atan}_2\left(\sqrt{\xi_{2,517}^2 + \xi_{2,518}^2}, \sqrt{\xi_{2,516}^2 + \xi_{2,519}^2}\right)$$

$$\xi_{2,546} = -\xi_{2,543} + \xi_{2,544}$$

$$\xi_{2,547} = -\xi_{2,543} - \xi_{2,544}$$

$$\xi_{2,548} = \xi_{2,542} + \xi_{2,543}$$

$$\xi_{2,549} = \begin{cases} 0 & \text{for } \xi_{2,520}^2 + \xi_{2,523}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,523}, \xi_{2,520}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,550} = \begin{cases} 0 & \text{for } \xi_{2,521}^2 + \xi_{2,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,521}, \xi_{2,522}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,551} = 2\operatorname{atan}_2\left(\sqrt{\xi_{2,521}^2 + \xi_{2,522}^2}, \sqrt{\xi_{2,520}^2 + \xi_{2,523}^2}\right)$$

$$\xi_{2,552} = -\xi_{2,549} + \xi_{2,550}$$

$$\xi_{2,553} = -\xi_{2,549} - \xi_{2,550}$$

$$\xi_{2,554} = \xi_{2,548} + \xi_{2,549}$$

$$\xi_{2,555} = \begin{cases} 0 & \text{for } \xi_{2,524}^2 + \xi_{2,527}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,527}, \xi_{2,524}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,556} = \begin{cases} 0 & \text{for } \xi_{2,525}^2 + \xi_{2,526}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,525}, \xi_{2,526}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,557} = 2\operatorname{atan}_2\left(\sqrt{\xi_{2,525}^2 + \xi_{2,526}^2}, \sqrt{\xi_{2,524}^2 + \xi_{2,527}^2}\right)$$

$$\xi_{2,558} = -\xi_{2,555} + \xi_{2,556}$$

$$\xi_{2,559} = -\xi_{2,555} - \xi_{2,556}$$

$$\xi_{2,560} = \xi_{2,554} + \xi_{2,555}$$

$$\xi_{2,561} = \begin{cases} 0 & \text{for } \xi_{2,117}^2 + \xi_{2,118}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{2,118}, \xi_{2,117}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,562} = \begin{cases} 0 & \text{for } \xi_{2,114}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{2,114}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,563} = 2\operatorname{atan}_2\left(\left|\xi_{2,114}\right|, \sqrt{\xi_{2,117}^2 + \xi_{2,118}^2}\right)$$

$$\xi_{2,564} = -\xi_{2,561} + \xi_{2,562}$$

$$\xi_{2,565} = -\xi_{2,561} - \xi_{2,562}$$

$$\xi_{2,566} = \xi_{2,560} + \xi_{2,561}$$

$$\xi_{2,567} = \begin{cases} 0 & \text{for } \xi_{2,112}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{2,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,568} = \begin{cases} 0 & \text{for } \xi_{2,113}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{2,113}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,569} = 2\operatorname{atan}_2\left(\left|\xi_{2,113}\right|, \left|\xi_{2,112}\right|\right)$$

$$\xi_{2,570} = -\xi_{2,567} + \xi_{2,568}$$

$$\xi_{2,571} = -\xi_{2,567} - \xi_{2,568}$$

$$\xi_{2,572} = \xi_{2,566} + \xi_{2,567}$$

$$\xi_{2,573} = \begin{cases} 0 & \text{for } \xi_{2,528}^2 + \xi_{2,531}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,531}, \xi_{2,528}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,574} = \begin{cases} 0 & \text{for } \xi_{2,529}^2 + \xi_{2,530}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,529}, \xi_{2,530}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,575} = 2\operatorname{atan}_2\left(\sqrt{\xi_{2,529}^2 + \xi_{2,530}^2}, \sqrt{\xi_{2,528}^2 + \xi_{2,531}^2}\right)$$

Exact shared-expression program 2

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{2,576} = -\xi_{2,573} + \xi_{2,574}$$

$$\xi_{2,577} = -\xi_{2,573} - \xi_{2,574}$$

$$\xi_{2,578} = \xi_{2,572} + \xi_{2,573}$$

$$\xi_{2,579} = \begin{cases} 0 & \text{for } \xi_{2,532}^2 + \xi_{2,535}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,535}, \xi_{2,532}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,580} = \begin{cases} 0 & \text{for } \xi_{2,533}^2 + \xi_{2,534}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{2,533}, \xi_{2,534}\right) & \text{otherwise} \end{cases}$$

$$\xi_{2,581} = 2\operatorname{atan}_2\left(\sqrt{\xi_{2,533}^2 + \xi_{2,534}^2}, \sqrt{\xi_{2,532}^2 + \xi_{2,535}^2}\right)$$

$$\xi_{2,582} = -\xi_{2,579} + \xi_{2,580}$$

$$\xi_{2,583} = -\xi_{2,579} - \xi_{2,580}$$

$$\xi_{2,584} = \xi_{2,578} + \xi_{2,579}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,0} = \left\lfloor -\frac{\theta}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{3,1} = \left\lfloor -\frac{\phi}{2\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{3,2} = -\frac{\pi \xi_{3,0}}{2} - \frac{\theta}{2}$$

$$\xi_{3,3} = -\frac{\pi \xi_{3,1}}{2} - \frac{\phi}{4}$$

$$\xi_{3,4} = \begin{cases} \xi_{3,2} & \text{for } |\xi_{3,2}| \geq |\xi_{3,3}| \\ \xi_{3,3} & \text{otherwise} \end{cases}$$

$$\xi_{3,5} = \xi_{3,2}$$

$$\xi_{3,6} = \begin{cases} \xi_{3,3} & \text{for } |\xi_{3,2}| \geq |\xi_{3,3}| \\ \xi_{3,2} & \text{otherwise} \end{cases}$$

$$\xi_{3,7} = \begin{cases} 1 & \text{for } |\xi_{3,2}| \geq |\xi_{3,3}| \\ \frac{1}{2} & \text{for } |\xi_{3,2}| \leq |\xi_{3,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{3,8} = \begin{cases} 0 & \text{for } |\xi_{3,2}| \geq |\xi_{3,3}| \\ \frac{1}{2} & \text{for } |\xi_{3,2}| \leq |\xi_{3,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,9} = \begin{cases} 0 & \text{for } |\xi_{3,2}| \geq |\xi_{3,3}| \\ \frac{1}{2} & \text{for } |\xi_{3,2}| \leq |\xi_{3,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{3,10} = \begin{cases} 0 & \text{for } |\xi_{3,2}| \geq |\xi_{3,3}| \\ -\frac{1}{2} & \text{for } |\xi_{3,2}| \leq |\xi_{3,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,11} = \begin{cases} -1 & \text{for } \xi_{3,4} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{3,12} = \begin{cases} -1 & \text{for } \xi_{3,5} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{3,13} = \begin{cases} 1 & \text{for } \xi_{3,11} = 1 \wedge \xi_{3,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,14} = \begin{cases} 0 & \text{for } \xi_{3,11} = 1 \wedge \xi_{3,12} = 1 \\ 1 & \text{for } \xi_{3,11} = 1 \wedge \xi_{3,12} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,15} = \begin{cases} 0 & \text{for } \xi_{3,11} = 1 \wedge (\xi_{3,11} = 1 \vee \xi_{3,12} = -1) \wedge (\xi_{3,11} = 1 \vee \xi_{3,12} = 1) \wedge (\xi_{3,12} = -1 \vee \xi_{3,12} = 1) \\ 1 & \text{for } \xi_{3,11} = -1 \wedge \xi_{3,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,16} = \begin{cases} 0 & \text{for } (\xi_{3,11} = -1 \vee \xi_{3,11} = 1) \wedge (\xi_{3,11} = 1 \vee \xi_{3,12} = 1) \wedge (\xi_{3,12} = -1 \vee \xi_{3,12} = 1) \wedge (\xi_{3,11} = -1 \vee \xi_{3,11} = 1 \vee \xi_{3,12} = -1) \wedge (\xi_{3,11} = -1 \vee \xi_{3,11} = 1 \vee \xi_{3,12} = 1) \wedge (\xi_{3,11} = -1 \vee \xi_{3,12} = -1 \vee \xi_{3,12} = 1) \wedge (\xi_{3,11} = 1 \vee \xi_{3,12} = -1 \vee \xi_{3,12} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{3,17} = \cos\left(\frac{\pi \xi_{3,0}}{2}\right)$$

$$\xi_{3,18} = -\sin\left(\frac{\pi \xi_{3,0}}{2}\right)$$

$$\xi_{3,19} = \xi_{3,17}^2$$

$$\xi_{3,20} = \xi_{3,17} \xi_{3,18}$$

$$\xi_{3,21} = -\xi_{3,18}^2$$

$$\xi_{3,22} = \cos\left(\frac{\pi \xi_{3,1}}{2}\right)$$

$$\xi_{3,23} = -\sin\left(\frac{\pi \xi_{3,1}}{2}\right)$$

$$\xi_{3,24} = \xi_{3,19} \xi_{3,22} - \xi_{3,21} \xi_{3,23}$$

$$\xi_{3,25} = \xi_{3,20} \xi_{3,22} - \xi_{3,20} \xi_{3,23}$$

$$\xi_{3,26} = \xi_{3,20} \xi_{3,22} + \xi_{3,20} \xi_{3,23}$$

$$\xi_{3,27} = \xi_{3,19} \xi_{3,23} + \xi_{3,21} \xi_{3,22}$$

$$\xi_{3,28} = \xi_{3,13} \left(-\xi_{3,10} \xi_{3,27} + \xi_{3,24} \xi_{3,7} - \xi_{3,25} \xi_{3,8} - \xi_{3,26} \xi_{3,9}\right) - \xi_{3,14} \left(\xi_{3,10} \xi_{3,26} + \xi_{3,24} \xi_{3,8} + \xi_{3,25} \xi_{3,7} - \xi_{3,27} \xi_{3,9}\right) - \xi_{3,15} \left(-\xi_{3,10} \xi_{3,25} + \xi_{3,24} \xi_{3,9} + \xi_{3,26} \xi_{3,7} + \xi_{3,27} \xi_{3,8}\right) - \xi_{3,16} \left(\xi_{3,10} \xi_{3,24} + \xi_{3,25} \xi_{3,9} - \xi_{3,26} \xi_{3,8} + \xi_{3,27} \xi_{3,7}\right)$$

$$\xi_{3,29} = \xi_{3,13} \left(\xi_{3,10} \xi_{3,26} + \xi_{3,24} \xi_{3,8} + \xi_{3,25} \xi_{3,7} - \xi_{3,27} \xi_{3,9}\right) + \xi_{3,14} \left(-\xi_{3,10} \xi_{3,27} + \xi_{3,24} \xi_{3,7} - \xi_{3,25} \xi_{3,8} - \xi_{3,26} \xi_{3,9}\right) - \xi_{3,15} \left(\xi_{3,10} \xi_{3,24} + \xi_{3,25} \xi_{3,9} - \xi_{3,26} \xi_{3,8} + \xi_{3,27} \xi_{3,7}\right) + \xi_{3,16} \left(-\xi_{3,10} \xi_{3,25} + \xi_{3,24} \xi_{3,9} + \xi_{3,26} \xi_{3,7} + \xi_{3,27} \xi_{3,8}\right)$$

$$\xi_{3,30} = \xi_{3,13} \left(-\xi_{3,10} \xi_{3,25} + \xi_{3,24} \xi_{3,9} + \xi_{3,26} \xi_{3,7} + \xi_{3,27} \xi_{3,8}\right) + \xi_{3,14} \left(\xi_{3,10} \xi_{3,24} + \xi_{3,25} \xi_{3,9} - \xi_{3,26} \xi_{3,8} + \xi_{3,27} \xi_{3,7}\right) + \xi_{3,15} \left(-\xi_{3,10} \xi_{3,27} + \xi_{3,24} \xi_{3,7} - \xi_{3,25} \xi_{3,8} - \xi_{3,26} \xi_{3,9}\right) - \xi_{3,16} \left(\xi_{3,10} \xi_{3,26} + \xi_{3,24} \xi_{3,8} + \xi_{3,25} \xi_{3,7} - \xi_{3,27} \xi_{3,9}\right)$$

$$\xi_{3,31} = \xi_{3,13} \left(\xi_{3,10} \xi_{3,24} + \xi_{3,25} \xi_{3,9} - \xi_{3,26} \xi_{3,8} + \xi_{3,27} \xi_{3,7}\right) - \xi_{3,14} \left(-\xi_{3,10} \xi_{3,25} + \xi_{3,24} \xi_{3,9} + \xi_{3,26} \xi_{3,7} + \xi_{3,27} \xi_{3,8}\right) + \xi_{3,15} \left(\xi_{3,10} \xi_{3,26} + \xi_{3,24} \xi_{3,8} + \xi_{3,25} \xi_{3,7} - \xi_{3,27} \xi_{3,9}\right) + \xi_{3,16} \left(-\xi_{3,10} \xi_{3,27} + \xi_{3,24} \xi_{3,7} - \xi_{3,25} \xi_{3,8} - \xi_{3,26} \xi_{3,9}\right)$$

$$\xi_{3,32} = -\xi_{3,10} \xi_{3,27} + \xi_{3,24} \xi_{3,7} - \xi_{3,25} \xi_{3,8} - \xi_{3,26} \xi_{3,9}$$

$$\xi_{3,33} = \xi_{3,10} \xi_{3,26} + \xi_{3,24} \xi_{3,8} + \xi_{3,25} \xi_{3,7} - \xi_{3,27} \xi_{3,9}$$

$$\xi_{3,34} = -\xi_{3,10} \xi_{3,25} + \xi_{3,24} \xi_{3,9} + \xi_{3,26} \xi_{3,7} + \xi_{3,27} \xi_{3,8}$$

$$\xi_{3,35} = \xi_{3,10} \xi_{3,24} + \xi_{3,25} \xi_{3,9} - \xi_{3,26} \xi_{3,8} + \xi_{3,27} \xi_{3,7}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,36} = -\xi_{3,10}\xi_{3,16} + \xi_{3,13}\xi_{3,7} - \xi_{3,14}\xi_{3,8} - \xi_{3,15}\xi_{3,9}$$

$$\xi_{3,37} = \xi_{3,10}\xi_{3,15} - \xi_{3,13}\xi_{3,8} - \xi_{3,14}\xi_{3,7} - \xi_{3,16}\xi_{3,9}$$

$$\xi_{3,38} = -\xi_{3,10}\xi_{3,14} - \xi_{3,13}\xi_{3,9} - \xi_{3,15}\xi_{3,7} + \xi_{3,16}\xi_{3,8}$$

$$\xi_{3,39} = -\xi_{3,10}\xi_{3,13} + \xi_{3,14}\xi_{3,9} - \xi_{3,15}\xi_{3,8} - \xi_{3,16}\xi_{3,7}$$

$$\xi_{3,40} = \xi_{3,11}\xi_{3,4}$$

$$\xi_{3,41} = \xi_{3,12}\xi_{3,5}$$

$$\xi_{3,42} = \xi_{3,11}\xi_{3,12}\xi_{3,6}$$

$$\xi_{3,43} = -\frac{\pi(2\xi_{3,0}+\xi_{3,1})}{2}$$

$$\xi_{3,44} = \begin{cases} 0 & \text{for } \left(\xi_{3,41} > \frac{\pi}{8} \vee \xi_{3,42} < 0\right) \wedge \xi_{3,41} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{3,45} = \frac{\pi}{8}$$

$$\xi_{3,46} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{3,41} > \frac{\pi}{8} \wedge \xi_{3,42} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{3,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,47} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{3,41} > \frac{\pi}{8} \wedge \xi_{3,42} < 0 \\ \frac{1}{2} & \text{for } \xi_{3,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,48} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{3,41} > \frac{\pi}{8} \vee \xi_{3,42} < 0\right) \wedge \xi_{3,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,49} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{3,41} > \frac{\pi}{8} \vee \xi_{3,42} < 0\right) \wedge \xi_{3,41} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{3,50} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{3,41} > \frac{\pi}{8} \vee \xi_{3,42} < 0\right) \wedge \xi_{3,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,51} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{3,41} > \frac{\pi}{8} \vee \xi_{3,42} < 0\right) \wedge \xi_{3,41} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{3,52} = \begin{cases} \frac{1}{2} & \text{for } \xi_{3,41} > \frac{\pi}{8} \wedge \xi_{3,42} < 0 \\ -\frac{1}{2} & \text{for } \xi_{3,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,53} = \left\lfloor \frac{2\xi_{3,40} - 2\xi_{3,44}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{3,54} = \left\lfloor \frac{2\xi_{3,41} - 2\xi_{3,45}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{3,55} = \left\lfloor \frac{2\xi_{3,42} - 2\xi_{3,46}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{3,56} = \xi_{3,40} - \xi_{3,44} - \frac{\pi\xi_{3,53}}{2}$$

$$\xi_{3,57} = \xi_{3,41} - \xi_{3,45} - \frac{\pi\xi_{3,54}}{2}$$

$$\xi_{3,58} = \xi_{3,42} - \xi_{3,46} - \frac{\pi\xi_{3,55}}{2}$$

$$\xi_{3,59} = \begin{cases} \xi_{3,56} & \text{for } (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,58}| \vee |\xi_{3,57}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,57}| \geq |\xi_{3,58}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \\ \xi_{3,57} & \text{for } (|\xi_{3,56}| \geq |\xi_{3,58}| \vee |\xi_{3,57}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,58}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,57}| \geq |\xi_{3,58}| \vee |\xi_{3,56}| \leq |\xi_{3,57}|) \wedge (|\xi_{3,56}| \leq |\xi_{3,57}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \\ \xi_{3,58} & \text{otherwise} \end{cases}$$

$$\xi_{3,60} = \begin{cases} \xi_{3,57} & \text{for } |\xi_{3,56}| \geq |\xi_{3,57}| \wedge |\xi_{3,57}| \geq |\xi_{3,58}| \\ \xi_{3,58} & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge |\xi_{3,57}| \leq |\xi_{3,58}| \\ \xi_{3,56} & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge |\xi_{3,56}| \leq |\xi_{3,57}| \\ \xi_{3,58} & \text{for } |\xi_{3,57}| \geq |\xi_{3,58}| \wedge |\xi_{3,56}| \leq |\xi_{3,58}| \\ \xi_{3,56} & \text{for } |\xi_{3,56}| \geq |\xi_{3,57}| \wedge |\xi_{3,56}| \leq |\xi_{3,58}| \\ \xi_{3,57} & \text{otherwise} \end{cases}$$

$$\xi_{3,61} = \begin{cases} \xi_{3,58} & \text{for } |\xi_{3,56}| \geq |\xi_{3,57}| \wedge |\xi_{3,57}| \geq |\xi_{3,58}| \\ \xi_{3,57} & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge |\xi_{3,57}| \leq |\xi_{3,58}| \\ \xi_{3,58} & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge |\xi_{3,56}| \leq |\xi_{3,57}| \\ \xi_{3,56} & \text{for } |\xi_{3,57}| \geq |\xi_{3,58}| \wedge |\xi_{3,56}| \leq |\xi_{3,58}| \\ \xi_{3,57} & \text{for } |\xi_{3,56}| \geq |\xi_{3,57}| \wedge |\xi_{3,56}| \leq |\xi_{3,58}| \\ \xi_{3,56} & \text{otherwise} \end{cases}$$

$$\xi_{3,62} = \begin{cases} 1 & \text{for } |\xi_{3,56}| \geq |\xi_{3,57}| \wedge |\xi_{3,57}| \geq |\xi_{3,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge |\xi_{3,57}| \leq |\xi_{3,58}| \\ 0 & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge |\xi_{3,56}| \leq |\xi_{3,57}| \\ \frac{1}{2} & \text{for } |\xi_{3,56}| \leq |\xi_{3,58}| \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,57}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,57}| \geq |\xi_{3,58}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{3,63} = \begin{cases} 0 & \text{for } |\xi_{3,56}| \geq |\xi_{3,57}| \wedge |\xi_{3,57}| \geq |\xi_{3,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge (|\xi_{3,56}| \geq |\xi_{3,58}| \vee |\xi_{3,56}| \leq |\xi_{3,57}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,58}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \leq |\xi_{3,57}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \\ \frac{1}{2} & \text{for } |\xi_{3,56}| \leq |\xi_{3,58}| \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,57}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,57}| \geq |\xi_{3,58}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,64} = \begin{cases} 0 & \text{for } (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,58}| \vee |\xi_{3,57}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,57}| \geq |\xi_{3,58}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{3,56}| \geq |\xi_{3,58}| \wedge |\xi_{3,56}| \leq |\xi_{3,57}| \\ \frac{1}{2} & \text{for } |\xi_{3,56}| \leq |\xi_{3,58}| \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,57}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,57}| \geq |\xi_{3,58}| \vee |\xi_{3,56}| \leq |\xi_{3,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{3,65} = \begin{cases} 0 & \text{for } (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,56}| \vee |\xi_{3,57}| \geq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \geq |\xi_{3,56}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,57}| \vee |\xi_{3,56}| \leq |\xi_{3,57}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,56}| \vee |\xi_{3,57}| \geq |\xi_{3,56}| \vee |\xi_{3,56}| \leq |\xi_{3,57}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,56}| \vee |\xi_{3,57}| \geq |\xi_{3,56}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \wedge (|\xi_{3,56}| \geq |\xi_{3,56}| \vee |\xi_{3,56}| \leq |\xi_{3,57}| \vee |\xi_{3,57}| \leq |\xi_{3,58}|) \\ \frac{1}{2} & \text{for } |\xi_{3,57}| \geq |\xi_{3,58}| \wedge |\xi_{3,56}| \leq |\xi_{3,58}| \\ -\frac{1}{2} & \text{for } |\xi_{3,56}| \geq |\xi_{3,57}| \wedge |\xi_{3,56}| \leq |\xi_{3,58}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,66} = \begin{cases} -1 & \text{for } \xi_{3,59} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{3,67} = \begin{cases} -1 & \text{for } \xi_{3,60} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{3,68} = \begin{cases} 1 & \text{for } \xi_{3,66} = 1 \wedge \xi_{3,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,69} = \begin{cases} 0 & \text{for } \xi_{3,66} = 1 \wedge \xi_{3,67} = 1 \\ 1 & \text{for } \xi_{3,66} = 1 \wedge \xi_{3,67} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,70} = \begin{cases} 0 & \text{for } \xi_{3,66} = 1 \wedge (\xi_{3,66} = 1 \vee \xi_{3,67} = -1) \wedge (\xi_{3,66} = 1 \vee \xi_{3,67} = 1) \wedge (\xi_{3,67} = -1 \vee \xi_{3,67} = 1) \\ 1 & \text{for } \xi_{3,66} = -1 \wedge \xi_{3,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,71} = \begin{cases} 0 & \text{for } (\xi_{3,66} = -1 \vee \xi_{3,66} = 1) \wedge (\xi_{3,66} = 1 \vee \xi_{3,67} = 1) \wedge (\xi_{3,67} = -1 \vee \xi_{3,67} = 1) \wedge (\xi_{3,66} = -1 \vee \xi_{3,66} = 1 \vee \xi_{3,67} = -1) \wedge (\xi_{3,66} = -1 \vee \xi_{3,66} = 1 \vee \xi_{3,67} = 1) \wedge (\xi_{3,66} = -1 \vee \xi_{3,67} = -1 \vee \xi_{3,67} = 1) \wedge (\xi_{3,66} = 1 \vee \xi_{3,67} = -1 \vee \xi_{3,67} = 1) \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,72} = \cos\left(\frac{\pi\xi_{3,53}}{2}\right)$$

$$\xi_{3,73} = -\sin\left(\frac{\pi\xi_{3,53}}{2}\right)$$

$$\xi_{3,74} = \cos\left(\frac{\pi\xi_{3,54}}{2}\right)$$

$$\xi_{3,75} = -\sin\left(\frac{\pi\xi_{3,54}}{2}\right)$$

$$\xi_{3,76} = \xi_{3,72}\xi_{3,74}$$

$$\xi_{3,77} = \xi_{3,73}\xi_{3,74}$$

$$\xi_{3,78} = \xi_{3,72}\xi_{3,75}$$

$$\xi_{3,79} = -\xi_{3,73}\xi_{3,75}$$

$$\xi_{3,80} = \cos\left(\frac{\pi\xi_{3,55}}{2}\right)$$

$$\xi_{3,81} = -\sin\left(\frac{\pi\xi_{3,55}}{2}\right)$$

$$\xi_{3,82} = \xi_{3,76}\xi_{3,80} - \xi_{3,79}\xi_{3,81}$$

$$\xi_{3,83} = \xi_{3,77}\xi_{3,80} - \xi_{3,78}\xi_{3,81}$$

$$\xi_{3,84} = \xi_{3,77}\xi_{3,81} + \xi_{3,78}\xi_{3,80}$$

$$\xi_{3,85} = \xi_{3,76}\xi_{3,81} + \xi_{3,79}\xi_{3,80}$$

$$\xi_{3,86} = \xi_{3,68}\left(\xi_{3,62}\xi_{3,82} - \xi_{3,63}\xi_{3,83} - \xi_{3,64}\xi_{3,84} - \xi_{3,65}\xi_{3,85}\right) - \xi_{3,69}\left(\xi_{3,62}\xi_{3,83} + \xi_{3,63}\xi_{3,82} - \xi_{3,64}\xi_{3,85} + \xi_{3,65}\xi_{3,84}\right) - \xi_{3,70}\left(\xi_{3,62}\xi_{3,84} + \xi_{3,63}\xi_{3,85} + \xi_{3,64}\xi_{3,82} - \xi_{3,65}\xi_{3,83}\right) - \xi_{3,71}\left(\xi_{3,62}\xi_{3,85} - \xi_{3,63}\xi_{3,84} + \xi_{3,64}\xi_{3,83} + \xi_{3,65}\xi_{3,82}\right)$$

$$\xi_{3,87} = \xi_{3,68}\left(\xi_{3,62}\xi_{3,83} + \xi_{3,63}\xi_{3,82} - \xi_{3,64}\xi_{3,85} + \xi_{3,65}\xi_{3,84}\right) + \xi_{3,69}\left(\xi_{3,62}\xi_{3,82} - \xi_{3,63}\xi_{3,83} - \xi_{3,64}\xi_{3,84} - \xi_{3,65}\xi_{3,85}\right) - \xi_{3,70}\left(\xi_{3,62}\xi_{3,85} - \xi_{3,63}\xi_{3,84} + \xi_{3,64}\xi_{3,83} + \xi_{3,65}\xi_{3,82}\right) + \xi_{3,71}\left(\xi_{3,62}\xi_{3,84} + \xi_{3,63}\xi_{3,85} + \xi_{3,64}\xi_{3,82} - \xi_{3,65}\xi_{3,83}\right)$$

$$\xi_{3,88} = \xi_{3,68}\left(\xi_{3,62}\xi_{3,84} + \xi_{3,63}\xi_{3,85} + \xi_{3,64}\xi_{3,82} - \xi_{3,65}\xi_{3,83}\right) + \xi_{3,69}\left(\xi_{3,62}\xi_{3,85} - \xi_{3,63}\xi_{3,84} + \xi_{3,64}\xi_{3,83} + \xi_{3,65}\xi_{3,82}\right) + \xi_{3,70}\left(\xi_{3,62}\xi_{3,82} - \xi_{3,63}\xi_{3,83} - \xi_{3,64}\xi_{3,84} - \xi_{3,65}\xi_{3,85}\right) - \xi_{3,71}\left(\xi_{3,62}\xi_{3,83} + \xi_{3,63}\xi_{3,82} - \xi_{3,64}\xi_{3,85} + \xi_{3,65}\xi_{3,84}\right)$$

$$\xi_{3,89} = \xi_{3,68}\left(\xi_{3,62}\xi_{3,85} - \xi_{3,63}\xi_{3,84} + \xi_{3,64}\xi_{3,83} + \xi_{3,65}\xi_{3,82}\right) - \xi_{3,69}\left(\xi_{3,62}\xi_{3,84} + \xi_{3,63}\xi_{3,85} + \xi_{3,64}\xi_{3,82} - \xi_{3,65}\xi_{3,83}\right) + \xi_{3,70}\left(\xi_{3,62}\xi_{3,83} + \xi_{3,63}\xi_{3,82} - \xi_{3,64}\xi_{3,85} + \xi_{3,65}\xi_{3,84}\right) + \xi_{3,71}\left(\xi_{3,62}\xi_{3,82} - \xi_{3,63}\xi_{3,83} - \xi_{3,64}\xi_{3,84} - \xi_{3,65}\xi_{3,85}\right)$$

$$\xi_{3,90} = \xi_{3,62}\xi_{3,82} - \xi_{3,63}\xi_{3,83} - \xi_{3,64}\xi_{3,84} - \xi_{3,65}\xi_{3,85}$$

$$\xi_{3,91} = \xi_{3,62}\xi_{3,83} + \xi_{3,63}\xi_{3,82} - \xi_{3,64}\xi_{3,85} + \xi_{3,65}\xi_{3,84}$$

$$\xi_{3,92} = \xi_{3,62}\xi_{3,84} + \xi_{3,63}\xi_{3,85} + \xi_{3,64}\xi_{3,82} - \xi_{3,65}\xi_{3,83}$$

$$\xi_{3,93} = \xi_{3,62}\xi_{3,85} - \xi_{3,63}\xi_{3,84} + \xi_{3,64}\xi_{3,83} + \xi_{3,65}\xi_{3,82}$$

$$\xi_{3,94} = \xi_{3,62}\xi_{3,68} - \xi_{3,63}\xi_{3,69} - \xi_{3,64}\xi_{3,70} - \xi_{3,65}\xi_{3,71}$$

$$\xi_{3,95} = -\xi_{3,62}\xi_{3,69} - \xi_{3,63}\xi_{3,68} - \xi_{3,64}\xi_{3,71} + \xi_{3,65}\xi_{3,70}$$

$$\xi_{3,96} = -\xi_{3,62}\xi_{3,70} + \xi_{3,63}\xi_{3,71} - \xi_{3,64}\xi_{3,68} - \xi_{3,65}\xi_{3,69}$$

$$\xi_{3,97} = -\xi_{3,62}\xi_{3,71} - \xi_{3,63}\xi_{3,70} + \xi_{3,64}\xi_{3,69} - \xi_{3,65}\xi_{3,68}$$

$$\xi_{3,98} = \xi_{3,59}\xi_{3,66}$$

$$\xi_{3,99} = \xi_{3,60}\xi_{3,67}$$

$$\xi_{3,100} = \xi_{3,61}\xi_{3,66}\xi_{3,67}$$

$$\xi_{3,101} = -\frac{\pi(\xi_{3,53}+\xi_{3,54}+\xi_{3,55})}{2}$$

$$\xi_{3,102} = \cos\left(2\xi_{3,98}\right)$$

$$\xi_{3,103} = \cos\left(2\xi_{3,99}\right)$$

$$\xi_{3,104} = \cos\left(2\xi_{3,100}\right)$$

$$\xi_{3,105} = \sqrt{\xi_{3,102}^2 - 2\xi_{3,102}\xi_{3,103}\xi_{3,104} + \xi_{3,103}^2 + \xi_{3,104}^2 - 1}$$

$$\xi_{3,106} = \xi_{3,102} - \xi_{3,103} + \xi_{3,104}$$

$$\xi_{3,107} = \sqrt{-\frac{\xi_{3,105}}{2} + \frac{\xi_{3,106}}{2} + \frac{1}{2}}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,108} = \sqrt{\frac{\xi_{3,105}}{2} - \frac{\xi_{3,106}}{2} + \frac{1}{2}}$$

$$\xi_{3,109} = \sqrt{-\frac{\xi_{3,105}}{2} - \frac{\xi_{3,106}}{2} + \frac{1}{2}}$$

$$\xi_{3,110} = \frac{(1-\xi_{3,103})(\xi_{3,102}+1)(\xi_{3,104}+1)}{2}$$

$$\xi_{3,111} = \begin{cases} -1 & \text{for } \xi_{3,100} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{3,112} = \begin{cases} 0 & \text{for } \xi_{3,107} = 0 \\ \frac{\sqrt{\xi_{3,110}}\xi_{3,111}}{\xi_{3,107}} & \text{otherwise} \end{cases}$$

$$\xi_{3,113} = \begin{cases} 1 & \text{for } \xi_{3,107} = 0 \\ \frac{\sqrt{\xi_{3,102}\xi_{3,103}\xi_{3,104}}}{\xi_{3,107}} & \text{otherwise} \end{cases}$$

$$\xi_{3,114} = 2\xi_{3,107}\xi_{3,108}\xi_{3,109}\xi_{3,113}$$

$$\xi_{3,115} = \sqrt{2}\xi_{3,113}\left(\xi_{3,107}^2\xi_{3,109} - \xi_{3,109}\right)$$

$$\xi_{3,116} = 2\xi_{3,107}\xi_{3,108}\xi_{3,109}\xi_{3,112}$$

$$\xi_{3,117} = \sqrt{2}\left(2\xi_{3,107}^2\xi_{3,109}\xi_{3,112} - \xi_{3,109}\xi_{3,112}\right)$$

$$\xi_{3,118} = \sqrt{2}\xi_{3,107}\xi_{3,108}\xi_{3,112}\xi_{3,113}$$

$$\xi_{3,119} = \sqrt{2}\xi_{3,107}\xi_{3,108}\left(-\xi_{3,109}^2 + \xi_{3,112}^2\right)$$

$$\xi_{3,120} = -\xi_{3,107}^2\xi_{3,109}^2 + \xi_{3,107}^2\xi_{3,112}^2 + \xi_{3,109}^2$$

$$\xi_{3,121} = \xi_{3,107}^2\xi_{3,109}^2 - \xi_{3,107}^2\xi_{3,112}^2 + \xi_{3,112}^2$$

$$\xi_{3,122} = 2\xi_{3,113}\left(\xi_{3,107}^2\xi_{3,112} - \frac{\xi_{3,112}}{2}\right)$$

$$\xi_{3,123} = -\sqrt{2}\xi_{3,107}\xi_{3,108}\xi_{3,112}\xi_{3,113}$$

$$\xi_{3,124} = -\xi_{3,112}\xi_{3,113}$$

$$\xi_{3,125} = \xi_{3,107}^2\xi_{3,109}^2 + \xi_{3,107}^2\xi_{3,112}^2 - \xi_{3,107}^2 - \xi_{3,109}^2 - \xi_{3,112}^2 + 1$$

$$\xi_{3,126} = -\sqrt{2}\left(\xi_{3,107}\xi_{3,108}\xi_{3,109}^2 + \xi_{3,107}\xi_{3,108}\xi_{3,112}^2 - \xi_{3,107}\xi_{3,108}\right)$$

$$\xi_{3,127} = \sqrt{2}\xi_{3,107}^2\xi_{3,109}\xi_{3,113}$$

$$\xi_{3,128} = \xi_{3,118}^2 - \xi_{3,121}^2 - \xi_{3,124}^2 - \xi_{3,127}^2$$

$$\xi_{3,129} = \xi_{3,118}\xi_{3,119} - \xi_{3,121}\xi_{3,122} - \xi_{3,124}\xi_{3,125}$$

$$\xi_{3,130} = \xi_{3,118}\xi_{3,120} - \xi_{3,121}\xi_{3,123} - \xi_{3,124}\xi_{3,126}$$

$$\xi_{3,131} = \xi_{3,114}^2 + \xi_{3,116}^2 + \xi_{3,119}^2 - \xi_{3,122}^2 - \xi_{3,125}^2$$

$$\xi_{3,132} = \xi_{3,114}\xi_{3,115} + \xi_{3,116}\xi_{3,117} + \xi_{3,119}\xi_{3,120} - \xi_{3,122}\xi_{3,123} - \xi_{3,125}\xi_{3,126}$$

$$\xi_{3,133} = \xi_{3,115}^2 + \xi_{3,117}^2 + \xi_{3,120}^2 - \xi_{3,123}^2 - \xi_{3,126}^2$$

$$\xi_{3,134} = \sin\left(2\xi_{3,100}\right)\sin\left(2\xi_{3,99}\right)\cos\left(2\xi_{3,98}\right)$$

$$\xi_{3,135} = \sin\left(2\xi_{3,100}\right)\sin\left(2\xi_{3,98}\right)\cos\left(2\xi_{3,99}\right)$$

$$\xi_{3,136} = \sin\left(2\xi_{3,98}\right)\sin\left(2\xi_{3,99}\right)\cos\left(2\xi_{3,100}\right)$$

$$\xi_{3,137} = \sin\left(2\xi_{3,98}\right)\cos\left(2\xi_{3,100}\right)\cos\left(2\xi_{3,99}\right)$$

$$\xi_{3,138} = \sin\left(2\xi_{3,99}\right)\cos\left(2\xi_{3,100}\right)\cos\left(2\xi_{3,98}\right)$$

$$\xi_{3,139} = \sin\left(2\xi_{3,100}\right)\cos\left(2\xi_{3,98}\right)\cos\left(2\xi_{3,99}\right)$$

$$\xi_{3,140} = \xi_{3,134}^2 - \xi_{3,137}^2$$

$$\xi_{3,141} = \xi_{3,135}^2 - \xi_{3,138}^2$$

$$\xi_{3,142} = \xi_{3,136}^2 - \xi_{3,139}^2$$

$$\xi_{3,143} = \frac{\xi_{3,128} - \xi_{3,141}}{\xi_{3,140} - \xi_{3,141}}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

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$$\xi_{3,144} = \frac{\xi_{3,129}}{\xi_{3,140}-\xi_{3,141}}$$

$$\xi_{3,145} = \frac{\xi_{3,130}}{\xi_{3,140}-\xi_{3,141}}$$

$$\xi_{3,146} = \frac{\xi_{3,131}-\xi_{3,141}}{\xi_{3,140}-\xi_{3,141}}$$

$$\xi_{3,147} = \frac{\xi_{3,132}}{\xi_{3,140}-\xi_{3,141}}$$

$$\xi_{3,148} = \frac{\xi_{3,133}-\xi_{3,141}}{\xi_{3,140}-\xi_{3,141}}$$

$$\xi_{3,149} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,143} & \text{otherwise} \end{cases}$$

$$\xi_{3,150} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,144} & \text{otherwise} \end{cases}$$

$$\xi_{3,151} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,145} & \text{otherwise} \end{cases}$$

$$\xi_{3,152} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,146} & \text{otherwise} \end{cases}$$

$$\xi_{3,153} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,147} & \text{otherwise} \end{cases}$$

$$\xi_{3,154} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,148} & \text{otherwise} \end{cases}$$

$$\xi_{3,155} = \frac{\xi_{3,128}-\xi_{3,142}}{\xi_{3,140}-\xi_{3,142}}$$

$$\xi_{3,156} = \frac{\xi_{3,129}}{\xi_{3,140}-\xi_{3,142}}$$

$$\xi_{3,157} = \frac{\xi_{3,130}}{\xi_{3,140}-\xi_{3,142}}$$

$$\xi_{3,158} = \frac{\xi_{3,131}-\xi_{3,142}}{\xi_{3,140}-\xi_{3,142}}$$

$$\xi_{3,159} = \frac{\xi_{3,132}}{\xi_{3,140}-\xi_{3,142}}$$

$$\xi_{3,160} = \frac{\xi_{3,133}-\xi_{3,142}}{\xi_{3,140}-\xi_{3,142}}$$

$$\xi_{3,161} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,142} \\ \xi_{3,155} & \text{otherwise} \end{cases}$$

$$\xi_{3,162} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,142} \\ \xi_{3,156} & \text{otherwise} \end{cases}$$

$$\xi_{3,163} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,142} \\ \xi_{3,157} & \text{otherwise} \end{cases}$$

$$\xi_{3,164} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,142} \\ \xi_{3,158} & \text{otherwise} \end{cases}$$

$$\xi_{3,165} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,142} \\ \xi_{3,159} & \text{otherwise} \end{cases}$$

$$\xi_{3,166} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,142} \\ \xi_{3,160} & \text{otherwise} \end{cases}$$

$$\xi_{3,167} = \xi_{3,149}\xi_{3,161} + \xi_{3,150}\xi_{3,162} + \xi_{3,151}\xi_{3,163}$$

$$\xi_{3,168} = \xi_{3,149}\xi_{3,162} + \xi_{3,150}\xi_{3,164} + \xi_{3,151}\xi_{3,165}$$

$$\xi_{3,169} = \xi_{3,149}\xi_{3,163} + \xi_{3,150}\xi_{3,165} + \xi_{3,151}\xi_{3,166}$$

$$\xi_{3,170} = \xi_{3,150}\xi_{3,161} + \xi_{3,152}\xi_{3,162} + \xi_{3,153}\xi_{3,163}$$

$$\xi_{3,171} = \xi_{3,150}\xi_{3,162} + \xi_{3,152}\xi_{3,164} + \xi_{3,153}\xi_{3,165}$$

$$\xi_{3,172} = \xi_{3,150}\xi_{3,163} + \xi_{3,152}\xi_{3,165} + \xi_{3,153}\xi_{3,166}$$

$$\xi_{3,173} = \xi_{3,151}\xi_{3,161} + \xi_{3,153}\xi_{3,162} + \xi_{3,154}\xi_{3,163}$$

$$\xi_{3,174} = \xi_{3,151}\xi_{3,162} + \xi_{3,153}\xi_{3,164} + \xi_{3,154}\xi_{3,165}$$

$$\xi_{3,175} = \xi_{3,151}\xi_{3,163} + \xi_{3,153}\xi_{3,165} + \xi_{3,154}\xi_{3,166}$$

$$\xi_{3,176} = \sqrt{\xi_{3,167}}$$

$$\xi_{3,177} = \frac{\xi_{3,167}}{\xi_{3,176}}$$

$$\xi_{3,178} = \frac{\xi_{3,170}}{\xi_{3,176}}$$

$$\xi_{3,179} = \frac{\xi_{3,173}}{\xi_{3,176}}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

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$$\xi_{3,180} = \sqrt{\xi_{3,171}}$$

$$\xi_{3,181} = \frac{\xi_{3,168}}{\xi_{3,180}}$$

$$\xi_{3,182} = \frac{\xi_{3,171}}{\xi_{3,180}}$$

$$\xi_{3,183} = \frac{\xi_{3,174}}{\xi_{3,180}}$$

$$\xi_{3,184} = \sqrt{\xi_{3,175}}$$

$$\xi_{3,185} = \frac{\xi_{3,169}}{\xi_{3,184}}$$

$$\xi_{3,186} = \frac{\xi_{3,172}}{\xi_{3,184}}$$

$$\xi_{3,187} = \frac{\xi_{3,175}}{\xi_{3,184}}$$

$$\xi_{3,188} = \begin{cases} \xi_{3,177} & \text{for } \xi_{3,167} \geq \xi_{3,171} \wedge \xi_{3,167} \geq \xi_{3,175} \\ \xi_{3,181} & \text{for } \xi_{3,171} \geq \xi_{3,175} \wedge \xi_{3,167} \leq \xi_{3,171} \\ \xi_{3,185} & \text{otherwise} \end{cases}$$

$$\xi_{3,189} = \begin{cases} \xi_{3,178} & \text{for } \xi_{3,167} \geq \xi_{3,171} \wedge \xi_{3,167} \geq \xi_{3,175} \\ \xi_{3,182} & \text{for } \xi_{3,171} \geq \xi_{3,175} \wedge \xi_{3,167} \leq \xi_{3,171} \\ \xi_{3,186} & \text{otherwise} \end{cases}$$

$$\xi_{3,190} = \begin{cases} \xi_{3,179} & \text{for } \xi_{3,167} \geq \xi_{3,171} \wedge \xi_{3,167} \geq \xi_{3,175} \\ \xi_{3,183} & \text{for } \xi_{3,171} \geq \xi_{3,175} \wedge \xi_{3,167} \leq \xi_{3,171} \\ \xi_{3,187} & \text{otherwise} \end{cases}$$

$$\xi_{3,191} = \frac{\xi_{3,128} - \xi_{3,140}}{-\xi_{3,140} + \xi_{3,141}}$$

$$\xi_{3,192} = \frac{\xi_{3,129}}{-\xi_{3,140} + \xi_{3,141}}$$

$$\xi_{3,193} = \frac{\xi_{3,130}}{-\xi_{3,140} + \xi_{3,141}}$$

$$\xi_{3,194} = \frac{\xi_{3,131} - \xi_{3,140}}{-\xi_{3,140} + \xi_{3,141}}$$

$$\xi_{3,195} = \frac{\xi_{3,132}}{-\xi_{3,140} + \xi_{3,141}}$$

$$\xi_{3,196} = \frac{\xi_{3,133} - \xi_{3,140}}{-\xi_{3,140} + \xi_{3,141}}$$

$$\xi_{3,197} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,191} & \text{otherwise} \end{cases}$$

$$\xi_{3,198} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,192} & \text{otherwise} \end{cases}$$

$$\xi_{3,199} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,193} & \text{otherwise} \end{cases}$$

$$\xi_{3,200} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,194} & \text{otherwise} \end{cases}$$

$$\xi_{3,201} = \begin{cases} 0 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,195} & \text{otherwise} \end{cases}$$

$$\xi_{3,202} = \begin{cases} 1 & \text{for } \xi_{3,140} = \xi_{3,141} \\ \xi_{3,196} & \text{otherwise} \end{cases}$$

$$\xi_{3,203} = \frac{\xi_{3,128} - \xi_{3,142}}{\xi_{3,141} - \xi_{3,142}}$$

$$\xi_{3,204} = \frac{\xi_{3,129}}{\xi_{3,141} - \xi_{3,142}}$$

$$\xi_{3,205} = \frac{\xi_{3,130}}{\xi_{3,141} - \xi_{3,142}}$$

$$\xi_{3,206} = \frac{\xi_{3,131} - \xi_{3,142}}{\xi_{3,141} - \xi_{3,142}}$$

$$\xi_{3,207} = \frac{\xi_{3,132}}{\xi_{3,141} - \xi_{3,142}}$$

$$\xi_{3,208} = \frac{\xi_{3,133} - \xi_{3,142}}{\xi_{3,141} - \xi_{3,142}}$$

$$\xi_{3,209} = \begin{cases} 1 & \text{for } \xi_{3,141} = \xi_{3,142} \\ \xi_{3,203} & \text{otherwise} \end{cases}$$

$$\xi_{3,210} = \begin{cases} 0 & \text{for } \xi_{3,141} = \xi_{3,142} \\ \xi_{3,204} & \text{otherwise} \end{cases}$$

$$\xi_{3,211} = \begin{cases} 0 & \text{for } \xi_{3,141} = \xi_{3,142} \\ \xi_{3,205} & \text{otherwise} \end{cases}$$

$$\xi_{3,212} = \begin{cases} 1 & \text{for } \xi_{3,141} = \xi_{3,142} \\ \xi_{3,206} & \text{otherwise} \end{cases}$$

$$\xi_{3,213} = \begin{cases} 0 & \text{for } \xi_{3,141} = \xi_{3,142} \\ \xi_{3,207} & \text{otherwise} \end{cases}$$

$$\xi_{3,214} = \begin{cases} 1 & \text{for } \xi_{3,141} = \xi_{3,142} \\ \xi_{3,208} & \text{otherwise} \end{cases}$$

$$\xi_{3,215} = \xi_{3,197}\xi_{3,209} + \xi_{3,198}\xi_{3,210} + \xi_{3,199}\xi_{3,211}$$

Exact shared-expression program 3

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$$\xi_{3,216} = \xi_{3,197}\xi_{3,210} + \xi_{3,198}\xi_{3,212} + \xi_{3,199}\xi_{3,213}$$

$$\xi_{3,217} = \xi_{3,197}\xi_{3,211} + \xi_{3,198}\xi_{3,213} + \xi_{3,199}\xi_{3,214}$$

$$\xi_{3,218} = \xi_{3,198}\xi_{3,209} + \xi_{3,200}\xi_{3,210} + \xi_{3,201}\xi_{3,211}$$

$$\xi_{3,219} = \xi_{3,198}\xi_{3,210} + \xi_{3,200}\xi_{3,212} + \xi_{3,201}\xi_{3,213}$$

$$\xi_{3,220} = \xi_{3,198}\xi_{3,211} + \xi_{3,200}\xi_{3,213} + \xi_{3,201}\xi_{3,214}$$

$$\xi_{3,221} = \xi_{3,199}\xi_{3,209} + \xi_{3,201}\xi_{3,210} + \xi_{3,202}\xi_{3,211}$$

$$\xi_{3,222} = \xi_{3,199}\xi_{3,210} + \xi_{3,201}\xi_{3,212} + \xi_{3,202}\xi_{3,213}$$

$$\xi_{3,223} = \xi_{3,199}\xi_{3,211} + \xi_{3,201}\xi_{3,213} + \xi_{3,202}\xi_{3,214}$$

$$\xi_{3,224} = \begin{cases} \xi_{3,188}^2 & \text{for } \xi_{3,140} = \xi_{3,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,225} = \begin{cases} \xi_{3,188}\xi_{3,189} & \text{for } \xi_{3,140} = \xi_{3,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,226} = \begin{cases} \xi_{3,188}\xi_{3,190} & \text{for } \xi_{3,140} = \xi_{3,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,227} = \begin{cases} \xi_{3,189}^2 & \text{for } \xi_{3,140} = \xi_{3,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,228} = \begin{cases} \xi_{3,189}\xi_{3,190} & \text{for } \xi_{3,140} = \xi_{3,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,229} = \begin{cases} \xi_{3,190}^2 & \text{for } \xi_{3,140} = \xi_{3,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{3,230} = \xi_{3,215} - \xi_{3,224}$$

$$\xi_{3,231} = \xi_{3,216} - \xi_{3,225}$$

$$\xi_{3,232} = \xi_{3,217} - \xi_{3,226}$$

$$\xi_{3,233} = \xi_{3,218} - \xi_{3,225}$$

$$\xi_{3,234} = \xi_{3,219} - \xi_{3,227}$$

$$\xi_{3,235} = \xi_{3,220} - \xi_{3,228}$$

$$\xi_{3,236} = \xi_{3,221} - \xi_{3,226}$$

$$\xi_{3,237} = \xi_{3,222} - \xi_{3,228}$$

$$\xi_{3,238} = \xi_{3,223} - \xi_{3,229}$$

$$\xi_{3,239} = \sqrt{\xi_{3,230}}$$

$$\xi_{3,240} = \frac{\xi_{3,230}}{\xi_{3,239}}$$

$$\xi_{3,241} = \frac{\xi_{3,233}}{\xi_{3,239}}$$

$$\xi_{3,242} = \frac{\xi_{3,236}}{\xi_{3,239}}$$

$$\xi_{3,243} = \sqrt{\xi_{3,234}}$$

$$\xi_{3,244} = \frac{\xi_{3,231}}{\xi_{3,243}}$$

$$\xi_{3,245} = \frac{\xi_{3,234}}{\xi_{3,243}}$$

$$\xi_{3,246} = \frac{\xi_{3,237}}{\xi_{3,243}}$$

$$\xi_{3,247} = \sqrt{\xi_{3,238}}$$

$$\xi_{3,248} = \frac{\xi_{3,232}}{\xi_{3,247}}$$

$$\xi_{3,249} = \frac{\xi_{3,235}}{\xi_{3,247}}$$

$$\xi_{3,250} = \frac{\xi_{3,238}}{\xi_{3,247}}$$

$$\xi_{3,251} = \begin{cases} \xi_{3,240} & \text{for } \xi_{3,230} \geq \xi_{3,234} \wedge \xi_{3,230} \geq \xi_{3,238} \\ \xi_{3,244} & \text{for } \xi_{3,234} \geq \xi_{3,238} \wedge \xi_{3,230} \leq \xi_{3,234} \\ \xi_{3,248} & \text{otherwise} \end{cases}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,252} = \begin{cases} \xi_{3,241} & \text{for } \xi_{3,230} \geq \xi_{3,234} \wedge \xi_{3,230} \geq \xi_{3,238} \\ \xi_{3,245} & \text{for } \xi_{3,234} \geq \xi_{3,238} \wedge \xi_{3,230} \leq \xi_{3,234} \\ \xi_{3,249} & \text{otherwise} \end{cases}$$

$$\xi_{3,253} = \begin{cases} \xi_{3,242} & \text{for } \xi_{3,230} \geq \xi_{3,234} \wedge \xi_{3,230} \geq \xi_{3,238} \\ \xi_{3,246} & \text{for } \xi_{3,234} \geq \xi_{3,238} \wedge \xi_{3,230} \leq \xi_{3,234} \\ \xi_{3,250} & \text{otherwise} \end{cases}$$

$$\xi_{3,254} = \xi_{3,189}\xi_{3,253} - \xi_{3,190}\xi_{3,252}$$

$$\xi_{3,255} = -\xi_{3,188}\xi_{3,253} + \xi_{3,190}\xi_{3,251}$$

$$\xi_{3,256} = \xi_{3,188}\xi_{3,252} - \xi_{3,189}\xi_{3,251}$$

$$\xi_{3,257} = \xi_{3,134}^2 + \xi_{3,137}^2$$

$$\xi_{3,258} = \xi_{3,135}^2 + \xi_{3,138}^2$$

$$\xi_{3,259} = \xi_{3,136}^2 + \xi_{3,139}^2$$

$$\xi_{3,260} = \frac{\xi_{3,121}\xi_{3,137}\xi_{3,188} + \xi_{3,189}(\xi_{3,114}\xi_{3,134} + \xi_{3,122}\xi_{3,137}) + \xi_{3,190}(\xi_{3,115}\xi_{3,134} + \xi_{3,123}\xi_{3,137})}{\xi_{3,257}}$$

$$\xi_{3,261} = \frac{\xi_{3,124}\xi_{3,137}\xi_{3,188} + \xi_{3,189}(\xi_{3,116}\xi_{3,134} + \xi_{3,125}\xi_{3,137}) + \xi_{3,190}(\xi_{3,117}\xi_{3,134} + \xi_{3,126}\xi_{3,137})}{\xi_{3,257}}$$

$$\xi_{3,262} = \frac{\xi_{3,119}\xi_{3,134}\xi_{3,189} + \xi_{3,120}\xi_{3,134}\xi_{3,190} + \xi_{3,188}(\xi_{3,118}\xi_{3,134} + \xi_{3,127}\xi_{3,137})}{\xi_{3,257}}$$

$$\xi_{3,263} = \frac{\xi_{3,121}\xi_{3,138}\xi_{3,251} + \xi_{3,252}(\xi_{3,114}\xi_{3,135} + \xi_{3,122}\xi_{3,138}) + \xi_{3,253}(\xi_{3,115}\xi_{3,135} + \xi_{3,123}\xi_{3,138})}{\xi_{3,258}}$$

$$\xi_{3,264} = \frac{\xi_{3,124}\xi_{3,138}\xi_{3,251} + \xi_{3,252}(\xi_{3,116}\xi_{3,135} + \xi_{3,125}\xi_{3,138}) + \xi_{3,253}(\xi_{3,117}\xi_{3,135} + \xi_{3,126}\xi_{3,138})}{\xi_{3,258}}$$

$$\xi_{3,265} = \frac{\xi_{3,119}\xi_{3,135}\xi_{3,252} + \xi_{3,120}\xi_{3,135}\xi_{3,253} + \xi_{3,251}(\xi_{3,118}\xi_{3,135} + \xi_{3,127}\xi_{3,138})}{\xi_{3,258}}$$

$$\xi_{3,266} = \frac{\xi_{3,121}\xi_{3,139}\xi_{3,254} + \xi_{3,255}(\xi_{3,114}\xi_{3,136} + \xi_{3,122}\xi_{3,139}) + \xi_{3,256}(\xi_{3,115}\xi_{3,136} + \xi_{3,123}\xi_{3,139})}{\xi_{3,259}}$$

$$\xi_{3,267} = \frac{\xi_{3,124}\xi_{3,139}\xi_{3,254} + \xi_{3,255}(\xi_{3,116}\xi_{3,136} + \xi_{3,125}\xi_{3,139}) + \xi_{3,256}(\xi_{3,117}\xi_{3,136} + \xi_{3,126}\xi_{3,139})}{\xi_{3,259}}$$

$$\xi_{3,268} = \frac{\xi_{3,119}\xi_{3,136}\xi_{3,255} + \xi_{3,120}\xi_{3,136}\xi_{3,256} + \xi_{3,254}(\xi_{3,118}\xi_{3,136} + \xi_{3,127}\xi_{3,139})}{\xi_{3,259}}$$

$$\xi_{3,269} = 1 - \xi_{3,260}^2$$

$$\xi_{3,270} = -\xi_{3,260}\xi_{3,261}$$

$$\xi_{3,271} = -\xi_{3,260}\xi_{3,262}$$

$$\xi_{3,272} = 1 - \xi_{3,261}^2$$

$$\xi_{3,273} = -\xi_{3,261}\xi_{3,262}$$

$$\xi_{3,274} = 1 - \xi_{3,262}^2$$

$$\xi_{3,275} = \xi_{3,269}^2 + \xi_{3,270}^2 + \xi_{3,271}^2$$

$$\xi_{3,276} = \xi_{3,270}^2 + \xi_{3,272}^2 + \xi_{3,273}^2$$

$$\xi_{3,277} = \xi_{3,271}^2 + \xi_{3,273}^2 + \xi_{3,274}^2$$

$$\xi_{3,278} = \sqrt{\xi_{3,275}}$$

$$\xi_{3,279} = \frac{\xi_{3,269}}{\xi_{3,278}}$$

$$\xi_{3,280} = \frac{\xi_{3,270}}{\xi_{3,278}}$$

$$\xi_{3,281} = \frac{\xi_{3,271}}{\xi_{3,278}}$$

$$\xi_{3,282} = \sqrt{\xi_{3,276}}$$

$$\xi_{3,283} = \frac{\xi_{3,272}}{\xi_{3,282}}$$

$$\xi_{3,284} = \frac{\xi_{3,270}}{\xi_{3,282}}$$

$$\xi_{3,285} = \frac{\xi_{3,273}}{\xi_{3,282}}$$

$$\xi_{3,286} = \sqrt{\xi_{3,277}}$$

$$\xi_{3,287} = \frac{\xi_{3,274}}{\xi_{3,286}}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,288} = \frac{\xi_{3,271}}{\xi_{3,286}}$$

$$\xi_{3,289} = \frac{\xi_{3,273}}{\xi_{3,286}}$$

$$\xi_{3,290} = \begin{cases} \xi_{3,279} & \text{for } \xi_{3,275} \geq \xi_{3,276} \wedge \xi_{3,275} \geq \xi_{3,277} \\ \xi_{3,284} & \text{for } \xi_{3,276} \geq \xi_{3,277} \wedge \xi_{3,275} \leq \xi_{3,276} \\ \xi_{3,288} & \text{otherwise} \end{cases}$$

$$\xi_{3,291} = \begin{cases} \xi_{3,280} & \text{for } \xi_{3,275} \geq \xi_{3,276} \wedge \xi_{3,275} \geq \xi_{3,277} \\ \xi_{3,283} & \text{for } \xi_{3,276} \geq \xi_{3,277} \wedge \xi_{3,275} \leq \xi_{3,276} \\ \xi_{3,289} & \text{otherwise} \end{cases}$$

$$\xi_{3,292} = \begin{cases} \xi_{3,281} & \text{for } \xi_{3,275} \geq \xi_{3,276} \wedge \xi_{3,275} \geq \xi_{3,277} \\ \xi_{3,285} & \text{for } \xi_{3,276} \geq \xi_{3,277} \wedge \xi_{3,275} \leq \xi_{3,276} \\ \xi_{3,287} & \text{otherwise} \end{cases}$$

$$\xi_{3,293} = \xi_{3,261}\xi_{3,292} - \xi_{3,262}\xi_{3,291}$$

$$\xi_{3,294} = -\xi_{3,260}\xi_{3,292} + \xi_{3,262}\xi_{3,290}$$

$$\xi_{3,295} = \xi_{3,260}\xi_{3,291} - \xi_{3,261}\xi_{3,290}$$

$$\xi_{3,296} = 1 - \xi_{3,263}^2$$

$$\xi_{3,297} = -\xi_{3,263}\xi_{3,264}$$

$$\xi_{3,298} = -\xi_{3,263}\xi_{3,265}$$

$$\xi_{3,299} = 1 - \xi_{3,264}^2$$

$$\xi_{3,300} = -\xi_{3,264}\xi_{3,265}$$

$$\xi_{3,301} = 1 - \xi_{3,265}^2$$

$$\xi_{3,302} = \xi_{3,296}^2 + \xi_{3,297}^2 + \xi_{3,298}^2$$

$$\xi_{3,303} = \xi_{3,297}^2 + \xi_{3,299}^2 + \xi_{3,300}^2$$

$$\xi_{3,304} = \xi_{3,298}^2 + \xi_{3,300}^2 + \xi_{3,301}^2$$

$$\xi_{3,305} = \sqrt{\xi_{3,302}}$$

$$\xi_{3,306} = \frac{\xi_{3,296}}{\xi_{3,305}}$$

$$\xi_{3,307} = \frac{\xi_{3,297}}{\xi_{3,305}}$$

$$\xi_{3,308} = \frac{\xi_{3,298}}{\xi_{3,305}}$$

$$\xi_{3,309} = \sqrt{\xi_{3,303}}$$

$$\xi_{3,310} = \frac{\xi_{3,299}}{\xi_{3,309}}$$

$$\xi_{3,311} = \frac{\xi_{3,297}}{\xi_{3,309}}$$

$$\xi_{3,312} = \frac{\xi_{3,300}}{\xi_{3,309}}$$

$$\xi_{3,313} = \sqrt{\xi_{3,304}}$$

$$\xi_{3,314} = \frac{\xi_{3,301}}{\xi_{3,313}}$$

$$\xi_{3,315} = \frac{\xi_{3,298}}{\xi_{3,313}}$$

$$\xi_{3,316} = \frac{\xi_{3,300}}{\xi_{3,313}}$$

$$\xi_{3,317} = \begin{cases} \xi_{3,306} & \text{for } \xi_{3,302} \geq \xi_{3,303} \wedge \xi_{3,302} \geq \xi_{3,304} \\ \xi_{3,311} & \text{for } \xi_{3,303} \geq \xi_{3,304} \wedge \xi_{3,302} \leq \xi_{3,303} \\ \xi_{3,315} & \text{otherwise} \end{cases}$$

$$\xi_{3,318} = \begin{cases} \xi_{3,307} & \text{for } \xi_{3,302} \geq \xi_{3,303} \wedge \xi_{3,302} \geq \xi_{3,304} \\ \xi_{3,310} & \text{for } \xi_{3,303} \geq \xi_{3,304} \wedge \xi_{3,302} \leq \xi_{3,303} \\ \xi_{3,316} & \text{otherwise} \end{cases}$$

$$\xi_{3,319} = \begin{cases} \xi_{3,308} & \text{for } \xi_{3,302} \geq \xi_{3,303} \wedge \xi_{3,302} \geq \xi_{3,304} \\ \xi_{3,312} & \text{for } \xi_{3,303} \geq \xi_{3,304} \wedge \xi_{3,302} \leq \xi_{3,303} \\ \xi_{3,314} & \text{otherwise} \end{cases}$$

$$\xi_{3,320} = \xi_{3,264}\xi_{3,319} - \xi_{3,265}\xi_{3,318}$$

$$\xi_{3,321} = -\xi_{3,263}\xi_{3,319} + \xi_{3,265}\xi_{3,317}$$

$$\xi_{3,322} = \xi_{3,263}\xi_{3,318} - \xi_{3,264}\xi_{3,317}$$

$$\xi_{3,323} = \xi_{3,261}\xi_{3,265} - \xi_{3,262}\xi_{3,264}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,324} = -\xi_{3,260}\xi_{3,265} + \xi_{3,262}\xi_{3,263}$$

$$\xi_{3,325} = \xi_{3,260}\xi_{3,264} - \xi_{3,261}\xi_{3,263}$$

$$\xi_{3,326} = 1 - \xi_{3,266}^2$$

$$\xi_{3,327} = -\xi_{3,266}\xi_{3,267}$$

$$\xi_{3,328} = -\xi_{3,266}\xi_{3,268}$$

$$\xi_{3,329} = 1 - \xi_{3,267}^2$$

$$\xi_{3,330} = -\xi_{3,267}\xi_{3,268}$$

$$\xi_{3,331} = 1 - \xi_{3,268}^2$$

$$\xi_{3,332} = \xi_{3,326}^2 + \xi_{3,327}^2 + \xi_{3,328}^2$$

$$\xi_{3,333} = \xi_{3,327}^2 + \xi_{3,329}^2 + \xi_{3,330}^2$$

$$\xi_{3,334} = \xi_{3,328}^2 + \xi_{3,330}^2 + \xi_{3,331}^2$$

$$\xi_{3,335} = \sqrt{\xi_{3,332}}$$

$$\xi_{3,336} = \frac{\xi_{3,326}}{\xi_{3,335}}$$

$$\xi_{3,337} = \frac{\xi_{3,327}}{\xi_{3,335}}$$

$$\xi_{3,338} = \frac{\xi_{3,328}}{\xi_{3,335}}$$

$$\xi_{3,339} = \sqrt{\xi_{3,333}}$$

$$\xi_{3,340} = \frac{\xi_{3,329}}{\xi_{3,339}}$$

$$\xi_{3,341} = \frac{\xi_{3,327}}{\xi_{3,339}}$$

$$\xi_{3,342} = \frac{\xi_{3,330}}{\xi_{3,339}}$$

$$\xi_{3,343} = \sqrt{\xi_{3,334}}$$

$$\xi_{3,344} = \frac{\xi_{3,331}}{\xi_{3,343}}$$

$$\xi_{3,345} = \frac{\xi_{3,328}}{\xi_{3,343}}$$

$$\xi_{3,346} = \frac{\xi_{3,330}}{\xi_{3,343}}$$

$$\xi_{3,347} = \begin{cases} \xi_{3,336} & \text{for } \xi_{3,332} \geq \xi_{3,333} \wedge \xi_{3,332} \geq \xi_{3,334} \\ \xi_{3,341} & \text{for } \xi_{3,333} \geq \xi_{3,334} \wedge \xi_{3,332} \leq \xi_{3,333} \\ \xi_{3,345} & \text{otherwise} \end{cases}$$

$$\xi_{3,348} = \begin{cases} \xi_{3,337} & \text{for } \xi_{3,332} \geq \xi_{3,333} \wedge \xi_{3,332} \geq \xi_{3,334} \\ \xi_{3,340} & \text{for } \xi_{3,333} \geq \xi_{3,334} \wedge \xi_{3,332} \leq \xi_{3,333} \\ \xi_{3,346} & \text{otherwise} \end{cases}$$

$$\xi_{3,349} = \begin{cases} \xi_{3,338} & \text{for } \xi_{3,332} \geq \xi_{3,333} \wedge \xi_{3,332} \geq \xi_{3,334} \\ \xi_{3,342} & \text{for } \xi_{3,333} \geq \xi_{3,334} \wedge \xi_{3,332} \leq \xi_{3,333} \\ \xi_{3,344} & \text{otherwise} \end{cases}$$

$$\xi_{3,350} = \xi_{3,267}\xi_{3,349} - \xi_{3,268}\xi_{3,348}$$

$$\xi_{3,351} = -\xi_{3,266}\xi_{3,349} + \xi_{3,268}\xi_{3,347}$$

$$\xi_{3,352} = \xi_{3,266}\xi_{3,348} - \xi_{3,267}\xi_{3,347}$$

$$\xi_{3,353} = -\xi_{3,261}\xi_{3,268} + \xi_{3,262}\xi_{3,267}$$

$$\xi_{3,354} = \xi_{3,260}\xi_{3,268} - \xi_{3,262}\xi_{3,266}$$

$$\xi_{3,355} = -\xi_{3,260}\xi_{3,267} + \xi_{3,261}\xi_{3,266}$$

$$\xi_{3,356} = \xi_{3,264}\xi_{3,268} - \xi_{3,265}\xi_{3,267}$$

$$\xi_{3,357} = -\xi_{3,263}\xi_{3,268} + \xi_{3,265}\xi_{3,266}$$

$$\xi_{3,358} = \xi_{3,263}\xi_{3,267} - \xi_{3,264}\xi_{3,266}$$

$$\xi_{3,359} = \begin{cases} 1 & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \\ \xi_{3,260} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \\ \xi_{3,320} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,260} & \text{for } \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,347} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,260} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,356} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,260} & \text{for } \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \end{cases}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,360} = \begin{cases} 0 & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \\ \xi_{3,290} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \\ \xi_{3,263} & \text{for } \xi_{3,259} = 0 \wedge (\xi_{3,257} = 0 \vee \xi_{3,259} = 0) \wedge (\xi_{3,257} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,259} = 0 \vee \xi_{3,257} \neq 0) \wedge (\xi_{3,259} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,258} \neq 0) \wedge \xi_{3,258} \neq 0 \\ \xi_{3,350} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,353} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,263} & \text{for } (\xi_{3,257} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} = 0 \vee \xi_{3,259} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,259} \neq 0) \wedge (\xi_{3,258} \neq 0 \vee \xi_{3,259} \neq 0) \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \end{cases}$$

$$\xi_{3,361} = \begin{cases} 0 & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \\ \xi_{3,261} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \\ \xi_{3,321} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,261} & \text{for } \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,348} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,261} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,357} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,261} & \text{for } \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \end{cases}$$

$$\xi_{3,362} = \begin{cases} 0 & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \\ \xi_{3,261} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \\ \xi_{3,321} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,261} & \text{for } \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,348} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,261} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,357} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,261} & \text{for } \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \end{cases}$$

$$\xi_{3,363} = \begin{cases} 1 & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \\ \xi_{3,291} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \\ \xi_{3,264} & \text{for } \xi_{3,259} = 0 \wedge (\xi_{3,257} = 0 \vee \xi_{3,259} = 0) \wedge (\xi_{3,257} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,259} = 0 \vee \xi_{3,257} \neq 0) \wedge (\xi_{3,259} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,258} \neq 0) \wedge \xi_{3,258} \neq 0 \\ \xi_{3,351} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,354} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,264} & \text{for } (\xi_{3,257} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} = 0 \vee \xi_{3,259} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,259} \neq 0) \wedge (\xi_{3,258} \neq 0 \vee \xi_{3,259} \neq 0) \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \end{cases}$$

$$\xi_{3,365} = \begin{cases} 0 & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \\ \xi_{3,262} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \\ \xi_{3,322} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,262} & \text{for } \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \\ \xi_{3,349} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,262} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,358} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,262} & \text{for } \xi_{3,257} \neq 0 \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \end{cases}$$

$$\xi_{3,366} = \begin{cases} 0 & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \\ \xi_{3,292} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,259} = 0 \wedge \xi_{3,257} \neq 0 \\ \xi_{3,265} & \text{for } \xi_{3,259} = 0 \wedge (\xi_{3,257} = 0 \vee \xi_{3,259} = 0) \wedge (\xi_{3,257} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,259} = 0 \vee \xi_{3,257} \neq 0) \wedge (\xi_{3,259} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,258} \neq 0) \wedge \xi_{3,258} \neq 0 \\ \xi_{3,352} & \text{for } \xi_{3,257} = 0 \wedge \xi_{3,258} = 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,355} & \text{for } \xi_{3,258} = 0 \wedge \xi_{3,257} \neq 0 \wedge \xi_{3,259} \neq 0 \\ \xi_{3,265} & \text{for } (\xi_{3,257} = 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} = 0 \vee \xi_{3,259} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,258} \neq 0) \wedge (\xi_{3,257} \neq 0 \vee \xi_{3,259} \neq 0) \wedge (\xi_{3,258} \neq 0 \vee \xi_{3,259} \neq 0) \wedge \xi_{3,258} \neq 0 \wedge \xi_{3,259} \neq 0 \end{cases}$$

$$\xi_{3,368} = \frac{\xi_{3,359}}{4} + \frac{\xi_{3,363}}{4} + \frac{\xi_{3,367}}{4} + \frac{1}{4}$$

$$\xi_{3,369} = \frac{\xi_{3,359}}{4} - \frac{\xi_{3,363}}{4} - \frac{\xi_{3,367}}{4} + \frac{1}{4}$$

$$\xi_{3,370} = -\frac{\xi_{3,359}}{4} + \frac{\xi_{3,363}}{4} - \frac{\xi_{3,367}}{4} + \frac{1}{4}$$

$$\xi_{3,371} = -\frac{\xi_{3,359}}{4} - \frac{\xi_{3,363}}{4} + \frac{\xi_{3,367}}{4} + \frac{1}{4}$$

$$\xi_{3,372} = \sqrt{\xi_{3,368}}$$

$$\xi_{3,373} = \frac{-\xi_{3,364} + \xi_{3,366}}{4\xi_{3,372}}$$

$$\xi_{3,374} = \frac{\xi_{3,361} - \xi_{3,365}}{4\xi_{3,372}}$$

$$\xi_{3,375} = \frac{-\xi_{3,360} + \xi_{3,362}}{4\xi_{3,372}}$$

$$\xi_{3,376} = \sqrt{\xi_{3,369}}$$

$$\xi_{3,377} = \frac{-\xi_{3,364} + \xi_{3,366}}{4\xi_{3,376}}$$

$$\xi_{3,378} = \frac{\xi_{3,360} + \xi_{3,362}}{4\xi_{3,376}}$$

$$\xi_{3,379} = \frac{\xi_{3,361} + \xi_{3,365}}{4\xi_{3,376}}$$

$$\xi_{3,380} = \sqrt{\xi_{3,370}}$$

$$\xi_{3,381} = \frac{\xi_{3,361} - \xi_{3,365}}{4\xi_{3,380}}$$

$$\xi_{3,382} = \frac{\xi_{3,360} + \xi_{3,362}}{4\xi_{3,380}}$$

$$\xi_{3,383} = \frac{\xi_{3,364} + \xi_{3,366}}{4\xi_{3,380}}$$

$$\xi_{3,384} = \sqrt{\xi_{3,371}}$$

$$\xi_{3,385} = \frac{-\xi_{3,360} + \xi_{3,362}}{4\xi_{3,384}}$$

$$\xi_{3,386} = \frac{\xi_{3,361} + \xi_{3,365}}{4\xi_{3,384}}$$

$$\xi_{3,387} = \frac{\xi_{3,364} + \xi_{3,366}}{4\xi_{3,384}}$$

$$\xi_{3,388} = \begin{cases} \xi_{3,372} & \text{for } \xi_{3,368} \geq \xi_{3,369} \wedge \xi_{3,368} \geq \xi_{3,370} \wedge \xi_{3,368} \geq \xi_{3,371} \\ \xi_{3,377} & \text{for } \xi_{3,369} \geq \xi_{3,370} \wedge \xi_{3,369} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,369} \\ \xi_{3,381} & \text{for } \xi_{3,370} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,370} \wedge \xi_{3,369} \leq \xi_{3,370} \\ \xi_{3,385} & \text{otherwise} \end{cases}$$

$$\xi_{3,389} = \begin{cases} \xi_{3,373} & \text{for } \xi_{3,368} \geq \xi_{3,369} \wedge \xi_{3,368} \geq \xi_{3,370} \wedge \xi_{3,368} \geq \xi_{3,371} \\ \xi_{3,376} & \text{for } \xi_{3,369} \geq \xi_{3,370} \wedge \xi_{3,369} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,369} \\ \xi_{3,382} & \text{for } \xi_{3,370} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,370} \wedge \xi_{3,369} \leq \xi_{3,370} \\ \xi_{3,386} & \text{otherwise} \end{cases}$$

$$\xi_{3,390} = \begin{cases} \xi_{3,374} & \text{for } \xi_{3,368} \geq \xi_{3,369} \wedge \xi_{3,368} \geq \xi_{3,370} \wedge \xi_{3,368} \geq \xi_{3,371} \\ \xi_{3,378} & \text{for } \xi_{3,369} \geq \xi_{3,370} \wedge \xi_{3,369} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,369} \\ \xi_{3,380} & \text{for } \xi_{3,370} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,370} \wedge \xi_{3,369} \leq \xi_{3,370} \\ \xi_{3,387} & \text{otherwise} \end{cases}$$

$$\xi_{3,391} = \begin{cases} \xi_{3,375} & \text{for } \xi_{3,368} \geq \xi_{3,369} \wedge \xi_{3,368} \geq \xi_{3,370} \wedge \xi_{3,368} \geq \xi_{3,371} \\ \xi_{3,379} & \text{for } \xi_{3,369} \geq \xi_{3,370} \wedge \xi_{3,369} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,369} \\ \xi_{3,383} & \text{for } \xi_{3,370} \geq \xi_{3,371} \wedge \xi_{3,368} \leq \xi_{3,370} \wedge \xi_{3,369} \leq \xi_{3,370} \\ \xi_{3,384} & \text{otherwise} \end{cases}$$

$$\xi_{3,392} = \frac{\xi_{3,188}}{4} + \frac{\xi_{3,252}}{4} + \frac{\xi_{3,256}}{4} + \frac{1}{4}$$

$$\xi_{3,393} = \frac{\xi_{3,188}}{4} - \frac{\xi_{3,252}}{4} - \frac{\xi_{3,256}}{4} + \frac{1}{4}$$

$$\xi_{3,394} = -\frac{\xi_{3,188}}{4} + \frac{\xi_{3,252}}{4} - \frac{\xi_{3,256}}{4} + \frac{1}{4}$$

$$\xi_{3,395} = -\frac{\xi_{3,188}}{4} - \frac{\xi_{3,252}}{4} + \frac{\xi_{3,256}}{4} + \frac{1}{4}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,396} = \sqrt{\xi_{3,392}}$$

$$\xi_{3,397} = \frac{\xi_{3,253} - \xi_{3,255}}{4\xi_{3,396}}$$

$$\xi_{3,398} = \frac{-\xi_{3,190} + \xi_{3,254}}{4\xi_{3,396}}$$

$$\xi_{3,399} = \frac{\xi_{3,189} - \xi_{3,251}}{4\xi_{3,396}}$$

$$\xi_{3,400} = \sqrt{\xi_{3,393}}$$

$$\xi_{3,401} = \frac{\xi_{3,253} - \xi_{3,255}}{4\xi_{3,400}}$$

$$\xi_{3,402} = \frac{\xi_{3,189} + \xi_{3,251}}{4\xi_{3,400}}$$

$$\xi_{3,403} = \frac{\xi_{3,190} + \xi_{3,254}}{4\xi_{3,400}}$$

$$\xi_{3,404} = \sqrt{\xi_{3,394}}$$

$$\xi_{3,405} = \frac{-\xi_{3,190} + \xi_{3,254}}{4\xi_{3,404}}$$

$$\xi_{3,406} = \frac{\xi_{3,189} + \xi_{3,251}}{4\xi_{3,404}}$$

$$\xi_{3,407} = \frac{\xi_{3,253} + \xi_{3,255}}{4\xi_{3,404}}$$

$$\xi_{3,408} = \sqrt{\xi_{3,395}}$$

$$\xi_{3,409} = \frac{\xi_{3,189} - \xi_{3,251}}{4\xi_{3,408}}$$

$$\xi_{3,410} = \frac{\xi_{3,190} + \xi_{3,254}}{4\xi_{3,408}}$$

$$\xi_{3,411} = \frac{\xi_{3,253} + \xi_{3,255}}{4\xi_{3,408}}$$

$$\xi_{3,412} = \begin{cases} \xi_{3,396} & \text{for } \xi_{3,392} \geq \xi_{3,393} \wedge \xi_{3,392} \geq \xi_{3,394} \wedge \xi_{3,392} \geq \xi_{3,395} \\ \xi_{3,401} & \text{for } \xi_{3,393} \geq \xi_{3,394} \wedge \xi_{3,393} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,393} \\ \xi_{3,405} & \text{for } \xi_{3,394} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,394} \wedge \xi_{3,393} \leq \xi_{3,394} \\ \xi_{3,409} & \text{otherwise} \end{cases}$$

$$\xi_{3,413} = \begin{cases} \xi_{3,397} & \text{for } \xi_{3,392} \geq \xi_{3,393} \wedge \xi_{3,392} \geq \xi_{3,394} \wedge \xi_{3,392} \geq \xi_{3,395} \\ \xi_{3,400} & \text{for } \xi_{3,393} \geq \xi_{3,394} \wedge \xi_{3,393} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,393} \\ \xi_{3,406} & \text{for } \xi_{3,394} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,394} \wedge \xi_{3,393} \leq \xi_{3,394} \\ \xi_{3,410} & \text{otherwise} \end{cases}$$

$$\xi_{3,414} = \begin{cases} \xi_{3,398} & \text{for } \xi_{3,392} \geq \xi_{3,393} \wedge \xi_{3,392} \geq \xi_{3,394} \wedge \xi_{3,392} \geq \xi_{3,395} \\ \xi_{3,402} & \text{for } \xi_{3,393} \geq \xi_{3,394} \wedge \xi_{3,393} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,393} \\ \xi_{3,404} & \text{for } \xi_{3,394} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,394} \wedge \xi_{3,393} \leq \xi_{3,394} \\ \xi_{3,411} & \text{otherwise} \end{cases}$$

$$\xi_{3,415} = \begin{cases} \xi_{3,399} & \text{for } \xi_{3,392} \geq \xi_{3,393} \wedge \xi_{3,392} \geq \xi_{3,394} \wedge \xi_{3,392} \geq \xi_{3,395} \\ \xi_{3,403} & \text{for } \xi_{3,393} \geq \xi_{3,394} \wedge \xi_{3,393} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,393} \\ \xi_{3,407} & \text{for } \xi_{3,394} \geq \xi_{3,395} \wedge \xi_{3,392} \leq \xi_{3,394} \wedge \xi_{3,393} \leq \xi_{3,394} \\ \xi_{3,408} & \text{otherwise} \end{cases}$$

$$\xi_{3,416} = \xi_{3,107}\left(\xi_{3,112} + i\xi_{3,113}\right)$$

$$\xi_{3,417} = -\frac{\sqrt{2}\xi_{3,107}\xi_{3,109}}{2} + \frac{\sqrt{2}i\xi_{3,108}\left(\xi_{3,112} + i\xi_{3,113}\right)}{2}$$

$$\xi_{3,418} = \frac{\sqrt{2}i\xi_{3,107}\xi_{3,109}}{2} - \frac{\sqrt{2}\xi_{3,108}\left(\xi_{3,112} + i\xi_{3,113}\right)}{2}$$

$$\xi_{3,419} = -\xi_{3,108}\xi_{3,109}$$

$$\xi_{3,420} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{3,107}\left(\xi_{3,112} + i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}i\xi_{3,108}\xi_{3,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{3,107}\left(\xi_{3,112} - i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}\xi_{3,108}\xi_{3,109}}{2}\right)}{2}$$

$$\xi_{3,421} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{3,107}\left(\xi_{3,112} + i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}i\xi_{3,108}\xi_{3,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{3,107}\left(\xi_{3,112} - i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}\xi_{3,108}\xi_{3,109}}{2}\right)}{2}$$

$$\xi_{3,422} = \frac{\sqrt{2}i\xi_{3,107}\xi_{3,109}}{2} - \frac{\sqrt{2}\xi_{3,108}\left(\xi_{3,112} - i\xi_{3,113}\right)}{2}$$

$$\xi_{3,423} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{3,107}\left(\xi_{3,112} - i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}i\xi_{3,108}\xi_{3,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{3,107}\left(\xi_{3,112} + i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}\xi_{3,108}\xi_{3,109}}{2}\right)}{2}$$

$$\xi_{3,424} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{3,107}\left(\xi_{3,112} - i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}i\xi_{3,108}\xi_{3,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{3,107}\left(\xi_{3,112} + i\xi_{3,113}\right)}{2} - \frac{\sqrt{2}\xi_{3,108}\xi_{3,109}}{2}\right)}{2}$$

$$\xi_{3,425} = -\frac{\sqrt{2}\xi_{3,107}\xi_{3,109}}{2} + \frac{\sqrt{2}i\xi_{3,108}\left(\xi_{3,112} - i\xi_{3,113}\right)}{2}$$

$$\xi_{3,426} = \xi_{3,107}\left(\xi_{3,112} - i\xi_{3,113}\right)$$

$$\xi_{3,427} = \left(\xi_{3,388} - i\xi_{3,391}\right)\left(\xi_{3,412} - i\xi_{3,415}\right)$$

$$\xi_{3,428} = \left(\xi_{3,388} - i\xi_{3,391}\right)\left(-i\xi_{3,413} - \xi_{3,414}\right)$$

$$\xi_{3,429} = \left(\xi_{3,412} - i\xi_{3,415}\right)\left(-i\xi_{3,389} - \xi_{3,390}\right)$$

$$\xi_{3,430} = \left(-i\xi_{3,389} - \xi_{3,390}\right)\left(-i\xi_{3,413} - \xi_{3,414}\right)$$

$$\xi_{3,431} = \left(\xi_{3,388} - i\xi_{3,391}\right)\left(-i\xi_{3,413} + \xi_{3,414}\right)$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,432} = (\xi_{3,388} - i\xi_{3,391})\,(\xi_{3,412} + i\xi_{3,415})$$

$$\xi_{3,433} = (-i\xi_{3,389} - \xi_{3,390})\,(-i\xi_{3,413} + \xi_{3,414})$$

$$\xi_{3,434} = (\xi_{3,412} + i\xi_{3,415})\,(-i\xi_{3,389} - \xi_{3,390})$$

$$\xi_{3,435} = (\xi_{3,412} - i\xi_{3,415})\,(-i\xi_{3,389} + \xi_{3,390})$$

$$\xi_{3,436} = (-i\xi_{3,389} + \xi_{3,390})\,(-i\xi_{3,413} - \xi_{3,414})$$

$$\xi_{3,437} = (\xi_{3,388} + i\xi_{3,391})\,(\xi_{3,412} - i\xi_{3,415})$$

$$\xi_{3,438} = (\xi_{3,388} + i\xi_{3,391})\,(-i\xi_{3,413} - \xi_{3,414})$$

$$\xi_{3,439} = (-i\xi_{3,389} + \xi_{3,390})\,(-i\xi_{3,413} + \xi_{3,414})$$

$$\xi_{3,440} = (\xi_{3,412} + i\xi_{3,415})\,(-i\xi_{3,389} + \xi_{3,390})$$

$$\xi_{3,441} = (\xi_{3,388} + i\xi_{3,391})\,(-i\xi_{3,413} + \xi_{3,414})$$

$$\xi_{3,442} = (\xi_{3,388} + i\xi_{3,391})\,(\xi_{3,412} + i\xi_{3,415})$$

$$\xi_{3,443} = \xi_{3,416}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}}\right) + \xi_{3,417}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}}\right) + \xi_{3,418}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}}\right) + \xi_{3,419}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}}\right)$$

$$\xi_{3,444} = \xi_{3,417}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}}\right) + \xi_{3,420}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}}\right) + \xi_{3,422}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}}\right) + \xi_{3,423}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}}\right)$$

$$\xi_{3,445} = \xi_{3,418}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}}\right) + \xi_{3,421}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}}\right) + \xi_{3,424}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}}\right) + \xi_{3,425}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}}\right)$$

$$\xi_{3,446} = \xi_{3,419}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}}\right) + \xi_{3,422}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}}\right) + \xi_{3,425}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}}\right) + \xi_{3,426}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}}\right)$$

$$\xi_{3,447} = \xi_{3,416}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}}\right) + \xi_{3,417}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}}\right) + \xi_{3,418}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}}\right) + \xi_{3,419}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}}\right)$$

$$\xi_{3,448} = \xi_{3,417}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}}\right) + \xi_{3,420}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}}\right) + \xi_{3,422}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}}\right) + \xi_{3,423}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}}\right)$$

$$\xi_{3,449} = \xi_{3,418}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}}\right) + \xi_{3,421}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}}\right) + \xi_{3,424}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}}\right) + \xi_{3,425}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}}\right)$$

$$\xi_{3,450} = \xi_{3,419}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,427}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,430}}\right) + \xi_{3,422}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,431}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,434}}\right) + \xi_{3,425}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,435}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,438}}\right) + \xi_{3,426}\left(-ie^{-i\xi_{3,100}}\sin\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,439}} + e^{-i\xi_{3,100}}\cos\left(\xi_{3,98} - \xi_{3,99}\right)\overline{\xi_{3,442}}\right)$$

$$\xi_{3,451} = \xi_{3,416}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}}\right) + \xi_{3,417}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}}\right) + \xi_{3,418}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}}\right) + \xi_{3,419}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}}\right)$$

$$\xi_{3,452} = \xi_{3,417}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}}\right) + \xi_{3,420}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}}\right) + \xi_{3,422}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}}\right) + \xi_{3,423}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}}\right)$$

$$\xi_{3,453} = \xi_{3,418}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}}\right) + \xi_{3,421}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}}\right) + \xi_{3,424}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}}\right) + \xi_{3,425}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}}\right)$$

$$\xi_{3,454} = \xi_{3,419}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}}\right) + \xi_{3,422}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}}\right) + \xi_{3,425}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}}\right) + \xi_{3,426}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}}\right)$$

$$\xi_{3,455} = \xi_{3,416}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}}\right) + \xi_{3,417}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}}\right) + \xi_{3,418}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}}\right) + \xi_{3,419}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}}\right)$$

$$\xi_{3,456} = \xi_{3,417}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}}\right) + \xi_{3,420}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}}\right) + \xi_{3,422}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}}\right) + \xi_{3,423}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}}\right)$$

$$\xi_{3,457} = \xi_{3,418}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}}\right) + \xi_{3,421}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}}\right) + \xi_{3,424}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}}\right) + \xi_{3,425}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}}\right)$$

$$\xi_{3,458} = \xi_{3,419}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,428}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,429}}\right) + \xi_{3,422}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,432}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,433}}\right) + \xi_{3,425}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,436}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,437}}\right) + \xi_{3,426}\left(-ie^{i\xi_{3,100}}\sin\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,440}} + e^{i\xi_{3,100}}\cos\left(\xi_{3,98} + \xi_{3,99}\right)\overline{\xi_{3,441}}\right)$$

$$\xi_{3,459} = \frac{\operatorname{re}\left(\xi_{3,443}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,450}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,452}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,457}\right)}{4}$$

$$\xi_{3,460} = -\frac{\operatorname{im}\left(\xi_{3,444}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,449}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,451}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,458}\right)}{4}$$

$$\xi_{3,461} = -\frac{\operatorname{re}\left(\xi_{3,444}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,449}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,451}\right)}{4} - \frac{\operatorname{re}\left(\xi_{3,458}\right)}{4}$$

$$\xi_{3,462} = -\frac{\operatorname{im}\left(\xi_{3,443}\right)}{4} + \frac{\operatorname{im}\left(\xi_{3,450}\right)}{4} + \frac{\operatorname{im}\left(\xi_{3,452}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,457}\right)}{4}$$

$$\xi_{3,463} = -\frac{\operatorname{im}\left(\xi_{3,445}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,448}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,454}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,455}\right)}{4}$$

$$\xi_{3,464} = -\frac{\operatorname{re}\left(\xi_{3,446}\right)}{4} - \frac{\operatorname{re}\left(\xi_{3,447}\right)}{4} - \frac{\operatorname{re}\left(\xi_{3,453}\right)}{4} - \frac{\operatorname{re}\left(\xi_{3,456}\right)}{4}$$

$$\xi_{3,465} = \frac{\operatorname{im}\left(\xi_{3,446}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,447}\right)}{4} - \frac{\operatorname{im}\left(\xi_{3,453}\right)}{4} + \frac{\operatorname{im}\left(\xi_{3,456}\right)}{4}$$

$$\xi_{3,466} = -\frac{\operatorname{re}\left(\xi_{3,445}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,448}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,454}\right)}{4} - \frac{\operatorname{re}\left(\xi_{3,455}\right)}{4}$$

$$\xi_{3,467} = -\frac{\operatorname{re}\left(\xi_{3,445}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,448}\right)}{4} - \frac{\operatorname{re}\left(\xi_{3,454}\right)}{4} + \frac{\operatorname{re}\left(\xi_{3,455}\right)}{4}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,468} = \frac{\operatorname{im}(\xi_{3,446})}{4} - \frac{\operatorname{im}(\xi_{3,447})}{4} + \frac{\operatorname{im}(\xi_{3,453})}{4} - \frac{\operatorname{im}(\xi_{3,456})}{4}$$

$$\xi_{3,469} = \frac{\operatorname{re}(\xi_{3,446})}{4} + \frac{\operatorname{re}(\xi_{3,447})}{4} - \frac{\operatorname{re}(\xi_{3,453})}{4} - \frac{\operatorname{re}(\xi_{3,456})}{4}$$

$$\xi_{3,470} = \frac{\operatorname{im}(\xi_{3,445})}{4} + \frac{\operatorname{im}(\xi_{3,448})}{4} - \frac{\operatorname{im}(\xi_{3,454})}{4} - \frac{\operatorname{im}(\xi_{3,455})}{4}$$

$$\xi_{3,471} = -\frac{\operatorname{im}(\xi_{3,443})}{4} + \frac{\operatorname{im}(\xi_{3,450})}{4} - \frac{\operatorname{im}(\xi_{3,452})}{4} + \frac{\operatorname{im}(\xi_{3,457})}{4}$$

$$\xi_{3,472} = -\frac{\operatorname{re}(\xi_{3,444})}{4} + \frac{\operatorname{re}(\xi_{3,449})}{4} - \frac{\operatorname{re}(\xi_{3,451})}{4} + \frac{\operatorname{re}(\xi_{3,458})}{4}$$

$$\xi_{3,473} = \frac{\operatorname{im}(\xi_{3,444})}{4} + \frac{\operatorname{im}(\xi_{3,449})}{4} - \frac{\operatorname{im}(\xi_{3,451})}{4} - \frac{\operatorname{im}(\xi_{3,458})}{4}$$

$$\xi_{3,474} = -\frac{\operatorname{re}(\xi_{3,443})}{4} - \frac{\operatorname{re}(\xi_{3,450})}{4} + \frac{\operatorname{re}(\xi_{3,452})}{4} + \frac{\operatorname{re}(\xi_{3,457})}{4}$$

$$\xi_{3,475} = \xi_{3,459}^2 + \xi_{3,463}^2 + \xi_{3,467}^2 + \xi_{3,471}^2$$

$$\xi_{3,476} = \xi_{3,460}^2 + \xi_{3,464}^2 + \xi_{3,468}^2 + \xi_{3,472}^2$$

$$\xi_{3,477} = \xi_{3,461}^2 + \xi_{3,465}^2 + \xi_{3,469}^2 + \xi_{3,473}^2$$

$$\xi_{3,478} = \xi_{3,462}^2 + \xi_{3,466}^2 + \xi_{3,470}^2 + \xi_{3,474}^2$$

$$\xi_{3,479} = \sqrt{\xi_{3,475}}$$

$$\xi_{3,480} = \frac{\xi_{3,459}}{\xi_{3,479}}$$

$$\xi_{3,481} = \frac{\xi_{3,463}}{\xi_{3,479}}$$

$$\xi_{3,482} = \frac{\xi_{3,467}}{\xi_{3,479}}$$

$$\xi_{3,483} = \frac{\xi_{3,471}}{\xi_{3,479}}$$

$$\xi_{3,484} = \sqrt{\xi_{3,476}}$$

$$\xi_{3,485} = \frac{\xi_{3,460}}{\xi_{3,484}}$$

$$\xi_{3,486} = \frac{\xi_{3,464}}{\xi_{3,484}}$$

$$\xi_{3,487} = \frac{\xi_{3,468}}{\xi_{3,484}}$$

$$\xi_{3,488} = \frac{\xi_{3,472}}{\xi_{3,484}}$$

$$\xi_{3,489} = \sqrt{\xi_{3,477}}$$

$$\xi_{3,490} = \frac{\xi_{3,461}}{\xi_{3,489}}$$

$$\xi_{3,491} = \frac{\xi_{3,465}}{\xi_{3,489}}$$

$$\xi_{3,492} = \frac{\xi_{3,469}}{\xi_{3,489}}$$

$$\xi_{3,493} = \frac{\xi_{3,473}}{\xi_{3,489}}$$

$$\xi_{3,494} = \sqrt{\xi_{3,478}}$$

$$\xi_{3,495} = \frac{\xi_{3,462}}{\xi_{3,494}}$$

$$\xi_{3,496} = \frac{\xi_{3,466}}{\xi_{3,494}}$$

$$\xi_{3,497} = \frac{\xi_{3,470}}{\xi_{3,494}}$$

$$\xi_{3,498} = \frac{\xi_{3,474}}{\xi_{3,494}}$$

$$\xi_{3,499} = \begin{cases} \xi_{3,480} & \text{for } \xi_{3,475} \geq \xi_{3,476} \wedge \xi_{3,475} \geq \xi_{3,477} \wedge \xi_{3,475} \geq \xi_{3,478} \\ \xi_{3,485} & \text{for } \xi_{3,476} \geq \xi_{3,477} \wedge \xi_{3,476} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,476} \\ \xi_{3,490} & \text{for } \xi_{3,477} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,477} \wedge \xi_{3,476} \leq \xi_{3,477} \\ \xi_{3,495} & \text{otherwise} \end{cases}$$

$$\xi_{3,500} = \begin{cases} \xi_{3,481} & \text{for } \xi_{3,475} \geq \xi_{3,476} \wedge \xi_{3,475} \geq \xi_{3,477} \wedge \xi_{3,475} \geq \xi_{3,478} \\ \xi_{3,486} & \text{for } \xi_{3,476} \geq \xi_{3,477} \wedge \xi_{3,476} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,476} \\ \xi_{3,491} & \text{for } \xi_{3,477} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,477} \wedge \xi_{3,476} \leq \xi_{3,477} \\ \xi_{3,496} & \text{otherwise} \end{cases}$$

$$\xi_{3,501} = \begin{cases} \xi_{3,482} & \text{for } \xi_{3,475} \geq \xi_{3,476} \wedge \xi_{3,475} \geq \xi_{3,477} \wedge \xi_{3,475} \geq \xi_{3,478} \\ \xi_{3,487} & \text{for } \xi_{3,476} \geq \xi_{3,477} \wedge \xi_{3,476} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,476} \\ \xi_{3,492} & \text{for } \xi_{3,477} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,477} \wedge \xi_{3,476} \leq \xi_{3,477} \\ \xi_{3,497} & \text{otherwise} \end{cases}$$

$$\xi_{3,502} = \begin{cases} \xi_{3,483} & \text{for } \xi_{3,475} \geq \xi_{3,476} \wedge \xi_{3,475} \geq \xi_{3,477} \wedge \xi_{3,475} \geq \xi_{3,478} \\ \xi_{3,488} & \text{for } \xi_{3,476} \geq \xi_{3,477} \wedge \xi_{3,476} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,476} \\ \xi_{3,493} & \text{for } \xi_{3,477} \geq \xi_{3,478} \wedge \xi_{3,475} \leq \xi_{3,477} \wedge \xi_{3,476} \leq \xi_{3,477} \\ \xi_{3,498} & \text{otherwise} \end{cases}$$

$$\xi_{3,503} = \xi_{3,459}\xi_{3,499} + \xi_{3,463}\xi_{3,500} + \xi_{3,467}\xi_{3,501} + \xi_{3,471}\xi_{3,502}$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,504} = \xi_{3,460}\xi_{3,499} + \xi_{3,464}\xi_{3,500} + \xi_{3,468}\xi_{3,501} + \xi_{3,472}\xi_{3,502}$$

$$\xi_{3,505} = \xi_{3,461}\xi_{3,499} + \xi_{3,465}\xi_{3,500} + \xi_{3,469}\xi_{3,501} + \xi_{3,473}\xi_{3,502}$$

$$\xi_{3,506} = \xi_{3,462}\xi_{3,499} + \xi_{3,466}\xi_{3,500} + \xi_{3,470}\xi_{3,501} + \xi_{3,474}\xi_{3,502}$$

$$\xi_{3,507} = \xi_{3,36}\xi_{3,47} - \xi_{3,37}\xi_{3,50} - \xi_{3,38}\xi_{3,51} - \xi_{3,39}\xi_{3,52}$$

$$\xi_{3,508} = \xi_{3,36}\xi_{3,50} + \xi_{3,37}\xi_{3,47} - \xi_{3,38}\xi_{3,52} + \xi_{3,39}\xi_{3,51}$$

$$\xi_{3,509} = \xi_{3,36}\xi_{3,51} + \xi_{3,37}\xi_{3,52} + \xi_{3,38}\xi_{3,47} - \xi_{3,39}\xi_{3,50}$$

$$\xi_{3,510} = \xi_{3,36}\xi_{3,52} - \xi_{3,37}\xi_{3,51} + \xi_{3,38}\xi_{3,50} + \xi_{3,39}\xi_{3,47}$$

$$\xi_{3,511} = \xi_{3,10}\xi_{3,50} + \xi_{3,49}\xi_{3,7} + \xi_{3,50}\xi_{3,8} + \xi_{3,50}\xi_{3,9}$$

$$\xi_{3,512} = -\xi_{3,10}\xi_{3,50} - \xi_{3,49}\xi_{3,8} + \xi_{3,50}\xi_{3,7} + \xi_{3,50}\xi_{3,9}$$

$$\xi_{3,513} = \xi_{3,10}\xi_{3,50} - \xi_{3,49}\xi_{3,9} + \xi_{3,50}\xi_{3,7} - \xi_{3,50}\xi_{3,8}$$

$$\xi_{3,514} = -\xi_{3,10}\xi_{3,49} + \xi_{3,50}\xi_{3,7} + \xi_{3,50}\xi_{3,8} - \xi_{3,50}\xi_{3,9}$$

$$\xi_{3,515} = \xi_{3,47}\left(\xi_{3,499}\xi_{3,94} + \xi_{3,500}\xi_{3,95} + \xi_{3,501}\xi_{3,96} + \xi_{3,502}\xi_{3,97}\right) - \xi_{3,47}\left(\xi_{3,499}\xi_{3,97} - \xi_{3,500}\xi_{3,96} + \xi_{3,501}\xi_{3,95} - \xi_{3,502}\xi_{3,94}\right) - \xi_{3,48}\left(\xi_{3,499}\xi_{3,95} - \xi_{3,500}\xi_{3,94} - \xi_{3,501}\xi_{3,97} + \xi_{3,502}\xi_{3,96}\right) - \xi_{3,49}\left(\xi_{3,499}\xi_{3,96} + \xi_{3,500}\xi_{3,97} - \xi_{3,501}\xi_{3,94} - \xi_{3,502}\xi_{3,95}\right)$$

$$\xi_{3,516} = \xi_{3,47}\left(\xi_{3,499}\xi_{3,95} - \xi_{3,500}\xi_{3,94} - \xi_{3,501}\xi_{3,97} + \xi_{3,502}\xi_{3,96}\right) + \xi_{3,47}\left(\xi_{3,499}\xi_{3,96} + \xi_{3,500}\xi_{3,97} - \xi_{3,501}\xi_{3,94} - \xi_{3,502}\xi_{3,95}\right) + \xi_{3,48}\left(\xi_{3,499}\xi_{3,94} + \xi_{3,500}\xi_{3,95} + \xi_{3,501}\xi_{3,96} + \xi_{3,502}\xi_{3,97}\right) - \xi_{3,49}\left(\xi_{3,499}\xi_{3,97} - \xi_{3,500}\xi_{3,96} + \xi_{3,501}\xi_{3,95} - \xi_{3,502}\xi_{3,94}\right)$$

$$\xi_{3,517} = -\xi_{3,47}\left(\xi_{3,499}\xi_{3,95} - \xi_{3,500}\xi_{3,94} - \xi_{3,501}\xi_{3,97} + \xi_{3,502}\xi_{3,96}\right) + \xi_{3,47}\left(\xi_{3,499}\xi_{3,96} + \xi_{3,500}\xi_{3,97} - \xi_{3,501}\xi_{3,94} - \xi_{3,502}\xi_{3,95}\right) + \xi_{3,48}\left(\xi_{3,499}\xi_{3,97} - \xi_{3,500}\xi_{3,96} + \xi_{3,501}\xi_{3,95} - \xi_{3,502}\xi_{3,94}\right) + \xi_{3,49}\left(\xi_{3,499}\xi_{3,94} + \xi_{3,500}\xi_{3,95} + \xi_{3,501}\xi_{3,96} + \xi_{3,502}\xi_{3,97}\right)$$

$$\xi_{3,518} = \xi_{3,47}\left(\xi_{3,499}\xi_{3,94} + \xi_{3,500}\xi_{3,95} + \xi_{3,501}\xi_{3,96} + \xi_{3,502}\xi_{3,97}\right) + \xi_{3,47}\left(\xi_{3,499}\xi_{3,97} - \xi_{3,500}\xi_{3,96} + \xi_{3,501}\xi_{3,95} - \xi_{3,502}\xi_{3,94}\right) - \xi_{3,48}\left(\xi_{3,499}\xi_{3,96} + \xi_{3,500}\xi_{3,97} - \xi_{3,501}\xi_{3,94} - \xi_{3,502}\xi_{3,95}\right) + \xi_{3,49}\left(\xi_{3,499}\xi_{3,95} - \xi_{3,500}\xi_{3,94} - \xi_{3,501}\xi_{3,97} + \xi_{3,502}\xi_{3,96}\right)$$

$$\xi_{3,519} = -\xi_{3,48}\left(-\xi_{3,503}\xi_{3,63} - \xi_{3,504}\xi_{3,62} + \xi_{3,505}\xi_{3,65} - \xi_{3,506}\xi_{3,64}\right) - \xi_{3,48}\left(-\xi_{3,503}\xi_{3,64} - \xi_{3,504}\xi_{3,65} - \xi_{3,505}\xi_{3,62} + \xi_{3,506}\xi_{3,63}\right) - \xi_{3,48}\left(-\xi_{3,503}\xi_{3,65} + \xi_{3,504}\xi_{3,64} - \xi_{3,505}\xi_{3,63} - \xi_{3,506}\xi_{3,62}\right) + \xi_{3,49}\left(\xi_{3,503}\xi_{3,62} - \xi_{3,504}\xi_{3,63} - \xi_{3,505}\xi_{3,64} - \xi_{3,506}\xi_{3,65}\right)$$

$$\xi_{3,520} = \xi_{3,48}\left(\xi_{3,503}\xi_{3,62} - \xi_{3,504}\xi_{3,63} - \xi_{3,505}\xi_{3,64} - \xi_{3,506}\xi_{3,65}\right) + \xi_{3,48}\left(-\xi_{3,503}\xi_{3,64} - \xi_{3,504}\xi_{3,65} - \xi_{3,505}\xi_{3,62} + \xi_{3,506}\xi_{3,63}\right) - \xi_{3,48}\left(-\xi_{3,503}\xi_{3,65} + \xi_{3,504}\xi_{3,64} - \xi_{3,505}\xi_{3,63} - \xi_{3,506}\xi_{3,62}\right) + \xi_{3,49}\left(-\xi_{3,503}\xi_{3,63} - \xi_{3,504}\xi_{3,62} + \xi_{3,505}\xi_{3,65} - \xi_{3,506}\xi_{3,64}\right)$$

$$\xi_{3,521} = \xi_{3,48}\left(\xi_{3,503}\xi_{3,62} - \xi_{3,504}\xi_{3,63} - \xi_{3,505}\xi_{3,64} - \xi_{3,506}\xi_{3,65}\right) - \xi_{3,48}\left(-\xi_{3,503}\xi_{3,63} - \xi_{3,504}\xi_{3,62} + \xi_{3,505}\xi_{3,65} - \xi_{3,506}\xi_{3,64}\right) + \xi_{3,48}\left(-\xi_{3,503}\xi_{3,65} + \xi_{3,504}\xi_{3,64} - \xi_{3,505}\xi_{3,63} - \xi_{3,506}\xi_{3,62}\right) + \xi_{3,49}\left(-\xi_{3,503}\xi_{3,64} - \xi_{3,504}\xi_{3,65} - \xi_{3,505}\xi_{3,62} + \xi_{3,506}\xi_{3,63}\right)$$

$$\xi_{3,522} = \xi_{3,48}\left(\xi_{3,503}\xi_{3,62} - \xi_{3,504}\xi_{3,63} - \xi_{3,505}\xi_{3,64} - \xi_{3,506}\xi_{3,65}\right) + \xi_{3,48}\left(-\xi_{3,503}\xi_{3,63} - \xi_{3,504}\xi_{3,62} + \xi_{3,505}\xi_{3,65} - \xi_{3,506}\xi_{3,64}\right) - \xi_{3,48}\left(-\xi_{3,503}\xi_{3,64} - \xi_{3,504}\xi_{3,65} - \xi_{3,505}\xi_{3,62} + \xi_{3,506}\xi_{3,63}\right) + \xi_{3,49}\left(-\xi_{3,503}\xi_{3,65} + \xi_{3,504}\xi_{3,64} - \xi_{3,505}\xi_{3,63} - \xi_{3,506}\xi_{3,62}\right)$$

$$\xi_{3,523} = \xi_{3,388}\left(\xi_{3,28}\xi_{3,86} - \xi_{3,29}\xi_{3,87} - \xi_{3,30}\xi_{3,88} - \xi_{3,31}\xi_{3,89}\right) + \xi_{3,389}\left(\xi_{3,28}\xi_{3,87} + \xi_{3,29}\xi_{3,86} + \xi_{3,30}\xi_{3,89} - \xi_{3,31}\xi_{3,88}\right) + \xi_{3,390}\left(\xi_{3,28}\xi_{3,88} - \xi_{3,29}\xi_{3,89} + \xi_{3,30}\xi_{3,86} + \xi_{3,31}\xi_{3,87}\right) + \xi_{3,391}\left(\xi_{3,28}\xi_{3,89} + \xi_{3,29}\xi_{3,88} - \xi_{3,30}\xi_{3,87} + \xi_{3,31}\xi_{3,86}\right)$$

$$\xi_{3,524} = \xi_{3,388}\left(\xi_{3,28}\xi_{3,87} + \xi_{3,29}\xi_{3,86} + \xi_{3,30}\xi_{3,89} - \xi_{3,31}\xi_{3,88}\right) - \xi_{3,389}\left(\xi_{3,28}\xi_{3,86} - \xi_{3,29}\xi_{3,87} - \xi_{3,30}\xi_{3,88} - \xi_{3,31}\xi_{3,89}\right) + \xi_{3,390}\left(\xi_{3,28}\xi_{3,89} + \xi_{3,29}\xi_{3,88} - \xi_{3,30}\xi_{3,87} + \xi_{3,31}\xi_{3,86}\right) - \xi_{3,391}\left(\xi_{3,28}\xi_{3,88} - \xi_{3,29}\xi_{3,89} + \xi_{3,30}\xi_{3,86} + \xi_{3,31}\xi_{3,87}\right)$$

$$\xi_{3,525} = \xi_{3,388}\left(\xi_{3,28}\xi_{3,88} - \xi_{3,29}\xi_{3,89} + \xi_{3,30}\xi_{3,86} + \xi_{3,31}\xi_{3,87}\right) - \xi_{3,389}\left(\xi_{3,28}\xi_{3,89} + \xi_{3,29}\xi_{3,88} - \xi_{3,30}\xi_{3,87} + \xi_{3,31}\xi_{3,86}\right) - \xi_{3,390}\left(\xi_{3,28}\xi_{3,86} - \xi_{3,29}\xi_{3,87} - \xi_{3,30}\xi_{3,88} - \xi_{3,31}\xi_{3,89}\right) + \xi_{3,391}\left(\xi_{3,28}\xi_{3,87} + \xi_{3,29}\xi_{3,86} + \xi_{3,30}\xi_{3,89} - \xi_{3,31}\xi_{3,88}\right)$$

$$\xi_{3,526} = \xi_{3,388}\left(\xi_{3,28}\xi_{3,89} + \xi_{3,29}\xi_{3,88} - \xi_{3,30}\xi_{3,87} + \xi_{3,31}\xi_{3,86}\right) + \xi_{3,389}\left(\xi_{3,28}\xi_{3,88} - \xi_{3,29}\xi_{3,89} + \xi_{3,30}\xi_{3,86} + \xi_{3,31}\xi_{3,87}\right) - \xi_{3,390}\left(\xi_{3,28}\xi_{3,87} + \xi_{3,29}\xi_{3,86} + \xi_{3,30}\xi_{3,89} - \xi_{3,31}\xi_{3,88}\right) - \xi_{3,391}\left(\xi_{3,28}\xi_{3,86} - \xi_{3,29}\xi_{3,87} - \xi_{3,30}\xi_{3,88} - \xi_{3,31}\xi_{3,89}\right)$$

$$\xi_{3,527} = \xi_{3,412}\left(\xi_{3,32}\xi_{3,90} - \xi_{3,33}\xi_{3,91} - \xi_{3,34}\xi_{3,92} - \xi_{3,35}\xi_{3,93}\right) + \xi_{3,413}\left(\xi_{3,32}\xi_{3,91} + \xi_{3,33}\xi_{3,90} + \xi_{3,34}\xi_{3,93} - \xi_{3,35}\xi_{3,92}\right) + \xi_{3,414}\left(\xi_{3,32}\xi_{3,92} - \xi_{3,33}\xi_{3,93} + \xi_{3,34}\xi_{3,90} + \xi_{3,35}\xi_{3,91}\right) + \xi_{3,415}\left(\xi_{3,32}\xi_{3,93} + \xi_{3,33}\xi_{3,92} - \xi_{3,34}\xi_{3,91} + \xi_{3,35}\xi_{3,90}\right)$$

$$\xi_{3,528} = \xi_{3,412}\left(\xi_{3,32}\xi_{3,91} + \xi_{3,33}\xi_{3,90} + \xi_{3,34}\xi_{3,93} - \xi_{3,35}\xi_{3,92}\right) - \xi_{3,413}\left(\xi_{3,32}\xi_{3,90} - \xi_{3,33}\xi_{3,91} - \xi_{3,34}\xi_{3,92} - \xi_{3,35}\xi_{3,93}\right) + \xi_{3,414}\left(\xi_{3,32}\xi_{3,93} + \xi_{3,33}\xi_{3,92} - \xi_{3,34}\xi_{3,91} + \xi_{3,35}\xi_{3,90}\right) - \xi_{3,415}\left(\xi_{3,32}\xi_{3,92} - \xi_{3,33}\xi_{3,93} + \xi_{3,34}\xi_{3,90} + \xi_{3,35}\xi_{3,91}\right)$$

$$\xi_{3,529} = \xi_{3,412}\left(\xi_{3,32}\xi_{3,92} - \xi_{3,33}\xi_{3,93} + \xi_{3,34}\xi_{3,90} + \xi_{3,35}\xi_{3,91}\right) - \xi_{3,413}\left(\xi_{3,32}\xi_{3,93} + \xi_{3,33}\xi_{3,92} - \xi_{3,34}\xi_{3,91} + \xi_{3,35}\xi_{3,90}\right) - \xi_{3,414}\left(\xi_{3,32}\xi_{3,90} - \xi_{3,33}\xi_{3,91} - \xi_{3,34}\xi_{3,92} - \xi_{3,35}\xi_{3,93}\right) + \xi_{3,415}\left(\xi_{3,32}\xi_{3,91} + \xi_{3,33}\xi_{3,90} + \xi_{3,34}\xi_{3,93} - \xi_{3,35}\xi_{3,92}\right)$$

$$\xi_{3,530} = \xi_{3,412}\left(\xi_{3,32}\xi_{3,93} + \xi_{3,33}\xi_{3,92} - \xi_{3,34}\xi_{3,91} + \xi_{3,35}\xi_{3,90}\right) + \xi_{3,413}\left(\xi_{3,32}\xi_{3,92} - \xi_{3,33}\xi_{3,93} + \xi_{3,34}\xi_{3,90} + \xi_{3,35}\xi_{3,91}\right) - \xi_{3,414}\left(\xi_{3,32}\xi_{3,91} + \xi_{3,33}\xi_{3,90} + \xi_{3,34}\xi_{3,93} - \xi_{3,35}\xi_{3,92}\right) - \xi_{3,415}\left(\xi_{3,32}\xi_{3,90} - \xi_{3,33}\xi_{3,91} - \xi_{3,34}\xi_{3,92} - \xi_{3,35}\xi_{3,93}\right)$$

$$\xi_{3,531} = \xi_{3,101} + \xi_{3,43}$$

$$\xi_{3,532} = \xi_{3,531} - \frac{\phi}{4}$$

$$\xi_{3,533} = \xi_{3,527}\cos\left(\frac{\phi}{4}\right) + \xi_{3,530}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{3,534} = \xi_{3,528}\cos\left(\frac{\phi}{4}\right) + \xi_{3,529}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{3,535} = -\xi_{3,528}\sin\left(\frac{\phi}{4}\right) + \xi_{3,529}\cos\left(\frac{\phi}{4}\right)$$

$$\xi_{3,536} = -\xi_{3,527}\sin\left(\frac{\phi}{4}\right) + \xi_{3,530}\cos\left(\frac{\phi}{4}\right)$$

$$\xi_{3,537} = \xi_{3,523}\cos\left(\frac{\phi}{4}\right) + \xi_{3,526}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{3,538} = \xi_{3,524}\cos\left(\frac{\phi}{4}\right) + \xi_{3,525}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{3,539} = -\xi_{3,524}\sin\left(\frac{\phi}{4}\right) + \xi_{3,525}\cos\left(\frac{\phi}{4}\right)$$

Exact shared-expression program 3

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,540} = -\xi_{3,523}\sin\left(\frac{\varphi}{4}\right) + \xi_{3,526}\cos\left(\frac{\varphi}{4}\right)$$

$$\xi_{3,541} = \begin{cases} 0 & \text{for } \xi_{3,507}^2 + \xi_{3,510}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,510}, \xi_{3,507}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,542} = \begin{cases} 0 & \text{for } \xi_{3,508}^2 + \xi_{3,509}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,508}, \xi_{3,509}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,543} = 2\operatorname{atan}_2\left(\sqrt{\xi_{3,508}^2 + \xi_{3,509}^2}, \sqrt{\xi_{3,507}^2 + \xi_{3,510}^2}\right)$$

$$\xi_{3,544} = -\xi_{3,541} + \xi_{3,542}$$

$$\xi_{3,545} = -\xi_{3,541} - \xi_{3,542}$$

$$\xi_{3,546} = \xi_{3,532} + \xi_{3,541}$$

$$\xi_{3,547} = \begin{cases} 0 & \text{for } \xi_{3,511}^2 + \xi_{3,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,514}, \xi_{3,511}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,548} = \begin{cases} 0 & \text{for } \xi_{3,512}^2 + \xi_{3,513}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,512}, \xi_{3,513}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,549} = 2\operatorname{atan}_2\left(\sqrt{\xi_{3,512}^2 + \xi_{3,513}^2}, \sqrt{\xi_{3,511}^2 + \xi_{3,514}^2}\right)$$

$$\xi_{3,550} = -\xi_{3,547} + \xi_{3,548}$$

$$\xi_{3,551} = -\xi_{3,547} - \xi_{3,548}$$

$$\xi_{3,552} = \xi_{3,546} + \xi_{3,547}$$

$$\xi_{3,553} = \begin{cases} 0 & \text{for } \xi_{3,515}^2 + \xi_{3,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,518}, \xi_{3,515}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,554} = \begin{cases} 0 & \text{for } \xi_{3,516}^2 + \xi_{3,517}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,516}, \xi_{3,517}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,555} = 2\operatorname{atan}_2\left(\sqrt{\xi_{3,516}^2 + \xi_{3,517}^2}, \sqrt{\xi_{3,515}^2 + \xi_{3,518}^2}\right)$$

$$\xi_{3,556} = -\xi_{3,553} + \xi_{3,554}$$

$$\xi_{3,557} = -\xi_{3,553} - \xi_{3,554}$$

$$\xi_{3,558} = \xi_{3,552} + \xi_{3,553}$$

$$\xi_{3,559} = \begin{cases} 0 & \text{for } \xi_{3,519}^2 + \xi_{3,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,522}, \xi_{3,519}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,560} = \begin{cases} 0 & \text{for } \xi_{3,520}^2 + \xi_{3,521}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,520}, \xi_{3,521}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,561} = 2\operatorname{atan}_2\left(\sqrt{\xi_{3,520}^2 + \xi_{3,521}^2}, \sqrt{\xi_{3,519}^2 + \xi_{3,522}^2}\right)$$

$$\xi_{3,562} = -\xi_{3,559} + \xi_{3,560}$$

$$\xi_{3,563} = -\xi_{3,559} - \xi_{3,560}$$

$$\xi_{3,564} = \xi_{3,558} + \xi_{3,559}$$

$$\xi_{3,565} = \begin{cases} 0 & \text{for } \xi_{3,112}^2 + \xi_{3,113}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{3,113}, \xi_{3,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,566} = \begin{cases} 0 & \text{for } \xi_{3,109}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{3,109}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,567} = 2\operatorname{atan}_2\left(\left|\xi_{3,109}\right|, \sqrt{\xi_{3,112}^2 + \xi_{3,113}^2}\right)$$

$$\xi_{3,568} = -\xi_{3,565} + \xi_{3,566}$$

$$\xi_{3,569} = -\xi_{3,565} - \xi_{3,566}$$

$$\xi_{3,570} = \xi_{3,564} + \xi_{3,565}$$

$$\xi_{3,571} = \begin{cases} 0 & \text{for } \xi_{3,107}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{3,107}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,572} = \begin{cases} 0 & \text{for } \xi_{3,108}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{3,108}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,573} = 2\operatorname{atan}_2\left(\left|\xi_{3,108}\right|, \left|\xi_{3,107}\right|\right)$$

$$\xi_{3,574} = -\xi_{3,571} + \xi_{3,572}$$

$$\xi_{3,575} = -\xi_{3,571} - \xi_{3,572}$$

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Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{3,576} = \xi_{3,570} + \xi_{3,571}$$

$$\xi_{3,577} = \begin{cases} 0 & \text{for } \xi_{3,537}^2 + \xi_{3,540}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,540}, \xi_{3,537}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,578} = \begin{cases} 0 & \text{for } \xi_{3,538}^2 + \xi_{3,539}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,538}, \xi_{3,539}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,579} = 2\operatorname{atan}_2\left(\sqrt{\xi_{3,538}^2 + \xi_{3,539}^2}, \sqrt{\xi_{3,537}^2 + \xi_{3,540}^2}\right)$$

$$\xi_{3,580} = -\xi_{3,577} + \xi_{3,578}$$

$$\xi_{3,581} = -\xi_{3,577} - \xi_{3,578}$$

$$\xi_{3,582} = \xi_{3,576} + \xi_{3,577}$$

$$\xi_{3,583} = \begin{cases} 0 & \text{for } \xi_{3,533}^2 + \xi_{3,536}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,536}, \xi_{3,533}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,584} = \begin{cases} 0 & \text{for } \xi_{3,534}^2 + \xi_{3,535}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{3,534}, \xi_{3,535}\right) & \text{otherwise} \end{cases}$$

$$\xi_{3,585} = 2\operatorname{atan}_2\left(\sqrt{\xi_{3,534}^2 + \xi_{3,535}^2}, \sqrt{\xi_{3,533}^2 + \xi_{3,536}^2}\right)$$

$$\xi_{3,586} = -\xi_{3,583} + \xi_{3,584}$$

$$\xi_{3,587} = -\xi_{3,583} - \xi_{3,584}$$

$$\xi_{3,588} = \xi_{3,582} + \xi_{3,583}$$

$$\xi_{3,589} = -\frac{\xi_{3,543}}{2} - \frac{\xi_{3,549}}{2} - \frac{\xi_{3,555}}{2} - \frac{\xi_{3,561}}{2} - \frac{\xi_{3,567}}{2} - \frac{\xi_{3,573}}{2} - \frac{\xi_{3,579}}{2} - \frac{\xi_{3,585}}{2} + \xi_{3,588}$$

Derived parameters for program 3

Exact output expressions used by the native gate labels.

$$v_{3,0} = \frac{2c_{3,543}+2c_{3,589}+\pi\left(-\frac{c_{3,545}}{w}+\frac{c_{3,544}+c_{3,545}}{w}+\frac{1}{2}\right)-\frac{3\pi}{2}}{\pi}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,0} = \left\lfloor -\frac{\theta}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{4,1} = \left\lfloor -\frac{\phi}{2\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{4,2} = -\frac{\pi \xi_{4,0}}{2} - \frac{\theta}{2}$$

$$\xi_{4,3} = -\frac{\pi \xi_{4,1}}{2} - \frac{\phi}{4}$$

$$\xi_{4,4} = \begin{cases} \xi_{4,2} & \text{for } |\xi_{4,2}| \geq |\xi_{4,3}| \\ \xi_{4,3} & \text{otherwise} \end{cases}$$

$$\xi_{4,5} = \xi_{4,2}$$

$$\xi_{4,6} = \begin{cases} \xi_{4,3} & \text{for } |\xi_{4,2}| \geq |\xi_{4,3}| \\ \xi_{4,2} & \text{otherwise} \end{cases}$$

$$\xi_{4,7} = \begin{cases} 1 & \text{for } |\xi_{4,2}| \geq |\xi_{4,3}| \\ \frac{1}{2} & \text{for } |\xi_{4,2}| \leq |\xi_{4,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{4,8} = \begin{cases} 0 & \text{for } |\xi_{4,2}| \geq |\xi_{4,3}| \\ \frac{1}{2} & \text{for } |\xi_{4,2}| \leq |\xi_{4,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,9} = \begin{cases} 0 & \text{for } |\xi_{4,2}| \geq |\xi_{4,3}| \\ \frac{1}{2} & \text{for } |\xi_{4,2}| \leq |\xi_{4,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{4,10} = \begin{cases} 0 & \text{for } |\xi_{4,2}| \geq |\xi_{4,3}| \\ -\frac{1}{2} & \text{for } |\xi_{4,2}| \leq |\xi_{4,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,11} = \begin{cases} -1 & \text{for } \xi_{4,4} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{4,12} = \begin{cases} -1 & \text{for } \xi_{4,5} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{4,13} = \begin{cases} 1 & \text{for } \xi_{4,11} = 1 \wedge \xi_{4,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,14} = \begin{cases} 0 & \text{for } \xi_{4,11} = 1 \wedge \xi_{4,12} = 1 \\ 1 & \text{for } \xi_{4,11} = 1 \wedge \xi_{4,12} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,15} = \begin{cases} 0 & \text{for } \xi_{4,11} = 1 \wedge (\xi_{4,11} = 1 \vee \xi_{4,12} = -1) \wedge (\xi_{4,11} = 1 \vee \xi_{4,12} = 1) \wedge (\xi_{4,12} = -1 \vee \xi_{4,12} = 1) \\ 1 & \text{for } \xi_{4,11} = -1 \wedge \xi_{4,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,16} = \begin{cases} 0 & \text{for } (\xi_{4,11} = -1 \vee \xi_{4,11} = 1) \wedge (\xi_{4,11} = 1 \vee \xi_{4,12} = 1) \wedge (\xi_{4,12} = -1 \vee \xi_{4,12} = 1) \wedge (\xi_{4,11} = -1 \vee \xi_{4,11} = 1 \vee \xi_{4,12} = -1) \wedge (\xi_{4,11} = -1 \vee \xi_{4,11} = 1 \vee \xi_{4,12} = 1) \wedge (\xi_{4,11} = -1 \vee \xi_{4,12} = -1 \vee \xi_{4,12} = 1) \wedge (\xi_{4,11} = 1 \vee \xi_{4,12} = -1 \vee \xi_{4,12} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{4,17} = \cos\left(\frac{\pi \xi_{4,0}}{2}\right)$$

$$\xi_{4,18} = -\sin\left(\frac{\pi \xi_{4,0}}{2}\right)$$

$$\xi_{4,19} = \xi_{4,17}^2$$

$$\xi_{4,20} = \xi_{4,17} \xi_{4,18}$$

$$\xi_{4,21} = -\xi_{4,18}^2$$

$$\xi_{4,22} = \cos\left(\frac{\pi \xi_{4,1}}{2}\right)$$

$$\xi_{4,23} = -\sin\left(\frac{\pi \xi_{4,1}}{2}\right)$$

$$\xi_{4,24} = \xi_{4,19} \xi_{4,22} - \xi_{4,21} \xi_{4,23}$$

$$\xi_{4,25} = \xi_{4,20} \xi_{4,22} - \xi_{4,20} \xi_{4,23}$$

$$\xi_{4,26} = \xi_{4,20} \xi_{4,22} + \xi_{4,20} \xi_{4,23}$$

$$\xi_{4,27} = \xi_{4,19} \xi_{4,23} + \xi_{4,21} \xi_{4,22}$$

$$\xi_{4,28} = \xi_{4,13} \left(-\xi_{4,10} \xi_{4,27} + \xi_{4,24} \xi_{4,7} - \xi_{4,25} \xi_{4,8} - \xi_{4,26} \xi_{4,9}\right) - \xi_{4,14} \left(\xi_{4,10} \xi_{4,26} + \xi_{4,24} \xi_{4,8} + \xi_{4,25} \xi_{4,7} - \xi_{4,27} \xi_{4,9}\right) - \xi_{4,15} \left(-\xi_{4,10} \xi_{4,25} + \xi_{4,24} \xi_{4,9} + \xi_{4,26} \xi_{4,7} + \xi_{4,27} \xi_{4,8}\right) - \xi_{4,16} \left(\xi_{4,10} \xi_{4,24} + \xi_{4,25} \xi_{4,9} - \xi_{4,26} \xi_{4,8} + \xi_{4,27} \xi_{4,7}\right)$$

$$\xi_{4,29} = \xi_{4,13} \left(\xi_{4,10} \xi_{4,26} + \xi_{4,24} \xi_{4,8} + \xi_{4,25} \xi_{4,7} - \xi_{4,27} \xi_{4,9}\right) + \xi_{4,14} \left(-\xi_{4,10} \xi_{4,27} + \xi_{4,24} \xi_{4,7} - \xi_{4,25} \xi_{4,8} - \xi_{4,26} \xi_{4,9}\right) - \xi_{4,15} \left(\xi_{4,10} \xi_{4,24} + \xi_{4,25} \xi_{4,9} - \xi_{4,26} \xi_{4,8} + \xi_{4,27} \xi_{4,7}\right) + \xi_{4,16} \left(-\xi_{4,10} \xi_{4,25} + \xi_{4,24} \xi_{4,9} + \xi_{4,26} \xi_{4,7} + \xi_{4,27} \xi_{4,8}\right)$$

$$\xi_{4,30} = \xi_{4,13} \left(-\xi_{4,10} \xi_{4,25} + \xi_{4,24} \xi_{4,9} + \xi_{4,26} \xi_{4,7} + \xi_{4,27} \xi_{4,8}\right) + \xi_{4,14} \left(\xi_{4,10} \xi_{4,24} + \xi_{4,25} \xi_{4,9} - \xi_{4,26} \xi_{4,8} + \xi_{4,27} \xi_{4,7}\right) + \xi_{4,15} \left(-\xi_{4,10} \xi_{4,27} + \xi_{4,24} \xi_{4,7} - \xi_{4,25} \xi_{4,8} - \xi_{4,26} \xi_{4,9}\right) - \xi_{4,16} \left(\xi_{4,10} \xi_{4,26} + \xi_{4,24} \xi_{4,8} + \xi_{4,25} \xi_{4,7} - \xi_{4,27} \xi_{4,9}\right)$$

$$\xi_{4,31} = \xi_{4,13} \left(\xi_{4,10} \xi_{4,24} + \xi_{4,25} \xi_{4,9} - \xi_{4,26} \xi_{4,8} + \xi_{4,27} \xi_{4,7}\right) - \xi_{4,14} \left(-\xi_{4,10} \xi_{4,25} + \xi_{4,24} \xi_{4,9} + \xi_{4,26} \xi_{4,7} + \xi_{4,27} \xi_{4,8}\right) + \xi_{4,15} \left(\xi_{4,10} \xi_{4,26} + \xi_{4,24} \xi_{4,8} + \xi_{4,25} \xi_{4,7} - \xi_{4,27} \xi_{4,9}\right) + \xi_{4,16} \left(-\xi_{4,10} \xi_{4,27} + \xi_{4,24} \xi_{4,7} - \xi_{4,25} \xi_{4,8} - \xi_{4,26} \xi_{4,9}\right)$$

$$\xi_{4,32} = -\xi_{4,10} \xi_{4,27} + \xi_{4,24} \xi_{4,7} - \xi_{4,25} \xi_{4,8} - \xi_{4,26} \xi_{4,9}$$

$$\xi_{4,33} = \xi_{4,10} \xi_{4,26} + \xi_{4,24} \xi_{4,8} + \xi_{4,25} \xi_{4,7} - \xi_{4,27} \xi_{4,9}$$

$$\xi_{4,34} = -\xi_{4,10} \xi_{4,25} + \xi_{4,24} \xi_{4,9} + \xi_{4,26} \xi_{4,7} + \xi_{4,27} \xi_{4,8}$$

$$\xi_{4,35} = \xi_{4,10} \xi_{4,24} + \xi_{4,25} \xi_{4,9} - \xi_{4,26} \xi_{4,8} + \xi_{4,27} \xi_{4,7}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,36} = -\xi_{4,10}\xi_{4,16} + \xi_{4,13}\xi_{4,7} - \xi_{4,14}\xi_{4,8} - \xi_{4,15}\xi_{4,9}$$

$$\xi_{4,37} = \xi_{4,10}\xi_{4,15} - \xi_{4,13}\xi_{4,8} - \xi_{4,14}\xi_{4,7} - \xi_{4,16}\xi_{4,9}$$

$$\xi_{4,38} = -\xi_{4,10}\xi_{4,14} - \xi_{4,13}\xi_{4,9} - \xi_{4,15}\xi_{4,7} + \xi_{4,16}\xi_{4,8}$$

$$\xi_{4,39} = -\xi_{4,10}\xi_{4,13} + \xi_{4,14}\xi_{4,9} - \xi_{4,15}\xi_{4,8} - \xi_{4,16}\xi_{4,7}$$

$$\xi_{4,40} = \xi_{4,11}\xi_{4,4}$$

$$\xi_{4,41} = \xi_{4,12}\xi_{4,5}$$

$$\xi_{4,42} = \xi_{4,11}\xi_{4,12}\xi_{4,6}$$

$$\xi_{4,43} = -\frac{\pi(2\xi_{4,0}+\xi_{4,1})}{2}$$

$$\xi_{4,44} = \begin{cases} 0 & \text{for } \left(\xi_{4,41} > \frac{\pi}{8} \vee \xi_{4,42} < 0\right) \wedge \xi_{4,41} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{4,45} = \frac{\pi}{8}$$

$$\xi_{4,46} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{4,41} > \frac{\pi}{8} \wedge \xi_{4,42} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{4,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,47} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{4,41} > \frac{\pi}{8} \wedge \xi_{4,42} < 0 \\ \frac{1}{2} & \text{for } \xi_{4,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,48} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{4,41} > \frac{\pi}{8} \vee \xi_{4,42} < 0\right) \wedge \xi_{4,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,49} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{4,41} > \frac{\pi}{8} \vee \xi_{4,42} < 0\right) \wedge \xi_{4,41} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{4,50} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{4,41} > \frac{\pi}{8} \vee \xi_{4,42} < 0\right) \wedge \xi_{4,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,51} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{4,41} > \frac{\pi}{8} \vee \xi_{4,42} < 0\right) \wedge \xi_{4,41} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{4,52} = \begin{cases} \frac{1}{2} & \text{for } \xi_{4,41} > \frac{\pi}{8} \wedge \xi_{4,42} < 0 \\ -\frac{1}{2} & \text{for } \xi_{4,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,53} = \left\lfloor \frac{2\xi_{4,40} - 2\xi_{4,44}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{4,54} = \left\lfloor \frac{2\xi_{4,41} - 2\xi_{4,45}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{4,55} = \left\lfloor \frac{2\xi_{4,42} - 2\xi_{4,46}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{4,56} = \xi_{4,40} - \xi_{4,44} - \frac{\pi\xi_{4,53}}{2}$$

$$\xi_{4,57} = \xi_{4,41} - \xi_{4,45} - \frac{\pi\xi_{4,54}}{2}$$

$$\xi_{4,58} = \xi_{4,42} - \xi_{4,46} - \frac{\pi\xi_{4,55}}{2}$$

$$\xi_{4,59} = \begin{cases} \xi_{4,56} & \text{for } (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,58}| \vee |\xi_{4,57}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,57}| \geq |\xi_{4,58}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \\ \xi_{4,57} & \text{for } (|\xi_{4,56}| \geq |\xi_{4,58}| \vee |\xi_{4,57}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,58}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,57}| \geq |\xi_{4,58}| \vee |\xi_{4,56}| \leq |\xi_{4,57}|) \wedge (|\xi_{4,56}| \leq |\xi_{4,57}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \\ \xi_{4,58} & \text{otherwise} \end{cases}$$

$$\xi_{4,60} = \begin{cases} \xi_{4,57} & \text{for } |\xi_{4,56}| \geq |\xi_{4,57}| \wedge |\xi_{4,57}| \geq |\xi_{4,58}| \\ \xi_{4,58} & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge |\xi_{4,57}| \leq |\xi_{4,58}| \\ \xi_{4,56} & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge |\xi_{4,56}| \leq |\xi_{4,57}| \\ \xi_{4,58} & \text{for } |\xi_{4,57}| \geq |\xi_{4,58}| \wedge |\xi_{4,56}| \leq |\xi_{4,58}| \\ \xi_{4,56} & \text{for } |\xi_{4,56}| \geq |\xi_{4,57}| \wedge |\xi_{4,56}| \leq |\xi_{4,58}| \\ \xi_{4,57} & \text{otherwise} \end{cases}$$

$$\xi_{4,61} = \begin{cases} \xi_{4,58} & \text{for } |\xi_{4,56}| \geq |\xi_{4,57}| \wedge |\xi_{4,57}| \geq |\xi_{4,58}| \\ \xi_{4,57} & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge |\xi_{4,57}| \leq |\xi_{4,58}| \\ \xi_{4,58} & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge |\xi_{4,56}| \leq |\xi_{4,57}| \\ \xi_{4,56} & \text{for } |\xi_{4,57}| \geq |\xi_{4,58}| \wedge |\xi_{4,56}| \leq |\xi_{4,58}| \\ \xi_{4,57} & \text{for } |\xi_{4,56}| \geq |\xi_{4,57}| \wedge |\xi_{4,56}| \leq |\xi_{4,58}| \\ \xi_{4,56} & \text{otherwise} \end{cases}$$

$$\xi_{4,62} = \begin{cases} 1 & \text{for } |\xi_{4,56}| \geq |\xi_{4,57}| \wedge |\xi_{4,57}| \geq |\xi_{4,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge |\xi_{4,57}| \leq |\xi_{4,58}| \\ 0 & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge |\xi_{4,56}| \leq |\xi_{4,57}| \\ \frac{1}{2} & \text{for } |\xi_{4,56}| \leq |\xi_{4,58}| \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,57}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,57}| \geq |\xi_{4,58}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{4,63} = \begin{cases} 0 & \text{for } |\xi_{4,56}| \geq |\xi_{4,57}| \wedge |\xi_{4,57}| \geq |\xi_{4,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge (|\xi_{4,56}| \geq |\xi_{4,58}| \vee |\xi_{4,56}| \leq |\xi_{4,57}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,58}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \leq |\xi_{4,57}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \\ \frac{1}{2} & \text{for } |\xi_{4,56}| \leq |\xi_{4,58}| \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,57}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,57}| \geq |\xi_{4,58}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,64} = \begin{cases} 0 & \text{for } (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,58}| \vee |\xi_{4,57}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,57}| \geq |\xi_{4,58}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{4,56}| \geq |\xi_{4,58}| \wedge |\xi_{4,56}| \leq |\xi_{4,57}| \\ \frac{1}{2} & \text{for } |\xi_{4,56}| \leq |\xi_{4,58}| \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,57}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,57}| \geq |\xi_{4,58}| \vee |\xi_{4,56}| \leq |\xi_{4,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{4,65} = \begin{cases} 0 & \text{for } (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,56}| \vee |\xi_{4,57}| \geq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \geq |\xi_{4,56}| \vee |\xi_{4,56}| \leq |\xi_{4,57}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,57}| \vee |\xi_{4,56}| \geq |\xi_{4,56}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,56}| \vee |\xi_{4,57}| \geq |\xi_{4,56}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,56}| \vee |\xi_{4,56}| \leq |\xi_{4,57}| \vee |\xi_{4,57}| \leq |\xi_{4,58}|) \wedge (|\xi_{4,56}| \geq |\xi_{4,56}| \vee |\xi_{4,56}| \leq |\xi_{4,57}| \vee |\xi_{4,58}| \leq |\xi_{4,58}|) \\ \frac{1}{2} & \text{for } |\xi_{4,57}| \geq |\xi_{4,58}| \wedge |\xi_{4,56}| \leq |\xi_{4,58}| \\ -\frac{1}{2} & \text{for } |\xi_{4,56}| \geq |\xi_{4,57}| \wedge |\xi_{4,56}| \leq |\xi_{4,58}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,66} = \begin{cases} -1 & \text{for } \xi_{4,59} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{4,67} = \begin{cases} -1 & \text{for } \xi_{4,60} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{4,68} = \begin{cases} 1 & \text{for } \xi_{4,66} = 1 \wedge \xi_{4,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,69} = \begin{cases} 0 & \text{for } \xi_{4,66} = 1 \wedge \xi_{4,67} = 1 \\ 1 & \text{for } \xi_{4,66} = 1 \wedge \xi_{4,67} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,70} = \begin{cases} 0 & \text{for } \xi_{4,66} = 1 \wedge (\xi_{4,66} = 1 \vee \xi_{4,67} = -1) \wedge (\xi_{4,66} = 1 \vee \xi_{4,67} = 1) \wedge (\xi_{4,67} = -1 \vee \xi_{4,67} = 1) \\ 1 & \text{for } \xi_{4,66} = -1 \wedge \xi_{4,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,71} = \begin{cases} 0 & \text{for } (\xi_{4,66} = -1 \vee \xi_{4,66} = 1) \wedge (\xi_{4,66} = 1 \vee \xi_{4,67} = 1) \wedge (\xi_{4,67} = -1 \vee \xi_{4,67} = 1) \wedge (\xi_{4,66} = -1 \vee \xi_{4,66} = 1 \vee \xi_{4,67} = -1) \wedge (\xi_{4,66} = -1 \vee \xi_{4,66} = 1 \vee \xi_{4,67} = 1) \wedge (\xi_{4,66} = -1 \vee \xi_{4,67} = -1 \vee \xi_{4,67} = 1) \wedge (\xi_{4,66} = 1 \vee \xi_{4,67} = -1 \vee \xi_{4,67} = 1) \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,72} = \cos\left(\frac{\pi\xi_{4,53}}{2}\right)$$

$$\xi_{4,73} = -\sin\left(\frac{\pi\xi_{4,53}}{2}\right)$$

$$\xi_{4,74} = \cos\left(\frac{\pi\xi_{4,54}}{2}\right)$$

$$\xi_{4,75} = -\sin\left(\frac{\pi\xi_{4,54}}{2}\right)$$

$$\xi_{4,76} = \xi_{4,72}\xi_{4,74}$$

$$\xi_{4,77} = \xi_{4,73}\xi_{4,74}$$

$$\xi_{4,78} = \xi_{4,72}\xi_{4,75}$$

$$\xi_{4,79} = -\xi_{4,73}\xi_{4,75}$$

$$\xi_{4,80} = \cos\left(\frac{\pi\xi_{4,55}}{2}\right)$$

$$\xi_{4,81} = -\sin\left(\frac{\pi\xi_{4,55}}{2}\right)$$

$$\xi_{4,82} = \xi_{4,76}\xi_{4,80} - \xi_{4,79}\xi_{4,81}$$

$$\xi_{4,83} = \xi_{4,77}\xi_{4,80} - \xi_{4,78}\xi_{4,81}$$

$$\xi_{4,84} = \xi_{4,77}\xi_{4,81} + \xi_{4,78}\xi_{4,80}$$

$$\xi_{4,85} = \xi_{4,76}\xi_{4,81} + \xi_{4,79}\xi_{4,80}$$

$$\xi_{4,86} = \xi_{4,68}\left(\xi_{4,62}\xi_{4,82} - \xi_{4,63}\xi_{4,83} - \xi_{4,64}\xi_{4,84} - \xi_{4,65}\xi_{4,85}\right) - \xi_{4,69}\left(\xi_{4,62}\xi_{4,83} + \xi_{4,63}\xi_{4,82} - \xi_{4,64}\xi_{4,85} + \xi_{4,65}\xi_{4,84}\right) - \xi_{4,70}\left(\xi_{4,62}\xi_{4,84} + \xi_{4,63}\xi_{4,85} + \xi_{4,64}\xi_{4,82} - \xi_{4,65}\xi_{4,83}\right) - \xi_{4,71}\left(\xi_{4,62}\xi_{4,85} - \xi_{4,63}\xi_{4,84} + \xi_{4,64}\xi_{4,83} + \xi_{4,65}\xi_{4,82}\right)$$

$$\xi_{4,87} = \xi_{4,68}\left(\xi_{4,62}\xi_{4,83} + \xi_{4,63}\xi_{4,82} - \xi_{4,64}\xi_{4,85} + \xi_{4,65}\xi_{4,84}\right) + \xi_{4,69}\left(\xi_{4,62}\xi_{4,82} - \xi_{4,63}\xi_{4,83} - \xi_{4,64}\xi_{4,84} - \xi_{4,65}\xi_{4,85}\right) - \xi_{4,70}\left(\xi_{4,62}\xi_{4,85} - \xi_{4,63}\xi_{4,84} + \xi_{4,64}\xi_{4,83} + \xi_{4,65}\xi_{4,82}\right) + \xi_{4,71}\left(\xi_{4,62}\xi_{4,84} + \xi_{4,63}\xi_{4,85} + \xi_{4,64}\xi_{4,82} - \xi_{4,65}\xi_{4,83}\right)$$

$$\xi_{4,88} = \xi_{4,68}\left(\xi_{4,62}\xi_{4,84} + \xi_{4,63}\xi_{4,85} + \xi_{4,64}\xi_{4,82} - \xi_{4,65}\xi_{4,83}\right) + \xi_{4,69}\left(\xi_{4,62}\xi_{4,85} - \xi_{4,63}\xi_{4,84} + \xi_{4,64}\xi_{4,83} + \xi_{4,65}\xi_{4,82}\right) + \xi_{4,70}\left(\xi_{4,62}\xi_{4,82} - \xi_{4,63}\xi_{4,83} - \xi_{4,64}\xi_{4,84} - \xi_{4,65}\xi_{4,85}\right) - \xi_{4,71}\left(\xi_{4,62}\xi_{4,83} + \xi_{4,63}\xi_{4,82} - \xi_{4,64}\xi_{4,85} + \xi_{4,65}\xi_{4,84}\right)$$

$$\xi_{4,89} = \xi_{4,68}\left(\xi_{4,62}\xi_{4,85} - \xi_{4,63}\xi_{4,84} + \xi_{4,64}\xi_{4,83} + \xi_{4,65}\xi_{4,82}\right) - \xi_{4,69}\left(\xi_{4,62}\xi_{4,84} + \xi_{4,63}\xi_{4,85} + \xi_{4,64}\xi_{4,82} - \xi_{4,65}\xi_{4,83}\right) + \xi_{4,70}\left(\xi_{4,62}\xi_{4,83} + \xi_{4,63}\xi_{4,82} - \xi_{4,64}\xi_{4,85} + \xi_{4,65}\xi_{4,84}\right) + \xi_{4,71}\left(\xi_{4,62}\xi_{4,82} - \xi_{4,63}\xi_{4,83} - \xi_{4,64}\xi_{4,84} - \xi_{4,65}\xi_{4,85}\right)$$

$$\xi_{4,90} = \xi_{4,62}\xi_{4,82} - \xi_{4,63}\xi_{4,83} - \xi_{4,64}\xi_{4,84} - \xi_{4,65}\xi_{4,85}$$

$$\xi_{4,91} = \xi_{4,62}\xi_{4,83} + \xi_{4,63}\xi_{4,82} - \xi_{4,64}\xi_{4,85} + \xi_{4,65}\xi_{4,84}$$

$$\xi_{4,92} = \xi_{4,62}\xi_{4,84} + \xi_{4,63}\xi_{4,85} + \xi_{4,64}\xi_{4,82} - \xi_{4,65}\xi_{4,83}$$

$$\xi_{4,93} = \xi_{4,62}\xi_{4,85} - \xi_{4,63}\xi_{4,84} + \xi_{4,64}\xi_{4,83} + \xi_{4,65}\xi_{4,82}$$

$$\xi_{4,94} = \xi_{4,62}\xi_{4,68} - \xi_{4,63}\xi_{4,69} - \xi_{4,64}\xi_{4,70} - \xi_{4,65}\xi_{4,71}$$

$$\xi_{4,95} = -\xi_{4,62}\xi_{4,69} - \xi_{4,63}\xi_{4,68} - \xi_{4,64}\xi_{4,71} + \xi_{4,65}\xi_{4,70}$$

$$\xi_{4,96} = -\xi_{4,62}\xi_{4,70} + \xi_{4,63}\xi_{4,71} - \xi_{4,64}\xi_{4,68} - \xi_{4,65}\xi_{4,69}$$

$$\xi_{4,97} = -\xi_{4,62}\xi_{4,71} - \xi_{4,63}\xi_{4,70} + \xi_{4,64}\xi_{4,69} - \xi_{4,65}\xi_{4,68}$$

$$\xi_{4,98} = \xi_{4,59}\xi_{4,66}$$

$$\xi_{4,99} = \xi_{4,60}\xi_{4,67}$$

$$\xi_{4,100} = \xi_{4,61}\xi_{4,66}\xi_{4,67}$$

$$\xi_{4,101} = -\frac{\pi(\xi_{4,53}+\xi_{4,54}+\xi_{4,55})}{2}$$

$$\xi_{4,102} = \cos\left(2\xi_{4,98}\right)$$

$$\xi_{4,103} = \cos\left(2\xi_{4,99}\right)$$

$$\xi_{4,104} = \cos\left(2\xi_{4,100}\right)$$

$$\xi_{4,105} = \sqrt{\xi_{4,102}^2 - 2\xi_{4,102}\xi_{4,103}\xi_{4,104} + \xi_{4,103}^2 + \xi_{4,104}^2 - 1}$$

$$\xi_{4,106} = \xi_{4,102} - \xi_{4,103} + \xi_{4,104}$$

$$\xi_{4,107} = \sqrt{-\frac{\xi_{4,105}}{2} + \frac{\xi_{4,106}}{2} + \frac{1}{2}}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,108} = \sqrt{\frac{\xi_{4,105}}{2} - \frac{\xi_{4,106}}{2} + \frac{1}{2}}$$

$$\xi_{4,109} = \sqrt{-\frac{\xi_{4,105}}{2} - \frac{\xi_{4,106}}{2} + \frac{1}{2}}$$

$$\xi_{4,110} = \frac{(1-\xi_{4,103})(\xi_{4,102}+1)(\xi_{4,104}+1)}{2}$$

$$\xi_{4,111} = \begin{cases} -1 & \text{for } \xi_{4,100} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{4,112} = \begin{cases} 0 & \text{for } \xi_{4,107} = 0 \\ \frac{\sqrt{\xi_{4,110}\xi_{4,111}}}{\xi_{4,107}} & \text{otherwise} \end{cases}$$

$$\xi_{4,113} = \begin{cases} 1 & \text{for } \xi_{4,107} = 0 \\ \frac{\sqrt{\xi_{4,102}\xi_{4,103}\xi_{4,104}}}{\xi_{4,107}} & \text{otherwise} \end{cases}$$

$$\xi_{4,114} = 2\xi_{4,107}\xi_{4,108}\xi_{4,109}\xi_{4,113}$$

$$\xi_{4,115} = \sqrt{2}\xi_{4,113}\left(\xi_{4,107}^2\xi_{4,109} - \xi_{4,109}\right)$$

$$\xi_{4,116} = 2\xi_{4,107}\xi_{4,108}\xi_{4,109}\xi_{4,112}$$

$$\xi_{4,117} = \sqrt{2}\left(2\xi_{4,107}^2\xi_{4,109}\xi_{4,112} - \xi_{4,109}\xi_{4,112}\right)$$

$$\xi_{4,118} = \sqrt{2}\xi_{4,107}\xi_{4,108}\xi_{4,112}\xi_{4,113}$$

$$\xi_{4,119} = \sqrt{2}\xi_{4,107}\xi_{4,108}\left(-\xi_{4,109}^2 + \xi_{4,112}^2\right)$$

$$\xi_{4,120} = -\xi_{4,107}^2\xi_{4,109}^2 + \xi_{4,107}^2\xi_{4,112}^2 + \xi_{4,109}^2$$

$$\xi_{4,121} = \xi_{4,107}^2\xi_{4,109}^2 - \xi_{4,107}^2\xi_{4,112}^2 + \xi_{4,112}^2$$

$$\xi_{4,122} = 2\xi_{4,113}\left(\xi_{4,107}^2\xi_{4,112} - \frac{\xi_{4,112}}{2}\right)$$

$$\xi_{4,123} = -\sqrt{2}\xi_{4,107}\xi_{4,108}\xi_{4,112}\xi_{4,113}$$

$$\xi_{4,124} = -\xi_{4,112}\xi_{4,113}$$

$$\xi_{4,125} = \xi_{4,107}^2\xi_{4,109}^2 + \xi_{4,107}^2\xi_{4,112}^2 - \xi_{4,107}^2 - \xi_{4,109}^2 - \xi_{4,112}^2 + 1$$

$$\xi_{4,126} = -\sqrt{2}\left(\xi_{4,107}\xi_{4,108}\xi_{4,109}^2 + \xi_{4,107}\xi_{4,108}\xi_{4,112}^2 - \xi_{4,107}\xi_{4,108}\right)$$

$$\xi_{4,127} = \sqrt{2}\xi_{4,107}^2\xi_{4,109}\xi_{4,113}$$

$$\xi_{4,128} = \xi_{4,118}^2 - \xi_{4,121}^2 - \xi_{4,124}^2 - \xi_{4,127}^2$$

$$\xi_{4,129} = \xi_{4,118}\xi_{4,119} - \xi_{4,121}\xi_{4,122} - \xi_{4,124}\xi_{4,125}$$

$$\xi_{4,130} = \xi_{4,118}\xi_{4,120} - \xi_{4,121}\xi_{4,123} - \xi_{4,124}\xi_{4,126}$$

$$\xi_{4,131} = \xi_{4,114}^2 + \xi_{4,116}^2 + \xi_{4,119}^2 - \xi_{4,122}^2 - \xi_{4,125}^2$$

$$\xi_{4,132} = \xi_{4,114}\xi_{4,115} + \xi_{4,116}\xi_{4,117} + \xi_{4,119}\xi_{4,120} - \xi_{4,122}\xi_{4,123} - \xi_{4,125}\xi_{4,126}$$

$$\xi_{4,133} = \xi_{4,115}^2 + \xi_{4,117}^2 + \xi_{4,120}^2 - \xi_{4,123}^2 - \xi_{4,126}^2$$

$$\xi_{4,134} = \sin\left(2\xi_{4,100}\right)\sin\left(2\xi_{4,99}\right)\cos\left(2\xi_{4,98}\right)$$

$$\xi_{4,135} = \sin\left(2\xi_{4,100}\right)\sin\left(2\xi_{4,98}\right)\cos\left(2\xi_{4,99}\right)$$

$$\xi_{4,136} = \sin\left(2\xi_{4,98}\right)\sin\left(2\xi_{4,99}\right)\cos\left(2\xi_{4,100}\right)$$

$$\xi_{4,137} = \sin\left(2\xi_{4,98}\right)\cos\left(2\xi_{4,100}\right)\cos\left(2\xi_{4,99}\right)$$

$$\xi_{4,138} = \sin\left(2\xi_{4,99}\right)\cos\left(2\xi_{4,100}\right)\cos\left(2\xi_{4,98}\right)$$

$$\xi_{4,139} = \sin\left(2\xi_{4,100}\right)\cos\left(2\xi_{4,98}\right)\cos\left(2\xi_{4,99}\right)$$

$$\xi_{4,140} = \xi_{4,134}^2 - \xi_{4,137}^2$$

$$\xi_{4,141} = \xi_{4,135}^2 - \xi_{4,138}^2$$

$$\xi_{4,142} = \xi_{4,136}^2 - \xi_{4,139}^2$$

$$\xi_{4,143} = \frac{\xi_{4,128} - \xi_{4,141}}{\xi_{4,140} - \xi_{4,141}}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,144} = \frac{\xi_{4,129}}{\xi_{4,140}-\xi_{4,141}}$$

$$\xi_{4,145} = \frac{\xi_{4,130}}{\xi_{4,140}-\xi_{4,141}}$$

$$\xi_{4,146} = \frac{\xi_{4,131}-\xi_{4,141}}{\xi_{4,140}-\xi_{4,141}}$$

$$\xi_{4,147} = \frac{\xi_{4,132}}{\xi_{4,140}-\xi_{4,141}}$$

$$\xi_{4,148} = \frac{\xi_{4,133}-\xi_{4,141}}{\xi_{4,140}-\xi_{4,141}}$$

$$\xi_{4,149} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,143} & \text{otherwise} \end{cases}$$

$$\xi_{4,150} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,144} & \text{otherwise} \end{cases}$$

$$\xi_{4,151} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,145} & \text{otherwise} \end{cases}$$

$$\xi_{4,152} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,146} & \text{otherwise} \end{cases}$$

$$\xi_{4,153} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,147} & \text{otherwise} \end{cases}$$

$$\xi_{4,154} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,148} & \text{otherwise} \end{cases}$$

$$\xi_{4,155} = \frac{\xi_{4,128}-\xi_{4,142}}{\xi_{4,140}-\xi_{4,142}}$$

$$\xi_{4,156} = \frac{\xi_{4,129}}{\xi_{4,140}-\xi_{4,142}}$$

$$\xi_{4,157} = \frac{\xi_{4,130}}{\xi_{4,140}-\xi_{4,142}}$$

$$\xi_{4,158} = \frac{\xi_{4,131}-\xi_{4,142}}{\xi_{4,140}-\xi_{4,142}}$$

$$\xi_{4,159} = \frac{\xi_{4,132}}{\xi_{4,140}-\xi_{4,142}}$$

$$\xi_{4,160} = \frac{\xi_{4,133}-\xi_{4,142}}{\xi_{4,140}-\xi_{4,142}}$$

$$\xi_{4,161} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,142} \\ \xi_{4,155} & \text{otherwise} \end{cases}$$

$$\xi_{4,162} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,142} \\ \xi_{4,156} & \text{otherwise} \end{cases}$$

$$\xi_{4,163} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,142} \\ \xi_{4,157} & \text{otherwise} \end{cases}$$

$$\xi_{4,164} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,142} \\ \xi_{4,158} & \text{otherwise} \end{cases}$$

$$\xi_{4,165} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,142} \\ \xi_{4,159} & \text{otherwise} \end{cases}$$

$$\xi_{4,166} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,142} \\ \xi_{4,160} & \text{otherwise} \end{cases}$$

$$\xi_{4,167} = \xi_{4,149}\xi_{4,161} + \xi_{4,150}\xi_{4,162} + \xi_{4,151}\xi_{4,163}$$

$$\xi_{4,168} = \xi_{4,149}\xi_{4,162} + \xi_{4,150}\xi_{4,164} + \xi_{4,151}\xi_{4,165}$$

$$\xi_{4,169} = \xi_{4,149}\xi_{4,163} + \xi_{4,150}\xi_{4,165} + \xi_{4,151}\xi_{4,166}$$

$$\xi_{4,170} = \xi_{4,150}\xi_{4,161} + \xi_{4,152}\xi_{4,162} + \xi_{4,153}\xi_{4,163}$$

$$\xi_{4,171} = \xi_{4,150}\xi_{4,162} + \xi_{4,152}\xi_{4,164} + \xi_{4,153}\xi_{4,165}$$

$$\xi_{4,172} = \xi_{4,150}\xi_{4,163} + \xi_{4,152}\xi_{4,165} + \xi_{4,153}\xi_{4,166}$$

$$\xi_{4,173} = \xi_{4,151}\xi_{4,161} + \xi_{4,153}\xi_{4,162} + \xi_{4,154}\xi_{4,163}$$

$$\xi_{4,174} = \xi_{4,151}\xi_{4,162} + \xi_{4,153}\xi_{4,164} + \xi_{4,154}\xi_{4,165}$$

$$\xi_{4,175} = \xi_{4,151}\xi_{4,163} + \xi_{4,153}\xi_{4,165} + \xi_{4,154}\xi_{4,166}$$

$$\xi_{4,176} = \sqrt{\xi_{4,167}}$$

$$\xi_{4,177} = \frac{\xi_{4,167}}{\xi_{4,176}}$$

$$\xi_{4,178} = \frac{\xi_{4,170}}{\xi_{4,176}}$$

$$\xi_{4,179} = \frac{\xi_{4,173}}{\xi_{4,176}}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,180} = \sqrt{\xi_{4,171}}$$

$$\xi_{4,181} = \frac{\xi_{4,168}}{\xi_{4,180}}$$

$$\xi_{4,182} = \frac{\xi_{4,171}}{\xi_{4,180}}$$

$$\xi_{4,183} = \frac{\xi_{4,174}}{\xi_{4,180}}$$

$$\xi_{4,184} = \sqrt{\xi_{4,175}}$$

$$\xi_{4,185} = \frac{\xi_{4,169}}{\xi_{4,184}}$$

$$\xi_{4,186} = \frac{\xi_{4,172}}{\xi_{4,184}}$$

$$\xi_{4,187} = \frac{\xi_{4,175}}{\xi_{4,184}}$$

$$\xi_{4,188} = \begin{cases} \xi_{4,177} & \text{for } \xi_{4,167} \geq \xi_{4,171} \wedge \xi_{4,167} \geq \xi_{4,175} \\ \xi_{4,181} & \text{for } \xi_{4,171} \geq \xi_{4,175} \wedge \xi_{4,167} \leq \xi_{4,171} \\ \xi_{4,185} & \text{otherwise} \end{cases}$$

$$\xi_{4,189} = \begin{cases} \xi_{4,178} & \text{for } \xi_{4,167} \geq \xi_{4,171} \wedge \xi_{4,167} \geq \xi_{4,175} \\ \xi_{4,182} & \text{for } \xi_{4,171} \geq \xi_{4,175} \wedge \xi_{4,167} \leq \xi_{4,171} \\ \xi_{4,186} & \text{otherwise} \end{cases}$$

$$\xi_{4,190} = \begin{cases} \xi_{4,179} & \text{for } \xi_{4,167} \geq \xi_{4,171} \wedge \xi_{4,167} \geq \xi_{4,175} \\ \xi_{4,183} & \text{for } \xi_{4,171} \geq \xi_{4,175} \wedge \xi_{4,167} \leq \xi_{4,171} \\ \xi_{4,187} & \text{otherwise} \end{cases}$$

$$\xi_{4,191} = \frac{\xi_{4,128} - \xi_{4,140}}{-\xi_{4,140} + \xi_{4,141}}$$

$$\xi_{4,192} = \frac{\xi_{4,129}}{-\xi_{4,140} + \xi_{4,141}}$$

$$\xi_{4,193} = \frac{\xi_{4,130}}{-\xi_{4,140} + \xi_{4,141}}$$

$$\xi_{4,194} = \frac{\xi_{4,131} - \xi_{4,140}}{-\xi_{4,140} + \xi_{4,141}}$$

$$\xi_{4,195} = \frac{\xi_{4,132}}{-\xi_{4,140} + \xi_{4,141}}$$

$$\xi_{4,196} = \frac{\xi_{4,133} - \xi_{4,140}}{-\xi_{4,140} + \xi_{4,141}}$$

$$\xi_{4,197} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,191} & \text{otherwise} \end{cases}$$

$$\xi_{4,198} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,192} & \text{otherwise} \end{cases}$$

$$\xi_{4,199} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,193} & \text{otherwise} \end{cases}$$

$$\xi_{4,200} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,194} & \text{otherwise} \end{cases}$$

$$\xi_{4,201} = \begin{cases} 0 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,195} & \text{otherwise} \end{cases}$$

$$\xi_{4,202} = \begin{cases} 1 & \text{for } \xi_{4,140} = \xi_{4,141} \\ \xi_{4,196} & \text{otherwise} \end{cases}$$

$$\xi_{4,203} = \frac{\xi_{4,128} - \xi_{4,142}}{\xi_{4,141} - \xi_{4,142}}$$

$$\xi_{4,204} = \frac{\xi_{4,129}}{\xi_{4,141} - \xi_{4,142}}$$

$$\xi_{4,205} = \frac{\xi_{4,130}}{\xi_{4,141} - \xi_{4,142}}$$

$$\xi_{4,206} = \frac{\xi_{4,131} - \xi_{4,142}}{\xi_{4,141} - \xi_{4,142}}$$

$$\xi_{4,207} = \frac{\xi_{4,132}}{\xi_{4,141} - \xi_{4,142}}$$

$$\xi_{4,208} = \frac{\xi_{4,133} - \xi_{4,142}}{\xi_{4,141} - \xi_{4,142}}$$

$$\xi_{4,209} = \begin{cases} 1 & \text{for } \xi_{4,141} = \xi_{4,142} \\ \xi_{4,203} & \text{otherwise} \end{cases}$$

$$\xi_{4,210} = \begin{cases} 0 & \text{for } \xi_{4,141} = \xi_{4,142} \\ \xi_{4,204} & \text{otherwise} \end{cases}$$

$$\xi_{4,211} = \begin{cases} 0 & \text{for } \xi_{4,141} = \xi_{4,142} \\ \xi_{4,205} & \text{otherwise} \end{cases}$$

$$\xi_{4,212} = \begin{cases} 1 & \text{for } \xi_{4,141} = \xi_{4,142} \\ \xi_{4,206} & \text{otherwise} \end{cases}$$

$$\xi_{4,213} = \begin{cases} 0 & \text{for } \xi_{4,141} = \xi_{4,142} \\ \xi_{4,207} & \text{otherwise} \end{cases}$$

$$\xi_{4,214} = \begin{cases} 1 & \text{for } \xi_{4,141} = \xi_{4,142} \\ \xi_{4,208} & \text{otherwise} \end{cases}$$

$$\xi_{4,215} = \xi_{4,197}\xi_{4,209} + \xi_{4,198}\xi_{4,210} + \xi_{4,199}\xi_{4,211}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

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$$\xi_{4,216} = \xi_{4,197}\xi_{4,210} + \xi_{4,198}\xi_{4,212} + \xi_{4,199}\xi_{4,213}$$

$$\xi_{4,217} = \xi_{4,197}\xi_{4,211} + \xi_{4,198}\xi_{4,213} + \xi_{4,199}\xi_{4,214}$$

$$\xi_{4,218} = \xi_{4,198}\xi_{4,209} + \xi_{4,200}\xi_{4,210} + \xi_{4,201}\xi_{4,211}$$

$$\xi_{4,219} = \xi_{4,198}\xi_{4,210} + \xi_{4,200}\xi_{4,212} + \xi_{4,201}\xi_{4,213}$$

$$\xi_{4,220} = \xi_{4,198}\xi_{4,211} + \xi_{4,200}\xi_{4,213} + \xi_{4,201}\xi_{4,214}$$

$$\xi_{4,221} = \xi_{4,199}\xi_{4,209} + \xi_{4,201}\xi_{4,210} + \xi_{4,202}\xi_{4,211}$$

$$\xi_{4,222} = \xi_{4,199}\xi_{4,210} + \xi_{4,201}\xi_{4,212} + \xi_{4,202}\xi_{4,213}$$

$$\xi_{4,223} = \xi_{4,199}\xi_{4,211} + \xi_{4,201}\xi_{4,213} + \xi_{4,202}\xi_{4,214}$$

$$\xi_{4,224} = \begin{cases} \xi_{4,188}^2 & \text{for } \xi_{4,140} = \xi_{4,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,225} = \begin{cases} \xi_{4,188}\xi_{4,189} & \text{for } \xi_{4,140} = \xi_{4,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,226} = \begin{cases} \xi_{4,188}\xi_{4,190} & \text{for } \xi_{4,140} = \xi_{4,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,227} = \begin{cases} \xi_{4,189}^2 & \text{for } \xi_{4,140} = \xi_{4,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,228} = \begin{cases} \xi_{4,189}\xi_{4,190} & \text{for } \xi_{4,140} = \xi_{4,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,229} = \begin{cases} \xi_{4,190}^2 & \text{for } \xi_{4,140} = \xi_{4,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{4,230} = \xi_{4,215} - \xi_{4,224}$$

$$\xi_{4,231} = \xi_{4,216} - \xi_{4,225}$$

$$\xi_{4,232} = \xi_{4,217} - \xi_{4,226}$$

$$\xi_{4,233} = \xi_{4,218} - \xi_{4,225}$$

$$\xi_{4,234} = \xi_{4,219} - \xi_{4,227}$$

$$\xi_{4,235} = \xi_{4,220} - \xi_{4,228}$$

$$\xi_{4,236} = \xi_{4,221} - \xi_{4,226}$$

$$\xi_{4,237} = \xi_{4,222} - \xi_{4,228}$$

$$\xi_{4,238} = \xi_{4,223} - \xi_{4,229}$$

$$\xi_{4,239} = \sqrt{\xi_{4,230}}$$

$$\xi_{4,240} = \frac{\xi_{4,230}}{\xi_{4,239}}$$

$$\xi_{4,241} = \frac{\xi_{4,233}}{\xi_{4,239}}$$

$$\xi_{4,242} = \frac{\xi_{4,236}}{\xi_{4,239}}$$

$$\xi_{4,243} = \sqrt{\xi_{4,234}}$$

$$\xi_{4,244} = \frac{\xi_{4,231}}{\xi_{4,243}}$$

$$\xi_{4,245} = \frac{\xi_{4,234}}{\xi_{4,243}}$$

$$\xi_{4,246} = \frac{\xi_{4,237}}{\xi_{4,243}}$$

$$\xi_{4,247} = \sqrt{\xi_{4,238}}$$

$$\xi_{4,248} = \frac{\xi_{4,232}}{\xi_{4,247}}$$

$$\xi_{4,249} = \frac{\xi_{4,235}}{\xi_{4,247}}$$

$$\xi_{4,250} = \frac{\xi_{4,238}}{\xi_{4,247}}$$

$$\xi_{4,251} = \begin{cases} \xi_{4,240} & \text{for } \xi_{4,230} \geq \xi_{4,234} \wedge \xi_{4,230} \geq \xi_{4,238} \\ \xi_{4,244} & \text{for } \xi_{4,234} \geq \xi_{4,238} \wedge \xi_{4,230} \leq \xi_{4,234} \\ \xi_{4,248} & \text{otherwise} \end{cases}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,252} = \begin{cases} \xi_{4,241} & \text{for } \xi_{4,230} \geq \xi_{4,234} \wedge \xi_{4,230} \geq \xi_{4,238} \\ \xi_{4,245} & \text{for } \xi_{4,234} \geq \xi_{4,238} \wedge \xi_{4,230} \leq \xi_{4,234} \\ \xi_{4,249} & \text{otherwise} \end{cases}$$

$$\xi_{4,253} = \begin{cases} \xi_{4,242} & \text{for } \xi_{4,230} \geq \xi_{4,234} \wedge \xi_{4,230} \geq \xi_{4,238} \\ \xi_{4,246} & \text{for } \xi_{4,234} \geq \xi_{4,238} \wedge \xi_{4,230} \leq \xi_{4,234} \\ \xi_{4,250} & \text{otherwise} \end{cases}$$

$$\xi_{4,254} = \xi_{4,189}\xi_{4,253} - \xi_{4,190}\xi_{4,252}$$

$$\xi_{4,255} = -\xi_{4,188}\xi_{4,253} + \xi_{4,190}\xi_{4,251}$$

$$\xi_{4,256} = \xi_{4,188}\xi_{4,252} - \xi_{4,189}\xi_{4,251}$$

$$\xi_{4,257} = \xi_{4,134}^2 + \xi_{4,137}^2$$

$$\xi_{4,258} = \xi_{4,135}^2 + \xi_{4,138}^2$$

$$\xi_{4,259} = \xi_{4,136}^2 + \xi_{4,139}^2$$

$$\xi_{4,260} = \frac{\xi_{4,121}\xi_{4,137}\xi_{4,188} + \xi_{4,189}(\xi_{4,114}\xi_{4,134} + \xi_{4,122}\xi_{4,137}) + \xi_{4,190}(\xi_{4,115}\xi_{4,134} + \xi_{4,123}\xi_{4,137})}{\xi_{4,257}}$$

$$\xi_{4,261} = \frac{\xi_{4,124}\xi_{4,137}\xi_{4,188} + \xi_{4,189}(\xi_{4,116}\xi_{4,134} + \xi_{4,125}\xi_{4,137}) + \xi_{4,190}(\xi_{4,117}\xi_{4,134} + \xi_{4,126}\xi_{4,137})}{\xi_{4,257}}$$

$$\xi_{4,262} = \frac{\xi_{4,119}\xi_{4,134}\xi_{4,189} + \xi_{4,120}\xi_{4,134}\xi_{4,190} + \xi_{4,188}(\xi_{4,118}\xi_{4,134} + \xi_{4,127}\xi_{4,137})}{\xi_{4,257}}$$

$$\xi_{4,263} = \frac{\xi_{4,121}\xi_{4,138}\xi_{4,251} + \xi_{4,252}(\xi_{4,114}\xi_{4,135} + \xi_{4,122}\xi_{4,138}) + \xi_{4,253}(\xi_{4,115}\xi_{4,135} + \xi_{4,123}\xi_{4,138})}{\xi_{4,258}}$$

$$\xi_{4,264} = \frac{\xi_{4,124}\xi_{4,138}\xi_{4,251} + \xi_{4,252}(\xi_{4,116}\xi_{4,135} + \xi_{4,125}\xi_{4,138}) + \xi_{4,253}(\xi_{4,117}\xi_{4,135} + \xi_{4,126}\xi_{4,138})}{\xi_{4,258}}$$

$$\xi_{4,265} = \frac{\xi_{4,119}\xi_{4,135}\xi_{4,252} + \xi_{4,120}\xi_{4,135}\xi_{4,253} + \xi_{4,251}(\xi_{4,118}\xi_{4,135} + \xi_{4,127}\xi_{4,138})}{\xi_{4,258}}$$

$$\xi_{4,266} = \frac{\xi_{4,121}\xi_{4,139}\xi_{4,254} + \xi_{4,255}(\xi_{4,114}\xi_{4,136} + \xi_{4,122}\xi_{4,139}) + \xi_{4,256}(\xi_{4,115}\xi_{4,136} + \xi_{4,123}\xi_{4,139})}{\xi_{4,259}}$$

$$\xi_{4,267} = \frac{\xi_{4,124}\xi_{4,139}\xi_{4,254} + \xi_{4,255}(\xi_{4,116}\xi_{4,136} + \xi_{4,125}\xi_{4,139}) + \xi_{4,256}(\xi_{4,117}\xi_{4,136} + \xi_{4,126}\xi_{4,139})}{\xi_{4,259}}$$

$$\xi_{4,268} = \frac{\xi_{4,119}\xi_{4,136}\xi_{4,255} + \xi_{4,120}\xi_{4,136}\xi_{4,256} + \xi_{4,254}(\xi_{4,118}\xi_{4,136} + \xi_{4,127}\xi_{4,139})}{\xi_{4,259}}$$

$$\xi_{4,269} = 1 - \xi_{4,260}^2$$

$$\xi_{4,270} = -\xi_{4,260}\xi_{4,261}$$

$$\xi_{4,271} = -\xi_{4,260}\xi_{4,262}$$

$$\xi_{4,272} = 1 - \xi_{4,261}^2$$

$$\xi_{4,273} = -\xi_{4,261}\xi_{4,262}$$

$$\xi_{4,274} = 1 - \xi_{4,262}^2$$

$$\xi_{4,275} = \xi_{4,269}^2 + \xi_{4,270}^2 + \xi_{4,271}^2$$

$$\xi_{4,276} = \xi_{4,270}^2 + \xi_{4,272}^2 + \xi_{4,273}^2$$

$$\xi_{4,277} = \xi_{4,271}^2 + \xi_{4,273}^2 + \xi_{4,274}^2$$

$$\xi_{4,278} = \sqrt{\xi_{4,275}}$$

$$\xi_{4,279} = \frac{\xi_{4,269}}{\xi_{4,278}}$$

$$\xi_{4,280} = \frac{\xi_{4,270}}{\xi_{4,278}}$$

$$\xi_{4,281} = \frac{\xi_{4,271}}{\xi_{4,278}}$$

$$\xi_{4,282} = \sqrt{\xi_{4,276}}$$

$$\xi_{4,283} = \frac{\xi_{4,272}}{\xi_{4,282}}$$

$$\xi_{4,284} = \frac{\xi_{4,270}}{\xi_{4,282}}$$

$$\xi_{4,285} = \frac{\xi_{4,273}}{\xi_{4,282}}$$

$$\xi_{4,286} = \sqrt{\xi_{4,277}}$$

$$\xi_{4,287} = \frac{\xi_{4,274}}{\xi_{4,286}}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,288} = \frac{\xi_{4,271}}{\xi_{4,286}}$$

$$\xi_{4,289} = \frac{\xi_{4,273}}{\xi_{4,286}}$$

$$\xi_{4,290} = \begin{cases} \xi_{4,279} & \text{for } \xi_{4,275} \geq \xi_{4,276} \wedge \xi_{4,275} \geq \xi_{4,277} \\ \xi_{4,284} & \text{for } \xi_{4,276} \geq \xi_{4,277} \wedge \xi_{4,275} \leq \xi_{4,276} \\ \xi_{4,288} & \text{otherwise} \end{cases}$$

$$\xi_{4,291} = \begin{cases} \xi_{4,280} & \text{for } \xi_{4,275} \geq \xi_{4,276} \wedge \xi_{4,275} \geq \xi_{4,277} \\ \xi_{4,283} & \text{for } \xi_{4,276} \geq \xi_{4,277} \wedge \xi_{4,275} \leq \xi_{4,276} \\ \xi_{4,289} & \text{otherwise} \end{cases}$$

$$\xi_{4,292} = \begin{cases} \xi_{4,281} & \text{for } \xi_{4,275} \geq \xi_{4,276} \wedge \xi_{4,275} \geq \xi_{4,277} \\ \xi_{4,285} & \text{for } \xi_{4,276} \geq \xi_{4,277} \wedge \xi_{4,275} \leq \xi_{4,276} \\ \xi_{4,287} & \text{otherwise} \end{cases}$$

$$\xi_{4,293} = \xi_{4,261}\xi_{4,292} - \xi_{4,262}\xi_{4,291}$$

$$\xi_{4,294} = -\xi_{4,260}\xi_{4,292} + \xi_{4,262}\xi_{4,290}$$

$$\xi_{4,295} = \xi_{4,260}\xi_{4,291} - \xi_{4,261}\xi_{4,290}$$

$$\xi_{4,296} = 1 - \xi_{4,263}^2$$

$$\xi_{4,297} = -\xi_{4,263}\xi_{4,264}$$

$$\xi_{4,298} = -\xi_{4,263}\xi_{4,265}$$

$$\xi_{4,299} = 1 - \xi_{4,264}^2$$

$$\xi_{4,300} = -\xi_{4,264}\xi_{4,265}$$

$$\xi_{4,301} = 1 - \xi_{4,265}^2$$

$$\xi_{4,302} = \xi_{4,296}^2 + \xi_{4,297}^2 + \xi_{4,298}^2$$

$$\xi_{4,303} = \xi_{4,297}^2 + \xi_{4,299}^2 + \xi_{4,300}^2$$

$$\xi_{4,304} = \xi_{4,298}^2 + \xi_{4,300}^2 + \xi_{4,301}^2$$

$$\xi_{4,305} = \sqrt{\xi_{4,302}}$$

$$\xi_{4,306} = \frac{\xi_{4,296}}{\xi_{4,305}}$$

$$\xi_{4,307} = \frac{\xi_{4,297}}{\xi_{4,305}}$$

$$\xi_{4,308} = \frac{\xi_{4,298}}{\xi_{4,305}}$$

$$\xi_{4,309} = \sqrt{\xi_{4,303}}$$

$$\xi_{4,310} = \frac{\xi_{4,299}}{\xi_{4,309}}$$

$$\xi_{4,311} = \frac{\xi_{4,297}}{\xi_{4,309}}$$

$$\xi_{4,312} = \frac{\xi_{4,300}}{\xi_{4,309}}$$

$$\xi_{4,313} = \sqrt{\xi_{4,304}}$$

$$\xi_{4,314} = \frac{\xi_{4,301}}{\xi_{4,313}}$$

$$\xi_{4,315} = \frac{\xi_{4,298}}{\xi_{4,313}}$$

$$\xi_{4,316} = \frac{\xi_{4,300}}{\xi_{4,313}}$$

$$\xi_{4,317} = \begin{cases} \xi_{4,306} & \text{for } \xi_{4,302} \geq \xi_{4,303} \wedge \xi_{4,302} \geq \xi_{4,304} \\ \xi_{4,311} & \text{for } \xi_{4,303} \geq \xi_{4,304} \wedge \xi_{4,302} \leq \xi_{4,303} \\ \xi_{4,315} & \text{otherwise} \end{cases}$$

$$\xi_{4,318} = \begin{cases} \xi_{4,307} & \text{for } \xi_{4,302} \geq \xi_{4,303} \wedge \xi_{4,302} \geq \xi_{4,304} \\ \xi_{4,310} & \text{for } \xi_{4,303} \geq \xi_{4,304} \wedge \xi_{4,302} \leq \xi_{4,303} \\ \xi_{4,316} & \text{otherwise} \end{cases}$$

$$\xi_{4,319} = \begin{cases} \xi_{4,308} & \text{for } \xi_{4,302} \geq \xi_{4,303} \wedge \xi_{4,302} \geq \xi_{4,304} \\ \xi_{4,312} & \text{for } \xi_{4,303} \geq \xi_{4,304} \wedge \xi_{4,302} \leq \xi_{4,303} \\ \xi_{4,314} & \text{otherwise} \end{cases}$$

$$\xi_{4,320} = \xi_{4,264}\xi_{4,319} - \xi_{4,265}\xi_{4,318}$$

$$\xi_{4,321} = -\xi_{4,263}\xi_{4,319} + \xi_{4,265}\xi_{4,317}$$

$$\xi_{4,322} = \xi_{4,263}\xi_{4,318} - \xi_{4,264}\xi_{4,317}$$

$$\xi_{4,323} = \xi_{4,261}\xi_{4,265} - \xi_{4,262}\xi_{4,264}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,324} = -\xi_{4,260}\xi_{4,265} + \xi_{4,262}\xi_{4,263}$$

$$\xi_{4,325} = \xi_{4,260}\xi_{4,264} - \xi_{4,261}\xi_{4,263}$$

$$\xi_{4,326} = 1 - \xi_{4,266}^2$$

$$\xi_{4,327} = -\xi_{4,266}\xi_{4,267}$$

$$\xi_{4,328} = -\xi_{4,266}\xi_{4,268}$$

$$\xi_{4,329} = 1 - \xi_{4,267}^2$$

$$\xi_{4,330} = -\xi_{4,267}\xi_{4,268}$$

$$\xi_{4,331} = 1 - \xi_{4,268}^2$$

$$\xi_{4,332} = \xi_{4,326}^2 + \xi_{4,327}^2 + \xi_{4,328}^2$$

$$\xi_{4,333} = \xi_{4,327}^2 + \xi_{4,329}^2 + \xi_{4,330}^2$$

$$\xi_{4,334} = \xi_{4,328}^2 + \xi_{4,330}^2 + \xi_{4,331}^2$$

$$\xi_{4,335} = \sqrt{\xi_{4,332}}$$

$$\xi_{4,336} = \frac{\xi_{4,326}}{\xi_{4,335}}$$

$$\xi_{4,337} = \frac{\xi_{4,327}}{\xi_{4,335}}$$

$$\xi_{4,338} = \frac{\xi_{4,328}}{\xi_{4,335}}$$

$$\xi_{4,339} = \sqrt{\xi_{4,333}}$$

$$\xi_{4,340} = \frac{\xi_{4,329}}{\xi_{4,339}}$$

$$\xi_{4,341} = \frac{\xi_{4,327}}{\xi_{4,339}}$$

$$\xi_{4,342} = \frac{\xi_{4,330}}{\xi_{4,339}}$$

$$\xi_{4,343} = \sqrt{\xi_{4,334}}$$

$$\xi_{4,344} = \frac{\xi_{4,331}}{\xi_{4,343}}$$

$$\xi_{4,345} = \frac{\xi_{4,328}}{\xi_{4,343}}$$

$$\xi_{4,346} = \frac{\xi_{4,330}}{\xi_{4,343}}$$

$$\xi_{4,347} = \begin{cases} \xi_{4,336} & \text{for } \xi_{4,332} \geq \xi_{4,333} \wedge \xi_{4,332} \geq \xi_{4,334} \\ \xi_{4,341} & \text{for } \xi_{4,333} \geq \xi_{4,334} \wedge \xi_{4,332} \leq \xi_{4,333} \\ \xi_{4,345} & \text{otherwise} \end{cases}$$

$$\xi_{4,348} = \begin{cases} \xi_{4,337} & \text{for } \xi_{4,332} \geq \xi_{4,333} \wedge \xi_{4,332} \geq \xi_{4,334} \\ \xi_{4,340} & \text{for } \xi_{4,333} \geq \xi_{4,334} \wedge \xi_{4,332} \leq \xi_{4,333} \\ \xi_{4,346} & \text{otherwise} \end{cases}$$

$$\xi_{4,349} = \begin{cases} \xi_{4,338} & \text{for } \xi_{4,332} \geq \xi_{4,333} \wedge \xi_{4,332} \geq \xi_{4,334} \\ \xi_{4,342} & \text{for } \xi_{4,333} \geq \xi_{4,334} \wedge \xi_{4,332} \leq \xi_{4,333} \\ \xi_{4,344} & \text{otherwise} \end{cases}$$

$$\xi_{4,350} = \xi_{4,267}\xi_{4,349} - \xi_{4,268}\xi_{4,348}$$

$$\xi_{4,351} = -\xi_{4,266}\xi_{4,349} + \xi_{4,268}\xi_{4,347}$$

$$\xi_{4,352} = \xi_{4,266}\xi_{4,348} - \xi_{4,267}\xi_{4,347}$$

$$\xi_{4,353} = -\xi_{4,261}\xi_{4,268} + \xi_{4,262}\xi_{4,267}$$

$$\xi_{4,354} = \xi_{4,260}\xi_{4,268} - \xi_{4,262}\xi_{4,266}$$

$$\xi_{4,355} = -\xi_{4,260}\xi_{4,267} + \xi_{4,261}\xi_{4,266}$$

$$\xi_{4,356} = \xi_{4,264}\xi_{4,268} - \xi_{4,265}\xi_{4,267}$$

$$\xi_{4,357} = -\xi_{4,263}\xi_{4,268} + \xi_{4,265}\xi_{4,266}$$

$$\xi_{4,358} = \xi_{4,263}\xi_{4,267} - \xi_{4,264}\xi_{4,266}$$

$$\xi_{4,359} = \begin{cases} 1 & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \\ \xi_{4,260} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \\ \xi_{4,320} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,258} \neq 0 \\ \xi_{4,260} & \text{for } \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,258} \neq 0 \\ \xi_{4,347} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,260} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,356} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,260} & \text{for } \xi_{4,257} \neq 0 \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \end{cases}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,360} = \begin{cases} 0 & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \\ \xi_{4,290} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \\ \xi_{4,263} & \text{for } \xi_{4,259} = 0 \wedge (\xi_{4,257} = 0 \vee \xi_{4,259} = 0) \wedge (\xi_{4,257} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,259} = 0 \vee \xi_{4,257} \neq 0) \wedge (\xi_{4,259} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,258} \neq 0) \wedge \xi_{4,258} \neq 0 \\ \xi_{4,350} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,353} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,263} & \text{for } (\xi_{4,257} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} = 0 \vee \xi_{4,259} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,259} \neq 0) \wedge (\xi_{4,258} \neq 0 \vee \xi_{4,259} \neq 0) \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \end{cases}$$

$$\xi_{4,362} = \begin{cases} 0 & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \\ \xi_{4,261} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \\ \xi_{4,321} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,258} \neq 0 \\ \xi_{4,261} & \text{for } \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,258} \neq 0 \\ \xi_{4,348} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,261} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,357} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,261} & \text{for } \xi_{4,257} \neq 0 \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \end{cases}$$

$$\xi_{4,363} = \begin{cases} 1 & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \\ \xi_{4,291} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \\ \xi_{4,264} & \text{for } \xi_{4,259} = 0 \wedge (\xi_{4,257} = 0 \vee \xi_{4,259} = 0) \wedge (\xi_{4,257} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,259} = 0 \vee \xi_{4,257} \neq 0) \wedge (\xi_{4,259} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,258} \neq 0) \wedge \xi_{4,258} \neq 0 \\ \xi_{4,351} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,354} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,264} & \text{for } (\xi_{4,257} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} = 0 \vee \xi_{4,259} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,259} \neq 0) \wedge (\xi_{4,258} \neq 0 \vee \xi_{4,259} \neq 0) \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \end{cases}$$

$$\xi_{4,365} = \begin{cases} 0 & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \\ \xi_{4,262} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \\ \xi_{4,322} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,258} \neq 0 \\ \xi_{4,262} & \text{for } \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,258} \neq 0 \\ \xi_{4,349} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,262} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,358} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,262} & \text{for } \xi_{4,257} \neq 0 \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \end{cases}$$

$$\xi_{4,366} = \begin{cases} 0 & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \\ \xi_{4,292} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,259} = 0 \wedge \xi_{4,257} \neq 0 \\ \xi_{4,265} & \text{for } \xi_{4,259} = 0 \wedge (\xi_{4,257} = 0 \vee \xi_{4,259} = 0) \wedge (\xi_{4,257} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,259} = 0 \vee \xi_{4,257} \neq 0) \wedge (\xi_{4,259} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,258} \neq 0) \wedge \xi_{4,258} \neq 0 \\ \xi_{4,352} & \text{for } \xi_{4,257} = 0 \wedge \xi_{4,258} = 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,355} & \text{for } \xi_{4,258} = 0 \wedge \xi_{4,257} \neq 0 \wedge \xi_{4,259} \neq 0 \\ \xi_{4,265} & \text{for } (\xi_{4,257} = 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} = 0 \vee \xi_{4,259} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,258} \neq 0) \wedge (\xi_{4,257} \neq 0 \vee \xi_{4,259} \neq 0) \wedge (\xi_{4,258} \neq 0 \vee \xi_{4,259} \neq 0) \wedge \xi_{4,258} \neq 0 \wedge \xi_{4,259} \neq 0 \end{cases}$$

$$\xi_{4,368} = \frac{\xi_{4,359}}{4} + \frac{\xi_{4,363}}{4} + \frac{\xi_{4,367}}{4} + \frac{1}{4}$$

$$\xi_{4,369} = \frac{\xi_{4,359}}{4} - \frac{\xi_{4,363}}{4} - \frac{\xi_{4,367}}{4} + \frac{1}{4}$$

$$\xi_{4,370} = -\frac{\xi_{4,359}}{4} + \frac{\xi_{4,363}}{4} - \frac{\xi_{4,367}}{4} + \frac{1}{4}$$

$$\xi_{4,371} = -\frac{\xi_{4,359}}{4} - \frac{\xi_{4,363}}{4} + \frac{\xi_{4,367}}{4} + \frac{1}{4}$$

$$\xi_{4,372} = \sqrt{\xi_{4,368}}$$

$$\xi_{4,373} = \frac{-\xi_{4,364}+\xi_{4,366}}{4\xi_{4,372}}$$

$$\xi_{4,374} = \frac{\xi_{4,361}-\xi_{4,365}}{4\xi_{4,372}}$$

$$\xi_{4,375} = \frac{-\xi_{4,360}+\xi_{4,362}}{4\xi_{4,372}}$$

$$\xi_{4,376} = \sqrt{\xi_{4,369}}$$

$$\xi_{4,377} = \frac{-\xi_{4,364}+\xi_{4,366}}{4\xi_{4,376}}$$

$$\xi_{4,378} = \frac{\xi_{4,360}+\xi_{4,362}}{4\xi_{4,376}}$$

$$\xi_{4,379} = \frac{\xi_{4,361}+\xi_{4,365}}{4\xi_{4,376}}$$

$$\xi_{4,380} = \sqrt{\xi_{4,370}}$$

$$\xi_{4,381} = \frac{\xi_{4,361}-\xi_{4,365}}{4\xi_{4,380}}$$

$$\xi_{4,382} = \frac{\xi_{4,360}+\xi_{4,362}}{4\xi_{4,380}}$$

$$\xi_{4,383} = \frac{\xi_{4,364}+\xi_{4,366}}{4\xi_{4,380}}$$

$$\xi_{4,384} = \sqrt{\xi_{4,371}}$$

$$\xi_{4,385} = \frac{-\xi_{4,360}+\xi_{4,362}}{4\xi_{4,384}}$$

$$\xi_{4,386} = \frac{\xi_{4,361}+\xi_{4,365}}{4\xi_{4,384}}$$

$$\xi_{4,387} = \frac{\xi_{4,364}+\xi_{4,366}}{4\xi_{4,384}}$$

$$\xi_{4,388} = \begin{cases} \xi_{4,372} & \text{for } \xi_{4,368} \geq \xi_{4,369} \wedge \xi_{4,368} \geq \xi_{4,370} \wedge \xi_{4,368} \geq \xi_{4,371} \\ \xi_{4,377} & \text{for } \xi_{4,369} \geq \xi_{4,370} \wedge \xi_{4,369} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,369} \\ \xi_{4,381} & \text{for } \xi_{4,370} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,370} \wedge \xi_{4,369} \leq \xi_{4,370} \\ \xi_{4,385} & \text{otherwise} \end{cases}$$

$$\xi_{4,389} = \begin{cases} \xi_{4,373} & \text{for } \xi_{4,368} \geq \xi_{4,369} \wedge \xi_{4,368} \geq \xi_{4,370} \wedge \xi_{4,368} \geq \xi_{4,371} \\ \xi_{4,376} & \text{for } \xi_{4,369} \geq \xi_{4,370} \wedge \xi_{4,369} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,369} \\ \xi_{4,382} & \text{for } \xi_{4,370} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,370} \wedge \xi_{4,369} \leq \xi_{4,370} \\ \xi_{4,386} & \text{otherwise} \end{cases}$$

$$\xi_{4,390} = \begin{cases} \xi_{4,374} & \text{for } \xi_{4,368} \geq \xi_{4,369} \wedge \xi_{4,368} \geq \xi_{4,370} \wedge \xi_{4,368} \geq \xi_{4,371} \\ \xi_{4,378} & \text{for } \xi_{4,369} \geq \xi_{4,370} \wedge \xi_{4,369} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,369} \\ \xi_{4,380} & \text{for } \xi_{4,370} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,370} \wedge \xi_{4,369} \leq \xi_{4,370} \\ \xi_{4,387} & \text{otherwise} \end{cases}$$

$$\xi_{4,391} = \begin{cases} \xi_{4,375} & \text{for } \xi_{4,368} \geq \xi_{4,369} \wedge \xi_{4,368} \geq \xi_{4,370} \wedge \xi_{4,368} \geq \xi_{4,371} \\ \xi_{4,379} & \text{for } \xi_{4,369} \geq \xi_{4,370} \wedge \xi_{4,369} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,369} \\ \xi_{4,383} & \text{for } \xi_{4,370} \geq \xi_{4,371} \wedge \xi_{4,368} \leq \xi_{4,370} \wedge \xi_{4,369} \leq \xi_{4,370} \\ \xi_{4,384} & \text{otherwise} \end{cases}$$

$$\xi_{4,392} = \frac{\xi_{4,188}}{4} + \frac{\xi_{4,252}}{4} + \frac{\xi_{4,256}}{4} + \frac{1}{4}$$

$$\xi_{4,393} = \frac{\xi_{4,188}}{4} - \frac{\xi_{4,252}}{4} - \frac{\xi_{4,256}}{4} + \frac{1}{4}$$

$$\xi_{4,394} = -\frac{\xi_{4,188}}{4} + \frac{\xi_{4,252}}{4} - \frac{\xi_{4,256}}{4} + \frac{1}{4}$$

$$\xi_{4,395} = -\frac{\xi_{4,188}}{4} - \frac{\xi_{4,252}}{4} + \frac{\xi_{4,256}}{4} + \frac{1}{4}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,396} = \sqrt{\xi_{4,392}}$$

$$\xi_{4,397} = \frac{\xi_{4,253} - \xi_{4,255}}{4\xi_{4,396}}$$

$$\xi_{4,398} = \frac{-\xi_{4,190} + \xi_{4,254}}{4\xi_{4,396}}$$

$$\xi_{4,399} = \frac{\xi_{4,189} - \xi_{4,251}}{4\xi_{4,396}}$$

$$\xi_{4,400} = \sqrt{\xi_{4,393}}$$

$$\xi_{4,401} = \frac{\xi_{4,253} - \xi_{4,255}}{4\xi_{4,400}}$$

$$\xi_{4,402} = \frac{\xi_{4,189} + \xi_{4,251}}{4\xi_{4,400}}$$

$$\xi_{4,403} = \frac{\xi_{4,190} + \xi_{4,254}}{4\xi_{4,400}}$$

$$\xi_{4,404} = \sqrt{\xi_{4,394}}$$

$$\xi_{4,405} = \frac{-\xi_{4,190} + \xi_{4,254}}{4\xi_{4,404}}$$

$$\xi_{4,406} = \frac{\xi_{4,189} + \xi_{4,251}}{4\xi_{4,404}}$$

$$\xi_{4,407} = \frac{\xi_{4,253} + \xi_{4,255}}{4\xi_{4,404}}$$

$$\xi_{4,408} = \sqrt{\xi_{4,395}}$$

$$\xi_{4,409} = \frac{\xi_{4,189} - \xi_{4,251}}{4\xi_{4,408}}$$

$$\xi_{4,410} = \frac{\xi_{4,190} + \xi_{4,254}}{4\xi_{4,408}}$$

$$\xi_{4,411} = \frac{\xi_{4,253} + \xi_{4,255}}{4\xi_{4,408}}$$

$$\xi_{4,412} = \begin{cases} \xi_{4,396} & \text{for } \xi_{4,392} \geq \xi_{4,393} \wedge \xi_{4,392} \geq \xi_{4,394} \wedge \xi_{4,392} \geq \xi_{4,395} \\ \xi_{4,401} & \text{for } \xi_{4,393} \geq \xi_{4,394} \wedge \xi_{4,393} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,393} \\ \xi_{4,405} & \text{for } \xi_{4,394} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,394} \wedge \xi_{4,393} \leq \xi_{4,394} \\ \xi_{4,409} & \text{otherwise} \end{cases}$$

$$\xi_{4,413} = \begin{cases} \xi_{4,397} & \text{for } \xi_{4,392} \geq \xi_{4,393} \wedge \xi_{4,392} \geq \xi_{4,394} \wedge \xi_{4,392} \geq \xi_{4,395} \\ \xi_{4,400} & \text{for } \xi_{4,393} \geq \xi_{4,394} \wedge \xi_{4,393} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,393} \\ \xi_{4,406} & \text{for } \xi_{4,394} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,394} \wedge \xi_{4,393} \leq \xi_{4,394} \\ \xi_{4,410} & \text{otherwise} \end{cases}$$

$$\xi_{4,414} = \begin{cases} \xi_{4,398} & \text{for } \xi_{4,392} \geq \xi_{4,393} \wedge \xi_{4,392} \geq \xi_{4,394} \wedge \xi_{4,392} \geq \xi_{4,395} \\ \xi_{4,402} & \text{for } \xi_{4,393} \geq \xi_{4,394} \wedge \xi_{4,393} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,393} \\ \xi_{4,404} & \text{for } \xi_{4,394} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,394} \wedge \xi_{4,393} \leq \xi_{4,394} \\ \xi_{4,411} & \text{otherwise} \end{cases}$$

$$\xi_{4,415} = \begin{cases} \xi_{4,399} & \text{for } \xi_{4,392} \geq \xi_{4,393} \wedge \xi_{4,392} \geq \xi_{4,394} \wedge \xi_{4,392} \geq \xi_{4,395} \\ \xi_{4,403} & \text{for } \xi_{4,393} \geq \xi_{4,394} \wedge \xi_{4,393} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,393} \\ \xi_{4,407} & \text{for } \xi_{4,394} \geq \xi_{4,395} \wedge \xi_{4,392} \leq \xi_{4,394} \wedge \xi_{4,393} \leq \xi_{4,394} \\ \xi_{4,408} & \text{otherwise} \end{cases}$$

$$\xi_{4,416} = \xi_{4,107}\left(\xi_{4,112} + i\xi_{4,113}\right)$$

$$\xi_{4,417} = -\frac{\sqrt{2}\xi_{4,107}\xi_{4,109}}{2} + \frac{\sqrt{2}i\xi_{4,108}(\xi_{4,112}+i\xi_{4,113})}{2}$$

$$\xi_{4,418} = \frac{\sqrt{2}i\xi_{4,107}\xi_{4,109}}{2} - \frac{\sqrt{2}\xi_{4,108}(\xi_{4,112}+i\xi_{4,113})}{2}$$

$$\xi_{4,419} = -\xi_{4,108}\xi_{4,109}$$

$$\xi_{4,420} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{4,107}(\xi_{4,112}+i\xi_{4,113})}{2} - \frac{\sqrt{2}i\xi_{4,108}\xi_{4,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{4,107}(\xi_{4,112}-i\xi_{4,113})}{2} - \frac{\sqrt{2}\xi_{4,108}\xi_{4,109}}{2}\right)}{2}$$

$$\xi_{4,421} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{4,107}(\xi_{4,112}+i\xi_{4,113})}{2} - \frac{\sqrt{2}i\xi_{4,108}\xi_{4,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{4,107}(\xi_{4,112}-i\xi_{4,113})}{2} - \frac{\sqrt{2}\xi_{4,108}\xi_{4,109}}{2}\right)}{2}$$

$$\xi_{4,422} = \frac{\sqrt{2}i\xi_{4,107}\xi_{4,109}}{2} - \frac{\sqrt{2}\xi_{4,108}(\xi_{4,112}-i\xi_{4,113})}{2}$$

$$\xi_{4,423} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{4,107}(\xi_{4,112}-i\xi_{4,113})}{2} - \frac{\sqrt{2}i\xi_{4,108}\xi_{4,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{4,107}(\xi_{4,112}+i\xi_{4,113})}{2} - \frac{\sqrt{2}\xi_{4,108}\xi_{4,109}}{2}\right)}{2}$$

$$\xi_{4,424} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{4,107}(\xi_{4,112}-i\xi_{4,113})}{2} - \frac{\sqrt{2}i\xi_{4,108}\xi_{4,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{4,107}(\xi_{4,112}+i\xi_{4,113})}{2} - \frac{\sqrt{2}\xi_{4,108}\xi_{4,109}}{2}\right)}{2}$$

$$\xi_{4,425} = -\frac{\sqrt{2}\xi_{4,107}\xi_{4,109}}{2} + \frac{\sqrt{2}i\xi_{4,108}(\xi_{4,112}-i\xi_{4,113})}{2}$$

$$\xi_{4,426} = \xi_{4,107}\left(\xi_{4,112} - i\xi_{4,113}\right)$$

$$\xi_{4,427} = \left(\xi_{4,388} - i\xi_{4,391}\right)\left(\xi_{4,412} - i\xi_{4,415}\right)$$

$$\xi_{4,428} = \left(\xi_{4,388} - i\xi_{4,391}\right)\left(-i\xi_{4,413} - \xi_{4,414}\right)$$

$$\xi_{4,429} = \left(\xi_{4,412} - i\xi_{4,415}\right)\left(-i\xi_{4,389} - \xi_{4,390}\right)$$

$$\xi_{4,430} = \left(-i\xi_{4,389} - \xi_{4,390}\right)\left(-i\xi_{4,413} - \xi_{4,414}\right)$$

$$\xi_{4,431} = \left(\xi_{4,388} - i\xi_{4,391}\right)\left(-i\xi_{4,413} + \xi_{4,414}\right)$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,432} = (\xi_{4,388} - i\xi_{4,391})\,(\xi_{4,412} + i\xi_{4,415})$$

$$\xi_{4,433} = (-i\xi_{4,389} - \xi_{4,390})\,(-i\xi_{4,413} + \xi_{4,414})$$

$$\xi_{4,434} = (\xi_{4,412} + i\xi_{4,415})\,(-i\xi_{4,389} - \xi_{4,390})$$

$$\xi_{4,435} = (\xi_{4,412} - i\xi_{4,415})\,(-i\xi_{4,389} + \xi_{4,390})$$

$$\xi_{4,436} = (-i\xi_{4,389} + \xi_{4,390})\,(-i\xi_{4,413} - \xi_{4,414})$$

$$\xi_{4,437} = (\xi_{4,388} + i\xi_{4,391})\,(\xi_{4,412} - i\xi_{4,415})$$

$$\xi_{4,438} = (\xi_{4,388} + i\xi_{4,391})\,(-i\xi_{4,413} - \xi_{4,414})$$

$$\xi_{4,439} = (-i\xi_{4,389} + \xi_{4,390})\,(-i\xi_{4,413} + \xi_{4,414})$$

$$\xi_{4,440} = (\xi_{4,412} + i\xi_{4,415})\,(-i\xi_{4,389} + \xi_{4,390})$$

$$\xi_{4,441} = (\xi_{4,388} + i\xi_{4,391})\,(-i\xi_{4,413} + \xi_{4,414})$$

$$\xi_{4,442} = (\xi_{4,388} + i\xi_{4,391})\,(\xi_{4,412} + i\xi_{4,415})$$

$$\xi_{4,443} = \xi_{4,416}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}}\right) + \xi_{4,417}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}}\right) + \xi_{4,418}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}}\right) + \xi_{4,419}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}}\right)$$

$$\xi_{4,444} = \xi_{4,417}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}}\right) + \xi_{4,420}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}}\right) + \xi_{4,422}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}}\right) + \xi_{4,423}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}}\right)$$

$$\xi_{4,445} = \xi_{4,418}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}}\right) + \xi_{4,421}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}}\right) + \xi_{4,424}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}}\right) + \xi_{4,425}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}}\right)$$

$$\xi_{4,446} = \xi_{4,419}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}}\right) + \xi_{4,422}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}}\right) + \xi_{4,425}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}}\right) + \xi_{4,426}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}}\right)$$

$$\xi_{4,447} = \xi_{4,416}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}}\right) + \xi_{4,417}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}}\right) + \xi_{4,418}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}}\right) + \xi_{4,419}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}}\right)$$

$$\xi_{4,448} = \xi_{4,417}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}}\right) + \xi_{4,420}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}}\right) + \xi_{4,422}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}}\right) + \xi_{4,423}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}}\right)$$

$$\xi_{4,449} = \xi_{4,418}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}}\right) + \xi_{4,421}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}}\right) + \xi_{4,424}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}}\right) + \xi_{4,425}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}}\right)$$

$$\xi_{4,450} = \xi_{4,419}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,427}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,430}}\right) + \xi_{4,422}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,431}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,434}}\right) + \xi_{4,425}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,435}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,438}}\right) + \xi_{4,426}\left(-ie^{-i\xi_{4,100}}\sin\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,439}} + e^{-i\xi_{4,100}}\cos\left(\xi_{4,98} - \xi_{4,99}\right)\overline{\xi_{4,442}}\right)$$

$$\xi_{4,451} = \xi_{4,416}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}}\right) + \xi_{4,417}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}}\right) + \xi_{4,418}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}}\right) + \xi_{4,419}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}}\right)$$

$$\xi_{4,452} = \xi_{4,417}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}}\right) + \xi_{4,420}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}}\right) + \xi_{4,422}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}}\right) + \xi_{4,423}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}}\right)$$

$$\xi_{4,453} = \xi_{4,418}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}}\right) + \xi_{4,421}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}}\right) + \xi_{4,424}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}}\right) + \xi_{4,425}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}}\right)$$

$$\xi_{4,454} = \xi_{4,419}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}}\right) + \xi_{4,422}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}}\right) + \xi_{4,425}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}}\right) + \xi_{4,426}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}}\right)$$

$$\xi_{4,455} = \xi_{4,416}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}}\right) + \xi_{4,417}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}}\right) + \xi_{4,418}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}}\right) + \xi_{4,419}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}}\right)$$

$$\xi_{4,456} = \xi_{4,417}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}}\right) + \xi_{4,420}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}}\right) + \xi_{4,422}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}}\right) + \xi_{4,423}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}}\right)$$

$$\xi_{4,457} = \xi_{4,418}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}}\right) + \xi_{4,421}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}}\right) + \xi_{4,424}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}}\right) + \xi_{4,425}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}}\right)$$

$$\xi_{4,458} = \xi_{4,419}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,428}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,429}}\right) + \xi_{4,422}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,432}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,433}}\right) + \xi_{4,425}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,436}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,437}}\right) + \xi_{4,426}\left(-ie^{i\xi_{4,100}}\sin\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,440}} + e^{i\xi_{4,100}}\cos\left(\xi_{4,98} + \xi_{4,99}\right)\overline{\xi_{4,441}}\right)$$

$$\xi_{4,459} = \frac{\operatorname{re}\left(\xi_{4,443}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,450}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,452}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,457}\right)}{4}$$

$$\xi_{4,460} = -\frac{\operatorname{im}\left(\xi_{4,444}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,449}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,451}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,458}\right)}{4}$$

$$\xi_{4,461} = -\frac{\operatorname{re}\left(\xi_{4,444}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,449}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,451}\right)}{4} - \frac{\operatorname{re}\left(\xi_{4,458}\right)}{4}$$

$$\xi_{4,462} = -\frac{\operatorname{im}\left(\xi_{4,443}\right)}{4} + \frac{\operatorname{im}\left(\xi_{4,450}\right)}{4} + \frac{\operatorname{im}\left(\xi_{4,452}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,457}\right)}{4}$$

$$\xi_{4,463} = -\frac{\operatorname{im}\left(\xi_{4,445}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,448}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,454}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,455}\right)}{4}$$

$$\xi_{4,464} = -\frac{\operatorname{re}\left(\xi_{4,446}\right)}{4} - \frac{\operatorname{re}\left(\xi_{4,447}\right)}{4} - \frac{\operatorname{re}\left(\xi_{4,453}\right)}{4} - \frac{\operatorname{re}\left(\xi_{4,456}\right)}{4}$$

$$\xi_{4,465} = \frac{\operatorname{im}\left(\xi_{4,446}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,447}\right)}{4} - \frac{\operatorname{im}\left(\xi_{4,453}\right)}{4} + \frac{\operatorname{im}\left(\xi_{4,456}\right)}{4}$$

$$\xi_{4,466} = -\frac{\operatorname{re}\left(\xi_{4,445}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,448}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,454}\right)}{4} - \frac{\operatorname{re}\left(\xi_{4,455}\right)}{4}$$

$$\xi_{4,467} = -\frac{\operatorname{re}\left(\xi_{4,445}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,448}\right)}{4} - \frac{\operatorname{re}\left(\xi_{4,454}\right)}{4} + \frac{\operatorname{re}\left(\xi_{4,455}\right)}{4}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,468} = \frac{\operatorname{im}(\xi_{4,446})}{4} - \frac{\operatorname{im}(\xi_{4,447})}{4} + \frac{\operatorname{im}(\xi_{4,453})}{4} - \frac{\operatorname{im}(\xi_{4,456})}{4}$$

$$\xi_{4,469} = \frac{\operatorname{re}(\xi_{4,446})}{4} + \frac{\operatorname{re}(\xi_{4,447})}{4} - \frac{\operatorname{re}(\xi_{4,453})}{4} - \frac{\operatorname{re}(\xi_{4,456})}{4}$$

$$\xi_{4,470} = \frac{\operatorname{im}(\xi_{4,445})}{4} + \frac{\operatorname{im}(\xi_{4,448})}{4} - \frac{\operatorname{im}(\xi_{4,454})}{4} - \frac{\operatorname{im}(\xi_{4,455})}{4}$$

$$\xi_{4,471} = -\frac{\operatorname{im}(\xi_{4,443})}{4} + \frac{\operatorname{im}(\xi_{4,450})}{4} - \frac{\operatorname{im}(\xi_{4,452})}{4} + \frac{\operatorname{im}(\xi_{4,457})}{4}$$

$$\xi_{4,472} = -\frac{\operatorname{re}(\xi_{4,444})}{4} + \frac{\operatorname{re}(\xi_{4,449})}{4} - \frac{\operatorname{re}(\xi_{4,451})}{4} + \frac{\operatorname{re}(\xi_{4,458})}{4}$$

$$\xi_{4,473} = \frac{\operatorname{im}(\xi_{4,444})}{4} + \frac{\operatorname{im}(\xi_{4,449})}{4} - \frac{\operatorname{im}(\xi_{4,451})}{4} - \frac{\operatorname{im}(\xi_{4,458})}{4}$$

$$\xi_{4,474} = -\frac{\operatorname{re}(\xi_{4,443})}{4} - \frac{\operatorname{re}(\xi_{4,450})}{4} + \frac{\operatorname{re}(\xi_{4,452})}{4} + \frac{\operatorname{re}(\xi_{4,457})}{4}$$

$$\xi_{4,475} = \xi_{4,459}^2 + \xi_{4,463}^2 + \xi_{4,467}^2 + \xi_{4,471}^2$$

$$\xi_{4,476} = \xi_{4,460}^2 + \xi_{4,464}^2 + \xi_{4,468}^2 + \xi_{4,472}^2$$

$$\xi_{4,477} = \xi_{4,461}^2 + \xi_{4,465}^2 + \xi_{4,469}^2 + \xi_{4,473}^2$$

$$\xi_{4,478} = \xi_{4,462}^2 + \xi_{4,466}^2 + \xi_{4,470}^2 + \xi_{4,474}^2$$

$$\xi_{4,479} = \sqrt{\xi_{4,475}}$$

$$\xi_{4,480} = \frac{\xi_{4,459}}{\xi_{4,479}}$$

$$\xi_{4,481} = \frac{\xi_{4,463}}{\xi_{4,479}}$$

$$\xi_{4,482} = \frac{\xi_{4,467}}{\xi_{4,479}}$$

$$\xi_{4,483} = \frac{\xi_{4,471}}{\xi_{4,479}}$$

$$\xi_{4,484} = \sqrt{\xi_{4,476}}$$

$$\xi_{4,485} = \frac{\xi_{4,460}}{\xi_{4,484}}$$

$$\xi_{4,486} = \frac{\xi_{4,464}}{\xi_{4,484}}$$

$$\xi_{4,487} = \frac{\xi_{4,468}}{\xi_{4,484}}$$

$$\xi_{4,488} = \frac{\xi_{4,472}}{\xi_{4,484}}$$

$$\xi_{4,489} = \sqrt{\xi_{4,477}}$$

$$\xi_{4,490} = \frac{\xi_{4,461}}{\xi_{4,489}}$$

$$\xi_{4,491} = \frac{\xi_{4,465}}{\xi_{4,489}}$$

$$\xi_{4,492} = \frac{\xi_{4,469}}{\xi_{4,489}}$$

$$\xi_{4,493} = \frac{\xi_{4,473}}{\xi_{4,489}}$$

$$\xi_{4,494} = \sqrt{\xi_{4,478}}$$

$$\xi_{4,495} = \frac{\xi_{4,462}}{\xi_{4,494}}$$

$$\xi_{4,496} = \frac{\xi_{4,466}}{\xi_{4,494}}$$

$$\xi_{4,497} = \frac{\xi_{4,470}}{\xi_{4,494}}$$

$$\xi_{4,498} = \frac{\xi_{4,474}}{\xi_{4,494}}$$

$$\xi_{4,499} = \begin{cases} \xi_{4,480} & \text{for } \xi_{4,475} \geq \xi_{4,476} \wedge \xi_{4,475} \geq \xi_{4,477} \wedge \xi_{4,475} \geq \xi_{4,478} \\ \xi_{4,485} & \text{for } \xi_{4,476} \geq \xi_{4,477} \wedge \xi_{4,476} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,476} \\ \xi_{4,490} & \text{for } \xi_{4,477} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,477} \wedge \xi_{4,476} \leq \xi_{4,477} \\ \xi_{4,495} & \text{otherwise} \end{cases}$$

$$\xi_{4,500} = \begin{cases} \xi_{4,481} & \text{for } \xi_{4,475} \geq \xi_{4,476} \wedge \xi_{4,475} \geq \xi_{4,477} \wedge \xi_{4,475} \geq \xi_{4,478} \\ \xi_{4,486} & \text{for } \xi_{4,476} \geq \xi_{4,477} \wedge \xi_{4,476} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,476} \\ \xi_{4,491} & \text{for } \xi_{4,477} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,477} \wedge \xi_{4,476} \leq \xi_{4,477} \\ \xi_{4,496} & \text{otherwise} \end{cases}$$

$$\xi_{4,501} = \begin{cases} \xi_{4,482} & \text{for } \xi_{4,475} \geq \xi_{4,476} \wedge \xi_{4,475} \geq \xi_{4,477} \wedge \xi_{4,475} \geq \xi_{4,478} \\ \xi_{4,487} & \text{for } \xi_{4,476} \geq \xi_{4,477} \wedge \xi_{4,476} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,476} \\ \xi_{4,492} & \text{for } \xi_{4,477} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,477} \wedge \xi_{4,476} \leq \xi_{4,477} \\ \xi_{4,497} & \text{otherwise} \end{cases}$$

$$\xi_{4,502} = \begin{cases} \xi_{4,483} & \text{for } \xi_{4,475} \geq \xi_{4,476} \wedge \xi_{4,475} \geq \xi_{4,477} \wedge \xi_{4,475} \geq \xi_{4,478} \\ \xi_{4,488} & \text{for } \xi_{4,476} \geq \xi_{4,477} \wedge \xi_{4,476} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,476} \\ \xi_{4,493} & \text{for } \xi_{4,477} \geq \xi_{4,478} \wedge \xi_{4,475} \leq \xi_{4,477} \wedge \xi_{4,476} \leq \xi_{4,477} \\ \xi_{4,498} & \text{otherwise} \end{cases}$$

$$\xi_{4,503} = \xi_{4,459}\xi_{4,499} + \xi_{4,463}\xi_{4,500} + \xi_{4,467}\xi_{4,501} + \xi_{4,471}\xi_{4,502}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,504} = \xi_{4,460}\xi_{4,499} + \xi_{4,464}\xi_{4,500} + \xi_{4,468}\xi_{4,501} + \xi_{4,472}\xi_{4,502}$$

$$\xi_{4,505} = \xi_{4,461}\xi_{4,499} + \xi_{4,465}\xi_{4,500} + \xi_{4,469}\xi_{4,501} + \xi_{4,473}\xi_{4,502}$$

$$\xi_{4,506} = \xi_{4,462}\xi_{4,499} + \xi_{4,466}\xi_{4,500} + \xi_{4,470}\xi_{4,501} + \xi_{4,474}\xi_{4,502}$$

$$\xi_{4,507} = \xi_{4,36}\xi_{4,47} - \xi_{4,37}\xi_{4,50} - \xi_{4,38}\xi_{4,51} - \xi_{4,39}\xi_{4,52}$$

$$\xi_{4,508} = \xi_{4,36}\xi_{4,50} + \xi_{4,37}\xi_{4,47} - \xi_{4,38}\xi_{4,52} + \xi_{4,39}\xi_{4,51}$$

$$\xi_{4,509} = \xi_{4,36}\xi_{4,51} + \xi_{4,37}\xi_{4,52} + \xi_{4,38}\xi_{4,47} - \xi_{4,39}\xi_{4,50}$$

$$\xi_{4,510} = \xi_{4,36}\xi_{4,52} - \xi_{4,37}\xi_{4,51} + \xi_{4,38}\xi_{4,50} + \xi_{4,39}\xi_{4,47}$$

$$\xi_{4,511} = \xi_{4,10}\xi_{4,50} + \xi_{4,49}\xi_{4,7} + \xi_{4,50}\xi_{4,8} + \xi_{4,50}\xi_{4,9}$$

$$\xi_{4,512} = -\xi_{4,10}\xi_{4,50} - \xi_{4,49}\xi_{4,8} + \xi_{4,50}\xi_{4,7} + \xi_{4,50}\xi_{4,9}$$

$$\xi_{4,513} = \xi_{4,10}\xi_{4,50} - \xi_{4,49}\xi_{4,9} + \xi_{4,50}\xi_{4,7} - \xi_{4,50}\xi_{4,8}$$

$$\xi_{4,514} = -\xi_{4,10}\xi_{4,49} + \xi_{4,50}\xi_{4,7} + \xi_{4,50}\xi_{4,8} - \xi_{4,50}\xi_{4,9}$$

$$\xi_{4,515} = \xi_{4,47}\left(\xi_{4,499}\xi_{4,94} + \xi_{4,500}\xi_{4,95} + \xi_{4,501}\xi_{4,96} + \xi_{4,502}\xi_{4,97}\right) - \xi_{4,47}\left(\xi_{4,499}\xi_{4,97} - \xi_{4,500}\xi_{4,96} + \xi_{4,501}\xi_{4,95} - \xi_{4,502}\xi_{4,94}\right) - \xi_{4,48}\left(\xi_{4,499}\xi_{4,95} - \xi_{4,500}\xi_{4,94} - \xi_{4,501}\xi_{4,97} + \xi_{4,502}\xi_{4,96}\right) - \xi_{4,49}\left(\xi_{4,499}\xi_{4,96} + \xi_{4,500}\xi_{4,97} - \xi_{4,501}\xi_{4,94} - \xi_{4,502}\xi_{4,95}\right)$$

$$\xi_{4,516} = \xi_{4,47}\left(\xi_{4,499}\xi_{4,95} - \xi_{4,500}\xi_{4,94} - \xi_{4,501}\xi_{4,97} + \xi_{4,502}\xi_{4,96}\right) + \xi_{4,47}\left(\xi_{4,499}\xi_{4,96} + \xi_{4,500}\xi_{4,97} - \xi_{4,501}\xi_{4,94} - \xi_{4,502}\xi_{4,95}\right) + \xi_{4,48}\left(\xi_{4,499}\xi_{4,94} + \xi_{4,500}\xi_{4,95} + \xi_{4,501}\xi_{4,96} + \xi_{4,502}\xi_{4,97}\right) - \xi_{4,49}\left(\xi_{4,499}\xi_{4,97} - \xi_{4,500}\xi_{4,96} + \xi_{4,501}\xi_{4,95} - \xi_{4,502}\xi_{4,94}\right)$$

$$\xi_{4,517} = -\xi_{4,47}\left(\xi_{4,499}\xi_{4,95} - \xi_{4,500}\xi_{4,94} - \xi_{4,501}\xi_{4,97} + \xi_{4,502}\xi_{4,96}\right) + \xi_{4,47}\left(\xi_{4,499}\xi_{4,96} + \xi_{4,500}\xi_{4,97} - \xi_{4,501}\xi_{4,94} - \xi_{4,502}\xi_{4,95}\right) + \xi_{4,48}\left(\xi_{4,499}\xi_{4,97} - \xi_{4,500}\xi_{4,96} + \xi_{4,501}\xi_{4,95} - \xi_{4,502}\xi_{4,94}\right) + \xi_{4,49}\left(\xi_{4,499}\xi_{4,94} + \xi_{4,500}\xi_{4,95} + \xi_{4,501}\xi_{4,96} + \xi_{4,502}\xi_{4,97}\right)$$

$$\xi_{4,518} = \xi_{4,47}\left(\xi_{4,499}\xi_{4,94} + \xi_{4,500}\xi_{4,95} + \xi_{4,501}\xi_{4,96} + \xi_{4,502}\xi_{4,97}\right) + \xi_{4,47}\left(\xi_{4,499}\xi_{4,97} - \xi_{4,500}\xi_{4,96} + \xi_{4,501}\xi_{4,95} - \xi_{4,502}\xi_{4,94}\right) - \xi_{4,48}\left(\xi_{4,499}\xi_{4,96} + \xi_{4,500}\xi_{4,97} - \xi_{4,501}\xi_{4,94} - \xi_{4,502}\xi_{4,95}\right) + \xi_{4,49}\left(\xi_{4,499}\xi_{4,95} - \xi_{4,500}\xi_{4,94} - \xi_{4,501}\xi_{4,97} + \xi_{4,502}\xi_{4,96}\right)$$

$$\xi_{4,519} = -\xi_{4,48}\left(-\xi_{4,503}\xi_{4,63} - \xi_{4,504}\xi_{4,62} + \xi_{4,505}\xi_{4,65} - \xi_{4,506}\xi_{4,64}\right) - \xi_{4,48}\left(-\xi_{4,503}\xi_{4,64} - \xi_{4,504}\xi_{4,65} - \xi_{4,505}\xi_{4,62} + \xi_{4,506}\xi_{4,63}\right) - \xi_{4,48}\left(-\xi_{4,503}\xi_{4,65} + \xi_{4,504}\xi_{4,64} - \xi_{4,505}\xi_{4,63} - \xi_{4,506}\xi_{4,62}\right) + \xi_{4,49}\left(\xi_{4,503}\xi_{4,62} - \xi_{4,504}\xi_{4,63} - \xi_{4,505}\xi_{4,64} - \xi_{4,506}\xi_{4,65}\right)$$

$$\xi_{4,520} = \xi_{4,48}\left(\xi_{4,503}\xi_{4,62} - \xi_{4,504}\xi_{4,63} - \xi_{4,505}\xi_{4,64} - \xi_{4,506}\xi_{4,65}\right) + \xi_{4,48}\left(-\xi_{4,503}\xi_{4,64} - \xi_{4,504}\xi_{4,65} - \xi_{4,505}\xi_{4,62} + \xi_{4,506}\xi_{4,63}\right) - \xi_{4,48}\left(-\xi_{4,503}\xi_{4,65} + \xi_{4,504}\xi_{4,64} - \xi_{4,505}\xi_{4,63} - \xi_{4,506}\xi_{4,62}\right) + \xi_{4,49}\left(-\xi_{4,503}\xi_{4,63} - \xi_{4,504}\xi_{4,62} + \xi_{4,505}\xi_{4,65} - \xi_{4,506}\xi_{4,64}\right)$$

$$\xi_{4,521} = \xi_{4,48}\left(\xi_{4,503}\xi_{4,62} - \xi_{4,504}\xi_{4,63} - \xi_{4,505}\xi_{4,64} - \xi_{4,506}\xi_{4,65}\right) - \xi_{4,48}\left(-\xi_{4,503}\xi_{4,63} - \xi_{4,504}\xi_{4,62} + \xi_{4,505}\xi_{4,65} - \xi_{4,506}\xi_{4,64}\right) + \xi_{4,48}\left(-\xi_{4,503}\xi_{4,65} + \xi_{4,504}\xi_{4,64} - \xi_{4,505}\xi_{4,63} - \xi_{4,506}\xi_{4,62}\right) + \xi_{4,49}\left(-\xi_{4,503}\xi_{4,64} - \xi_{4,504}\xi_{4,65} - \xi_{4,505}\xi_{4,62} + \xi_{4,506}\xi_{4,63}\right)$$

$$\xi_{4,522} = \xi_{4,48}\left(\xi_{4,503}\xi_{4,62} - \xi_{4,504}\xi_{4,63} - \xi_{4,505}\xi_{4,64} - \xi_{4,506}\xi_{4,65}\right) + \xi_{4,48}\left(-\xi_{4,503}\xi_{4,63} - \xi_{4,504}\xi_{4,62} + \xi_{4,505}\xi_{4,65} - \xi_{4,506}\xi_{4,64}\right) - \xi_{4,48}\left(-\xi_{4,503}\xi_{4,64} - \xi_{4,504}\xi_{4,65} - \xi_{4,505}\xi_{4,62} + \xi_{4,506}\xi_{4,63}\right) + \xi_{4,49}\left(-\xi_{4,503}\xi_{4,65} + \xi_{4,504}\xi_{4,64} - \xi_{4,505}\xi_{4,63} - \xi_{4,506}\xi_{4,62}\right)$$

$$\xi_{4,523} = \xi_{4,388}\left(\xi_{4,28}\xi_{4,86} - \xi_{4,29}\xi_{4,87} - \xi_{4,30}\xi_{4,88} - \xi_{4,31}\xi_{4,89}\right) + \xi_{4,389}\left(\xi_{4,28}\xi_{4,87} + \xi_{4,29}\xi_{4,86} + \xi_{4,30}\xi_{4,89} - \xi_{4,31}\xi_{4,88}\right) + \xi_{4,390}\left(\xi_{4,28}\xi_{4,88} - \xi_{4,29}\xi_{4,89} + \xi_{4,30}\xi_{4,86} + \xi_{4,31}\xi_{4,87}\right) + \xi_{4,391}\left(\xi_{4,28}\xi_{4,89} + \xi_{4,29}\xi_{4,88} - \xi_{4,30}\xi_{4,87} + \xi_{4,31}\xi_{4,86}\right)$$

$$\xi_{4,524} = \xi_{4,388}\left(\xi_{4,28}\xi_{4,87} + \xi_{4,29}\xi_{4,86} + \xi_{4,30}\xi_{4,89} - \xi_{4,31}\xi_{4,88}\right) - \xi_{4,389}\left(\xi_{4,28}\xi_{4,86} - \xi_{4,29}\xi_{4,87} - \xi_{4,30}\xi_{4,88} - \xi_{4,31}\xi_{4,89}\right) + \xi_{4,390}\left(\xi_{4,28}\xi_{4,89} + \xi_{4,29}\xi_{4,88} - \xi_{4,30}\xi_{4,87} + \xi_{4,31}\xi_{4,86}\right) - \xi_{4,391}\left(\xi_{4,28}\xi_{4,88} - \xi_{4,29}\xi_{4,89} + \xi_{4,30}\xi_{4,86} + \xi_{4,31}\xi_{4,87}\right)$$

$$\xi_{4,525} = \xi_{4,388}\left(\xi_{4,28}\xi_{4,88} - \xi_{4,29}\xi_{4,89} + \xi_{4,30}\xi_{4,86} + \xi_{4,31}\xi_{4,87}\right) - \xi_{4,389}\left(\xi_{4,28}\xi_{4,89} + \xi_{4,29}\xi_{4,88} - \xi_{4,30}\xi_{4,87} + \xi_{4,31}\xi_{4,86}\right) - \xi_{4,390}\left(\xi_{4,28}\xi_{4,86} - \xi_{4,29}\xi_{4,87} - \xi_{4,30}\xi_{4,88} - \xi_{4,31}\xi_{4,89}\right) + \xi_{4,391}\left(\xi_{4,28}\xi_{4,87} + \xi_{4,29}\xi_{4,86} + \xi_{4,30}\xi_{4,89} - \xi_{4,31}\xi_{4,88}\right)$$

$$\xi_{4,526} = \xi_{4,388}\left(\xi_{4,28}\xi_{4,89} + \xi_{4,29}\xi_{4,88} - \xi_{4,30}\xi_{4,87} + \xi_{4,31}\xi_{4,86}\right) + \xi_{4,389}\left(\xi_{4,28}\xi_{4,88} - \xi_{4,29}\xi_{4,89} + \xi_{4,30}\xi_{4,86} + \xi_{4,31}\xi_{4,87}\right) - \xi_{4,390}\left(\xi_{4,28}\xi_{4,87} + \xi_{4,29}\xi_{4,86} + \xi_{4,30}\xi_{4,89} - \xi_{4,31}\xi_{4,88}\right) - \xi_{4,391}\left(\xi_{4,28}\xi_{4,86} - \xi_{4,29}\xi_{4,87} - \xi_{4,30}\xi_{4,88} - \xi_{4,31}\xi_{4,89}\right)$$

$$\xi_{4,527} = \xi_{4,412}\left(\xi_{4,32}\xi_{4,90} - \xi_{4,33}\xi_{4,91} - \xi_{4,34}\xi_{4,92} - \xi_{4,35}\xi_{4,93}\right) + \xi_{4,413}\left(\xi_{4,32}\xi_{4,91} + \xi_{4,33}\xi_{4,90} + \xi_{4,34}\xi_{4,93} - \xi_{4,35}\xi_{4,92}\right) + \xi_{4,414}\left(\xi_{4,32}\xi_{4,92} - \xi_{4,33}\xi_{4,93} + \xi_{4,34}\xi_{4,90} + \xi_{4,35}\xi_{4,91}\right) + \xi_{4,415}\left(\xi_{4,32}\xi_{4,93} + \xi_{4,33}\xi_{4,92} - \xi_{4,34}\xi_{4,91} + \xi_{4,35}\xi_{4,90}\right)$$

$$\xi_{4,528} = \xi_{4,412}\left(\xi_{4,32}\xi_{4,91} + \xi_{4,33}\xi_{4,90} + \xi_{4,34}\xi_{4,93} - \xi_{4,35}\xi_{4,92}\right) - \xi_{4,413}\left(\xi_{4,32}\xi_{4,90} - \xi_{4,33}\xi_{4,91} - \xi_{4,34}\xi_{4,92} - \xi_{4,35}\xi_{4,93}\right) + \xi_{4,414}\left(\xi_{4,32}\xi_{4,93} + \xi_{4,33}\xi_{4,92} - \xi_{4,34}\xi_{4,91} + \xi_{4,35}\xi_{4,90}\right) - \xi_{4,415}\left(\xi_{4,32}\xi_{4,92} - \xi_{4,33}\xi_{4,93} + \xi_{4,34}\xi_{4,90} + \xi_{4,35}\xi_{4,91}\right)$$

$$\xi_{4,529} = \xi_{4,412}\left(\xi_{4,32}\xi_{4,92} - \xi_{4,33}\xi_{4,93} + \xi_{4,34}\xi_{4,90} + \xi_{4,35}\xi_{4,91}\right) - \xi_{4,413}\left(\xi_{4,32}\xi_{4,93} + \xi_{4,33}\xi_{4,92} - \xi_{4,34}\xi_{4,91} + \xi_{4,35}\xi_{4,90}\right) - \xi_{4,414}\left(\xi_{4,32}\xi_{4,90} - \xi_{4,33}\xi_{4,91} - \xi_{4,34}\xi_{4,92} - \xi_{4,35}\xi_{4,93}\right) + \xi_{4,415}\left(\xi_{4,32}\xi_{4,91} + \xi_{4,33}\xi_{4,90} + \xi_{4,34}\xi_{4,93} - \xi_{4,35}\xi_{4,92}\right)$$

$$\xi_{4,530} = \xi_{4,412}\left(\xi_{4,32}\xi_{4,93} + \xi_{4,33}\xi_{4,92} - \xi_{4,34}\xi_{4,91} + \xi_{4,35}\xi_{4,90}\right) + \xi_{4,413}\left(\xi_{4,32}\xi_{4,92} - \xi_{4,33}\xi_{4,93} + \xi_{4,34}\xi_{4,90} + \xi_{4,35}\xi_{4,91}\right) - \xi_{4,414}\left(\xi_{4,32}\xi_{4,91} + \xi_{4,33}\xi_{4,90} + \xi_{4,34}\xi_{4,93} - \xi_{4,35}\xi_{4,92}\right) - \xi_{4,415}\left(\xi_{4,32}\xi_{4,90} - \xi_{4,33}\xi_{4,91} - \xi_{4,34}\xi_{4,92} - \xi_{4,35}\xi_{4,93}\right)$$

$$\xi_{4,531} = \xi_{4,101} + \xi_{4,43}$$

$$\xi_{4,532} = \xi_{4,531} - \frac{\phi}{4}$$

$$\xi_{4,533} = \xi_{4,527}\cos\left(\frac{\phi}{4}\right) + \xi_{4,530}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{4,534} = \xi_{4,528}\cos\left(\frac{\phi}{4}\right) + \xi_{4,529}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{4,535} = -\xi_{4,528}\sin\left(\frac{\phi}{4}\right) + \xi_{4,529}\cos\left(\frac{\phi}{4}\right)$$

$$\xi_{4,536} = -\xi_{4,527}\sin\left(\frac{\phi}{4}\right) + \xi_{4,530}\cos\left(\frac{\phi}{4}\right)$$

$$\xi_{4,537} = \xi_{4,523}\cos\left(\frac{\phi}{4}\right) + \xi_{4,526}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{4,538} = \xi_{4,524}\cos\left(\frac{\phi}{4}\right) + \xi_{4,525}\sin\left(\frac{\phi}{4}\right)$$

$$\xi_{4,539} = -\xi_{4,524}\sin\left(\frac{\phi}{4}\right) + \xi_{4,525}\cos\left(\frac{\phi}{4}\right)$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,540} = -\xi_{4,523}\sin\left(\frac{\varphi}{4}\right) + \xi_{4,526}\cos\left(\frac{\varphi}{4}\right)$$

$$\xi_{4,541} = \begin{cases} 0 & \text{for } \xi_{4,507}^2 + \xi_{4,510}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,510}, \xi_{4,507}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,542} = \begin{cases} 0 & \text{for } \xi_{4,508}^2 + \xi_{4,509}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,508}, \xi_{4,509}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,543} = 2\operatorname{atan}_2\left(\sqrt{\xi_{4,508}^2 + \xi_{4,509}^2}, \sqrt{\xi_{4,507}^2 + \xi_{4,510}^2}\right)$$

$$\xi_{4,544} = -\xi_{4,541} + \xi_{4,542}$$

$$\xi_{4,545} = -\xi_{4,541} - \xi_{4,542}$$

$$\xi_{4,546} = \xi_{4,532} + \xi_{4,541}$$

$$\xi_{4,547} = \begin{cases} 0 & \text{for } \xi_{4,511}^2 + \xi_{4,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,514}, \xi_{4,511}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,548} = \begin{cases} 0 & \text{for } \xi_{4,512}^2 + \xi_{4,513}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,512}, \xi_{4,513}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,549} = 2\operatorname{atan}_2\left(\sqrt{\xi_{4,512}^2 + \xi_{4,513}^2}, \sqrt{\xi_{4,511}^2 + \xi_{4,514}^2}\right)$$

$$\xi_{4,550} = -\xi_{4,547} + \xi_{4,548}$$

$$\xi_{4,551} = -\xi_{4,547} - \xi_{4,548}$$

$$\xi_{4,552} = \xi_{4,546} + \xi_{4,547}$$

$$\xi_{4,553} = \begin{cases} 0 & \text{for } \xi_{4,515}^2 + \xi_{4,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,518}, \xi_{4,515}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,554} = \begin{cases} 0 & \text{for } \xi_{4,516}^2 + \xi_{4,517}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,516}, \xi_{4,517}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,555} = 2\operatorname{atan}_2\left(\sqrt{\xi_{4,516}^2 + \xi_{4,517}^2}, \sqrt{\xi_{4,515}^2 + \xi_{4,518}^2}\right)$$

$$\xi_{4,556} = -\xi_{4,553} + \xi_{4,554}$$

$$\xi_{4,557} = -\xi_{4,553} - \xi_{4,554}$$

$$\xi_{4,558} = \xi_{4,552} + \xi_{4,553}$$

$$\xi_{4,559} = \begin{cases} 0 & \text{for } \xi_{4,519}^2 + \xi_{4,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,522}, \xi_{4,519}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,560} = \begin{cases} 0 & \text{for } \xi_{4,520}^2 + \xi_{4,521}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,520}, \xi_{4,521}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,561} = 2\operatorname{atan}_2\left(\sqrt{\xi_{4,520}^2 + \xi_{4,521}^2}, \sqrt{\xi_{4,519}^2 + \xi_{4,522}^2}\right)$$

$$\xi_{4,562} = -\xi_{4,559} + \xi_{4,560}$$

$$\xi_{4,563} = -\xi_{4,559} - \xi_{4,560}$$

$$\xi_{4,564} = \xi_{4,558} + \xi_{4,559}$$

$$\xi_{4,565} = \begin{cases} 0 & \text{for } \xi_{4,112}^2 + \xi_{4,113}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{4,113}, \xi_{4,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,566} = \begin{cases} 0 & \text{for } \xi_{4,109}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{4,109}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,567} = 2\operatorname{atan}_2\left(\left|\xi_{4,109}\right|, \sqrt{\xi_{4,112}^2 + \xi_{4,113}^2}\right)$$

$$\xi_{4,568} = -\xi_{4,565} + \xi_{4,566}$$

$$\xi_{4,569} = -\xi_{4,565} - \xi_{4,566}$$

$$\xi_{4,570} = \xi_{4,564} + \xi_{4,565}$$

$$\xi_{4,571} = \begin{cases} 0 & \text{for } \xi_{4,107}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{4,107}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,572} = \begin{cases} 0 & \text{for } \xi_{4,108}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{4,108}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,573} = 2\operatorname{atan}_2\left(\left|\xi_{4,108}\right|, \left|\xi_{4,107}\right|\right)$$

$$\xi_{4,574} = -\xi_{4,571} + \xi_{4,572}$$

$$\xi_{4,575} = -\xi_{4,571} - \xi_{4,572}$$

Exact shared-expression program 4

Original parameters: phi, theta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{4,576} = \xi_{4,570} + \xi_{4,571}$$

$$\xi_{4,577} = \begin{cases} 0 & \text{for } \xi_{4,537}^2 + \xi_{4,540}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,540}, \xi_{4,537}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,578} = \begin{cases} 0 & \text{for } \xi_{4,538}^2 + \xi_{4,539}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,538}, \xi_{4,539}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,579} = 2\operatorname{atan}_2\left(\sqrt{\xi_{4,538}^2 + \xi_{4,539}^2}, \sqrt{\xi_{4,537}^2 + \xi_{4,540}^2}\right)$$

$$\xi_{4,580} = -\xi_{4,577} + \xi_{4,578}$$

$$\xi_{4,581} = -\xi_{4,577} - \xi_{4,578}$$

$$\xi_{4,582} = \xi_{4,576} + \xi_{4,577}$$

$$\xi_{4,583} = \begin{cases} 0 & \text{for } \xi_{4,533}^2 + \xi_{4,536}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,536}, \xi_{4,533}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,584} = \begin{cases} 0 & \text{for } \xi_{4,534}^2 + \xi_{4,535}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{4,534}, \xi_{4,535}\right) & \text{otherwise} \end{cases}$$

$$\xi_{4,585} = 2\operatorname{atan}_2\left(\sqrt{\xi_{4,534}^2 + \xi_{4,535}^2}, \sqrt{\xi_{4,533}^2 + \xi_{4,536}^2}\right)$$

$$\xi_{4,586} = -\xi_{4,583} + \xi_{4,584}$$

$$\xi_{4,587} = -\xi_{4,583} - \xi_{4,584}$$

$$\xi_{4,588} = \xi_{4,582} + \xi_{4,583}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,0} = \left\lfloor -\frac{\theta}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{5,1} = \left\lfloor -\frac{\phi}{2\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{5,2} = -\frac{\pi \xi_{5,0}}{2} - \frac{\theta}{2}$$

$$\xi_{5,3} = -\frac{\pi \xi_{5,1}}{2} - \frac{\phi}{4}$$

$$\xi_{5,4} = \begin{cases} \xi_{5,2} & \text{for } |\xi_{5,2}| \geq |\xi_{5,3}| \\ \xi_{5,3} & \text{otherwise} \end{cases}$$

$$\xi_{5,5} = \xi_{5,2}$$

$$\xi_{5,6} = \begin{cases} \xi_{5,3} & \text{for } |\xi_{5,2}| \geq |\xi_{5,3}| \\ \xi_{5,2} & \text{otherwise} \end{cases}$$

$$\xi_{5,7} = \begin{cases} 1 & \text{for } |\xi_{5,2}| \geq |\xi_{5,3}| \\ \frac{1}{2} & \text{for } |\xi_{5,2}| \leq |\xi_{5,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{5,8} = \begin{cases} 0 & \text{for } |\xi_{5,2}| \geq |\xi_{5,3}| \\ \frac{1}{2} & \text{for } |\xi_{5,2}| \leq |\xi_{5,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,9} = \begin{cases} 0 & \text{for } |\xi_{5,2}| \geq |\xi_{5,3}| \\ \frac{1}{2} & \text{for } |\xi_{5,2}| \leq |\xi_{5,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{5,10} = \begin{cases} 0 & \text{for } |\xi_{5,2}| \geq |\xi_{5,3}| \\ -\frac{1}{2} & \text{for } |\xi_{5,2}| \leq |\xi_{5,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,11} = \begin{cases} -1 & \text{for } \xi_{5,4} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{5,12} = \begin{cases} -1 & \text{for } \xi_{5,5} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{5,13} = \begin{cases} 1 & \text{for } \xi_{5,11} = 1 \wedge \xi_{5,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,14} = \begin{cases} 0 & \text{for } \xi_{5,11} = 1 \wedge \xi_{5,12} = 1 \\ 1 & \text{for } \xi_{5,11} = 1 \wedge \xi_{5,12} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,15} = \begin{cases} 0 & \text{for } \xi_{5,11} = 1 \wedge (\xi_{5,11} = 1 \vee \xi_{5,12} = -1) \wedge (\xi_{5,11} = 1 \vee \xi_{5,12} = 1) \wedge (\xi_{5,12} = -1 \vee \xi_{5,12} = 1) \\ 1 & \text{for } \xi_{5,11} = -1 \wedge \xi_{5,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,16} = \begin{cases} 0 & \text{for } (\xi_{5,11} = -1 \vee \xi_{5,11} = 1) \wedge (\xi_{5,11} = 1 \vee \xi_{5,12} = 1) \wedge (\xi_{5,12} = -1 \vee \xi_{5,12} = 1) \wedge (\xi_{5,11} = -1 \vee \xi_{5,11} = 1 \vee \xi_{5,12} = -1) \wedge (\xi_{5,11} = -1 \vee \xi_{5,11} = 1 \vee \xi_{5,12} = 1) \wedge (\xi_{5,11} = -1 \vee \xi_{5,12} = -1 \vee \xi_{5,12} = 1) \wedge (\xi_{5,11} = 1 \vee \xi_{5,12} = -1 \vee \xi_{5,12} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{5,17} = \cos\left(\frac{\pi \xi_{5,0}}{2}\right)$$

$$\xi_{5,18} = -\sin\left(\frac{\pi \xi_{5,0}}{2}\right)$$

$$\xi_{5,19} = \xi_{5,17}^2$$

$$\xi_{5,20} = \xi_{5,17}\xi_{5,18}$$

$$\xi_{5,21} = -\xi_{5,18}^2$$

$$\xi_{5,22} = \cos\left(\frac{\pi \xi_{5,1}}{2}\right)$$

$$\xi_{5,23} = -\sin\left(\frac{\pi \xi_{5,1}}{2}\right)$$

$$\xi_{5,24} = \xi_{5,19}\xi_{5,22} - \xi_{5,21}\xi_{5,23}$$

$$\xi_{5,25} = \xi_{5,20}\xi_{5,22} - \xi_{5,20}\xi_{5,23}$$

$$\xi_{5,26} = \xi_{5,20}\xi_{5,22} + \xi_{5,20}\xi_{5,23}$$

$$\xi_{5,27} = \xi_{5,19}\xi_{5,23} + \xi_{5,21}\xi_{5,22}$$

$$\xi_{5,28} = \xi_{5,13}\left(-\xi_{5,10}\xi_{5,27} + \xi_{5,24}\xi_{5,7} - \xi_{5,25}\xi_{5,8} - \xi_{5,26}\xi_{5,9}\right) - \xi_{5,14}\left(\xi_{5,10}\xi_{5,26} + \xi_{5,24}\xi_{5,8} + \xi_{5,25}\xi_{5,7} - \xi_{5,27}\xi_{5,9}\right) - \xi_{5,15}\left(-\xi_{5,10}\xi_{5,25} + \xi_{5,24}\xi_{5,9} + \xi_{5,26}\xi_{5,7} + \xi_{5,27}\xi_{5,8}\right) - \xi_{5,16}\left(\xi_{5,10}\xi_{5,24} + \xi_{5,25}\xi_{5,9} - \xi_{5,26}\xi_{5,8} + \xi_{5,27}\xi_{5,7}\right)$$

$$\xi_{5,29} = \xi_{5,13}\left(\xi_{5,10}\xi_{5,26} + \xi_{5,24}\xi_{5,8} + \xi_{5,25}\xi_{5,7} - \xi_{5,27}\xi_{5,9}\right) + \xi_{5,14}\left(-\xi_{5,10}\xi_{5,27} + \xi_{5,24}\xi_{5,7} - \xi_{5,25}\xi_{5,8} - \xi_{5,26}\xi_{5,9}\right) - \xi_{5,15}\left(\xi_{5,10}\xi_{5,24} + \xi_{5,25}\xi_{5,9} - \xi_{5,26}\xi_{5,8} + \xi_{5,27}\xi_{5,7}\right) + \xi_{5,16}\left(-\xi_{5,10}\xi_{5,25} + \xi_{5,24}\xi_{5,9} + \xi_{5,26}\xi_{5,7} + \xi_{5,27}\xi_{5,8}\right)$$

$$\xi_{5,30} = \xi_{5,13}\left(-\xi_{5,10}\xi_{5,25} + \xi_{5,24}\xi_{5,9} + \xi_{5,26}\xi_{5,7} + \xi_{5,27}\xi_{5,8}\right) + \xi_{5,14}\left(\xi_{5,10}\xi_{5,24} + \xi_{5,25}\xi_{5,9} - \xi_{5,26}\xi_{5,8} + \xi_{5,27}\xi_{5,7}\right) + \xi_{5,15}\left(-\xi_{5,10}\xi_{5,27} + \xi_{5,24}\xi_{5,7} - \xi_{5,25}\xi_{5,8} - \xi_{5,26}\xi_{5,9}\right) - \xi_{5,16}\left(\xi_{5,10}\xi_{5,26} + \xi_{5,24}\xi_{5,8} + \xi_{5,25}\xi_{5,7} - \xi_{5,27}\xi_{5,9}\right)$$

$$\xi_{5,31} = \xi_{5,13}\left(\xi_{5,10}\xi_{5,24} + \xi_{5,25}\xi_{5,9} - \xi_{5,26}\xi_{5,8} + \xi_{5,27}\xi_{5,7}\right) - \xi_{5,14}\left(-\xi_{5,10}\xi_{5,25} + \xi_{5,24}\xi_{5,9} + \xi_{5,26}\xi_{5,7} + \xi_{5,27}\xi_{5,8}\right) + \xi_{5,15}\left(\xi_{5,10}\xi_{5,26} + \xi_{5,24}\xi_{5,8} + \xi_{5,25}\xi_{5,7} - \xi_{5,27}\xi_{5,9}\right) + \xi_{5,16}\left(-\xi_{5,10}\xi_{5,27} + \xi_{5,24}\xi_{5,7} - \xi_{5,25}\xi_{5,8} - \xi_{5,26}\xi_{5,9}\right)$$

$$\xi_{5,32} = -\xi_{5,10}\xi_{5,27} + \xi_{5,24}\xi_{5,7} - \xi_{5,25}\xi_{5,8} - \xi_{5,26}\xi_{5,9}$$

$$\xi_{5,33} = \xi_{5,10}\xi_{5,26} + \xi_{5,24}\xi_{5,8} + \xi_{5,25}\xi_{5,7} - \xi_{5,27}\xi_{5,9}$$

$$\xi_{5,34} = -\xi_{5,10}\xi_{5,25} + \xi_{5,24}\xi_{5,9} + \xi_{5,26}\xi_{5,7} + \xi_{5,27}\xi_{5,8}$$

$$\xi_{5,35} = \xi_{5,10}\xi_{5,24} + \xi_{5,25}\xi_{5,9} - \xi_{5,26}\xi_{5,8} + \xi_{5,27}\xi_{5,7}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,36} = -\xi_{5,10}\xi_{5,16} + \xi_{5,13}\xi_{5,7} - \xi_{5,14}\xi_{5,8} - \xi_{5,15}\xi_{5,9}$$

$$\xi_{5,37} = \xi_{5,10}\xi_{5,15} - \xi_{5,13}\xi_{5,8} - \xi_{5,14}\xi_{5,7} - \xi_{5,16}\xi_{5,9}$$

$$\xi_{5,38} = -\xi_{5,10}\xi_{5,14} - \xi_{5,13}\xi_{5,9} - \xi_{5,15}\xi_{5,7} + \xi_{5,16}\xi_{5,8}$$

$$\xi_{5,39} = -\xi_{5,10}\xi_{5,13} + \xi_{5,14}\xi_{5,9} - \xi_{5,15}\xi_{5,8} - \xi_{5,16}\xi_{5,7}$$

$$\xi_{5,40} = \xi_{5,11}\xi_{5,4}$$

$$\xi_{5,41} = \xi_{5,12}\xi_{5,5}$$

$$\xi_{5,42} = \xi_{5,11}\xi_{5,12}\xi_{5,6}$$

$$\xi_{5,43} = -\frac{\pi(2\xi_{5,0}+\xi_{5,1})}{2}$$

$$\xi_{5,44} = \begin{cases} 0 & \text{for } \left(\xi_{5,41} > \frac{\pi}{8} \vee \xi_{5,42} < 0\right) \wedge \xi_{5,41} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{5,45} = \frac{\pi}{8}$$

$$\xi_{5,46} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{5,41} > \frac{\pi}{8} \wedge \xi_{5,42} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{5,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,47} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{5,41} > \frac{\pi}{8} \wedge \xi_{5,42} < 0 \\ \frac{1}{2} & \text{for } \xi_{5,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,48} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{5,41} > \frac{\pi}{8} \vee \xi_{5,42} < 0\right) \wedge \xi_{5,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,49} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{5,41} > \frac{\pi}{8} \vee \xi_{5,42} < 0\right) \wedge \xi_{5,41} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{5,50} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{5,41} > \frac{\pi}{8} \vee \xi_{5,42} < 0\right) \wedge \xi_{5,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,51} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{5,41} > \frac{\pi}{8} \vee \xi_{5,42} < 0\right) \wedge \xi_{5,41} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{5,52} = \begin{cases} \frac{1}{2} & \text{for } \xi_{5,41} > \frac{\pi}{8} \wedge \xi_{5,42} < 0 \\ -\frac{1}{2} & \text{for } \xi_{5,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,53} = \left\lfloor \frac{2\xi_{5,40} - 2\xi_{5,44}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{5,54} = \left\lfloor \frac{2\xi_{5,41} - 2\xi_{5,45}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{5,55} = \left\lfloor \frac{2\xi_{5,42} - 2\xi_{5,46}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{5,56} = \xi_{5,40} - \xi_{5,44} - \frac{\pi\xi_{5,53}}{2}$$

$$\xi_{5,57} = \xi_{5,41} - \xi_{5,45} - \frac{\pi\xi_{5,54}}{2}$$

$$\xi_{5,58} = \xi_{5,42} - \xi_{5,46} - \frac{\pi\xi_{5,55}}{2}$$

$$\xi_{5,59} = \begin{cases} \xi_{5,56} & \text{for } (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,56}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,57}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,57}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,57}| \geq |\xi_{5,58}| \vee |\xi_{5,57}| \leq |\xi_{5,58}|) \\ \xi_{5,57} & \text{for } (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,57}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,57}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,57}|) \wedge (|\xi_{5,56}| \leq |\xi_{5,57}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \\ \xi_{5,58} & \text{otherwise} \end{cases}$$

$$\xi_{5,60} = \begin{cases} \xi_{5,57} & \text{for } |\xi_{5,56}| \geq |\xi_{5,57}| \wedge |\xi_{5,57}| \geq |\xi_{5,58}| \\ \xi_{5,58} & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge |\xi_{5,57}| \leq |\xi_{5,58}| \\ \xi_{5,56} & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge |\xi_{5,56}| \leq |\xi_{5,57}| \\ \xi_{5,58} & \text{for } |\xi_{5,57}| \geq |\xi_{5,58}| \wedge |\xi_{5,56}| \leq |\xi_{5,58}| \\ \xi_{5,56} & \text{for } |\xi_{5,56}| \geq |\xi_{5,57}| \wedge |\xi_{5,56}| \leq |\xi_{5,58}| \\ \xi_{5,57} & \text{otherwise} \end{cases}$$

$$\xi_{5,61} = \begin{cases} \xi_{5,58} & \text{for } |\xi_{5,56}| \geq |\xi_{5,57}| \wedge |\xi_{5,57}| \geq |\xi_{5,58}| \\ \xi_{5,57} & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge |\xi_{5,57}| \leq |\xi_{5,58}| \\ \xi_{5,58} & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge |\xi_{5,56}| \leq |\xi_{5,57}| \\ \xi_{5,56} & \text{for } |\xi_{5,57}| \geq |\xi_{5,58}| \wedge |\xi_{5,56}| \leq |\xi_{5,58}| \\ \xi_{5,57} & \text{for } |\xi_{5,56}| \geq |\xi_{5,57}| \wedge |\xi_{5,56}| \leq |\xi_{5,58}| \\ \xi_{5,56} & \text{otherwise} \end{cases}$$

$$\xi_{5,62} = \begin{cases} 1 & \text{for } |\xi_{5,56}| \geq |\xi_{5,57}| \wedge |\xi_{5,57}| \geq |\xi_{5,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge |\xi_{5,57}| \leq |\xi_{5,58}| \\ 0 & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge |\xi_{5,56}| \leq |\xi_{5,57}| \\ \frac{1}{2} & \text{for } |\xi_{5,56}| \leq |\xi_{5,58}| \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,57}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,57}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{5,63} = \begin{cases} 0 & \text{for } |\xi_{5,56}| \geq |\xi_{5,57}| \wedge |\xi_{5,57}| \geq |\xi_{5,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,57}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,57}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \leq |\xi_{5,57}| \vee |\xi_{5,57}| \leq |\xi_{5,58}|) \\ \frac{1}{2} & \text{for } |\xi_{5,56}| \leq |\xi_{5,58}| \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,57}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,57}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,64} = \begin{cases} 0 & \text{for } (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,56}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,57}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,57}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,57}| \geq |\xi_{5,58}| \vee |\xi_{5,57}| \leq |\xi_{5,58}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{5,56}| \geq |\xi_{5,58}| \wedge |\xi_{5,56}| \leq |\xi_{5,57}| \\ \frac{1}{2} & \text{for } |\xi_{5,56}| \leq |\xi_{5,58}| \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,57}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,57}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{5,65} = \begin{cases} 0 & \text{for } (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,56}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,57}| \geq |\xi_{5,58}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,57}| \vee |\xi_{5,56}| \leq |\xi_{5,57}|) \wedge (|\xi_{5,56}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \wedge (|\xi_{5,57}| \geq |\xi_{5,58}| \vee |\xi_{5,56}| \leq |\xi_{5,58}|) \\ \frac{1}{2} & \text{for } |\xi_{5,57}| \geq |\xi_{5,58}| \wedge |\xi_{5,56}| \leq |\xi_{5,58}| \\ -\frac{1}{2} & \text{for } |\xi_{5,56}| \geq |\xi_{5,57}| \wedge |\xi_{5,56}| \leq |\xi_{5,58}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,66} = \begin{cases} -1 & \text{for } \xi_{5,59} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{5,67} = \begin{cases} -1 & \text{for } \xi_{5,60} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{5,68} = \begin{cases} 1 & \text{for } \xi_{5,66} = 1 \wedge \xi_{5,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,69} = \begin{cases} 0 & \text{for } \xi_{5,66} = 1 \wedge \xi_{5,67} = 1 \\ 1 & \text{for } \xi_{5,66} = 1 \wedge \xi_{5,67} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,70} = \begin{cases} 0 & \text{for } \xi_{5,66} = 1 \wedge (\xi_{5,66} = 1 \vee \xi_{5,67} = -1) \wedge (\xi_{5,66} = 1 \vee \xi_{5,67} = 1) \wedge (\xi_{5,67} = -1 \vee \xi_{5,67} = 1) \\ 1 & \text{for } \xi_{5,66} = -1 \wedge \xi_{5,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,71} = \begin{cases} 0 & \text{for } (\xi_{5,66} = -1 \vee \xi_{5,66} = 1) \wedge (\xi_{5,66} = 1 \vee \xi_{5,67} = 1) \wedge (\xi_{5,67} = -1 \vee \xi_{5,67} = 1) \wedge (\xi_{5,66} = -1 \vee \xi_{5,66} = 1 \vee \xi_{5,67} = -1) \wedge (\xi_{5,66} = -1 \vee \xi_{5,66} = 1 \vee \xi_{5,67} = 1) \wedge (\xi_{5,66} = -1 \vee \xi_{5,67} = -1 \vee \xi_{5,67} = 1) \wedge (\xi_{5,66} = 1 \vee \xi_{5,67} = -1 \vee \xi_{5,67} = 1) \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,72} = \cos\left(\frac{\pi\xi_{5,53}}{2}\right)$$

$$\xi_{5,73} = -\sin\left(\frac{\pi\xi_{5,53}}{2}\right)$$

$$\xi_{5,74} = \cos\left(\frac{\pi\xi_{5,54}}{2}\right)$$

$$\xi_{5,75} = -\sin\left(\frac{\pi\xi_{5,54}}{2}\right)$$

$$\xi_{5,76} = \xi_{5,72}\xi_{5,74}$$

$$\xi_{5,77} = \xi_{5,73}\xi_{5,74}$$

$$\xi_{5,78} = \xi_{5,72}\xi_{5,75}$$

$$\xi_{5,79} = -\xi_{5,73}\xi_{5,75}$$

$$\xi_{5,80} = \cos\left(\frac{\pi\xi_{5,55}}{2}\right)$$

$$\xi_{5,81} = -\sin\left(\frac{\pi\xi_{5,55}}{2}\right)$$

$$\xi_{5,82} = \xi_{5,76}\xi_{5,80} - \xi_{5,79}\xi_{5,81}$$

$$\xi_{5,83} = \xi_{5,77}\xi_{5,80} - \xi_{5,78}\xi_{5,81}$$

$$\xi_{5,84} = \xi_{5,77}\xi_{5,81} + \xi_{5,78}\xi_{5,80}$$

$$\xi_{5,85} = \xi_{5,76}\xi_{5,81} + \xi_{5,79}\xi_{5,80}$$

$$\xi_{5,86} = \xi_{5,68}\left(\xi_{5,62}\xi_{5,82} - \xi_{5,63}\xi_{5,83} - \xi_{5,64}\xi_{5,84} - \xi_{5,65}\xi_{5,85}\right) - \xi_{5,69}\left(\xi_{5,62}\xi_{5,83} + \xi_{5,63}\xi_{5,82} - \xi_{5,64}\xi_{5,85} + \xi_{5,65}\xi_{5,84}\right) - \xi_{5,70}\left(\xi_{5,62}\xi_{5,84} + \xi_{5,63}\xi_{5,85} + \xi_{5,64}\xi_{5,82} - \xi_{5,65}\xi_{5,83}\right) - \xi_{5,71}\left(\xi_{5,62}\xi_{5,85} - \xi_{5,63}\xi_{5,84} + \xi_{5,64}\xi_{5,83} + \xi_{5,65}\xi_{5,82}\right)$$

$$\xi_{5,87} = \xi_{5,68}\left(\xi_{5,62}\xi_{5,83} + \xi_{5,63}\xi_{5,82} - \xi_{5,64}\xi_{5,85} + \xi_{5,65}\xi_{5,84}\right) + \xi_{5,69}\left(\xi_{5,62}\xi_{5,82} - \xi_{5,63}\xi_{5,83} - \xi_{5,64}\xi_{5,84} - \xi_{5,65}\xi_{5,85}\right) - \xi_{5,70}\left(\xi_{5,62}\xi_{5,85} - \xi_{5,63}\xi_{5,84} + \xi_{5,64}\xi_{5,83} + \xi_{5,65}\xi_{5,82}\right) + \xi_{5,71}\left(\xi_{5,62}\xi_{5,84} + \xi_{5,63}\xi_{5,85} + \xi_{5,64}\xi_{5,82} - \xi_{5,65}\xi_{5,83}\right)$$

$$\xi_{5,88} = \xi_{5,68}\left(\xi_{5,62}\xi_{5,84} + \xi_{5,63}\xi_{5,85} + \xi_{5,64}\xi_{5,82} - \xi_{5,65}\xi_{5,83}\right) + \xi_{5,69}\left(\xi_{5,62}\xi_{5,85} - \xi_{5,63}\xi_{5,84} + \xi_{5,64}\xi_{5,83} + \xi_{5,65}\xi_{5,82}\right) + \xi_{5,70}\left(\xi_{5,62}\xi_{5,82} - \xi_{5,63}\xi_{5,83} - \xi_{5,64}\xi_{5,84} - \xi_{5,65}\xi_{5,85}\right) - \xi_{5,71}\left(\xi_{5,62}\xi_{5,83} + \xi_{5,63}\xi_{5,82} - \xi_{5,64}\xi_{5,85} + \xi_{5,65}\xi_{5,84}\right)$$

$$\xi_{5,89} = \xi_{5,68}\left(\xi_{5,62}\xi_{5,85} - \xi_{5,63}\xi_{5,84} + \xi_{5,64}\xi_{5,83} + \xi_{5,65}\xi_{5,82}\right) - \xi_{5,69}\left(\xi_{5,62}\xi_{5,84} + \xi_{5,63}\xi_{5,85} + \xi_{5,64}\xi_{5,82} - \xi_{5,65}\xi_{5,83}\right) + \xi_{5,70}\left(\xi_{5,62}\xi_{5,83} + \xi_{5,63}\xi_{5,82} - \xi_{5,64}\xi_{5,85} + \xi_{5,65}\xi_{5,84}\right) + \xi_{5,71}\left(\xi_{5,62}\xi_{5,82} - \xi_{5,63}\xi_{5,83} - \xi_{5,64}\xi_{5,84} - \xi_{5,65}\xi_{5,85}\right)$$

$$\xi_{5,90} = \xi_{5,62}\xi_{5,82} - \xi_{5,63}\xi_{5,83} - \xi_{5,64}\xi_{5,84} - \xi_{5,65}\xi_{5,85}$$

$$\xi_{5,91} = \xi_{5,62}\xi_{5,83} + \xi_{5,63}\xi_{5,82} - \xi_{5,64}\xi_{5,85} + \xi_{5,65}\xi_{5,84}$$

$$\xi_{5,92} = \xi_{5,62}\xi_{5,84} + \xi_{5,63}\xi_{5,85} + \xi_{5,64}\xi_{5,82} - \xi_{5,65}\xi_{5,83}$$

$$\xi_{5,93} = \xi_{5,62}\xi_{5,85} - \xi_{5,63}\xi_{5,84} + \xi_{5,64}\xi_{5,83} + \xi_{5,65}\xi_{5,82}$$

$$\xi_{5,94} = \xi_{5,62}\xi_{5,68} - \xi_{5,63}\xi_{5,69} - \xi_{5,64}\xi_{5,70} - \xi_{5,65}\xi_{5,71}$$

$$\xi_{5,95} = -\xi_{5,62}\xi_{5,69} - \xi_{5,63}\xi_{5,68} - \xi_{5,64}\xi_{5,71} + \xi_{5,65}\xi_{5,70}$$

$$\xi_{5,96} = -\xi_{5,62}\xi_{5,70} + \xi_{5,63}\xi_{5,71} - \xi_{5,64}\xi_{5,68} - \xi_{5,65}\xi_{5,69}$$

$$\xi_{5,97} = -\xi_{5,62}\xi_{5,71} - \xi_{5,63}\xi_{5,70} + \xi_{5,64}\xi_{5,69} - \xi_{5,65}\xi_{5,68}$$

$$\xi_{5,98} = \xi_{5,59}\xi_{5,66}$$

$$\xi_{5,99} = \xi_{5,60}\xi_{5,67}$$

$$\xi_{5,100} = \xi_{5,61}\xi_{5,66}\xi_{5,67}$$

$$\xi_{5,101} = -\frac{\pi(\xi_{5,53}+\xi_{5,54}+\xi_{5,55})}{2}$$

$$\xi_{5,102} = \cos\left(2\xi_{5,98}\right)$$

$$\xi_{5,103} = \cos\left(2\xi_{5,99}\right)$$

$$\xi_{5,104} = \cos\left(2\xi_{5,100}\right)$$

$$\xi_{5,105} = \sqrt{\xi_{5,102}^2 - 2\xi_{5,102}\xi_{5,103}\xi_{5,104} + \xi_{5,103}^2 + \xi_{5,104}^2 - 1}$$

$$\xi_{5,106} = \xi_{5,102} - \xi_{5,103} + \xi_{5,104}$$

$$\xi_{5,107} = \sqrt{-\frac{\xi_{5,105}}{2} + \frac{\xi_{5,106}}{2} + \frac{1}{2}}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,108} = \sqrt{\frac{\xi_{5,105}}{2} - \frac{\xi_{5,106}}{2} + \frac{1}{2}}$$

$$\xi_{5,109} = \sqrt{-\frac{\xi_{5,105}}{2} - \frac{\xi_{5,106}}{2} + \frac{1}{2}}$$

$$\xi_{5,110} = \frac{(1-\xi_{5,103})(\xi_{5,102}+1)(\xi_{5,104}+1)}{2}$$

$$\xi_{5,111} = \begin{cases} -1 & \text{for } \xi_{5,100} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{5,112} = \begin{cases} 0 & \text{for } \xi_{5,107} = 0 \\ \frac{\sqrt{\xi_{5,110}\xi_{5,111}}}{\xi_{5,107}} & \text{otherwise} \end{cases}$$

$$\xi_{5,113} = \begin{cases} 1 & \text{for } \xi_{5,107} = 0 \\ \frac{\sqrt{\xi_{5,102}\xi_{5,103}\xi_{5,104}}}{\xi_{5,107}} & \text{otherwise} \end{cases}$$

$$\xi_{5,114} = 2\xi_{5,107}\xi_{5,108}\xi_{5,109}\xi_{5,113}$$

$$\xi_{5,115} = \sqrt{2}\xi_{5,113}\left(\xi_{5,107}^2\xi_{5,109} - \xi_{5,109}\right)$$

$$\xi_{5,116} = 2\xi_{5,107}\xi_{5,108}\xi_{5,109}\xi_{5,112}$$

$$\xi_{5,117} = \sqrt{2}\left(2\xi_{5,107}^2\xi_{5,109}\xi_{5,112} - \xi_{5,109}\xi_{5,112}\right)$$

$$\xi_{5,118} = \sqrt{2}\xi_{5,107}\xi_{5,108}\xi_{5,112}\xi_{5,113}$$

$$\xi_{5,119} = \sqrt{2}\xi_{5,107}\xi_{5,108}\left(-\xi_{5,109}^2 + \xi_{5,112}^2\right)$$

$$\xi_{5,120} = -\xi_{5,107}^2\xi_{5,109}^2 + \xi_{5,107}^2\xi_{5,112}^2 + \xi_{5,109}^2$$

$$\xi_{5,121} = \xi_{5,107}^2\xi_{5,109}^2 - \xi_{5,107}^2\xi_{5,112}^2 + \xi_{5,112}^2$$

$$\xi_{5,122} = 2\xi_{5,113}\left(\xi_{5,107}^2\xi_{5,112} - \frac{\xi_{5,112}}{2}\right)$$

$$\xi_{5,123} = -\sqrt{2}\xi_{5,107}\xi_{5,108}\xi_{5,112}\xi_{5,113}$$

$$\xi_{5,124} = -\xi_{5,112}\xi_{5,113}$$

$$\xi_{5,125} = \xi_{5,107}^2\xi_{5,109}^2 + \xi_{5,107}^2\xi_{5,112}^2 - \xi_{5,107}^2 - \xi_{5,109}^2 - \xi_{5,112}^2 + 1$$

$$\xi_{5,126} = -\sqrt{2}\left(\xi_{5,107}\xi_{5,108}\xi_{5,109}^2 + \xi_{5,107}\xi_{5,108}\xi_{5,112}^2 - \xi_{5,107}\xi_{5,108}\right)$$

$$\xi_{5,127} = \sqrt{2}\xi_{5,107}^2\xi_{5,109}\xi_{5,113}$$

$$\xi_{5,128} = \xi_{5,118}^2 - \xi_{5,121}^2 - \xi_{5,124}^2 - \xi_{5,127}^2$$

$$\xi_{5,129} = \xi_{5,118}\xi_{5,119} - \xi_{5,121}\xi_{5,122} - \xi_{5,124}\xi_{5,125}$$

$$\xi_{5,130} = \xi_{5,118}\xi_{5,120} - \xi_{5,121}\xi_{5,123} - \xi_{5,124}\xi_{5,126}$$

$$\xi_{5,131} = \xi_{5,114}^2 + \xi_{5,116}^2 + \xi_{5,119}^2 - \xi_{5,122}^2 - \xi_{5,125}^2$$

$$\xi_{5,132} = \xi_{5,114}\xi_{5,115} + \xi_{5,116}\xi_{5,117} + \xi_{5,119}\xi_{5,120} - \xi_{5,122}\xi_{5,123} - \xi_{5,125}\xi_{5,126}$$

$$\xi_{5,133} = \xi_{5,115}^2 + \xi_{5,117}^2 + \xi_{5,120}^2 - \xi_{5,123}^2 - \xi_{5,126}^2$$

$$\xi_{5,134} = \sin\left(2\xi_{5,100}\right)\sin\left(2\xi_{5,99}\right)\cos\left(2\xi_{5,98}\right)$$

$$\xi_{5,135} = \sin\left(2\xi_{5,100}\right)\sin\left(2\xi_{5,98}\right)\cos\left(2\xi_{5,99}\right)$$

$$\xi_{5,136} = \sin\left(2\xi_{5,98}\right)\sin\left(2\xi_{5,99}\right)\cos\left(2\xi_{5,100}\right)$$

$$\xi_{5,137} = \sin\left(2\xi_{5,98}\right)\cos\left(2\xi_{5,100}\right)\cos\left(2\xi_{5,99}\right)$$

$$\xi_{5,138} = \sin\left(2\xi_{5,99}\right)\cos\left(2\xi_{5,100}\right)\cos\left(2\xi_{5,98}\right)$$

$$\xi_{5,139} = \sin\left(2\xi_{5,100}\right)\cos\left(2\xi_{5,98}\right)\cos\left(2\xi_{5,99}\right)$$

$$\xi_{5,140} = \xi_{5,134}^2 - \xi_{5,137}^2$$

$$\xi_{5,141} = \xi_{5,135}^2 - \xi_{5,138}^2$$

$$\xi_{5,142} = \xi_{5,136}^2 - \xi_{5,139}^2$$

$$\xi_{5,143} = \frac{\xi_{5,128} - \xi_{5,141}}{\xi_{5,140} - \xi_{5,141}}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,144} = \frac{\xi_{5,129}}{\xi_{5,140}-\xi_{5,141}}$$

$$\xi_{5,145} = \frac{\xi_{5,130}}{\xi_{5,140}-\xi_{5,141}}$$

$$\xi_{5,146} = \frac{\xi_{5,131}-\xi_{5,141}}{\xi_{5,140}-\xi_{5,141}}$$

$$\xi_{5,147} = \frac{\xi_{5,132}}{\xi_{5,140}-\xi_{5,141}}$$

$$\xi_{5,148} = \frac{\xi_{5,133}-\xi_{5,141}}{\xi_{5,140}-\xi_{5,141}}$$

$$\xi_{5,149} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,143} & \text{otherwise} \end{cases}$$

$$\xi_{5,150} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,144} & \text{otherwise} \end{cases}$$

$$\xi_{5,151} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,145} & \text{otherwise} \end{cases}$$

$$\xi_{5,152} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,146} & \text{otherwise} \end{cases}$$

$$\xi_{5,153} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,147} & \text{otherwise} \end{cases}$$

$$\xi_{5,154} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,148} & \text{otherwise} \end{cases}$$

$$\xi_{5,155} = \frac{\xi_{5,128}-\xi_{5,142}}{\xi_{5,140}-\xi_{5,142}}$$

$$\xi_{5,156} = \frac{\xi_{5,129}}{\xi_{5,140}-\xi_{5,142}}$$

$$\xi_{5,157} = \frac{\xi_{5,130}}{\xi_{5,140}-\xi_{5,142}}$$

$$\xi_{5,158} = \frac{\xi_{5,131}-\xi_{5,142}}{\xi_{5,140}-\xi_{5,142}}$$

$$\xi_{5,159} = \frac{\xi_{5,132}}{\xi_{5,140}-\xi_{5,142}}$$

$$\xi_{5,160} = \frac{\xi_{5,133}-\xi_{5,142}}{\xi_{5,140}-\xi_{5,142}}$$

$$\xi_{5,161} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,142} \\ \xi_{5,155} & \text{otherwise} \end{cases}$$

$$\xi_{5,162} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,142} \\ \xi_{5,156} & \text{otherwise} \end{cases}$$

$$\xi_{5,163} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,142} \\ \xi_{5,157} & \text{otherwise} \end{cases}$$

$$\xi_{5,164} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,142} \\ \xi_{5,158} & \text{otherwise} \end{cases}$$

$$\xi_{5,165} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,142} \\ \xi_{5,159} & \text{otherwise} \end{cases}$$

$$\xi_{5,166} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,142} \\ \xi_{5,160} & \text{otherwise} \end{cases}$$

$$\xi_{5,167} = \xi_{5,149}\xi_{5,161} + \xi_{5,150}\xi_{5,162} + \xi_{5,151}\xi_{5,163}$$

$$\xi_{5,168} = \xi_{5,149}\xi_{5,162} + \xi_{5,150}\xi_{5,164} + \xi_{5,151}\xi_{5,165}$$

$$\xi_{5,169} = \xi_{5,149}\xi_{5,163} + \xi_{5,150}\xi_{5,165} + \xi_{5,151}\xi_{5,166}$$

$$\xi_{5,170} = \xi_{5,150}\xi_{5,161} + \xi_{5,152}\xi_{5,162} + \xi_{5,153}\xi_{5,163}$$

$$\xi_{5,171} = \xi_{5,150}\xi_{5,162} + \xi_{5,152}\xi_{5,164} + \xi_{5,153}\xi_{5,165}$$

$$\xi_{5,172} = \xi_{5,150}\xi_{5,163} + \xi_{5,152}\xi_{5,165} + \xi_{5,153}\xi_{5,166}$$

$$\xi_{5,173} = \xi_{5,151}\xi_{5,161} + \xi_{5,153}\xi_{5,162} + \xi_{5,154}\xi_{5,163}$$

$$\xi_{5,174} = \xi_{5,151}\xi_{5,162} + \xi_{5,153}\xi_{5,164} + \xi_{5,154}\xi_{5,165}$$

$$\xi_{5,175} = \xi_{5,151}\xi_{5,163} + \xi_{5,153}\xi_{5,165} + \xi_{5,154}\xi_{5,166}$$

$$\xi_{5,176} = \sqrt{\xi_{5,167}}$$

$$\xi_{5,177} = \frac{\xi_{5,167}}{\xi_{5,176}}$$

$$\xi_{5,178} = \frac{\xi_{5,170}}{\xi_{5,176}}$$

$$\xi_{5,179} = \frac{\xi_{5,173}}{\xi_{5,176}}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,180} = \sqrt{\xi_{5,171}}$$

$$\xi_{5,181} = \frac{\xi_{5,168}}{\xi_{5,180}}$$

$$\xi_{5,182} = \frac{\xi_{5,171}}{\xi_{5,180}}$$

$$\xi_{5,183} = \frac{\xi_{5,174}}{\xi_{5,180}}$$

$$\xi_{5,184} = \sqrt{\xi_{5,175}}$$

$$\xi_{5,185} = \frac{\xi_{5,169}}{\xi_{5,184}}$$

$$\xi_{5,186} = \frac{\xi_{5,172}}{\xi_{5,184}}$$

$$\xi_{5,187} = \frac{\xi_{5,175}}{\xi_{5,184}}$$

$$\xi_{5,188} = \begin{cases} \xi_{5,177} & \text{for } \xi_{5,167} \geq \xi_{5,171} \wedge \xi_{5,167} \geq \xi_{5,175} \\ \xi_{5,181} & \text{for } \xi_{5,171} \geq \xi_{5,175} \wedge \xi_{5,167} \leq \xi_{5,171} \\ \xi_{5,185} & \text{otherwise} \end{cases}$$

$$\xi_{5,189} = \begin{cases} \xi_{5,178} & \text{for } \xi_{5,167} \geq \xi_{5,171} \wedge \xi_{5,167} \geq \xi_{5,175} \\ \xi_{5,182} & \text{for } \xi_{5,171} \geq \xi_{5,175} \wedge \xi_{5,167} \leq \xi_{5,171} \\ \xi_{5,186} & \text{otherwise} \end{cases}$$

$$\xi_{5,190} = \begin{cases} \xi_{5,179} & \text{for } \xi_{5,167} \geq \xi_{5,171} \wedge \xi_{5,167} \geq \xi_{5,175} \\ \xi_{5,183} & \text{for } \xi_{5,171} \geq \xi_{5,175} \wedge \xi_{5,167} \leq \xi_{5,171} \\ \xi_{5,187} & \text{otherwise} \end{cases}$$

$$\xi_{5,191} = \frac{\xi_{5,128} - \xi_{5,140}}{-\xi_{5,140} + \xi_{5,141}}$$

$$\xi_{5,192} = \frac{\xi_{5,129}}{-\xi_{5,140} + \xi_{5,141}}$$

$$\xi_{5,193} = \frac{\xi_{5,130}}{-\xi_{5,140} + \xi_{5,141}}$$

$$\xi_{5,194} = \frac{\xi_{5,131} - \xi_{5,140}}{-\xi_{5,140} + \xi_{5,141}}$$

$$\xi_{5,195} = \frac{\xi_{5,132}}{-\xi_{5,140} + \xi_{5,141}}$$

$$\xi_{5,196} = \frac{\xi_{5,133} - \xi_{5,140}}{-\xi_{5,140} + \xi_{5,141}}$$

$$\xi_{5,197} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,191} & \text{otherwise} \end{cases}$$

$$\xi_{5,198} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,192} & \text{otherwise} \end{cases}$$

$$\xi_{5,199} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,193} & \text{otherwise} \end{cases}$$

$$\xi_{5,200} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,194} & \text{otherwise} \end{cases}$$

$$\xi_{5,201} = \begin{cases} 0 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,195} & \text{otherwise} \end{cases}$$

$$\xi_{5,202} = \begin{cases} 1 & \text{for } \xi_{5,140} = \xi_{5,141} \\ \xi_{5,196} & \text{otherwise} \end{cases}$$

$$\xi_{5,203} = \frac{\xi_{5,128} - \xi_{5,142}}{\xi_{5,141} - \xi_{5,142}}$$

$$\xi_{5,204} = \frac{\xi_{5,129}}{\xi_{5,141} - \xi_{5,142}}$$

$$\xi_{5,205} = \frac{\xi_{5,130}}{\xi_{5,141} - \xi_{5,142}}$$

$$\xi_{5,206} = \frac{\xi_{5,131} - \xi_{5,142}}{\xi_{5,141} - \xi_{5,142}}$$

$$\xi_{5,207} = \frac{\xi_{5,132}}{\xi_{5,141} - \xi_{5,142}}$$

$$\xi_{5,208} = \frac{\xi_{5,133} - \xi_{5,142}}{\xi_{5,141} - \xi_{5,142}}$$

$$\xi_{5,209} = \begin{cases} 1 & \text{for } \xi_{5,141} = \xi_{5,142} \\ \xi_{5,203} & \text{otherwise} \end{cases}$$

$$\xi_{5,210} = \begin{cases} 0 & \text{for } \xi_{5,141} = \xi_{5,142} \\ \xi_{5,204} & \text{otherwise} \end{cases}$$

$$\xi_{5,211} = \begin{cases} 0 & \text{for } \xi_{5,141} = \xi_{5,142} \\ \xi_{5,205} & \text{otherwise} \end{cases}$$

$$\xi_{5,212} = \begin{cases} 1 & \text{for } \xi_{5,141} = \xi_{5,142} \\ \xi_{5,206} & \text{otherwise} \end{cases}$$

$$\xi_{5,213} = \begin{cases} 0 & \text{for } \xi_{5,141} = \xi_{5,142} \\ \xi_{5,207} & \text{otherwise} \end{cases}$$

$$\xi_{5,214} = \begin{cases} 1 & \text{for } \xi_{5,141} = \xi_{5,142} \\ \xi_{5,208} & \text{otherwise} \end{cases}$$

$$\xi_{5,215} = \xi_{5,197}\xi_{5,209} + \xi_{5,198}\xi_{5,210} + \xi_{5,199}\xi_{5,211}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,216} = \xi_{5,197}\xi_{5,210} + \xi_{5,198}\xi_{5,212} + \xi_{5,199}\xi_{5,213}$$

$$\xi_{5,217} = \xi_{5,197}\xi_{5,211} + \xi_{5,198}\xi_{5,213} + \xi_{5,199}\xi_{5,214}$$

$$\xi_{5,218} = \xi_{5,198}\xi_{5,209} + \xi_{5,200}\xi_{5,210} + \xi_{5,201}\xi_{5,211}$$

$$\xi_{5,219} = \xi_{5,198}\xi_{5,210} + \xi_{5,200}\xi_{5,212} + \xi_{5,201}\xi_{5,213}$$

$$\xi_{5,220} = \xi_{5,198}\xi_{5,211} + \xi_{5,200}\xi_{5,213} + \xi_{5,201}\xi_{5,214}$$

$$\xi_{5,221} = \xi_{5,199}\xi_{5,209} + \xi_{5,201}\xi_{5,210} + \xi_{5,202}\xi_{5,211}$$

$$\xi_{5,222} = \xi_{5,199}\xi_{5,210} + \xi_{5,201}\xi_{5,212} + \xi_{5,202}\xi_{5,213}$$

$$\xi_{5,223} = \xi_{5,199}\xi_{5,211} + \xi_{5,201}\xi_{5,213} + \xi_{5,202}\xi_{5,214}$$

$$\xi_{5,224} = \begin{cases} \xi_{5,188}^2 & \text{for } \xi_{5,140} = \xi_{5,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,225} = \begin{cases} \xi_{5,188}\xi_{5,189} & \text{for } \xi_{5,140} = \xi_{5,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,226} = \begin{cases} \xi_{5,188}\xi_{5,190} & \text{for } \xi_{5,140} = \xi_{5,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,227} = \begin{cases} \xi_{5,189}^2 & \text{for } \xi_{5,140} = \xi_{5,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,228} = \begin{cases} \xi_{5,189}\xi_{5,190} & \text{for } \xi_{5,140} = \xi_{5,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,229} = \begin{cases} \xi_{5,190}^2 & \text{for } \xi_{5,140} = \xi_{5,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{5,230} = \xi_{5,215} - \xi_{5,224}$$

$$\xi_{5,231} = \xi_{5,216} - \xi_{5,225}$$

$$\xi_{5,232} = \xi_{5,217} - \xi_{5,226}$$

$$\xi_{5,233} = \xi_{5,218} - \xi_{5,225}$$

$$\xi_{5,234} = \xi_{5,219} - \xi_{5,227}$$

$$\xi_{5,235} = \xi_{5,220} - \xi_{5,228}$$

$$\xi_{5,236} = \xi_{5,221} - \xi_{5,226}$$

$$\xi_{5,237} = \xi_{5,222} - \xi_{5,228}$$

$$\xi_{5,238} = \xi_{5,223} - \xi_{5,229}$$

$$\xi_{5,239} = \sqrt{\xi_{5,230}}$$

$$\xi_{5,240} = \frac{\xi_{5,230}}{\xi_{5,239}}$$

$$\xi_{5,241} = \frac{\xi_{5,233}}{\xi_{5,239}}$$

$$\xi_{5,242} = \frac{\xi_{5,236}}{\xi_{5,239}}$$

$$\xi_{5,243} = \sqrt{\xi_{5,234}}$$

$$\xi_{5,244} = \frac{\xi_{5,231}}{\xi_{5,243}}$$

$$\xi_{5,245} = \frac{\xi_{5,234}}{\xi_{5,243}}$$

$$\xi_{5,246} = \frac{\xi_{5,237}}{\xi_{5,243}}$$

$$\xi_{5,247} = \sqrt{\xi_{5,238}}$$

$$\xi_{5,248} = \frac{\xi_{5,232}}{\xi_{5,247}}$$

$$\xi_{5,249} = \frac{\xi_{5,235}}{\xi_{5,247}}$$

$$\xi_{5,250} = \frac{\xi_{5,238}}{\xi_{5,247}}$$

$$\xi_{5,251} = \begin{cases} \xi_{5,240} & \text{for } \xi_{5,230} \geq \xi_{5,234} \wedge \xi_{5,230} \geq \xi_{5,238} \\ \xi_{5,244} & \text{for } \xi_{5,234} \geq \xi_{5,238} \wedge \xi_{5,230} \leq \xi_{5,234} \\ \xi_{5,248} & \text{otherwise} \end{cases}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,252} = \begin{cases} \xi_{5,241} & \text{for } \xi_{5,230} \geq \xi_{5,234} \wedge \xi_{5,230} \geq \xi_{5,238} \\ \xi_{5,245} & \text{for } \xi_{5,234} \geq \xi_{5,238} \wedge \xi_{5,230} \leq \xi_{5,234} \\ \xi_{5,249} & \text{otherwise} \end{cases}$$

$$\xi_{5,253} = \begin{cases} \xi_{5,242} & \text{for } \xi_{5,230} \geq \xi_{5,234} \wedge \xi_{5,230} \geq \xi_{5,238} \\ \xi_{5,246} & \text{for } \xi_{5,234} \geq \xi_{5,238} \wedge \xi_{5,230} \leq \xi_{5,234} \\ \xi_{5,250} & \text{otherwise} \end{cases}$$

$$\xi_{5,254} = \xi_{5,189}\xi_{5,253} - \xi_{5,190}\xi_{5,252}$$

$$\xi_{5,255} = -\xi_{5,188}\xi_{5,253} + \xi_{5,190}\xi_{5,251}$$

$$\xi_{5,256} = \xi_{5,188}\xi_{5,252} - \xi_{5,189}\xi_{5,251}$$

$$\xi_{5,257} = \xi_{5,134}^2 + \xi_{5,137}^2$$

$$\xi_{5,258} = \xi_{5,135}^2 + \xi_{5,138}^2$$

$$\xi_{5,259} = \xi_{5,136}^2 + \xi_{5,139}^2$$

$$\xi_{5,260} = \frac{\xi_{5,121}\xi_{5,137}\xi_{5,188} + \xi_{5,189}(\xi_{5,114}\xi_{5,134} + \xi_{5,122}\xi_{5,137}) + \xi_{5,190}(\xi_{5,115}\xi_{5,134} + \xi_{5,123}\xi_{5,137})}{\xi_{5,257}}$$

$$\xi_{5,261} = \frac{\xi_{5,124}\xi_{5,137}\xi_{5,188} + \xi_{5,189}(\xi_{5,116}\xi_{5,134} + \xi_{5,125}\xi_{5,137}) + \xi_{5,190}(\xi_{5,117}\xi_{5,134} + \xi_{5,126}\xi_{5,137})}{\xi_{5,257}}$$

$$\xi_{5,262} = \frac{\xi_{5,119}\xi_{5,134}\xi_{5,189} + \xi_{5,120}\xi_{5,134}\xi_{5,190} + \xi_{5,188}(\xi_{5,118}\xi_{5,134} + \xi_{5,127}\xi_{5,137})}{\xi_{5,257}}$$

$$\xi_{5,263} = \frac{\xi_{5,121}\xi_{5,138}\xi_{5,251} + \xi_{5,252}(\xi_{5,114}\xi_{5,135} + \xi_{5,122}\xi_{5,138}) + \xi_{5,253}(\xi_{5,115}\xi_{5,135} + \xi_{5,123}\xi_{5,138})}{\xi_{5,258}}$$

$$\xi_{5,264} = \frac{\xi_{5,124}\xi_{5,138}\xi_{5,251} + \xi_{5,252}(\xi_{5,116}\xi_{5,135} + \xi_{5,125}\xi_{5,138}) + \xi_{5,253}(\xi_{5,117}\xi_{5,135} + \xi_{5,126}\xi_{5,138})}{\xi_{5,258}}$$

$$\xi_{5,265} = \frac{\xi_{5,119}\xi_{5,135}\xi_{5,252} + \xi_{5,120}\xi_{5,135}\xi_{5,253} + \xi_{5,251}(\xi_{5,118}\xi_{5,135} + \xi_{5,127}\xi_{5,138})}{\xi_{5,258}}$$

$$\xi_{5,266} = \frac{\xi_{5,121}\xi_{5,139}\xi_{5,254} + \xi_{5,255}(\xi_{5,114}\xi_{5,136} + \xi_{5,122}\xi_{5,139}) + \xi_{5,256}(\xi_{5,115}\xi_{5,136} + \xi_{5,123}\xi_{5,139})}{\xi_{5,259}}$$

$$\xi_{5,267} = \frac{\xi_{5,124}\xi_{5,139}\xi_{5,254} + \xi_{5,255}(\xi_{5,116}\xi_{5,136} + \xi_{5,125}\xi_{5,139}) + \xi_{5,256}(\xi_{5,117}\xi_{5,136} + \xi_{5,126}\xi_{5,139})}{\xi_{5,259}}$$

$$\xi_{5,268} = \frac{\xi_{5,119}\xi_{5,136}\xi_{5,255} + \xi_{5,120}\xi_{5,136}\xi_{5,256} + \xi_{5,254}(\xi_{5,118}\xi_{5,136} + \xi_{5,127}\xi_{5,139})}{\xi_{5,259}}$$

$$\xi_{5,269} = 1 - \xi_{5,260}^2$$

$$\xi_{5,270} = -\xi_{5,260}\xi_{5,261}$$

$$\xi_{5,271} = -\xi_{5,260}\xi_{5,262}$$

$$\xi_{5,272} = 1 - \xi_{5,261}^2$$

$$\xi_{5,273} = -\xi_{5,261}\xi_{5,262}$$

$$\xi_{5,274} = 1 - \xi_{5,262}^2$$

$$\xi_{5,275} = \xi_{5,269}^2 + \xi_{5,270}^2 + \xi_{5,271}^2$$

$$\xi_{5,276} = \xi_{5,270}^2 + \xi_{5,272}^2 + \xi_{5,273}^2$$

$$\xi_{5,277} = \xi_{5,271}^2 + \xi_{5,273}^2 + \xi_{5,274}^2$$

$$\xi_{5,278} = \sqrt{\xi_{5,275}}$$

$$\xi_{5,279} = \frac{\xi_{5,269}}{\xi_{5,278}}$$

$$\xi_{5,280} = \frac{\xi_{5,270}}{\xi_{5,278}}$$

$$\xi_{5,281} = \frac{\xi_{5,271}}{\xi_{5,278}}$$

$$\xi_{5,282} = \sqrt{\xi_{5,276}}$$

$$\xi_{5,283} = \frac{\xi_{5,272}}{\xi_{5,282}}$$

$$\xi_{5,284} = \frac{\xi_{5,270}}{\xi_{5,282}}$$

$$\xi_{5,285} = \frac{\xi_{5,273}}{\xi_{5,282}}$$

$$\xi_{5,286} = \sqrt{\xi_{5,277}}$$

$$\xi_{5,287} = \frac{\xi_{5,274}}{\xi_{5,286}}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,288} = \frac{\xi_{5,271}}{\xi_{5,286}}$$

$$\xi_{5,289} = \frac{\xi_{5,273}}{\xi_{5,286}}$$

$$\xi_{5,290} = \begin{cases} \xi_{5,279} & \text{for } \xi_{5,275} \geq \xi_{5,276} \wedge \xi_{5,275} \geq \xi_{5,277} \\ \xi_{5,284} & \text{for } \xi_{5,276} \geq \xi_{5,277} \wedge \xi_{5,275} \leq \xi_{5,276} \\ \xi_{5,288} & \text{otherwise} \end{cases}$$

$$\xi_{5,291} = \begin{cases} \xi_{5,280} & \text{for } \xi_{5,275} \geq \xi_{5,276} \wedge \xi_{5,275} \geq \xi_{5,277} \\ \xi_{5,283} & \text{for } \xi_{5,276} \geq \xi_{5,277} \wedge \xi_{5,275} \leq \xi_{5,276} \\ \xi_{5,289} & \text{otherwise} \end{cases}$$

$$\xi_{5,292} = \begin{cases} \xi_{5,281} & \text{for } \xi_{5,275} \geq \xi_{5,276} \wedge \xi_{5,275} \geq \xi_{5,277} \\ \xi_{5,285} & \text{for } \xi_{5,276} \geq \xi_{5,277} \wedge \xi_{5,275} \leq \xi_{5,276} \\ \xi_{5,287} & \text{otherwise} \end{cases}$$

$$\xi_{5,293} = \xi_{5,261}\xi_{5,292} - \xi_{5,262}\xi_{5,291}$$

$$\xi_{5,294} = -\xi_{5,260}\xi_{5,292} + \xi_{5,262}\xi_{5,290}$$

$$\xi_{5,295} = \xi_{5,260}\xi_{5,291} - \xi_{5,261}\xi_{5,290}$$

$$\xi_{5,296} = 1 - \xi_{5,263}^2$$

$$\xi_{5,297} = -\xi_{5,263}\xi_{5,264}$$

$$\xi_{5,298} = -\xi_{5,263}\xi_{5,265}$$

$$\xi_{5,299} = 1 - \xi_{5,264}^2$$

$$\xi_{5,300} = -\xi_{5,264}\xi_{5,265}$$

$$\xi_{5,301} = 1 - \xi_{5,265}^2$$

$$\xi_{5,302} = \xi_{5,296}^2 + \xi_{5,297}^2 + \xi_{5,298}^2$$

$$\xi_{5,303} = \xi_{5,297}^2 + \xi_{5,299}^2 + \xi_{5,300}^2$$

$$\xi_{5,304} = \xi_{5,298}^2 + \xi_{5,300}^2 + \xi_{5,301}^2$$

$$\xi_{5,305} = \sqrt{\xi_{5,302}}$$

$$\xi_{5,306} = \frac{\xi_{5,296}}{\xi_{5,305}}$$

$$\xi_{5,307} = \frac{\xi_{5,297}}{\xi_{5,305}}$$

$$\xi_{5,308} = \frac{\xi_{5,298}}{\xi_{5,305}}$$

$$\xi_{5,309} = \sqrt{\xi_{5,303}}$$

$$\xi_{5,310} = \frac{\xi_{5,299}}{\xi_{5,309}}$$

$$\xi_{5,311} = \frac{\xi_{5,297}}{\xi_{5,309}}$$

$$\xi_{5,312} = \frac{\xi_{5,300}}{\xi_{5,309}}$$

$$\xi_{5,313} = \sqrt{\xi_{5,304}}$$

$$\xi_{5,314} = \frac{\xi_{5,301}}{\xi_{5,313}}$$

$$\xi_{5,315} = \frac{\xi_{5,298}}{\xi_{5,313}}$$

$$\xi_{5,316} = \frac{\xi_{5,300}}{\xi_{5,313}}$$

$$\xi_{5,317} = \begin{cases} \xi_{5,306} & \text{for } \xi_{5,302} \geq \xi_{5,303} \wedge \xi_{5,302} \geq \xi_{5,304} \\ \xi_{5,311} & \text{for } \xi_{5,303} \geq \xi_{5,304} \wedge \xi_{5,302} \leq \xi_{5,303} \\ \xi_{5,315} & \text{otherwise} \end{cases}$$

$$\xi_{5,318} = \begin{cases} \xi_{5,307} & \text{for } \xi_{5,302} \geq \xi_{5,303} \wedge \xi_{5,302} \geq \xi_{5,304} \\ \xi_{5,310} & \text{for } \xi_{5,303} \geq \xi_{5,304} \wedge \xi_{5,302} \leq \xi_{5,303} \\ \xi_{5,316} & \text{otherwise} \end{cases}$$

$$\xi_{5,319} = \begin{cases} \xi_{5,308} & \text{for } \xi_{5,302} \geq \xi_{5,303} \wedge \xi_{5,302} \geq \xi_{5,304} \\ \xi_{5,312} & \text{for } \xi_{5,303} \geq \xi_{5,304} \wedge \xi_{5,302} \leq \xi_{5,303} \\ \xi_{5,314} & \text{otherwise} \end{cases}$$

$$\xi_{5,320} = \xi_{5,264}\xi_{5,319} - \xi_{5,265}\xi_{5,318}$$

$$\xi_{5,321} = -\xi_{5,263}\xi_{5,319} + \xi_{5,265}\xi_{5,317}$$

$$\xi_{5,322} = \xi_{5,263}\xi_{5,318} - \xi_{5,264}\xi_{5,317}$$

$$\xi_{5,323} = \xi_{5,261}\xi_{5,265} - \xi_{5,262}\xi_{5,264}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,324} = -\xi_{5,260}\xi_{5,265} + \xi_{5,262}\xi_{5,263}$$

$$\xi_{5,325} = \xi_{5,260}\xi_{5,264} - \xi_{5,261}\xi_{5,263}$$

$$\xi_{5,326} = 1 - \xi_{5,266}^2$$

$$\xi_{5,327} = -\xi_{5,266}\xi_{5,267}$$

$$\xi_{5,328} = -\xi_{5,266}\xi_{5,268}$$

$$\xi_{5,329} = 1 - \xi_{5,267}^2$$

$$\xi_{5,330} = -\xi_{5,267}\xi_{5,268}$$

$$\xi_{5,331} = 1 - \xi_{5,268}^2$$

$$\xi_{5,332} = \xi_{5,326}^2 + \xi_{5,327}^2 + \xi_{5,328}^2$$

$$\xi_{5,333} = \xi_{5,327}^2 + \xi_{5,329}^2 + \xi_{5,330}^2$$

$$\xi_{5,334} = \xi_{5,328}^2 + \xi_{5,330}^2 + \xi_{5,331}^2$$

$$\xi_{5,335} = \sqrt{\xi_{5,332}}$$

$$\xi_{5,336} = \frac{\xi_{5,326}}{\xi_{5,335}}$$

$$\xi_{5,337} = \frac{\xi_{5,327}}{\xi_{5,335}}$$

$$\xi_{5,338} = \frac{\xi_{5,328}}{\xi_{5,335}}$$

$$\xi_{5,339} = \sqrt{\xi_{5,333}}$$

$$\xi_{5,340} = \frac{\xi_{5,329}}{\xi_{5,339}}$$

$$\xi_{5,341} = \frac{\xi_{5,327}}{\xi_{5,339}}$$

$$\xi_{5,342} = \frac{\xi_{5,330}}{\xi_{5,339}}$$

$$\xi_{5,343} = \sqrt{\xi_{5,334}}$$

$$\xi_{5,344} = \frac{\xi_{5,331}}{\xi_{5,343}}$$

$$\xi_{5,345} = \frac{\xi_{5,328}}{\xi_{5,343}}$$

$$\xi_{5,346} = \frac{\xi_{5,330}}{\xi_{5,343}}$$

$$\xi_{5,347} = \begin{cases} \xi_{5,336} & \text{for } \xi_{5,332} \geq \xi_{5,333} \wedge \xi_{5,332} \geq \xi_{5,334} \\ \xi_{5,341} & \text{for } \xi_{5,333} \geq \xi_{5,334} \wedge \xi_{5,332} \leq \xi_{5,333} \\ \xi_{5,345} & \text{otherwise} \end{cases}$$

$$\xi_{5,348} = \begin{cases} \xi_{5,337} & \text{for } \xi_{5,332} \geq \xi_{5,333} \wedge \xi_{5,332} \geq \xi_{5,334} \\ \xi_{5,340} & \text{for } \xi_{5,333} \geq \xi_{5,334} \wedge \xi_{5,332} \leq \xi_{5,333} \\ \xi_{5,346} & \text{otherwise} \end{cases}$$

$$\xi_{5,349} = \begin{cases} \xi_{5,338} & \text{for } \xi_{5,332} \geq \xi_{5,333} \wedge \xi_{5,332} \geq \xi_{5,334} \\ \xi_{5,342} & \text{for } \xi_{5,333} \geq \xi_{5,334} \wedge \xi_{5,332} \leq \xi_{5,333} \\ \xi_{5,344} & \text{otherwise} \end{cases}$$

$$\xi_{5,350} = \xi_{5,267}\xi_{5,349} - \xi_{5,268}\xi_{5,348}$$

$$\xi_{5,351} = -\xi_{5,266}\xi_{5,349} + \xi_{5,268}\xi_{5,347}$$

$$\xi_{5,352} = \xi_{5,266}\xi_{5,348} - \xi_{5,267}\xi_{5,347}$$

$$\xi_{5,353} = -\xi_{5,261}\xi_{5,268} + \xi_{5,262}\xi_{5,267}$$

$$\xi_{5,354} = \xi_{5,260}\xi_{5,268} - \xi_{5,262}\xi_{5,266}$$

$$\xi_{5,355} = -\xi_{5,260}\xi_{5,267} + \xi_{5,261}\xi_{5,266}$$

$$\xi_{5,356} = \xi_{5,264}\xi_{5,268} - \xi_{5,265}\xi_{5,267}$$

$$\xi_{5,357} = -\xi_{5,263}\xi_{5,268} + \xi_{5,265}\xi_{5,266}$$

$$\xi_{5,358} = \xi_{5,263}\xi_{5,267} - \xi_{5,264}\xi_{5,266}$$

$$\xi_{5,359} = \begin{cases} 1 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,260} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,320} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,260} & \text{for } \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,347} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,260} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,356} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,260} & \text{for } \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,360} = \begin{cases} 0 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,290} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,263} & \text{for } \xi_{5,259} = 0 \wedge (\xi_{5,257} = 0 \vee \xi_{5,259} = 0) \wedge (\xi_{5,257} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,259} = 0 \vee \xi_{5,257} \neq 0) \wedge (\xi_{5,259} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,258} \neq 0) \wedge \xi_{5,258} \neq 0 \\ \xi_{5,350} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,353} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,263} & \text{for } (\xi_{5,257} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} = 0 \vee \xi_{5,259} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,259} \neq 0) \wedge (\xi_{5,258} \neq 0 \vee \xi_{5,259} \neq 0) \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,361} = \begin{cases} 0 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,261} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,321} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,261} & \text{for } \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,348} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,261} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,357} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,261} & \text{for } \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,362} = \begin{cases} 0 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,261} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,321} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,261} & \text{for } \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,348} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,261} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,357} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,261} & \text{for } \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,363} = \begin{cases} 1 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,291} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,264} & \text{for } \xi_{5,259} = 0 \wedge (\xi_{5,257} = 0 \vee \xi_{5,259} = 0) \wedge (\xi_{5,257} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,259} = 0 \vee \xi_{5,257} \neq 0) \wedge (\xi_{5,259} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,258} \neq 0) \wedge \xi_{5,258} \neq 0 \\ \xi_{5,351} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,354} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,264} & \text{for } (\xi_{5,257} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} = 0 \vee \xi_{5,259} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,259} \neq 0) \wedge (\xi_{5,258} \neq 0 \vee \xi_{5,259} \neq 0) \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,364} = \begin{cases} 0 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,262} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,322} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,262} & \text{for } \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,349} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,262} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,358} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,262} & \text{for } \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,365} = \begin{cases} 0 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,262} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,322} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,262} & \text{for } \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,349} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,262} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,358} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,262} & \text{for } \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,366} = \begin{cases} 0 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,292} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,265} & \text{for } \xi_{5,259} = 0 \wedge (\xi_{5,257} = 0 \vee \xi_{5,259} = 0) \wedge (\xi_{5,257} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,259} = 0 \vee \xi_{5,257} \neq 0) \wedge (\xi_{5,259} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,258} \neq 0) \wedge \xi_{5,258} \neq 0 \\ \xi_{5,352} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,355} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,265} & \text{for } (\xi_{5,257} = 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} = 0 \vee \xi_{5,259} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,258} \neq 0) \wedge (\xi_{5,257} \neq 0 \vee \xi_{5,259} \neq 0) \wedge (\xi_{5,258} \neq 0 \vee \xi_{5,259} \neq 0) \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,367} = \begin{cases} 1 & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \\ \xi_{5,291} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \\ \xi_{5,264} & \text{for } \xi_{5,259} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \\ \xi_{5,351} & \text{for } \xi_{5,257} = 0 \wedge \xi_{5,258} = 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,354} & \text{for } \xi_{5,258} = 0 \wedge \xi_{5,257} \neq 0 \wedge \xi_{5,259} \neq 0 \\ \xi_{5,264} & \text{for } \xi_{5,257} \neq 0 \wedge \xi_{5,258} \neq 0 \wedge \xi_{5,259} \neq 0 \end{cases}$$

$$\xi_{5,368} = \frac{\xi_{5,359}}{4} + \frac{\xi_{5,363}}{4} + \frac{\xi_{5,367}}{4} + \frac{1}{4}$$

$$\xi_{5,369} = \frac{\xi_{5,359}}{4} - \frac{\xi_{5,363}}{4} - \frac{\xi_{5,367}}{4} + \frac{1}{4}$$

$$\xi_{5,370} = -\frac{\xi_{5,359}}{4} + \frac{\xi_{5,363}}{4} - \frac{\xi_{5,367}}{4} + \frac{1}{4}$$

$$\xi_{5,371} = -\frac{\xi_{5,359}}{4} - \frac{\xi_{5,363}}{4} + \frac{\xi_{5,367}}{4} + \frac{1}{4}$$

$$\xi_{5,372} = \sqrt{\xi_{5,368}}$$

$$\xi_{5,373} = \frac{-\xi_{5,364}+\xi_{5,366}}{4\xi_{5,372}}$$

$$\xi_{5,374} = \frac{\xi_{5,361}-\xi_{5,365}}{4\xi_{5,372}}$$

$$\xi_{5,375} = \frac{-\xi_{5,360}+\xi_{5,362}}{4\xi_{5,372}}$$

$$\xi_{5,376} = \sqrt{\xi_{5,369}}$$

$$\xi_{5,377} = \frac{-\xi_{5,364}+\xi_{5,366}}{4\xi_{5,376}}$$

$$\xi_{5,378} = \frac{\xi_{5,360}+\xi_{5,362}}{4\xi_{5,376}}$$

$$\xi_{5,379} = \frac{\xi_{5,361}+\xi_{5,365}}{4\xi_{5,376}}$$

$$\xi_{5,380} = \sqrt{\xi_{5,370}}$$

$$\xi_{5,381} = \frac{\xi_{5,361}-\xi_{5,365}}{4\xi_{5,380}}$$

$$\xi_{5,382} = \frac{\xi_{5,360}+\xi_{5,362}}{4\xi_{5,380}}$$

$$\xi_{5,383} = \frac{\xi_{5,364}+\xi_{5,366}}{4\xi_{5,380}}$$

$$\xi_{5,384} = \sqrt{\xi_{5,371}}$$

$$\xi_{5,385} = \frac{-\xi_{5,360}+\xi_{5,362}}{4\xi_{5,384}}$$

$$\xi_{5,386} = \frac{\xi_{5,361}+\xi_{5,365}}{4\xi_{5,384}}$$

$$\xi_{5,387} = \frac{\xi_{5,364}+\xi_{5,366}}{4\xi_{5,384}}$$

$$\xi_{5,388} = \begin{cases} \xi_{5,372} & \text{for } \xi_{5,368} \geq \xi_{5,369} \wedge \xi_{5,368} \geq \xi_{5,370} \wedge \xi_{5,368} \geq \xi_{5,371} \\ \xi_{5,377} & \text{for } \xi_{5,369} \geq \xi_{5,370} \wedge \xi_{5,369} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,369} \\ \xi_{5,381} & \text{for } \xi_{5,370} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,370} \wedge \xi_{5,369} \leq \xi_{5,370} \\ \xi_{5,385} & \text{otherwise} \end{cases}$$

$$\xi_{5,389} = \begin{cases} \xi_{5,373} & \text{for } \xi_{5,368} \geq \xi_{5,369} \wedge \xi_{5,368} \geq \xi_{5,370} \wedge \xi_{5,368} \geq \xi_{5,371} \\ \xi_{5,376} & \text{for } \xi_{5,369} \geq \xi_{5,370} \wedge \xi_{5,369} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,369} \\ \xi_{5,382} & \text{for } \xi_{5,370} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,370} \wedge \xi_{5,369} \leq \xi_{5,370} \\ \xi_{5,386} & \text{otherwise} \end{cases}$$

$$\xi_{5,390} = \begin{cases} \xi_{5,374} & \text{for } \xi_{5,368} \geq \xi_{5,369} \wedge \xi_{5,368} \geq \xi_{5,370} \wedge \xi_{5,368} \geq \xi_{5,371} \\ \xi_{5,378} & \text{for } \xi_{5,369} \geq \xi_{5,370} \wedge \xi_{5,369} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,369} \\ \xi_{5,380} & \text{for } \xi_{5,370} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,370} \wedge \xi_{5,369} \leq \xi_{5,370} \\ \xi_{5,387} & \text{otherwise} \end{cases}$$

$$\xi_{5,391} = \begin{cases} \xi_{5,375} & \text{for } \xi_{5,368} \geq \xi_{5,369} \wedge \xi_{5,368} \geq \xi_{5,370} \wedge \xi_{5,368} \geq \xi_{5,371} \\ \xi_{5,379} & \text{for } \xi_{5,369} \geq \xi_{5,370} \wedge \xi_{5,369} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,369} \\ \xi_{5,383} & \text{for } \xi_{5,370} \geq \xi_{5,371} \wedge \xi_{5,368} \leq \xi_{5,370} \wedge \xi_{5,369} \leq \xi_{5,370} \\ \xi_{5,384} & \text{otherwise} \end{cases}$$

$$\xi_{5,392} = \frac{\xi_{5,188}}{4} + \frac{\xi_{5,252}}{4} + \frac{\xi_{5,256}}{4} + \frac{1}{4}$$

$$\xi_{5,393} = \frac{\xi_{5,188}}{4} - \frac{\xi_{5,252}}{4} - \frac{\xi_{5,256}}{4} + \frac{1}{4}$$

$$\xi_{5,394} = -\frac{\xi_{5,188}}{4} + \frac{\xi_{5,252}}{4} - \frac{\xi_{5,256}}{4} + \frac{1}{4}$$

$$\xi_{5,395} = -\frac{\xi_{5,188}}{4} - \frac{\xi_{5,252}}{4} + \frac{\xi_{5,256}}{4} + \frac{1}{4}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,396} = \sqrt{\xi_{5,392}}$$

$$\xi_{5,397} = \frac{\xi_{5,253} - \xi_{5,255}}{4\xi_{5,396}}$$

$$\xi_{5,398} = \frac{-\xi_{5,190} + \xi_{5,254}}{4\xi_{5,396}}$$

$$\xi_{5,399} = \frac{\xi_{5,189} - \xi_{5,251}}{4\xi_{5,396}}$$

$$\xi_{5,400} = \sqrt{\xi_{5,393}}$$

$$\xi_{5,401} = \frac{\xi_{5,253} - \xi_{5,255}}{4\xi_{5,400}}$$

$$\xi_{5,402} = \frac{\xi_{5,189} + \xi_{5,251}}{4\xi_{5,400}}$$

$$\xi_{5,403} = \frac{\xi_{5,190} + \xi_{5,254}}{4\xi_{5,400}}$$

$$\xi_{5,404} = \sqrt{\xi_{5,394}}$$

$$\xi_{5,405} = \frac{-\xi_{5,190} + \xi_{5,254}}{4\xi_{5,404}}$$

$$\xi_{5,406} = \frac{\xi_{5,189} + \xi_{5,251}}{4\xi_{5,404}}$$

$$\xi_{5,407} = \frac{\xi_{5,253} + \xi_{5,255}}{4\xi_{5,404}}$$

$$\xi_{5,408} = \sqrt{\xi_{5,395}}$$

$$\xi_{5,409} = \frac{\xi_{5,189} - \xi_{5,251}}{4\xi_{5,408}}$$

$$\xi_{5,410} = \frac{\xi_{5,190} + \xi_{5,254}}{4\xi_{5,408}}$$

$$\xi_{5,411} = \frac{\xi_{5,253} + \xi_{5,255}}{4\xi_{5,408}}$$

$$\xi_{5,412} = \begin{cases} \xi_{5,396} & \text{for } \xi_{5,392} \geq \xi_{5,393} \wedge \xi_{5,392} \geq \xi_{5,394} \wedge \xi_{5,392} \geq \xi_{5,395} \\ \xi_{5,401} & \text{for } \xi_{5,393} \geq \xi_{5,394} \wedge \xi_{5,393} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,393} \\ \xi_{5,405} & \text{for } \xi_{5,394} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,394} \wedge \xi_{5,393} \leq \xi_{5,394} \\ \xi_{5,409} & \text{otherwise} \end{cases}$$

$$\xi_{5,413} = \begin{cases} \xi_{5,397} & \text{for } \xi_{5,392} \geq \xi_{5,393} \wedge \xi_{5,392} \geq \xi_{5,394} \wedge \xi_{5,392} \geq \xi_{5,395} \\ \xi_{5,400} & \text{for } \xi_{5,393} \geq \xi_{5,394} \wedge \xi_{5,393} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,393} \\ \xi_{5,406} & \text{for } \xi_{5,394} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,394} \wedge \xi_{5,393} \leq \xi_{5,394} \\ \xi_{5,410} & \text{otherwise} \end{cases}$$

$$\xi_{5,414} = \begin{cases} \xi_{5,398} & \text{for } \xi_{5,392} \geq \xi_{5,393} \wedge \xi_{5,392} \geq \xi_{5,394} \wedge \xi_{5,392} \geq \xi_{5,395} \\ \xi_{5,402} & \text{for } \xi_{5,393} \geq \xi_{5,394} \wedge \xi_{5,393} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,393} \\ \xi_{5,404} & \text{for } \xi_{5,394} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,394} \wedge \xi_{5,393} \leq \xi_{5,394} \\ \xi_{5,411} & \text{otherwise} \end{cases}$$

$$\xi_{5,415} = \begin{cases} \xi_{5,399} & \text{for } \xi_{5,392} \geq \xi_{5,393} \wedge \xi_{5,392} \geq \xi_{5,394} \wedge \xi_{5,392} \geq \xi_{5,395} \\ \xi_{5,403} & \text{for } \xi_{5,393} \geq \xi_{5,394} \wedge \xi_{5,393} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,393} \\ \xi_{5,407} & \text{for } \xi_{5,394} \geq \xi_{5,395} \wedge \xi_{5,392} \leq \xi_{5,394} \wedge \xi_{5,393} \leq \xi_{5,394} \\ \xi_{5,408} & \text{otherwise} \end{cases}$$

$$\xi_{5,416} = \xi_{5,107}\left(\xi_{5,112} + i\xi_{5,113}\right)$$

$$\xi_{5,417} = -\frac{\sqrt{2}\xi_{5,107}\xi_{5,109}}{2} + \frac{\sqrt{2}i\xi_{5,108}\left(\xi_{5,112} + i\xi_{5,113}\right)}{2}$$

$$\xi_{5,418} = \frac{\sqrt{2}i\xi_{5,107}\xi_{5,109}}{2} - \frac{\sqrt{2}\xi_{5,108}\left(\xi_{5,112} + i\xi_{5,113}\right)}{2}$$

$$\xi_{5,419} = -\xi_{5,108}\xi_{5,109}$$

$$\xi_{5,420} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{5,107}\left(\xi_{5,112} + i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}i\xi_{5,108}\xi_{5,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{5,107}\left(\xi_{5,112} - i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}\xi_{5,108}\xi_{5,109}}{2}\right)}{2}$$

$$\xi_{5,421} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{5,107}\left(\xi_{5,112} + i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}i\xi_{5,108}\xi_{5,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{5,107}\left(\xi_{5,112} - i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}\xi_{5,108}\xi_{5,109}}{2}\right)}{2}$$

$$\xi_{5,422} = \frac{\sqrt{2}i\xi_{5,107}\xi_{5,109}}{2} - \frac{\sqrt{2}\xi_{5,108}\left(\xi_{5,112} - i\xi_{5,113}\right)}{2}$$

$$\xi_{5,423} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{5,107}\left(\xi_{5,112} - i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}i\xi_{5,108}\xi_{5,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{5,107}\left(\xi_{5,112} + i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}\xi_{5,108}\xi_{5,109}}{2}\right)}{2}$$

$$\xi_{5,424} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{5,107}\left(\xi_{5,112} - i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}i\xi_{5,108}\xi_{5,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{5,107}\left(\xi_{5,112} + i\xi_{5,113}\right)}{2} - \frac{\sqrt{2}\xi_{5,108}\xi_{5,109}}{2}\right)}{2}$$

$$\xi_{5,425} = -\frac{\sqrt{2}\xi_{5,107}\xi_{5,109}}{2} + \frac{\sqrt{2}i\xi_{5,108}\left(\xi_{5,112} - i\xi_{5,113}\right)}{2}$$

$$\xi_{5,426} = \xi_{5,107}\left(\xi_{5,112} - i\xi_{5,113}\right)$$

$$\xi_{5,427} = \left(\xi_{5,388} - i\xi_{5,391}\right)\left(\xi_{5,412} - i\xi_{5,415}\right)$$

$$\xi_{5,428} = \left(\xi_{5,388} - i\xi_{5,391}\right)\left(-i\xi_{5,413} - \xi_{5,414}\right)$$

$$\xi_{5,429} = \left(\xi_{5,412} - i\xi_{5,415}\right)\left(-i\xi_{5,389} - \xi_{5,390}\right)$$

$$\xi_{5,430} = \left(-i\xi_{5,389} - \xi_{5,390}\right)\left(-i\xi_{5,413} - \xi_{5,414}\right)$$

$$\xi_{5,431} = \left(\xi_{5,388} - i\xi_{5,391}\right)\left(-i\xi_{5,413} + \xi_{5,414}\right)$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,432} = (\xi_{5,388} - i\xi_{5,391})\,(\xi_{5,412} + i\xi_{5,415})$$

$$\xi_{5,433} = (-i\xi_{5,389} - \xi_{5,390})\,(-i\xi_{5,413} + \xi_{5,414})$$

$$\xi_{5,434} = (\xi_{5,412} + i\xi_{5,415})\,(-i\xi_{5,389} - \xi_{5,390})$$

$$\xi_{5,435} = (\xi_{5,412} - i\xi_{5,415})\,(-i\xi_{5,389} + \xi_{5,390})$$

$$\xi_{5,436} = (-i\xi_{5,389} + \xi_{5,390})\,(-i\xi_{5,413} - \xi_{5,414})$$

$$\xi_{5,437} = (\xi_{5,388} + i\xi_{5,391})\,(\xi_{5,412} - i\xi_{5,415})$$

$$\xi_{5,438} = (\xi_{5,388} + i\xi_{5,391})\,(-i\xi_{5,413} - \xi_{5,414})$$

$$\xi_{5,439} = (-i\xi_{5,389} + \xi_{5,390})\,(-i\xi_{5,413} + \xi_{5,414})$$

$$\xi_{5,440} = (\xi_{5,412} + i\xi_{5,415})\,(-i\xi_{5,389} + \xi_{5,390})$$

$$\xi_{5,441} = (\xi_{5,388} + i\xi_{5,391})\,(-i\xi_{5,413} + \xi_{5,414})$$

$$\xi_{5,442} = (\xi_{5,388} + i\xi_{5,391})\,(\xi_{5,412} + i\xi_{5,415})$$

$$\xi_{5,443} = \xi_{5,416}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}}\right) + \xi_{5,417}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}}\right) + \xi_{5,418}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}}\right) + \xi_{5,419}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}}\right)$$

$$\xi_{5,444} = \xi_{5,417}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}}\right) + \xi_{5,420}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}}\right) + \xi_{5,422}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}}\right) + \xi_{5,423}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}}\right)$$

$$\xi_{5,445} = \xi_{5,418}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}}\right) + \xi_{5,421}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}}\right) + \xi_{5,424}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}}\right) + \xi_{5,425}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}}\right)$$

$$\xi_{5,446} = \xi_{5,419}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}}\right) + \xi_{5,422}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}}\right) + \xi_{5,425}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}}\right) + \xi_{5,426}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}}\right)$$

$$\xi_{5,447} = \xi_{5,416}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}}\right) + \xi_{5,417}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}}\right) + \xi_{5,418}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}}\right) + \xi_{5,419}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}}\right)$$

$$\xi_{5,448} = \xi_{5,417}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}}\right) + \xi_{5,420}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}}\right) + \xi_{5,422}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}}\right) + \xi_{5,423}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}}\right)$$

$$\xi_{5,449} = \xi_{5,418}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}}\right) + \xi_{5,421}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}}\right) + \xi_{5,424}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}}\right) + \xi_{5,425}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}}\right)$$

$$\xi_{5,450} = \xi_{5,419}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,427}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,430}}\right) + \xi_{5,422}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,431}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,434}}\right) + \xi_{5,425}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,435}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,438}}\right) + \xi_{5,426}\left(-ie^{-i\xi_{5,100}}\sin\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,439}} + e^{-i\xi_{5,100}}\cos\left(\xi_{5,98}-\xi_{5,99}\right)\overline{\xi_{5,442}}\right)$$

$$\xi_{5,451} = \xi_{5,416}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}}\right) + \xi_{5,417}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}}\right) + \xi_{5,418}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}}\right) + \xi_{5,419}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}}\right)$$

$$\xi_{5,452} = \xi_{5,417}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}}\right) + \xi_{5,420}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}}\right) + \xi_{5,422}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}}\right) + \xi_{5,423}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}}\right)$$

$$\xi_{5,453} = \xi_{5,418}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}}\right) + \xi_{5,421}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}}\right) + \xi_{5,424}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}}\right) + \xi_{5,425}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}}\right)$$

$$\xi_{5,454} = \xi_{5,419}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}}\right) + \xi_{5,422}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}}\right) + \xi_{5,425}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}}\right) + \xi_{5,426}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}}\right)$$

$$\xi_{5,455} = \xi_{5,416}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}}\right) + \xi_{5,417}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}}\right) + \xi_{5,418}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}}\right) + \xi_{5,419}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}}\right)$$

$$\xi_{5,456} = \xi_{5,417}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}}\right) + \xi_{5,420}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}}\right) + \xi_{5,422}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}}\right) + \xi_{5,423}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}}\right)$$

$$\xi_{5,457} = \xi_{5,418}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}}\right) + \xi_{5,421}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}}\right) + \xi_{5,424}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}}\right) + \xi_{5,425}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}}\right)$$

$$\xi_{5,458} = \xi_{5,419}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,428}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,429}}\right) + \xi_{5,422}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,432}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,433}}\right) + \xi_{5,425}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,436}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,437}}\right) + \xi_{5,426}\left(-ie^{i\xi_{5,100}}\sin\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,440}} + e^{i\xi_{5,100}}\cos\left(\xi_{5,98}+\xi_{5,99}\right)\overline{\xi_{5,441}}\right)$$

$$\xi_{5,459} = \frac{\operatorname{re}(\xi_{5,443})}{4} + \frac{\operatorname{re}(\xi_{5,450})}{4} + \frac{\operatorname{re}(\xi_{5,452})}{4} + \frac{\operatorname{re}(\xi_{5,457})}{4}$$

$$\xi_{5,460} = -\frac{\operatorname{im}(\xi_{5,444})}{4} - \frac{\operatorname{im}(\xi_{5,449})}{4} - \frac{\operatorname{im}(\xi_{5,451})}{4} - \frac{\operatorname{im}(\xi_{5,458})}{4}$$

$$\xi_{5,461} = -\frac{\operatorname{re}(\xi_{5,444})}{4} + \frac{\operatorname{re}(\xi_{5,449})}{4} + \frac{\operatorname{re}(\xi_{5,451})}{4} - \frac{\operatorname{re}(\xi_{5,458})}{4}$$

$$\xi_{5,462} = -\frac{\operatorname{im}(\xi_{5,443})}{4} + \frac{\operatorname{im}(\xi_{5,450})}{4} + \frac{\operatorname{im}(\xi_{5,452})}{4} - \frac{\operatorname{im}(\xi_{5,457})}{4}$$

$$\xi_{5,463} = -\frac{\operatorname{im}(\xi_{5,445})}{4} - \frac{\operatorname{im}(\xi_{5,448})}{4} - \frac{\operatorname{im}(\xi_{5,454})}{4} - \frac{\operatorname{im}(\xi_{5,455})}{4}$$

$$\xi_{5,464} = -\frac{\operatorname{re}(\xi_{5,446})}{4} - \frac{\operatorname{re}(\xi_{5,447})}{4} - \frac{\operatorname{re}(\xi_{5,453})}{4} - \frac{\operatorname{re}(\xi_{5,456})}{4}$$

$$\xi_{5,465} = \frac{\operatorname{im}(\xi_{5,446})}{4} - \frac{\operatorname{im}(\xi_{5,447})}{4} - \frac{\operatorname{im}(\xi_{5,453})}{4} + \frac{\operatorname{im}(\xi_{5,456})}{4}$$

$$\xi_{5,466} = -\frac{\operatorname{re}(\xi_{5,445})}{4} + \frac{\operatorname{re}(\xi_{5,448})}{4} + \frac{\operatorname{re}(\xi_{5,454})}{4} - \frac{\operatorname{re}(\xi_{5,455})}{4}$$

$$\xi_{5,467} = -\frac{\operatorname{re}(\xi_{5,445})}{4} + \frac{\operatorname{re}(\xi_{5,448})}{4} - \frac{\operatorname{re}(\xi_{5,454})}{4} + \frac{\operatorname{re}(\xi_{5,455})}{4}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,468} = \frac{\operatorname{im}(\xi_{5,446})}{4} - \frac{\operatorname{im}(\xi_{5,447})}{4} + \frac{\operatorname{im}(\xi_{5,453})}{4} - \frac{\operatorname{im}(\xi_{5,456})}{4}$$

$$\xi_{5,469} = \frac{\operatorname{re}(\xi_{5,446})}{4} + \frac{\operatorname{re}(\xi_{5,447})}{4} - \frac{\operatorname{re}(\xi_{5,453})}{4} - \frac{\operatorname{re}(\xi_{5,456})}{4}$$

$$\xi_{5,470} = \frac{\operatorname{im}(\xi_{5,445})}{4} + \frac{\operatorname{im}(\xi_{5,448})}{4} - \frac{\operatorname{im}(\xi_{5,454})}{4} - \frac{\operatorname{im}(\xi_{5,455})}{4}$$

$$\xi_{5,471} = -\frac{\operatorname{im}(\xi_{5,443})}{4} + \frac{\operatorname{im}(\xi_{5,450})}{4} - \frac{\operatorname{im}(\xi_{5,452})}{4} + \frac{\operatorname{im}(\xi_{5,457})}{4}$$

$$\xi_{5,472} = -\frac{\operatorname{re}(\xi_{5,444})}{4} + \frac{\operatorname{re}(\xi_{5,449})}{4} - \frac{\operatorname{re}(\xi_{5,451})}{4} + \frac{\operatorname{re}(\xi_{5,458})}{4}$$

$$\xi_{5,473} = \frac{\operatorname{im}(\xi_{5,444})}{4} + \frac{\operatorname{im}(\xi_{5,449})}{4} - \frac{\operatorname{im}(\xi_{5,451})}{4} - \frac{\operatorname{im}(\xi_{5,458})}{4}$$

$$\xi_{5,474} = -\frac{\operatorname{re}(\xi_{5,443})}{4} - \frac{\operatorname{re}(\xi_{5,450})}{4} + \frac{\operatorname{re}(\xi_{5,452})}{4} + \frac{\operatorname{re}(\xi_{5,457})}{4}$$

$$\xi_{5,475} = \xi_{5,459}^2 + \xi_{5,463}^2 + \xi_{5,467}^2 + \xi_{5,471}^2$$

$$\xi_{5,476} = \xi_{5,460}^2 + \xi_{5,464}^2 + \xi_{5,468}^2 + \xi_{5,472}^2$$

$$\xi_{5,477} = \xi_{5,461}^2 + \xi_{5,465}^2 + \xi_{5,469}^2 + \xi_{5,473}^2$$

$$\xi_{5,478} = \xi_{5,462}^2 + \xi_{5,466}^2 + \xi_{5,470}^2 + \xi_{5,474}^2$$

$$\xi_{5,479} = \sqrt{\xi_{5,475}}$$

$$\xi_{5,480} = \frac{\xi_{5,459}}{\xi_{5,479}}$$

$$\xi_{5,481} = \frac{\xi_{5,463}}{\xi_{5,479}}$$

$$\xi_{5,482} = \frac{\xi_{5,467}}{\xi_{5,479}}$$

$$\xi_{5,483} = \frac{\xi_{5,471}}{\xi_{5,479}}$$

$$\xi_{5,484} = \sqrt{\xi_{5,476}}$$

$$\xi_{5,485} = \frac{\xi_{5,460}}{\xi_{5,484}}$$

$$\xi_{5,486} = \frac{\xi_{5,464}}{\xi_{5,484}}$$

$$\xi_{5,487} = \frac{\xi_{5,468}}{\xi_{5,484}}$$

$$\xi_{5,488} = \frac{\xi_{5,472}}{\xi_{5,484}}$$

$$\xi_{5,489} = \sqrt{\xi_{5,477}}$$

$$\xi_{5,490} = \frac{\xi_{5,461}}{\xi_{5,489}}$$

$$\xi_{5,491} = \frac{\xi_{5,465}}{\xi_{5,489}}$$

$$\xi_{5,492} = \frac{\xi_{5,469}}{\xi_{5,489}}$$

$$\xi_{5,493} = \frac{\xi_{5,473}}{\xi_{5,489}}$$

$$\xi_{5,494} = \sqrt{\xi_{5,478}}$$

$$\xi_{5,495} = \frac{\xi_{5,462}}{\xi_{5,494}}$$

$$\xi_{5,496} = \frac{\xi_{5,466}}{\xi_{5,494}}$$

$$\xi_{5,497} = \frac{\xi_{5,470}}{\xi_{5,494}}$$

$$\xi_{5,498} = \frac{\xi_{5,474}}{\xi_{5,494}}$$

$$\xi_{5,499} = \begin{cases} \xi_{5,480} & \text{for } \xi_{5,475} \geq \xi_{5,476} \wedge \xi_{5,475} \geq \xi_{5,477} \wedge \xi_{5,475} \geq \xi_{5,478} \\ \xi_{5,485} & \text{for } \xi_{5,476} \geq \xi_{5,477} \wedge \xi_{5,476} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,476} \\ \xi_{5,490} & \text{for } \xi_{5,477} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,477} \wedge \xi_{5,476} \leq \xi_{5,477} \\ \xi_{5,495} & \text{otherwise} \end{cases}$$

$$\xi_{5,500} = \begin{cases} \xi_{5,481} & \text{for } \xi_{5,475} \geq \xi_{5,476} \wedge \xi_{5,475} \geq \xi_{5,477} \wedge \xi_{5,475} \geq \xi_{5,478} \\ \xi_{5,486} & \text{for } \xi_{5,476} \geq \xi_{5,477} \wedge \xi_{5,476} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,476} \\ \xi_{5,491} & \text{for } \xi_{5,477} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,477} \wedge \xi_{5,476} \leq \xi_{5,477} \\ \xi_{5,496} & \text{otherwise} \end{cases}$$

$$\xi_{5,501} = \begin{cases} \xi_{5,482} & \text{for } \xi_{5,475} \geq \xi_{5,476} \wedge \xi_{5,475} \geq \xi_{5,477} \wedge \xi_{5,475} \geq \xi_{5,478} \\ \xi_{5,487} & \text{for } \xi_{5,476} \geq \xi_{5,477} \wedge \xi_{5,476} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,476} \\ \xi_{5,492} & \text{for } \xi_{5,477} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,477} \wedge \xi_{5,476} \leq \xi_{5,477} \\ \xi_{5,497} & \text{otherwise} \end{cases}$$

$$\xi_{5,502} = \begin{cases} \xi_{5,483} & \text{for } \xi_{5,475} \geq \xi_{5,476} \wedge \xi_{5,475} \geq \xi_{5,477} \wedge \xi_{5,475} \geq \xi_{5,478} \\ \xi_{5,488} & \text{for } \xi_{5,476} \geq \xi_{5,477} \wedge \xi_{5,476} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,476} \\ \xi_{5,493} & \text{for } \xi_{5,477} \geq \xi_{5,478} \wedge \xi_{5,475} \leq \xi_{5,477} \wedge \xi_{5,476} \leq \xi_{5,477} \\ \xi_{5,498} & \text{otherwise} \end{cases}$$

$$\xi_{5,503} = \xi_{5,459}\xi_{5,499} + \xi_{5,463}\xi_{5,500} + \xi_{5,467}\xi_{5,501} + \xi_{5,471}\xi_{5,502}$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,504} = \xi_{5,460}\xi_{5,499} + \xi_{5,464}\xi_{5,500} + \xi_{5,468}\xi_{5,501} + \xi_{5,472}\xi_{5,502}$$

$$\xi_{5,505} = \xi_{5,461}\xi_{5,499} + \xi_{5,465}\xi_{5,500} + \xi_{5,469}\xi_{5,501} + \xi_{5,473}\xi_{5,502}$$

$$\xi_{5,506} = \xi_{5,462}\xi_{5,499} + \xi_{5,466}\xi_{5,500} + \xi_{5,470}\xi_{5,501} + \xi_{5,474}\xi_{5,502}$$

$$\xi_{5,507} = \xi_{5,36}\xi_{5,47} - \xi_{5,37}\xi_{5,50} - \xi_{5,38}\xi_{5,51} - \xi_{5,39}\xi_{5,52}$$

$$\xi_{5,508} = \xi_{5,36}\xi_{5,50} + \xi_{5,37}\xi_{5,47} - \xi_{5,38}\xi_{5,52} + \xi_{5,39}\xi_{5,51}$$

$$\xi_{5,509} = \xi_{5,36}\xi_{5,51} + \xi_{5,37}\xi_{5,52} + \xi_{5,38}\xi_{5,47} - \xi_{5,39}\xi_{5,50}$$

$$\xi_{5,510} = \xi_{5,36}\xi_{5,52} - \xi_{5,37}\xi_{5,51} + \xi_{5,38}\xi_{5,50} + \xi_{5,39}\xi_{5,47}$$

$$\xi_{5,511} = \xi_{5,10}\xi_{5,50} + \xi_{5,49}\xi_{5,7} + \xi_{5,50}\xi_{5,8} + \xi_{5,50}\xi_{5,9}$$

$$\xi_{5,512} = -\xi_{5,10}\xi_{5,50} - \xi_{5,49}\xi_{5,8} + \xi_{5,50}\xi_{5,7} + \xi_{5,50}\xi_{5,9}$$

$$\xi_{5,513} = \xi_{5,10}\xi_{5,50} - \xi_{5,49}\xi_{5,9} + \xi_{5,50}\xi_{5,7} - \xi_{5,50}\xi_{5,8}$$

$$\xi_{5,514} = -\xi_{5,10}\xi_{5,49} + \xi_{5,50}\xi_{5,7} + \xi_{5,50}\xi_{5,8} - \xi_{5,50}\xi_{5,9}$$

$$\xi_{5,515} = \xi_{5,47}\left(\xi_{5,499}\xi_{5,94} + \xi_{5,500}\xi_{5,95} + \xi_{5,501}\xi_{5,96} + \xi_{5,502}\xi_{5,97}\right) - \xi_{5,47}\left(\xi_{5,499}\xi_{5,97} - \xi_{5,500}\xi_{5,96} + \xi_{5,501}\xi_{5,95} - \xi_{5,502}\xi_{5,94}\right) - \xi_{5,48}\left(\xi_{5,499}\xi_{5,95} - \xi_{5,500}\xi_{5,94} - \xi_{5,501}\xi_{5,97} + \xi_{5,502}\xi_{5,96}\right) - \xi_{5,49}\left(\xi_{5,499}\xi_{5,96} + \xi_{5,500}\xi_{5,97} - \xi_{5,501}\xi_{5,94} - \xi_{5,502}\xi_{5,95}\right)$$

$$\xi_{5,516} = \xi_{5,47}\left(\xi_{5,499}\xi_{5,95} - \xi_{5,500}\xi_{5,94} - \xi_{5,501}\xi_{5,97} + \xi_{5,502}\xi_{5,96}\right) + \xi_{5,47}\left(\xi_{5,499}\xi_{5,96} + \xi_{5,500}\xi_{5,97} - \xi_{5,501}\xi_{5,94} - \xi_{5,502}\xi_{5,95}\right) + \xi_{5,48}\left(\xi_{5,499}\xi_{5,94} + \xi_{5,500}\xi_{5,95} + \xi_{5,501}\xi_{5,96} + \xi_{5,502}\xi_{5,97}\right) - \xi_{5,49}\left(\xi_{5,499}\xi_{5,97} - \xi_{5,500}\xi_{5,96} + \xi_{5,501}\xi_{5,95} - \xi_{5,502}\xi_{5,94}\right)$$

$$\xi_{5,517} = -\xi_{5,47}\left(\xi_{5,499}\xi_{5,95} - \xi_{5,500}\xi_{5,94} - \xi_{5,501}\xi_{5,97} + \xi_{5,502}\xi_{5,96}\right) + \xi_{5,47}\left(\xi_{5,499}\xi_{5,96} + \xi_{5,500}\xi_{5,97} - \xi_{5,501}\xi_{5,94} - \xi_{5,502}\xi_{5,95}\right) + \xi_{5,48}\left(\xi_{5,499}\xi_{5,97} - \xi_{5,500}\xi_{5,96} + \xi_{5,501}\xi_{5,95} - \xi_{5,502}\xi_{5,94}\right) + \xi_{5,49}\left(\xi_{5,499}\xi_{5,94} + \xi_{5,500}\xi_{5,95} + \xi_{5,501}\xi_{5,96} + \xi_{5,502}\xi_{5,97}\right)$$

$$\xi_{5,518} = \xi_{5,47}\left(\xi_{5,499}\xi_{5,94} + \xi_{5,500}\xi_{5,95} + \xi_{5,501}\xi_{5,96} + \xi_{5,502}\xi_{5,97}\right) + \xi_{5,47}\left(\xi_{5,499}\xi_{5,97} - \xi_{5,500}\xi_{5,96} + \xi_{5,501}\xi_{5,95} - \xi_{5,502}\xi_{5,94}\right) - \xi_{5,48}\left(\xi_{5,499}\xi_{5,96} + \xi_{5,500}\xi_{5,97} - \xi_{5,501}\xi_{5,94} - \xi_{5,502}\xi_{5,95}\right) + \xi_{5,49}\left(\xi_{5,499}\xi_{5,95} - \xi_{5,500}\xi_{5,94} - \xi_{5,501}\xi_{5,97} + \xi_{5,502}\xi_{5,96}\right)$$

$$\xi_{5,519} = -\xi_{5,48}\left(-\xi_{5,503}\xi_{5,63} - \xi_{5,504}\xi_{5,62} + \xi_{5,505}\xi_{5,65} - \xi_{5,506}\xi_{5,64}\right) - \xi_{5,48}\left(-\xi_{5,503}\xi_{5,64} - \xi_{5,504}\xi_{5,65} - \xi_{5,505}\xi_{5,62} + \xi_{5,506}\xi_{5,63}\right) - \xi_{5,48}\left(-\xi_{5,503}\xi_{5,65} + \xi_{5,504}\xi_{5,64} - \xi_{5,505}\xi_{5,63} - \xi_{5,506}\xi_{5,62}\right) + \xi_{5,49}\left(\xi_{5,503}\xi_{5,62} - \xi_{5,504}\xi_{5,63} - \xi_{5,505}\xi_{5,64} - \xi_{5,506}\xi_{5,65}\right)$$

$$\xi_{5,520} = \xi_{5,48}\left(\xi_{5,503}\xi_{5,62} - \xi_{5,504}\xi_{5,63} - \xi_{5,505}\xi_{5,64} - \xi_{5,506}\xi_{5,65}\right) + \xi_{5,48}\left(-\xi_{5,503}\xi_{5,64} - \xi_{5,504}\xi_{5,65} - \xi_{5,505}\xi_{5,62} + \xi_{5,506}\xi_{5,63}\right) - \xi_{5,48}\left(-\xi_{5,503}\xi_{5,65} + \xi_{5,504}\xi_{5,64} - \xi_{5,505}\xi_{5,63} - \xi_{5,506}\xi_{5,62}\right) + \xi_{5,49}\left(-\xi_{5,503}\xi_{5,63} - \xi_{5,504}\xi_{5,62} + \xi_{5,505}\xi_{5,65} - \xi_{5,506}\xi_{5,64}\right)$$

$$\xi_{5,521} = \xi_{5,48}\left(\xi_{5,503}\xi_{5,62} - \xi_{5,504}\xi_{5,63} - \xi_{5,505}\xi_{5,64} - \xi_{5,506}\xi_{5,65}\right) - \xi_{5,48}\left(-\xi_{5,503}\xi_{5,63} - \xi_{5,504}\xi_{5,62} + \xi_{5,505}\xi_{5,65} - \xi_{5,506}\xi_{5,64}\right) + \xi_{5,48}\left(-\xi_{5,503}\xi_{5,65} + \xi_{5,504}\xi_{5,64} - \xi_{5,505}\xi_{5,63} - \xi_{5,506}\xi_{5,62}\right) + \xi_{5,49}\left(-\xi_{5,503}\xi_{5,64} - \xi_{5,504}\xi_{5,65} - \xi_{5,505}\xi_{5,62} + \xi_{5,506}\xi_{5,63}\right)$$

$$\xi_{5,522} = \xi_{5,48}\left(\xi_{5,503}\xi_{5,62} - \xi_{5,504}\xi_{5,63} - \xi_{5,505}\xi_{5,64} - \xi_{5,506}\xi_{5,65}\right) + \xi_{5,48}\left(-\xi_{5,503}\xi_{5,63} - \xi_{5,504}\xi_{5,62} + \xi_{5,505}\xi_{5,65} - \xi_{5,506}\xi_{5,64}\right) - \xi_{5,48}\left(-\xi_{5,503}\xi_{5,64} - \xi_{5,504}\xi_{5,65} - \xi_{5,505}\xi_{5,62} + \xi_{5,506}\xi_{5,63}\right) + \xi_{5,49}\left(-\xi_{5,503}\xi_{5,65} + \xi_{5,504}\xi_{5,64} - \xi_{5,505}\xi_{5,63} - \xi_{5,506}\xi_{5,62}\right)$$

$$\xi_{5,523} = \xi_{5,388}\left(\xi_{5,28}\xi_{5,86} - \xi_{5,29}\xi_{5,87} - \xi_{5,30}\xi_{5,88} - \xi_{5,31}\xi_{5,89}\right) + \xi_{5,389}\left(\xi_{5,28}\xi_{5,87} + \xi_{5,29}\xi_{5,86} + \xi_{5,30}\xi_{5,89} - \xi_{5,31}\xi_{5,88}\right) + \xi_{5,390}\left(\xi_{5,28}\xi_{5,88} - \xi_{5,29}\xi_{5,89} + \xi_{5,30}\xi_{5,86} + \xi_{5,31}\xi_{5,87}\right) + \xi_{5,391}\left(\xi_{5,28}\xi_{5,89} + \xi_{5,29}\xi_{5,88} - \xi_{5,30}\xi_{5,87} + \xi_{5,31}\xi_{5,86}\right)$$

$$\xi_{5,524} = \xi_{5,388}\left(\xi_{5,28}\xi_{5,87} + \xi_{5,29}\xi_{5,86} + \xi_{5,30}\xi_{5,89} - \xi_{5,31}\xi_{5,88}\right) - \xi_{5,389}\left(\xi_{5,28}\xi_{5,86} - \xi_{5,29}\xi_{5,87} - \xi_{5,30}\xi_{5,88} - \xi_{5,31}\xi_{5,89}\right) + \xi_{5,390}\left(\xi_{5,28}\xi_{5,89} + \xi_{5,29}\xi_{5,88} - \xi_{5,30}\xi_{5,87} + \xi_{5,31}\xi_{5,86}\right) - \xi_{5,391}\left(\xi_{5,28}\xi_{5,88} - \xi_{5,29}\xi_{5,89} + \xi_{5,30}\xi_{5,86} + \xi_{5,31}\xi_{5,87}\right)$$

$$\xi_{5,525} = \xi_{5,388}\left(\xi_{5,28}\xi_{5,88} - \xi_{5,29}\xi_{5,89} + \xi_{5,30}\xi_{5,86} + \xi_{5,31}\xi_{5,87}\right) - \xi_{5,389}\left(\xi_{5,28}\xi_{5,89} + \xi_{5,29}\xi_{5,88} - \xi_{5,30}\xi_{5,87} + \xi_{5,31}\xi_{5,86}\right) - \xi_{5,390}\left(\xi_{5,28}\xi_{5,86} - \xi_{5,29}\xi_{5,87} - \xi_{5,30}\xi_{5,88} - \xi_{5,31}\xi_{5,89}\right) + \xi_{5,391}\left(\xi_{5,28}\xi_{5,87} + \xi_{5,29}\xi_{5,86} + \xi_{5,30}\xi_{5,89} - \xi_{5,31}\xi_{5,88}\right)$$

$$\xi_{5,526} = \xi_{5,388}\left(\xi_{5,28}\xi_{5,89} + \xi_{5,29}\xi_{5,88} - \xi_{5,30}\xi_{5,87} + \xi_{5,31}\xi_{5,86}\right) + \xi_{5,389}\left(\xi_{5,28}\xi_{5,88} - \xi_{5,29}\xi_{5,89} + \xi_{5,30}\xi_{5,86} + \xi_{5,31}\xi_{5,87}\right) - \xi_{5,390}\left(\xi_{5,28}\xi_{5,87} + \xi_{5,29}\xi_{5,86} + \xi_{5,30}\xi_{5,89} - \xi_{5,31}\xi_{5,88}\right) - \xi_{5,391}\left(\xi_{5,28}\xi_{5,86} - \xi_{5,29}\xi_{5,87} - \xi_{5,30}\xi_{5,88} - \xi_{5,31}\xi_{5,89}\right)$$

$$\xi_{5,527} = \xi_{5,412}\left(\xi_{5,32}\xi_{5,90} - \xi_{5,33}\xi_{5,91} - \xi_{5,34}\xi_{5,92} - \xi_{5,35}\xi_{5,93}\right) + \xi_{5,413}\left(\xi_{5,32}\xi_{5,91} + \xi_{5,33}\xi_{5,90} + \xi_{5,34}\xi_{5,93} - \xi_{5,35}\xi_{5,92}\right) + \xi_{5,414}\left(\xi_{5,32}\xi_{5,92} - \xi_{5,33}\xi_{5,93} + \xi_{5,34}\xi_{5,90} + \xi_{5,35}\xi_{5,91}\right) + \xi_{5,415}\left(\xi_{5,32}\xi_{5,93} + \xi_{5,33}\xi_{5,92} - \xi_{5,34}\xi_{5,91} + \xi_{5,35}\xi_{5,90}\right)$$

$$\xi_{5,528} = \xi_{5,412}\left(\xi_{5,32}\xi_{5,91} + \xi_{5,33}\xi_{5,90} + \xi_{5,34}\xi_{5,93} - \xi_{5,35}\xi_{5,92}\right) - \xi_{5,413}\left(\xi_{5,32}\xi_{5,90} - \xi_{5,33}\xi_{5,91} - \xi_{5,34}\xi_{5,92} - \xi_{5,35}\xi_{5,93}\right) + \xi_{5,414}\left(\xi_{5,32}\xi_{5,93} + \xi_{5,33}\xi_{5,92} - \xi_{5,34}\xi_{5,91} + \xi_{5,35}\xi_{5,90}\right) - \xi_{5,415}\left(\xi_{5,32}\xi_{5,92} - \xi_{5,33}\xi_{5,93} + \xi_{5,34}\xi_{5,90} + \xi_{5,35}\xi_{5,91}\right)$$

$$\xi_{5,529} = \xi_{5,412}\left(\xi_{5,32}\xi_{5,92} - \xi_{5,33}\xi_{5,93} + \xi_{5,34}\xi_{5,90} + \xi_{5,35}\xi_{5,91}\right) - \xi_{5,413}\left(\xi_{5,32}\xi_{5,93} + \xi_{5,33}\xi_{5,92} - \xi_{5,34}\xi_{5,91} + \xi_{5,35}\xi_{5,90}\right) - \xi_{5,414}\left(\xi_{5,32}\xi_{5,90} - \xi_{5,33}\xi_{5,91} - \xi_{5,34}\xi_{5,92} - \xi_{5,35}\xi_{5,93}\right) + \xi_{5,415}\left(\xi_{5,32}\xi_{5,91} + \xi_{5,33}\xi_{5,90} + \xi_{5,34}\xi_{5,93} - \xi_{5,35}\xi_{5,92}\right)$$

$$\xi_{5,530} = \xi_{5,412}\left(\xi_{5,32}\xi_{5,93} + \xi_{5,33}\xi_{5,92} - \xi_{5,34}\xi_{5,91} + \xi_{5,35}\xi_{5,90}\right) + \xi_{5,413}\left(\xi_{5,32}\xi_{5,92} - \xi_{5,33}\xi_{5,93} + \xi_{5,34}\xi_{5,90} + \xi_{5,35}\xi_{5,91}\right) - \xi_{5,414}\left(\xi_{5,32}\xi_{5,91} + \xi_{5,33}\xi_{5,90} + \xi_{5,34}\xi_{5,93} - \xi_{5,35}\xi_{5,92}\right) - \xi_{5,415}\left(\xi_{5,32}\xi_{5,90} - \xi_{5,33}\xi_{5,91} - \xi_{5,34}\xi_{5,92} - \xi_{5,35}\xi_{5,93}\right)$$

$$\xi_{5,531} = \xi_{5,101} + \xi_{5,43}$$

$$\xi_{5,532} = \xi_{5,531} - \gamma - \frac{\phi}{4}$$

$$\xi_{5,533} = \xi_{5,511}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) - \xi_{5,514}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{5,534} = \xi_{5,512}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{5,513}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{5,535} = -\xi_{5,512}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{5,513}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{5,536} = \xi_{5,511}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{5,514}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{5,537} = \xi_{5,507}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{5,510}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{5,538} = \xi_{5,508}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) - \xi_{5,509}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{5,539} = \xi_{5,508}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{5,509}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,540} = -\xi_{5,507}\sin\left(\frac{\chi}{4}+\frac{\zeta}{4}\right)+\xi_{5,510}\cos\left(\frac{\chi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{5,541} = \xi_{5,527}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{5,530}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{5,542} = \xi_{5,528}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{5,529}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{5,543} = -\xi_{5,528}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{5,529}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{5,544} = -\xi_{5,527}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{5,530}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{5,545} = \xi_{5,523}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{5,526}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{5,546} = \xi_{5,524}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{5,525}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{5,547} = -\xi_{5,524}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{5,525}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{5,548} = -\xi_{5,523}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{5,526}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{5,549} = \begin{cases} 0 & \text{for } \xi_{5,537}^2+\xi_{5,540}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,540},\xi_{5,537}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,550} = \begin{cases} 0 & \text{for } \xi_{5,538}^2+\xi_{5,539}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,538},\xi_{5,539}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,551} = 2\operatorname{atan}_2\left(\sqrt{\xi_{5,538}^2+\xi_{5,539}^2},\sqrt{\xi_{5,537}^2+\xi_{5,540}^2}\right)$$

$$\xi_{5,552} = -\xi_{5,549}+\xi_{5,550}$$

$$\xi_{5,553} = -\xi_{5,549}-\xi_{5,550}$$

$$\xi_{5,554} = \xi_{5,532}+\xi_{5,549}$$

$$\xi_{5,555} = \begin{cases} 0 & \text{for } \xi_{5,533}^2+\xi_{5,536}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,536},\xi_{5,533}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,556} = \begin{cases} 0 & \text{for } \xi_{5,534}^2+\xi_{5,535}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,534},\xi_{5,535}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,557} = 2\operatorname{atan}_2\left(\sqrt{\xi_{5,534}^2+\xi_{5,535}^2},\sqrt{\xi_{5,533}^2+\xi_{5,536}^2}\right)$$

$$\xi_{5,558} = -\xi_{5,555}+\xi_{5,556}$$

$$\xi_{5,559} = -\xi_{5,555}-\xi_{5,556}$$

$$\xi_{5,560} = \xi_{5,554}+\xi_{5,555}$$

$$\xi_{5,561} = \begin{cases} 0 & \text{for } \xi_{5,515}^2+\xi_{5,518}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,518},\xi_{5,515}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,562} = \begin{cases} 0 & \text{for } \xi_{5,516}^2+\xi_{5,517}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,516},\xi_{5,517}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,563} = 2\operatorname{atan}_2\left(\sqrt{\xi_{5,516}^2+\xi_{5,517}^2},\sqrt{\xi_{5,515}^2+\xi_{5,518}^2}\right)$$

$$\xi_{5,564} = -\xi_{5,561}+\xi_{5,562}$$

$$\xi_{5,565} = -\xi_{5,561}-\xi_{5,562}$$

$$\xi_{5,566} = \xi_{5,560}+\xi_{5,561}$$

$$\xi_{5,567} = \begin{cases} 0 & \text{for } \xi_{5,519}^2+\xi_{5,522}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,522},\xi_{5,519}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,568} = \begin{cases} 0 & \text{for } \xi_{5,520}^2+\xi_{5,521}^2=0 \\ \operatorname{atan}_2\left(-\xi_{5,520},\xi_{5,521}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,569} = 2\operatorname{atan}_2\left(\sqrt{\xi_{5,520}^2+\xi_{5,521}^2},\sqrt{\xi_{5,519}^2+\xi_{5,522}^2}\right)$$

$$\xi_{5,570} = -\xi_{5,567}+\xi_{5,568}$$

$$\xi_{5,571} = -\xi_{5,567}-\xi_{5,568}$$

$$\xi_{5,572} = \xi_{5,566}+\xi_{5,567}$$

$$\xi_{5,573} = \begin{cases} 0 & \text{for } \xi_{5,112}^2+\xi_{5,113}^2=0 \\ \operatorname{atan}_2\left(\xi_{5,113},\xi_{5,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,574} = \begin{cases} 0 & \text{for } \xi_{5,109}^2=0 \\ \operatorname{atan}_2\left(\xi_{5,109},0\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,575} = 2\operatorname{atan}_2\left(\left|\xi_{5,109}\right|,\sqrt{\xi_{5,112}^2+\xi_{5,113}^2}\right)$$

Exact shared-expression program 5

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{5,576} = -\xi_{5,573} + \xi_{5,574}$$

$$\xi_{5,577} = -\xi_{5,573} - \xi_{5,574}$$

$$\xi_{5,578} = \xi_{5,572} + \xi_{5,573}$$

$$\xi_{5,579} = \begin{cases} 0 & \text{for } \xi_{5,107}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{5,107}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,580} = \begin{cases} 0 & \text{for } \xi_{5,108}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{5,108}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,581} = 2\operatorname{atan}_2\left(\left|\xi_{5,108}\right|, \left|\xi_{5,107}\right|\right)$$

$$\xi_{5,582} = -\xi_{5,579} + \xi_{5,580}$$

$$\xi_{5,583} = -\xi_{5,579} - \xi_{5,580}$$

$$\xi_{5,584} = \xi_{5,578} + \xi_{5,579}$$

$$\xi_{5,585} = \begin{cases} 0 & \text{for } \xi_{5,545}^2 + \xi_{5,548}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{5,548}, \xi_{5,545}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,586} = \begin{cases} 0 & \text{for } \xi_{5,546}^2 + \xi_{5,547}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{5,546}, \xi_{5,547}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,587} = 2\operatorname{atan}_2\left(\sqrt{\xi_{5,546}^2 + \xi_{5,547}^2}, \sqrt{\xi_{5,545}^2 + \xi_{5,548}^2}\right)$$

$$\xi_{5,588} = -\xi_{5,585} + \xi_{5,586}$$

$$\xi_{5,589} = -\xi_{5,585} - \xi_{5,586}$$

$$\xi_{5,590} = \xi_{5,584} + \xi_{5,585}$$

$$\xi_{5,591} = \begin{cases} 0 & \text{for } \xi_{5,541}^2 + \xi_{5,544}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{5,544}, \xi_{5,541}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,592} = \begin{cases} 0 & \text{for } \xi_{5,542}^2 + \xi_{5,543}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{5,542}, \xi_{5,543}\right) & \text{otherwise} \end{cases}$$

$$\xi_{5,593} = 2\operatorname{atan}_2\left(\sqrt{\xi_{5,542}^2 + \xi_{5,543}^2}, \sqrt{\xi_{5,541}^2 + \xi_{5,544}^2}\right)$$

$$\xi_{5,594} = -\xi_{5,591} + \xi_{5,592}$$

$$\xi_{5,595} = -\xi_{5,591} - \xi_{5,592}$$

$$\xi_{5,596} = \xi_{5,590} + \xi_{5,591}$$

$$\xi_{5,597} = -\frac{\xi_{5,551}}{2} - \frac{\xi_{5,557}}{2} - \frac{\xi_{5,563}}{2} - \frac{\xi_{5,569}}{2} - \frac{\xi_{5,575}}{2} - \frac{\xi_{5,581}}{2} - \frac{\xi_{5,587}}{2} - \frac{\xi_{5,593}}{2} + \xi_{5,596}$$

Derived parameters for program 5

Exact output expressions used by the native gate labels.

$$v_{5,0} = \frac{2\xi_{5,551}+2\xi_{5,597}+\pi\left(-\frac{\xi_{5,553}}{\pi}+\frac{\xi_{5,552}+\xi_{5,553}}{\pi}+\frac{1}{2}\right)-\frac{3\pi}{2}}{\pi}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,0} = \left\lfloor -\frac{\theta}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{6,1} = \left\lfloor -\frac{\phi}{2\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{6,2} = -\frac{\pi \xi_{6,0}}{2} - \frac{\theta}{2}$$

$$\xi_{6,3} = -\frac{\pi \xi_{6,1}}{2} - \frac{\phi}{4}$$

$$\xi_{6,4} = \begin{cases} \xi_{6,2} & \text{for } |\xi_{6,2}| \geq |\xi_{6,3}| \\ \xi_{6,3} & \text{otherwise} \end{cases}$$

$$\xi_{6,5} = \xi_{6,2}$$

$$\xi_{6,6} = \begin{cases} \xi_{6,3} & \text{for } |\xi_{6,2}| \geq |\xi_{6,3}| \\ \xi_{6,2} & \text{otherwise} \end{cases}$$

$$\xi_{6,7} = \begin{cases} 1 & \text{for } |\xi_{6,2}| \geq |\xi_{6,3}| \\ \frac{1}{2} & \text{for } |\xi_{6,2}| \leq |\xi_{6,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{6,8} = \begin{cases} 0 & \text{for } |\xi_{6,2}| \geq |\xi_{6,3}| \\ \frac{1}{2} & \text{for } |\xi_{6,2}| \leq |\xi_{6,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,9} = \begin{cases} 0 & \text{for } |\xi_{6,2}| \geq |\xi_{6,3}| \\ \frac{1}{2} & \text{for } |\xi_{6,2}| \leq |\xi_{6,3}| \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{6,10} = \begin{cases} 0 & \text{for } |\xi_{6,2}| \geq |\xi_{6,3}| \\ -\frac{1}{2} & \text{for } |\xi_{6,2}| \leq |\xi_{6,3}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,11} = \begin{cases} -1 & \text{for } \xi_{6,4} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{6,12} = \begin{cases} -1 & \text{for } \xi_{6,5} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{6,13} = \begin{cases} 1 & \text{for } \xi_{6,11} = 1 \wedge \xi_{6,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,14} = \begin{cases} 0 & \text{for } \xi_{6,11} = 1 \wedge \xi_{6,12} = 1 \\ 1 & \text{for } \xi_{6,11} = 1 \wedge \xi_{6,12} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,15} = \begin{cases} 0 & \text{for } \xi_{6,11} = 1 \wedge (\xi_{6,11} = 1 \vee \xi_{6,12} = -1) \wedge (\xi_{6,11} = 1 \vee \xi_{6,12} = 1) \wedge (\xi_{6,12} = -1 \vee \xi_{6,12} = 1) \\ 1 & \text{for } \xi_{6,11} = -1 \wedge \xi_{6,12} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,16} = \begin{cases} 0 & \text{for } (\xi_{6,11} = -1 \vee \xi_{6,11} = 1) \wedge (\xi_{6,11} = 1 \vee \xi_{6,12} = 1) \wedge (\xi_{6,12} = -1 \vee \xi_{6,12} = 1) \wedge (\xi_{6,11} = -1 \vee \xi_{6,11} = 1 \vee \xi_{6,12} = -1) \wedge (\xi_{6,11} = -1 \vee \xi_{6,11} = 1 \vee \xi_{6,12} = 1) \wedge (\xi_{6,11} = -1 \vee \xi_{6,12} = -1 \vee \xi_{6,12} = 1) \wedge (\xi_{6,11} = 1 \vee \xi_{6,12} = -1 \vee \xi_{6,12} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{6,17} = \cos\left(\frac{\pi \xi_{6,0}}{2}\right)$$

$$\xi_{6,18} = -\sin\left(\frac{\pi \xi_{6,0}}{2}\right)$$

$$\xi_{6,19} = \xi_{6,17}^2$$

$$\xi_{6,20} = \xi_{6,17} \xi_{6,18}$$

$$\xi_{6,21} = -\xi_{6,18}^2$$

$$\xi_{6,22} = \cos\left(\frac{\pi \xi_{6,1}}{2}\right)$$

$$\xi_{6,23} = -\sin\left(\frac{\pi \xi_{6,1}}{2}\right)$$

$$\xi_{6,24} = \xi_{6,19} \xi_{6,22} - \xi_{6,21} \xi_{6,23}$$

$$\xi_{6,25} = \xi_{6,20} \xi_{6,22} - \xi_{6,20} \xi_{6,23}$$

$$\xi_{6,26} = \xi_{6,20} \xi_{6,22} + \xi_{6,20} \xi_{6,23}$$

$$\xi_{6,27} = \xi_{6,19} \xi_{6,23} + \xi_{6,21} \xi_{6,22}$$

$$\xi_{6,28} = \xi_{6,13} \left(-\xi_{6,10} \xi_{6,27} + \xi_{6,24} \xi_{6,7} - \xi_{6,25} \xi_{6,8} - \xi_{6,26} \xi_{6,9}\right) - \xi_{6,14} \left(\xi_{6,10} \xi_{6,26} + \xi_{6,24} \xi_{6,8} + \xi_{6,25} \xi_{6,7} - \xi_{6,27} \xi_{6,9}\right) - \xi_{6,15} \left(-\xi_{6,10} \xi_{6,25} + \xi_{6,24} \xi_{6,9} + \xi_{6,26} \xi_{6,7} + \xi_{6,27} \xi_{6,8}\right) - \xi_{6,16} \left(\xi_{6,10} \xi_{6,24} + \xi_{6,25} \xi_{6,9} - \xi_{6,26} \xi_{6,8} + \xi_{6,27} \xi_{6,7}\right)$$

$$\xi_{6,29} = \xi_{6,13} \left(\xi_{6,10} \xi_{6,26} + \xi_{6,24} \xi_{6,8} + \xi_{6,25} \xi_{6,7} - \xi_{6,27} \xi_{6,9}\right) + \xi_{6,14} \left(-\xi_{6,10} \xi_{6,27} + \xi_{6,24} \xi_{6,7} - \xi_{6,25} \xi_{6,8} - \xi_{6,26} \xi_{6,9}\right) - \xi_{6,15} \left(\xi_{6,10} \xi_{6,24} + \xi_{6,25} \xi_{6,9} - \xi_{6,26} \xi_{6,8} + \xi_{6,27} \xi_{6,7}\right) + \xi_{6,16} \left(-\xi_{6,10} \xi_{6,25} + \xi_{6,24} \xi_{6,9} + \xi_{6,26} \xi_{6,7} + \xi_{6,27} \xi_{6,8}\right)$$

$$\xi_{6,30} = \xi_{6,13} \left(-\xi_{6,10} \xi_{6,25} + \xi_{6,24} \xi_{6,9} + \xi_{6,26} \xi_{6,7} + \xi_{6,27} \xi_{6,8}\right) + \xi_{6,14} \left(\xi_{6,10} \xi_{6,24} + \xi_{6,25} \xi_{6,9} - \xi_{6,26} \xi_{6,8} + \xi_{6,27} \xi_{6,7}\right) + \xi_{6,15} \left(-\xi_{6,10} \xi_{6,27} + \xi_{6,24} \xi_{6,7} - \xi_{6,25} \xi_{6,8} - \xi_{6,26} \xi_{6,9}\right) - \xi_{6,16} \left(\xi_{6,10} \xi_{6,26} + \xi_{6,24} \xi_{6,8} + \xi_{6,25} \xi_{6,7} - \xi_{6,27} \xi_{6,9}\right)$$

$$\xi_{6,31} = \xi_{6,13} \left(\xi_{6,10} \xi_{6,24} + \xi_{6,25} \xi_{6,9} - \xi_{6,26} \xi_{6,8} + \xi_{6,27} \xi_{6,7}\right) - \xi_{6,14} \left(-\xi_{6,10} \xi_{6,25} + \xi_{6,24} \xi_{6,9} + \xi_{6,26} \xi_{6,7} + \xi_{6,27} \xi_{6,8}\right) + \xi_{6,15} \left(\xi_{6,10} \xi_{6,26} + \xi_{6,24} \xi_{6,8} + \xi_{6,25} \xi_{6,7} - \xi_{6,27} \xi_{6,9}\right) + \xi_{6,16} \left(-\xi_{6,10} \xi_{6,27} + \xi_{6,24} \xi_{6,7} - \xi_{6,25} \xi_{6,8} - \xi_{6,26} \xi_{6,9}\right)$$

$$\xi_{6,32} = -\xi_{6,10} \xi_{6,27} + \xi_{6,24} \xi_{6,7} - \xi_{6,25} \xi_{6,8} - \xi_{6,26} \xi_{6,9}$$

$$\xi_{6,33} = \xi_{6,10} \xi_{6,26} + \xi_{6,24} \xi_{6,8} + \xi_{6,25} \xi_{6,7} - \xi_{6,27} \xi_{6,9}$$

$$\xi_{6,34} = -\xi_{6,10} \xi_{6,25} + \xi_{6,24} \xi_{6,9} + \xi_{6,26} \xi_{6,7} + \xi_{6,27} \xi_{6,8}$$

$$\xi_{6,35} = \xi_{6,10} \xi_{6,24} + \xi_{6,25} \xi_{6,9} - \xi_{6,26} \xi_{6,8} + \xi_{6,27} \xi_{6,7}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,36} = -\xi_{6,10}\xi_{6,16} + \xi_{6,13}\xi_{6,7} - \xi_{6,14}\xi_{6,8} - \xi_{6,15}\xi_{6,9}$$

$$\xi_{6,37} = \xi_{6,10}\xi_{6,15} - \xi_{6,13}\xi_{6,8} - \xi_{6,14}\xi_{6,7} - \xi_{6,16}\xi_{6,9}$$

$$\xi_{6,38} = -\xi_{6,10}\xi_{6,14} - \xi_{6,13}\xi_{6,9} - \xi_{6,15}\xi_{6,7} + \xi_{6,16}\xi_{6,8}$$

$$\xi_{6,39} = -\xi_{6,10}\xi_{6,13} + \xi_{6,14}\xi_{6,9} - \xi_{6,15}\xi_{6,8} - \xi_{6,16}\xi_{6,7}$$

$$\xi_{6,40} = \xi_{6,11}\xi_{6,4}$$

$$\xi_{6,41} = \xi_{6,12}\xi_{6,5}$$

$$\xi_{6,42} = \xi_{6,11}\xi_{6,12}\xi_{6,6}$$

$$\xi_{6,43} = -\frac{\pi(2\xi_{6,0}+\xi_{6,1})}{2}$$

$$\xi_{6,44} = \begin{cases} 0 & \text{for } \left(\xi_{6,41} > \frac{\pi}{8} \vee \xi_{6,42} < 0\right) \wedge \xi_{6,41} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{6,45} = \frac{\pi}{8}$$

$$\xi_{6,46} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{6,41} > \frac{\pi}{8} \wedge \xi_{6,42} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{6,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,47} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{6,41} > \frac{\pi}{8} \wedge \xi_{6,42} < 0 \\ \frac{1}{2} & \text{for } \xi_{6,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,48} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{6,41} > \frac{\pi}{8} \vee \xi_{6,42} < 0\right) \wedge \xi_{6,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,49} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{6,41} > \frac{\pi}{8} \vee \xi_{6,42} < 0\right) \wedge \xi_{6,41} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{6,50} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{6,41} > \frac{\pi}{8} \vee \xi_{6,42} < 0\right) \wedge \xi_{6,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,51} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{6,41} > \frac{\pi}{8} \vee \xi_{6,42} < 0\right) \wedge \xi_{6,41} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{6,52} = \begin{cases} \frac{1}{2} & \text{for } \xi_{6,41} > \frac{\pi}{8} \wedge \xi_{6,42} < 0 \\ -\frac{1}{2} & \text{for } \xi_{6,41} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,53} = \left\lfloor \frac{2\xi_{6,40} - 2\xi_{6,44}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{6,54} = \left\lfloor \frac{2\xi_{6,41} - 2\xi_{6,45}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{6,55} = \left\lfloor \frac{2\xi_{6,42} - 2\xi_{6,46}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{6,56} = \xi_{6,40} - \xi_{6,44} - \frac{\pi\xi_{6,53}}{2}$$

$$\xi_{6,57} = \xi_{6,41} - \xi_{6,45} - \frac{\pi\xi_{6,54}}{2}$$

$$\xi_{6,58} = \xi_{6,42} - \xi_{6,46} - \frac{\pi\xi_{6,55}}{2}$$

$$\xi_{6,59} = \begin{cases} \xi_{6,56} & \text{for } (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,58}| \vee |\xi_{6,57}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,57}| \geq |\xi_{6,58}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \\ \xi_{6,57} & \text{for } (|\xi_{6,56}| \geq |\xi_{6,58}| \vee |\xi_{6,57}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,58}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,57}| \geq |\xi_{6,58}| \vee |\xi_{6,56}| \leq |\xi_{6,57}|) \wedge (|\xi_{6,56}| \leq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \\ \xi_{6,58} & \text{otherwise} \end{cases}$$

$$\xi_{6,60} = \begin{cases} \xi_{6,57} & \text{for } |\xi_{6,56}| \geq |\xi_{6,57}| \wedge |\xi_{6,57}| \geq |\xi_{6,58}| \\ \xi_{6,58} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,57}| \leq |\xi_{6,58}| \\ \xi_{6,56} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,56}| \leq |\xi_{6,57}| \\ \xi_{6,58} & \text{for } |\xi_{6,57}| \geq |\xi_{6,58}| \wedge |\xi_{6,56}| \leq |\xi_{6,58}| \\ \xi_{6,56} & \text{for } |\xi_{6,56}| \geq |\xi_{6,57}| \wedge |\xi_{6,56}| \leq |\xi_{6,58}| \\ \xi_{6,57} & \text{otherwise} \end{cases}$$

$$\xi_{6,61} = \begin{cases} \xi_{6,58} & \text{for } |\xi_{6,56}| \geq |\xi_{6,57}| \wedge |\xi_{6,57}| \geq |\xi_{6,58}| \\ \xi_{6,57} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,57}| \leq |\xi_{6,58}| \\ \xi_{6,58} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,56}| \leq |\xi_{6,57}| \\ \xi_{6,56} & \text{for } |\xi_{6,57}| \geq |\xi_{6,58}| \wedge |\xi_{6,56}| \leq |\xi_{6,58}| \\ \xi_{6,57} & \text{for } |\xi_{6,56}| \geq |\xi_{6,57}| \wedge |\xi_{6,56}| \leq |\xi_{6,58}| \\ \xi_{6,56} & \text{otherwise} \end{cases}$$

$$\xi_{6,62} = \begin{cases} 1 & \text{for } |\xi_{6,56}| \geq |\xi_{6,57}| \wedge |\xi_{6,57}| \geq |\xi_{6,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,57}| \leq |\xi_{6,58}| \\ 0 & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,56}| \leq |\xi_{6,57}| \\ \frac{1}{2} & \text{for } |\xi_{6,56}| \leq |\xi_{6,58}| \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,57}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,57}| \geq |\xi_{6,58}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{6,63} = \begin{cases} 0 & \text{for } |\xi_{6,56}| \geq |\xi_{6,57}| \wedge |\xi_{6,57}| \geq |\xi_{6,58}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge (|\xi_{6,56}| \geq |\xi_{6,58}| \vee |\xi_{6,56}| \leq |\xi_{6,57}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,58}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \leq |\xi_{6,57}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \\ \frac{1}{2} & \text{for } |\xi_{6,56}| \leq |\xi_{6,58}| \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,57}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,57}| \geq |\xi_{6,58}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,64} = \begin{cases} 0 & \text{for } (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,58}| \vee |\xi_{6,57}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,57}| \geq |\xi_{6,58}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,56}| \leq |\xi_{6,57}| \\ \frac{1}{2} & \text{for } |\xi_{6,56}| \leq |\xi_{6,58}| \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,57}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,57}| \geq |\xi_{6,58}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{6,65} = \begin{cases} 0 & \text{for } (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,56}| \vee |\xi_{6,57}| \geq |\xi_{6,56}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \geq |\xi_{6,56}| \vee |\xi_{6,56}| \leq |\xi_{6,57}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,57}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,56}| \vee |\xi_{6,57}| \geq |\xi_{6,56}| \vee |\xi_{6,56}| \leq |\xi_{6,57}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,56}| \vee |\xi_{6,57}| \geq |\xi_{6,56}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,56}| \vee |\xi_{6,56}| \leq |\xi_{6,57}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,57}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{6,56}| \geq |\xi_{6,58}| \wedge |\xi_{6,56}| \leq |\xi_{6,57}| \\ \frac{1}{2} & \text{for } |\xi_{6,56}| \leq |\xi_{6,58}| \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,57}| \geq |\xi_{6,58}|) \wedge (|\xi_{6,56}| \geq |\xi_{6,57}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \wedge (|\xi_{6,57}| \geq |\xi_{6,58}| \vee |\xi_{6,56}| \leq |\xi_{6,58}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{6,66} = \begin{cases} -1 & \text{for } \xi_{6,59} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{6,67} = \begin{cases} -1 & \text{for } \xi_{6,60} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{6,68} = \begin{cases} 1 & \text{for } \xi_{6,66} = 1 \wedge \xi_{6,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,69} = \begin{cases} 0 & \text{for } \xi_{6,66} = 1 \wedge \xi_{6,67} = 1 \\ 1 & \text{for } \xi_{6,66} = 1 \wedge \xi_{6,67} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,70} = \begin{cases} 0 & \text{for } \xi_{6,66} = 1 \wedge (\xi_{6,66} = 1 \vee \xi_{6,67} = -1) \wedge (\xi_{6,66} = 1 \vee \xi_{6,67} = 1) \wedge (\xi_{6,67} = -1 \vee \xi_{6,67} = 1) \\ 1 & \text{for } \xi_{6,66} = -1 \wedge \xi_{6,67} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,71} = \begin{cases} 0 & \text{for } (\xi_{6,66} = -1 \vee \xi_{6,66} = 1) \wedge (\xi_{6,66} = 1 \vee \xi_{6,67} = 1) \wedge (\xi_{6,67} = -1 \vee \xi_{6,67} = 1) \wedge (\xi_{6,66} = -1 \vee \xi_{6,66} = 1 \vee \xi_{6,67} = -1) \wedge (\xi_{6,66} = -1 \vee \xi_{6,66} = 1 \vee \xi_{6,67} = 1) \wedge (\xi_{6,66} = -1 \vee \xi_{6,67} = -1 \vee \xi_{6,67} = 1) \wedge (\xi_{6,66} = 1 \vee \xi_{6,67} = -1 \vee \xi_{6,67} = 1) \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,72} = \cos\left(\frac{\pi\xi_{6,53}}{2}\right)$$

$$\xi_{6,73} = -\sin\left(\frac{\pi\xi_{6,53}}{2}\right)$$

$$\xi_{6,74} = \cos\left(\frac{\pi\xi_{6,54}}{2}\right)$$

$$\xi_{6,75} = -\sin\left(\frac{\pi\xi_{6,54}}{2}\right)$$

$$\xi_{6,76} = \xi_{6,72}\xi_{6,74}$$

$$\xi_{6,77} = \xi_{6,73}\xi_{6,74}$$

$$\xi_{6,78} = \xi_{6,72}\xi_{6,75}$$

$$\xi_{6,79} = -\xi_{6,73}\xi_{6,75}$$

$$\xi_{6,80} = \cos\left(\frac{\pi\xi_{6,55}}{2}\right)$$

$$\xi_{6,81} = -\sin\left(\frac{\pi\xi_{6,55}}{2}\right)$$

$$\xi_{6,82} = \xi_{6,76}\xi_{6,80} - \xi_{6,79}\xi_{6,81}$$

$$\xi_{6,83} = \xi_{6,77}\xi_{6,80} - \xi_{6,78}\xi_{6,81}$$

$$\xi_{6,84} = \xi_{6,77}\xi_{6,81} + \xi_{6,78}\xi_{6,80}$$

$$\xi_{6,85} = \xi_{6,76}\xi_{6,81} + \xi_{6,79}\xi_{6,80}$$

$$\xi_{6,86} = \xi_{6,68}\left(\xi_{6,62}\xi_{6,82} - \xi_{6,63}\xi_{6,83} - \xi_{6,64}\xi_{6,84} - \xi_{6,65}\xi_{6,85}\right) - \xi_{6,69}\left(\xi_{6,62}\xi_{6,83} + \xi_{6,63}\xi_{6,82} - \xi_{6,64}\xi_{6,85} + \xi_{6,65}\xi_{6,84}\right) - \xi_{6,70}\left(\xi_{6,62}\xi_{6,84} + \xi_{6,63}\xi_{6,85} + \xi_{6,64}\xi_{6,82} - \xi_{6,65}\xi_{6,83}\right) - \xi_{6,71}\left(\xi_{6,62}\xi_{6,85} - \xi_{6,63}\xi_{6,84} + \xi_{6,64}\xi_{6,83} + \xi_{6,65}\xi_{6,82}\right)$$

$$\xi_{6,87} = \xi_{6,68}\left(\xi_{6,62}\xi_{6,83} + \xi_{6,63}\xi_{6,82} - \xi_{6,64}\xi_{6,85} + \xi_{6,65}\xi_{6,84}\right) + \xi_{6,69}\left(\xi_{6,62}\xi_{6,82} - \xi_{6,63}\xi_{6,83} - \xi_{6,64}\xi_{6,84} - \xi_{6,65}\xi_{6,85}\right) - \xi_{6,70}\left(\xi_{6,62}\xi_{6,85} - \xi_{6,63}\xi_{6,84} + \xi_{6,64}\xi_{6,83} + \xi_{6,65}\xi_{6,82}\right) + \xi_{6,71}\left(\xi_{6,62}\xi_{6,84} + \xi_{6,63}\xi_{6,85} + \xi_{6,64}\xi_{6,82} - \xi_{6,65}\xi_{6,83}\right)$$

$$\xi_{6,88} = \xi_{6,68}\left(\xi_{6,62}\xi_{6,84} + \xi_{6,63}\xi_{6,85} + \xi_{6,64}\xi_{6,82} - \xi_{6,65}\xi_{6,83}\right) + \xi_{6,69}\left(\xi_{6,62}\xi_{6,85} - \xi_{6,63}\xi_{6,84} + \xi_{6,64}\xi_{6,83} + \xi_{6,65}\xi_{6,82}\right) + \xi_{6,70}\left(\xi_{6,62}\xi_{6,82} - \xi_{6,63}\xi_{6,83} - \xi_{6,64}\xi_{6,84} - \xi_{6,65}\xi_{6,85}\right) - \xi_{6,71}\left(\xi_{6,62}\xi_{6,83} + \xi_{6,63}\xi_{6,82} - \xi_{6,64}\xi_{6,85} + \xi_{6,65}\xi_{6,84}\right)$$

$$\xi_{6,89} = \xi_{6,68}\left(\xi_{6,62}\xi_{6,85} - \xi_{6,63}\xi_{6,84} + \xi_{6,64}\xi_{6,83} + \xi_{6,65}\xi_{6,82}\right) - \xi_{6,69}\left(\xi_{6,62}\xi_{6,84} + \xi_{6,63}\xi_{6,85} + \xi_{6,64}\xi_{6,82} - \xi_{6,65}\xi_{6,83}\right) + \xi_{6,70}\left(\xi_{6,62}\xi_{6,83} + \xi_{6,63}\xi_{6,82} - \xi_{6,64}\xi_{6,85} + \xi_{6,65}\xi_{6,84}\right) + \xi_{6,71}\left(\xi_{6,62}\xi_{6,82} - \xi_{6,63}\xi_{6,83} - \xi_{6,64}\xi_{6,84} - \xi_{6,65}\xi_{6,85}\right)$$

$$\xi_{6,90} = \xi_{6,62}\xi_{6,82} - \xi_{6,63}\xi_{6,83} - \xi_{6,64}\xi_{6,84} - \xi_{6,65}\xi_{6,85}$$

$$\xi_{6,91} = \xi_{6,62}\xi_{6,83} + \xi_{6,63}\xi_{6,82} - \xi_{6,64}\xi_{6,85} + \xi_{6,65}\xi_{6,84}$$

$$\xi_{6,92} = \xi_{6,62}\xi_{6,84} + \xi_{6,63}\xi_{6,85} + \xi_{6,64}\xi_{6,82} - \xi_{6,65}\xi_{6,83}$$

$$\xi_{6,93} = \xi_{6,62}\xi_{6,85} - \xi_{6,63}\xi_{6,84} + \xi_{6,64}\xi_{6,83} + \xi_{6,65}\xi_{6,82}$$

$$\xi_{6,94} = \xi_{6,62}\xi_{6,68} - \xi_{6,63}\xi_{6,69} - \xi_{6,64}\xi_{6,70} - \xi_{6,65}\xi_{6,71}$$

$$\xi_{6,95} = -\xi_{6,62}\xi_{6,69} - \xi_{6,63}\xi_{6,68} - \xi_{6,64}\xi_{6,71} + \xi_{6,65}\xi_{6,70}$$

$$\xi_{6,96} = -\xi_{6,62}\xi_{6,70} + \xi_{6,63}\xi_{6,71} - \xi_{6,64}\xi_{6,68} - \xi_{6,65}\xi_{6,69}$$

$$\xi_{6,97} = -\xi_{6,62}\xi_{6,71} - \xi_{6,63}\xi_{6,70} + \xi_{6,64}\xi_{6,69} - \xi_{6,65}\xi_{6,68}$$

$$\xi_{6,98} = \xi_{6,59}\xi_{6,66}$$

$$\xi_{6,99} = \xi_{6,60}\xi_{6,67}$$

$$\xi_{6,100} = \xi_{6,61}\xi_{6,66}\xi_{6,67}$$

$$\xi_{6,101} = -\frac{\pi(\xi_{6,53}+\xi_{6,54}+\xi_{6,55})}{2}$$

$$\xi_{6,102} = \cos\left(2\xi_{6,98}\right)$$

$$\xi_{6,103} = \cos\left(2\xi_{6,99}\right)$$

$$\xi_{6,104} = \cos\left(2\xi_{6,100}\right)$$

$$\xi_{6,105} = \sqrt{\xi_{6,102}^2 - 2\xi_{6,102}\xi_{6,103}\xi_{6,104} + \xi_{6,103}^2 + \xi_{6,104}^2 - 1}$$

$$\xi_{6,106} = \xi_{6,102} - \xi_{6,103} + \xi_{6,104}$$

$$\xi_{6,107} = \sqrt{-\frac{\xi_{6,105}}{2} + \frac{\xi_{6,106}}{2} + \frac{1}{2}}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,108} = \sqrt{\frac{\xi_{6,105}}{2} - \frac{\xi_{6,106}}{2} + \frac{1}{2}}$$

$$\xi_{6,109} = \sqrt{-\frac{\xi_{6,105}}{2} - \frac{\xi_{6,106}}{2} + \frac{1}{2}}$$

$$\xi_{6,110} = \frac{(1-\xi_{6,103})(\xi_{6,102}+1)(\xi_{6,104}+1)}{2}$$

$$\xi_{6,111} = \begin{cases} -1 & \text{for } \xi_{6,100} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{6,112} = \begin{cases} 0 & \text{for } \xi_{6,107} = 0 \\ \frac{\sqrt{\xi_{6,110}\xi_{6,111}}}{\xi_{6,107}} & \text{otherwise} \end{cases}$$

$$\xi_{6,113} = \begin{cases} 1 & \text{for } \xi_{6,107} = 0 \\ \frac{\sqrt{\xi_{6,102}\xi_{6,103}\xi_{6,104}}}{\xi_{6,107}} & \text{otherwise} \end{cases}$$

$$\xi_{6,114} = 2\xi_{6,107}\xi_{6,108}\xi_{6,109}\xi_{6,113}$$

$$\xi_{6,115} = \sqrt{2}\xi_{6,113}\left(\xi_{6,107}^2\xi_{6,109} - \xi_{6,109}\right)$$

$$\xi_{6,116} = 2\xi_{6,107}\xi_{6,108}\xi_{6,109}\xi_{6,112}$$

$$\xi_{6,117} = \sqrt{2}\left(2\xi_{6,107}^2\xi_{6,109}\xi_{6,112} - \xi_{6,109}\xi_{6,112}\right)$$

$$\xi_{6,118} = \sqrt{2}\xi_{6,107}\xi_{6,108}\xi_{6,112}\xi_{6,113}$$

$$\xi_{6,119} = \sqrt{2}\xi_{6,107}\xi_{6,108}\left(-\xi_{6,109}^2 + \xi_{6,112}^2\right)$$

$$\xi_{6,120} = -\xi_{6,107}^2\xi_{6,109}^2 + \xi_{6,107}^2\xi_{6,112}^2 + \xi_{6,109}^2$$

$$\xi_{6,121} = \xi_{6,107}^2\xi_{6,109}^2 - \xi_{6,107}^2\xi_{6,112}^2 + \xi_{6,112}^2$$

$$\xi_{6,122} = 2\xi_{6,113}\left(\xi_{6,107}^2\xi_{6,112} - \frac{\xi_{6,112}}{2}\right)$$

$$\xi_{6,123} = -\sqrt{2}\xi_{6,107}\xi_{6,108}\xi_{6,112}\xi_{6,113}$$

$$\xi_{6,124} = -\xi_{6,112}\xi_{6,113}$$

$$\xi_{6,125} = \xi_{6,107}^2\xi_{6,109}^2 + \xi_{6,107}^2\xi_{6,112}^2 - \xi_{6,107}^2 - \xi_{6,109}^2 - \xi_{6,112}^2 + 1$$

$$\xi_{6,126} = -\sqrt{2}\left(\xi_{6,107}\xi_{6,108}\xi_{6,109}^2 + \xi_{6,107}\xi_{6,108}\xi_{6,112}^2 - \xi_{6,107}\xi_{6,108}\right)$$

$$\xi_{6,127} = \sqrt{2}\xi_{6,107}^2\xi_{6,109}\xi_{6,113}$$

$$\xi_{6,128} = \xi_{6,118}^2 - \xi_{6,121}^2 - \xi_{6,124}^2 - \xi_{6,127}^2$$

$$\xi_{6,129} = \xi_{6,118}\xi_{6,119} - \xi_{6,121}\xi_{6,122} - \xi_{6,124}\xi_{6,125}$$

$$\xi_{6,130} = \xi_{6,118}\xi_{6,120} - \xi_{6,121}\xi_{6,123} - \xi_{6,124}\xi_{6,126}$$

$$\xi_{6,131} = \xi_{6,114}^2 + \xi_{6,116}^2 + \xi_{6,119}^2 - \xi_{6,122}^2 - \xi_{6,125}^2$$

$$\xi_{6,132} = \xi_{6,114}\xi_{6,115} + \xi_{6,116}\xi_{6,117} + \xi_{6,119}\xi_{6,120} - \xi_{6,122}\xi_{6,123} - \xi_{6,125}\xi_{6,126}$$

$$\xi_{6,133} = \xi_{6,115}^2 + \xi_{6,117}^2 + \xi_{6,120}^2 - \xi_{6,123}^2 - \xi_{6,126}^2$$

$$\xi_{6,134} = \sin\left(2\xi_{6,100}\right)\sin\left(2\xi_{6,99}\right)\cos\left(2\xi_{6,98}\right)$$

$$\xi_{6,135} = \sin\left(2\xi_{6,100}\right)\sin\left(2\xi_{6,98}\right)\cos\left(2\xi_{6,99}\right)$$

$$\xi_{6,136} = \sin\left(2\xi_{6,98}\right)\sin\left(2\xi_{6,99}\right)\cos\left(2\xi_{6,100}\right)$$

$$\xi_{6,137} = \sin\left(2\xi_{6,98}\right)\cos\left(2\xi_{6,100}\right)\cos\left(2\xi_{6,99}\right)$$

$$\xi_{6,138} = \sin\left(2\xi_{6,99}\right)\cos\left(2\xi_{6,100}\right)\cos\left(2\xi_{6,98}\right)$$

$$\xi_{6,139} = \sin\left(2\xi_{6,100}\right)\cos\left(2\xi_{6,98}\right)\cos\left(2\xi_{6,99}\right)$$

$$\xi_{6,140} = \xi_{6,134}^2 - \xi_{6,137}^2$$

$$\xi_{6,141} = \xi_{6,135}^2 - \xi_{6,138}^2$$

$$\xi_{6,142} = \xi_{6,136}^2 - \xi_{6,139}^2$$

$$\xi_{6,143} = \frac{\xi_{6,128} - \xi_{6,141}}{\xi_{6,140} - \xi_{6,141}}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,144} = \frac{\xi_{6,129}}{\xi_{6,140}-\xi_{6,141}}$$

$$\xi_{6,145} = \frac{\xi_{6,130}}{\xi_{6,140}-\xi_{6,141}}$$

$$\xi_{6,146} = \frac{\xi_{6,131}-\xi_{6,141}}{\xi_{6,140}-\xi_{6,141}}$$

$$\xi_{6,147} = \frac{\xi_{6,132}}{\xi_{6,140}-\xi_{6,141}}$$

$$\xi_{6,148} = \frac{\xi_{6,133}-\xi_{6,141}}{\xi_{6,140}-\xi_{6,141}}$$

$$\xi_{6,149} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,143} & \text{otherwise} \end{cases}$$

$$\xi_{6,150} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,144} & \text{otherwise} \end{cases}$$

$$\xi_{6,151} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,145} & \text{otherwise} \end{cases}$$

$$\xi_{6,152} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,146} & \text{otherwise} \end{cases}$$

$$\xi_{6,153} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,147} & \text{otherwise} \end{cases}$$

$$\xi_{6,154} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,148} & \text{otherwise} \end{cases}$$

$$\xi_{6,155} = \frac{\xi_{6,128}-\xi_{6,142}}{\xi_{6,140}-\xi_{6,142}}$$

$$\xi_{6,156} = \frac{\xi_{6,129}}{\xi_{6,140}-\xi_{6,142}}$$

$$\xi_{6,157} = \frac{\xi_{6,130}}{\xi_{6,140}-\xi_{6,142}}$$

$$\xi_{6,158} = \frac{\xi_{6,131}-\xi_{6,142}}{\xi_{6,140}-\xi_{6,142}}$$

$$\xi_{6,159} = \frac{\xi_{6,132}}{\xi_{6,140}-\xi_{6,142}}$$

$$\xi_{6,160} = \frac{\xi_{6,133}-\xi_{6,142}}{\xi_{6,140}-\xi_{6,142}}$$

$$\xi_{6,161} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,142} \\ \xi_{6,155} & \text{otherwise} \end{cases}$$

$$\xi_{6,162} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,142} \\ \xi_{6,156} & \text{otherwise} \end{cases}$$

$$\xi_{6,163} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,142} \\ \xi_{6,157} & \text{otherwise} \end{cases}$$

$$\xi_{6,164} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,142} \\ \xi_{6,158} & \text{otherwise} \end{cases}$$

$$\xi_{6,165} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,142} \\ \xi_{6,159} & \text{otherwise} \end{cases}$$

$$\xi_{6,166} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,142} \\ \xi_{6,160} & \text{otherwise} \end{cases}$$

$$\xi_{6,167} = \xi_{6,149}\xi_{6,161} + \xi_{6,150}\xi_{6,162} + \xi_{6,151}\xi_{6,163}$$

$$\xi_{6,168} = \xi_{6,149}\xi_{6,162} + \xi_{6,150}\xi_{6,164} + \xi_{6,151}\xi_{6,165}$$

$$\xi_{6,169} = \xi_{6,149}\xi_{6,163} + \xi_{6,150}\xi_{6,165} + \xi_{6,151}\xi_{6,166}$$

$$\xi_{6,170} = \xi_{6,150}\xi_{6,161} + \xi_{6,152}\xi_{6,162} + \xi_{6,153}\xi_{6,163}$$

$$\xi_{6,171} = \xi_{6,150}\xi_{6,162} + \xi_{6,152}\xi_{6,164} + \xi_{6,153}\xi_{6,165}$$

$$\xi_{6,172} = \xi_{6,150}\xi_{6,163} + \xi_{6,152}\xi_{6,165} + \xi_{6,153}\xi_{6,166}$$

$$\xi_{6,173} = \xi_{6,151}\xi_{6,161} + \xi_{6,153}\xi_{6,162} + \xi_{6,154}\xi_{6,163}$$

$$\xi_{6,174} = \xi_{6,151}\xi_{6,162} + \xi_{6,153}\xi_{6,164} + \xi_{6,154}\xi_{6,165}$$

$$\xi_{6,175} = \xi_{6,151}\xi_{6,163} + \xi_{6,153}\xi_{6,165} + \xi_{6,154}\xi_{6,166}$$

$$\xi_{6,176} = \sqrt{\xi_{6,167}}$$

$$\xi_{6,177} = \frac{\xi_{6,167}}{\xi_{6,176}}$$

$$\xi_{6,178} = \frac{\xi_{6,170}}{\xi_{6,176}}$$

$$\xi_{6,179} = \frac{\xi_{6,173}}{\xi_{6,176}}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,180} = \sqrt{\xi_{6,171}}$$

$$\xi_{6,181} = \frac{\xi_{6,168}}{\xi_{6,180}}$$

$$\xi_{6,182} = \frac{\xi_{6,171}}{\xi_{6,180}}$$

$$\xi_{6,183} = \frac{\xi_{6,174}}{\xi_{6,180}}$$

$$\xi_{6,184} = \sqrt{\xi_{6,175}}$$

$$\xi_{6,185} = \frac{\xi_{6,169}}{\xi_{6,184}}$$

$$\xi_{6,186} = \frac{\xi_{6,172}}{\xi_{6,184}}$$

$$\xi_{6,187} = \frac{\xi_{6,175}}{\xi_{6,184}}$$

$$\xi_{6,188} = \begin{cases} \xi_{6,177} & \text{for } \xi_{6,167} \geq \xi_{6,171} \wedge \xi_{6,167} \geq \xi_{6,175} \\ \xi_{6,181} & \text{for } \xi_{6,171} \geq \xi_{6,175} \wedge \xi_{6,167} \leq \xi_{6,171} \\ \xi_{6,185} & \text{otherwise} \end{cases}$$

$$\xi_{6,189} = \begin{cases} \xi_{6,178} & \text{for } \xi_{6,167} \geq \xi_{6,171} \wedge \xi_{6,167} \geq \xi_{6,175} \\ \xi_{6,182} & \text{for } \xi_{6,171} \geq \xi_{6,175} \wedge \xi_{6,167} \leq \xi_{6,171} \\ \xi_{6,186} & \text{otherwise} \end{cases}$$

$$\xi_{6,190} = \begin{cases} \xi_{6,179} & \text{for } \xi_{6,167} \geq \xi_{6,171} \wedge \xi_{6,167} \geq \xi_{6,175} \\ \xi_{6,183} & \text{for } \xi_{6,171} \geq \xi_{6,175} \wedge \xi_{6,167} \leq \xi_{6,171} \\ \xi_{6,187} & \text{otherwise} \end{cases}$$

$$\xi_{6,191} = \frac{\xi_{6,128} - \xi_{6,140}}{-\xi_{6,140} + \xi_{6,141}}$$

$$\xi_{6,192} = \frac{\xi_{6,129}}{-\xi_{6,140} + \xi_{6,141}}$$

$$\xi_{6,193} = \frac{\xi_{6,130}}{-\xi_{6,140} + \xi_{6,141}}$$

$$\xi_{6,194} = \frac{\xi_{6,131} - \xi_{6,140}}{-\xi_{6,140} + \xi_{6,141}}$$

$$\xi_{6,195} = \frac{\xi_{6,132}}{-\xi_{6,140} + \xi_{6,141}}$$

$$\xi_{6,196} = \frac{\xi_{6,133} - \xi_{6,140}}{-\xi_{6,140} + \xi_{6,141}}$$

$$\xi_{6,197} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,191} & \text{otherwise} \end{cases}$$

$$\xi_{6,198} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,192} & \text{otherwise} \end{cases}$$

$$\xi_{6,199} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,193} & \text{otherwise} \end{cases}$$

$$\xi_{6,200} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,194} & \text{otherwise} \end{cases}$$

$$\xi_{6,201} = \begin{cases} 0 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,195} & \text{otherwise} \end{cases}$$

$$\xi_{6,202} = \begin{cases} 1 & \text{for } \xi_{6,140} = \xi_{6,141} \\ \xi_{6,196} & \text{otherwise} \end{cases}$$

$$\xi_{6,203} = \frac{\xi_{6,128} - \xi_{6,142}}{\xi_{6,141} - \xi_{6,142}}$$

$$\xi_{6,204} = \frac{\xi_{6,129}}{\xi_{6,141} - \xi_{6,142}}$$

$$\xi_{6,205} = \frac{\xi_{6,130}}{\xi_{6,141} - \xi_{6,142}}$$

$$\xi_{6,206} = \frac{\xi_{6,131} - \xi_{6,142}}{\xi_{6,141} - \xi_{6,142}}$$

$$\xi_{6,207} = \frac{\xi_{6,132}}{\xi_{6,141} - \xi_{6,142}}$$

$$\xi_{6,208} = \frac{\xi_{6,133} - \xi_{6,142}}{\xi_{6,141} - \xi_{6,142}}$$

$$\xi_{6,209} = \begin{cases} 1 & \text{for } \xi_{6,141} = \xi_{6,142} \\ \xi_{6,203} & \text{otherwise} \end{cases}$$

$$\xi_{6,210} = \begin{cases} 0 & \text{for } \xi_{6,141} = \xi_{6,142} \\ \xi_{6,204} & \text{otherwise} \end{cases}$$

$$\xi_{6,211} = \begin{cases} 0 & \text{for } \xi_{6,141} = \xi_{6,142} \\ \xi_{6,205} & \text{otherwise} \end{cases}$$

$$\xi_{6,212} = \begin{cases} 1 & \text{for } \xi_{6,141} = \xi_{6,142} \\ \xi_{6,206} & \text{otherwise} \end{cases}$$

$$\xi_{6,213} = \begin{cases} 0 & \text{for } \xi_{6,141} = \xi_{6,142} \\ \xi_{6,207} & \text{otherwise} \end{cases}$$

$$\xi_{6,214} = \begin{cases} 1 & \text{for } \xi_{6,141} = \xi_{6,142} \\ \xi_{6,208} & \text{otherwise} \end{cases}$$

$$\xi_{6,215} = \xi_{6,197}\xi_{6,209} + \xi_{6,198}\xi_{6,210} + \xi_{6,199}\xi_{6,211}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,216} = \xi_{6,197}\xi_{6,210} + \xi_{6,198}\xi_{6,212} + \xi_{6,199}\xi_{6,213}$$

$$\xi_{6,217} = \xi_{6,197}\xi_{6,211} + \xi_{6,198}\xi_{6,213} + \xi_{6,199}\xi_{6,214}$$

$$\xi_{6,218} = \xi_{6,198}\xi_{6,209} + \xi_{6,200}\xi_{6,210} + \xi_{6,201}\xi_{6,211}$$

$$\xi_{6,219} = \xi_{6,198}\xi_{6,210} + \xi_{6,200}\xi_{6,212} + \xi_{6,201}\xi_{6,213}$$

$$\xi_{6,220} = \xi_{6,198}\xi_{6,211} + \xi_{6,200}\xi_{6,213} + \xi_{6,201}\xi_{6,214}$$

$$\xi_{6,221} = \xi_{6,199}\xi_{6,209} + \xi_{6,201}\xi_{6,210} + \xi_{6,202}\xi_{6,211}$$

$$\xi_{6,222} = \xi_{6,199}\xi_{6,210} + \xi_{6,201}\xi_{6,212} + \xi_{6,202}\xi_{6,213}$$

$$\xi_{6,223} = \xi_{6,199}\xi_{6,211} + \xi_{6,201}\xi_{6,213} + \xi_{6,202}\xi_{6,214}$$

$$\xi_{6,224} = \begin{cases} \xi_{6,188}^2 & \text{for } \xi_{6,140} = \xi_{6,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,225} = \begin{cases} \xi_{6,188}\xi_{6,189} & \text{for } \xi_{6,140} = \xi_{6,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,226} = \begin{cases} \xi_{6,188}\xi_{6,190} & \text{for } \xi_{6,140} = \xi_{6,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,227} = \begin{cases} \xi_{6,189}^2 & \text{for } \xi_{6,140} = \xi_{6,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,228} = \begin{cases} \xi_{6,189}\xi_{6,190} & \text{for } \xi_{6,140} = \xi_{6,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,229} = \begin{cases} \xi_{6,190}^2 & \text{for } \xi_{6,140} = \xi_{6,141} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{6,230} = \xi_{6,215} - \xi_{6,224}$$

$$\xi_{6,231} = \xi_{6,216} - \xi_{6,225}$$

$$\xi_{6,232} = \xi_{6,217} - \xi_{6,226}$$

$$\xi_{6,233} = \xi_{6,218} - \xi_{6,225}$$

$$\xi_{6,234} = \xi_{6,219} - \xi_{6,227}$$

$$\xi_{6,235} = \xi_{6,220} - \xi_{6,228}$$

$$\xi_{6,236} = \xi_{6,221} - \xi_{6,226}$$

$$\xi_{6,237} = \xi_{6,222} - \xi_{6,228}$$

$$\xi_{6,238} = \xi_{6,223} - \xi_{6,229}$$

$$\xi_{6,239} = \sqrt{\xi_{6,230}}$$

$$\xi_{6,240} = \frac{\xi_{6,230}}{\xi_{6,239}}$$

$$\xi_{6,241} = \frac{\xi_{6,233}}{\xi_{6,239}}$$

$$\xi_{6,242} = \frac{\xi_{6,236}}{\xi_{6,239}}$$

$$\xi_{6,243} = \sqrt{\xi_{6,234}}$$

$$\xi_{6,244} = \frac{\xi_{6,231}}{\xi_{6,243}}$$

$$\xi_{6,245} = \frac{\xi_{6,234}}{\xi_{6,243}}$$

$$\xi_{6,246} = \frac{\xi_{6,237}}{\xi_{6,243}}$$

$$\xi_{6,247} = \sqrt{\xi_{6,238}}$$

$$\xi_{6,248} = \frac{\xi_{6,232}}{\xi_{6,247}}$$

$$\xi_{6,249} = \frac{\xi_{6,235}}{\xi_{6,247}}$$

$$\xi_{6,250} = \frac{\xi_{6,238}}{\xi_{6,247}}$$

$$\xi_{6,251} = \begin{cases} \xi_{6,240} & \text{for } \xi_{6,230} \geq \xi_{6,234} \wedge \xi_{6,230} \geq \xi_{6,238} \\ \xi_{6,244} & \text{for } \xi_{6,234} \geq \xi_{6,238} \wedge \xi_{6,230} \leq \xi_{6,234} \\ \xi_{6,248} & \text{otherwise} \end{cases}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,252} = \begin{cases} \xi_{6,241} & \text{for } \xi_{6,230} \geq \xi_{6,234} \wedge \xi_{6,230} \geq \xi_{6,238} \\ \xi_{6,245} & \text{for } \xi_{6,234} \geq \xi_{6,238} \wedge \xi_{6,230} \leq \xi_{6,234} \\ \xi_{6,249} & \text{otherwise} \end{cases}$$

$$\xi_{6,253} = \begin{cases} \xi_{6,242} & \text{for } \xi_{6,230} \geq \xi_{6,234} \wedge \xi_{6,230} \geq \xi_{6,238} \\ \xi_{6,246} & \text{for } \xi_{6,234} \geq \xi_{6,238} \wedge \xi_{6,230} \leq \xi_{6,234} \\ \xi_{6,250} & \text{otherwise} \end{cases}$$

$$\xi_{6,254} = \xi_{6,189}\xi_{6,253} - \xi_{6,190}\xi_{6,252}$$

$$\xi_{6,255} = -\xi_{6,188}\xi_{6,253} + \xi_{6,190}\xi_{6,251}$$

$$\xi_{6,256} = \xi_{6,188}\xi_{6,252} - \xi_{6,189}\xi_{6,251}$$

$$\xi_{6,257} = \xi_{6,134}^2 + \xi_{6,137}^2$$

$$\xi_{6,258} = \xi_{6,135}^2 + \xi_{6,138}^2$$

$$\xi_{6,259} = \xi_{6,136}^2 + \xi_{6,139}^2$$

$$\xi_{6,260} = \frac{\xi_{6,121}\xi_{6,137}\xi_{6,188}+\xi_{6,189}(\xi_{6,114}\xi_{6,134}+\xi_{6,122}\xi_{6,137})+\xi_{6,190}(\xi_{6,115}\xi_{6,134}+\xi_{6,123}\xi_{6,137})}{\xi_{6,257}}$$

$$\xi_{6,261} = \frac{\xi_{6,124}\xi_{6,137}\xi_{6,188}+\xi_{6,189}(\xi_{6,116}\xi_{6,134}+\xi_{6,125}\xi_{6,137})+\xi_{6,190}(\xi_{6,117}\xi_{6,134}+\xi_{6,126}\xi_{6,137})}{\xi_{6,257}}$$

$$\xi_{6,262} = \frac{\xi_{6,119}\xi_{6,134}\xi_{6,189}+\xi_{6,120}\xi_{6,134}\xi_{6,190}+\xi_{6,188}(\xi_{6,118}\xi_{6,134}+\xi_{6,127}\xi_{6,137})}{\xi_{6,257}}$$

$$\xi_{6,263} = \frac{\xi_{6,121}\xi_{6,138}\xi_{6,251}+\xi_{6,252}(\xi_{6,114}\xi_{6,135}+\xi_{6,122}\xi_{6,138})+\xi_{6,253}(\xi_{6,115}\xi_{6,135}+\xi_{6,123}\xi_{6,138})}{\xi_{6,258}}$$

$$\xi_{6,264} = \frac{\xi_{6,124}\xi_{6,138}\xi_{6,251}+\xi_{6,252}(\xi_{6,116}\xi_{6,135}+\xi_{6,125}\xi_{6,138})+\xi_{6,253}(\xi_{6,117}\xi_{6,135}+\xi_{6,126}\xi_{6,138})}{\xi_{6,258}}$$

$$\xi_{6,265} = \frac{\xi_{6,119}\xi_{6,135}\xi_{6,252}+\xi_{6,120}\xi_{6,135}\xi_{6,253}+\xi_{6,251}(\xi_{6,118}\xi_{6,135}+\xi_{6,127}\xi_{6,138})}{\xi_{6,258}}$$

$$\xi_{6,266} = \frac{\xi_{6,121}\xi_{6,139}\xi_{6,254}+\xi_{6,255}(\xi_{6,114}\xi_{6,136}+\xi_{6,122}\xi_{6,139})+\xi_{6,256}(\xi_{6,115}\xi_{6,136}+\xi_{6,123}\xi_{6,139})}{\xi_{6,259}}$$

$$\xi_{6,267} = \frac{\xi_{6,124}\xi_{6,139}\xi_{6,254}+\xi_{6,255}(\xi_{6,116}\xi_{6,136}+\xi_{6,125}\xi_{6,139})+\xi_{6,256}(\xi_{6,117}\xi_{6,136}+\xi_{6,126}\xi_{6,139})}{\xi_{6,259}}$$

$$\xi_{6,268} = \frac{\xi_{6,119}\xi_{6,136}\xi_{6,255}+\xi_{6,120}\xi_{6,136}\xi_{6,256}+\xi_{6,254}(\xi_{6,118}\xi_{6,136}+\xi_{6,127}\xi_{6,139})}{\xi_{6,259}}$$

$$\xi_{6,269} = 1 - \xi_{6,260}^2$$

$$\xi_{6,270} = -\xi_{6,260}\xi_{6,261}$$

$$\xi_{6,271} = -\xi_{6,260}\xi_{6,262}$$

$$\xi_{6,272} = 1 - \xi_{6,261}^2$$

$$\xi_{6,273} = -\xi_{6,261}\xi_{6,262}$$

$$\xi_{6,274} = 1 - \xi_{6,262}^2$$

$$\xi_{6,275} = \xi_{6,269}^2 + \xi_{6,270}^2 + \xi_{6,271}^2$$

$$\xi_{6,276} = \xi_{6,270}^2 + \xi_{6,272}^2 + \xi_{6,273}^2$$

$$\xi_{6,277} = \xi_{6,271}^2 + \xi_{6,273}^2 + \xi_{6,274}^2$$

$$\xi_{6,278} = \sqrt{\xi_{6,275}}$$

$$\xi_{6,279} = \frac{\xi_{6,269}}{\xi_{6,278}}$$

$$\xi_{6,280} = \frac{\xi_{6,270}}{\xi_{6,278}}$$

$$\xi_{6,281} = \frac{\xi_{6,271}}{\xi_{6,278}}$$

$$\xi_{6,282} = \sqrt{\xi_{6,276}}$$

$$\xi_{6,283} = \frac{\xi_{6,272}}{\xi_{6,282}}$$

$$\xi_{6,284} = \frac{\xi_{6,270}}{\xi_{6,282}}$$

$$\xi_{6,285} = \frac{\xi_{6,273}}{\xi_{6,282}}$$

$$\xi_{6,286} = \sqrt{\xi_{6,277}}$$

$$\xi_{6,287} = \frac{\xi_{6,274}}{\xi_{6,286}}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,288} = \frac{\xi_{6,271}}{\xi_{6,286}}$$

$$\xi_{6,289} = \frac{\xi_{6,273}}{\xi_{6,286}}$$

$$\xi_{6,290} = \begin{cases} \xi_{6,279} & \text{for } \xi_{6,275} \geq \xi_{6,276} \wedge \xi_{6,275} \geq \xi_{6,277} \\ \xi_{6,284} & \text{for } \xi_{6,276} \geq \xi_{6,277} \wedge \xi_{6,275} \leq \xi_{6,276} \\ \xi_{6,288} & \text{otherwise} \end{cases}$$

$$\xi_{6,291} = \begin{cases} \xi_{6,280} & \text{for } \xi_{6,275} \geq \xi_{6,276} \wedge \xi_{6,275} \geq \xi_{6,277} \\ \xi_{6,283} & \text{for } \xi_{6,276} \geq \xi_{6,277} \wedge \xi_{6,275} \leq \xi_{6,276} \\ \xi_{6,289} & \text{otherwise} \end{cases}$$

$$\xi_{6,292} = \begin{cases} \xi_{6,281} & \text{for } \xi_{6,275} \geq \xi_{6,276} \wedge \xi_{6,275} \geq \xi_{6,277} \\ \xi_{6,285} & \text{for } \xi_{6,276} \geq \xi_{6,277} \wedge \xi_{6,275} \leq \xi_{6,276} \\ \xi_{6,287} & \text{otherwise} \end{cases}$$

$$\xi_{6,293} = \xi_{6,261}\xi_{6,292} - \xi_{6,262}\xi_{6,291}$$

$$\xi_{6,294} = -\xi_{6,260}\xi_{6,292} + \xi_{6,262}\xi_{6,290}$$

$$\xi_{6,295} = \xi_{6,260}\xi_{6,291} - \xi_{6,261}\xi_{6,290}$$

$$\xi_{6,296} = 1 - \xi_{6,263}^2$$

$$\xi_{6,297} = -\xi_{6,263}\xi_{6,264}$$

$$\xi_{6,298} = -\xi_{6,263}\xi_{6,265}$$

$$\xi_{6,299} = 1 - \xi_{6,264}^2$$

$$\xi_{6,300} = -\xi_{6,264}\xi_{6,265}$$

$$\xi_{6,301} = 1 - \xi_{6,265}^2$$

$$\xi_{6,302} = \xi_{6,296}^2 + \xi_{6,297}^2 + \xi_{6,298}^2$$

$$\xi_{6,303} = \xi_{6,297}^2 + \xi_{6,299}^2 + \xi_{6,300}^2$$

$$\xi_{6,304} = \xi_{6,298}^2 + \xi_{6,300}^2 + \xi_{6,301}^2$$

$$\xi_{6,305} = \sqrt{\xi_{6,302}}$$

$$\xi_{6,306} = \frac{\xi_{6,296}}{\xi_{6,305}}$$

$$\xi_{6,307} = \frac{\xi_{6,297}}{\xi_{6,305}}$$

$$\xi_{6,308} = \frac{\xi_{6,298}}{\xi_{6,305}}$$

$$\xi_{6,309} = \sqrt{\xi_{6,303}}$$

$$\xi_{6,310} = \frac{\xi_{6,299}}{\xi_{6,309}}$$

$$\xi_{6,311} = \frac{\xi_{6,297}}{\xi_{6,309}}$$

$$\xi_{6,312} = \frac{\xi_{6,300}}{\xi_{6,309}}$$

$$\xi_{6,313} = \sqrt{\xi_{6,304}}$$

$$\xi_{6,314} = \frac{\xi_{6,301}}{\xi_{6,313}}$$

$$\xi_{6,315} = \frac{\xi_{6,298}}{\xi_{6,313}}$$

$$\xi_{6,316} = \frac{\xi_{6,300}}{\xi_{6,313}}$$

$$\xi_{6,317} = \begin{cases} \xi_{6,306} & \text{for } \xi_{6,302} \geq \xi_{6,303} \wedge \xi_{6,302} \geq \xi_{6,304} \\ \xi_{6,311} & \text{for } \xi_{6,303} \geq \xi_{6,304} \wedge \xi_{6,302} \leq \xi_{6,303} \\ \xi_{6,315} & \text{otherwise} \end{cases}$$

$$\xi_{6,318} = \begin{cases} \xi_{6,307} & \text{for } \xi_{6,302} \geq \xi_{6,303} \wedge \xi_{6,302} \geq \xi_{6,304} \\ \xi_{6,310} & \text{for } \xi_{6,303} \geq \xi_{6,304} \wedge \xi_{6,302} \leq \xi_{6,303} \\ \xi_{6,316} & \text{otherwise} \end{cases}$$

$$\xi_{6,319} = \begin{cases} \xi_{6,308} & \text{for } \xi_{6,302} \geq \xi_{6,303} \wedge \xi_{6,302} \geq \xi_{6,304} \\ \xi_{6,312} & \text{for } \xi_{6,303} \geq \xi_{6,304} \wedge \xi_{6,302} \leq \xi_{6,303} \\ \xi_{6,314} & \text{otherwise} \end{cases}$$

$$\xi_{6,320} = \xi_{6,264}\xi_{6,319} - \xi_{6,265}\xi_{6,318}$$

$$\xi_{6,321} = -\xi_{6,263}\xi_{6,319} + \xi_{6,265}\xi_{6,317}$$

$$\xi_{6,322} = \xi_{6,263}\xi_{6,318} - \xi_{6,264}\xi_{6,317}$$

$$\xi_{6,323} = \xi_{6,261}\xi_{6,265} - \xi_{6,262}\xi_{6,264}$$

Exact shared-expression program 6

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$$\xi_{6,324} = -\xi_{6,260}\xi_{6,265} + \xi_{6,262}\xi_{6,263}$$

$$\xi_{6,325} = \xi_{6,260}\xi_{6,264} - \xi_{6,261}\xi_{6,263}$$

$$\xi_{6,326} = 1 - \xi_{6,266}^2$$

$$\xi_{6,327} = -\xi_{6,266}\xi_{6,267}$$

$$\xi_{6,328} = -\xi_{6,266}\xi_{6,268}$$

$$\xi_{6,329} = 1 - \xi_{6,267}^2$$

$$\xi_{6,330} = -\xi_{6,267}\xi_{6,268}$$

$$\xi_{6,331} = 1 - \xi_{6,268}^2$$

$$\xi_{6,332} = \xi_{6,326}^2 + \xi_{6,327}^2 + \xi_{6,328}^2$$

$$\xi_{6,333} = \xi_{6,327}^2 + \xi_{6,329}^2 + \xi_{6,330}^2$$

$$\xi_{6,334} = \xi_{6,328}^2 + \xi_{6,330}^2 + \xi_{6,331}^2$$

$$\xi_{6,335} = \sqrt{\xi_{6,332}}$$

$$\xi_{6,336} = \frac{\xi_{6,326}}{\xi_{6,335}}$$

$$\xi_{6,337} = \frac{\xi_{6,327}}{\xi_{6,335}}$$

$$\xi_{6,338} = \frac{\xi_{6,328}}{\xi_{6,335}}$$

$$\xi_{6,339} = \sqrt{\xi_{6,333}}$$

$$\xi_{6,340} = \frac{\xi_{6,329}}{\xi_{6,339}}$$

$$\xi_{6,341} = \frac{\xi_{6,327}}{\xi_{6,339}}$$

$$\xi_{6,342} = \frac{\xi_{6,330}}{\xi_{6,339}}$$

$$\xi_{6,343} = \sqrt{\xi_{6,334}}$$

$$\xi_{6,344} = \frac{\xi_{6,331}}{\xi_{6,343}}$$

$$\xi_{6,345} = \frac{\xi_{6,328}}{\xi_{6,343}}$$

$$\xi_{6,346} = \frac{\xi_{6,330}}{\xi_{6,343}}$$

$$\xi_{6,347} = \begin{cases} \xi_{6,336} & \text{for } \xi_{6,332} \geq \xi_{6,333} \wedge \xi_{6,332} \geq \xi_{6,334} \\ \xi_{6,341} & \text{for } \xi_{6,333} \geq \xi_{6,334} \wedge \xi_{6,332} \leq \xi_{6,333} \\ \xi_{6,345} & \text{otherwise} \end{cases}$$

$$\xi_{6,348} = \begin{cases} \xi_{6,337} & \text{for } \xi_{6,332} \geq \xi_{6,333} \wedge \xi_{6,332} \geq \xi_{6,334} \\ \xi_{6,340} & \text{for } \xi_{6,333} \geq \xi_{6,334} \wedge \xi_{6,332} \leq \xi_{6,333} \\ \xi_{6,346} & \text{otherwise} \end{cases}$$

$$\xi_{6,349} = \begin{cases} \xi_{6,338} & \text{for } \xi_{6,332} \geq \xi_{6,333} \wedge \xi_{6,332} \geq \xi_{6,334} \\ \xi_{6,342} & \text{for } \xi_{6,333} \geq \xi_{6,334} \wedge \xi_{6,332} \leq \xi_{6,333} \\ \xi_{6,344} & \text{otherwise} \end{cases}$$

$$\xi_{6,350} = \xi_{6,267}\xi_{6,349} - \xi_{6,268}\xi_{6,348}$$

$$\xi_{6,351} = -\xi_{6,266}\xi_{6,349} + \xi_{6,268}\xi_{6,347}$$

$$\xi_{6,352} = \xi_{6,266}\xi_{6,348} - \xi_{6,267}\xi_{6,347}$$

$$\xi_{6,353} = -\xi_{6,261}\xi_{6,268} + \xi_{6,262}\xi_{6,267}$$

$$\xi_{6,354} = \xi_{6,260}\xi_{6,268} - \xi_{6,262}\xi_{6,266}$$

$$\xi_{6,355} = -\xi_{6,260}\xi_{6,267} + \xi_{6,261}\xi_{6,266}$$

$$\xi_{6,356} = \xi_{6,264}\xi_{6,268} - \xi_{6,265}\xi_{6,267}$$

$$\xi_{6,357} = -\xi_{6,263}\xi_{6,268} + \xi_{6,265}\xi_{6,266}$$

$$\xi_{6,358} = \xi_{6,263}\xi_{6,267} - \xi_{6,264}\xi_{6,266}$$

$$\xi_{6,359} = \begin{cases} 1 & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \\ \xi_{6,260} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \\ \xi_{6,320} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,258} \neq 0 \\ \xi_{6,260} & \text{for } \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,258} \neq 0 \\ \xi_{6,347} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,260} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,356} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,260} & \text{for } \xi_{6,257} \neq 0 \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \end{cases}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,360} = \begin{cases} 0 & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \\ \xi_{6,290} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \\ \xi_{6,263} & \text{for } \xi_{6,259} = 0 \wedge (\xi_{6,257} = 0 \vee \xi_{6,259} = 0) \wedge (\xi_{6,257} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,259} = 0 \vee \xi_{6,257} \neq 0) \wedge (\xi_{6,259} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,258} \neq 0) \wedge \xi_{6,258} \neq 0 \\ \xi_{6,350} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,353} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,263} & \text{for } (\xi_{6,257} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} = 0 \vee \xi_{6,259} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,259} \neq 0) \wedge (\xi_{6,258} \neq 0 \vee \xi_{6,259} \neq 0) \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \end{cases}$$

$$\xi_{6,362} = \begin{cases} 0 & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \\ \xi_{6,261} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \\ \xi_{6,321} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,258} \neq 0 \\ \xi_{6,261} & \text{for } \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,258} \neq 0 \\ \xi_{6,348} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,261} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,357} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,261} & \text{for } \xi_{6,257} \neq 0 \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \end{cases}$$

$$\xi_{6,363} = \begin{cases} 1 & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \\ \xi_{6,291} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \\ \xi_{6,264} & \text{for } \xi_{6,259} = 0 \wedge (\xi_{6,257} = 0 \vee \xi_{6,259} = 0) \wedge (\xi_{6,257} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,259} = 0 \vee \xi_{6,257} \neq 0) \wedge (\xi_{6,259} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,258} \neq 0) \wedge \xi_{6,258} \neq 0 \\ \xi_{6,351} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,354} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,264} & \text{for } (\xi_{6,257} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} = 0 \vee \xi_{6,259} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,259} \neq 0) \wedge (\xi_{6,258} \neq 0 \vee \xi_{6,259} \neq 0) \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \end{cases}$$

$$\xi_{6,365} = \begin{cases} 0 & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \\ \xi_{6,262} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \\ \xi_{6,322} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,258} \neq 0 \\ \xi_{6,262} & \text{for } \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,258} \neq 0 \\ \xi_{6,349} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,262} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,358} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,262} & \text{for } \xi_{6,257} \neq 0 \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \end{cases}$$

$$\xi_{6,366} = \begin{cases} 0 & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \\ \xi_{6,292} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,259} = 0 \wedge \xi_{6,257} \neq 0 \\ \xi_{6,265} & \text{for } \xi_{6,259} = 0 \wedge (\xi_{6,257} = 0 \vee \xi_{6,259} = 0) \wedge (\xi_{6,257} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,259} = 0 \vee \xi_{6,257} \neq 0) \wedge (\xi_{6,259} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,258} \neq 0) \wedge \xi_{6,258} \neq 0 \\ \xi_{6,352} & \text{for } \xi_{6,257} = 0 \wedge \xi_{6,258} = 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,355} & \text{for } \xi_{6,258} = 0 \wedge \xi_{6,257} \neq 0 \wedge \xi_{6,259} \neq 0 \\ \xi_{6,265} & \text{for } (\xi_{6,257} = 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} = 0 \vee \xi_{6,259} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,258} \neq 0) \wedge (\xi_{6,257} \neq 0 \vee \xi_{6,259} \neq 0) \wedge (\xi_{6,258} \neq 0 \vee \xi_{6,259} \neq 0) \wedge \xi_{6,258} \neq 0 \wedge \xi_{6,259} \neq 0 \end{cases}$$

$$\xi_{6,368} = \frac{\xi_{6,359}}{4} + \frac{\xi_{6,363}}{4} + \frac{\xi_{6,367}}{4} + \frac{1}{4}$$

$$\xi_{6,369} = \frac{\xi_{6,359}}{4} - \frac{\xi_{6,363}}{4} - \frac{\xi_{6,367}}{4} + \frac{1}{4}$$

$$\xi_{6,370} = -\frac{\xi_{6,359}}{4} + \frac{\xi_{6,363}}{4} - \frac{\xi_{6,367}}{4} + \frac{1}{4}$$

$$\xi_{6,371} = -\frac{\xi_{6,359}}{4} - \frac{\xi_{6,363}}{4} + \frac{\xi_{6,367}}{4} + \frac{1}{4}$$

$$\xi_{6,372} = \sqrt{\xi_{6,368}}$$

$$\xi_{6,373} = \frac{-\xi_{6,364} + \xi_{6,366}}{4\xi_{6,372}}$$

$$\xi_{6,374} = \frac{\xi_{6,361} - \xi_{6,365}}{4\xi_{6,372}}$$

$$\xi_{6,375} = \frac{-\xi_{6,360} + \xi_{6,362}}{4\xi_{6,372}}$$

$$\xi_{6,376} = \sqrt{\xi_{6,369}}$$

$$\xi_{6,377} = \frac{-\xi_{6,364} + \xi_{6,366}}{4\xi_{6,376}}$$

$$\xi_{6,378} = \frac{\xi_{6,360} + \xi_{6,362}}{4\xi_{6,376}}$$

$$\xi_{6,379} = \frac{\xi_{6,361} + \xi_{6,365}}{4\xi_{6,376}}$$

$$\xi_{6,380} = \sqrt{\xi_{6,370}}$$

$$\xi_{6,381} = \frac{\xi_{6,361} - \xi_{6,365}}{4\xi_{6,380}}$$

$$\xi_{6,382} = \frac{\xi_{6,360} + \xi_{6,362}}{4\xi_{6,380}}$$

$$\xi_{6,383} = \frac{\xi_{6,364} + \xi_{6,366}}{4\xi_{6,380}}$$

$$\xi_{6,384} = \sqrt{\xi_{6,371}}$$

$$\xi_{6,385} = \frac{-\xi_{6,360} + \xi_{6,362}}{4\xi_{6,384}}$$

$$\xi_{6,386} = \frac{\xi_{6,361} + \xi_{6,365}}{4\xi_{6,384}}$$

$$\xi_{6,387} = \frac{\xi_{6,364} + \xi_{6,366}}{4\xi_{6,384}}$$

$$\xi_{6,388} = \begin{cases} \xi_{6,372} & \text{for } \xi_{6,368} \geq \xi_{6,369} \wedge \xi_{6,368} \geq \xi_{6,370} \wedge \xi_{6,368} \geq \xi_{6,371} \\ \xi_{6,377} & \text{for } \xi_{6,369} \geq \xi_{6,370} \wedge \xi_{6,369} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,369} \\ \xi_{6,381} & \text{for } \xi_{6,370} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,370} \wedge \xi_{6,369} \leq \xi_{6,370} \\ \xi_{6,385} & \text{otherwise} \end{cases}$$

$$\xi_{6,389} = \begin{cases} \xi_{6,373} & \text{for } \xi_{6,368} \geq \xi_{6,369} \wedge \xi_{6,368} \geq \xi_{6,370} \wedge \xi_{6,368} \geq \xi_{6,371} \\ \xi_{6,376} & \text{for } \xi_{6,369} \geq \xi_{6,370} \wedge \xi_{6,369} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,369} \\ \xi_{6,382} & \text{for } \xi_{6,370} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,370} \wedge \xi_{6,369} \leq \xi_{6,370} \\ \xi_{6,386} & \text{otherwise} \end{cases}$$

$$\xi_{6,390} = \begin{cases} \xi_{6,374} & \text{for } \xi_{6,368} \geq \xi_{6,369} \wedge \xi_{6,368} \geq \xi_{6,370} \wedge \xi_{6,368} \geq \xi_{6,371} \\ \xi_{6,378} & \text{for } \xi_{6,369} \geq \xi_{6,370} \wedge \xi_{6,369} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,369} \\ \xi_{6,380} & \text{for } \xi_{6,370} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,370} \wedge \xi_{6,369} \leq \xi_{6,370} \\ \xi_{6,387} & \text{otherwise} \end{cases}$$

$$\xi_{6,391} = \begin{cases} \xi_{6,375} & \text{for } \xi_{6,368} \geq \xi_{6,369} \wedge \xi_{6,368} \geq \xi_{6,370} \wedge \xi_{6,368} \geq \xi_{6,371} \\ \xi_{6,379} & \text{for } \xi_{6,369} \geq \xi_{6,370} \wedge \xi_{6,369} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,369} \\ \xi_{6,383} & \text{for } \xi_{6,370} \geq \xi_{6,371} \wedge \xi_{6,368} \leq \xi_{6,370} \wedge \xi_{6,369} \leq \xi_{6,370} \\ \xi_{6,384} & \text{otherwise} \end{cases}$$

$$\xi_{6,392} = \frac{\xi_{6,188}}{4} + \frac{\xi_{6,252}}{4} + \frac{\xi_{6,256}}{4} + \frac{1}{4}$$

$$\xi_{6,393} = \frac{\xi_{6,188}}{4} - \frac{\xi_{6,252}}{4} - \frac{\xi_{6,256}}{4} + \frac{1}{4}$$

$$\xi_{6,394} = -\frac{\xi_{6,188}}{4} + \frac{\xi_{6,252}}{4} - \frac{\xi_{6,256}}{4} + \frac{1}{4}$$

$$\xi_{6,395} = -\frac{\xi_{6,188}}{4} - \frac{\xi_{6,252}}{4} + \frac{\xi_{6,256}}{4} + \frac{1}{4}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,396} = \sqrt{\xi_{6,392}}$$

$$\xi_{6,397} = \frac{\xi_{6,253} - \xi_{6,255}}{4\xi_{6,396}}$$

$$\xi_{6,398} = \frac{-\xi_{6,190} + \xi_{6,254}}{4\xi_{6,396}}$$

$$\xi_{6,399} = \frac{\xi_{6,189} - \xi_{6,251}}{4\xi_{6,396}}$$

$$\xi_{6,400} = \sqrt{\xi_{6,393}}$$

$$\xi_{6,401} = \frac{\xi_{6,253} - \xi_{6,255}}{4\xi_{6,400}}$$

$$\xi_{6,402} = \frac{\xi_{6,189} + \xi_{6,251}}{4\xi_{6,400}}$$

$$\xi_{6,403} = \frac{\xi_{6,190} + \xi_{6,254}}{4\xi_{6,400}}$$

$$\xi_{6,404} = \sqrt{\xi_{6,394}}$$

$$\xi_{6,405} = \frac{-\xi_{6,190} + \xi_{6,254}}{4\xi_{6,404}}$$

$$\xi_{6,406} = \frac{\xi_{6,189} + \xi_{6,251}}{4\xi_{6,404}}$$

$$\xi_{6,407} = \frac{\xi_{6,253} + \xi_{6,255}}{4\xi_{6,404}}$$

$$\xi_{6,408} = \sqrt{\xi_{6,395}}$$

$$\xi_{6,409} = \frac{\xi_{6,189} - \xi_{6,251}}{4\xi_{6,408}}$$

$$\xi_{6,410} = \frac{\xi_{6,190} + \xi_{6,254}}{4\xi_{6,408}}$$

$$\xi_{6,411} = \frac{\xi_{6,253} + \xi_{6,255}}{4\xi_{6,408}}$$

$$\xi_{6,412} = \begin{cases} \xi_{6,396} & \text{for } \xi_{6,392} \geq \xi_{6,393} \wedge \xi_{6,392} \geq \xi_{6,394} \wedge \xi_{6,392} \geq \xi_{6,395} \\ \xi_{6,401} & \text{for } \xi_{6,393} \geq \xi_{6,394} \wedge \xi_{6,393} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,393} \\ \xi_{6,405} & \text{for } \xi_{6,394} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,394} \wedge \xi_{6,393} \leq \xi_{6,394} \\ \xi_{6,409} & \text{otherwise} \end{cases}$$

$$\xi_{6,413} = \begin{cases} \xi_{6,397} & \text{for } \xi_{6,392} \geq \xi_{6,393} \wedge \xi_{6,392} \geq \xi_{6,394} \wedge \xi_{6,392} \geq \xi_{6,395} \\ \xi_{6,400} & \text{for } \xi_{6,393} \geq \xi_{6,394} \wedge \xi_{6,393} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,393} \\ \xi_{6,406} & \text{for } \xi_{6,394} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,394} \wedge \xi_{6,393} \leq \xi_{6,394} \\ \xi_{6,410} & \text{otherwise} \end{cases}$$

$$\xi_{6,414} = \begin{cases} \xi_{6,398} & \text{for } \xi_{6,392} \geq \xi_{6,393} \wedge \xi_{6,392} \geq \xi_{6,394} \wedge \xi_{6,392} \geq \xi_{6,395} \\ \xi_{6,402} & \text{for } \xi_{6,393} \geq \xi_{6,394} \wedge \xi_{6,393} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,393} \\ \xi_{6,404} & \text{for } \xi_{6,394} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,394} \wedge \xi_{6,393} \leq \xi_{6,394} \\ \xi_{6,411} & \text{otherwise} \end{cases}$$

$$\xi_{6,415} = \begin{cases} \xi_{6,399} & \text{for } \xi_{6,392} \geq \xi_{6,393} \wedge \xi_{6,392} \geq \xi_{6,394} \wedge \xi_{6,392} \geq \xi_{6,395} \\ \xi_{6,403} & \text{for } \xi_{6,393} \geq \xi_{6,394} \wedge \xi_{6,393} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,393} \\ \xi_{6,407} & \text{for } \xi_{6,394} \geq \xi_{6,395} \wedge \xi_{6,392} \leq \xi_{6,394} \wedge \xi_{6,393} \leq \xi_{6,394} \\ \xi_{6,408} & \text{otherwise} \end{cases}$$

$$\xi_{6,416} = \xi_{6,107}\left(\xi_{6,112} + i\xi_{6,113}\right)$$

$$\xi_{6,417} = -\frac{\sqrt{2}\xi_{6,107}\xi_{6,109}}{2} + \frac{\sqrt{2}i\xi_{6,108}\left(\xi_{6,112} + i\xi_{6,113}\right)}{2}$$

$$\xi_{6,418} = \frac{\sqrt{2}i\xi_{6,107}\xi_{6,109}}{2} - \frac{\sqrt{2}\xi_{6,108}\left(\xi_{6,112} + i\xi_{6,113}\right)}{2}$$

$$\xi_{6,419} = -\xi_{6,108}\xi_{6,109}$$

$$\xi_{6,420} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{6,107}\left(\xi_{6,112} + i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}i\xi_{6,108}\xi_{6,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{6,107}\left(\xi_{6,112} - i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}\xi_{6,108}\xi_{6,109}}{2}\right)}{2}$$

$$\xi_{6,421} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{6,107}\left(\xi_{6,112} + i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}i\xi_{6,108}\xi_{6,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{6,107}\left(\xi_{6,112} - i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}\xi_{6,108}\xi_{6,109}}{2}\right)}{2}$$

$$\xi_{6,422} = \frac{\sqrt{2}i\xi_{6,107}\xi_{6,109}}{2} - \frac{\sqrt{2}\xi_{6,108}\left(\xi_{6,112} - i\xi_{6,113}\right)}{2}$$

$$\xi_{6,423} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{6,107}\left(\xi_{6,112} - i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}i\xi_{6,108}\xi_{6,109}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{6,107}\left(\xi_{6,112} + i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}\xi_{6,108}\xi_{6,109}}{2}\right)}{2}$$

$$\xi_{6,424} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{6,107}\left(\xi_{6,112} - i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}i\xi_{6,108}\xi_{6,109}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{6,107}\left(\xi_{6,112} + i\xi_{6,113}\right)}{2} - \frac{\sqrt{2}\xi_{6,108}\xi_{6,109}}{2}\right)}{2}$$

$$\xi_{6,425} = -\frac{\sqrt{2}\xi_{6,107}\xi_{6,109}}{2} + \frac{\sqrt{2}i\xi_{6,108}\left(\xi_{6,112} - i\xi_{6,113}\right)}{2}$$

$$\xi_{6,426} = \xi_{6,107}\left(\xi_{6,112} - i\xi_{6,113}\right)$$

$$\xi_{6,427} = \left(\xi_{6,388} - i\xi_{6,391}\right)\left(\xi_{6,412} - i\xi_{6,415}\right)$$

$$\xi_{6,428} = \left(\xi_{6,388} - i\xi_{6,391}\right)\left(-i\xi_{6,413} - \xi_{6,414}\right)$$

$$\xi_{6,429} = \left(\xi_{6,412} - i\xi_{6,415}\right)\left(-i\xi_{6,389} - \xi_{6,390}\right)$$

$$\xi_{6,430} = \left(-i\xi_{6,389} - \xi_{6,390}\right)\left(-i\xi_{6,413} - \xi_{6,414}\right)$$

$$\xi_{6,431} = \left(\xi_{6,388} - i\xi_{6,391}\right)\left(-i\xi_{6,413} + \xi_{6,414}\right)$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,432} = (\xi_{6,388} - i\xi_{6,391})\,(\xi_{6,412} + i\xi_{6,415})$$

$$\xi_{6,433} = (-i\xi_{6,389} - \xi_{6,390})\,(-i\xi_{6,413} + \xi_{6,414})$$

$$\xi_{6,434} = (\xi_{6,412} + i\xi_{6,415})\,(-i\xi_{6,389} - \xi_{6,390})$$

$$\xi_{6,435} = (\xi_{6,412} - i\xi_{6,415})\,(-i\xi_{6,389} + \xi_{6,390})$$

$$\xi_{6,436} = (-i\xi_{6,389} + \xi_{6,390})\,(-i\xi_{6,413} - \xi_{6,414})$$

$$\xi_{6,437} = (\xi_{6,388} + i\xi_{6,391})\,(\xi_{6,412} - i\xi_{6,415})$$

$$\xi_{6,438} = (\xi_{6,388} + i\xi_{6,391})\,(-i\xi_{6,413} - \xi_{6,414})$$

$$\xi_{6,439} = (-i\xi_{6,389} + \xi_{6,390})\,(-i\xi_{6,413} + \xi_{6,414})$$

$$\xi_{6,440} = (\xi_{6,412} + i\xi_{6,415})\,(-i\xi_{6,389} + \xi_{6,390})$$

$$\xi_{6,441} = (\xi_{6,388} + i\xi_{6,391})\,(-i\xi_{6,413} + \xi_{6,414})$$

$$\xi_{6,442} = (\xi_{6,388} + i\xi_{6,391})\,(\xi_{6,412} + i\xi_{6,415})$$

$$\xi_{6,443} = \xi_{6,416}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}}\right) + \xi_{6,417}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}}\right) + \xi_{6,418}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}}\right) + \xi_{6,419}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}}\right)$$

$$\xi_{6,444} = \xi_{6,417}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}}\right) + \xi_{6,420}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}}\right) + \xi_{6,422}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}}\right) + \xi_{6,423}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}}\right)$$

$$\xi_{6,445} = \xi_{6,418}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}}\right) + \xi_{6,421}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}}\right) + \xi_{6,424}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}}\right) + \xi_{6,425}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}}\right)$$

$$\xi_{6,446} = \xi_{6,419}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}}\right) + \xi_{6,422}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}}\right) + \xi_{6,425}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}}\right) + \xi_{6,426}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}}\right)$$

$$\xi_{6,447} = \xi_{6,416}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}}\right) + \xi_{6,417}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}}\right) + \xi_{6,418}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}}\right) + \xi_{6,419}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}}\right)$$

$$\xi_{6,448} = \xi_{6,417}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}}\right) + \xi_{6,420}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}}\right) + \xi_{6,422}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}}\right) + \xi_{6,423}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}}\right)$$

$$\xi_{6,449} = \xi_{6,418}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}}\right) + \xi_{6,421}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}}\right) + \xi_{6,424}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}}\right) + \xi_{6,425}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}}\right) + \xi_{6,426}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}}\right)$$

$$\xi_{6,450} = \xi_{6,419}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,427}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,430}}\right) + \xi_{6,422}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,431}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,434}}\right) + \xi_{6,425}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,435}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,438}}\right) + \xi_{6,426}\left(-ie^{-i\xi_{6,100}}\sin(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,439}} + e^{-i\xi_{6,100}}\cos(\xi_{6,98}-\xi_{6,99})\overline{\xi_{6,442}}\right)$$

$$\xi_{6,451} = \xi_{6,416}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}}\right) + \xi_{6,417}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}}\right) + \xi_{6,418}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}}\right) + \xi_{6,419}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}}\right)$$

$$\xi_{6,452} = \xi_{6,417}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}}\right) + \xi_{6,420}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}}\right) + \xi_{6,422}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}}\right) + \xi_{6,423}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}}\right)$$

$$\xi_{6,453} = \xi_{6,418}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}}\right) + \xi_{6,421}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}}\right) + \xi_{6,424}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}}\right) + \xi_{6,425}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}}\right)$$

$$\xi_{6,454} = \xi_{6,419}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}}\right) + \xi_{6,422}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}}\right) + \xi_{6,425}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}}\right) + \xi_{6,426}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}}\right)$$

$$\xi_{6,455} = \xi_{6,416}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}}\right) + \xi_{6,417}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}}\right) + \xi_{6,418}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}}\right) + \xi_{6,419}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}}\right)$$

$$\xi_{6,456} = \xi_{6,417}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}}\right) + \xi_{6,420}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}}\right) + \xi_{6,422}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}}\right) + \xi_{6,423}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}}\right)$$

$$\xi_{6,457} = \xi_{6,418}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}}\right) + \xi_{6,421}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}}\right) + \xi_{6,424}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}}\right) + \xi_{6,425}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}}\right)$$

$$\xi_{6,458} = \xi_{6,419}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,428}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,429}}\right) + \xi_{6,422}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,432}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,433}}\right) + \xi_{6,425}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,436}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,437}}\right) + \xi_{6,426}\left(-ie^{i\xi_{6,100}}\sin(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,440}} + e^{i\xi_{6,100}}\cos(\xi_{6,98}+\xi_{6,99})\overline{\xi_{6,441}}\right)$$

$$\xi_{6,459} = \frac{\operatorname{re}(\xi_{6,443})}{4} + \frac{\operatorname{re}(\xi_{6,450})}{4} + \frac{\operatorname{re}(\xi_{6,452})}{4} + \frac{\operatorname{re}(\xi_{6,457})}{4}$$

$$\xi_{6,460} = -\frac{\operatorname{im}(\xi_{6,444})}{4} - \frac{\operatorname{im}(\xi_{6,449})}{4} - \frac{\operatorname{im}(\xi_{6,451})}{4} - \frac{\operatorname{im}(\xi_{6,458})}{4}$$

$$\xi_{6,461} = -\frac{\operatorname{re}(\xi_{6,444})}{4} + \frac{\operatorname{re}(\xi_{6,449})}{4} + \frac{\operatorname{re}(\xi_{6,451})}{4} - \frac{\operatorname{re}(\xi_{6,458})}{4}$$

$$\xi_{6,462} = -\frac{\operatorname{im}(\xi_{6,443})}{4} + \frac{\operatorname{im}(\xi_{6,450})}{4} + \frac{\operatorname{im}(\xi_{6,452})}{4} - \frac{\operatorname{im}(\xi_{6,457})}{4}$$

$$\xi_{6,463} = -\frac{\operatorname{im}(\xi_{6,445})}{4} - \frac{\operatorname{im}(\xi_{6,448})}{4} - \frac{\operatorname{im}(\xi_{6,454})}{4} - \frac{\operatorname{im}(\xi_{6,455})}{4}$$

$$\xi_{6,464} = -\frac{\operatorname{re}(\xi_{6,446})}{4} - \frac{\operatorname{re}(\xi_{6,447})}{4} - \frac{\operatorname{re}(\xi_{6,453})}{4} - \frac{\operatorname{re}(\xi_{6,456})}{4}$$

$$\xi_{6,465} = \frac{\operatorname{im}(\xi_{6,446})}{4} - \frac{\operatorname{im}(\xi_{6,447})}{4} - \frac{\operatorname{im}(\xi_{6,453})}{4} + \frac{\operatorname{im}(\xi_{6,456})}{4}$$

$$\xi_{6,466} = -\frac{\operatorname{re}(\xi_{6,445})}{4} + \frac{\operatorname{re}(\xi_{6,448})}{4} + \frac{\operatorname{re}(\xi_{6,454})}{4} - \frac{\operatorname{re}(\xi_{6,455})}{4}$$

$$\xi_{6,467} = -\frac{\operatorname{re}(\xi_{6,445})}{4} + \frac{\operatorname{re}(\xi_{6,448})}{4} - \frac{\operatorname{re}(\xi_{6,454})}{4} + \frac{\operatorname{re}(\xi_{6,455})}{4}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,468} = \frac{\operatorname{im}(\xi_{6,446})}{4} - \frac{\operatorname{im}(\xi_{6,447})}{4} + \frac{\operatorname{im}(\xi_{6,453})}{4} - \frac{\operatorname{im}(\xi_{6,456})}{4}$$

$$\xi_{6,469} = \frac{\operatorname{re}(\xi_{6,446})}{4} + \frac{\operatorname{re}(\xi_{6,447})}{4} - \frac{\operatorname{re}(\xi_{6,453})}{4} - \frac{\operatorname{re}(\xi_{6,456})}{4}$$

$$\xi_{6,470} = \frac{\operatorname{im}(\xi_{6,445})}{4} + \frac{\operatorname{im}(\xi_{6,448})}{4} - \frac{\operatorname{im}(\xi_{6,454})}{4} - \frac{\operatorname{im}(\xi_{6,455})}{4}$$

$$\xi_{6,471} = -\frac{\operatorname{im}(\xi_{6,443})}{4} + \frac{\operatorname{im}(\xi_{6,450})}{4} - \frac{\operatorname{im}(\xi_{6,452})}{4} + \frac{\operatorname{im}(\xi_{6,457})}{4}$$

$$\xi_{6,472} = -\frac{\operatorname{re}(\xi_{6,444})}{4} + \frac{\operatorname{re}(\xi_{6,449})}{4} - \frac{\operatorname{re}(\xi_{6,451})}{4} + \frac{\operatorname{re}(\xi_{6,458})}{4}$$

$$\xi_{6,473} = \frac{\operatorname{im}(\xi_{6,444})}{4} + \frac{\operatorname{im}(\xi_{6,449})}{4} - \frac{\operatorname{im}(\xi_{6,451})}{4} - \frac{\operatorname{im}(\xi_{6,458})}{4}$$

$$\xi_{6,474} = -\frac{\operatorname{re}(\xi_{6,443})}{4} - \frac{\operatorname{re}(\xi_{6,450})}{4} + \frac{\operatorname{re}(\xi_{6,452})}{4} + \frac{\operatorname{re}(\xi_{6,457})}{4}$$

$$\xi_{6,475} = \xi_{6,459}^2 + \xi_{6,463}^2 + \xi_{6,467}^2 + \xi_{6,471}^2$$

$$\xi_{6,476} = \xi_{6,460}^2 + \xi_{6,464}^2 + \xi_{6,468}^2 + \xi_{6,472}^2$$

$$\xi_{6,477} = \xi_{6,461}^2 + \xi_{6,465}^2 + \xi_{6,469}^2 + \xi_{6,473}^2$$

$$\xi_{6,478} = \xi_{6,462}^2 + \xi_{6,466}^2 + \xi_{6,470}^2 + \xi_{6,474}^2$$

$$\xi_{6,479} = \sqrt{\xi_{6,475}}$$

$$\xi_{6,480} = \frac{\xi_{6,459}}{\xi_{6,479}}$$

$$\xi_{6,481} = \frac{\xi_{6,463}}{\xi_{6,479}}$$

$$\xi_{6,482} = \frac{\xi_{6,467}}{\xi_{6,479}}$$

$$\xi_{6,483} = \frac{\xi_{6,471}}{\xi_{6,479}}$$

$$\xi_{6,484} = \sqrt{\xi_{6,476}}$$

$$\xi_{6,485} = \frac{\xi_{6,460}}{\xi_{6,484}}$$

$$\xi_{6,486} = \frac{\xi_{6,464}}{\xi_{6,484}}$$

$$\xi_{6,487} = \frac{\xi_{6,468}}{\xi_{6,484}}$$

$$\xi_{6,488} = \frac{\xi_{6,472}}{\xi_{6,484}}$$

$$\xi_{6,489} = \sqrt{\xi_{6,477}}$$

$$\xi_{6,490} = \frac{\xi_{6,461}}{\xi_{6,489}}$$

$$\xi_{6,491} = \frac{\xi_{6,465}}{\xi_{6,489}}$$

$$\xi_{6,492} = \frac{\xi_{6,469}}{\xi_{6,489}}$$

$$\xi_{6,493} = \frac{\xi_{6,473}}{\xi_{6,489}}$$

$$\xi_{6,494} = \sqrt{\xi_{6,478}}$$

$$\xi_{6,495} = \frac{\xi_{6,462}}{\xi_{6,494}}$$

$$\xi_{6,496} = \frac{\xi_{6,466}}{\xi_{6,494}}$$

$$\xi_{6,497} = \frac{\xi_{6,470}}{\xi_{6,494}}$$

$$\xi_{6,498} = \frac{\xi_{6,474}}{\xi_{6,494}}$$

$$\xi_{6,499} = \begin{cases} \xi_{6,480} & \text{for } \xi_{6,475} \geq \xi_{6,476} \wedge \xi_{6,475} \geq \xi_{6,477} \wedge \xi_{6,475} \geq \xi_{6,478} \\ \xi_{6,485} & \text{for } \xi_{6,476} \geq \xi_{6,477} \wedge \xi_{6,476} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,476} \\ \xi_{6,490} & \text{for } \xi_{6,477} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,477} \wedge \xi_{6,476} \leq \xi_{6,477} \\ \xi_{6,495} & \text{otherwise} \end{cases}$$

$$\xi_{6,500} = \begin{cases} \xi_{6,481} & \text{for } \xi_{6,475} \geq \xi_{6,476} \wedge \xi_{6,475} \geq \xi_{6,477} \wedge \xi_{6,475} \geq \xi_{6,478} \\ \xi_{6,486} & \text{for } \xi_{6,476} \geq \xi_{6,477} \wedge \xi_{6,476} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,476} \\ \xi_{6,491} & \text{for } \xi_{6,477} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,477} \wedge \xi_{6,476} \leq \xi_{6,477} \\ \xi_{6,496} & \text{otherwise} \end{cases}$$

$$\xi_{6,501} = \begin{cases} \xi_{6,482} & \text{for } \xi_{6,475} \geq \xi_{6,476} \wedge \xi_{6,475} \geq \xi_{6,477} \wedge \xi_{6,475} \geq \xi_{6,478} \\ \xi_{6,487} & \text{for } \xi_{6,476} \geq \xi_{6,477} \wedge \xi_{6,476} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,476} \\ \xi_{6,492} & \text{for } \xi_{6,477} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,477} \wedge \xi_{6,476} \leq \xi_{6,477} \\ \xi_{6,497} & \text{otherwise} \end{cases}$$

$$\xi_{6,502} = \begin{cases} \xi_{6,483} & \text{for } \xi_{6,475} \geq \xi_{6,476} \wedge \xi_{6,475} \geq \xi_{6,477} \wedge \xi_{6,475} \geq \xi_{6,478} \\ \xi_{6,488} & \text{for } \xi_{6,476} \geq \xi_{6,477} \wedge \xi_{6,476} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,476} \\ \xi_{6,493} & \text{for } \xi_{6,477} \geq \xi_{6,478} \wedge \xi_{6,475} \leq \xi_{6,477} \wedge \xi_{6,476} \leq \xi_{6,477} \\ \xi_{6,498} & \text{otherwise} \end{cases}$$

$$\xi_{6,503} = \xi_{6,459}\xi_{6,499} + \xi_{6,463}\xi_{6,500} + \xi_{6,467}\xi_{6,501} + \xi_{6,471}\xi_{6,502}$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,504} = \xi_{6,460}\xi_{6,499} + \xi_{6,464}\xi_{6,500} + \xi_{6,468}\xi_{6,501} + \xi_{6,472}\xi_{6,502}$$

$$\xi_{6,505} = \xi_{6,461}\xi_{6,499} + \xi_{6,465}\xi_{6,500} + \xi_{6,469}\xi_{6,501} + \xi_{6,473}\xi_{6,502}$$

$$\xi_{6,506} = \xi_{6,462}\xi_{6,499} + \xi_{6,466}\xi_{6,500} + \xi_{6,470}\xi_{6,501} + \xi_{6,474}\xi_{6,502}$$

$$\xi_{6,507} = \xi_{6,36}\xi_{6,47} - \xi_{6,37}\xi_{6,50} - \xi_{6,38}\xi_{6,51} - \xi_{6,39}\xi_{6,52}$$

$$\xi_{6,508} = \xi_{6,36}\xi_{6,50} + \xi_{6,37}\xi_{6,47} - \xi_{6,38}\xi_{6,52} + \xi_{6,39}\xi_{6,51}$$

$$\xi_{6,509} = \xi_{6,36}\xi_{6,51} + \xi_{6,37}\xi_{6,52} + \xi_{6,38}\xi_{6,47} - \xi_{6,39}\xi_{6,50}$$

$$\xi_{6,510} = \xi_{6,36}\xi_{6,52} - \xi_{6,37}\xi_{6,51} + \xi_{6,38}\xi_{6,50} + \xi_{6,39}\xi_{6,47}$$

$$\xi_{6,511} = \xi_{6,10}\xi_{6,50} + \xi_{6,49}\xi_{6,7} + \xi_{6,50}\xi_{6,8} + \xi_{6,50}\xi_{6,9}$$

$$\xi_{6,512} = -\xi_{6,10}\xi_{6,50} - \xi_{6,49}\xi_{6,8} + \xi_{6,50}\xi_{6,7} + \xi_{6,50}\xi_{6,9}$$

$$\xi_{6,513} = \xi_{6,10}\xi_{6,50} - \xi_{6,49}\xi_{6,9} + \xi_{6,50}\xi_{6,7} - \xi_{6,50}\xi_{6,8}$$

$$\xi_{6,514} = -\xi_{6,10}\xi_{6,49} + \xi_{6,50}\xi_{6,7} + \xi_{6,50}\xi_{6,8} - \xi_{6,50}\xi_{6,9}$$

$$\xi_{6,515} = \xi_{6,47}\left(\xi_{6,499}\xi_{6,94} + \xi_{6,500}\xi_{6,95} + \xi_{6,501}\xi_{6,96} + \xi_{6,502}\xi_{6,97}\right) - \xi_{6,47}\left(\xi_{6,499}\xi_{6,97} - \xi_{6,500}\xi_{6,96} + \xi_{6,501}\xi_{6,95} - \xi_{6,502}\xi_{6,94}\right) - \xi_{6,48}\left(\xi_{6,499}\xi_{6,95} - \xi_{6,500}\xi_{6,94} - \xi_{6,501}\xi_{6,97} + \xi_{6,502}\xi_{6,96}\right) - \xi_{6,49}\left(\xi_{6,499}\xi_{6,96} + \xi_{6,500}\xi_{6,97} - \xi_{6,501}\xi_{6,94} - \xi_{6,502}\xi_{6,95}\right)$$

$$\xi_{6,516} = \xi_{6,47}\left(\xi_{6,499}\xi_{6,95} - \xi_{6,500}\xi_{6,94} - \xi_{6,501}\xi_{6,97} + \xi_{6,502}\xi_{6,96}\right) + \xi_{6,47}\left(\xi_{6,499}\xi_{6,96} + \xi_{6,500}\xi_{6,97} - \xi_{6,501}\xi_{6,94} - \xi_{6,502}\xi_{6,95}\right) + \xi_{6,48}\left(\xi_{6,499}\xi_{6,94} + \xi_{6,500}\xi_{6,95} + \xi_{6,501}\xi_{6,96} + \xi_{6,502}\xi_{6,97}\right) - \xi_{6,49}\left(\xi_{6,499}\xi_{6,97} - \xi_{6,500}\xi_{6,96} + \xi_{6,501}\xi_{6,95} - \xi_{6,502}\xi_{6,94}\right)$$

$$\xi_{6,517} = -\xi_{6,47}\left(\xi_{6,499}\xi_{6,95} - \xi_{6,500}\xi_{6,94} - \xi_{6,501}\xi_{6,97} + \xi_{6,502}\xi_{6,96}\right) + \xi_{6,47}\left(\xi_{6,499}\xi_{6,96} + \xi_{6,500}\xi_{6,97} - \xi_{6,501}\xi_{6,94} - \xi_{6,502}\xi_{6,95}\right) + \xi_{6,48}\left(\xi_{6,499}\xi_{6,97} - \xi_{6,500}\xi_{6,96} + \xi_{6,501}\xi_{6,95} - \xi_{6,502}\xi_{6,94}\right) + \xi_{6,49}\left(\xi_{6,499}\xi_{6,94} + \xi_{6,500}\xi_{6,95} + \xi_{6,501}\xi_{6,96} + \xi_{6,502}\xi_{6,97}\right)$$

$$\xi_{6,518} = \xi_{6,47}\left(\xi_{6,499}\xi_{6,94} + \xi_{6,500}\xi_{6,95} + \xi_{6,501}\xi_{6,96} + \xi_{6,502}\xi_{6,97}\right) + \xi_{6,47}\left(\xi_{6,499}\xi_{6,97} - \xi_{6,500}\xi_{6,96} + \xi_{6,501}\xi_{6,95} - \xi_{6,502}\xi_{6,94}\right) - \xi_{6,48}\left(\xi_{6,499}\xi_{6,96} + \xi_{6,500}\xi_{6,97} - \xi_{6,501}\xi_{6,94} - \xi_{6,502}\xi_{6,95}\right) + \xi_{6,49}\left(\xi_{6,499}\xi_{6,95} - \xi_{6,500}\xi_{6,94} - \xi_{6,501}\xi_{6,97} + \xi_{6,502}\xi_{6,96}\right)$$

$$\xi_{6,519} = -\xi_{6,48}\left(-\xi_{6,503}\xi_{6,63} - \xi_{6,504}\xi_{6,62} + \xi_{6,505}\xi_{6,65} - \xi_{6,506}\xi_{6,64}\right) - \xi_{6,48}\left(-\xi_{6,503}\xi_{6,64} - \xi_{6,504}\xi_{6,65} - \xi_{6,505}\xi_{6,62} + \xi_{6,506}\xi_{6,63}\right) - \xi_{6,48}\left(-\xi_{6,503}\xi_{6,65} + \xi_{6,504}\xi_{6,64} - \xi_{6,505}\xi_{6,63} - \xi_{6,506}\xi_{6,62}\right) + \xi_{6,49}\left(\xi_{6,503}\xi_{6,62} - \xi_{6,504}\xi_{6,63} - \xi_{6,505}\xi_{6,64} - \xi_{6,506}\xi_{6,65}\right)$$

$$\xi_{6,520} = \xi_{6,48}\left(\xi_{6,503}\xi_{6,62} - \xi_{6,504}\xi_{6,63} - \xi_{6,505}\xi_{6,64} - \xi_{6,506}\xi_{6,65}\right) + \xi_{6,48}\left(-\xi_{6,503}\xi_{6,64} - \xi_{6,504}\xi_{6,65} - \xi_{6,505}\xi_{6,62} + \xi_{6,506}\xi_{6,63}\right) - \xi_{6,48}\left(-\xi_{6,503}\xi_{6,65} + \xi_{6,504}\xi_{6,64} - \xi_{6,505}\xi_{6,63} - \xi_{6,506}\xi_{6,62}\right) + \xi_{6,49}\left(-\xi_{6,503}\xi_{6,63} - \xi_{6,504}\xi_{6,62} + \xi_{6,505}\xi_{6,65} - \xi_{6,506}\xi_{6,64}\right)$$

$$\xi_{6,521} = \xi_{6,48}\left(\xi_{6,503}\xi_{6,62} - \xi_{6,504}\xi_{6,63} - \xi_{6,505}\xi_{6,64} - \xi_{6,506}\xi_{6,65}\right) - \xi_{6,48}\left(-\xi_{6,503}\xi_{6,63} - \xi_{6,504}\xi_{6,62} + \xi_{6,505}\xi_{6,65} - \xi_{6,506}\xi_{6,64}\right) + \xi_{6,48}\left(-\xi_{6,503}\xi_{6,65} + \xi_{6,504}\xi_{6,64} - \xi_{6,505}\xi_{6,63} - \xi_{6,506}\xi_{6,62}\right) + \xi_{6,49}\left(-\xi_{6,503}\xi_{6,64} - \xi_{6,504}\xi_{6,65} - \xi_{6,505}\xi_{6,62} + \xi_{6,506}\xi_{6,63}\right)$$

$$\xi_{6,522} = \xi_{6,48}\left(\xi_{6,503}\xi_{6,62} - \xi_{6,504}\xi_{6,63} - \xi_{6,505}\xi_{6,64} - \xi_{6,506}\xi_{6,65}\right) + \xi_{6,48}\left(-\xi_{6,503}\xi_{6,63} - \xi_{6,504}\xi_{6,62} + \xi_{6,505}\xi_{6,65} - \xi_{6,506}\xi_{6,64}\right) - \xi_{6,48}\left(-\xi_{6,503}\xi_{6,64} - \xi_{6,504}\xi_{6,65} - \xi_{6,505}\xi_{6,62} + \xi_{6,506}\xi_{6,63}\right) + \xi_{6,49}\left(-\xi_{6,503}\xi_{6,65} + \xi_{6,504}\xi_{6,64} - \xi_{6,505}\xi_{6,63} - \xi_{6,506}\xi_{6,62}\right)$$

$$\xi_{6,523} = \xi_{6,388}\left(\xi_{6,28}\xi_{6,86} - \xi_{6,29}\xi_{6,87} - \xi_{6,30}\xi_{6,88} - \xi_{6,31}\xi_{6,89}\right) + \xi_{6,389}\left(\xi_{6,28}\xi_{6,87} + \xi_{6,29}\xi_{6,86} + \xi_{6,30}\xi_{6,89} - \xi_{6,31}\xi_{6,88}\right) + \xi_{6,390}\left(\xi_{6,28}\xi_{6,88} - \xi_{6,29}\xi_{6,89} + \xi_{6,30}\xi_{6,86} + \xi_{6,31}\xi_{6,87}\right) + \xi_{6,391}\left(\xi_{6,28}\xi_{6,89} + \xi_{6,29}\xi_{6,88} - \xi_{6,30}\xi_{6,87} + \xi_{6,31}\xi_{6,86}\right)$$

$$\xi_{6,524} = \xi_{6,388}\left(\xi_{6,28}\xi_{6,87} + \xi_{6,29}\xi_{6,86} + \xi_{6,30}\xi_{6,89} - \xi_{6,31}\xi_{6,88}\right) - \xi_{6,389}\left(\xi_{6,28}\xi_{6,86} - \xi_{6,29}\xi_{6,87} - \xi_{6,30}\xi_{6,88} - \xi_{6,31}\xi_{6,89}\right) + \xi_{6,390}\left(\xi_{6,28}\xi_{6,89} + \xi_{6,29}\xi_{6,88} - \xi_{6,30}\xi_{6,87} + \xi_{6,31}\xi_{6,86}\right) - \xi_{6,391}\left(\xi_{6,28}\xi_{6,88} - \xi_{6,29}\xi_{6,89} + \xi_{6,30}\xi_{6,86} + \xi_{6,31}\xi_{6,87}\right)$$

$$\xi_{6,525} = \xi_{6,388}\left(\xi_{6,28}\xi_{6,88} - \xi_{6,29}\xi_{6,89} + \xi_{6,30}\xi_{6,86} + \xi_{6,31}\xi_{6,87}\right) - \xi_{6,389}\left(\xi_{6,28}\xi_{6,89} + \xi_{6,29}\xi_{6,88} - \xi_{6,30}\xi_{6,87} + \xi_{6,31}\xi_{6,86}\right) - \xi_{6,390}\left(\xi_{6,28}\xi_{6,86} - \xi_{6,29}\xi_{6,87} - \xi_{6,30}\xi_{6,88} - \xi_{6,31}\xi_{6,89}\right) + \xi_{6,391}\left(\xi_{6,28}\xi_{6,87} + \xi_{6,29}\xi_{6,86} + \xi_{6,30}\xi_{6,89} - \xi_{6,31}\xi_{6,88}\right)$$

$$\xi_{6,526} = \xi_{6,388}\left(\xi_{6,28}\xi_{6,89} + \xi_{6,29}\xi_{6,88} - \xi_{6,30}\xi_{6,87} + \xi_{6,31}\xi_{6,86}\right) + \xi_{6,389}\left(\xi_{6,28}\xi_{6,88} - \xi_{6,29}\xi_{6,89} + \xi_{6,30}\xi_{6,86} + \xi_{6,31}\xi_{6,87}\right) - \xi_{6,390}\left(\xi_{6,28}\xi_{6,87} + \xi_{6,29}\xi_{6,86} + \xi_{6,30}\xi_{6,89} - \xi_{6,31}\xi_{6,88}\right) - \xi_{6,391}\left(\xi_{6,28}\xi_{6,86} - \xi_{6,29}\xi_{6,87} - \xi_{6,30}\xi_{6,88} - \xi_{6,31}\xi_{6,89}\right)$$

$$\xi_{6,527} = \xi_{6,412}\left(\xi_{6,32}\xi_{6,90} - \xi_{6,33}\xi_{6,91} - \xi_{6,34}\xi_{6,92} - \xi_{6,35}\xi_{6,93}\right) + \xi_{6,413}\left(\xi_{6,32}\xi_{6,91} + \xi_{6,33}\xi_{6,90} + \xi_{6,34}\xi_{6,93} - \xi_{6,35}\xi_{6,92}\right) + \xi_{6,414}\left(\xi_{6,32}\xi_{6,92} - \xi_{6,33}\xi_{6,93} + \xi_{6,34}\xi_{6,90} + \xi_{6,35}\xi_{6,91}\right) + \xi_{6,415}\left(\xi_{6,32}\xi_{6,93} + \xi_{6,33}\xi_{6,92} - \xi_{6,34}\xi_{6,91} + \xi_{6,35}\xi_{6,90}\right)$$

$$\xi_{6,528} = \xi_{6,412}\left(\xi_{6,32}\xi_{6,91} + \xi_{6,33}\xi_{6,90} + \xi_{6,34}\xi_{6,93} - \xi_{6,35}\xi_{6,92}\right) - \xi_{6,413}\left(\xi_{6,32}\xi_{6,90} - \xi_{6,33}\xi_{6,91} - \xi_{6,34}\xi_{6,92} - \xi_{6,35}\xi_{6,93}\right) + \xi_{6,414}\left(\xi_{6,32}\xi_{6,93} + \xi_{6,33}\xi_{6,92} - \xi_{6,34}\xi_{6,91} + \xi_{6,35}\xi_{6,90}\right) - \xi_{6,415}\left(\xi_{6,32}\xi_{6,92} - \xi_{6,33}\xi_{6,93} + \xi_{6,34}\xi_{6,90} + \xi_{6,35}\xi_{6,91}\right)$$

$$\xi_{6,529} = \xi_{6,412}\left(\xi_{6,32}\xi_{6,92} - \xi_{6,33}\xi_{6,93} + \xi_{6,34}\xi_{6,90} + \xi_{6,35}\xi_{6,91}\right) - \xi_{6,413}\left(\xi_{6,32}\xi_{6,93} + \xi_{6,33}\xi_{6,92} - \xi_{6,34}\xi_{6,91} + \xi_{6,35}\xi_{6,90}\right) - \xi_{6,414}\left(\xi_{6,32}\xi_{6,90} - \xi_{6,33}\xi_{6,91} - \xi_{6,34}\xi_{6,92} - \xi_{6,35}\xi_{6,93}\right) + \xi_{6,415}\left(\xi_{6,32}\xi_{6,91} + \xi_{6,33}\xi_{6,90} + \xi_{6,34}\xi_{6,93} - \xi_{6,35}\xi_{6,92}\right)$$

$$\xi_{6,530} = \xi_{6,412}\left(\xi_{6,32}\xi_{6,93} + \xi_{6,33}\xi_{6,92} - \xi_{6,34}\xi_{6,91} + \xi_{6,35}\xi_{6,90}\right) + \xi_{6,413}\left(\xi_{6,32}\xi_{6,92} - \xi_{6,33}\xi_{6,93} + \xi_{6,34}\xi_{6,90} + \xi_{6,35}\xi_{6,91}\right) - \xi_{6,414}\left(\xi_{6,32}\xi_{6,91} + \xi_{6,33}\xi_{6,90} + \xi_{6,34}\xi_{6,93} - \xi_{6,35}\xi_{6,92}\right) - \xi_{6,415}\left(\xi_{6,32}\xi_{6,90} - \xi_{6,33}\xi_{6,91} - \xi_{6,34}\xi_{6,92} - \xi_{6,35}\xi_{6,93}\right)$$

$$\xi_{6,531} = \xi_{6,101} + \xi_{6,43}$$

$$\xi_{6,532} = \xi_{6,531} - \gamma - \frac{\phi}{4}$$

$$\xi_{6,533} = \xi_{6,511}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) - \xi_{6,514}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{6,534} = \xi_{6,512}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{6,513}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{6,535} = -\xi_{6,512}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{6,513}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{6,536} = \xi_{6,511}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{6,514}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{6,537} = \xi_{6,507}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{6,510}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{6,538} = \xi_{6,508}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) - \xi_{6,509}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

$$\xi_{6,539} = \xi_{6,508}\sin\left(\frac{\chi}{4} + \frac{\zeta}{4}\right) + \xi_{6,509}\cos\left(\frac{\chi}{4} + \frac{\zeta}{4}\right)$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,540} = -\xi_{6,507}\sin\left(\frac{\chi}{4}+\frac{\zeta}{4}\right)+\xi_{6,510}\cos\left(\frac{\chi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{6,541} = \xi_{6,527}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{6,530}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{6,542} = \xi_{6,528}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{6,529}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{6,543} = -\xi_{6,528}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{6,529}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{6,544} = -\xi_{6,527}\sin\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)+\xi_{6,530}\cos\left(\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}-\frac{\zeta}{4}\right)$$

$$\xi_{6,545} = \xi_{6,523}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{6,526}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{6,546} = \xi_{6,524}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{6,525}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{6,547} = -\xi_{6,524}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{6,525}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{6,548} = -\xi_{6,523}\sin\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)+\xi_{6,526}\cos\left(-\frac{\chi}{4}+\frac{\gamma}{2}+\frac{\phi}{4}+\frac{\zeta}{4}\right)$$

$$\xi_{6,549} = \begin{cases} 0 & \text{for } \xi_{6,537}^2+\xi_{6,540}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,540},\xi_{6,537}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,550} = \begin{cases} 0 & \text{for } \xi_{6,538}^2+\xi_{6,539}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,538},\xi_{6,539}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,551} = 2\operatorname{atan}_2\left(\sqrt{\xi_{6,538}^2+\xi_{6,539}^2},\sqrt{\xi_{6,537}^2+\xi_{6,540}^2}\right)$$

$$\xi_{6,552} = -\xi_{6,549}+\xi_{6,550}$$

$$\xi_{6,553} = -\xi_{6,549}-\xi_{6,550}$$

$$\xi_{6,554} = \xi_{6,532}+\xi_{6,549}$$

$$\xi_{6,555} = \begin{cases} 0 & \text{for } \xi_{6,533}^2+\xi_{6,536}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,536},\xi_{6,533}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,556} = \begin{cases} 0 & \text{for } \xi_{6,534}^2+\xi_{6,535}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,534},\xi_{6,535}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,557} = 2\operatorname{atan}_2\left(\sqrt{\xi_{6,534}^2+\xi_{6,535}^2},\sqrt{\xi_{6,533}^2+\xi_{6,536}^2}\right)$$

$$\xi_{6,558} = -\xi_{6,555}+\xi_{6,556}$$

$$\xi_{6,559} = -\xi_{6,555}-\xi_{6,556}$$

$$\xi_{6,560} = \xi_{6,554}+\xi_{6,555}$$

$$\xi_{6,561} = \begin{cases} 0 & \text{for } \xi_{6,515}^2+\xi_{6,518}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,518},\xi_{6,515}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,562} = \begin{cases} 0 & \text{for } \xi_{6,516}^2+\xi_{6,517}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,516},\xi_{6,517}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,563} = 2\operatorname{atan}_2\left(\sqrt{\xi_{6,516}^2+\xi_{6,517}^2},\sqrt{\xi_{6,515}^2+\xi_{6,518}^2}\right)$$

$$\xi_{6,564} = -\xi_{6,561}+\xi_{6,562}$$

$$\xi_{6,565} = -\xi_{6,561}-\xi_{6,562}$$

$$\xi_{6,566} = \xi_{6,560}+\xi_{6,561}$$

$$\xi_{6,567} = \begin{cases} 0 & \text{for } \xi_{6,519}^2+\xi_{6,522}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,522},\xi_{6,519}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,568} = \begin{cases} 0 & \text{for } \xi_{6,520}^2+\xi_{6,521}^2=0 \\ \operatorname{atan}_2\left(-\xi_{6,520},\xi_{6,521}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,569} = 2\operatorname{atan}_2\left(\sqrt{\xi_{6,520}^2+\xi_{6,521}^2},\sqrt{\xi_{6,519}^2+\xi_{6,522}^2}\right)$$

$$\xi_{6,570} = -\xi_{6,567}+\xi_{6,568}$$

$$\xi_{6,571} = -\xi_{6,567}-\xi_{6,568}$$

$$\xi_{6,572} = \xi_{6,566}+\xi_{6,567}$$

$$\xi_{6,573} = \begin{cases} 0 & \text{for } \xi_{6,112}^2+\xi_{6,113}^2=0 \\ \operatorname{atan}_2\left(\xi_{6,113},\xi_{6,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,574} = \begin{cases} 0 & \text{for } \xi_{6,109}^2=0 \\ \operatorname{atan}_2\left(\xi_{6,109},0\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,575} = 2\operatorname{atan}_2\left(\left|\xi_{6,109}\right|,\sqrt{\xi_{6,112}^2+\xi_{6,113}^2}\right)$$

Exact shared-expression program 6

Original parameters: chi, gamma, phi, theta, zeta. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{6,576} = -\xi_{6,573} + \xi_{6,574}$$

$$\xi_{6,577} = -\xi_{6,573} - \xi_{6,574}$$

$$\xi_{6,578} = \xi_{6,572} + \xi_{6,573}$$

$$\xi_{6,579} = \begin{cases} 0 & \text{for } \xi_{6,107}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{6,107}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,580} = \begin{cases} 0 & \text{for } \xi_{6,108}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{6,108}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,581} = 2\operatorname{atan}_2\left(\left|\xi_{6,108}\right|, \left|\xi_{6,107}\right|\right)$$

$$\xi_{6,582} = -\xi_{6,579} + \xi_{6,580}$$

$$\xi_{6,583} = -\xi_{6,579} - \xi_{6,580}$$

$$\xi_{6,584} = \xi_{6,578} + \xi_{6,579}$$

$$\xi_{6,585} = \begin{cases} 0 & \text{for } \xi_{6,545}^2 + \xi_{6,548}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{6,548}, \xi_{6,545}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,586} = \begin{cases} 0 & \text{for } \xi_{6,546}^2 + \xi_{6,547}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{6,546}, \xi_{6,547}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,587} = 2\operatorname{atan}_2\left(\sqrt{\xi_{6,546}^2 + \xi_{6,547}^2}, \sqrt{\xi_{6,545}^2 + \xi_{6,548}^2}\right)$$

$$\xi_{6,588} = -\xi_{6,585} + \xi_{6,586}$$

$$\xi_{6,589} = -\xi_{6,585} - \xi_{6,586}$$

$$\xi_{6,590} = \xi_{6,584} + \xi_{6,585}$$

$$\xi_{6,591} = \begin{cases} 0 & \text{for } \xi_{6,541}^2 + \xi_{6,544}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{6,544}, \xi_{6,541}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,592} = \begin{cases} 0 & \text{for } \xi_{6,542}^2 + \xi_{6,543}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{6,542}, \xi_{6,543}\right) & \text{otherwise} \end{cases}$$

$$\xi_{6,593} = 2\operatorname{atan}_2\left(\sqrt{\xi_{6,542}^2 + \xi_{6,543}^2}, \sqrt{\xi_{6,541}^2 + \xi_{6,544}^2}\right)$$

$$\xi_{6,594} = -\xi_{6,591} + \xi_{6,592}$$

$$\xi_{6,595} = -\xi_{6,591} - \xi_{6,592}$$

$$\xi_{6,596} = \xi_{6,590} + \xi_{6,591}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,0} = \left\lfloor \frac{1}{2} - \frac{t}{2} \right\rfloor$$

$$\xi_{7,1} = -\frac{\pi \xi_{7,0}}{2} - \frac{\pi t}{4}$$

$$\xi_{7,2} = \xi_{7,1}$$

$$\xi_{7,3} = 1$$

$$\xi_{7,4} = 0$$

$$\xi_{7,5} = 0$$

$$\xi_{7,6} = 0$$

$$\xi_{7,7} = \begin{cases} -1 & \text{for } \xi_{7,2} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{7,8} = \begin{cases} 1 & \text{for } \xi_{7,7} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,9} = 0$$

$$\xi_{7,10} = 0$$

$$\xi_{7,11} = \begin{cases} 0 & \text{for } \xi_{7,7} = 1 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{7,12} = \cos\left(\frac{\pi \xi_{7,0}}{2}\right)$$

$$\xi_{7,13} = -\sin\left(\frac{\pi \xi_{7,0}}{2}\right)$$

$$\xi_{7,14} = \xi_{7,12}^2$$

$$\xi_{7,15} = \xi_{7,12}\xi_{7,13}$$

$$\xi_{7,16} = -\xi_{7,13}^2$$

$$\xi_{7,17} = \xi_{7,12}\xi_{7,14} - \xi_{7,13}\xi_{7,16}$$

$$\xi_{7,18} = \xi_{7,12}\xi_{7,15} - \xi_{7,13}\xi_{7,15}$$

$$\xi_{7,19} = \xi_{7,12}\xi_{7,15} + \xi_{7,13}\xi_{7,15}$$

$$\xi_{7,20} = \xi_{7,12}\xi_{7,16} + \xi_{7,13}\xi_{7,14}$$

$$\xi_{7,21} = -\xi_{7,10}\left(\xi_{7,17}\xi_{7,5} - \xi_{7,18}\xi_{7,6} + \xi_{7,19}\xi_{7,3} + \xi_{7,20}\xi_{7,4}\right) - \xi_{7,11}\left(\xi_{7,17}\xi_{7,6} + \xi_{7,18}\xi_{7,5} - \xi_{7,19}\xi_{7,4} + \xi_{7,20}\xi_{7,3}\right) + \xi_{7,8}\left(\xi_{7,17}\xi_{7,3} - \xi_{7,18}\xi_{7,4} - \xi_{7,19}\xi_{7,5} - \xi_{7,20}\xi_{7,6}\right) - \xi_{7,9}\left(\xi_{7,17}\xi_{7,4} + \xi_{7,18}\xi_{7,3} + \xi_{7,19}\xi_{7,6} - \xi_{7,20}\xi_{7,5}\right)$$

$$\xi_{7,22} = -\xi_{7,10}\left(\xi_{7,17}\xi_{7,6} + \xi_{7,18}\xi_{7,5} - \xi_{7,19}\xi_{7,4} + \xi_{7,20}\xi_{7,3}\right) + \xi_{7,11}\left(\xi_{7,17}\xi_{7,5} - \xi_{7,18}\xi_{7,6} + \xi_{7,19}\xi_{7,3} + \xi_{7,20}\xi_{7,4}\right) + \xi_{7,8}\left(\xi_{7,17}\xi_{7,4} + \xi_{7,18}\xi_{7,3} + \xi_{7,19}\xi_{7,6} - \xi_{7,20}\xi_{7,5}\right) + \xi_{7,9}\left(\xi_{7,17}\xi_{7,3} - \xi_{7,18}\xi_{7,4} - \xi_{7,19}\xi_{7,5} - \xi_{7,20}\xi_{7,6}\right)$$

$$\xi_{7,23} = \xi_{7,10}\left(\xi_{7,17}\xi_{7,3} - \xi_{7,18}\xi_{7,4} - \xi_{7,19}\xi_{7,5} - \xi_{7,20}\xi_{7,6}\right) - \xi_{7,11}\left(\xi_{7,17}\xi_{7,4} + \xi_{7,18}\xi_{7,3} + \xi_{7,19}\xi_{7,6} - \xi_{7,20}\xi_{7,5}\right) + \xi_{7,8}\left(\xi_{7,17}\xi_{7,5} - \xi_{7,18}\xi_{7,6} + \xi_{7,19}\xi_{7,3} + \xi_{7,20}\xi_{7,4}\right) + \xi_{7,9}\left(\xi_{7,17}\xi_{7,6} + \xi_{7,18}\xi_{7,5} - \xi_{7,19}\xi_{7,4} + \xi_{7,20}\xi_{7,3}\right)$$

$$\xi_{7,24} = \xi_{7,10}\left(\xi_{7,17}\xi_{7,4} + \xi_{7,18}\xi_{7,3} + \xi_{7,19}\xi_{7,6} - \xi_{7,20}\xi_{7,5}\right) + \xi_{7,11}\left(\xi_{7,17}\xi_{7,3} - \xi_{7,18}\xi_{7,4} - \xi_{7,19}\xi_{7,5} - \xi_{7,20}\xi_{7,6}\right) + \xi_{7,8}\left(\xi_{7,17}\xi_{7,6} + \xi_{7,18}\xi_{7,5} - \xi_{7,19}\xi_{7,4} + \xi_{7,20}\xi_{7,3}\right) - \xi_{7,9}\left(\xi_{7,17}\xi_{7,5} - \xi_{7,18}\xi_{7,6} + \xi_{7,19}\xi_{7,3} + \xi_{7,20}\xi_{7,4}\right)$$

$$\xi_{7,25} = \xi_{7,17}\xi_{7,3} - \xi_{7,18}\xi_{7,4} - \xi_{7,19}\xi_{7,5} - \xi_{7,20}\xi_{7,6}$$

$$\xi_{7,26} = \xi_{7,17}\xi_{7,4} + \xi_{7,18}\xi_{7,3} + \xi_{7,19}\xi_{7,6} - \xi_{7,20}\xi_{7,5}$$

$$\xi_{7,27} = \xi_{7,17}\xi_{7,5} - \xi_{7,18}\xi_{7,6} + \xi_{7,19}\xi_{7,3} + \xi_{7,20}\xi_{7,4}$$

$$\xi_{7,28} = \xi_{7,17}\xi_{7,6} + \xi_{7,18}\xi_{7,5} - \xi_{7,19}\xi_{7,4} + \xi_{7,20}\xi_{7,3}$$

$$\xi_{7,29} = -\xi_{7,10}\xi_{7,5} - \xi_{7,11}\xi_{7,6} + \xi_{7,3}\xi_{7,8} - \xi_{7,4}\xi_{7,9}$$

$$\xi_{7,30} = \xi_{7,10}\xi_{7,6} - \xi_{7,11}\xi_{7,5} - \xi_{7,3}\xi_{7,9} - \xi_{7,4}\xi_{7,8}$$

$$\xi_{7,31} = -\xi_{7,10}\xi_{7,3} + \xi_{7,11}\xi_{7,4} - \xi_{7,5}\xi_{7,8} - \xi_{7,6}\xi_{7,9}$$

$$\xi_{7,32} = -\xi_{7,10}\xi_{7,4} - \xi_{7,11}\xi_{7,3} + \xi_{7,5}\xi_{7,9} - \xi_{7,6}\xi_{7,8}$$

$$\xi_{7,33} = \xi_{7,2}\xi_{7,7}$$

$$\xi_{7,34} = \xi_{7,2}\xi_{7,7}^2$$

$$\xi_{7,35} = -\frac{3\pi \xi_{7,0}}{2}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,36} = \begin{cases} 0 & \text{for } \left(\xi_{7,33} > \frac{\pi}{8} \vee \xi_{7,34} < 0\right) \wedge \xi_{7,33} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{7,37} = \frac{\pi}{8}$$

$$\xi_{7,38} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{7,33} > \frac{\pi}{8} \wedge \xi_{7,34} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{7,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,39} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{7,33} > \frac{\pi}{8} \wedge \xi_{7,34} < 0 \\ \frac{1}{2} & \text{for } \xi_{7,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,40} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{7,33} > \frac{\pi}{8} \vee \xi_{7,34} < 0\right) \wedge \xi_{7,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,41} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{7,33} > \frac{\pi}{8} \vee \xi_{7,34} < 0\right) \wedge \xi_{7,33} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{7,42} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{7,33} > \frac{\pi}{8} \vee \xi_{7,34} < 0\right) \wedge \xi_{7,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,43} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{7,33} > \frac{\pi}{8} \vee \xi_{7,34} < 0\right) \wedge \xi_{7,33} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{7,44} = \begin{cases} \frac{1}{2} & \text{for } \xi_{7,33} > \frac{\pi}{8} \wedge \xi_{7,34} < 0 \\ -\frac{1}{2} & \text{for } \xi_{7,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,45} = \left\lfloor \frac{2\xi_{7,33} - 2\xi_{7,36}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{7,46} = \left\lfloor \frac{2\xi_{7,33} - 2\xi_{7,37}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{7,47} = \left\lfloor \frac{2\xi_{7,34} - 2\xi_{7,38}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{7,48} = \xi_{7,33} - \xi_{7,36} - \frac{\pi\xi_{7,45}}{2}$$

$$\xi_{7,49} = \xi_{7,33} - \xi_{7,37} - \frac{\pi\xi_{7,46}}{2}$$

$$\xi_{7,50} = \xi_{7,34} - \xi_{7,38} - \frac{\pi\xi_{7,47}}{2}$$

$$\xi_{7,51} = \begin{cases} \xi_{7,48} & \text{for } (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,49}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \\ \xi_{7,49} & \text{for } (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,49}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,49}|) \wedge (|\xi_{7,48}| \leq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \\ \xi_{7,50} & \text{otherwise} \end{cases}$$

$$\xi_{7,52} = \begin{cases} \xi_{7,49} & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,49}| \geq |\xi_{7,50}| \\ \xi_{7,50} & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge |\xi_{7,49}| \leq |\xi_{7,50}| \\ \xi_{7,48} & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge |\xi_{7,48}| \leq |\xi_{7,49}| \\ \xi_{7,50} & \text{for } |\xi_{7,49}| \geq |\xi_{7,50}| \wedge |\xi_{7,48}| \leq |\xi_{7,50}| \\ \xi_{7,48} & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,48}| \leq |\xi_{7,50}| \\ \xi_{7,49} & \text{otherwise} \end{cases}$$

$$\xi_{7,53} = \begin{cases} \xi_{7,50} & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,49}| \geq |\xi_{7,50}| \\ \xi_{7,49} & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge |\xi_{7,49}| \leq |\xi_{7,50}| \\ \xi_{7,50} & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge |\xi_{7,48}| \leq |\xi_{7,49}| \\ \xi_{7,48} & \text{for } |\xi_{7,49}| \geq |\xi_{7,50}| \wedge |\xi_{7,48}| \leq |\xi_{7,50}| \\ \xi_{7,49} & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,48}| \leq |\xi_{7,50}| \\ \xi_{7,48} & \text{otherwise} \end{cases}$$

$$\xi_{7,54} = \begin{cases} 1 & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,49}| \geq |\xi_{7,50}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge |\xi_{7,49}| \leq |\xi_{7,50}| \\ 0 & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge |\xi_{7,48}| \leq |\xi_{7,49}| \\ \frac{1}{2} & \text{for } |\xi_{7,48}| \leq |\xi_{7,50}| \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,49}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,49}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{7,55} = \begin{cases} 0 & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,49}| \geq |\xi_{7,50}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,49}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \leq |\xi_{7,49}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \\ \frac{1}{2} & \text{for } |\xi_{7,48}| \leq |\xi_{7,50}| \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,49}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,49}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,56} = \begin{cases} 0 & \text{for } (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,49}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{7,48}| \geq |\xi_{7,50}| \wedge |\xi_{7,48}| \leq |\xi_{7,49}| \\ \frac{1}{2} & \text{for } |\xi_{7,48}| \leq |\xi_{7,50}| \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,49}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,49}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{7,57} = \begin{cases} 0 & \text{for } (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,49}| \vee |\xi_{7,48}| \geq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,49}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,49}| \leq |\xi_{7,50}|) \wedge (|\xi_{7,48}| \geq |\xi_{7,50}| \vee |\xi_{7,48}| \leq |\xi_{7,49}| \vee |\xi_{7,48}| \leq |\xi_{7,50}|) \\ \frac{1}{2} & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,49}| \leq |\xi_{7,50}| \\ -\frac{1}{2} & \text{for } |\xi_{7,48}| \geq |\xi_{7,49}| \wedge |\xi_{7,49}| \leq |\xi_{7,50}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,58} = \begin{cases} -1 & \text{for } \xi_{7,51} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{7,59} = \begin{cases} -1 & \text{for } \xi_{7,52} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{7,60} = \begin{cases} 1 & \text{for } \xi_{7,58} = 1 \wedge \xi_{7,59} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,61} = \begin{cases} 0 & \text{for } \xi_{7,58} = 1 \wedge \xi_{7,59} = 1 \\ 1 & \text{for } \xi_{7,58} = 1 \wedge \xi_{7,59} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,62} = \begin{cases} 0 & \text{for } \xi_{7,58} = 1 \wedge (\xi_{7,58} = 1 \vee \xi_{7,59} = -1) \wedge (\xi_{7,58} = 1 \vee \xi_{7,59} = 1) \wedge (\xi_{7,59} = -1 \vee \xi_{7,59} = 1) \\ 1 & \text{for } \xi_{7,58} = -1 \wedge \xi_{7,59} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,63} = \begin{cases} 0 & \text{for } (\xi_{7,58} = -1 \vee \xi_{7,58} = 1) \wedge (\xi_{7,58} = 1 \vee \xi_{7,59} = 1) \wedge (\xi_{7,59} = -1 \vee \xi_{7,59} = 1) \wedge (\xi_{7,58} = -1 \vee \xi_{7,58} = 1 \vee \xi_{7,59} = -1) \wedge (\xi_{7,58} = -1 \vee \xi_{7,58} = 1 \vee \xi_{7,59} = 1) \wedge (\xi_{7,58} = -1 \vee \xi_{7,59} = -1 \vee \xi_{7,59} = 1) \wedge (\xi_{7,58} = 1 \vee \xi_{7,59} = -1 \vee \xi_{7,59} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{7,64} = \cos\left(\frac{\pi\xi_{7,45}}{2}\right)$$

$$\xi_{7,65} = -\sin\left(\frac{\pi\xi_{7,45}}{2}\right)$$

$$\xi_{7,66} = \cos\left(\frac{\pi\xi_{7,46}}{2}\right)$$

$$\xi_{7,67} = -\sin\left(\frac{\pi\xi_{7,46}}{2}\right)$$

$$\xi_{7,68} = \xi_{7,64}\xi_{7,66}$$

$$\xi_{7,69} = \xi_{7,65}\xi_{7,66}$$

$$\xi_{7,70} = \xi_{7,64}\xi_{7,67}$$

$$\xi_{7,71} = -\xi_{7,65}\xi_{7,67}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,72} = \cos\left(\frac{\pi\xi_{7,47}}{2}\right)$$

$$\xi_{7,73} = -\sin\left(\frac{\pi\xi_{7,47}}{2}\right)$$

$$\xi_{7,74} = \xi_{7,68}\xi_{7,72} - \xi_{7,71}\xi_{7,73}$$

$$\xi_{7,75} = \xi_{7,69}\xi_{7,72} - \xi_{7,70}\xi_{7,73}$$

$$\xi_{7,76} = \xi_{7,69}\xi_{7,73} + \xi_{7,70}\xi_{7,72}$$

$$\xi_{7,77} = \xi_{7,68}\xi_{7,73} + \xi_{7,71}\xi_{7,72}$$

$$\xi_{7,78} = \xi_{7,60}\left(\xi_{7,54}\xi_{7,74} - \xi_{7,55}\xi_{7,75} - \xi_{7,56}\xi_{7,76} - \xi_{7,57}\xi_{7,77}\right) - \xi_{7,61}\left(\xi_{7,54}\xi_{7,75} + \xi_{7,55}\xi_{7,74} - \xi_{7,56}\xi_{7,77} + \xi_{7,57}\xi_{7,76}\right) - \xi_{7,62}\left(\xi_{7,54}\xi_{7,76} + \xi_{7,55}\xi_{7,77} + \xi_{7,56}\xi_{7,74} - \xi_{7,57}\xi_{7,75}\right) - \xi_{7,63}\left(\xi_{7,54}\xi_{7,77} - \xi_{7,55}\xi_{7,76} + \xi_{7,56}\xi_{7,75} + \xi_{7,57}\xi_{7,74}\right)$$

$$\xi_{7,79} = \xi_{7,60}\left(\xi_{7,54}\xi_{7,75} + \xi_{7,55}\xi_{7,74} - \xi_{7,56}\xi_{7,77} + \xi_{7,57}\xi_{7,76}\right) + \xi_{7,61}\left(\xi_{7,54}\xi_{7,74} - \xi_{7,55}\xi_{7,75} - \xi_{7,56}\xi_{7,76} - \xi_{7,57}\xi_{7,77}\right) - \xi_{7,62}\left(\xi_{7,54}\xi_{7,77} - \xi_{7,55}\xi_{7,76} + \xi_{7,56}\xi_{7,75} + \xi_{7,57}\xi_{7,74}\right) + \xi_{7,63}\left(\xi_{7,54}\xi_{7,76} + \xi_{7,55}\xi_{7,77} + \xi_{7,56}\xi_{7,74} - \xi_{7,57}\xi_{7,75}\right)$$

$$\xi_{7,80} = \xi_{7,60}\left(\xi_{7,54}\xi_{7,76} + \xi_{7,55}\xi_{7,77} + \xi_{7,56}\xi_{7,74} - \xi_{7,57}\xi_{7,75}\right) + \xi_{7,61}\left(\xi_{7,54}\xi_{7,77} - \xi_{7,55}\xi_{7,76} + \xi_{7,56}\xi_{7,75} + \xi_{7,57}\xi_{7,74}\right) + \xi_{7,62}\left(\xi_{7,54}\xi_{7,74} - \xi_{7,55}\xi_{7,75} - \xi_{7,56}\xi_{7,76} - \xi_{7,57}\xi_{7,77}\right) - \xi_{7,63}\left(\xi_{7,54}\xi_{7,75} + \xi_{7,55}\xi_{7,74} - \xi_{7,56}\xi_{7,77} + \xi_{7,57}\xi_{7,76}\right)$$

$$\xi_{7,81} = \xi_{7,60}\left(\xi_{7,54}\xi_{7,77} - \xi_{7,55}\xi_{7,76} + \xi_{7,56}\xi_{7,75} + \xi_{7,57}\xi_{7,74}\right) - \xi_{7,61}\left(\xi_{7,54}\xi_{7,76} + \xi_{7,55}\xi_{7,77} + \xi_{7,56}\xi_{7,74} - \xi_{7,57}\xi_{7,75}\right) + \xi_{7,62}\left(\xi_{7,54}\xi_{7,75} + \xi_{7,55}\xi_{7,74} - \xi_{7,56}\xi_{7,77} + \xi_{7,57}\xi_{7,76}\right) + \xi_{7,63}\left(\xi_{7,54}\xi_{7,74} - \xi_{7,55}\xi_{7,75} - \xi_{7,56}\xi_{7,76} - \xi_{7,57}\xi_{7,77}\right)$$

$$\xi_{7,82} = \xi_{7,54}\xi_{7,74} - \xi_{7,55}\xi_{7,75} - \xi_{7,56}\xi_{7,76} - \xi_{7,57}\xi_{7,77}$$

$$\xi_{7,83} = \xi_{7,54}\xi_{7,75} + \xi_{7,55}\xi_{7,74} - \xi_{7,56}\xi_{7,77} + \xi_{7,57}\xi_{7,76}$$

$$\xi_{7,84} = \xi_{7,54}\xi_{7,76} + \xi_{7,55}\xi_{7,77} + \xi_{7,56}\xi_{7,74} - \xi_{7,57}\xi_{7,75}$$

$$\xi_{7,85} = \xi_{7,54}\xi_{7,77} - \xi_{7,55}\xi_{7,76} + \xi_{7,56}\xi_{7,75} + \xi_{7,57}\xi_{7,74}$$

$$\xi_{7,86} = \xi_{7,54}\xi_{7,60} - \xi_{7,55}\xi_{7,61} - \xi_{7,56}\xi_{7,62} - \xi_{7,57}\xi_{7,63}$$

$$\xi_{7,87} = -\xi_{7,54}\xi_{7,61} - \xi_{7,55}\xi_{7,60} - \xi_{7,56}\xi_{7,63} + \xi_{7,57}\xi_{7,62}$$

$$\xi_{7,88} = -\xi_{7,54}\xi_{7,62} + \xi_{7,55}\xi_{7,63} - \xi_{7,56}\xi_{7,60} - \xi_{7,57}\xi_{7,61}$$

$$\xi_{7,89} = -\xi_{7,54}\xi_{7,63} - \xi_{7,55}\xi_{7,62} + \xi_{7,56}\xi_{7,61} - \xi_{7,57}\xi_{7,60}$$

$$\xi_{7,90} = \xi_{7,51}\xi_{7,58}$$

$$\xi_{7,91} = \xi_{7,52}\xi_{7,59}$$

$$\xi_{7,92} = \xi_{7,53}\xi_{7,58}\xi_{7,59}$$

$$\xi_{7,93} = -\frac{\pi(\xi_{7,45}+\xi_{7,46}+\xi_{7,47})}{2}$$

$$\xi_{7,94} = \cos\left(2\xi_{7,90}\right)$$

$$\xi_{7,95} = \cos\left(2\xi_{7,91}\right)$$

$$\xi_{7,96} = \cos\left(2\xi_{7,92}\right)$$

$$\xi_{7,97} = \sqrt{\xi_{7,94}^2 - 2\xi_{7,94}\xi_{7,95}\xi_{7,96} + \xi_{7,95}^2 + \xi_{7,96}^2 - 1}$$

$$\xi_{7,98} = \xi_{7,94} - \xi_{7,95} + \xi_{7,96}$$

$$\xi_{7,99} = \sqrt{-\frac{\xi_{7,97}}{2} + \frac{\xi_{7,98}}{2} + \frac{1}{2}}$$

$$\xi_{7,100} = \sqrt{\frac{\xi_{7,97}}{2} - \frac{\xi_{7,98}}{2} + \frac{1}{2}}$$

$$\xi_{7,101} = \sqrt{-\frac{\xi_{7,97}}{2} - \frac{\xi_{7,98}}{2} + \frac{1}{2}}$$

$$\xi_{7,102} = \frac{(1-\xi_{7,95})(\xi_{7,94}+1)(\xi_{7,96}+1)}{2}$$

$$\xi_{7,103} = \begin{cases} -1 & \text{for } \xi_{7,92} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{7,104} = \begin{cases} 0 & \text{for } \xi_{7,99} = 0 \\ \frac{\sqrt{\xi_{7,102}}\xi_{7,103}}{\xi_{7,99}} & \text{otherwise} \end{cases}$$

$$\xi_{7,105} = \begin{cases} 1 & \text{for } \xi_{7,99} = 0 \\ \frac{\sqrt{\xi_{7,94}\xi_{7,95}\xi_{7,96}}}{\xi_{7,99}} & \text{otherwise} \end{cases}$$

$$\xi_{7,106} = 2\xi_{7,100}\xi_{7,101}\xi_{7,105}\xi_{7,99}$$

$$\xi_{7,107} = \sqrt{2}\xi_{7,105}\left(\xi_{7,101}\xi_{7,99}^2 - \xi_{7,101}\right)$$

Exact shared-expression program 7

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$$\xi_{7,108} = 2\xi_{7,100}\xi_{7,101}\xi_{7,104}\xi_{7,99}$$

$$\xi_{7,109} = \sqrt{2}\left(2\xi_{7,101}\xi_{7,104}\xi_{7,99}^2 - \xi_{7,101}\xi_{7,104}\right)$$

$$\xi_{7,110} = \sqrt{2}\xi_{7,100}\xi_{7,104}\xi_{7,105}\xi_{7,99}$$

$$\xi_{7,111} = \sqrt{2}\xi_{7,100}\xi_{7,99}\left(-\xi_{7,101}^2 + \xi_{7,104}^2\right)$$

$$\xi_{7,112} = -\xi_{7,101}^2\xi_{7,99}^2 + \xi_{7,101}^2 + \xi_{7,104}^2\xi_{7,99}^2$$

$$\xi_{7,113} = \xi_{7,101}^2\xi_{7,99}^2 - \xi_{7,104}^2\xi_{7,99}^2 + \xi_{7,104}^2$$

$$\xi_{7,114} = 2\xi_{7,105}\left(\xi_{7,104}\xi_{7,99}^2 - \frac{\xi_{7,104}}{2}\right)$$

$$\xi_{7,115} = -\sqrt{2}\xi_{7,100}\xi_{7,104}\xi_{7,105}\xi_{7,99}$$

$$\xi_{7,116} = -\xi_{7,104}\xi_{7,105}$$

$$\xi_{7,117} = \xi_{7,101}^2\xi_{7,99}^2 - \xi_{7,101}^2 + \xi_{7,104}^2\xi_{7,99}^2 - \xi_{7,104}^2 - \xi_{7,99}^2 + 1$$

$$\xi_{7,118} = -\sqrt{2}\left(\xi_{7,100}\xi_{7,101}^2\xi_{7,99} + \xi_{7,100}\xi_{7,104}^2\xi_{7,99} - \xi_{7,100}\xi_{7,99}\right)$$

$$\xi_{7,119} = \sqrt{2}\xi_{7,101}\xi_{7,105}\xi_{7,99}^2$$

$$\xi_{7,120} = \xi_{7,110}^2 - \xi_{7,113}^2 - \xi_{7,116}^2 - \xi_{7,119}^2$$

$$\xi_{7,121} = \xi_{7,110}\xi_{7,111} - \xi_{7,113}\xi_{7,114} - \xi_{7,116}\xi_{7,117}$$

$$\xi_{7,122} = \xi_{7,110}\xi_{7,112} - \xi_{7,113}\xi_{7,115} - \xi_{7,116}\xi_{7,118}$$

$$\xi_{7,123} = \xi_{7,106}^2 + \xi_{7,108}^2 + \xi_{7,111}^2 - \xi_{7,114}^2 - \xi_{7,117}^2$$

$$\xi_{7,124} = \xi_{7,106}\xi_{7,107} + \xi_{7,108}\xi_{7,109} + \xi_{7,111}\xi_{7,112} - \xi_{7,114}\xi_{7,115} - \xi_{7,117}\xi_{7,118}$$

$$\xi_{7,125} = \xi_{7,107}^2 + \xi_{7,109}^2 + \xi_{7,112}^2 - \xi_{7,115}^2 - \xi_{7,118}^2$$

$$\xi_{7,126} = \sin\left(2\xi_{7,91}\right)\sin\left(2\xi_{7,92}\right)\cos\left(2\xi_{7,90}\right)$$

$$\xi_{7,127} = \sin\left(2\xi_{7,90}\right)\sin\left(2\xi_{7,92}\right)\cos\left(2\xi_{7,91}\right)$$

$$\xi_{7,128} = \sin\left(2\xi_{7,90}\right)\sin\left(2\xi_{7,91}\right)\cos\left(2\xi_{7,92}\right)$$

$$\xi_{7,129} = \sin\left(2\xi_{7,90}\right)\cos\left(2\xi_{7,91}\right)\cos\left(2\xi_{7,92}\right)$$

$$\xi_{7,130} = \sin\left(2\xi_{7,91}\right)\cos\left(2\xi_{7,90}\right)\cos\left(2\xi_{7,92}\right)$$

$$\xi_{7,131} = \sin\left(2\xi_{7,92}\right)\cos\left(2\xi_{7,90}\right)\cos\left(2\xi_{7,91}\right)$$

$$\xi_{7,132} = \xi_{7,126}^2 - \xi_{7,129}^2$$

$$\xi_{7,133} = \xi_{7,127}^2 - \xi_{7,130}^2$$

$$\xi_{7,134} = \xi_{7,128}^2 - \xi_{7,131}^2$$

$$\xi_{7,135} = \frac{\xi_{7,120} - \xi_{7,133}}{\xi_{7,132} - \xi_{7,133}}$$

$$\xi_{7,136} = \frac{\xi_{7,121}}{\xi_{7,132} - \xi_{7,133}}$$

$$\xi_{7,137} = \frac{\xi_{7,122}}{\xi_{7,132} - \xi_{7,133}}$$

$$\xi_{7,138} = \frac{\xi_{7,123} - \xi_{7,133}}{\xi_{7,132} - \xi_{7,133}}$$

$$\xi_{7,139} = \frac{\xi_{7,124}}{\xi_{7,132} - \xi_{7,133}}$$

$$\xi_{7,140} = \frac{\xi_{7,125} - \xi_{7,133}}{\xi_{7,132} - \xi_{7,133}}$$

$$\xi_{7,141} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,135} & \text{otherwise} \end{cases}$$

$$\xi_{7,142} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,136} & \text{otherwise} \end{cases}$$

$$\xi_{7,143} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,137} & \text{otherwise} \end{cases}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,144} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,138} & \text{otherwise} \end{cases}$$

$$\xi_{7,145} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,139} & \text{otherwise} \end{cases}$$

$$\xi_{7,146} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,140} & \text{otherwise} \end{cases}$$

$$\xi_{7,147} = \frac{\xi_{7,120} - \xi_{7,134}}{\xi_{7,132} - \xi_{7,134}}$$

$$\xi_{7,148} = \frac{\xi_{7,121}}{\xi_{7,132} - \xi_{7,134}}$$

$$\xi_{7,149} = \frac{\xi_{7,122}}{\xi_{7,132} - \xi_{7,134}}$$

$$\xi_{7,150} = \frac{\xi_{7,123} - \xi_{7,134}}{\xi_{7,132} - \xi_{7,134}}$$

$$\xi_{7,151} = \frac{\xi_{7,124}}{\xi_{7,132} - \xi_{7,134}}$$

$$\xi_{7,152} = \frac{\xi_{7,125} - \xi_{7,134}}{\xi_{7,132} - \xi_{7,134}}$$

$$\xi_{7,153} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,134} \\ \xi_{7,147} & \text{otherwise} \end{cases}$$

$$\xi_{7,154} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,134} \\ \xi_{7,148} & \text{otherwise} \end{cases}$$

$$\xi_{7,155} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,134} \\ \xi_{7,149} & \text{otherwise} \end{cases}$$

$$\xi_{7,156} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,134} \\ \xi_{7,150} & \text{otherwise} \end{cases}$$

$$\xi_{7,157} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,134} \\ \xi_{7,151} & \text{otherwise} \end{cases}$$

$$\xi_{7,158} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,134} \\ \xi_{7,152} & \text{otherwise} \end{cases}$$

$$\xi_{7,159} = \xi_{7,141}\xi_{7,153} + \xi_{7,142}\xi_{7,154} + \xi_{7,143}\xi_{7,155}$$

$$\xi_{7,160} = \xi_{7,141}\xi_{7,154} + \xi_{7,142}\xi_{7,156} + \xi_{7,143}\xi_{7,157}$$

$$\xi_{7,161} = \xi_{7,141}\xi_{7,155} + \xi_{7,142}\xi_{7,157} + \xi_{7,143}\xi_{7,158}$$

$$\xi_{7,162} = \xi_{7,142}\xi_{7,153} + \xi_{7,144}\xi_{7,154} + \xi_{7,145}\xi_{7,155}$$

$$\xi_{7,163} = \xi_{7,142}\xi_{7,154} + \xi_{7,144}\xi_{7,156} + \xi_{7,145}\xi_{7,157}$$

$$\xi_{7,164} = \xi_{7,142}\xi_{7,155} + \xi_{7,144}\xi_{7,157} + \xi_{7,145}\xi_{7,158}$$

$$\xi_{7,165} = \xi_{7,143}\xi_{7,153} + \xi_{7,145}\xi_{7,154} + \xi_{7,146}\xi_{7,155}$$

$$\xi_{7,166} = \xi_{7,143}\xi_{7,154} + \xi_{7,145}\xi_{7,156} + \xi_{7,146}\xi_{7,157}$$

$$\xi_{7,167} = \xi_{7,143}\xi_{7,155} + \xi_{7,145}\xi_{7,157} + \xi_{7,146}\xi_{7,158}$$

$$\xi_{7,168} = \sqrt{\xi_{7,159}}$$

$$\xi_{7,169} = \frac{\xi_{7,159}}{\xi_{7,168}}$$

$$\xi_{7,170} = \frac{\xi_{7,162}}{\xi_{7,168}}$$

$$\xi_{7,171} = \frac{\xi_{7,165}}{\xi_{7,168}}$$

$$\xi_{7,172} = \sqrt{\xi_{7,163}}$$

$$\xi_{7,173} = \frac{\xi_{7,160}}{\xi_{7,172}}$$

$$\xi_{7,174} = \frac{\xi_{7,163}}{\xi_{7,172}}$$

$$\xi_{7,175} = \frac{\xi_{7,166}}{\xi_{7,172}}$$

$$\xi_{7,176} = \sqrt{\xi_{7,167}}$$

$$\xi_{7,177} = \frac{\xi_{7,161}}{\xi_{7,176}}$$

$$\xi_{7,178} = \frac{\xi_{7,164}}{\xi_{7,176}}$$

$$\xi_{7,179} = \frac{\xi_{7,167}}{\xi_{7,176}}$$

Exact shared-expression program 7

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$$\xi_{7,180} = \begin{cases} \xi_{7,169} & \text{for } \xi_{7,159} \geq \xi_{7,163} \wedge \xi_{7,159} \geq \xi_{7,167} \\ \xi_{7,173} & \text{for } \xi_{7,163} \geq \xi_{7,167} \wedge \xi_{7,159} \leq \xi_{7,163} \\ \xi_{7,177} & \text{otherwise} \end{cases}$$

$$\xi_{7,181} = \begin{cases} \xi_{7,170} & \text{for } \xi_{7,159} \geq \xi_{7,163} \wedge \xi_{7,159} \geq \xi_{7,167} \\ \xi_{7,174} & \text{for } \xi_{7,163} \geq \xi_{7,167} \wedge \xi_{7,159} \leq \xi_{7,163} \\ \xi_{7,178} & \text{otherwise} \end{cases}$$

$$\xi_{7,182} = \begin{cases} \xi_{7,171} & \text{for } \xi_{7,159} \geq \xi_{7,163} \wedge \xi_{7,159} \geq \xi_{7,167} \\ \xi_{7,175} & \text{for } \xi_{7,163} \geq \xi_{7,167} \wedge \xi_{7,159} \leq \xi_{7,163} \\ \xi_{7,179} & \text{otherwise} \end{cases}$$

$$\xi_{7,183} = \frac{\xi_{7,120} - \xi_{7,132}}{-\xi_{7,132} + \xi_{7,133}}$$

$$\xi_{7,184} = \frac{\xi_{7,121}}{-\xi_{7,132} + \xi_{7,133}}$$

$$\xi_{7,185} = \frac{\xi_{7,122}}{-\xi_{7,132} + \xi_{7,133}}$$

$$\xi_{7,186} = \frac{\xi_{7,123} - \xi_{7,132}}{-\xi_{7,132} + \xi_{7,133}}$$

$$\xi_{7,187} = \frac{\xi_{7,124}}{-\xi_{7,132} + \xi_{7,133}}$$

$$\xi_{7,188} = \frac{\xi_{7,125} - \xi_{7,132}}{-\xi_{7,132} + \xi_{7,133}}$$

$$\xi_{7,189} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,183} & \text{otherwise} \end{cases}$$

$$\xi_{7,190} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,184} & \text{otherwise} \end{cases}$$

$$\xi_{7,191} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,185} & \text{otherwise} \end{cases}$$

$$\xi_{7,192} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,186} & \text{otherwise} \end{cases}$$

$$\xi_{7,193} = \begin{cases} 0 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,187} & \text{otherwise} \end{cases}$$

$$\xi_{7,194} = \begin{cases} 1 & \text{for } \xi_{7,132} = \xi_{7,133} \\ \xi_{7,188} & \text{otherwise} \end{cases}$$

$$\xi_{7,195} = \frac{\xi_{7,120} - \xi_{7,134}}{\xi_{7,133} - \xi_{7,134}}$$

$$\xi_{7,196} = \frac{\xi_{7,121}}{\xi_{7,133} - \xi_{7,134}}$$

$$\xi_{7,197} = \frac{\xi_{7,122}}{\xi_{7,133} - \xi_{7,134}}$$

$$\xi_{7,198} = \frac{\xi_{7,123} - \xi_{7,134}}{\xi_{7,133} - \xi_{7,134}}$$

$$\xi_{7,199} = \frac{\xi_{7,124}}{\xi_{7,133} - \xi_{7,134}}$$

$$\xi_{7,200} = \frac{\xi_{7,125} - \xi_{7,134}}{\xi_{7,133} - \xi_{7,134}}$$

$$\xi_{7,201} = \begin{cases} 1 & \text{for } \xi_{7,133} = \xi_{7,134} \\ \xi_{7,195} & \text{otherwise} \end{cases}$$

$$\xi_{7,202} = \begin{cases} 0 & \text{for } \xi_{7,133} = \xi_{7,134} \\ \xi_{7,196} & \text{otherwise} \end{cases}$$

$$\xi_{7,203} = \begin{cases} 0 & \text{for } \xi_{7,133} = \xi_{7,134} \\ \xi_{7,197} & \text{otherwise} \end{cases}$$

$$\xi_{7,204} = \begin{cases} 1 & \text{for } \xi_{7,133} = \xi_{7,134} \\ \xi_{7,198} & \text{otherwise} \end{cases}$$

$$\xi_{7,205} = \begin{cases} 0 & \text{for } \xi_{7,133} = \xi_{7,134} \\ \xi_{7,199} & \text{otherwise} \end{cases}$$

$$\xi_{7,206} = \begin{cases} 1 & \text{for } \xi_{7,133} = \xi_{7,134} \\ \xi_{7,200} & \text{otherwise} \end{cases}$$

$$\xi_{7,207} = \xi_{7,189}\xi_{7,201} + \xi_{7,190}\xi_{7,202} + \xi_{7,191}\xi_{7,203}$$

$$\xi_{7,208} = \xi_{7,189}\xi_{7,202} + \xi_{7,190}\xi_{7,204} + \xi_{7,191}\xi_{7,205}$$

$$\xi_{7,209} = \xi_{7,189}\xi_{7,203} + \xi_{7,190}\xi_{7,205} + \xi_{7,191}\xi_{7,206}$$

$$\xi_{7,210} = \xi_{7,190}\xi_{7,201} + \xi_{7,192}\xi_{7,202} + \xi_{7,193}\xi_{7,203}$$

$$\xi_{7,211} = \xi_{7,190}\xi_{7,202} + \xi_{7,192}\xi_{7,204} + \xi_{7,193}\xi_{7,205}$$

$$\xi_{7,212} = \xi_{7,190}\xi_{7,203} + \xi_{7,192}\xi_{7,205} + \xi_{7,193}\xi_{7,206}$$

$$\xi_{7,213} = \xi_{7,191}\xi_{7,201} + \xi_{7,193}\xi_{7,202} + \xi_{7,194}\xi_{7,203}$$

$$\xi_{7,214} = \xi_{7,191}\xi_{7,202} + \xi_{7,193}\xi_{7,204} + \xi_{7,194}\xi_{7,205}$$

$$\xi_{7,215} = \xi_{7,191}\xi_{7,203} + \xi_{7,193}\xi_{7,205} + \xi_{7,194}\xi_{7,206}$$

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$$\xi_{7,216} = \begin{cases} \xi_{7,180}^2 & \text{for } \xi_{7,132} = \xi_{7,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,217} = \begin{cases} \xi_{7,180}\xi_{7,181} & \text{for } \xi_{7,132} = \xi_{7,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,218} = \begin{cases} \xi_{7,180}\xi_{7,182} & \text{for } \xi_{7,132} = \xi_{7,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,219} = \begin{cases} \xi_{7,181}^2 & \text{for } \xi_{7,132} = \xi_{7,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,220} = \begin{cases} \xi_{7,181}\xi_{7,182} & \text{for } \xi_{7,132} = \xi_{7,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,221} = \begin{cases} \xi_{7,182}^2 & \text{for } \xi_{7,132} = \xi_{7,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{7,222} = \xi_{7,207} - \xi_{7,216}$$

$$\xi_{7,223} = \xi_{7,208} - \xi_{7,217}$$

$$\xi_{7,224} = \xi_{7,209} - \xi_{7,218}$$

$$\xi_{7,225} = \xi_{7,210} - \xi_{7,217}$$

$$\xi_{7,226} = \xi_{7,211} - \xi_{7,219}$$

$$\xi_{7,227} = \xi_{7,212} - \xi_{7,220}$$

$$\xi_{7,228} = \xi_{7,213} - \xi_{7,218}$$

$$\xi_{7,229} = \xi_{7,214} - \xi_{7,220}$$

$$\xi_{7,230} = \xi_{7,215} - \xi_{7,221}$$

$$\xi_{7,231} = \sqrt{\xi_{7,222}}$$

$$\xi_{7,232} = \frac{\xi_{7,222}}{\xi_{7,231}}$$

$$\xi_{7,233} = \frac{\xi_{7,225}}{\xi_{7,231}}$$

$$\xi_{7,234} = \frac{\xi_{7,228}}{\xi_{7,231}}$$

$$\xi_{7,235} = \sqrt{\xi_{7,226}}$$

$$\xi_{7,236} = \frac{\xi_{7,223}}{\xi_{7,235}}$$

$$\xi_{7,237} = \frac{\xi_{7,226}}{\xi_{7,235}}$$

$$\xi_{7,238} = \frac{\xi_{7,229}}{\xi_{7,235}}$$

$$\xi_{7,239} = \sqrt{\xi_{7,230}}$$

$$\xi_{7,240} = \frac{\xi_{7,224}}{\xi_{7,239}}$$

$$\xi_{7,241} = \frac{\xi_{7,227}}{\xi_{7,239}}$$

$$\xi_{7,242} = \frac{\xi_{7,230}}{\xi_{7,239}}$$

$$\xi_{7,243} = \begin{cases} \xi_{7,232} & \text{for } \xi_{7,222} \geq \xi_{7,226} \wedge \xi_{7,222} \geq \xi_{7,230} \\ \xi_{7,236} & \text{for } \xi_{7,226} \geq \xi_{7,230} \wedge \xi_{7,222} \leq \xi_{7,226} \\ \xi_{7,240} & \text{otherwise} \end{cases}$$

$$\xi_{7,244} = \begin{cases} \xi_{7,233} & \text{for } \xi_{7,222} \geq \xi_{7,226} \wedge \xi_{7,222} \geq \xi_{7,230} \\ \xi_{7,237} & \text{for } \xi_{7,226} \geq \xi_{7,230} \wedge \xi_{7,222} \leq \xi_{7,226} \\ \xi_{7,241} & \text{otherwise} \end{cases}$$

$$\xi_{7,245} = \begin{cases} \xi_{7,234} & \text{for } \xi_{7,222} \geq \xi_{7,226} \wedge \xi_{7,222} \geq \xi_{7,230} \\ \xi_{7,238} & \text{for } \xi_{7,226} \geq \xi_{7,230} \wedge \xi_{7,222} \leq \xi_{7,226} \\ \xi_{7,242} & \text{otherwise} \end{cases}$$

$$\xi_{7,246} = \xi_{7,181}\xi_{7,245} - \xi_{7,182}\xi_{7,244}$$

$$\xi_{7,247} = -\xi_{7,180}\xi_{7,245} + \xi_{7,182}\xi_{7,243}$$

$$\xi_{7,248} = \xi_{7,180}\xi_{7,244} - \xi_{7,181}\xi_{7,243}$$

$$\xi_{7,249} = \xi_{7,126}^2 + \xi_{7,129}^2$$

$$\xi_{7,250} = \xi_{7,127}^2 + \xi_{7,130}^2$$

$$\xi_{7,251} = \xi_{7,128}^2 + \xi_{7,131}^2$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,252} = \frac{\xi_{7,113}\xi_{7,129}\xi_{7,180}+\xi_{7,181}(\xi_{7,106}\xi_{7,126}+\xi_{7,114}\xi_{7,129})+\xi_{7,182}(\xi_{7,107}\xi_{7,126}+\xi_{7,115}\xi_{7,129})}{\xi_{7,249}}$$

$$\xi_{7,253} = \frac{\xi_{7,116}\xi_{7,129}\xi_{7,180}+\xi_{7,181}(\xi_{7,108}\xi_{7,126}+\xi_{7,117}\xi_{7,129})+\xi_{7,182}(\xi_{7,109}\xi_{7,126}+\xi_{7,118}\xi_{7,129})}{\xi_{7,249}}$$

$$\xi_{7,254} = \frac{\xi_{7,111}\xi_{7,126}\xi_{7,181}+\xi_{7,112}\xi_{7,126}\xi_{7,182}+\xi_{7,180}(\xi_{7,110}\xi_{7,126}+\xi_{7,119}\xi_{7,129})}{\xi_{7,249}}$$

$$\xi_{7,255} = \frac{\xi_{7,113}\xi_{7,130}\xi_{7,243}+\xi_{7,244}(\xi_{7,106}\xi_{7,127}+\xi_{7,114}\xi_{7,130})+\xi_{7,245}(\xi_{7,107}\xi_{7,127}+\xi_{7,115}\xi_{7,130})}{\xi_{7,250}}$$

$$\xi_{7,256} = \frac{\xi_{7,116}\xi_{7,130}\xi_{7,243}+\xi_{7,244}(\xi_{7,108}\xi_{7,127}+\xi_{7,117}\xi_{7,130})+\xi_{7,245}(\xi_{7,109}\xi_{7,127}+\xi_{7,118}\xi_{7,130})}{\xi_{7,250}}$$

$$\xi_{7,257} = \frac{\xi_{7,111}\xi_{7,127}\xi_{7,244}+\xi_{7,112}\xi_{7,127}\xi_{7,245}+\xi_{7,243}(\xi_{7,110}\xi_{7,127}+\xi_{7,119}\xi_{7,130})}{\xi_{7,250}}$$

$$\xi_{7,258} = \frac{\xi_{7,113}\xi_{7,131}\xi_{7,246}+\xi_{7,247}(\xi_{7,106}\xi_{7,128}+\xi_{7,114}\xi_{7,131})+\xi_{7,248}(\xi_{7,107}\xi_{7,128}+\xi_{7,115}\xi_{7,131})}{\xi_{7,251}}$$

$$\xi_{7,259} = \frac{\xi_{7,116}\xi_{7,131}\xi_{7,246}+\xi_{7,247}(\xi_{7,108}\xi_{7,128}+\xi_{7,117}\xi_{7,131})+\xi_{7,248}(\xi_{7,109}\xi_{7,128}+\xi_{7,118}\xi_{7,131})}{\xi_{7,251}}$$

$$\xi_{7,260} = \frac{\xi_{7,111}\xi_{7,128}\xi_{7,247}+\xi_{7,112}\xi_{7,128}\xi_{7,248}+\xi_{7,246}(\xi_{7,110}\xi_{7,128}+\xi_{7,119}\xi_{7,131})}{\xi_{7,251}}$$

$$\xi_{7,261} = 1 - \xi_{7,252}^2$$

$$\xi_{7,262} = -\xi_{7,252}\xi_{7,253}$$

$$\xi_{7,263} = -\xi_{7,252}\xi_{7,254}$$

$$\xi_{7,264} = 1 - \xi_{7,253}^2$$

$$\xi_{7,265} = -\xi_{7,253}\xi_{7,254}$$

$$\xi_{7,266} = 1 - \xi_{7,254}^2$$

$$\xi_{7,267} = \xi_{7,261}^2 + \xi_{7,262}^2 + \xi_{7,263}^2$$

$$\xi_{7,268} = \xi_{7,262}^2 + \xi_{7,264}^2 + \xi_{7,265}^2$$

$$\xi_{7,269} = \xi_{7,263}^2 + \xi_{7,265}^2 + \xi_{7,266}^2$$

$$\xi_{7,270} = \sqrt{\xi_{7,267}}$$

$$\xi_{7,271} = \frac{\xi_{7,261}}{\xi_{7,270}}$$

$$\xi_{7,272} = \frac{\xi_{7,262}}{\xi_{7,270}}$$

$$\xi_{7,273} = \frac{\xi_{7,263}}{\xi_{7,270}}$$

$$\xi_{7,274} = \sqrt{\xi_{7,268}}$$

$$\xi_{7,275} = \frac{\xi_{7,264}}{\xi_{7,274}}$$

$$\xi_{7,276} = \frac{\xi_{7,262}}{\xi_{7,274}}$$

$$\xi_{7,277} = \frac{\xi_{7,265}}{\xi_{7,274}}$$

$$\xi_{7,278} = \sqrt{\xi_{7,269}}$$

$$\xi_{7,279} = \frac{\xi_{7,266}}{\xi_{7,278}}$$

$$\xi_{7,280} = \frac{\xi_{7,263}}{\xi_{7,278}}$$

$$\xi_{7,281} = \frac{\xi_{7,265}}{\xi_{7,278}}$$

$$\xi_{7,282} = \begin{cases} \xi_{7,271} & \text{for } \xi_{7,267} \geq \xi_{7,268} \wedge \xi_{7,267} \geq \xi_{7,269} \\ \xi_{7,276} & \text{for } \xi_{7,268} \geq \xi_{7,269} \wedge \xi_{7,267} \leq \xi_{7,268} \\ \xi_{7,280} & \text{otherwise} \end{cases}$$

$$\xi_{7,283} = \begin{cases} \xi_{7,272} & \text{for } \xi_{7,267} \geq \xi_{7,268} \wedge \xi_{7,267} \geq \xi_{7,269} \\ \xi_{7,275} & \text{for } \xi_{7,268} \geq \xi_{7,269} \wedge \xi_{7,267} \leq \xi_{7,268} \\ \xi_{7,281} & \text{otherwise} \end{cases}$$

$$\xi_{7,284} = \begin{cases} \xi_{7,273} & \text{for } \xi_{7,267} \geq \xi_{7,268} \wedge \xi_{7,267} \geq \xi_{7,269} \\ \xi_{7,277} & \text{for } \xi_{7,268} \geq \xi_{7,269} \wedge \xi_{7,267} \leq \xi_{7,268} \\ \xi_{7,279} & \text{otherwise} \end{cases}$$

$$\xi_{7,285} = \xi_{7,253}\xi_{7,284} - \xi_{7,254}\xi_{7,283}$$

$$\xi_{7,286} = -\xi_{7,252}\xi_{7,284} + \xi_{7,254}\xi_{7,282}$$

$$\xi_{7,287} = \xi_{7,252}\xi_{7,283} - \xi_{7,253}\xi_{7,282}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,288} = 1 - \xi_{7,255}^2$$

$$\xi_{7,289} = -\xi_{7,255}\xi_{7,256}$$

$$\xi_{7,290} = -\xi_{7,255}\xi_{7,257}$$

$$\xi_{7,291} = 1 - \xi_{7,256}^2$$

$$\xi_{7,292} = -\xi_{7,256}\xi_{7,257}$$

$$\xi_{7,293} = 1 - \xi_{7,257}^2$$

$$\xi_{7,294} = \xi_{7,288}^2 + \xi_{7,289}^2 + \xi_{7,290}^2$$

$$\xi_{7,295} = \xi_{7,289}^2 + \xi_{7,291}^2 + \xi_{7,292}^2$$

$$\xi_{7,296} = \xi_{7,290}^2 + \xi_{7,292}^2 + \xi_{7,293}^2$$

$$\xi_{7,297} = \sqrt{\xi_{7,294}}$$

$$\xi_{7,298} = \frac{\xi_{7,288}}{\xi_{7,297}}$$

$$\xi_{7,299} = \frac{\xi_{7,289}}{\xi_{7,297}}$$

$$\xi_{7,300} = \frac{\xi_{7,290}}{\xi_{7,297}}$$

$$\xi_{7,301} = \sqrt{\xi_{7,295}}$$

$$\xi_{7,302} = \frac{\xi_{7,291}}{\xi_{7,301}}$$

$$\xi_{7,303} = \frac{\xi_{7,289}}{\xi_{7,301}}$$

$$\xi_{7,304} = \frac{\xi_{7,292}}{\xi_{7,301}}$$

$$\xi_{7,305} = \sqrt{\xi_{7,296}}$$

$$\xi_{7,306} = \frac{\xi_{7,293}}{\xi_{7,305}}$$

$$\xi_{7,307} = \frac{\xi_{7,290}}{\xi_{7,305}}$$

$$\xi_{7,308} = \frac{\xi_{7,292}}{\xi_{7,305}}$$

$$\xi_{7,309} = \begin{cases} \xi_{7,298} & \text{for } \xi_{7,294} \geq \xi_{7,295} \wedge \xi_{7,294} \geq \xi_{7,296} \\ \xi_{7,303} & \text{for } \xi_{7,295} \geq \xi_{7,296} \wedge \xi_{7,294} \leq \xi_{7,295} \\ \xi_{7,307} & \text{otherwise} \end{cases}$$

$$\xi_{7,310} = \begin{cases} \xi_{7,299} & \text{for } \xi_{7,294} \geq \xi_{7,295} \wedge \xi_{7,294} \geq \xi_{7,296} \\ \xi_{7,302} & \text{for } \xi_{7,295} \geq \xi_{7,296} \wedge \xi_{7,294} \leq \xi_{7,295} \\ \xi_{7,308} & \text{otherwise} \end{cases}$$

$$\xi_{7,311} = \begin{cases} \xi_{7,300} & \text{for } \xi_{7,294} \geq \xi_{7,295} \wedge \xi_{7,294} \geq \xi_{7,296} \\ \xi_{7,304} & \text{for } \xi_{7,295} \geq \xi_{7,296} \wedge \xi_{7,294} \leq \xi_{7,295} \\ \xi_{7,306} & \text{otherwise} \end{cases}$$

$$\xi_{7,312} = \xi_{7,256}\xi_{7,311} - \xi_{7,257}\xi_{7,310}$$

$$\xi_{7,313} = -\xi_{7,255}\xi_{7,311} + \xi_{7,257}\xi_{7,309}$$

$$\xi_{7,314} = \xi_{7,255}\xi_{7,310} - \xi_{7,256}\xi_{7,309}$$

$$\xi_{7,315} = \xi_{7,253}\xi_{7,257} - \xi_{7,254}\xi_{7,256}$$

$$\xi_{7,316} = -\xi_{7,252}\xi_{7,257} + \xi_{7,254}\xi_{7,255}$$

$$\xi_{7,317} = \xi_{7,252}\xi_{7,256} - \xi_{7,253}\xi_{7,255}$$

$$\xi_{7,318} = 1 - \xi_{7,258}^2$$

$$\xi_{7,319} = -\xi_{7,258}\xi_{7,259}$$

$$\xi_{7,320} = -\xi_{7,258}\xi_{7,260}$$

$$\xi_{7,321} = 1 - \xi_{7,259}^2$$

$$\xi_{7,322} = -\xi_{7,259}\xi_{7,260}$$

$$\xi_{7,323} = 1 - \xi_{7,260}^2$$

Exact shared-expression program 7

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$$\xi_{7,324} = \xi_{7,318}^2 + \xi_{7,319}^2 + \xi_{7,320}^2$$

$$\xi_{7,325} = \xi_{7,319}^2 + \xi_{7,321}^2 + \xi_{7,322}^2$$

$$\xi_{7,326} = \xi_{7,320}^2 + \xi_{7,322}^2 + \xi_{7,323}^2$$

$$\xi_{7,327} = \sqrt{\xi_{7,324}}$$

$$\xi_{7,328} = \frac{\xi_{7,318}}{\xi_{7,327}}$$

$$\xi_{7,329} = \frac{\xi_{7,319}}{\xi_{7,327}}$$

$$\xi_{7,330} = \frac{\xi_{7,320}}{\xi_{7,327}}$$

$$\xi_{7,331} = \sqrt{\xi_{7,325}}$$

$$\xi_{7,332} = \frac{\xi_{7,321}}{\xi_{7,331}}$$

$$\xi_{7,333} = \frac{\xi_{7,319}}{\xi_{7,331}}$$

$$\xi_{7,334} = \frac{\xi_{7,322}}{\xi_{7,331}}$$

$$\xi_{7,335} = \sqrt{\xi_{7,326}}$$

$$\xi_{7,336} = \frac{\xi_{7,323}}{\xi_{7,335}}$$

$$\xi_{7,337} = \frac{\xi_{7,320}}{\xi_{7,335}}$$

$$\xi_{7,338} = \frac{\xi_{7,322}}{\xi_{7,335}}$$

$$\xi_{7,339} = \begin{cases} \xi_{7,328} & \text{for } \xi_{7,324} \geq \xi_{7,325} \wedge \xi_{7,324} \geq \xi_{7,326} \\ \xi_{7,333} & \text{for } \xi_{7,325} \geq \xi_{7,326} \wedge \xi_{7,324} \leq \xi_{7,325} \\ \xi_{7,337} & \text{otherwise} \end{cases}$$

$$\xi_{7,340} = \begin{cases} \xi_{7,329} & \text{for } \xi_{7,324} \geq \xi_{7,325} \wedge \xi_{7,324} \geq \xi_{7,326} \\ \xi_{7,332} & \text{for } \xi_{7,325} \geq \xi_{7,326} \wedge \xi_{7,324} \leq \xi_{7,325} \\ \xi_{7,338} & \text{otherwise} \end{cases}$$

$$\xi_{7,341} = \begin{cases} \xi_{7,330} & \text{for } \xi_{7,324} \geq \xi_{7,325} \wedge \xi_{7,324} \geq \xi_{7,326} \\ \xi_{7,334} & \text{for } \xi_{7,325} \geq \xi_{7,326} \wedge \xi_{7,324} \leq \xi_{7,325} \\ \xi_{7,336} & \text{otherwise} \end{cases}$$

$$\xi_{7,342} = \xi_{7,259}\xi_{7,341} - \xi_{7,260}\xi_{7,340}$$

$$\xi_{7,343} = -\xi_{7,258}\xi_{7,341} + \xi_{7,260}\xi_{7,339}$$

$$\xi_{7,344} = \xi_{7,258}\xi_{7,340} - \xi_{7,259}\xi_{7,339}$$

$$\xi_{7,345} = -\xi_{7,253}\xi_{7,260} + \xi_{7,254}\xi_{7,259}$$

$$\xi_{7,346} = \xi_{7,252}\xi_{7,260} - \xi_{7,254}\xi_{7,258}$$

$$\xi_{7,347} = -\xi_{7,252}\xi_{7,259} + \xi_{7,253}\xi_{7,258}$$

$$\xi_{7,348} = \xi_{7,256}\xi_{7,260} - \xi_{7,257}\xi_{7,259}$$

$$\xi_{7,349} = -\xi_{7,255}\xi_{7,260} + \xi_{7,257}\xi_{7,258}$$

$$\xi_{7,350} = \xi_{7,255}\xi_{7,259} - \xi_{7,256}\xi_{7,258}$$

$$\xi_{7,351} = \begin{cases} 1 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,252} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,312} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,252} & \text{for } \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,339} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,252} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,348} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,252} & \text{for } \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

$$\xi_{7,352} = \begin{cases} 0 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,282} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,255} & \text{for } \xi_{7,251} = 0 \wedge (\xi_{7,249} = 0 \vee \xi_{7,251} = 0) \wedge (\xi_{7,249} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,251} = 0 \vee \xi_{7,249} \neq 0) \wedge (\xi_{7,251} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,250} \neq 0) \wedge \xi_{7,250} \neq 0 \\ \xi_{7,342} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,345} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,255} & \text{for } (\xi_{7,249} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} = 0 \vee \xi_{7,251} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,251} \neq 0) \wedge (\xi_{7,250} \neq 0 \vee \xi_{7,251} \neq 0) \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

$$\xi_{7,353} = \begin{cases} 0 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,253} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,313} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,253} & \text{for } \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,340} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,253} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,349} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,253} & \text{for } \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

$$\xi_{7,354} = \begin{cases} 0 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,253} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,313} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,253} & \text{for } \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,340} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,253} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,349} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,253} & \text{for } \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

$$\xi_{7,355} = \begin{cases} 1 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,283} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,256} & \text{for } \xi_{7,251} = 0 \wedge (\xi_{7,249} = 0 \vee \xi_{7,251} = 0) \wedge (\xi_{7,249} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,251} = 0 \vee \xi_{7,249} \neq 0) \wedge (\xi_{7,251} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,250} \neq 0) \wedge \xi_{7,250} \neq 0 \\ \xi_{7,343} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,346} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,256} & \text{for } (\xi_{7,249} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} = 0 \vee \xi_{7,251} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,251} \neq 0) \wedge (\xi_{7,250} \neq 0 \vee \xi_{7,251} \neq 0) \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

$$\xi_{7,356} = \begin{cases} 0 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,254} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,314} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,254} & \text{for } \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,341} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,254} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,350} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,254} & \text{for } \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

$$\xi_{7,357} = \begin{cases} 0 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,254} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,314} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,254} & \text{for } \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \\ \xi_{7,341} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,254} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,350} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,254} & \text{for } \xi_{7,249} \neq 0 \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

$$\xi_{7,358} = \begin{cases} 0 & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \\ \xi_{7,284} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,251} = 0 \wedge \xi_{7,249} \neq 0 \\ \xi_{7,257} & \text{for } \xi_{7,251} = 0 \wedge (\xi_{7,249} = 0 \vee \xi_{7,251} = 0) \wedge (\xi_{7,249} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,251} = 0 \vee \xi_{7,249} \neq 0) \wedge (\xi_{7,251} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,250} \neq 0) \wedge \xi_{7,250} \neq 0 \\ \xi_{7,344} & \text{for } \xi_{7,249} = 0 \wedge \xi_{7,250} = 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,347} & \text{for } \xi_{7,250} = 0 \wedge \xi_{7,249} \neq 0 \wedge \xi_{7,251} \neq 0 \\ \xi_{7,257} & \text{for } (\xi_{7,249} = 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} = 0 \vee \xi_{7,251} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,250} \neq 0) \wedge (\xi_{7,249} \neq 0 \vee \xi_{7,251} \neq 0) \wedge (\xi_{7,250} \neq 0 \vee \xi_{7,251} \neq 0) \wedge \xi_{7,250} \neq 0 \wedge \xi_{7,251} \neq 0 \end{cases}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,360} = \frac{\xi_{7,351}}{4} + \frac{\xi_{7,355}}{4} + \frac{\xi_{7,359}}{4} + \frac{1}{4}$$

$$\xi_{7,361} = \frac{\xi_{7,351}}{4} - \frac{\xi_{7,355}}{4} - \frac{\xi_{7,359}}{4} + \frac{1}{4}$$

$$\xi_{7,362} = -\frac{\xi_{7,351}}{4} + \frac{\xi_{7,355}}{4} - \frac{\xi_{7,359}}{4} + \frac{1}{4}$$

$$\xi_{7,363} = -\frac{\xi_{7,351}}{4} - \frac{\xi_{7,355}}{4} + \frac{\xi_{7,359}}{4} + \frac{1}{4}$$

$$\xi_{7,364} = \sqrt{\xi_{7,360}}$$

$$\xi_{7,365} = \frac{-\xi_{7,356}+\xi_{7,358}}{4\xi_{7,364}}$$

$$\xi_{7,366} = \frac{\xi_{7,353}-\xi_{7,357}}{4\xi_{7,364}}$$

$$\xi_{7,367} = \frac{-\xi_{7,352}+\xi_{7,354}}{4\xi_{7,364}}$$

$$\xi_{7,368} = \sqrt{\xi_{7,361}}$$

$$\xi_{7,369} = \frac{-\xi_{7,356}+\xi_{7,358}}{4\xi_{7,368}}$$

$$\xi_{7,370} = \frac{\xi_{7,352}+\xi_{7,354}}{4\xi_{7,368}}$$

$$\xi_{7,371} = \frac{\xi_{7,353}+\xi_{7,357}}{4\xi_{7,368}}$$

$$\xi_{7,372} = \sqrt{\xi_{7,362}}$$

$$\xi_{7,373} = \frac{\xi_{7,353}-\xi_{7,357}}{4\xi_{7,372}}$$

$$\xi_{7,374} = \frac{\xi_{7,352}+\xi_{7,354}}{4\xi_{7,372}}$$

$$\xi_{7,375} = \frac{\xi_{7,356}+\xi_{7,358}}{4\xi_{7,372}}$$

$$\xi_{7,376} = \sqrt{\xi_{7,363}}$$

$$\xi_{7,377} = \frac{-\xi_{7,352}+\xi_{7,354}}{4\xi_{7,376}}$$

$$\xi_{7,378} = \frac{\xi_{7,353}+\xi_{7,357}}{4\xi_{7,376}}$$

$$\xi_{7,379} = \frac{\xi_{7,356}+\xi_{7,358}}{4\xi_{7,376}}$$

$$\xi_{7,380} = \begin{cases} \xi_{7,364} & \text{for } \xi_{7,360} \geq \xi_{7,361} \wedge \xi_{7,360} \geq \xi_{7,362} \wedge \xi_{7,360} \geq \xi_{7,363} \\ \xi_{7,369} & \text{for } \xi_{7,361} \geq \xi_{7,362} \wedge \xi_{7,361} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,361} \\ \xi_{7,373} & \text{for } \xi_{7,362} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,362} \wedge \xi_{7,361} \leq \xi_{7,362} \\ \xi_{7,377} & \text{otherwise} \end{cases}$$

$$\xi_{7,381} = \begin{cases} \xi_{7,365} & \text{for } \xi_{7,360} \geq \xi_{7,361} \wedge \xi_{7,360} \geq \xi_{7,362} \wedge \xi_{7,360} \geq \xi_{7,363} \\ \xi_{7,368} & \text{for } \xi_{7,361} \geq \xi_{7,362} \wedge \xi_{7,361} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,361} \\ \xi_{7,374} & \text{for } \xi_{7,362} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,362} \wedge \xi_{7,361} \leq \xi_{7,362} \\ \xi_{7,378} & \text{otherwise} \end{cases}$$

$$\xi_{7,382} = \begin{cases} \xi_{7,366} & \text{for } \xi_{7,360} \geq \xi_{7,361} \wedge \xi_{7,360} \geq \xi_{7,362} \wedge \xi_{7,360} \geq \xi_{7,363} \\ \xi_{7,370} & \text{for } \xi_{7,361} \geq \xi_{7,362} \wedge \xi_{7,361} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,361} \\ \xi_{7,372} & \text{for } \xi_{7,362} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,362} \wedge \xi_{7,361} \leq \xi_{7,362} \\ \xi_{7,379} & \text{otherwise} \end{cases}$$

$$\xi_{7,383} = \begin{cases} \xi_{7,367} & \text{for } \xi_{7,360} \geq \xi_{7,361} \wedge \xi_{7,360} \geq \xi_{7,362} \wedge \xi_{7,360} \geq \xi_{7,363} \\ \xi_{7,371} & \text{for } \xi_{7,361} \geq \xi_{7,362} \wedge \xi_{7,361} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,361} \\ \xi_{7,375} & \text{for } \xi_{7,362} \geq \xi_{7,363} \wedge \xi_{7,360} \leq \xi_{7,362} \wedge \xi_{7,361} \leq \xi_{7,362} \\ \xi_{7,376} & \text{otherwise} \end{cases}$$

$$\xi_{7,384} = \frac{\xi_{7,180}}{4} + \frac{\xi_{7,244}}{4} + \frac{\xi_{7,248}}{4} + \frac{1}{4}$$

$$\xi_{7,385} = \frac{\xi_{7,180}}{4} - \frac{\xi_{7,244}}{4} - \frac{\xi_{7,248}}{4} + \frac{1}{4}$$

$$\xi_{7,386} = -\frac{\xi_{7,180}}{4} + \frac{\xi_{7,244}}{4} - \frac{\xi_{7,248}}{4} + \frac{1}{4}$$

$$\xi_{7,387} = -\frac{\xi_{7,180}}{4} - \frac{\xi_{7,244}}{4} + \frac{\xi_{7,248}}{4} + \frac{1}{4}$$

$$\xi_{7,388} = \sqrt{\xi_{7,384}}$$

$$\xi_{7,389} = \frac{\xi_{7,245}-\xi_{7,247}}{4\xi_{7,388}}$$

$$\xi_{7,390} = \frac{-\xi_{7,182}+\xi_{7,246}}{4\xi_{7,388}}$$

$$\xi_{7,391} = \frac{\xi_{7,181}-\xi_{7,243}}{4\xi_{7,388}}$$

$$\xi_{7,392} = \sqrt{\xi_{7,385}}$$

$$\xi_{7,393} = \frac{\xi_{7,245}-\xi_{7,247}}{4\xi_{7,392}}$$

$$\xi_{7,394} = \frac{\xi_{7,181}+\xi_{7,243}}{4\xi_{7,392}}$$

$$\xi_{7,395} = \frac{\xi_{7,182}+\xi_{7,246}}{4\xi_{7,392}}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,396} = \sqrt{\xi_{7,386}}$$

$$\xi_{7,397} = \frac{-\xi_{7,182}+\xi_{7,246}}{4\xi_{7,396}}$$

$$\xi_{7,398} = \frac{\xi_{7,181}+\xi_{7,243}}{4\xi_{7,396}}$$

$$\xi_{7,399} = \frac{\xi_{7,245}+\xi_{7,247}}{4\xi_{7,396}}$$

$$\xi_{7,400} = \sqrt{\xi_{7,387}}$$

$$\xi_{7,401} = \frac{\xi_{7,181}-\xi_{7,243}}{4\xi_{7,400}}$$

$$\xi_{7,402} = \frac{\xi_{7,182}+\xi_{7,246}}{4\xi_{7,400}}$$

$$\xi_{7,403} = \frac{\xi_{7,245}+\xi_{7,247}}{4\xi_{7,400}}$$

$$\xi_{7,404} = \begin{cases} \xi_{7,388} & \text{for } \xi_{7,384} \geq \xi_{7,385} \wedge \xi_{7,384} \geq \xi_{7,386} \wedge \xi_{7,384} \geq \xi_{7,387} \\ \xi_{7,393} & \text{for } \xi_{7,385} \geq \xi_{7,386} \wedge \xi_{7,385} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,385} \\ \xi_{7,397} & \text{for } \xi_{7,386} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,386} \wedge \xi_{7,385} \leq \xi_{7,386} \\ \xi_{7,401} & \text{otherwise} \end{cases}$$

$$\xi_{7,405} = \begin{cases} \xi_{7,389} & \text{for } \xi_{7,384} \geq \xi_{7,385} \wedge \xi_{7,384} \geq \xi_{7,386} \wedge \xi_{7,384} \geq \xi_{7,387} \\ \xi_{7,392} & \text{for } \xi_{7,385} \geq \xi_{7,386} \wedge \xi_{7,385} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,385} \\ \xi_{7,398} & \text{for } \xi_{7,386} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,386} \wedge \xi_{7,385} \leq \xi_{7,386} \\ \xi_{7,402} & \text{otherwise} \end{cases}$$

$$\xi_{7,406} = \begin{cases} \xi_{7,390} & \text{for } \xi_{7,384} \geq \xi_{7,385} \wedge \xi_{7,384} \geq \xi_{7,386} \wedge \xi_{7,384} \geq \xi_{7,387} \\ \xi_{7,394} & \text{for } \xi_{7,385} \geq \xi_{7,386} \wedge \xi_{7,385} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,385} \\ \xi_{7,396} & \text{for } \xi_{7,386} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,386} \wedge \xi_{7,385} \leq \xi_{7,386} \\ \xi_{7,403} & \text{otherwise} \end{cases}$$

$$\xi_{7,407} = \begin{cases} \xi_{7,391} & \text{for } \xi_{7,384} \geq \xi_{7,385} \wedge \xi_{7,384} \geq \xi_{7,386} \wedge \xi_{7,384} \geq \xi_{7,387} \\ \xi_{7,395} & \text{for } \xi_{7,385} \geq \xi_{7,386} \wedge \xi_{7,385} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,385} \\ \xi_{7,399} & \text{for } \xi_{7,386} \geq \xi_{7,387} \wedge \xi_{7,384} \leq \xi_{7,386} \wedge \xi_{7,385} \leq \xi_{7,386} \\ \xi_{7,400} & \text{otherwise} \end{cases}$$

$$\xi_{7,408} = \xi_{7,99}\left(\xi_{7,104} + i\xi_{7,105}\right)$$

$$\xi_{7,409} = \frac{\sqrt{2}i\xi_{7,100}\left(\xi_{7,104}+i\xi_{7,105}\right)}{2} - \frac{\sqrt{2}\xi_{7,101}\xi_{7,99}}{2}$$

$$\xi_{7,410} = -\frac{\sqrt{2}\xi_{7,100}\left(\xi_{7,104}+i\xi_{7,105}\right)}{2} + \frac{\sqrt{2}i\xi_{7,101}\xi_{7,99}}{2}$$

$$\xi_{7,411} = -\xi_{7,100}\xi_{7,101}$$

$$\xi_{7,412} = \frac{\sqrt{2}i\left(-\frac{\sqrt{2}\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}i\xi_{7,99}\left(\xi_{7,104}-i\xi_{7,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}\left(-\frac{\sqrt{2}i\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}\xi_{7,99}\left(\xi_{7,104}+i\xi_{7,105}\right)}{2}\right)}{2}$$

$$\xi_{7,413} = \frac{\sqrt{2}\left(-\frac{\sqrt{2}\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}i\xi_{7,99}\left(\xi_{7,104}-i\xi_{7,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}i\left(-\frac{\sqrt{2}i\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}\xi_{7,99}\left(\xi_{7,104}+i\xi_{7,105}\right)}{2}\right)}{2}$$

$$\xi_{7,414} = -\frac{\sqrt{2}\xi_{7,100}\left(\xi_{7,104}-i\xi_{7,105}\right)}{2} + \frac{\sqrt{2}i\xi_{7,101}\xi_{7,99}}{2}$$

$$\xi_{7,415} = \frac{\sqrt{2}\left(-\frac{\sqrt{2}\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}i\xi_{7,99}\left(\xi_{7,104}+i\xi_{7,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}i\left(-\frac{\sqrt{2}i\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}\xi_{7,99}\left(\xi_{7,104}-i\xi_{7,105}\right)}{2}\right)}{2}$$

$$\xi_{7,416} = \frac{\sqrt{2}i\left(-\frac{\sqrt{2}\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}i\xi_{7,99}\left(\xi_{7,104}+i\xi_{7,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}\left(-\frac{\sqrt{2}i\xi_{7,100}\xi_{7,101}}{2} + \frac{\sqrt{2}\xi_{7,99}\left(\xi_{7,104}-i\xi_{7,105}\right)}{2}\right)}{2}$$

$$\xi_{7,417} = \frac{\sqrt{2}i\xi_{7,100}\left(\xi_{7,104}-i\xi_{7,105}\right)}{2} - \frac{\sqrt{2}\xi_{7,101}\xi_{7,99}}{2}$$

$$\xi_{7,418} = \xi_{7,99}\left(\xi_{7,104} - i\xi_{7,105}\right)$$

$$\xi_{7,419} = \left(\xi_{7,380} - i\xi_{7,383}\right)\left(\xi_{7,404} - i\xi_{7,407}\right)$$

$$\xi_{7,420} = \left(\xi_{7,380} - i\xi_{7,383}\right)\left(-i\xi_{7,405} - \xi_{7,406}\right)$$

$$\xi_{7,421} = \left(\xi_{7,404} - i\xi_{7,407}\right)\left(-i\xi_{7,381} - \xi_{7,382}\right)$$

$$\xi_{7,422} = \left(-i\xi_{7,381} - \xi_{7,382}\right)\left(-i\xi_{7,405} - \xi_{7,406}\right)$$

$$\xi_{7,423} = \left(\xi_{7,380} - i\xi_{7,383}\right)\left(-i\xi_{7,405} + \xi_{7,406}\right)$$

$$\xi_{7,424} = \left(\xi_{7,380} - i\xi_{7,383}\right)\left(\xi_{7,404} + i\xi_{7,407}\right)$$

$$\xi_{7,425} = \left(-i\xi_{7,381} - \xi_{7,382}\right)\left(-i\xi_{7,405} + \xi_{7,406}\right)$$

$$\xi_{7,426} = \left(\xi_{7,404} + i\xi_{7,407}\right)\left(-i\xi_{7,381} - \xi_{7,382}\right)$$

$$\xi_{7,427} = \left(\xi_{7,404} - i\xi_{7,407}\right)\left(-i\xi_{7,381} + \xi_{7,382}\right)$$

$$\xi_{7,428} = \left(-i\xi_{7,381} + \xi_{7,382}\right)\left(-i\xi_{7,405} - \xi_{7,406}\right)$$

$$\xi_{7,429} = \left(\xi_{7,380} + i\xi_{7,383}\right)\left(\xi_{7,404} - i\xi_{7,407}\right)$$

$$\xi_{7,430} = \left(\xi_{7,380} + i\xi_{7,383}\right)\left(-i\xi_{7,405} - \xi_{7,406}\right)$$

$$\xi_{7,431} = \left(-i\xi_{7,381} + \xi_{7,382}\right)\left(-i\xi_{7,405} + \xi_{7,406}\right)$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,432} = (\xi_{7,404} + i\xi_{7,407})\,(-i\xi_{7,381} + \xi_{7,382})$$

$$\xi_{7,433} = (\xi_{7,380} + i\xi_{7,383})\,(-i\xi_{7,405} + \xi_{7,406})$$

$$\xi_{7,434} = (\xi_{7,380} + i\xi_{7,383})\,(\xi_{7,404} + i\xi_{7,407})$$

$$\xi_{7,435} = \xi_{7,408}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}}\right) + \xi_{7,409}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}}\right) + \xi_{7,410}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}}\right) + \xi_{7,411}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}}\right)$$

$$\xi_{7,436} = \xi_{7,409}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}}\right) + \xi_{7,412}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}}\right) + \xi_{7,414}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}}\right) + \xi_{7,415}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}}\right)$$

$$\xi_{7,437} = \xi_{7,410}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}}\right) + \xi_{7,413}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}}\right) + \xi_{7,416}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}}\right) + \xi_{7,417}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}}\right)$$

$$\xi_{7,438} = \xi_{7,411}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}}\right) + \xi_{7,414}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}}\right) + \xi_{7,417}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}}\right) + \xi_{7,418}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}}\right)$$

$$\xi_{7,439} = \xi_{7,408}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}}\right) + \xi_{7,409}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}}\right) + \xi_{7,410}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}}\right) + \xi_{7,411}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}}\right)$$

$$\xi_{7,440} = \xi_{7,409}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}}\right) + \xi_{7,412}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}}\right) + \xi_{7,414}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}}\right) + \xi_{7,415}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}}\right)$$

$$\xi_{7,441} = \xi_{7,410}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}}\right) + \xi_{7,413}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}}\right) + \xi_{7,416}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}}\right) + \xi_{7,417}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}}\right)$$

$$\xi_{7,442} = \xi_{7,411}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,419}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,422}}\right) + \xi_{7,414}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,423}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,426}}\right) + \xi_{7,417}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,427}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,430}}\right) + \xi_{7,418}\left(-ie^{-i\xi_{7,92}}\sin\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,431}} + e^{-i\xi_{7,92}}\cos\left(\xi_{7,90}-\xi_{7,91}\right)\overline{\xi_{7,434}}\right)$$

$$\xi_{7,443} = \xi_{7,408}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}}\right) + \xi_{7,409}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}}\right) + \xi_{7,410}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}}\right) + \xi_{7,411}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,433}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,432}}\right)$$

$$\xi_{7,444} = \xi_{7,409}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}}\right) + \xi_{7,412}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}}\right) + \xi_{7,414}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,433}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,432}}\right) + \xi_{7,415}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}}\right)$$

$$\xi_{7,445} = \xi_{7,410}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}}\right) + \xi_{7,413}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}}\right) + \xi_{7,416}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}}\right) + \xi_{7,417}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,433}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,432}}\right)$$

$$\xi_{7,446} = \xi_{7,411}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}}\right) + \xi_{7,414}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}}\right) + \xi_{7,417}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}}\right) + \xi_{7,418}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,433}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,432}}\right)$$

$$\xi_{7,447} = \xi_{7,408}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}}\right) + \xi_{7,409}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}}\right) + \xi_{7,410}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}}\right) + \xi_{7,411}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,432}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,433}}\right)$$

$$\xi_{7,448} = \xi_{7,409}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}}\right) + \xi_{7,412}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}}\right) + \xi_{7,414}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}}\right) + \xi_{7,415}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}}\right)$$

$$\xi_{7,449} = \xi_{7,410}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}}\right) + \xi_{7,413}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}}\right) + \xi_{7,416}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}}\right) + \xi_{7,417}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,432}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,433}}\right)$$

$$\xi_{7,450} = \xi_{7,411}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,420}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,421}}\right) + \xi_{7,414}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,424}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,425}}\right) + \xi_{7,417}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,428}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,429}}\right) + \xi_{7,418}\left(-ie^{i\xi_{7,92}}\sin\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,432}} + e^{i\xi_{7,92}}\cos\left(\xi_{7,90}+\xi_{7,91}\right)\overline{\xi_{7,433}}\right)$$

$$\xi_{7,451} = \frac{\operatorname{re}(\xi_{7,435})}{4} + \frac{\operatorname{re}(\xi_{7,442})}{4} + \frac{\operatorname{re}(\xi_{7,444})}{4} + \frac{\operatorname{re}(\xi_{7,449})}{4}$$

$$\xi_{7,452} = -\frac{\operatorname{im}(\xi_{7,436})}{4} - \frac{\operatorname{im}(\xi_{7,441})}{4} - \frac{\operatorname{im}(\xi_{7,443})}{4} - \frac{\operatorname{im}(\xi_{7,450})}{4}$$

$$\xi_{7,453} = -\frac{\operatorname{re}(\xi_{7,436})}{4} + \frac{\operatorname{re}(\xi_{7,441})}{4} + \frac{\operatorname{re}(\xi_{7,443})}{4} - \frac{\operatorname{re}(\xi_{7,450})}{4}$$

$$\xi_{7,454} = -\frac{\operatorname{im}(\xi_{7,435})}{4} + \frac{\operatorname{im}(\xi_{7,442})}{4} + \frac{\operatorname{im}(\xi_{7,444})}{4} - \frac{\operatorname{im}(\xi_{7,449})}{4}$$

$$\xi_{7,455} = -\frac{\operatorname{im}(\xi_{7,437})}{4} - \frac{\operatorname{im}(\xi_{7,440})}{4} - \frac{\operatorname{im}(\xi_{7,446})}{4} - \frac{\operatorname{im}(\xi_{7,447})}{4}$$

$$\xi_{7,456} = -\frac{\operatorname{re}(\xi_{7,438})}{4} - \frac{\operatorname{re}(\xi_{7,439})}{4} - \frac{\operatorname{re}(\xi_{7,445})}{4} - \frac{\operatorname{re}(\xi_{7,448})}{4}$$

$$\xi_{7,457} = \frac{\operatorname{im}(\xi_{7,438})}{4} - \frac{\operatorname{im}(\xi_{7,439})}{4} - \frac{\operatorname{im}(\xi_{7,445})}{4} + \frac{\operatorname{im}(\xi_{7,448})}{4}$$

$$\xi_{7,458} = -\frac{\operatorname{re}(\xi_{7,437})}{4} + \frac{\operatorname{re}(\xi_{7,440})}{4} + \frac{\operatorname{re}(\xi_{7,446})}{4} - \frac{\operatorname{re}(\xi_{7,447})}{4}$$

$$\xi_{7,459} = -\frac{\operatorname{re}(\xi_{7,437})}{4} + \frac{\operatorname{re}(\xi_{7,440})}{4} - \frac{\operatorname{re}(\xi_{7,446})}{4} + \frac{\operatorname{re}(\xi_{7,447})}{4}$$

$$\xi_{7,460} = \frac{\operatorname{im}(\xi_{7,438})}{4} - \frac{\operatorname{im}(\xi_{7,439})}{4} + \frac{\operatorname{im}(\xi_{7,445})}{4} - \frac{\operatorname{im}(\xi_{7,448})}{4}$$

$$\xi_{7,461} = \frac{\operatorname{re}(\xi_{7,438})}{4} + \frac{\operatorname{re}(\xi_{7,439})}{4} - \frac{\operatorname{re}(\xi_{7,445})}{4} - \frac{\operatorname{re}(\xi_{7,448})}{4}$$

$$\xi_{7,462} = \frac{\operatorname{im}(\xi_{7,437})}{4} + \frac{\operatorname{im}(\xi_{7,440})}{4} - \frac{\operatorname{im}(\xi_{7,446})}{4} - \frac{\operatorname{im}(\xi_{7,447})}{4}$$

$$\xi_{7,463} = -\frac{\operatorname{im}(\xi_{7,435})}{4} + \frac{\operatorname{im}(\xi_{7,442})}{4} - \frac{\operatorname{im}(\xi_{7,444})}{4} + \frac{\operatorname{im}(\xi_{7,449})}{4}$$

$$\xi_{7,464} = -\frac{\operatorname{re}(\xi_{7,436})}{4} + \frac{\operatorname{re}(\xi_{7,441})}{4} - \frac{\operatorname{re}(\xi_{7,443})}{4} + \frac{\operatorname{re}(\xi_{7,450})}{4}$$

$$\xi_{7,465} = \frac{\operatorname{im}(\xi_{7,436})}{4} + \frac{\operatorname{im}(\xi_{7,441})}{4} - \frac{\operatorname{im}(\xi_{7,443})}{4} - \frac{\operatorname{im}(\xi_{7,450})}{4}$$

$$\xi_{7,466} = -\frac{\operatorname{re}(\xi_{7,435})}{4} - \frac{\operatorname{re}(\xi_{7,442})}{4} + \frac{\operatorname{re}(\xi_{7,444})}{4} + \frac{\operatorname{re}(\xi_{7,449})}{4}$$

$$\xi_{7,467} = \xi_{7,451}^2 + \xi_{7,455}^2 + \xi_{7,459}^2 + \xi_{7,463}^2$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,468} = \xi_{7,452}^2 + \xi_{7,456}^2 + \xi_{7,460}^2 + \xi_{7,464}^2$$

$$\xi_{7,469} = \xi_{7,453}^2 + \xi_{7,457}^2 + \xi_{7,461}^2 + \xi_{7,465}^2$$

$$\xi_{7,470} = \xi_{7,454}^2 + \xi_{7,458}^2 + \xi_{7,462}^2 + \xi_{7,466}^2$$

$$\xi_{7,471} = \sqrt{\xi_{7,467}}$$

$$\xi_{7,472} = \frac{\xi_{7,451}}{\xi_{7,471}}$$

$$\xi_{7,473} = \frac{\xi_{7,455}}{\xi_{7,471}}$$

$$\xi_{7,474} = \frac{\xi_{7,459}}{\xi_{7,471}}$$

$$\xi_{7,475} = \frac{\xi_{7,463}}{\xi_{7,471}}$$

$$\xi_{7,476} = \sqrt{\xi_{7,468}}$$

$$\xi_{7,477} = \frac{\xi_{7,452}}{\xi_{7,476}}$$

$$\xi_{7,478} = \frac{\xi_{7,456}}{\xi_{7,476}}$$

$$\xi_{7,479} = \frac{\xi_{7,460}}{\xi_{7,476}}$$

$$\xi_{7,480} = \frac{\xi_{7,464}}{\xi_{7,476}}$$

$$\xi_{7,481} = \sqrt{\xi_{7,469}}$$

$$\xi_{7,482} = \frac{\xi_{7,453}}{\xi_{7,481}}$$

$$\xi_{7,483} = \frac{\xi_{7,457}}{\xi_{7,481}}$$

$$\xi_{7,484} = \frac{\xi_{7,461}}{\xi_{7,481}}$$

$$\xi_{7,485} = \frac{\xi_{7,465}}{\xi_{7,481}}$$

$$\xi_{7,486} = \sqrt{\xi_{7,470}}$$

$$\xi_{7,487} = \frac{\xi_{7,454}}{\xi_{7,486}}$$

$$\xi_{7,488} = \frac{\xi_{7,458}}{\xi_{7,486}}$$

$$\xi_{7,489} = \frac{\xi_{7,462}}{\xi_{7,486}}$$

$$\xi_{7,490} = \frac{\xi_{7,466}}{\xi_{7,486}}$$

$$\xi_{7,491} = \begin{cases} \xi_{7,472} & \text{for } \xi_{7,467} \geq \xi_{7,468} \wedge \xi_{7,467} \geq \xi_{7,469} \wedge \xi_{7,467} \geq \xi_{7,470} \\ \xi_{7,477} & \text{for } \xi_{7,468} \geq \xi_{7,469} \wedge \xi_{7,468} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,468} \\ \xi_{7,482} & \text{for } \xi_{7,469} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,469} \wedge \xi_{7,468} \leq \xi_{7,469} \\ \xi_{7,487} & \text{otherwise} \end{cases}$$

$$\xi_{7,492} = \begin{cases} \xi_{7,473} & \text{for } \xi_{7,467} \geq \xi_{7,468} \wedge \xi_{7,467} \geq \xi_{7,469} \wedge \xi_{7,467} \geq \xi_{7,470} \\ \xi_{7,478} & \text{for } \xi_{7,468} \geq \xi_{7,469} \wedge \xi_{7,468} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,468} \\ \xi_{7,483} & \text{for } \xi_{7,469} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,469} \wedge \xi_{7,468} \leq \xi_{7,469} \\ \xi_{7,488} & \text{otherwise} \end{cases}$$

$$\xi_{7,493} = \begin{cases} \xi_{7,474} & \text{for } \xi_{7,467} \geq \xi_{7,468} \wedge \xi_{7,467} \geq \xi_{7,469} \wedge \xi_{7,467} \geq \xi_{7,470} \\ \xi_{7,479} & \text{for } \xi_{7,468} \geq \xi_{7,469} \wedge \xi_{7,468} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,468} \\ \xi_{7,484} & \text{for } \xi_{7,469} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,469} \wedge \xi_{7,468} \leq \xi_{7,469} \\ \xi_{7,489} & \text{otherwise} \end{cases}$$

$$\xi_{7,494} = \begin{cases} \xi_{7,475} & \text{for } \xi_{7,467} \geq \xi_{7,468} \wedge \xi_{7,467} \geq \xi_{7,469} \wedge \xi_{7,467} \geq \xi_{7,470} \\ \xi_{7,480} & \text{for } \xi_{7,468} \geq \xi_{7,469} \wedge \xi_{7,468} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,468} \\ \xi_{7,485} & \text{for } \xi_{7,469} \geq \xi_{7,470} \wedge \xi_{7,467} \leq \xi_{7,469} \wedge \xi_{7,468} \leq \xi_{7,469} \\ \xi_{7,490} & \text{otherwise} \end{cases}$$

$$\xi_{7,495} = \xi_{7,451}\xi_{7,491} + \xi_{7,455}\xi_{7,492} + \xi_{7,459}\xi_{7,493} + \xi_{7,463}\xi_{7,494}$$

$$\xi_{7,496} = \xi_{7,452}\xi_{7,491} + \xi_{7,456}\xi_{7,492} + \xi_{7,460}\xi_{7,493} + \xi_{7,464}\xi_{7,494}$$

$$\xi_{7,497} = \xi_{7,453}\xi_{7,491} + \xi_{7,457}\xi_{7,492} + \xi_{7,461}\xi_{7,493} + \xi_{7,465}\xi_{7,494}$$

$$\xi_{7,498} = \xi_{7,454}\xi_{7,491} + \xi_{7,458}\xi_{7,492} + \xi_{7,462}\xi_{7,493} + \xi_{7,466}\xi_{7,494}$$

$$\xi_{7,499} = \xi_{7,29}\xi_{7,39} - \xi_{7,30}\xi_{7,42} - \xi_{7,31}\xi_{7,43} - \xi_{7,32}\xi_{7,44}$$

$$\xi_{7,500} = \xi_{7,29}\xi_{7,42} + \xi_{7,30}\xi_{7,39} - \xi_{7,31}\xi_{7,44} + \xi_{7,32}\xi_{7,43}$$

$$\xi_{7,501} = \xi_{7,29}\xi_{7,43} + \xi_{7,30}\xi_{7,44} + \xi_{7,31}\xi_{7,39} - \xi_{7,32}\xi_{7,42}$$

$$\xi_{7,502} = \xi_{7,29}\xi_{7,44} - \xi_{7,30}\xi_{7,43} + \xi_{7,31}\xi_{7,42} + \xi_{7,32}\xi_{7,39}$$

$$\xi_{7,503} = \xi_{7,3}\xi_{7,41} + \xi_{7,4}\xi_{7,42} + \xi_{7,42}\xi_{7,5} + \xi_{7,42}\xi_{7,6}$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,504} = \xi_{7,3}\xi_{7,42} - \xi_{7,4}\xi_{7,41} + \xi_{7,42}\xi_{7,5} - \xi_{7,42}\xi_{7,6}$$

$$\xi_{7,505} = \xi_{7,3}\xi_{7,42} - \xi_{7,4}\xi_{7,42} - \xi_{7,41}\xi_{7,5} + \xi_{7,42}\xi_{7,6}$$

$$\xi_{7,506} = \xi_{7,3}\xi_{7,42} + \xi_{7,4}\xi_{7,42} - \xi_{7,41}\xi_{7,6} - \xi_{7,42}\xi_{7,5}$$

$$\xi_{7,507} = \xi_{7,39}\left(\xi_{7,491}\xi_{7,86} + \xi_{7,492}\xi_{7,87} + \xi_{7,493}\xi_{7,88} + \xi_{7,494}\xi_{7,89}\right) - \xi_{7,39}\left(\xi_{7,491}\xi_{7,89} - \xi_{7,492}\xi_{7,88} + \xi_{7,493}\xi_{7,87} - \xi_{7,494}\xi_{7,86}\right) - \xi_{7,40}\left(\xi_{7,491}\xi_{7,87} - \xi_{7,492}\xi_{7,86} - \xi_{7,493}\xi_{7,89} + \xi_{7,494}\xi_{7,88}\right) - \xi_{7,41}\left(\xi_{7,491}\xi_{7,88} + \xi_{7,492}\xi_{7,89} - \xi_{7,493}\xi_{7,86} - \xi_{7,494}\xi_{7,87}\right)$$

$$\xi_{7,508} = \xi_{7,39}\left(\xi_{7,491}\xi_{7,87} - \xi_{7,492}\xi_{7,86} - \xi_{7,493}\xi_{7,89} + \xi_{7,494}\xi_{7,88}\right) + \xi_{7,39}\left(\xi_{7,491}\xi_{7,88} + \xi_{7,492}\xi_{7,89} - \xi_{7,493}\xi_{7,86} - \xi_{7,494}\xi_{7,87}\right) + \xi_{7,40}\left(\xi_{7,491}\xi_{7,86} + \xi_{7,492}\xi_{7,87} + \xi_{7,493}\xi_{7,88} + \xi_{7,494}\xi_{7,89}\right) - \xi_{7,41}\left(\xi_{7,491}\xi_{7,89} - \xi_{7,492}\xi_{7,88} + \xi_{7,493}\xi_{7,87} - \xi_{7,494}\xi_{7,86}\right)$$

$$\xi_{7,509} = -\xi_{7,39}\left(\xi_{7,491}\xi_{7,87} - \xi_{7,492}\xi_{7,86} - \xi_{7,493}\xi_{7,89} + \xi_{7,494}\xi_{7,88}\right) + \xi_{7,39}\left(\xi_{7,491}\xi_{7,88} + \xi_{7,492}\xi_{7,89} - \xi_{7,493}\xi_{7,86} - \xi_{7,494}\xi_{7,87}\right) + \xi_{7,40}\left(\xi_{7,491}\xi_{7,89} - \xi_{7,492}\xi_{7,88} + \xi_{7,493}\xi_{7,87} - \xi_{7,494}\xi_{7,86}\right) + \xi_{7,41}\left(\xi_{7,491}\xi_{7,86} + \xi_{7,492}\xi_{7,87} + \xi_{7,493}\xi_{7,88} + \xi_{7,494}\xi_{7,89}\right)$$

$$\xi_{7,510} = \xi_{7,39}\left(\xi_{7,491}\xi_{7,86} + \xi_{7,492}\xi_{7,87} + \xi_{7,493}\xi_{7,88} + \xi_{7,494}\xi_{7,89}\right) + \xi_{7,39}\left(\xi_{7,491}\xi_{7,89} - \xi_{7,492}\xi_{7,88} + \xi_{7,493}\xi_{7,87} - \xi_{7,494}\xi_{7,86}\right) - \xi_{7,40}\left(\xi_{7,491}\xi_{7,88} + \xi_{7,492}\xi_{7,89} - \xi_{7,493}\xi_{7,86} - \xi_{7,494}\xi_{7,87}\right) + \xi_{7,41}\left(\xi_{7,491}\xi_{7,87} - \xi_{7,492}\xi_{7,86} - \xi_{7,493}\xi_{7,89} + \xi_{7,494}\xi_{7,88}\right)$$

$$\xi_{7,511} = -\xi_{7,40}\left(-\xi_{7,495}\xi_{7,55} - \xi_{7,496}\xi_{7,54} + \xi_{7,497}\xi_{7,57} - \xi_{7,498}\xi_{7,56}\right) - \xi_{7,40}\left(-\xi_{7,495}\xi_{7,56} - \xi_{7,496}\xi_{7,57} - \xi_{7,497}\xi_{7,54} + \xi_{7,498}\xi_{7,55}\right) - \xi_{7,40}\left(-\xi_{7,495}\xi_{7,57} + \xi_{7,496}\xi_{7,56} - \xi_{7,497}\xi_{7,55} - \xi_{7,498}\xi_{7,54}\right) + \xi_{7,41}\left(\xi_{7,495}\xi_{7,54} - \xi_{7,496}\xi_{7,55} - \xi_{7,497}\xi_{7,56} - \xi_{7,498}\xi_{7,57}\right)$$

$$\xi_{7,512} = \xi_{7,40}\left(\xi_{7,495}\xi_{7,54} - \xi_{7,496}\xi_{7,55} - \xi_{7,497}\xi_{7,56} - \xi_{7,498}\xi_{7,57}\right) + \xi_{7,40}\left(-\xi_{7,495}\xi_{7,56} - \xi_{7,496}\xi_{7,57} - \xi_{7,497}\xi_{7,54} + \xi_{7,498}\xi_{7,55}\right) - \xi_{7,40}\left(-\xi_{7,495}\xi_{7,57} + \xi_{7,496}\xi_{7,56} - \xi_{7,497}\xi_{7,55} - \xi_{7,498}\xi_{7,54}\right) + \xi_{7,41}\left(-\xi_{7,495}\xi_{7,55} - \xi_{7,496}\xi_{7,54} + \xi_{7,497}\xi_{7,57} - \xi_{7,498}\xi_{7,56}\right)$$

$$\xi_{7,513} = \xi_{7,40}\left(\xi_{7,495}\xi_{7,54} - \xi_{7,496}\xi_{7,55} - \xi_{7,497}\xi_{7,56} - \xi_{7,498}\xi_{7,57}\right) - \xi_{7,40}\left(-\xi_{7,495}\xi_{7,55} - \xi_{7,496}\xi_{7,54} + \xi_{7,497}\xi_{7,57} - \xi_{7,498}\xi_{7,56}\right) + \xi_{7,40}\left(-\xi_{7,495}\xi_{7,57} + \xi_{7,496}\xi_{7,56} - \xi_{7,497}\xi_{7,55} - \xi_{7,498}\xi_{7,54}\right) + \xi_{7,41}\left(-\xi_{7,495}\xi_{7,56} - \xi_{7,496}\xi_{7,57} - \xi_{7,497}\xi_{7,54} + \xi_{7,498}\xi_{7,55}\right)$$

$$\xi_{7,514} = \xi_{7,40}\left(\xi_{7,495}\xi_{7,54} - \xi_{7,496}\xi_{7,55} - \xi_{7,497}\xi_{7,56} - \xi_{7,498}\xi_{7,57}\right) + \xi_{7,40}\left(-\xi_{7,495}\xi_{7,55} - \xi_{7,496}\xi_{7,54} + \xi_{7,497}\xi_{7,57} - \xi_{7,498}\xi_{7,56}\right) - \xi_{7,40}\left(-\xi_{7,495}\xi_{7,56} - \xi_{7,496}\xi_{7,57} - \xi_{7,497}\xi_{7,54} + \xi_{7,498}\xi_{7,55}\right) + \xi_{7,41}\left(-\xi_{7,495}\xi_{7,57} + \xi_{7,496}\xi_{7,56} - \xi_{7,497}\xi_{7,55} - \xi_{7,498}\xi_{7,54}\right)$$

$$\xi_{7,515} = \xi_{7,380}\left(\xi_{7,21}\xi_{7,78} - \xi_{7,22}\xi_{7,79} - \xi_{7,23}\xi_{7,80} - \xi_{7,24}\xi_{7,81}\right) + \xi_{7,381}\left(\xi_{7,21}\xi_{7,79} + \xi_{7,22}\xi_{7,78} + \xi_{7,23}\xi_{7,81} - \xi_{7,24}\xi_{7,80}\right) + \xi_{7,382}\left(\xi_{7,21}\xi_{7,80} - \xi_{7,22}\xi_{7,81} + \xi_{7,23}\xi_{7,78} + \xi_{7,24}\xi_{7,79}\right) + \xi_{7,383}\left(\xi_{7,21}\xi_{7,81} + \xi_{7,22}\xi_{7,80} - \xi_{7,23}\xi_{7,79} + \xi_{7,24}\xi_{7,78}\right)$$

$$\xi_{7,516} = \xi_{7,380}\left(\xi_{7,21}\xi_{7,79} + \xi_{7,22}\xi_{7,78} + \xi_{7,23}\xi_{7,81} - \xi_{7,24}\xi_{7,80}\right) - \xi_{7,381}\left(\xi_{7,21}\xi_{7,78} - \xi_{7,22}\xi_{7,79} - \xi_{7,23}\xi_{7,80} - \xi_{7,24}\xi_{7,81}\right) + \xi_{7,382}\left(\xi_{7,21}\xi_{7,81} + \xi_{7,22}\xi_{7,80} - \xi_{7,23}\xi_{7,79} + \xi_{7,24}\xi_{7,78}\right) - \xi_{7,383}\left(\xi_{7,21}\xi_{7,80} - \xi_{7,22}\xi_{7,81} + \xi_{7,23}\xi_{7,78} + \xi_{7,24}\xi_{7,79}\right)$$

$$\xi_{7,517} = \xi_{7,380}\left(\xi_{7,21}\xi_{7,80} - \xi_{7,22}\xi_{7,81} + \xi_{7,23}\xi_{7,78} + \xi_{7,24}\xi_{7,79}\right) - \xi_{7,381}\left(\xi_{7,21}\xi_{7,81} + \xi_{7,22}\xi_{7,80} - \xi_{7,23}\xi_{7,79} + \xi_{7,24}\xi_{7,78}\right) - \xi_{7,382}\left(\xi_{7,21}\xi_{7,78} - \xi_{7,22}\xi_{7,79} - \xi_{7,23}\xi_{7,80} - \xi_{7,24}\xi_{7,81}\right) + \xi_{7,383}\left(\xi_{7,21}\xi_{7,79} + \xi_{7,22}\xi_{7,78} + \xi_{7,23}\xi_{7,81} - \xi_{7,24}\xi_{7,80}\right)$$

$$\xi_{7,518} = \xi_{7,380}\left(\xi_{7,21}\xi_{7,81} + \xi_{7,22}\xi_{7,80} - \xi_{7,23}\xi_{7,79} + \xi_{7,24}\xi_{7,78}\right) + \xi_{7,381}\left(\xi_{7,21}\xi_{7,80} - \xi_{7,22}\xi_{7,81} + \xi_{7,23}\xi_{7,78} + \xi_{7,24}\xi_{7,79}\right) - \xi_{7,382}\left(\xi_{7,21}\xi_{7,79} + \xi_{7,22}\xi_{7,78} + \xi_{7,23}\xi_{7,81} - \xi_{7,24}\xi_{7,80}\right) - \xi_{7,383}\left(\xi_{7,21}\xi_{7,78} - \xi_{7,22}\xi_{7,79} - \xi_{7,23}\xi_{7,80} - \xi_{7,24}\xi_{7,81}\right)$$

$$\xi_{7,519} = \xi_{7,404}\left(\xi_{7,25}\xi_{7,82} - \xi_{7,26}\xi_{7,83} - \xi_{7,27}\xi_{7,84} - \xi_{7,28}\xi_{7,85}\right) + \xi_{7,405}\left(\xi_{7,25}\xi_{7,83} + \xi_{7,26}\xi_{7,82} + \xi_{7,27}\xi_{7,85} - \xi_{7,28}\xi_{7,84}\right) + \xi_{7,406}\left(\xi_{7,25}\xi_{7,84} - \xi_{7,26}\xi_{7,85} + \xi_{7,27}\xi_{7,82} + \xi_{7,28}\xi_{7,83}\right) + \xi_{7,407}\left(\xi_{7,25}\xi_{7,85} + \xi_{7,26}\xi_{7,84} - \xi_{7,27}\xi_{7,83} + \xi_{7,28}\xi_{7,82}\right)$$

$$\xi_{7,520} = \xi_{7,404}\left(\xi_{7,25}\xi_{7,83} + \xi_{7,26}\xi_{7,82} + \xi_{7,27}\xi_{7,85} - \xi_{7,28}\xi_{7,84}\right) - \xi_{7,405}\left(\xi_{7,25}\xi_{7,82} - \xi_{7,26}\xi_{7,83} - \xi_{7,27}\xi_{7,84} - \xi_{7,28}\xi_{7,85}\right) + \xi_{7,406}\left(\xi_{7,25}\xi_{7,85} + \xi_{7,26}\xi_{7,84} - \xi_{7,27}\xi_{7,83} + \xi_{7,28}\xi_{7,82}\right) - \xi_{7,407}\left(\xi_{7,25}\xi_{7,84} - \xi_{7,26}\xi_{7,85} + \xi_{7,27}\xi_{7,82} + \xi_{7,28}\xi_{7,83}\right)$$

$$\xi_{7,521} = \xi_{7,404}\left(\xi_{7,25}\xi_{7,84} - \xi_{7,26}\xi_{7,85} + \xi_{7,27}\xi_{7,82} + \xi_{7,28}\xi_{7,83}\right) - \xi_{7,405}\left(\xi_{7,25}\xi_{7,85} + \xi_{7,26}\xi_{7,84} - \xi_{7,27}\xi_{7,83} + \xi_{7,28}\xi_{7,82}\right) - \xi_{7,406}\left(\xi_{7,25}\xi_{7,82} - \xi_{7,26}\xi_{7,83} - \xi_{7,27}\xi_{7,84} - \xi_{7,28}\xi_{7,85}\right) + \xi_{7,407}\left(\xi_{7,25}\xi_{7,83} + \xi_{7,26}\xi_{7,82} + \xi_{7,27}\xi_{7,85} - \xi_{7,28}\xi_{7,84}\right)$$

$$\xi_{7,522} = \xi_{7,404}\left(\xi_{7,25}\xi_{7,85} + \xi_{7,26}\xi_{7,84} - \xi_{7,27}\xi_{7,83} + \xi_{7,28}\xi_{7,82}\right) + \xi_{7,405}\left(\xi_{7,25}\xi_{7,84} - \xi_{7,26}\xi_{7,85} + \xi_{7,27}\xi_{7,82} + \xi_{7,28}\xi_{7,83}\right) - \xi_{7,406}\left(\xi_{7,25}\xi_{7,83} + \xi_{7,26}\xi_{7,82} + \xi_{7,27}\xi_{7,85} - \xi_{7,28}\xi_{7,84}\right) - \xi_{7,407}\left(\xi_{7,25}\xi_{7,82} - \xi_{7,26}\xi_{7,83} - \xi_{7,27}\xi_{7,84} - \xi_{7,28}\xi_{7,85}\right)$$

$$\xi_{7,523} = \xi_{7,35} + \xi_{7,93}$$

$$\xi_{7,524} = \xi_{7,523} + \frac{\pi t\left(4 global_{shift} + 1\right)}{4}$$

$$\xi_{7,525} = \begin{cases} 0 & \text{for } \xi_{7,499}^2 + \xi_{7,502}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,502}, \xi_{7,499}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,526} = \begin{cases} 0 & \text{for } \xi_{7,500}^2 + \xi_{7,501}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,500}, \xi_{7,501}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,527} = 2\operatorname{atan}_2\left(\sqrt{\xi_{7,500}^2 + \xi_{7,501}^2}, \sqrt{\xi_{7,499}^2 + \xi_{7,502}^2}\right)$$

$$\xi_{7,528} = -\xi_{7,525} + \xi_{7,526}$$

$$\xi_{7,529} = -\xi_{7,525} - \xi_{7,526}$$

$$\xi_{7,530} = \xi_{7,524} + \xi_{7,525}$$

$$\xi_{7,531} = \begin{cases} 0 & \text{for } \xi_{7,503}^2 + \xi_{7,506}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,506}, \xi_{7,503}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,532} = \begin{cases} 0 & \text{for } \xi_{7,504}^2 + \xi_{7,505}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,504}, \xi_{7,505}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,533} = 2\operatorname{atan}_2\left(\sqrt{\xi_{7,504}^2 + \xi_{7,505}^2}, \sqrt{\xi_{7,503}^2 + \xi_{7,506}^2}\right)$$

$$\xi_{7,534} = -\xi_{7,531} + \xi_{7,532}$$

$$\xi_{7,535} = -\xi_{7,531} - \xi_{7,532}$$

$$\xi_{7,536} = \xi_{7,530} + \xi_{7,531}$$

$$\xi_{7,537} = \begin{cases} 0 & \text{for } \xi_{7,507}^2 + \xi_{7,510}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,510}, \xi_{7,507}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,538} = \begin{cases} 0 & \text{for } \xi_{7,508}^2 + \xi_{7,509}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,508}, \xi_{7,509}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,539} = 2\operatorname{atan}_2\left(\sqrt{\xi_{7,508}^2 + \xi_{7,509}^2}, \sqrt{\xi_{7,507}^2 + \xi_{7,510}^2}\right)$$

Exact shared-expression program 7

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{7,540} = -\xi_{7,537} + \xi_{7,538}$$

$$\xi_{7,541} = -\xi_{7,537} - \xi_{7,538}$$

$$\xi_{7,542} = \xi_{7,536} + \xi_{7,537}$$

$$\xi_{7,543} = \begin{cases} 0 & \text{for } \xi_{7,511}^2 + \xi_{7,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,514}, \xi_{7,511}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,544} = \begin{cases} 0 & \text{for } \xi_{7,512}^2 + \xi_{7,513}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,512}, \xi_{7,513}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,545} = 2\operatorname{atan}_2\left(\sqrt{\xi_{7,512}^2 + \xi_{7,513}^2}, \sqrt{\xi_{7,511}^2 + \xi_{7,514}^2}\right)$$

$$\xi_{7,546} = -\xi_{7,543} + \xi_{7,544}$$

$$\xi_{7,547} = -\xi_{7,543} - \xi_{7,544}$$

$$\xi_{7,548} = \xi_{7,542} + \xi_{7,543}$$

$$\xi_{7,549} = \begin{cases} 0 & \text{for } \xi_{7,104}^2 + \xi_{7,105}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{7,105}, \xi_{7,104}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,550} = \begin{cases} 0 & \text{for } \xi_{7,101}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{7,101}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,551} = 2\operatorname{atan}_2\left(\left|\xi_{7,101}\right|, \sqrt{\xi_{7,104}^2 + \xi_{7,105}^2}\right)$$

$$\xi_{7,552} = -\xi_{7,549} + \xi_{7,550}$$

$$\xi_{7,553} = -\xi_{7,549} - \xi_{7,550}$$

$$\xi_{7,554} = \xi_{7,548} + \xi_{7,549}$$

$$\xi_{7,555} = \begin{cases} 0 & \text{for } \xi_{7,99}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{7,99}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,556} = \begin{cases} 0 & \text{for } \xi_{7,100}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{7,100}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,557} = 2\operatorname{atan}_2\left(\left|\xi_{7,100}\right|, \left|\xi_{7,99}\right|\right)$$

$$\xi_{7,558} = -\xi_{7,555} + \xi_{7,556}$$

$$\xi_{7,559} = -\xi_{7,555} - \xi_{7,556}$$

$$\xi_{7,560} = \xi_{7,554} + \xi_{7,555}$$

$$\xi_{7,561} = \begin{cases} 0 & \text{for } \xi_{7,515}^2 + \xi_{7,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,518}, \xi_{7,515}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,562} = \begin{cases} 0 & \text{for } \xi_{7,516}^2 + \xi_{7,517}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,516}, \xi_{7,517}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,563} = 2\operatorname{atan}_2\left(\sqrt{\xi_{7,516}^2 + \xi_{7,517}^2}, \sqrt{\xi_{7,515}^2 + \xi_{7,518}^2}\right)$$

$$\xi_{7,564} = -\xi_{7,561} + \xi_{7,562}$$

$$\xi_{7,565} = -\xi_{7,561} - \xi_{7,562}$$

$$\xi_{7,566} = \xi_{7,560} + \xi_{7,561}$$

$$\xi_{7,567} = \begin{cases} 0 & \text{for } \xi_{7,519}^2 + \xi_{7,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,522}, \xi_{7,519}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,568} = \begin{cases} 0 & \text{for } \xi_{7,520}^2 + \xi_{7,521}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{7,520}, \xi_{7,521}\right) & \text{otherwise} \end{cases}$$

$$\xi_{7,569} = 2\operatorname{atan}_2\left(\sqrt{\xi_{7,520}^2 + \xi_{7,521}^2}, \sqrt{\xi_{7,519}^2 + \xi_{7,522}^2}\right)$$

$$\xi_{7,570} = -\xi_{7,567} + \xi_{7,568}$$

$$\xi_{7,571} = -\xi_{7,567} - \xi_{7,568}$$

$$\xi_{7,572} = \xi_{7,566} + \xi_{7,567}$$

$$\xi_{7,573} = -\frac{\xi_{7,527}}{2} - \frac{\xi_{7,533}}{2} - \frac{\xi_{7,539}}{2} - \frac{\xi_{7,545}}{2} - \frac{\xi_{7,551}}{2} - \frac{\xi_{7,557}}{2} - \frac{\xi_{7,563}}{2} - \frac{\xi_{7,569}}{2} + \xi_{7,572}$$

Derived parameters for program 7

Exact output expressions used by the native gate labels.

$$v_{7,0} = \frac{2\xi_{7,527}+2\xi_{7,573}+\pi\left(-\frac{\xi_{7,529}}{\pi}+\frac{\xi_{7,528}+\xi_{7,529}}{\pi}+\frac{1}{2}\right)-\frac{3\pi}{2}}{\pi}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,0} = \left\lfloor \frac{1}{2} - \frac{t}{2} \right\rfloor$$

$$\xi_{8,1} = -\frac{\pi \xi_{8,0}}{2} - \frac{\pi t}{4}$$

$$\xi_{8,2} = \xi_{8,1}$$

$$\xi_{8,3} = 1$$

$$\xi_{8,4} = 0$$

$$\xi_{8,5} = 0$$

$$\xi_{8,6} = 0$$

$$\xi_{8,7} = \begin{cases} -1 & \text{for } \xi_{8,2} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{8,8} = \begin{cases} 1 & \text{for } \xi_{8,7} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,9} = 0$$

$$\xi_{8,10} = 0$$

$$\xi_{8,11} = \begin{cases} 0 & \text{for } \xi_{8,7} = 1 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{8,12} = \cos\left(\frac{\pi \xi_{8,0}}{2}\right)$$

$$\xi_{8,13} = -\sin\left(\frac{\pi \xi_{8,0}}{2}\right)$$

$$\xi_{8,14} = \xi_{8,12}^2$$

$$\xi_{8,15} = \xi_{8,12}\xi_{8,13}$$

$$\xi_{8,16} = -\xi_{8,13}^2$$

$$\xi_{8,17} = \xi_{8,12}\xi_{8,14} - \xi_{8,13}\xi_{8,16}$$

$$\xi_{8,18} = \xi_{8,12}\xi_{8,15} - \xi_{8,13}\xi_{8,15}$$

$$\xi_{8,19} = \xi_{8,12}\xi_{8,15} + \xi_{8,13}\xi_{8,15}$$

$$\xi_{8,20} = \xi_{8,12}\xi_{8,16} + \xi_{8,13}\xi_{8,14}$$

$$\xi_{8,21} = -\xi_{8,10}\left(\xi_{8,17}\xi_{8,5} - \xi_{8,18}\xi_{8,6} + \xi_{8,19}\xi_{8,3} + \xi_{8,20}\xi_{8,4}\right) - \xi_{8,11}\left(\xi_{8,17}\xi_{8,6} + \xi_{8,18}\xi_{8,5} - \xi_{8,19}\xi_{8,4} + \xi_{8,20}\xi_{8,3}\right) + \xi_{8,8}\left(\xi_{8,17}\xi_{8,3} - \xi_{8,18}\xi_{8,4} - \xi_{8,19}\xi_{8,5} - \xi_{8,20}\xi_{8,6}\right) - \xi_{8,9}\left(\xi_{8,17}\xi_{8,4} + \xi_{8,18}\xi_{8,3} + \xi_{8,19}\xi_{8,6} - \xi_{8,20}\xi_{8,5}\right)$$

$$\xi_{8,22} = -\xi_{8,10}\left(\xi_{8,17}\xi_{8,6} + \xi_{8,18}\xi_{8,5} - \xi_{8,19}\xi_{8,4} + \xi_{8,20}\xi_{8,3}\right) + \xi_{8,11}\left(\xi_{8,17}\xi_{8,5} - \xi_{8,18}\xi_{8,6} + \xi_{8,19}\xi_{8,3} + \xi_{8,20}\xi_{8,4}\right) + \xi_{8,8}\left(\xi_{8,17}\xi_{8,4} + \xi_{8,18}\xi_{8,3} + \xi_{8,19}\xi_{8,6} - \xi_{8,20}\xi_{8,5}\right) + \xi_{8,9}\left(\xi_{8,17}\xi_{8,3} - \xi_{8,18}\xi_{8,4} - \xi_{8,19}\xi_{8,5} - \xi_{8,20}\xi_{8,6}\right)$$

$$\xi_{8,23} = \xi_{8,10}\left(\xi_{8,17}\xi_{8,3} - \xi_{8,18}\xi_{8,4} - \xi_{8,19}\xi_{8,5} - \xi_{8,20}\xi_{8,6}\right) - \xi_{8,11}\left(\xi_{8,17}\xi_{8,4} + \xi_{8,18}\xi_{8,3} + \xi_{8,19}\xi_{8,6} - \xi_{8,20}\xi_{8,5}\right) + \xi_{8,8}\left(\xi_{8,17}\xi_{8,5} - \xi_{8,18}\xi_{8,6} + \xi_{8,19}\xi_{8,3} + \xi_{8,20}\xi_{8,4}\right) + \xi_{8,9}\left(\xi_{8,17}\xi_{8,6} + \xi_{8,18}\xi_{8,5} - \xi_{8,19}\xi_{8,4} + \xi_{8,20}\xi_{8,3}\right)$$

$$\xi_{8,24} = \xi_{8,10}\left(\xi_{8,17}\xi_{8,4} + \xi_{8,18}\xi_{8,3} + \xi_{8,19}\xi_{8,6} - \xi_{8,20}\xi_{8,5}\right) + \xi_{8,11}\left(\xi_{8,17}\xi_{8,3} - \xi_{8,18}\xi_{8,4} - \xi_{8,19}\xi_{8,5} - \xi_{8,20}\xi_{8,6}\right) + \xi_{8,8}\left(\xi_{8,17}\xi_{8,6} + \xi_{8,18}\xi_{8,5} - \xi_{8,19}\xi_{8,4} + \xi_{8,20}\xi_{8,3}\right) - \xi_{8,9}\left(\xi_{8,17}\xi_{8,5} - \xi_{8,18}\xi_{8,6} + \xi_{8,19}\xi_{8,3} + \xi_{8,20}\xi_{8,4}\right)$$

$$\xi_{8,25} = \xi_{8,17}\xi_{8,3} - \xi_{8,18}\xi_{8,4} - \xi_{8,19}\xi_{8,5} - \xi_{8,20}\xi_{8,6}$$

$$\xi_{8,26} = \xi_{8,17}\xi_{8,4} + \xi_{8,18}\xi_{8,3} + \xi_{8,19}\xi_{8,6} - \xi_{8,20}\xi_{8,5}$$

$$\xi_{8,27} = \xi_{8,17}\xi_{8,5} - \xi_{8,18}\xi_{8,6} + \xi_{8,19}\xi_{8,3} + \xi_{8,20}\xi_{8,4}$$

$$\xi_{8,28} = \xi_{8,17}\xi_{8,6} + \xi_{8,18}\xi_{8,5} - \xi_{8,19}\xi_{8,4} + \xi_{8,20}\xi_{8,3}$$

$$\xi_{8,29} = -\xi_{8,10}\xi_{8,5} - \xi_{8,11}\xi_{8,6} + \xi_{8,3}\xi_{8,8} - \xi_{8,4}\xi_{8,9}$$

$$\xi_{8,30} = \xi_{8,10}\xi_{8,6} - \xi_{8,11}\xi_{8,5} - \xi_{8,3}\xi_{8,9} - \xi_{8,4}\xi_{8,8}$$

$$\xi_{8,31} = -\xi_{8,10}\xi_{8,3} + \xi_{8,11}\xi_{8,4} - \xi_{8,5}\xi_{8,8} - \xi_{8,6}\xi_{8,9}$$

$$\xi_{8,32} = -\xi_{8,10}\xi_{8,4} - \xi_{8,11}\xi_{8,3} + \xi_{8,5}\xi_{8,9} - \xi_{8,6}\xi_{8,8}$$

$$\xi_{8,33} = \xi_{8,2}\xi_{8,7}$$

$$\xi_{8,34} = \xi_{8,2}\xi_{8,7}^2$$

$$\xi_{8,35} = -\frac{3\pi \xi_{8,0}}{2}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,36} = \begin{cases} 0 & \text{for } \left(\xi_{8,33} > \frac{\pi}{8} \vee \xi_{8,34} < 0\right) \wedge \xi_{8,33} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{8,37} = \frac{\pi}{8}$$

$$\xi_{8,38} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{8,33} > \frac{\pi}{8} \wedge \xi_{8,34} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{8,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,39} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{8,33} > \frac{\pi}{8} \wedge \xi_{8,34} < 0 \\ \frac{1}{2} & \text{for } \xi_{8,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,40} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{8,33} > \frac{\pi}{8} \vee \xi_{8,34} < 0\right) \wedge \xi_{8,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,41} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{8,33} > \frac{\pi}{8} \vee \xi_{8,34} < 0\right) \wedge \xi_{8,33} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{8,42} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{8,33} > \frac{\pi}{8} \vee \xi_{8,34} < 0\right) \wedge \xi_{8,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,43} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{8,33} > \frac{\pi}{8} \vee \xi_{8,34} < 0\right) \wedge \xi_{8,33} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{8,44} = \begin{cases} \frac{1}{2} & \text{for } \xi_{8,33} > \frac{\pi}{8} \wedge \xi_{8,34} < 0 \\ -\frac{1}{2} & \text{for } \xi_{8,33} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,45} = \left\lfloor \frac{2\xi_{8,33} - 2\xi_{8,36}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{8,46} = \left\lfloor \frac{2\xi_{8,33} - 2\xi_{8,37}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{8,47} = \left\lfloor \frac{2\xi_{8,34} - 2\xi_{8,38}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{8,48} = \xi_{8,33} - \xi_{8,36} - \frac{\pi\xi_{8,45}}{2}$$

$$\xi_{8,49} = \xi_{8,33} - \xi_{8,37} - \frac{\pi\xi_{8,46}}{2}$$

$$\xi_{8,50} = \xi_{8,34} - \xi_{8,38} - \frac{\pi\xi_{8,47}}{2}$$

$$\xi_{8,51} = \begin{cases} \xi_{8,48} & \text{for } (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,48}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \leq |\xi_{8,50}|) \\ \xi_{8,49} & \text{for } (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \leq |\xi_{8,49}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \\ \xi_{8,50} & \text{otherwise} \end{cases}$$

$$\xi_{8,52} = \begin{cases} \xi_{8,49} & \text{for } |\xi_{8,48}| \geq |\xi_{8,49}| \wedge |\xi_{8,49}| \geq |\xi_{8,50}| \\ \xi_{8,50} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,49}| \leq |\xi_{8,50}| \\ \xi_{8,48} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,48}| \leq |\xi_{8,49}| \\ \xi_{8,50} & \text{for } |\xi_{8,49}| \geq |\xi_{8,50}| \wedge |\xi_{8,48}| \leq |\xi_{8,50}| \\ \xi_{8,48} & \text{for } |\xi_{8,48}| \geq |\xi_{8,49}| \wedge |\xi_{8,48}| \leq |\xi_{8,50}| \\ \xi_{8,49} & \text{otherwise} \end{cases}$$

$$\xi_{8,53} = \begin{cases} \xi_{8,50} & \text{for } |\xi_{8,48}| \geq |\xi_{8,49}| \wedge |\xi_{8,49}| \geq |\xi_{8,50}| \\ \xi_{8,49} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,49}| \leq |\xi_{8,50}| \\ \xi_{8,50} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,48}| \leq |\xi_{8,49}| \\ \xi_{8,48} & \text{for } |\xi_{8,49}| \geq |\xi_{8,50}| \wedge |\xi_{8,48}| \leq |\xi_{8,50}| \\ \xi_{8,49} & \text{for } |\xi_{8,48}| \geq |\xi_{8,49}| \wedge |\xi_{8,48}| \leq |\xi_{8,50}| \\ \xi_{8,48} & \text{otherwise} \end{cases}$$

$$\xi_{8,54} = \begin{cases} 1 & \text{for } |\xi_{8,48}| \geq |\xi_{8,49}| \wedge |\xi_{8,49}| \geq |\xi_{8,50}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,49}| \leq |\xi_{8,50}| \\ 0 & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,48}| \leq |\xi_{8,49}| \\ \frac{1}{2} & \text{for } |\xi_{8,48}| \leq |\xi_{8,50}| \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{8,55} = \begin{cases} 0 & \text{for } |\xi_{8,48}| \geq |\xi_{8,49}| \wedge |\xi_{8,49}| \geq |\xi_{8,50}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,49}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \leq |\xi_{8,49}| \vee |\xi_{8,49}| \leq |\xi_{8,50}|) \\ \frac{1}{2} & \text{for } |\xi_{8,48}| \leq |\xi_{8,50}| \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,56} = \begin{cases} 0 & \text{for } (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,48}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \leq |\xi_{8,50}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,48}| \leq |\xi_{8,49}| \\ \frac{1}{2} & \text{for } |\xi_{8,48}| \leq |\xi_{8,50}| \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{8,57} = \begin{cases} 0 & \text{for } (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,48}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \leq |\xi_{8,50}| \vee |\xi_{8,48}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,49}| \vee |\xi_{8,49}| \leq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \geq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \leq |\xi_{8,50}| \vee |\xi_{8,48}| \geq |\xi_{8,50}|) \wedge (|\xi_{8,48}| \geq |\xi_{8,50}| \vee |\xi_{8,49}| \leq |\xi_{8,50}| \vee |\xi_{8,48}| \leq |\xi_{8,50}|) \\ \frac{1}{2} & \text{for } |\xi_{8,48}| \geq |\xi_{8,50}| \wedge |\xi_{8,48}| \leq |\xi_{8,50}| \\ -\frac{1}{2} & \text{for } |\xi_{8,48}| \geq |\xi_{8,48}| \wedge |\xi_{8,48}| \leq |\xi_{8,50}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,58} = \begin{cases} -1 & \text{for } \xi_{8,51} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{8,59} = \begin{cases} -1 & \text{for } \xi_{8,52} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{8,60} = \begin{cases} 1 & \text{for } \xi_{8,58} = 1 \wedge \xi_{8,59} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,61} = \begin{cases} 0 & \text{for } \xi_{8,58} = 1 \wedge \xi_{8,59} = 1 \\ 1 & \text{for } \xi_{8,58} = 1 \wedge \xi_{8,59} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,62} = \begin{cases} 0 & \text{for } \xi_{8,58} = 1 \wedge (\xi_{8,58} = 1 \vee \xi_{8,59} = -1) \wedge (\xi_{8,58} = 1 \vee \xi_{8,59} = 1) \wedge (\xi_{8,59} = -1 \vee \xi_{8,59} = 1) \\ 1 & \text{for } \xi_{8,58} = -1 \wedge \xi_{8,59} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,63} = \begin{cases} 0 & \text{for } (\xi_{8,58} = -1 \vee \xi_{8,58} = 1) \wedge (\xi_{8,58} = 1 \vee \xi_{8,59} = 1) \wedge (\xi_{8,59} = -1 \vee \xi_{8,59} = 1) \wedge (\xi_{8,58} = -1 \vee \xi_{8,58} = 1 \vee \xi_{8,59} = -1) \wedge (\xi_{8,58} = -1 \vee \xi_{8,58} = 1 \vee \xi_{8,59} = 1) \wedge (\xi_{8,58} = -1 \vee \xi_{8,59} = -1 \vee \xi_{8,59} = 1) \wedge (\xi_{8,58} = 1 \vee \xi_{8,59} = -1 \vee \xi_{8,59} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{8,64} = \cos\left(\frac{\pi\xi_{8,45}}{2}\right)$$

$$\xi_{8,65} = -\sin\left(\frac{\pi\xi_{8,45}}{2}\right)$$

$$\xi_{8,66} = \cos\left(\frac{\pi\xi_{8,46}}{2}\right)$$

$$\xi_{8,67} = -\sin\left(\frac{\pi\xi_{8,46}}{2}\right)$$

$$\xi_{8,68} = \xi_{8,64}\xi_{8,66}$$

$$\xi_{8,69} = \xi_{8,65}\xi_{8,66}$$

$$\xi_{8,70} = \xi_{8,64}\xi_{8,67}$$

$$\xi_{8,71} = -\xi_{8,65}\xi_{8,67}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,72} = \cos\left(\frac{\pi\xi_{8,47}}{2}\right)$$

$$\xi_{8,73} = -\sin\left(\frac{\pi\xi_{8,47}}{2}\right)$$

$$\xi_{8,74} = \xi_{8,68}\xi_{8,72} - \xi_{8,71}\xi_{8,73}$$

$$\xi_{8,75} = \xi_{8,69}\xi_{8,72} - \xi_{8,70}\xi_{8,73}$$

$$\xi_{8,76} = \xi_{8,69}\xi_{8,73} + \xi_{8,70}\xi_{8,72}$$

$$\xi_{8,77} = \xi_{8,68}\xi_{8,73} + \xi_{8,71}\xi_{8,72}$$

$$\xi_{8,78} = \xi_{8,60}\left(\xi_{8,54}\xi_{8,74} - \xi_{8,55}\xi_{8,75} - \xi_{8,56}\xi_{8,76} - \xi_{8,57}\xi_{8,77}\right) - \xi_{8,61}\left(\xi_{8,54}\xi_{8,75} + \xi_{8,55}\xi_{8,74} - \xi_{8,56}\xi_{8,77} + \xi_{8,57}\xi_{8,76}\right) - \xi_{8,62}\left(\xi_{8,54}\xi_{8,76} + \xi_{8,55}\xi_{8,77} + \xi_{8,56}\xi_{8,74} - \xi_{8,57}\xi_{8,75}\right) - \xi_{8,63}\left(\xi_{8,54}\xi_{8,77} - \xi_{8,55}\xi_{8,76} + \xi_{8,56}\xi_{8,75} + \xi_{8,57}\xi_{8,74}\right)$$

$$\xi_{8,79} = \xi_{8,60}\left(\xi_{8,54}\xi_{8,75} + \xi_{8,55}\xi_{8,74} - \xi_{8,56}\xi_{8,77} + \xi_{8,57}\xi_{8,76}\right) + \xi_{8,61}\left(\xi_{8,54}\xi_{8,74} - \xi_{8,55}\xi_{8,75} - \xi_{8,56}\xi_{8,76} - \xi_{8,57}\xi_{8,77}\right) - \xi_{8,62}\left(\xi_{8,54}\xi_{8,77} - \xi_{8,55}\xi_{8,76} + \xi_{8,56}\xi_{8,75} + \xi_{8,57}\xi_{8,74}\right) + \xi_{8,63}\left(\xi_{8,54}\xi_{8,76} + \xi_{8,55}\xi_{8,77} + \xi_{8,56}\xi_{8,74} - \xi_{8,57}\xi_{8,75}\right)$$

$$\xi_{8,80} = \xi_{8,60}\left(\xi_{8,54}\xi_{8,76} + \xi_{8,55}\xi_{8,77} + \xi_{8,56}\xi_{8,74} - \xi_{8,57}\xi_{8,75}\right) + \xi_{8,61}\left(\xi_{8,54}\xi_{8,77} - \xi_{8,55}\xi_{8,76} + \xi_{8,56}\xi_{8,75} + \xi_{8,57}\xi_{8,74}\right) + \xi_{8,62}\left(\xi_{8,54}\xi_{8,74} - \xi_{8,55}\xi_{8,75} - \xi_{8,56}\xi_{8,76} - \xi_{8,57}\xi_{8,77}\right) - \xi_{8,63}\left(\xi_{8,54}\xi_{8,75} + \xi_{8,55}\xi_{8,74} - \xi_{8,56}\xi_{8,77} + \xi_{8,57}\xi_{8,76}\right)$$

$$\xi_{8,81} = \xi_{8,60}\left(\xi_{8,54}\xi_{8,77} - \xi_{8,55}\xi_{8,76} + \xi_{8,56}\xi_{8,75} + \xi_{8,57}\xi_{8,74}\right) - \xi_{8,61}\left(\xi_{8,54}\xi_{8,76} + \xi_{8,55}\xi_{8,77} + \xi_{8,56}\xi_{8,74} - \xi_{8,57}\xi_{8,75}\right) + \xi_{8,62}\left(\xi_{8,54}\xi_{8,75} + \xi_{8,55}\xi_{8,74} - \xi_{8,56}\xi_{8,77} + \xi_{8,57}\xi_{8,76}\right) + \xi_{8,63}\left(\xi_{8,54}\xi_{8,74} - \xi_{8,55}\xi_{8,75} - \xi_{8,56}\xi_{8,76} - \xi_{8,57}\xi_{8,77}\right)$$

$$\xi_{8,82} = \xi_{8,54}\xi_{8,74} - \xi_{8,55}\xi_{8,75} - \xi_{8,56}\xi_{8,76} - \xi_{8,57}\xi_{8,77}$$

$$\xi_{8,83} = \xi_{8,54}\xi_{8,75} + \xi_{8,55}\xi_{8,74} - \xi_{8,56}\xi_{8,77} + \xi_{8,57}\xi_{8,76}$$

$$\xi_{8,84} = \xi_{8,54}\xi_{8,76} + \xi_{8,55}\xi_{8,77} + \xi_{8,56}\xi_{8,74} - \xi_{8,57}\xi_{8,75}$$

$$\xi_{8,85} = \xi_{8,54}\xi_{8,77} - \xi_{8,55}\xi_{8,76} + \xi_{8,56}\xi_{8,75} + \xi_{8,57}\xi_{8,74}$$

$$\xi_{8,86} = \xi_{8,54}\xi_{8,60} - \xi_{8,55}\xi_{8,61} - \xi_{8,56}\xi_{8,62} - \xi_{8,57}\xi_{8,63}$$

$$\xi_{8,87} = -\xi_{8,54}\xi_{8,61} - \xi_{8,55}\xi_{8,60} - \xi_{8,56}\xi_{8,63} + \xi_{8,57}\xi_{8,62}$$

$$\xi_{8,88} = -\xi_{8,54}\xi_{8,62} + \xi_{8,55}\xi_{8,63} - \xi_{8,56}\xi_{8,60} - \xi_{8,57}\xi_{8,61}$$

$$\xi_{8,89} = -\xi_{8,54}\xi_{8,63} - \xi_{8,55}\xi_{8,62} + \xi_{8,56}\xi_{8,61} - \xi_{8,57}\xi_{8,60}$$

$$\xi_{8,90} = \xi_{8,51}\xi_{8,58}$$

$$\xi_{8,91} = \xi_{8,52}\xi_{8,59}$$

$$\xi_{8,92} = \xi_{8,53}\xi_{8,58}\xi_{8,59}$$

$$\xi_{8,93} = -\frac{\pi(\xi_{8,45}+\xi_{8,46}+\xi_{8,47})}{2}$$

$$\xi_{8,94} = \cos\left(2\xi_{8,90}\right)$$

$$\xi_{8,95} = \cos\left(2\xi_{8,91}\right)$$

$$\xi_{8,96} = \cos\left(2\xi_{8,92}\right)$$

$$\xi_{8,97} = \sqrt{\xi_{8,94}^2 - 2\xi_{8,94}\xi_{8,95}\xi_{8,96} + \xi_{8,95}^2 + \xi_{8,96}^2 - 1}$$

$$\xi_{8,98} = \xi_{8,94} - \xi_{8,95} + \xi_{8,96}$$

$$\xi_{8,99} = \sqrt{-\frac{\xi_{8,97}}{2} + \frac{\xi_{8,98}}{2} + \frac{1}{2}}$$

$$\xi_{8,100} = \sqrt{\frac{\xi_{8,97}}{2} - \frac{\xi_{8,98}}{2} + \frac{1}{2}}$$

$$\xi_{8,101} = \sqrt{-\frac{\xi_{8,97}}{2} - \frac{\xi_{8,98}}{2} + \frac{1}{2}}$$

$$\xi_{8,102} = \frac{(1-\xi_{8,95})(\xi_{8,94}+1)(\xi_{8,96}+1)}{2}$$

$$\xi_{8,103} = \begin{cases} -1 & \text{for } \xi_{8,92} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{8,104} = \begin{cases} 0 & \text{for } \xi_{8,99} = 0 \\ \frac{\sqrt{\xi_{8,102}}\xi_{8,103}}{\xi_{8,99}} & \text{otherwise} \end{cases}$$

$$\xi_{8,105} = \begin{cases} 1 & \text{for } \xi_{8,99} = 0 \\ \frac{\sqrt{\xi_{8,94}\xi_{8,95}\xi_{8,96}}}{\xi_{8,99}} & \text{otherwise} \end{cases}$$

$$\xi_{8,106} = 2\xi_{8,100}\xi_{8,101}\xi_{8,105}\xi_{8,99}$$

$$\xi_{8,107} = \sqrt{2}\xi_{8,105}\left(\xi_{8,101}\xi_{8,99}^2 - \xi_{8,101}\right)$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,108} = 2\xi_{8,100}\xi_{8,101}\xi_{8,104}\xi_{8,99}$$

$$\xi_{8,109} = \sqrt{2}\left(2\xi_{8,101}\xi_{8,104}\xi_{8,99}^2 - \xi_{8,101}\xi_{8,104}\right)$$

$$\xi_{8,110} = \sqrt{2}\xi_{8,100}\xi_{8,104}\xi_{8,105}\xi_{8,99}$$

$$\xi_{8,111} = \sqrt{2}\xi_{8,100}\xi_{8,99}\left(-\xi_{8,101}^2 + \xi_{8,104}^2\right)$$

$$\xi_{8,112} = -\xi_{8,101}^2\xi_{8,99}^2 + \xi_{8,101}^2 + \xi_{8,104}^2\xi_{8,99}$$

$$\xi_{8,113} = \xi_{8,101}^2\xi_{8,99}^2 - \xi_{8,104}^2\xi_{8,99}^2 + \xi_{8,104}^2$$

$$\xi_{8,114} = 2\xi_{8,105}\left(\xi_{8,104}\xi_{8,99}^2 - \frac{\xi_{8,104}}{2}\right)$$

$$\xi_{8,115} = -\sqrt{2}\xi_{8,100}\xi_{8,104}\xi_{8,105}\xi_{8,99}$$

$$\xi_{8,116} = -\xi_{8,104}\xi_{8,105}$$

$$\xi_{8,117} = \xi_{8,101}^2\xi_{8,99}^2 - \xi_{8,101}^2 + \xi_{8,104}^2\xi_{8,99}^2 - \xi_{8,104}^2 - \xi_{8,99}^2 + 1$$

$$\xi_{8,118} = -\sqrt{2}\left(\xi_{8,100}\xi_{8,101}^2\xi_{8,99} + \xi_{8,100}\xi_{8,104}^2\xi_{8,99} - \xi_{8,100}\xi_{8,99}\right)$$

$$\xi_{8,119} = \sqrt{2}\xi_{8,101}\xi_{8,105}\xi_{8,99}^2$$

$$\xi_{8,120} = \xi_{8,110}^2 - \xi_{8,113}^2 - \xi_{8,116}^2 - \xi_{8,119}^2$$

$$\xi_{8,121} = \xi_{8,110}\xi_{8,111} - \xi_{8,113}\xi_{8,114} - \xi_{8,116}\xi_{8,117}$$

$$\xi_{8,122} = \xi_{8,110}\xi_{8,112} - \xi_{8,113}\xi_{8,115} - \xi_{8,116}\xi_{8,118}$$

$$\xi_{8,123} = \xi_{8,106}^2 + \xi_{8,108}^2 + \xi_{8,111}^2 - \xi_{8,114}^2 - \xi_{8,117}^2$$

$$\xi_{8,124} = \xi_{8,106}\xi_{8,107} + \xi_{8,108}\xi_{8,109} + \xi_{8,111}\xi_{8,112} - \xi_{8,114}\xi_{8,115} - \xi_{8,117}\xi_{8,118}$$

$$\xi_{8,125} = \xi_{8,107}^2 + \xi_{8,109}^2 + \xi_{8,112}^2 - \xi_{8,115}^2 - \xi_{8,118}^2$$

$$\xi_{8,126} = \sin\left(2\xi_{8,91}\right)\sin\left(2\xi_{8,92}\right)\cos\left(2\xi_{8,90}\right)$$

$$\xi_{8,127} = \sin\left(2\xi_{8,90}\right)\sin\left(2\xi_{8,92}\right)\cos\left(2\xi_{8,91}\right)$$

$$\xi_{8,128} = \sin\left(2\xi_{8,90}\right)\sin\left(2\xi_{8,91}\right)\cos\left(2\xi_{8,92}\right)$$

$$\xi_{8,129} = \sin\left(2\xi_{8,90}\right)\cos\left(2\xi_{8,91}\right)\cos\left(2\xi_{8,92}\right)$$

$$\xi_{8,130} = \sin\left(2\xi_{8,91}\right)\cos\left(2\xi_{8,90}\right)\cos\left(2\xi_{8,92}\right)$$

$$\xi_{8,131} = \sin\left(2\xi_{8,92}\right)\cos\left(2\xi_{8,90}\right)\cos\left(2\xi_{8,91}\right)$$

$$\xi_{8,132} = \xi_{8,126}^2 - \xi_{8,129}^2$$

$$\xi_{8,133} = \xi_{8,127}^2 - \xi_{8,130}^2$$

$$\xi_{8,134} = \xi_{8,128}^2 - \xi_{8,131}^2$$

$$\xi_{8,135} = \frac{\xi_{8,120} - \xi_{8,133}}{\xi_{8,132} - \xi_{8,133}}$$

$$\xi_{8,136} = \frac{\xi_{8,121}}{\xi_{8,132} - \xi_{8,133}}$$

$$\xi_{8,137} = \frac{\xi_{8,122}}{\xi_{8,132} - \xi_{8,133}}$$

$$\xi_{8,138} = \frac{\xi_{8,123} - \xi_{8,133}}{\xi_{8,132} - \xi_{8,133}}$$

$$\xi_{8,139} = \frac{\xi_{8,124}}{\xi_{8,132} - \xi_{8,133}}$$

$$\xi_{8,140} = \frac{\xi_{8,125} - \xi_{8,133}}{\xi_{8,132} - \xi_{8,133}}$$

$$\xi_{8,141} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,135} & \text{otherwise} \end{cases}$$

$$\xi_{8,142} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,136} & \text{otherwise} \end{cases}$$

$$\xi_{8,143} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,137} & \text{otherwise} \end{cases}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,144} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,138} & \text{otherwise} \end{cases}$$

$$\xi_{8,145} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,139} & \text{otherwise} \end{cases}$$

$$\xi_{8,146} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,140} & \text{otherwise} \end{cases}$$

$$\xi_{8,147} = \frac{\xi_{8,120} - \xi_{8,134}}{\xi_{8,132} - \xi_{8,134}}$$

$$\xi_{8,148} = \frac{\xi_{8,121}}{\xi_{8,132} - \xi_{8,134}}$$

$$\xi_{8,149} = \frac{\xi_{8,122}}{\xi_{8,132} - \xi_{8,134}}$$

$$\xi_{8,150} = \frac{\xi_{8,123} - \xi_{8,134}}{\xi_{8,132} - \xi_{8,134}}$$

$$\xi_{8,151} = \frac{\xi_{8,124}}{\xi_{8,132} - \xi_{8,134}}$$

$$\xi_{8,152} = \frac{\xi_{8,125} - \xi_{8,134}}{\xi_{8,132} - \xi_{8,134}}$$

$$\xi_{8,153} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,134} \\ \xi_{8,147} & \text{otherwise} \end{cases}$$

$$\xi_{8,154} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,134} \\ \xi_{8,148} & \text{otherwise} \end{cases}$$

$$\xi_{8,155} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,134} \\ \xi_{8,149} & \text{otherwise} \end{cases}$$

$$\xi_{8,156} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,134} \\ \xi_{8,150} & \text{otherwise} \end{cases}$$

$$\xi_{8,157} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,134} \\ \xi_{8,151} & \text{otherwise} \end{cases}$$

$$\xi_{8,158} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,134} \\ \xi_{8,152} & \text{otherwise} \end{cases}$$

$$\xi_{8,159} = \xi_{8,141}\xi_{8,153} + \xi_{8,142}\xi_{8,154} + \xi_{8,143}\xi_{8,155}$$

$$\xi_{8,160} = \xi_{8,141}\xi_{8,154} + \xi_{8,142}\xi_{8,156} + \xi_{8,143}\xi_{8,157}$$

$$\xi_{8,161} = \xi_{8,141}\xi_{8,155} + \xi_{8,142}\xi_{8,157} + \xi_{8,143}\xi_{8,158}$$

$$\xi_{8,162} = \xi_{8,142}\xi_{8,153} + \xi_{8,144}\xi_{8,154} + \xi_{8,145}\xi_{8,155}$$

$$\xi_{8,163} = \xi_{8,142}\xi_{8,154} + \xi_{8,144}\xi_{8,156} + \xi_{8,145}\xi_{8,157}$$

$$\xi_{8,164} = \xi_{8,142}\xi_{8,155} + \xi_{8,144}\xi_{8,157} + \xi_{8,145}\xi_{8,158}$$

$$\xi_{8,165} = \xi_{8,143}\xi_{8,153} + \xi_{8,145}\xi_{8,154} + \xi_{8,146}\xi_{8,155}$$

$$\xi_{8,166} = \xi_{8,143}\xi_{8,154} + \xi_{8,145}\xi_{8,156} + \xi_{8,146}\xi_{8,157}$$

$$\xi_{8,167} = \xi_{8,143}\xi_{8,155} + \xi_{8,145}\xi_{8,157} + \xi_{8,146}\xi_{8,158}$$

$$\xi_{8,168} = \sqrt{\xi_{8,159}}$$

$$\xi_{8,169} = \frac{\xi_{8,159}}{\xi_{8,168}}$$

$$\xi_{8,170} = \frac{\xi_{8,162}}{\xi_{8,168}}$$

$$\xi_{8,171} = \frac{\xi_{8,165}}{\xi_{8,168}}$$

$$\xi_{8,172} = \sqrt{\xi_{8,163}}$$

$$\xi_{8,173} = \frac{\xi_{8,160}}{\xi_{8,172}}$$

$$\xi_{8,174} = \frac{\xi_{8,163}}{\xi_{8,172}}$$

$$\xi_{8,175} = \frac{\xi_{8,166}}{\xi_{8,172}}$$

$$\xi_{8,176} = \sqrt{\xi_{8,167}}$$

$$\xi_{8,177} = \frac{\xi_{8,161}}{\xi_{8,176}}$$

$$\xi_{8,178} = \frac{\xi_{8,164}}{\xi_{8,176}}$$

$$\xi_{8,179} = \frac{\xi_{8,167}}{\xi_{8,176}}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,180} = \begin{cases} \xi_{8,169} & \text{for } \xi_{8,159} \geq \xi_{8,163} \wedge \xi_{8,159} \geq \xi_{8,167} \\ \xi_{8,173} & \text{for } \xi_{8,163} \geq \xi_{8,167} \wedge \xi_{8,159} \leq \xi_{8,163} \\ \xi_{8,177} & \text{otherwise} \end{cases}$$

$$\xi_{8,181} = \begin{cases} \xi_{8,170} & \text{for } \xi_{8,159} \geq \xi_{8,163} \wedge \xi_{8,159} \geq \xi_{8,167} \\ \xi_{8,174} & \text{for } \xi_{8,163} \geq \xi_{8,167} \wedge \xi_{8,159} \leq \xi_{8,163} \\ \xi_{8,178} & \text{otherwise} \end{cases}$$

$$\xi_{8,182} = \begin{cases} \xi_{8,171} & \text{for } \xi_{8,159} \geq \xi_{8,163} \wedge \xi_{8,159} \geq \xi_{8,167} \\ \xi_{8,175} & \text{for } \xi_{8,163} \geq \xi_{8,167} \wedge \xi_{8,159} \leq \xi_{8,163} \\ \xi_{8,179} & \text{otherwise} \end{cases}$$

$$\xi_{8,183} = \frac{\xi_{8,120} - \xi_{8,132}}{-\xi_{8,132} + \xi_{8,133}}$$

$$\xi_{8,184} = \frac{\xi_{8,121}}{-\xi_{8,132} + \xi_{8,133}}$$

$$\xi_{8,185} = \frac{\xi_{8,122}}{-\xi_{8,132} + \xi_{8,133}}$$

$$\xi_{8,186} = \frac{\xi_{8,123} - \xi_{8,132}}{-\xi_{8,132} + \xi_{8,133}}$$

$$\xi_{8,187} = \frac{\xi_{8,124}}{-\xi_{8,132} + \xi_{8,133}}$$

$$\xi_{8,188} = \frac{\xi_{8,125} - \xi_{8,132}}{-\xi_{8,132} + \xi_{8,133}}$$

$$\xi_{8,189} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,183} & \text{otherwise} \end{cases}$$

$$\xi_{8,190} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,184} & \text{otherwise} \end{cases}$$

$$\xi_{8,191} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,185} & \text{otherwise} \end{cases}$$

$$\xi_{8,192} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,186} & \text{otherwise} \end{cases}$$

$$\xi_{8,193} = \begin{cases} 0 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,187} & \text{otherwise} \end{cases}$$

$$\xi_{8,194} = \begin{cases} 1 & \text{for } \xi_{8,132} = \xi_{8,133} \\ \xi_{8,188} & \text{otherwise} \end{cases}$$

$$\xi_{8,195} = \frac{\xi_{8,120} - \xi_{8,134}}{\xi_{8,133} - \xi_{8,134}}$$

$$\xi_{8,196} = \frac{\xi_{8,121}}{\xi_{8,133} - \xi_{8,134}}$$

$$\xi_{8,197} = \frac{\xi_{8,122}}{\xi_{8,133} - \xi_{8,134}}$$

$$\xi_{8,198} = \frac{\xi_{8,123} - \xi_{8,134}}{\xi_{8,133} - \xi_{8,134}}$$

$$\xi_{8,199} = \frac{\xi_{8,124}}{\xi_{8,133} - \xi_{8,134}}$$

$$\xi_{8,200} = \frac{\xi_{8,125} - \xi_{8,134}}{\xi_{8,133} - \xi_{8,134}}$$

$$\xi_{8,201} = \begin{cases} 1 & \text{for } \xi_{8,133} = \xi_{8,134} \\ \xi_{8,195} & \text{otherwise} \end{cases}$$

$$\xi_{8,202} = \begin{cases} 0 & \text{for } \xi_{8,133} = \xi_{8,134} \\ \xi_{8,196} & \text{otherwise} \end{cases}$$

$$\xi_{8,203} = \begin{cases} 0 & \text{for } \xi_{8,133} = \xi_{8,134} \\ \xi_{8,197} & \text{otherwise} \end{cases}$$

$$\xi_{8,204} = \begin{cases} 1 & \text{for } \xi_{8,133} = \xi_{8,134} \\ \xi_{8,198} & \text{otherwise} \end{cases}$$

$$\xi_{8,205} = \begin{cases} 0 & \text{for } \xi_{8,133} = \xi_{8,134} \\ \xi_{8,199} & \text{otherwise} \end{cases}$$

$$\xi_{8,206} = \begin{cases} 1 & \text{for } \xi_{8,133} = \xi_{8,134} \\ \xi_{8,200} & \text{otherwise} \end{cases}$$

$$\xi_{8,207} = \xi_{8,189}\xi_{8,201} + \xi_{8,190}\xi_{8,202} + \xi_{8,191}\xi_{8,203}$$

$$\xi_{8,208} = \xi_{8,189}\xi_{8,202} + \xi_{8,190}\xi_{8,204} + \xi_{8,191}\xi_{8,205}$$

$$\xi_{8,209} = \xi_{8,189}\xi_{8,203} + \xi_{8,190}\xi_{8,205} + \xi_{8,191}\xi_{8,206}$$

$$\xi_{8,210} = \xi_{8,190}\xi_{8,201} + \xi_{8,192}\xi_{8,202} + \xi_{8,193}\xi_{8,203}$$

$$\xi_{8,211} = \xi_{8,190}\xi_{8,202} + \xi_{8,192}\xi_{8,204} + \xi_{8,193}\xi_{8,205}$$

$$\xi_{8,212} = \xi_{8,190}\xi_{8,203} + \xi_{8,192}\xi_{8,205} + \xi_{8,193}\xi_{8,206}$$

$$\xi_{8,213} = \xi_{8,191}\xi_{8,201} + \xi_{8,193}\xi_{8,202} + \xi_{8,194}\xi_{8,203}$$

$$\xi_{8,214} = \xi_{8,191}\xi_{8,202} + \xi_{8,193}\xi_{8,204} + \xi_{8,194}\xi_{8,205}$$

$$\xi_{8,215} = \xi_{8,191}\xi_{8,203} + \xi_{8,193}\xi_{8,205} + \xi_{8,194}\xi_{8,206}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

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$$\xi_{8,216} = \begin{cases} \xi_{8,180}^2 & \text{for } \xi_{8,132} = \xi_{8,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,217} = \begin{cases} \xi_{8,180}\xi_{8,181} & \text{for } \xi_{8,132} = \xi_{8,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,218} = \begin{cases} \xi_{8,180}\xi_{8,182} & \text{for } \xi_{8,132} = \xi_{8,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,219} = \begin{cases} \xi_{8,181}^2 & \text{for } \xi_{8,132} = \xi_{8,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,220} = \begin{cases} \xi_{8,181}\xi_{8,182} & \text{for } \xi_{8,132} = \xi_{8,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,221} = \begin{cases} \xi_{8,182}^2 & \text{for } \xi_{8,132} = \xi_{8,133} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{8,222} = \xi_{8,207} - \xi_{8,216}$$

$$\xi_{8,223} = \xi_{8,208} - \xi_{8,217}$$

$$\xi_{8,224} = \xi_{8,209} - \xi_{8,218}$$

$$\xi_{8,225} = \xi_{8,210} - \xi_{8,217}$$

$$\xi_{8,226} = \xi_{8,211} - \xi_{8,219}$$

$$\xi_{8,227} = \xi_{8,212} - \xi_{8,220}$$

$$\xi_{8,228} = \xi_{8,213} - \xi_{8,218}$$

$$\xi_{8,229} = \xi_{8,214} - \xi_{8,220}$$

$$\xi_{8,230} = \xi_{8,215} - \xi_{8,221}$$

$$\xi_{8,231} = \sqrt{\xi_{8,222}}$$

$$\xi_{8,232} = \frac{\xi_{8,222}}{\xi_{8,231}}$$

$$\xi_{8,233} = \frac{\xi_{8,225}}{\xi_{8,231}}$$

$$\xi_{8,234} = \frac{\xi_{8,228}}{\xi_{8,231}}$$

$$\xi_{8,235} = \sqrt{\xi_{8,226}}$$

$$\xi_{8,236} = \frac{\xi_{8,223}}{\xi_{8,235}}$$

$$\xi_{8,237} = \frac{\xi_{8,226}}{\xi_{8,235}}$$

$$\xi_{8,238} = \frac{\xi_{8,229}}{\xi_{8,235}}$$

$$\xi_{8,239} = \sqrt{\xi_{8,230}}$$

$$\xi_{8,240} = \frac{\xi_{8,224}}{\xi_{8,239}}$$

$$\xi_{8,241} = \frac{\xi_{8,227}}{\xi_{8,239}}$$

$$\xi_{8,242} = \frac{\xi_{8,230}}{\xi_{8,239}}$$

$$\xi_{8,243} = \begin{cases} \xi_{8,232} & \text{for } \xi_{8,222} \geq \xi_{8,226} \wedge \xi_{8,222} \geq \xi_{8,230} \\ \xi_{8,236} & \text{for } \xi_{8,226} \geq \xi_{8,230} \wedge \xi_{8,222} \leq \xi_{8,226} \\ \xi_{8,240} & \text{otherwise} \end{cases}$$

$$\xi_{8,244} = \begin{cases} \xi_{8,233} & \text{for } \xi_{8,222} \geq \xi_{8,226} \wedge \xi_{8,222} \geq \xi_{8,230} \\ \xi_{8,237} & \text{for } \xi_{8,226} \geq \xi_{8,230} \wedge \xi_{8,222} \leq \xi_{8,226} \\ \xi_{8,241} & \text{otherwise} \end{cases}$$

$$\xi_{8,245} = \begin{cases} \xi_{8,234} & \text{for } \xi_{8,222} \geq \xi_{8,226} \wedge \xi_{8,222} \geq \xi_{8,230} \\ \xi_{8,238} & \text{for } \xi_{8,226} \geq \xi_{8,230} \wedge \xi_{8,222} \leq \xi_{8,226} \\ \xi_{8,242} & \text{otherwise} \end{cases}$$

$$\xi_{8,246} = \xi_{8,181}\xi_{8,245} - \xi_{8,182}\xi_{8,244}$$

$$\xi_{8,247} = -\xi_{8,180}\xi_{8,245} + \xi_{8,182}\xi_{8,243}$$

$$\xi_{8,248} = \xi_{8,180}\xi_{8,244} - \xi_{8,181}\xi_{8,243}$$

$$\xi_{8,249} = \xi_{8,126}^2 + \xi_{8,129}^2$$

$$\xi_{8,250} = \xi_{8,127}^2 + \xi_{8,130}^2$$

$$\xi_{8,251} = \xi_{8,128}^2 + \xi_{8,131}^2$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,252} = \frac{\xi_{8,113}\xi_{8,129}\xi_{8,180}+\xi_{8,181}(\xi_{8,106}\xi_{8,126}+\xi_{8,114}\xi_{8,129})+\xi_{8,182}(\xi_{8,107}\xi_{8,126}+\xi_{8,115}\xi_{8,129})}{\xi_{8,249}}$$

$$\xi_{8,253} = \frac{\xi_{8,116}\xi_{8,129}\xi_{8,180}+\xi_{8,181}(\xi_{8,108}\xi_{8,126}+\xi_{8,117}\xi_{8,129})+\xi_{8,182}(\xi_{8,109}\xi_{8,126}+\xi_{8,118}\xi_{8,129})}{\xi_{8,249}}$$

$$\xi_{8,254} = \frac{\xi_{8,111}\xi_{8,126}\xi_{8,181}+\xi_{8,112}\xi_{8,126}\xi_{8,182}+\xi_{8,180}(\xi_{8,110}\xi_{8,126}+\xi_{8,119}\xi_{8,129})}{\xi_{8,249}}$$

$$\xi_{8,255} = \frac{\xi_{8,113}\xi_{8,130}\xi_{8,243}+\xi_{8,244}(\xi_{8,106}\xi_{8,127}+\xi_{8,114}\xi_{8,130})+\xi_{8,245}(\xi_{8,107}\xi_{8,127}+\xi_{8,115}\xi_{8,130})}{\xi_{8,250}}$$

$$\xi_{8,256} = \frac{\xi_{8,116}\xi_{8,130}\xi_{8,243}+\xi_{8,244}(\xi_{8,108}\xi_{8,127}+\xi_{8,117}\xi_{8,130})+\xi_{8,245}(\xi_{8,109}\xi_{8,127}+\xi_{8,118}\xi_{8,130})}{\xi_{8,250}}$$

$$\xi_{8,257} = \frac{\xi_{8,111}\xi_{8,127}\xi_{8,244}+\xi_{8,112}\xi_{8,127}\xi_{8,245}+\xi_{8,243}(\xi_{8,110}\xi_{8,127}+\xi_{8,119}\xi_{8,130})}{\xi_{8,250}}$$

$$\xi_{8,258} = \frac{\xi_{8,113}\xi_{8,131}\xi_{8,246}+\xi_{8,247}(\xi_{8,106}\xi_{8,128}+\xi_{8,114}\xi_{8,131})+\xi_{8,248}(\xi_{8,107}\xi_{8,128}+\xi_{8,115}\xi_{8,131})}{\xi_{8,251}}$$

$$\xi_{8,259} = \frac{\xi_{8,116}\xi_{8,131}\xi_{8,246}+\xi_{8,247}(\xi_{8,108}\xi_{8,128}+\xi_{8,117}\xi_{8,131})+\xi_{8,248}(\xi_{8,109}\xi_{8,128}+\xi_{8,118}\xi_{8,131})}{\xi_{8,251}}$$

$$\xi_{8,260} = \frac{\xi_{8,111}\xi_{8,128}\xi_{8,247}+\xi_{8,112}\xi_{8,128}\xi_{8,248}+\xi_{8,246}(\xi_{8,110}\xi_{8,128}+\xi_{8,119}\xi_{8,131})}{\xi_{8,251}}$$

$$\xi_{8,261} = 1 - \xi_{8,252}^2$$

$$\xi_{8,262} = -\xi_{8,252}\xi_{8,253}$$

$$\xi_{8,263} = -\xi_{8,252}\xi_{8,254}$$

$$\xi_{8,264} = 1 - \xi_{8,253}^2$$

$$\xi_{8,265} = -\xi_{8,253}\xi_{8,254}$$

$$\xi_{8,266} = 1 - \xi_{8,254}^2$$

$$\xi_{8,267} = \xi_{8,261}^2 + \xi_{8,262}^2 + \xi_{8,263}^2$$

$$\xi_{8,268} = \xi_{8,262}^2 + \xi_{8,264}^2 + \xi_{8,265}^2$$

$$\xi_{8,269} = \xi_{8,263}^2 + \xi_{8,265}^2 + \xi_{8,266}^2$$

$$\xi_{8,270} = \sqrt{\xi_{8,267}}$$

$$\xi_{8,271} = \frac{\xi_{8,261}}{\xi_{8,270}}$$

$$\xi_{8,272} = \frac{\xi_{8,262}}{\xi_{8,270}}$$

$$\xi_{8,273} = \frac{\xi_{8,263}}{\xi_{8,270}}$$

$$\xi_{8,274} = \sqrt{\xi_{8,268}}$$

$$\xi_{8,275} = \frac{\xi_{8,264}}{\xi_{8,274}}$$

$$\xi_{8,276} = \frac{\xi_{8,262}}{\xi_{8,274}}$$

$$\xi_{8,277} = \frac{\xi_{8,265}}{\xi_{8,274}}$$

$$\xi_{8,278} = \sqrt{\xi_{8,269}}$$

$$\xi_{8,279} = \frac{\xi_{8,266}}{\xi_{8,278}}$$

$$\xi_{8,280} = \frac{\xi_{8,263}}{\xi_{8,278}}$$

$$\xi_{8,281} = \frac{\xi_{8,265}}{\xi_{8,278}}$$

$$\xi_{8,282} = \begin{cases} \xi_{8,271} & \text{for } \xi_{8,267} \geq \xi_{8,268} \wedge \xi_{8,267} \geq \xi_{8,269} \\ \xi_{8,276} & \text{for } \xi_{8,268} \geq \xi_{8,269} \wedge \xi_{8,267} \leq \xi_{8,268} \\ \xi_{8,280} & \text{otherwise} \end{cases}$$

$$\xi_{8,283} = \begin{cases} \xi_{8,272} & \text{for } \xi_{8,267} \geq \xi_{8,268} \wedge \xi_{8,267} \geq \xi_{8,269} \\ \xi_{8,275} & \text{for } \xi_{8,268} \geq \xi_{8,269} \wedge \xi_{8,267} \leq \xi_{8,268} \\ \xi_{8,281} & \text{otherwise} \end{cases}$$

$$\xi_{8,284} = \begin{cases} \xi_{8,273} & \text{for } \xi_{8,267} \geq \xi_{8,268} \wedge \xi_{8,267} \geq \xi_{8,269} \\ \xi_{8,277} & \text{for } \xi_{8,268} \geq \xi_{8,269} \wedge \xi_{8,267} \leq \xi_{8,268} \\ \xi_{8,279} & \text{otherwise} \end{cases}$$

$$\xi_{8,285} = \xi_{8,253}\xi_{8,284} - \xi_{8,254}\xi_{8,283}$$

$$\xi_{8,286} = -\xi_{8,252}\xi_{8,284} + \xi_{8,254}\xi_{8,282}$$

$$\xi_{8,287} = \xi_{8,252}\xi_{8,283} - \xi_{8,253}\xi_{8,282}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,288} = 1 - \xi_{8,255}^2$$

$$\xi_{8,289} = -\xi_{8,255}\xi_{8,256}$$

$$\xi_{8,290} = -\xi_{8,255}\xi_{8,257}$$

$$\xi_{8,291} = 1 - \xi_{8,256}^2$$

$$\xi_{8,292} = -\xi_{8,256}\xi_{8,257}$$

$$\xi_{8,293} = 1 - \xi_{8,257}^2$$

$$\xi_{8,294} = \xi_{8,288}^2 + \xi_{8,289}^2 + \xi_{8,290}^2$$

$$\xi_{8,295} = \xi_{8,289}^2 + \xi_{8,291}^2 + \xi_{8,292}^2$$

$$\xi_{8,296} = \xi_{8,290}^2 + \xi_{8,292}^2 + \xi_{8,293}^2$$

$$\xi_{8,297} = \sqrt{\xi_{8,294}}$$

$$\xi_{8,298} = \frac{\xi_{8,288}}{\xi_{8,297}}$$

$$\xi_{8,299} = \frac{\xi_{8,289}}{\xi_{8,297}}$$

$$\xi_{8,300} = \frac{\xi_{8,290}}{\xi_{8,297}}$$

$$\xi_{8,301} = \sqrt{\xi_{8,295}}$$

$$\xi_{8,302} = \frac{\xi_{8,291}}{\xi_{8,301}}$$

$$\xi_{8,303} = \frac{\xi_{8,289}}{\xi_{8,301}}$$

$$\xi_{8,304} = \frac{\xi_{8,292}}{\xi_{8,301}}$$

$$\xi_{8,305} = \sqrt{\xi_{8,296}}$$

$$\xi_{8,306} = \frac{\xi_{8,293}}{\xi_{8,305}}$$

$$\xi_{8,307} = \frac{\xi_{8,290}}{\xi_{8,305}}$$

$$\xi_{8,308} = \frac{\xi_{8,292}}{\xi_{8,305}}$$

$$\xi_{8,309} = \begin{cases} \xi_{8,298} & \text{for } \xi_{8,294} \geq \xi_{8,295} \wedge \xi_{8,294} \geq \xi_{8,296} \\ \xi_{8,303} & \text{for } \xi_{8,295} \geq \xi_{8,296} \wedge \xi_{8,294} \leq \xi_{8,295} \\ \xi_{8,307} & \text{otherwise} \end{cases}$$

$$\xi_{8,310} = \begin{cases} \xi_{8,299} & \text{for } \xi_{8,294} \geq \xi_{8,295} \wedge \xi_{8,294} \geq \xi_{8,296} \\ \xi_{8,302} & \text{for } \xi_{8,295} \geq \xi_{8,296} \wedge \xi_{8,294} \leq \xi_{8,295} \\ \xi_{8,308} & \text{otherwise} \end{cases}$$

$$\xi_{8,311} = \begin{cases} \xi_{8,300} & \text{for } \xi_{8,294} \geq \xi_{8,295} \wedge \xi_{8,294} \geq \xi_{8,296} \\ \xi_{8,304} & \text{for } \xi_{8,295} \geq \xi_{8,296} \wedge \xi_{8,294} \leq \xi_{8,295} \\ \xi_{8,306} & \text{otherwise} \end{cases}$$

$$\xi_{8,312} = \xi_{8,256}\xi_{8,311} - \xi_{8,257}\xi_{8,310}$$

$$\xi_{8,313} = -\xi_{8,255}\xi_{8,311} + \xi_{8,257}\xi_{8,309}$$

$$\xi_{8,314} = \xi_{8,255}\xi_{8,310} - \xi_{8,256}\xi_{8,309}$$

$$\xi_{8,315} = \xi_{8,253}\xi_{8,257} - \xi_{8,254}\xi_{8,256}$$

$$\xi_{8,316} = -\xi_{8,252}\xi_{8,257} + \xi_{8,254}\xi_{8,255}$$

$$\xi_{8,317} = \xi_{8,252}\xi_{8,256} - \xi_{8,253}\xi_{8,255}$$

$$\xi_{8,318} = 1 - \xi_{8,258}^2$$

$$\xi_{8,319} = -\xi_{8,258}\xi_{8,259}$$

$$\xi_{8,320} = -\xi_{8,258}\xi_{8,260}$$

$$\xi_{8,321} = 1 - \xi_{8,259}^2$$

$$\xi_{8,322} = -\xi_{8,259}\xi_{8,260}$$

$$\xi_{8,323} = 1 - \xi_{8,260}^2$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

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$$\xi_{8,324} = \xi_{8,318}^2 + \xi_{8,319}^2 + \xi_{8,320}^2$$

$$\xi_{8,325} = \xi_{8,319}^2 + \xi_{8,321}^2 + \xi_{8,322}^2$$

$$\xi_{8,326} = \xi_{8,320}^2 + \xi_{8,322}^2 + \xi_{8,323}^2$$

$$\xi_{8,327} = \sqrt{\xi_{8,324}}$$

$$\xi_{8,328} = \frac{\xi_{8,318}}{\xi_{8,327}}$$

$$\xi_{8,329} = \frac{\xi_{8,319}}{\xi_{8,327}}$$

$$\xi_{8,330} = \frac{\xi_{8,320}}{\xi_{8,327}}$$

$$\xi_{8,331} = \sqrt{\xi_{8,325}}$$

$$\xi_{8,332} = \frac{\xi_{8,321}}{\xi_{8,331}}$$

$$\xi_{8,333} = \frac{\xi_{8,319}}{\xi_{8,331}}$$

$$\xi_{8,334} = \frac{\xi_{8,322}}{\xi_{8,331}}$$

$$\xi_{8,335} = \sqrt{\xi_{8,326}}$$

$$\xi_{8,336} = \frac{\xi_{8,323}}{\xi_{8,335}}$$

$$\xi_{8,337} = \frac{\xi_{8,320}}{\xi_{8,335}}$$

$$\xi_{8,338} = \frac{\xi_{8,322}}{\xi_{8,335}}$$

$$\xi_{8,339} = \begin{cases} \xi_{8,328} & \text{for } \xi_{8,324} \geq \xi_{8,325} \wedge \xi_{8,324} \geq \xi_{8,326} \\ \xi_{8,333} & \text{for } \xi_{8,325} \geq \xi_{8,326} \wedge \xi_{8,324} \leq \xi_{8,325} \\ \xi_{8,337} & \text{otherwise} \end{cases}$$

$$\xi_{8,340} = \begin{cases} \xi_{8,329} & \text{for } \xi_{8,324} \geq \xi_{8,325} \wedge \xi_{8,324} \geq \xi_{8,326} \\ \xi_{8,332} & \text{for } \xi_{8,325} \geq \xi_{8,326} \wedge \xi_{8,324} \leq \xi_{8,325} \\ \xi_{8,338} & \text{otherwise} \end{cases}$$

$$\xi_{8,341} = \begin{cases} \xi_{8,330} & \text{for } \xi_{8,324} \geq \xi_{8,325} \wedge \xi_{8,324} \geq \xi_{8,326} \\ \xi_{8,334} & \text{for } \xi_{8,325} \geq \xi_{8,326} \wedge \xi_{8,324} \leq \xi_{8,325} \\ \xi_{8,336} & \text{otherwise} \end{cases}$$

$$\xi_{8,342} = \xi_{8,250}\xi_{8,341} - \xi_{8,260}\xi_{8,340}$$

$$\xi_{8,343} = -\xi_{8,258}\xi_{8,341} + \xi_{8,260}\xi_{8,339}$$

$$\xi_{8,344} = \xi_{8,258}\xi_{8,340} - \xi_{8,250}\xi_{8,339}$$

$$\xi_{8,345} = -\xi_{8,253}\xi_{8,260} + \xi_{8,254}\xi_{8,259}$$

$$\xi_{8,346} = \xi_{8,252}\xi_{8,260} - \xi_{8,254}\xi_{8,258}$$

$$\xi_{8,347} = -\xi_{8,252}\xi_{8,259} + \xi_{8,253}\xi_{8,258}$$

$$\xi_{8,348} = \xi_{8,256}\xi_{8,260} - \xi_{8,257}\xi_{8,259}$$

$$\xi_{8,349} = -\xi_{8,255}\xi_{8,260} + \xi_{8,257}\xi_{8,258}$$

$$\xi_{8,350} = \xi_{8,255}\xi_{8,259} - \xi_{8,256}\xi_{8,258}$$

$$\xi_{8,351} = \begin{cases} 1 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \\ \xi_{8,252} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,312} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,250} \neq 0 \\ \xi_{8,252} & \text{for } \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,250} \neq 0 \\ \xi_{8,339} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,252} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,348} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,252} & \text{for } \xi_{8,249} \neq 0 \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

$$\xi_{8,352} = \begin{cases} 0 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \\ \xi_{8,282} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,255} & \text{for } \xi_{8,251} = 0 \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} = 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,249} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge \xi_{8,250} \neq 0 \\ \xi_{8,342} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,345} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,255} & \text{for } (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,250} \neq 0 \vee \xi_{8,251} \neq 0) \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

$$\xi_{8,353} = \begin{cases} 0 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,255} = 0 \\ \xi_{8,283} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,256} & \text{for } \xi_{8,251} = 0 \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} = 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,249} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge \xi_{8,250} \neq 0 \\ \xi_{8,343} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,346} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,256} & \text{for } (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,250} \neq 0 \vee \xi_{8,251} \neq 0) \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

$$\xi_{8,354} = \begin{cases} 0 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \\ \xi_{8,253} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,313} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,250} \neq 0 \\ \xi_{8,253} & \text{for } \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,250} \neq 0 \\ \xi_{8,340} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,253} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,349} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,253} & \text{for } \xi_{8,249} \neq 0 \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

$$\xi_{8,355} = \begin{cases} 1 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \\ \xi_{8,283} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,256} & \text{for } \xi_{8,251} = 0 \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} = 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,249} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge \xi_{8,250} \neq 0 \\ \xi_{8,343} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,346} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,256} & \text{for } (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,250} \neq 0 \vee \xi_{8,251} \neq 0) \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

$$\xi_{8,356} = \begin{cases} 0 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,254} = 0 \\ \xi_{8,284} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,257} & \text{for } \xi_{8,251} = 0 \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} = 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,249} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge \xi_{8,250} \neq 0 \\ \xi_{8,344} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,347} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,257} & \text{for } (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,250} \neq 0 \vee \xi_{8,251} \neq 0) \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

$$\xi_{8,357} = \begin{cases} 0 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \\ \xi_{8,254} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,314} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,250} \neq 0 \\ \xi_{8,254} & \text{for } \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,250} \neq 0 \\ \xi_{8,341} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,254} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,350} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,254} & \text{for } \xi_{8,249} \neq 0 \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

$$\xi_{8,358} = \begin{cases} 0 & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \\ \xi_{8,284} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,251} = 0 \wedge \xi_{8,249} \neq 0 \\ \xi_{8,257} & \text{for } \xi_{8,251} = 0 \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} = 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,249} \neq 0) \wedge (\xi_{8,251} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge \xi_{8,250} \neq 0 \\ \xi_{8,344} & \text{for } \xi_{8,249} = 0 \wedge \xi_{8,250} = 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,347} & \text{for } \xi_{8,250} = 0 \wedge \xi_{8,249} \neq 0 \wedge \xi_{8,251} \neq 0 \\ \xi_{8,257} & \text{for } (\xi_{8,249} = 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} = 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,250} \neq 0) \wedge (\xi_{8,249} \neq 0 \vee \xi_{8,251} \neq 0) \wedge (\xi_{8,250} \neq 0 \vee \xi_{8,251} \neq 0) \wedge \xi_{8,250} \neq 0 \wedge \xi_{8,251} \neq 0 \end{cases}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,360} = \frac{\xi_{8,351}}{4} + \frac{\xi_{8,355}}{4} + \frac{\xi_{8,359}}{4} + \frac{1}{4}$$

$$\xi_{8,361} = \frac{\xi_{8,351}}{4} - \frac{\xi_{8,355}}{4} - \frac{\xi_{8,359}}{4} + \frac{1}{4}$$

$$\xi_{8,362} = -\frac{\xi_{8,351}}{4} + \frac{\xi_{8,355}}{4} - \frac{\xi_{8,359}}{4} + \frac{1}{4}$$

$$\xi_{8,363} = -\frac{\xi_{8,351}}{4} - \frac{\xi_{8,355}}{4} + \frac{\xi_{8,359}}{4} + \frac{1}{4}$$

$$\xi_{8,364} = \sqrt{\xi_{8,360}}$$

$$\xi_{8,365} = \frac{-\xi_{8,356}+\xi_{8,358}}{4\xi_{8,364}}$$

$$\xi_{8,366} = \frac{\xi_{8,353}-\xi_{8,357}}{4\xi_{8,364}}$$

$$\xi_{8,367} = \frac{-\xi_{8,352}+\xi_{8,354}}{4\xi_{8,364}}$$

$$\xi_{8,368} = \sqrt{\xi_{8,361}}$$

$$\xi_{8,369} = \frac{-\xi_{8,356}+\xi_{8,358}}{4\xi_{8,368}}$$

$$\xi_{8,370} = \frac{\xi_{8,352}+\xi_{8,354}}{4\xi_{8,368}}$$

$$\xi_{8,371} = \frac{\xi_{8,353}+\xi_{8,357}}{4\xi_{8,368}}$$

$$\xi_{8,372} = \sqrt{\xi_{8,362}}$$

$$\xi_{8,373} = \frac{\xi_{8,353}-\xi_{8,357}}{4\xi_{8,372}}$$

$$\xi_{8,374} = \frac{\xi_{8,352}+\xi_{8,354}}{4\xi_{8,372}}$$

$$\xi_{8,375} = \frac{\xi_{8,356}+\xi_{8,358}}{4\xi_{8,372}}$$

$$\xi_{8,376} = \sqrt{\xi_{8,363}}$$

$$\xi_{8,377} = \frac{-\xi_{8,352}+\xi_{8,354}}{4\xi_{8,376}}$$

$$\xi_{8,378} = \frac{\xi_{8,353}+\xi_{8,357}}{4\xi_{8,376}}$$

$$\xi_{8,379} = \frac{\xi_{8,356}+\xi_{8,358}}{4\xi_{8,376}}$$

$$\xi_{8,380} = \begin{cases} \xi_{8,364} & \text{for } \xi_{8,360} \geq \xi_{8,361} \wedge \xi_{8,360} \geq \xi_{8,362} \wedge \xi_{8,360} \geq \xi_{8,363} \\ \xi_{8,369} & \text{for } \xi_{8,361} \geq \xi_{8,362} \wedge \xi_{8,361} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,361} \\ \xi_{8,373} & \text{for } \xi_{8,362} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,362} \wedge \xi_{8,361} \leq \xi_{8,362} \\ \xi_{8,377} & \text{otherwise} \end{cases}$$

$$\xi_{8,381} = \begin{cases} \xi_{8,365} & \text{for } \xi_{8,360} \geq \xi_{8,361} \wedge \xi_{8,360} \geq \xi_{8,362} \wedge \xi_{8,360} \geq \xi_{8,363} \\ \xi_{8,368} & \text{for } \xi_{8,361} \geq \xi_{8,362} \wedge \xi_{8,361} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,361} \\ \xi_{8,374} & \text{for } \xi_{8,362} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,362} \wedge \xi_{8,361} \leq \xi_{8,362} \\ \xi_{8,378} & \text{otherwise} \end{cases}$$

$$\xi_{8,382} = \begin{cases} \xi_{8,366} & \text{for } \xi_{8,360} \geq \xi_{8,361} \wedge \xi_{8,360} \geq \xi_{8,362} \wedge \xi_{8,360} \geq \xi_{8,363} \\ \xi_{8,370} & \text{for } \xi_{8,361} \geq \xi_{8,362} \wedge \xi_{8,361} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,361} \\ \xi_{8,372} & \text{for } \xi_{8,362} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,362} \wedge \xi_{8,361} \leq \xi_{8,362} \\ \xi_{8,379} & \text{otherwise} \end{cases}$$

$$\xi_{8,383} = \begin{cases} \xi_{8,367} & \text{for } \xi_{8,360} \geq \xi_{8,361} \wedge \xi_{8,360} \geq \xi_{8,362} \wedge \xi_{8,360} \geq \xi_{8,363} \\ \xi_{8,371} & \text{for } \xi_{8,361} \geq \xi_{8,362} \wedge \xi_{8,361} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,361} \\ \xi_{8,375} & \text{for } \xi_{8,362} \geq \xi_{8,363} \wedge \xi_{8,360} \leq \xi_{8,362} \wedge \xi_{8,361} \leq \xi_{8,362} \\ \xi_{8,376} & \text{otherwise} \end{cases}$$

$$\xi_{8,384} = \frac{\xi_{8,180}}{4} + \frac{\xi_{8,244}}{4} + \frac{\xi_{8,248}}{4} + \frac{1}{4}$$

$$\xi_{8,385} = \frac{\xi_{8,180}}{4} - \frac{\xi_{8,244}}{4} - \frac{\xi_{8,248}}{4} + \frac{1}{4}$$

$$\xi_{8,386} = -\frac{\xi_{8,180}}{4} + \frac{\xi_{8,244}}{4} - \frac{\xi_{8,248}}{4} + \frac{1}{4}$$

$$\xi_{8,387} = -\frac{\xi_{8,180}}{4} - \frac{\xi_{8,244}}{4} + \frac{\xi_{8,248}}{4} + \frac{1}{4}$$

$$\xi_{8,388} = \sqrt{\xi_{8,384}}$$

$$\xi_{8,389} = \frac{\xi_{8,245}-\xi_{8,247}}{4\xi_{8,388}}$$

$$\xi_{8,390} = \frac{-\xi_{8,182}+\xi_{8,246}}{4\xi_{8,388}}$$

$$\xi_{8,391} = \frac{\xi_{8,181}-\xi_{8,243}}{4\xi_{8,388}}$$

$$\xi_{8,392} = \sqrt{\xi_{8,385}}$$

$$\xi_{8,393} = \frac{\xi_{8,245}-\xi_{8,247}}{4\xi_{8,392}}$$

$$\xi_{8,394} = \frac{\xi_{8,181}+\xi_{8,243}}{4\xi_{8,392}}$$

$$\xi_{8,395} = \frac{\xi_{8,182}+\xi_{8,246}}{4\xi_{8,392}}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,396} = \sqrt{\xi_{8,386}}$$

$$\xi_{8,397} = \frac{-\xi_{8,182}+\xi_{8,246}}{4\xi_{8,396}}$$

$$\xi_{8,398} = \frac{\xi_{8,181}+\xi_{8,243}}{4\xi_{8,396}}$$

$$\xi_{8,399} = \frac{\xi_{8,245}+\xi_{8,247}}{4\xi_{8,396}}$$

$$\xi_{8,400} = \sqrt{\xi_{8,387}}$$

$$\xi_{8,401} = \frac{\xi_{8,181}-\xi_{8,243}}{4\xi_{8,400}}$$

$$\xi_{8,402} = \frac{\xi_{8,182}+\xi_{8,246}}{4\xi_{8,400}}$$

$$\xi_{8,403} = \frac{\xi_{8,245}+\xi_{8,247}}{4\xi_{8,400}}$$

$$\xi_{8,404} = \begin{cases} \xi_{8,388} & \text{for } \xi_{8,384} \geq \xi_{8,385} \wedge \xi_{8,384} \geq \xi_{8,386} \wedge \xi_{8,384} \geq \xi_{8,387} \\ \xi_{8,393} & \text{for } \xi_{8,385} \geq \xi_{8,386} \wedge \xi_{8,385} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,385} \\ \xi_{8,397} & \text{for } \xi_{8,386} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,386} \wedge \xi_{8,385} \leq \xi_{8,386} \\ \xi_{8,401} & \text{otherwise} \end{cases}$$

$$\xi_{8,405} = \begin{cases} \xi_{8,389} & \text{for } \xi_{8,384} \geq \xi_{8,385} \wedge \xi_{8,384} \geq \xi_{8,386} \wedge \xi_{8,384} \geq \xi_{8,387} \\ \xi_{8,392} & \text{for } \xi_{8,385} \geq \xi_{8,386} \wedge \xi_{8,385} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,385} \\ \xi_{8,398} & \text{for } \xi_{8,386} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,386} \wedge \xi_{8,385} \leq \xi_{8,386} \\ \xi_{8,402} & \text{otherwise} \end{cases}$$

$$\xi_{8,406} = \begin{cases} \xi_{8,390} & \text{for } \xi_{8,384} \geq \xi_{8,385} \wedge \xi_{8,384} \geq \xi_{8,386} \wedge \xi_{8,384} \geq \xi_{8,387} \\ \xi_{8,394} & \text{for } \xi_{8,385} \geq \xi_{8,386} \wedge \xi_{8,385} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,385} \\ \xi_{8,396} & \text{for } \xi_{8,386} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,386} \wedge \xi_{8,385} \leq \xi_{8,386} \\ \xi_{8,403} & \text{otherwise} \end{cases}$$

$$\xi_{8,407} = \begin{cases} \xi_{8,391} & \text{for } \xi_{8,384} \geq \xi_{8,385} \wedge \xi_{8,384} \geq \xi_{8,386} \wedge \xi_{8,384} \geq \xi_{8,387} \\ \xi_{8,395} & \text{for } \xi_{8,385} \geq \xi_{8,386} \wedge \xi_{8,385} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,385} \\ \xi_{8,399} & \text{for } \xi_{8,386} \geq \xi_{8,387} \wedge \xi_{8,384} \leq \xi_{8,386} \wedge \xi_{8,385} \leq \xi_{8,386} \\ \xi_{8,400} & \text{otherwise} \end{cases}$$

$$\xi_{8,408} = \xi_{8,99}\left(\xi_{8,104} + i\xi_{8,105}\right)$$

$$\xi_{8,409} = \frac{\sqrt{2}i\xi_{8,100}\left(\xi_{8,104}+i\xi_{8,105}\right)}{2} - \frac{\sqrt{2}\xi_{8,101}\xi_{8,99}}{2}$$

$$\xi_{8,410} = -\frac{\sqrt{2}\xi_{8,100}\left(\xi_{8,104}+i\xi_{8,105}\right)}{2} + \frac{\sqrt{2}i\xi_{8,101}\xi_{8,99}}{2}$$

$$\xi_{8,411} = -\xi_{8,100}\xi_{8,101}$$

$$\xi_{8,412} = \frac{\sqrt{2}i\left(-\frac{\sqrt{2}\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}i\xi_{8,99}\left(\xi_{8,104}-i\xi_{8,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}\left(-\frac{\sqrt{2}i\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}\xi_{8,99}\left(\xi_{8,104}+i\xi_{8,105}\right)}{2}\right)}{2}$$

$$\xi_{8,413} = \frac{\sqrt{2}\left(-\frac{\sqrt{2}\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}i\xi_{8,99}\left(\xi_{8,104}-i\xi_{8,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}i\left(-\frac{\sqrt{2}i\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}\xi_{8,99}\left(\xi_{8,104}+i\xi_{8,105}\right)}{2}\right)}{2}$$

$$\xi_{8,414} = -\frac{\sqrt{2}\xi_{8,100}\left(\xi_{8,104}-i\xi_{8,105}\right)}{2} + \frac{\sqrt{2}i\xi_{8,101}\xi_{8,99}}{2}$$

$$\xi_{8,415} = \frac{\sqrt{2}\left(-\frac{\sqrt{2}\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}i\xi_{8,99}\left(\xi_{8,104}+i\xi_{8,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}i\left(-\frac{\sqrt{2}i\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}\xi_{8,99}\left(\xi_{8,104}-i\xi_{8,105}\right)}{2}\right)}{2}$$

$$\xi_{8,416} = \frac{\sqrt{2}i\left(-\frac{\sqrt{2}i\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}i\xi_{8,99}\left(\xi_{8,104}+i\xi_{8,105}\right)}{2}\right)}{2} + \frac{\sqrt{2}\left(-\frac{\sqrt{2}i\xi_{8,100}\xi_{8,101}}{2} + \frac{\sqrt{2}\xi_{8,99}\left(\xi_{8,104}-i\xi_{8,105}\right)}{2}\right)}{2}$$

$$\xi_{8,417} = \frac{\sqrt{2}i\xi_{8,100}\left(\xi_{8,104}-i\xi_{8,105}\right)}{2} - \frac{\sqrt{2}\xi_{8,101}\xi_{8,99}}{2}$$

$$\xi_{8,418} = \xi_{8,99}\left(\xi_{8,104} - i\xi_{8,105}\right)$$

$$\xi_{8,419} = \left(\xi_{8,380} - i\xi_{8,383}\right)\left(\xi_{8,404} - i\xi_{8,407}\right)$$

$$\xi_{8,420} = \left(\xi_{8,380} - i\xi_{8,383}\right)\left(-i\xi_{8,405} - \xi_{8,406}\right)$$

$$\xi_{8,421} = \left(\xi_{8,404} - i\xi_{8,407}\right)\left(-i\xi_{8,381} - \xi_{8,382}\right)$$

$$\xi_{8,422} = \left(-i\xi_{8,381} - \xi_{8,382}\right)\left(-i\xi_{8,405} - \xi_{8,406}\right)$$

$$\xi_{8,423} = \left(\xi_{8,380} - i\xi_{8,383}\right)\left(-i\xi_{8,405} + \xi_{8,406}\right)$$

$$\xi_{8,424} = \left(\xi_{8,380} - i\xi_{8,383}\right)\left(\xi_{8,404} + i\xi_{8,407}\right)$$

$$\xi_{8,425} = \left(-i\xi_{8,381} - \xi_{8,382}\right)\left(-i\xi_{8,405} + \xi_{8,406}\right)$$

$$\xi_{8,426} = \left(\xi_{8,404} + i\xi_{8,407}\right)\left(-i\xi_{8,381} - \xi_{8,382}\right)$$

$$\xi_{8,427} = \left(\xi_{8,404} - i\xi_{8,407}\right)\left(-i\xi_{8,381} + \xi_{8,382}\right)$$

$$\xi_{8,428} = \left(-i\xi_{8,381} + \xi_{8,382}\right)\left(-i\xi_{8,405} - \xi_{8,406}\right)$$

$$\xi_{8,429} = \left(\xi_{8,380} + i\xi_{8,383}\right)\left(\xi_{8,404} - i\xi_{8,407}\right)$$

$$\xi_{8,430} = \left(\xi_{8,380} + i\xi_{8,383}\right)\left(-i\xi_{8,405} - \xi_{8,406}\right)$$

$$\xi_{8,431} = \left(-i\xi_{8,381} + \xi_{8,382}\right)\left(-i\xi_{8,405} + \xi_{8,406}\right)$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,432} = \left(\xi_{8,404} + i\xi_{8,407}\right)\left(-i\xi_{8,381} + \xi_{8,382}\right)$$

$$\xi_{8,433} = \left(\xi_{8,380} + i\xi_{8,383}\right)\left(-i\xi_{8,405} + \xi_{8,406}\right)$$

$$\xi_{8,434} = \left(\xi_{8,380} + i\xi_{8,383}\right)\left(\xi_{8,404} + i\xi_{8,407}\right)$$

$$\xi_{8,435} = \xi_{8,408}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}}\right) + \xi_{8,409}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}}\right) + \xi_{8,410}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}}\right) + \xi_{8,411}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}}\right)$$

$$\xi_{8,436} = \xi_{8,409}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}}\right) + \xi_{8,412}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}}\right) + \xi_{8,414}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}}\right) + \xi_{8,415}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}}\right)$$

$$\xi_{8,437} = \xi_{8,410}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}}\right) + \xi_{8,413}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}}\right) + \xi_{8,416}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}}\right) + \xi_{8,417}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}}\right)$$

$$\xi_{8,438} = \xi_{8,411}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}}\right) + \xi_{8,414}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}}\right) + \xi_{8,417}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}}\right) + \xi_{8,418}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}}\right)$$

$$\xi_{8,439} = \xi_{8,408}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}}\right) + \xi_{8,409}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}}\right) + \xi_{8,410}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}}\right) + \xi_{8,411}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}}\right)$$

$$\xi_{8,440} = \xi_{8,409}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}}\right) + \xi_{8,412}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}}\right) + \xi_{8,414}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}}\right) + \xi_{8,415}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}}\right)$$

$$\xi_{8,441} = \xi_{8,410}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}}\right) + \xi_{8,413}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}}\right) + \xi_{8,416}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}}\right) + \xi_{8,417}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}}\right)$$

$$\xi_{8,442} = \xi_{8,411}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,419}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,422}}\right) + \xi_{8,414}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,423}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,426}}\right) + \xi_{8,417}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,427}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,430}}\right) + \xi_{8,418}\left(-ie^{-i\xi_{8,92}}\sin\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,431}} + e^{-i\xi_{8,92}}\cos\left(\xi_{8,90}-\xi_{8,91}\right)\overline{\xi_{8,434}}\right)$$

$$\xi_{8,443} = \xi_{8,408}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}}\right) + \xi_{8,409}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}}\right) + \xi_{8,410}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}}\right) + \xi_{8,411}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}}\right)$$

$$\xi_{8,444} = \xi_{8,409}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}}\right) + \xi_{8,412}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}}\right) + \xi_{8,414}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}}\right) + \xi_{8,415}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}}\right)$$

$$\xi_{8,445} = \xi_{8,410}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}}\right) + \xi_{8,413}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}}\right) + \xi_{8,416}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}}\right) + \xi_{8,417}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}}\right)$$

$$\xi_{8,446} = \xi_{8,411}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}}\right) + \xi_{8,414}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}}\right) + \xi_{8,417}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}}\right) + \xi_{8,418}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}}\right)$$

$$\xi_{8,447} = \xi_{8,408}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}}\right) + \xi_{8,409}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}}\right) + \xi_{8,410}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}}\right) + \xi_{8,411}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}}\right)$$

$$\xi_{8,448} = \xi_{8,409}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}}\right) + \xi_{8,412}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}}\right) + \xi_{8,414}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}}\right) + \xi_{8,415}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}}\right)$$

$$\xi_{8,449} = \xi_{8,410}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}}\right) + \xi_{8,413}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}}\right) + \xi_{8,416}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}}\right) + \xi_{8,417}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}}\right)$$

$$\xi_{8,450} = \xi_{8,411}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,420}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,421}}\right) + \xi_{8,414}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,424}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,425}}\right) + \xi_{8,417}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,428}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,429}}\right) + \xi_{8,418}\left(-ie^{i\xi_{8,92}}\sin\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,432}} + e^{i\xi_{8,92}}\cos\left(\xi_{8,90}+\xi_{8,91}\right)\overline{\xi_{8,433}}\right)$$

$$\xi_{8,451} = \frac{\operatorname{re}\left(\xi_{8,435}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,442}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,444}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,449}\right)}{4}$$

$$\xi_{8,452} = -\frac{\operatorname{im}\left(\xi_{8,436}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,441}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,443}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,450}\right)}{4}$$

$$\xi_{8,453} = -\frac{\operatorname{re}\left(\xi_{8,436}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,441}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,443}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,450}\right)}{4}$$

$$\xi_{8,454} = -\frac{\operatorname{im}\left(\xi_{8,435}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,442}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,444}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,449}\right)}{4}$$

$$\xi_{8,455} = -\frac{\operatorname{im}\left(\xi_{8,437}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,440}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,446}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,447}\right)}{4}$$

$$\xi_{8,456} = -\frac{\operatorname{re}\left(\xi_{8,438}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,439}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,445}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,448}\right)}{4}$$

$$\xi_{8,457} = \frac{\operatorname{im}\left(\xi_{8,438}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,439}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,445}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,448}\right)}{4}$$

$$\xi_{8,458} = -\frac{\operatorname{re}\left(\xi_{8,437}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,440}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,446}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,447}\right)}{4}$$

$$\xi_{8,459} = -\frac{\operatorname{re}\left(\xi_{8,437}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,440}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,446}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,447}\right)}{4}$$

$$\xi_{8,460} = \frac{\operatorname{im}\left(\xi_{8,438}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,439}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,445}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,448}\right)}{4}$$

$$\xi_{8,461} = \frac{\operatorname{re}\left(\xi_{8,438}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,439}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,445}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,448}\right)}{4}$$

$$\xi_{8,462} = \frac{\operatorname{im}\left(\xi_{8,437}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,440}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,446}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,447}\right)}{4}$$

$$\xi_{8,463} = -\frac{\operatorname{im}\left(\xi_{8,435}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,442}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,444}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,449}\right)}{4}$$

$$\xi_{8,464} = -\frac{\operatorname{re}\left(\xi_{8,436}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,441}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,443}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,450}\right)}{4}$$

$$\xi_{8,465} = \frac{\operatorname{im}\left(\xi_{8,436}\right)}{4} + \frac{\operatorname{im}\left(\xi_{8,441}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,443}\right)}{4} - \frac{\operatorname{im}\left(\xi_{8,450}\right)}{4}$$

$$\xi_{8,466} = -\frac{\operatorname{re}\left(\xi_{8,435}\right)}{4} - \frac{\operatorname{re}\left(\xi_{8,442}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,444}\right)}{4} + \frac{\operatorname{re}\left(\xi_{8,449}\right)}{4}$$

$$\xi_{8,467} = \xi_{8,451}^2 + \xi_{8,455}^2 + \xi_{8,459}^2 + \xi_{8,463}^2$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,468} = \xi_{8,452}^2 + \xi_{8,456}^2 + \xi_{8,460}^2 + \xi_{8,464}^2$$

$$\xi_{8,469} = \xi_{8,453}^2 + \xi_{8,457}^2 + \xi_{8,461}^2 + \xi_{8,465}^2$$

$$\xi_{8,470} = \xi_{8,454}^2 + \xi_{8,458}^2 + \xi_{8,462}^2 + \xi_{8,466}^2$$

$$\xi_{8,471} = \sqrt{\xi_{8,467}}$$

$$\xi_{8,472} = \frac{\xi_{8,451}}{\xi_{8,471}}$$

$$\xi_{8,473} = \frac{\xi_{8,455}}{\xi_{8,471}}$$

$$\xi_{8,474} = \frac{\xi_{8,459}}{\xi_{8,471}}$$

$$\xi_{8,475} = \frac{\xi_{8,463}}{\xi_{8,471}}$$

$$\xi_{8,476} = \sqrt{\xi_{8,468}}$$

$$\xi_{8,477} = \frac{\xi_{8,452}}{\xi_{8,476}}$$

$$\xi_{8,478} = \frac{\xi_{8,456}}{\xi_{8,476}}$$

$$\xi_{8,479} = \frac{\xi_{8,460}}{\xi_{8,476}}$$

$$\xi_{8,480} = \frac{\xi_{8,464}}{\xi_{8,476}}$$

$$\xi_{8,481} = \sqrt{\xi_{8,469}}$$

$$\xi_{8,482} = \frac{\xi_{8,453}}{\xi_{8,481}}$$

$$\xi_{8,483} = \frac{\xi_{8,457}}{\xi_{8,481}}$$

$$\xi_{8,484} = \frac{\xi_{8,461}}{\xi_{8,481}}$$

$$\xi_{8,485} = \frac{\xi_{8,465}}{\xi_{8,481}}$$

$$\xi_{8,486} = \sqrt{\xi_{8,470}}$$

$$\xi_{8,487} = \frac{\xi_{8,454}}{\xi_{8,486}}$$

$$\xi_{8,488} = \frac{\xi_{8,458}}{\xi_{8,486}}$$

$$\xi_{8,489} = \frac{\xi_{8,462}}{\xi_{8,486}}$$

$$\xi_{8,490} = \frac{\xi_{8,466}}{\xi_{8,486}}$$

$$\xi_{8,491} = \begin{cases} \xi_{8,472} & \text{for } \xi_{8,467} \geq \xi_{8,468} \wedge \xi_{8,467} \geq \xi_{8,469} \wedge \xi_{8,467} \geq \xi_{8,470} \\ \xi_{8,477} & \text{for } \xi_{8,468} \geq \xi_{8,469} \wedge \xi_{8,468} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,468} \\ \xi_{8,482} & \text{for } \xi_{8,469} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,469} \wedge \xi_{8,468} \leq \xi_{8,469} \\ \xi_{8,487} & \text{otherwise} \end{cases}$$

$$\xi_{8,492} = \begin{cases} \xi_{8,473} & \text{for } \xi_{8,467} \geq \xi_{8,468} \wedge \xi_{8,467} \geq \xi_{8,469} \wedge \xi_{8,467} \geq \xi_{8,470} \\ \xi_{8,478} & \text{for } \xi_{8,468} \geq \xi_{8,469} \wedge \xi_{8,468} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,468} \\ \xi_{8,483} & \text{for } \xi_{8,469} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,469} \wedge \xi_{8,468} \leq \xi_{8,469} \\ \xi_{8,488} & \text{otherwise} \end{cases}$$

$$\xi_{8,493} = \begin{cases} \xi_{8,474} & \text{for } \xi_{8,467} \geq \xi_{8,468} \wedge \xi_{8,467} \geq \xi_{8,469} \wedge \xi_{8,467} \geq \xi_{8,470} \\ \xi_{8,479} & \text{for } \xi_{8,468} \geq \xi_{8,469} \wedge \xi_{8,468} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,468} \\ \xi_{8,484} & \text{for } \xi_{8,469} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,469} \wedge \xi_{8,468} \leq \xi_{8,469} \\ \xi_{8,489} & \text{otherwise} \end{cases}$$

$$\xi_{8,494} = \begin{cases} \xi_{8,475} & \text{for } \xi_{8,467} \geq \xi_{8,468} \wedge \xi_{8,467} \geq \xi_{8,469} \wedge \xi_{8,467} \geq \xi_{8,470} \\ \xi_{8,480} & \text{for } \xi_{8,468} \geq \xi_{8,469} \wedge \xi_{8,468} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,468} \\ \xi_{8,485} & \text{for } \xi_{8,469} \geq \xi_{8,470} \wedge \xi_{8,467} \leq \xi_{8,469} \wedge \xi_{8,468} \leq \xi_{8,469} \\ \xi_{8,490} & \text{otherwise} \end{cases}$$

$$\xi_{8,495} = \xi_{8,451}\xi_{8,491} + \xi_{8,455}\xi_{8,492} + \xi_{8,459}\xi_{8,493} + \xi_{8,463}\xi_{8,494}$$

$$\xi_{8,496} = \xi_{8,452}\xi_{8,491} + \xi_{8,456}\xi_{8,492} + \xi_{8,460}\xi_{8,493} + \xi_{8,464}\xi_{8,494}$$

$$\xi_{8,497} = \xi_{8,453}\xi_{8,491} + \xi_{8,457}\xi_{8,492} + \xi_{8,461}\xi_{8,493} + \xi_{8,465}\xi_{8,494}$$

$$\xi_{8,498} = \xi_{8,454}\xi_{8,491} + \xi_{8,458}\xi_{8,492} + \xi_{8,462}\xi_{8,493} + \xi_{8,466}\xi_{8,494}$$

$$\xi_{8,499} = \xi_{8,29}\xi_{8,39} - \xi_{8,30}\xi_{8,42} - \xi_{8,31}\xi_{8,43} - \xi_{8,32}\xi_{8,44}$$

$$\xi_{8,500} = \xi_{8,29}\xi_{8,42} + \xi_{8,30}\xi_{8,39} - \xi_{8,31}\xi_{8,44} + \xi_{8,32}\xi_{8,43}$$

$$\xi_{8,501} = \xi_{8,29}\xi_{8,43} + \xi_{8,30}\xi_{8,44} + \xi_{8,31}\xi_{8,39} - \xi_{8,32}\xi_{8,42}$$

$$\xi_{8,502} = \xi_{8,29}\xi_{8,44} - \xi_{8,30}\xi_{8,43} + \xi_{8,31}\xi_{8,42} + \xi_{8,32}\xi_{8,39}$$

$$\xi_{8,503} = \xi_{8,3}\xi_{8,41} + \xi_{8,4}\xi_{8,42} + \xi_{8,42}\xi_{8,5} + \xi_{8,42}\xi_{8,6}$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,504} = \xi_{8,3}\xi_{8,42} - \xi_{8,4}\xi_{8,41} + \xi_{8,42}\xi_{8,5} - \xi_{8,42}\xi_{8,6}$$

$$\xi_{8,505} = \xi_{8,3}\xi_{8,42} - \xi_{8,4}\xi_{8,42} - \xi_{8,41}\xi_{8,5} + \xi_{8,42}\xi_{8,6}$$

$$\xi_{8,506} = \xi_{8,3}\xi_{8,42} + \xi_{8,4}\xi_{8,42} - \xi_{8,41}\xi_{8,6} - \xi_{8,42}\xi_{8,5}$$

$$\xi_{8,507} = \xi_{8,39}\left(\xi_{8,491}\xi_{8,86} + \xi_{8,492}\xi_{8,87} + \xi_{8,493}\xi_{8,88} + \xi_{8,494}\xi_{8,89}\right) - \xi_{8,39}\left(\xi_{8,491}\xi_{8,89} - \xi_{8,492}\xi_{8,88} + \xi_{8,493}\xi_{8,87} - \xi_{8,494}\xi_{8,86}\right) - \xi_{8,40}\left(\xi_{8,491}\xi_{8,87} - \xi_{8,492}\xi_{8,86} - \xi_{8,493}\xi_{8,89} + \xi_{8,494}\xi_{8,88}\right) - \xi_{8,41}\left(\xi_{8,491}\xi_{8,88} + \xi_{8,492}\xi_{8,89} - \xi_{8,493}\xi_{8,86} - \xi_{8,494}\xi_{8,87}\right)$$

$$\xi_{8,508} = \xi_{8,39}\left(\xi_{8,491}\xi_{8,87} - \xi_{8,492}\xi_{8,86} - \xi_{8,493}\xi_{8,89} + \xi_{8,494}\xi_{8,88}\right) + \xi_{8,39}\left(\xi_{8,491}\xi_{8,88} + \xi_{8,492}\xi_{8,89} - \xi_{8,493}\xi_{8,86} - \xi_{8,494}\xi_{8,87}\right) + \xi_{8,40}\left(\xi_{8,491}\xi_{8,86} + \xi_{8,492}\xi_{8,87} + \xi_{8,493}\xi_{8,88} + \xi_{8,494}\xi_{8,89}\right) - \xi_{8,41}\left(\xi_{8,491}\xi_{8,89} - \xi_{8,492}\xi_{8,88} + \xi_{8,493}\xi_{8,87} - \xi_{8,494}\xi_{8,86}\right)$$

$$\xi_{8,509} = -\xi_{8,39}\left(\xi_{8,491}\xi_{8,87} - \xi_{8,492}\xi_{8,86} - \xi_{8,493}\xi_{8,89} + \xi_{8,494}\xi_{8,88}\right) + \xi_{8,39}\left(\xi_{8,491}\xi_{8,88} + \xi_{8,492}\xi_{8,89} - \xi_{8,493}\xi_{8,86} - \xi_{8,494}\xi_{8,87}\right) + \xi_{8,40}\left(\xi_{8,491}\xi_{8,89} - \xi_{8,492}\xi_{8,88} + \xi_{8,493}\xi_{8,87} - \xi_{8,494}\xi_{8,86}\right) + \xi_{8,41}\left(\xi_{8,491}\xi_{8,86} + \xi_{8,492}\xi_{8,87} + \xi_{8,493}\xi_{8,88} + \xi_{8,494}\xi_{8,89}\right)$$

$$\xi_{8,510} = \xi_{8,39}\left(\xi_{8,491}\xi_{8,86} + \xi_{8,492}\xi_{8,87} + \xi_{8,493}\xi_{8,88} + \xi_{8,494}\xi_{8,89}\right) + \xi_{8,39}\left(\xi_{8,491}\xi_{8,89} - \xi_{8,492}\xi_{8,88} + \xi_{8,493}\xi_{8,87} - \xi_{8,494}\xi_{8,86}\right) - \xi_{8,40}\left(\xi_{8,491}\xi_{8,88} + \xi_{8,492}\xi_{8,89} - \xi_{8,493}\xi_{8,86} - \xi_{8,494}\xi_{8,87}\right) + \xi_{8,41}\left(\xi_{8,491}\xi_{8,87} - \xi_{8,492}\xi_{8,86} - \xi_{8,493}\xi_{8,89} + \xi_{8,494}\xi_{8,88}\right)$$

$$\xi_{8,511} = -\xi_{8,40}\left(-\xi_{8,495}\xi_{8,55} - \xi_{8,496}\xi_{8,54} + \xi_{8,497}\xi_{8,57} - \xi_{8,498}\xi_{8,56}\right) - \xi_{8,40}\left(-\xi_{8,495}\xi_{8,56} - \xi_{8,496}\xi_{8,57} - \xi_{8,497}\xi_{8,54} + \xi_{8,498}\xi_{8,55}\right) - \xi_{8,40}\left(-\xi_{8,495}\xi_{8,57} + \xi_{8,496}\xi_{8,56} - \xi_{8,497}\xi_{8,55} - \xi_{8,498}\xi_{8,54}\right) + \xi_{8,41}\left(\xi_{8,495}\xi_{8,54} - \xi_{8,496}\xi_{8,55} - \xi_{8,497}\xi_{8,56} - \xi_{8,498}\xi_{8,57}\right)$$

$$\xi_{8,512} = \xi_{8,40}\left(\xi_{8,495}\xi_{8,54} - \xi_{8,496}\xi_{8,55} - \xi_{8,497}\xi_{8,56} - \xi_{8,498}\xi_{8,57}\right) + \xi_{8,40}\left(-\xi_{8,495}\xi_{8,56} - \xi_{8,496}\xi_{8,57} - \xi_{8,497}\xi_{8,54} + \xi_{8,498}\xi_{8,55}\right) - \xi_{8,40}\left(-\xi_{8,495}\xi_{8,57} + \xi_{8,496}\xi_{8,56} - \xi_{8,497}\xi_{8,55} - \xi_{8,498}\xi_{8,54}\right) + \xi_{8,41}\left(-\xi_{8,495}\xi_{8,55} - \xi_{8,496}\xi_{8,54} + \xi_{8,497}\xi_{8,57} - \xi_{8,498}\xi_{8,56}\right)$$

$$\xi_{8,513} = \xi_{8,40}\left(\xi_{8,495}\xi_{8,54} - \xi_{8,496}\xi_{8,55} - \xi_{8,497}\xi_{8,56} - \xi_{8,498}\xi_{8,57}\right) - \xi_{8,40}\left(-\xi_{8,495}\xi_{8,55} - \xi_{8,496}\xi_{8,54} + \xi_{8,497}\xi_{8,57} - \xi_{8,498}\xi_{8,56}\right) + \xi_{8,40}\left(-\xi_{8,495}\xi_{8,57} + \xi_{8,496}\xi_{8,56} - \xi_{8,497}\xi_{8,55} - \xi_{8,498}\xi_{8,54}\right) + \xi_{8,41}\left(-\xi_{8,495}\xi_{8,56} - \xi_{8,496}\xi_{8,57} - \xi_{8,497}\xi_{8,54} + \xi_{8,498}\xi_{8,55}\right)$$

$$\xi_{8,514} = \xi_{8,40}\left(\xi_{8,495}\xi_{8,54} - \xi_{8,496}\xi_{8,55} - \xi_{8,497}\xi_{8,56} - \xi_{8,498}\xi_{8,57}\right) + \xi_{8,40}\left(-\xi_{8,495}\xi_{8,55} - \xi_{8,496}\xi_{8,54} + \xi_{8,497}\xi_{8,57} - \xi_{8,498}\xi_{8,56}\right) - \xi_{8,40}\left(-\xi_{8,495}\xi_{8,56} - \xi_{8,496}\xi_{8,57} - \xi_{8,497}\xi_{8,54} + \xi_{8,498}\xi_{8,55}\right) + \xi_{8,41}\left(-\xi_{8,495}\xi_{8,57} + \xi_{8,496}\xi_{8,56} - \xi_{8,497}\xi_{8,55} - \xi_{8,498}\xi_{8,54}\right)$$

$$\xi_{8,515} = \xi_{8,380}\left(\xi_{8,21}\xi_{8,78} - \xi_{8,22}\xi_{8,79} - \xi_{8,23}\xi_{8,80} - \xi_{8,24}\xi_{8,81}\right) + \xi_{8,381}\left(\xi_{8,21}\xi_{8,79} + \xi_{8,22}\xi_{8,78} + \xi_{8,23}\xi_{8,81} - \xi_{8,24}\xi_{8,80}\right) + \xi_{8,382}\left(\xi_{8,21}\xi_{8,80} - \xi_{8,22}\xi_{8,81} + \xi_{8,23}\xi_{8,78} + \xi_{8,24}\xi_{8,79}\right) + \xi_{8,383}\left(\xi_{8,21}\xi_{8,81} + \xi_{8,22}\xi_{8,80} - \xi_{8,23}\xi_{8,79} + \xi_{8,24}\xi_{8,78}\right)$$

$$\xi_{8,516} = \xi_{8,380}\left(\xi_{8,21}\xi_{8,79} + \xi_{8,22}\xi_{8,78} + \xi_{8,23}\xi_{8,81} - \xi_{8,24}\xi_{8,80}\right) - \xi_{8,381}\left(\xi_{8,21}\xi_{8,78} - \xi_{8,22}\xi_{8,79} - \xi_{8,23}\xi_{8,80} - \xi_{8,24}\xi_{8,81}\right) + \xi_{8,382}\left(\xi_{8,21}\xi_{8,81} + \xi_{8,22}\xi_{8,80} - \xi_{8,23}\xi_{8,79} + \xi_{8,24}\xi_{8,78}\right) - \xi_{8,383}\left(\xi_{8,21}\xi_{8,80} - \xi_{8,22}\xi_{8,81} + \xi_{8,23}\xi_{8,78} + \xi_{8,24}\xi_{8,79}\right)$$

$$\xi_{8,517} = \xi_{8,380}\left(\xi_{8,21}\xi_{8,80} - \xi_{8,22}\xi_{8,81} + \xi_{8,23}\xi_{8,78} + \xi_{8,24}\xi_{8,79}\right) - \xi_{8,381}\left(\xi_{8,21}\xi_{8,81} + \xi_{8,22}\xi_{8,80} - \xi_{8,23}\xi_{8,79} + \xi_{8,24}\xi_{8,78}\right) - \xi_{8,382}\left(\xi_{8,21}\xi_{8,78} - \xi_{8,22}\xi_{8,79} - \xi_{8,23}\xi_{8,80} - \xi_{8,24}\xi_{8,81}\right) + \xi_{8,383}\left(\xi_{8,21}\xi_{8,79} + \xi_{8,22}\xi_{8,78} + \xi_{8,23}\xi_{8,81} - \xi_{8,24}\xi_{8,80}\right)$$

$$\xi_{8,518} = \xi_{8,380}\left(\xi_{8,21}\xi_{8,81} + \xi_{8,22}\xi_{8,80} - \xi_{8,23}\xi_{8,79} + \xi_{8,24}\xi_{8,78}\right) + \xi_{8,381}\left(\xi_{8,21}\xi_{8,80} + \xi_{8,22}\xi_{8,81} + \xi_{8,23}\xi_{8,78} + \xi_{8,24}\xi_{8,79}\right) - \xi_{8,382}\left(\xi_{8,21}\xi_{8,79} + \xi_{8,22}\xi_{8,78} + \xi_{8,23}\xi_{8,81} - \xi_{8,24}\xi_{8,80}\right) - \xi_{8,383}\left(\xi_{8,21}\xi_{8,78} - \xi_{8,22}\xi_{8,79} - \xi_{8,23}\xi_{8,80} - \xi_{8,24}\xi_{8,81}\right)$$

$$\xi_{8,519} = \xi_{8,404}\left(\xi_{8,25}\xi_{8,82} - \xi_{8,26}\xi_{8,83} - \xi_{8,27}\xi_{8,84} - \xi_{8,28}\xi_{8,85}\right) + \xi_{8,405}\left(\xi_{8,25}\xi_{8,83} + \xi_{8,26}\xi_{8,82} + \xi_{8,27}\xi_{8,85} - \xi_{8,28}\xi_{8,84}\right) + \xi_{8,406}\left(\xi_{8,25}\xi_{8,84} - \xi_{8,26}\xi_{8,85} + \xi_{8,27}\xi_{8,82} + \xi_{8,28}\xi_{8,83}\right) + \xi_{8,407}\left(\xi_{8,25}\xi_{8,85} + \xi_{8,26}\xi_{8,84} - \xi_{8,27}\xi_{8,83} + \xi_{8,28}\xi_{8,82}\right)$$

$$\xi_{8,520} = \xi_{8,404}\left(\xi_{8,25}\xi_{8,83} + \xi_{8,26}\xi_{8,82} + \xi_{8,27}\xi_{8,85} - \xi_{8,28}\xi_{8,84}\right) - \xi_{8,405}\left(\xi_{8,25}\xi_{8,82} - \xi_{8,26}\xi_{8,83} - \xi_{8,27}\xi_{8,84} - \xi_{8,28}\xi_{8,85}\right) + \xi_{8,406}\left(\xi_{8,25}\xi_{8,85} + \xi_{8,26}\xi_{8,84} - \xi_{8,27}\xi_{8,83} + \xi_{8,28}\xi_{8,82}\right) - \xi_{8,407}\left(\xi_{8,25}\xi_{8,84} - \xi_{8,26}\xi_{8,85} + \xi_{8,27}\xi_{8,82} + \xi_{8,28}\xi_{8,83}\right)$$

$$\xi_{8,521} = \xi_{8,404}\left(\xi_{8,25}\xi_{8,84} - \xi_{8,26}\xi_{8,85} + \xi_{8,27}\xi_{8,82} + \xi_{8,28}\xi_{8,83}\right) - \xi_{8,405}\left(\xi_{8,25}\xi_{8,85} + \xi_{8,26}\xi_{8,84} - \xi_{8,27}\xi_{8,83} + \xi_{8,28}\xi_{8,82}\right) - \xi_{8,406}\left(\xi_{8,25}\xi_{8,82} - \xi_{8,26}\xi_{8,83} - \xi_{8,27}\xi_{8,84} - \xi_{8,28}\xi_{8,85}\right) + \xi_{8,407}\left(\xi_{8,25}\xi_{8,83} + \xi_{8,26}\xi_{8,82} + \xi_{8,27}\xi_{8,85} - \xi_{8,28}\xi_{8,84}\right)$$

$$\xi_{8,522} = \xi_{8,404}\left(\xi_{8,25}\xi_{8,85} + \xi_{8,26}\xi_{8,84} - \xi_{8,27}\xi_{8,83} + \xi_{8,28}\xi_{8,82}\right) + \xi_{8,405}\left(\xi_{8,25}\xi_{8,84} - \xi_{8,26}\xi_{8,85} + \xi_{8,27}\xi_{8,82} + \xi_{8,28}\xi_{8,83}\right) - \xi_{8,406}\left(\xi_{8,25}\xi_{8,83} + \xi_{8,26}\xi_{8,82} + \xi_{8,27}\xi_{8,85} - \xi_{8,28}\xi_{8,84}\right) - \xi_{8,407}\left(\xi_{8,25}\xi_{8,82} - \xi_{8,26}\xi_{8,83} - \xi_{8,27}\xi_{8,84} - \xi_{8,28}\xi_{8,85}\right)$$

$$\xi_{8,523} = \xi_{8,35} + \xi_{8,93}$$

$$\xi_{8,524} = \xi_{8,523} + \frac{\pi t\left(4 g\mathrm{lobal}_{shift} + 1\right)}{4}$$

$$\xi_{8,525} = \begin{cases} 0 & \text{for } \xi_{8,499}^2 + \xi_{8,502}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,502}, \xi_{8,499}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,526} = \begin{cases} 0 & \text{for } \xi_{8,500}^2 + \xi_{8,501}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,500}, \xi_{8,501}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,527} = 2\operatorname{atan}_2\left(\sqrt{\xi_{8,500}^2 + \xi_{8,501}^2}, \sqrt{\xi_{8,499}^2 + \xi_{8,502}^2}\right)$$

$$\xi_{8,528} = -\xi_{8,525} + \xi_{8,526}$$

$$\xi_{8,529} = -\xi_{8,525} - \xi_{8,526}$$

$$\xi_{8,530} = \xi_{8,524} + \xi_{8,525}$$

$$\xi_{8,531} = \begin{cases} 0 & \text{for } \xi_{8,503}^2 + \xi_{8,506}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,506}, \xi_{8,503}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,532} = \begin{cases} 0 & \text{for } \xi_{8,504}^2 + \xi_{8,505}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,504}, \xi_{8,505}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,533} = 2\operatorname{atan}_2\left(\sqrt{\xi_{8,504}^2 + \xi_{8,505}^2}, \sqrt{\xi_{8,503}^2 + \xi_{8,506}^2}\right)$$

$$\xi_{8,534} = -\xi_{8,531} + \xi_{8,532}$$

$$\xi_{8,535} = -\xi_{8,531} - \xi_{8,532}$$

$$\xi_{8,536} = \xi_{8,530} + \xi_{8,531}$$

$$\xi_{8,537} = \begin{cases} 0 & \text{for } \xi_{8,507}^2 + \xi_{8,510}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,510}, \xi_{8,507}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,538} = \begin{cases} 0 & \text{for } \xi_{8,508}^2 + \xi_{8,509}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,508}, \xi_{8,509}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,539} = 2\operatorname{atan}_2\left(\sqrt{\xi_{8,508}^2 + \xi_{8,509}^2}, \sqrt{\xi_{8,507}^2 + \xi_{8,510}^2}\right)$$

Exact shared-expression program 8

Original parameters: global_shift, t. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{8,540} = -\xi_{8,537} + \xi_{8,538}$$

$$\xi_{8,541} = -\xi_{8,537} - \xi_{8,538}$$

$$\xi_{8,542} = \xi_{8,536} + \xi_{8,537}$$

$$\xi_{8,543} = \begin{cases} 0 & \text{for } \xi_{8,511}^2 + \xi_{8,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,514}, \xi_{8,511}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,544} = \begin{cases} 0 & \text{for } \xi_{8,512}^2 + \xi_{8,513}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,512}, \xi_{8,513}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,545} = 2\operatorname{atan}_2\left(\sqrt{\xi_{8,512}^2 + \xi_{8,513}^2}, \sqrt{\xi_{8,511}^2 + \xi_{8,514}^2}\right)$$

$$\xi_{8,546} = -\xi_{8,543} + \xi_{8,544}$$

$$\xi_{8,547} = -\xi_{8,543} - \xi_{8,544}$$

$$\xi_{8,548} = \xi_{8,542} + \xi_{8,543}$$

$$\xi_{8,549} = \begin{cases} 0 & \text{for } \xi_{8,104}^2 + \xi_{8,105}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{8,105}, \xi_{8,104}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,550} = \begin{cases} 0 & \text{for } \xi_{8,101}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{8,101}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,551} = 2\operatorname{atan}_2\left(\left|\xi_{8,101}\right|, \sqrt{\xi_{8,104}^2 + \xi_{8,105}^2}\right)$$

$$\xi_{8,552} = -\xi_{8,549} + \xi_{8,550}$$

$$\xi_{8,553} = -\xi_{8,549} - \xi_{8,550}$$

$$\xi_{8,554} = \xi_{8,548} + \xi_{8,549}$$

$$\xi_{8,555} = \begin{cases} 0 & \text{for } \xi_{8,99}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{8,99}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,556} = \begin{cases} 0 & \text{for } \xi_{8,100}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{8,100}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,557} = 2\operatorname{atan}_2\left(\left|\xi_{8,100}\right|, \left|\xi_{8,99}\right|\right)$$

$$\xi_{8,558} = -\xi_{8,555} + \xi_{8,556}$$

$$\xi_{8,559} = -\xi_{8,555} - \xi_{8,556}$$

$$\xi_{8,560} = \xi_{8,554} + \xi_{8,555}$$

$$\xi_{8,561} = \begin{cases} 0 & \text{for } \xi_{8,515}^2 + \xi_{8,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,518}, \xi_{8,515}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,562} = \begin{cases} 0 & \text{for } \xi_{8,516}^2 + \xi_{8,517}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,516}, \xi_{8,517}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,563} = 2\operatorname{atan}_2\left(\sqrt{\xi_{8,516}^2 + \xi_{8,517}^2}, \sqrt{\xi_{8,515}^2 + \xi_{8,518}^2}\right)$$

$$\xi_{8,564} = -\xi_{8,561} + \xi_{8,562}$$

$$\xi_{8,565} = -\xi_{8,561} - \xi_{8,562}$$

$$\xi_{8,566} = \xi_{8,560} + \xi_{8,561}$$

$$\xi_{8,567} = \begin{cases} 0 & \text{for } \xi_{8,519}^2 + \xi_{8,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,522}, \xi_{8,519}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,568} = \begin{cases} 0 & \text{for } \xi_{8,520}^2 + \xi_{8,521}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{8,520}, \xi_{8,521}\right) & \text{otherwise} \end{cases}$$

$$\xi_{8,569} = 2\operatorname{atan}_2\left(\sqrt{\xi_{8,520}^2 + \xi_{8,521}^2}, \sqrt{\xi_{8,519}^2 + \xi_{8,522}^2}\right)$$

$$\xi_{8,570} = -\xi_{8,567} + \xi_{8,568}$$

$$\xi_{8,571} = -\xi_{8,567} - \xi_{8,568}$$

$$\xi_{8,572} = \xi_{8,566} + \xi_{8,567}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,0} = \left\lfloor \frac{1}{2} - a \right\rfloor$$

$$\xi_{9,1} = \left\lfloor \frac{1}{2} - b \right\rfloor$$

$$\xi_{9,2} = \left\lfloor \frac{1}{2} - c \right\rfloor$$

$$\xi_{9,3} = -\frac{\pi \xi_{9,0}}{2} - \frac{\pi a}{2}$$

$$\xi_{9,4} = -\frac{\pi \xi_{9,1}}{2} - \frac{\pi b}{2}$$

$$\xi_{9,5} = -\frac{\pi \xi_{9,2}}{2} - \frac{\pi c}{2}$$

$$\xi_{9,6} = \begin{cases} \xi_{9,3} & \text{for } (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \\ \xi_{9,4} & \text{for } (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,4}|) \wedge (|\xi_{9,3}| \leq |\xi_{9,4}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \\ \xi_{9,5} & \text{otherwise} \end{cases}$$

$$\xi_{9,7} = \begin{cases} \xi_{9,4} & \text{for } |\xi_{9,3}| \geq |\xi_{9,4}| \wedge |\xi_{9,4}| \geq |\xi_{9,5}| \\ \xi_{9,5} & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge |\xi_{9,4}| \leq |\xi_{9,5}| \\ \xi_{9,3} & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge |\xi_{9,3}| \leq |\xi_{9,4}| \\ \xi_{9,5} & \text{for } |\xi_{9,4}| \geq |\xi_{9,5}| \wedge |\xi_{9,3}| \leq |\xi_{9,5}| \\ \xi_{9,3} & \text{for } |\xi_{9,3}| \geq |\xi_{9,4}| \wedge |\xi_{9,3}| \leq |\xi_{9,5}| \\ \xi_{9,4} & \text{otherwise} \end{cases}$$

$$\xi_{9,8} = \begin{cases} \xi_{9,5} & \text{for } |\xi_{9,3}| \geq |\xi_{9,4}| \wedge |\xi_{9,4}| \geq |\xi_{9,5}| \\ \xi_{9,4} & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge |\xi_{9,4}| \leq |\xi_{9,5}| \\ \xi_{9,5} & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge |\xi_{9,3}| \leq |\xi_{9,4}| \\ \xi_{9,3} & \text{for } |\xi_{9,4}| \geq |\xi_{9,5}| \wedge |\xi_{9,3}| \leq |\xi_{9,5}| \\ \xi_{9,4} & \text{for } |\xi_{9,3}| \geq |\xi_{9,4}| \wedge |\xi_{9,3}| \leq |\xi_{9,5}| \\ \xi_{9,3} & \text{otherwise} \end{cases}$$

$$\xi_{9,9} = \begin{cases} 1 & \text{for } |\xi_{9,3}| \geq |\xi_{9,4}| \wedge |\xi_{9,4}| \geq |\xi_{9,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge |\xi_{9,4}| \leq |\xi_{9,5}| \\ 0 & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge |\xi_{9,3}| \leq |\xi_{9,4}| \\ \frac{1}{2} & \text{for } |\xi_{9,3}| \leq |\xi_{9,5}| \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,4}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{9,10} = \begin{cases} 0 & \text{for } |\xi_{9,3}| \geq |\xi_{9,4}| \wedge |\xi_{9,4}| \geq |\xi_{9,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,4}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \leq |\xi_{9,4}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \\ \frac{1}{2} & \text{for } |\xi_{9,3}| \leq |\xi_{9,5}| \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,4}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,11} = \begin{cases} 0 & \text{for } (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{9,3}| \geq |\xi_{9,5}| \wedge |\xi_{9,3}| \leq |\xi_{9,4}| \\ \frac{1}{2} & \text{for } |\xi_{9,3}| \leq |\xi_{9,5}| \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,4}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{9,12} = \begin{cases} 0 & \text{for } (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \geq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,4}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,3}| \geq |\xi_{9,4}| \vee |\xi_{9,3}| \geq |\xi_{9,5}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \wedge (|\xi_{9,4}| \geq |\xi_{9,5}| \vee |\xi_{9,3}| \leq |\xi_{9,4}| \vee |\xi_{9,4}| \leq |\xi_{9,5}|) \\ \frac{1}{2} & \text{for } |\xi_{9,4}| \geq |\xi_{9,5}| \wedge |\xi_{9,3}| \leq |\xi_{9,5}| \\ -\frac{1}{2} & \text{for } |\xi_{9,3}| \geq |\xi_{9,4}| \wedge |\xi_{9,3}| \leq |\xi_{9,5}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,13} = \begin{cases} -1 & \text{for } \xi_{9,6} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{9,14} = \begin{cases} -1 & \text{for } \xi_{9,7} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{9,15} = \begin{cases} 1 & \text{for } \xi_{9,13} = 1 \wedge \xi_{9,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,16} = \begin{cases} 0 & \text{for } \xi_{9,13} = 1 \wedge \xi_{9,14} = 1 \\ 1 & \text{for } \xi_{9,13} = 1 \wedge \xi_{9,14} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,17} = \begin{cases} 0 & \text{for } \xi_{9,13} = 1 \wedge (\xi_{9,13} = 1 \vee \xi_{9,14} = -1) \wedge (\xi_{9,13} = 1 \vee \xi_{9,14} = 1) \wedge (\xi_{9,14} = -1 \vee \xi_{9,14} = 1) \\ 1 & \text{for } \xi_{9,13} = -1 \wedge \xi_{9,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,18} = \begin{cases} 0 & \text{for } (\xi_{9,13} = -1 \vee \xi_{9,13} = 1) \wedge (\xi_{9,13} = 1 \vee \xi_{9,14} = 1) \wedge (\xi_{9,14} = -1 \vee \xi_{9,14} = 1) \wedge (\xi_{9,13} = -1 \vee \xi_{9,13} = 1 \vee \xi_{9,14} = -1) \wedge (\xi_{9,13} = -1 \vee \xi_{9,13} = 1 \vee \xi_{9,14} = 1) \wedge (\xi_{9,13} = -1 \vee \xi_{9,14} = -1 \vee \xi_{9,14} = 1) \wedge (\xi_{9,13} = 1 \vee \xi_{9,14} = -1 \vee \xi_{9,14} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{9,19} = \cos\left(\frac{\pi \xi_{9,0}}{2}\right)$$

$$\xi_{9,20} = -\sin\left(\frac{\pi \xi_{9,0}}{2}\right)$$

$$\xi_{9,21} = \cos\left(\frac{\pi \xi_{9,1}}{2}\right)$$

$$\xi_{9,22} = -\sin\left(\frac{\pi \xi_{9,1}}{2}\right)$$

$$\xi_{9,23} = \xi_{9,19}\xi_{9,21}$$

$$\xi_{9,24} = \xi_{9,20}\xi_{9,21}$$

$$\xi_{9,25} = \xi_{9,19}\xi_{9,22}$$

$$\xi_{9,26} = -\xi_{9,20}\xi_{9,22}$$

$$\xi_{9,27} = \cos\left(\frac{\pi \xi_{9,2}}{2}\right)$$

$$\xi_{9,28} = -\sin\left(\frac{\pi \xi_{9,2}}{2}\right)$$

$$\xi_{9,29} = \xi_{9,23}\xi_{9,27} - \xi_{9,26}\xi_{9,28}$$

$$\xi_{9,30} = \xi_{9,24}\xi_{9,27} - \xi_{9,25}\xi_{9,28}$$

$$\xi_{9,31} = \xi_{9,24}\xi_{9,28} + \xi_{9,25}\xi_{9,27}$$

$$\xi_{9,32} = \xi_{9,23}\xi_{9,28} + \xi_{9,26}\xi_{9,27}$$

$$\xi_{9,33} = \xi_{9,15}\left(-\xi_{9,10}\xi_{9,30} - \xi_{9,11}\xi_{9,31} - \xi_{9,12}\xi_{9,32} + \xi_{9,29}\xi_{9,9}\right) - \xi_{9,16}\left(\xi_{9,10}\xi_{9,29} - \xi_{9,11}\xi_{9,32} + \xi_{9,12}\xi_{9,31} + \xi_{9,30}\xi_{9,9}\right) - \xi_{9,17}\left(\xi_{9,10}\xi_{9,32} + \xi_{9,11}\xi_{9,29} - \xi_{9,12}\xi_{9,30} + \xi_{9,31}\xi_{9,9}\right) - \xi_{9,18}\left(-\xi_{9,10}\xi_{9,31} + \xi_{9,11}\xi_{9,30} + \xi_{9,12}\xi_{9,29} + \xi_{9,32}\xi_{9,9}\right)$$

$$\xi_{9,34} = \xi_{9,15}\left(\xi_{9,10}\xi_{9,29} - \xi_{9,11}\xi_{9,32} + \xi_{9,12}\xi_{9,31} + \xi_{9,30}\xi_{9,9}\right) + \xi_{9,16}\left(-\xi_{9,10}\xi_{9,30} - \xi_{9,11}\xi_{9,31} - \xi_{9,12}\xi_{9,32} + \xi_{9,29}\xi_{9,9}\right) - \xi_{9,17}\left(-\xi_{9,10}\xi_{9,31} + \xi_{9,11}\xi_{9,30} + \xi_{9,12}\xi_{9,29} + \xi_{9,32}\xi_{9,9}\right) + \xi_{9,18}\left(\xi_{9,10}\xi_{9,32} + \xi_{9,11}\xi_{9,29} - \xi_{9,12}\xi_{9,30} + \xi_{9,31}\xi_{9,9}\right)$$

$$\xi_{9,35} = \xi_{9,15}\left(\xi_{9,10}\xi_{9,32} + \xi_{9,11}\xi_{9,29} - \xi_{9,12}\xi_{9,30} + \xi_{9,31}\xi_{9,9}\right) + \xi_{9,16}\left(-\xi_{9,10}\xi_{9,31} + \xi_{9,11}\xi_{9,30} + \xi_{9,12}\xi_{9,29} + \xi_{9,32}\xi_{9,9}\right) + \xi_{9,17}\left(-\xi_{9,10}\xi_{9,30} - \xi_{9,11}\xi_{9,31} - \xi_{9,12}\xi_{9,32} + \xi_{9,29}\xi_{9,9}\right) - \xi_{9,18}\left(\xi_{9,10}\xi_{9,29} - \xi_{9,11}\xi_{9,32} + \xi_{9,12}\xi_{9,31} + \xi_{9,30}\xi_{9,9}\right)$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,36} = \xi_{9,15}\left(-\xi_{9,10}\xi_{9,31} + \xi_{9,11}\xi_{9,30} + \xi_{9,12}\xi_{9,29} + \xi_{9,32}\xi_{9,9}\right) - \xi_{9,16}\left(\xi_{9,10}\xi_{9,32} + \xi_{9,11}\xi_{9,29} - \xi_{9,12}\xi_{9,30} + \xi_{9,31}\xi_{9,9}\right) + \xi_{9,17}\left(\xi_{9,10}\xi_{9,29} - \xi_{9,11}\xi_{9,32} + \xi_{9,12}\xi_{9,31} + \xi_{9,30}\xi_{9,9}\right) + \xi_{9,18}\left(-\xi_{9,10}\xi_{9,30} - \xi_{9,11}\xi_{9,31} - \xi_{9,12}\xi_{9,32} + \xi_{9,29}\xi_{9,9}\right)$$

$$\xi_{9,37} = -\xi_{9,10}\xi_{9,30} - \xi_{9,11}\xi_{9,31} - \xi_{9,12}\xi_{9,32} + \xi_{9,29}\xi_{9,9}$$

$$\xi_{9,38} = \xi_{9,10}\xi_{9,29} - \xi_{9,11}\xi_{9,32} + \xi_{9,12}\xi_{9,31} + \xi_{9,30}\xi_{9,9}$$

$$\xi_{9,39} = \xi_{9,10}\xi_{9,32} + \xi_{9,11}\xi_{9,29} - \xi_{9,12}\xi_{9,30} + \xi_{9,31}\xi_{9,9}$$

$$\xi_{9,40} = -\xi_{9,10}\xi_{9,31} + \xi_{9,11}\xi_{9,30} + \xi_{9,12}\xi_{9,29} + \xi_{9,32}\xi_{9,9}$$

$$\xi_{9,41} = -\xi_{9,10}\xi_{9,16} - \xi_{9,11}\xi_{9,17} - \xi_{9,12}\xi_{9,18} + \xi_{9,15}\xi_{9,9}$$

$$\xi_{9,42} = -\xi_{9,10}\xi_{9,15} - \xi_{9,11}\xi_{9,18} + \xi_{9,12}\xi_{9,17} - \xi_{9,16}\xi_{9,9}$$

$$\xi_{9,43} = \xi_{9,10}\xi_{9,18} - \xi_{9,11}\xi_{9,15} - \xi_{9,12}\xi_{9,16} - \xi_{9,17}\xi_{9,9}$$

$$\xi_{9,44} = -\xi_{9,10}\xi_{9,17} + \xi_{9,11}\xi_{9,16} - \xi_{9,12}\xi_{9,15} - \xi_{9,18}\xi_{9,9}$$

$$\xi_{9,45} = \xi_{9,13}\xi_{9,6}$$

$$\xi_{9,46} = \xi_{9,14}\xi_{9,7}$$

$$\xi_{9,47} = \xi_{9,13}\xi_{9,14}\xi_{9,8}$$

$$\xi_{9,48} = -\frac{\pi(\xi_{9,0}+\xi_{9,1}+\xi_{9,2})}{2}$$

$$\xi_{9,49} = \begin{cases} 0 & \text{for } \left(\xi_{9,46} > \frac{\pi}{8} \vee \xi_{9,47} < 0\right) \wedge \xi_{9,46} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{9,50} = \frac{\pi}{8}$$

$$\xi_{9,51} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{9,46} > \frac{\pi}{8} \wedge \xi_{9,47} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{9,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,52} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{9,46} > \frac{\pi}{8} \wedge \xi_{9,47} < 0 \\ \frac{1}{2} & \text{for } \xi_{9,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,53} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{9,46} > \frac{\pi}{8} \vee \xi_{9,47} < 0\right) \wedge \xi_{9,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,54} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{9,46} > \frac{\pi}{8} \vee \xi_{9,47} < 0\right) \wedge \xi_{9,46} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{9,55} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{9,46} > \frac{\pi}{8} \vee \xi_{9,47} < 0\right) \wedge \xi_{9,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,56} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{9,46} > \frac{\pi}{8} \vee \xi_{9,47} < 0\right) \wedge \xi_{9,46} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{9,57} = \begin{cases} \frac{1}{2} & \text{for } \xi_{9,46} > \frac{\pi}{8} \wedge \xi_{9,47} < 0 \\ -\frac{1}{2} & \text{for } \xi_{9,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,58} = \left\lfloor \frac{2\xi_{9,45} - 2\xi_{9,49}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{9,59} = \left\lfloor \frac{2\xi_{9,46} - 2\xi_{9,50}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{9,60} = \left\lfloor \frac{2\xi_{9,47} - 2\xi_{9,51}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{9,61} = \xi_{9,45} - \xi_{9,49} - \frac{\pi\xi_{9,58}}{2}$$

$$\xi_{9,62} = \xi_{9,46} - \xi_{9,50} - \frac{\pi\xi_{9,59}}{2}$$

$$\xi_{9,63} = \xi_{9,47} - \xi_{9,51} - \frac{\pi\xi_{9,60}}{2}$$

$$\xi_{9,64} = \begin{cases} \xi_{9,61} & \text{for } \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,61}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,62}| \leq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,63}| \vee |\xi_{9,62}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,62}| \geq |\xi_{9,63}| \vee |\xi_{9,62}| \leq |\xi_{9,63}|\right) \\ \xi_{9,62} & \text{for } \left(|\xi_{9,61}| \geq |\xi_{9,63}| \vee |\xi_{9,62}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,63}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,62}| \geq |\xi_{9,63}| \vee |\xi_{9,61}| \leq |\xi_{9,62}|\right) \wedge \left(|\xi_{9,61}| \leq |\xi_{9,62}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \\ \xi_{9,63} & \text{otherwise} \end{cases}$$

$$\xi_{9,65} = \begin{cases} \xi_{9,62} & \text{for } |\xi_{9,61}| \geq |\xi_{9,62}| \wedge |\xi_{9,62}| \geq |\xi_{9,63}| \\ \xi_{9,63} & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge |\xi_{9,62}| \leq |\xi_{9,63}| \\ \xi_{9,61} & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge |\xi_{9,61}| \leq |\xi_{9,62}| \\ \xi_{9,63} & \text{for } |\xi_{9,62}| \geq |\xi_{9,63}| \wedge |\xi_{9,61}| \leq |\xi_{9,63}| \\ \xi_{9,61} & \text{for } |\xi_{9,61}| \geq |\xi_{9,62}| \wedge |\xi_{9,61}| \leq |\xi_{9,63}| \\ \xi_{9,62} & \text{otherwise} \end{cases}$$

$$\xi_{9,66} = \begin{cases} \xi_{9,63} & \text{for } |\xi_{9,61}| \geq |\xi_{9,62}| \wedge |\xi_{9,62}| \geq |\xi_{9,63}| \\ \xi_{9,62} & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge |\xi_{9,62}| \leq |\xi_{9,63}| \\ \xi_{9,63} & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge |\xi_{9,61}| \leq |\xi_{9,62}| \\ \xi_{9,61} & \text{for } |\xi_{9,62}| \geq |\xi_{9,63}| \wedge |\xi_{9,61}| \leq |\xi_{9,63}| \\ \xi_{9,62} & \text{for } |\xi_{9,61}| \geq |\xi_{9,62}| \wedge |\xi_{9,61}| \leq |\xi_{9,63}| \\ \xi_{9,61} & \text{otherwise} \end{cases}$$

$$\xi_{9,67} = \begin{cases} 1 & \text{for } |\xi_{9,61}| \geq |\xi_{9,62}| \wedge |\xi_{9,62}| \geq |\xi_{9,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge |\xi_{9,62}| \leq |\xi_{9,63}| \\ 0 & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge |\xi_{9,61}| \leq |\xi_{9,62}| \\ \frac{1}{2} & \text{for } |\xi_{9,61}| \leq |\xi_{9,63}| \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,62}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,62}| \geq |\xi_{9,63}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{9,68} = \begin{cases} 0 & \text{for } |\xi_{9,61}| \geq |\xi_{9,62}| \wedge |\xi_{9,62}| \geq |\xi_{9,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge \left(|\xi_{9,61}| \geq |\xi_{9,63}| \vee |\xi_{9,61}| \leq |\xi_{9,62}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,63}| \vee |\xi_{9,62}| \leq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \leq |\xi_{9,62}| \vee |\xi_{9,62}| \leq |\xi_{9,63}|\right) \\ \frac{1}{2} & \text{for } |\xi_{9,61}| \leq |\xi_{9,63}| \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,62}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,62}| \geq |\xi_{9,63}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,69} = \begin{cases} 0 & \text{for } \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,61}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,62}| \leq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,63}| \vee |\xi_{9,62}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,62}| \geq |\xi_{9,63}| \vee |\xi_{9,62}| \leq |\xi_{9,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{9,61}| \geq |\xi_{9,63}| \wedge |\xi_{9,61}| \leq |\xi_{9,62}| \\ \frac{1}{2} & \text{for } |\xi_{9,61}| \leq |\xi_{9,63}| \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,62}| \geq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,61}| \geq |\xi_{9,62}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \wedge \left(|\xi_{9,62}| \geq |\xi_{9,63}| \vee |\xi_{9,61}| \leq |\xi_{9,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{9,70} = \begin{cases} 0 & \text{for } \left(|\xi_{9,64}| \geq |\xi_{9,65}| \vee |\xi_{9,64}| \geq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,64}| \geq |\xi_{9,65}| \vee |\xi_{9,65}| \geq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,64}| \geq |\xi_{9,65}| \vee |\xi_{9,64}| \leq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,64}| \geq |\xi_{9,65}| \vee |\xi_{9,65}| \leq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,64}| \geq |\xi_{9,66}| \vee |\xi_{9,65}| \geq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,64}| \geq |\xi_{9,66}| \vee |\xi_{9,65}| \leq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,65}| \geq |\xi_{9,64}| \vee |\xi_{9,65}| \leq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,65}| \geq |\xi_{9,66}| \vee |\xi_{9,64}| \leq |\xi_{9,66}|\right) \wedge \left(|\xi_{9,65}| \geq |\xi_{9,66}| \vee |\xi_{9,65}| \leq |\xi_{9,66}|\right) \\ \frac{1}{2} & \text{for } |\xi_{9,64}| \geq |\xi_{9,66}| \wedge |\xi_{9,64}| \leq |\xi_{9,66}| \\ -\frac{1}{2} & \text{for } |\xi_{9,64}| \geq |\xi_{9,66}| \wedge |\xi_{9,64}| \leq |\xi_{9,66}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,71} = \begin{cases} -1 & \text{for } \xi_{9,64} < 0 \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,72} = \begin{cases} -1 & \text{for } \xi_{9,65} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{9,73} = \begin{cases} 1 & \text{for } \xi_{9,71} = 1 \wedge \xi_{9,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,74} = \begin{cases} 0 & \text{for } \xi_{9,71} = 1 \wedge \xi_{9,72} = 1 \\ 1 & \text{for } \xi_{9,71} = 1 \wedge \xi_{9,72} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,75} = \begin{cases} 0 & \text{for } \xi_{9,71} = 1 \wedge (\xi_{9,71} = 1 \vee \xi_{9,72} = -1) \wedge (\xi_{9,71} = 1 \vee \xi_{9,72} = 1) \wedge (\xi_{9,72} = -1 \vee \xi_{9,72} = 1) \\ 1 & \text{for } \xi_{9,71} = -1 \wedge \xi_{9,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,76} = \begin{cases} 0 & \text{for } (\xi_{9,71} = -1 \vee \xi_{9,71} = 1) \wedge (\xi_{9,71} = 1 \vee \xi_{9,72} = 1) \wedge (\xi_{9,72} = -1 \vee \xi_{9,72} = 1) \wedge (\xi_{9,71} = -1 \vee \xi_{9,71} = 1 \vee \xi_{9,72} = -1) \wedge (\xi_{9,71} = -1 \vee \xi_{9,71} = 1 \vee \xi_{9,72} = 1) \wedge (\xi_{9,71} = -1 \vee \xi_{9,72} = -1 \vee \xi_{9,72} = 1) \wedge (\xi_{9,71} = 1 \vee \xi_{9,72} = -1 \vee \xi_{9,72} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{9,77} = \cos\left(\frac{\pi\xi_{9,58}}{2}\right)$$

$$\xi_{9,78} = -\sin\left(\frac{\pi\xi_{9,58}}{2}\right)$$

$$\xi_{9,79} = \cos\left(\frac{\pi\xi_{9,59}}{2}\right)$$

$$\xi_{9,80} = -\sin\left(\frac{\pi\xi_{9,59}}{2}\right)$$

$$\xi_{9,81} = \xi_{9,77}\xi_{9,79}$$

$$\xi_{9,82} = \xi_{9,78}\xi_{9,79}$$

$$\xi_{9,83} = \xi_{9,77}\xi_{9,80}$$

$$\xi_{9,84} = -\xi_{9,78}\xi_{9,80}$$

$$\xi_{9,85} = \cos\left(\frac{\pi\xi_{9,60}}{2}\right)$$

$$\xi_{9,86} = -\sin\left(\frac{\pi\xi_{9,60}}{2}\right)$$

$$\xi_{9,87} = \xi_{9,81}\xi_{9,85} - \xi_{9,84}\xi_{9,86}$$

$$\xi_{9,88} = \xi_{9,82}\xi_{9,85} - \xi_{9,83}\xi_{9,86}$$

$$\xi_{9,89} = \xi_{9,82}\xi_{9,86} + \xi_{9,83}\xi_{9,85}$$

$$\xi_{9,90} = \xi_{9,81}\xi_{9,86} + \xi_{9,84}\xi_{9,85}$$

$$\xi_{9,91} = \xi_{9,73}\left(\xi_{9,67}\xi_{9,87} - \xi_{9,68}\xi_{9,88} - \xi_{9,69}\xi_{9,89} - \xi_{9,70}\xi_{9,90}\right) - \xi_{9,74}\left(\xi_{9,67}\xi_{9,88} + \xi_{9,68}\xi_{9,87} - \xi_{9,69}\xi_{9,90} + \xi_{9,70}\xi_{9,89}\right) - \xi_{9,75}\left(\xi_{9,67}\xi_{9,89} + \xi_{9,68}\xi_{9,90} + \xi_{9,69}\xi_{9,87} - \xi_{9,70}\xi_{9,88}\right) - \xi_{9,76}\left(\xi_{9,67}\xi_{9,90} - \xi_{9,68}\xi_{9,89} + \xi_{9,69}\xi_{9,88} + \xi_{9,70}\xi_{9,87}\right)$$

$$\xi_{9,92} = \xi_{9,73}\left(\xi_{9,67}\xi_{9,88} + \xi_{9,68}\xi_{9,87} - \xi_{9,69}\xi_{9,90} + \xi_{9,70}\xi_{9,89}\right) + \xi_{9,74}\left(\xi_{9,67}\xi_{9,87} - \xi_{9,68}\xi_{9,88} - \xi_{9,69}\xi_{9,89} - \xi_{9,70}\xi_{9,90}\right) - \xi_{9,75}\left(\xi_{9,67}\xi_{9,90} - \xi_{9,68}\xi_{9,89} + \xi_{9,69}\xi_{9,88} + \xi_{9,70}\xi_{9,87}\right) + \xi_{9,76}\left(\xi_{9,67}\xi_{9,89} + \xi_{9,68}\xi_{9,90} + \xi_{9,69}\xi_{9,87} - \xi_{9,70}\xi_{9,88}\right)$$

$$\xi_{9,93} = \xi_{9,73}\left(\xi_{9,67}\xi_{9,89} + \xi_{9,68}\xi_{9,90} + \xi_{9,69}\xi_{9,87} - \xi_{9,70}\xi_{9,88}\right) + \xi_{9,74}\left(\xi_{9,67}\xi_{9,90} - \xi_{9,68}\xi_{9,89} + \xi_{9,69}\xi_{9,88} + \xi_{9,70}\xi_{9,87}\right) + \xi_{9,75}\left(\xi_{9,67}\xi_{9,87} - \xi_{9,68}\xi_{9,88} - \xi_{9,69}\xi_{9,89} - \xi_{9,70}\xi_{9,90}\right) - \xi_{9,76}\left(\xi_{9,67}\xi_{9,88} + \xi_{9,68}\xi_{9,87} - \xi_{9,69}\xi_{9,90} + \xi_{9,70}\xi_{9,89}\right)$$

$$\xi_{9,94} = \xi_{9,73}\left(\xi_{9,67}\xi_{9,90} - \xi_{9,68}\xi_{9,89} + \xi_{9,69}\xi_{9,88} + \xi_{9,70}\xi_{9,87}\right) - \xi_{9,74}\left(\xi_{9,67}\xi_{9,89} + \xi_{9,68}\xi_{9,90} + \xi_{9,69}\xi_{9,87} - \xi_{9,70}\xi_{9,88}\right) + \xi_{9,75}\left(\xi_{9,67}\xi_{9,88} + \xi_{9,68}\xi_{9,87} - \xi_{9,69}\xi_{9,90} + \xi_{9,70}\xi_{9,89}\right) + \xi_{9,76}\left(\xi_{9,67}\xi_{9,87} - \xi_{9,68}\xi_{9,88} - \xi_{9,69}\xi_{9,89} - \xi_{9,70}\xi_{9,90}\right)$$

$$\xi_{9,95} = \xi_{9,67}\xi_{9,87} - \xi_{9,68}\xi_{9,88} - \xi_{9,69}\xi_{9,89} - \xi_{9,70}\xi_{9,90}$$

$$\xi_{9,96} = \xi_{9,67}\xi_{9,88} + \xi_{9,68}\xi_{9,87} - \xi_{9,69}\xi_{9,90} + \xi_{9,70}\xi_{9,89}$$

$$\xi_{9,97} = \xi_{9,67}\xi_{9,89} + \xi_{9,68}\xi_{9,90} + \xi_{9,69}\xi_{9,87} - \xi_{9,70}\xi_{9,88}$$

$$\xi_{9,98} = \xi_{9,67}\xi_{9,90} - \xi_{9,68}\xi_{9,89} + \xi_{9,69}\xi_{9,88} + \xi_{9,70}\xi_{9,87}$$

$$\xi_{9,99} = \xi_{9,67}\xi_{9,73} - \xi_{9,68}\xi_{9,74} - \xi_{9,69}\xi_{9,75} - \xi_{9,70}\xi_{9,76}$$

$$\xi_{9,100} = -\xi_{9,67}\xi_{9,74} - \xi_{9,68}\xi_{9,73} - \xi_{9,69}\xi_{9,76} + \xi_{9,70}\xi_{9,75}$$

$$\xi_{9,101} = -\xi_{9,67}\xi_{9,75} + \xi_{9,68}\xi_{9,76} - \xi_{9,69}\xi_{9,73} - \xi_{9,70}\xi_{9,74}$$

$$\xi_{9,102} = -\xi_{9,67}\xi_{9,76} - \xi_{9,68}\xi_{9,75} + \xi_{9,69}\xi_{9,74} - \xi_{9,70}\xi_{9,73}$$

$$\xi_{9,103} = \xi_{9,64}\xi_{9,71}$$

$$\xi_{9,104} = \xi_{9,65}\xi_{9,72}$$

$$\xi_{9,105} = \xi_{9,66}\xi_{9,71}\xi_{9,72}$$

$$\xi_{9,106} = -\frac{\pi(\xi_{9,58}+\xi_{9,59}+\xi_{9,60})}{2}$$

$$\xi_{9,107} = \cos\left(2\xi_{9,103}\right)$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{9,108} = \cos\left(2\xi_{9,104}\right)$$

$$\xi_{9,109} = \cos\left(2\xi_{9,105}\right)$$

$$\xi_{9,110} = \sqrt{\xi_{9,107}^2 - 2\xi_{9,107}\xi_{9,108}\xi_{9,109} + \xi_{9,108}^2 + \xi_{9,109}^2 - 1}$$

$$\xi_{9,111} = \xi_{9,107} - \xi_{9,108} + \xi_{9,109}$$

$$\xi_{9,112} = \sqrt{-\frac{\xi_{9,110}}{2} + \frac{\xi_{9,111}}{2} + \frac{1}{2}}$$

$$\xi_{9,113} = \sqrt{\frac{\xi_{9,110}}{2} - \frac{\xi_{9,111}}{2} + \frac{1}{2}}$$

$$\xi_{9,114} = \sqrt{-\frac{\xi_{9,110}}{2} - \frac{\xi_{9,111}}{2} + \frac{1}{2}}$$

$$\xi_{9,115} = \frac{(1-\xi_{9,108})(\xi_{9,107}+1)(\xi_{9,109}+1)}{2}$$

$$\xi_{9,116} = \begin{cases} -1 & \text{for } \xi_{9,105} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{9,117} = \begin{cases} 0 & \text{for } \xi_{9,112} = 0 \\ \frac{\sqrt{\xi_{9,115}\xi_{9,116}}}{\xi_{9,112}} & \text{otherwise} \end{cases}$$

$$\xi_{9,118} = \begin{cases} 1 & \text{for } \xi_{9,112} = 0 \\ \frac{\sqrt{\xi_{9,107}\xi_{9,108}\xi_{9,109}}}{\xi_{9,112}} & \text{otherwise} \end{cases}$$

$$\xi_{9,119} = 2\xi_{9,112}\xi_{9,113}\xi_{9,114}\xi_{9,118}$$

$$\xi_{9,120} = \sqrt{2}\xi_{9,118}\left(\xi_{9,112}^2\xi_{9,114} - \xi_{9,114}\right)$$

$$\xi_{9,121} = 2\xi_{9,112}\xi_{9,113}\xi_{9,114}\xi_{9,117}$$

$$\xi_{9,122} = \sqrt{2}\left(2\xi_{9,112}^2\xi_{9,114}\xi_{9,117} - \xi_{9,114}\xi_{9,117}\right)$$

$$\xi_{9,123} = \sqrt{2}\xi_{9,112}\xi_{9,113}\xi_{9,117}\xi_{9,118}$$

$$\xi_{9,124} = \sqrt{2}\xi_{9,112}\xi_{9,113}\left(-\xi_{9,114}^2 + \xi_{9,117}^2\right)$$

$$\xi_{9,125} = -\xi_{9,112}^2\xi_{9,114}^2 + \xi_{9,112}^2\xi_{9,117}^2 + \xi_{9,114}^2$$

$$\xi_{9,126} = \xi_{9,112}^2\xi_{9,114}^2 - \xi_{9,112}^2\xi_{9,117}^2 + \xi_{9,117}^2$$

$$\xi_{9,127} = 2\xi_{9,118}\left(\xi_{9,112}^2\xi_{9,117} - \frac{\xi_{9,117}}{2}\right)$$

$$\xi_{9,128} = -\sqrt{2}\xi_{9,112}\xi_{9,113}\xi_{9,117}\xi_{9,118}$$

$$\xi_{9,129} = -\xi_{9,117}\xi_{9,118}$$

$$\xi_{9,130} = \xi_{9,112}^2\xi_{9,114}^2 + \xi_{9,112}^2\xi_{9,117}^2 - \xi_{9,112}^2 - \xi_{9,114}^2 - \xi_{9,117}^2 + 1$$

$$\xi_{9,131} = -\sqrt{2}\left(\xi_{9,112}\xi_{9,113}\xi_{9,114}^2 + \xi_{9,112}\xi_{9,113}\xi_{9,117}^2 - \xi_{9,112}\xi_{9,113}\right)$$

$$\xi_{9,132} = \sqrt{2}\xi_{9,112}^2\xi_{9,114}\xi_{9,118}$$

$$\xi_{9,133} = \xi_{9,123}^2 - \xi_{9,126}^2 - \xi_{9,129}^2 - \xi_{9,132}^2$$

$$\xi_{9,134} = \xi_{9,123}\xi_{9,124} - \xi_{9,126}\xi_{9,127} - \xi_{9,129}\xi_{9,130}$$

$$\xi_{9,135} = \xi_{9,123}\xi_{9,125} - \xi_{9,126}\xi_{9,128} - \xi_{9,129}\xi_{9,131}$$

$$\xi_{9,136} = \xi_{9,119}^2 + \xi_{9,121}^2 + \xi_{9,124}^2 - \xi_{9,127}^2 - \xi_{9,130}^2$$

$$\xi_{9,137} = \xi_{9,119}\xi_{9,120} + \xi_{9,121}\xi_{9,122} + \xi_{9,124}\xi_{9,125} - \xi_{9,127}\xi_{9,128} - \xi_{9,130}\xi_{9,131}$$

$$\xi_{9,138} = \xi_{9,120}^2 + \xi_{9,122}^2 + \xi_{9,125}^2 - \xi_{9,128}^2 - \xi_{9,131}^2$$

$$\xi_{9,139} = \sin\left(2\xi_{9,104}\right)\sin\left(2\xi_{9,105}\right)\cos\left(2\xi_{9,103}\right)$$

$$\xi_{9,140} = \sin\left(2\xi_{9,103}\right)\sin\left(2\xi_{9,105}\right)\cos\left(2\xi_{9,104}\right)$$

$$\xi_{9,141} = \sin\left(2\xi_{9,103}\right)\sin\left(2\xi_{9,104}\right)\cos\left(2\xi_{9,105}\right)$$

$$\xi_{9,142} = \sin\left(2\xi_{9,103}\right)\cos\left(2\xi_{9,104}\right)\cos\left(2\xi_{9,105}\right)$$

$$\xi_{9,143} = \sin\left(2\xi_{9,104}\right)\cos\left(2\xi_{9,103}\right)\cos\left(2\xi_{9,105}\right)$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{9,144} = \sin\left(2\xi_{9,105}\right)\cos\left(2\xi_{9,103}\right)\cos\left(2\xi_{9,104}\right)$$

$$\xi_{9,145} = \xi_{9,139}^2 - \xi_{9,142}^2$$

$$\xi_{9,146} = \xi_{9,140}^2 - \xi_{9,143}^2$$

$$\xi_{9,147} = \xi_{9,141}^2 - \xi_{9,144}^2$$

$$\xi_{9,148} = \frac{\xi_{9,133} - \xi_{9,146}}{\xi_{9,145} - \xi_{9,146}}$$

$$\xi_{9,149} = \frac{\xi_{9,134}}{\xi_{9,145} - \xi_{9,146}}$$

$$\xi_{9,150} = \frac{\xi_{9,135}}{\xi_{9,145} - \xi_{9,146}}$$

$$\xi_{9,151} = \frac{\xi_{9,136} - \xi_{9,146}}{\xi_{9,145} - \xi_{9,146}}$$

$$\xi_{9,152} = \frac{\xi_{9,137}}{\xi_{9,145} - \xi_{9,146}}$$

$$\xi_{9,153} = \frac{\xi_{9,138} - \xi_{9,146}}{\xi_{9,145} - \xi_{9,146}}$$

$$\xi_{9,154} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,148} & \text{otherwise} \end{cases}$$

$$\xi_{9,155} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,149} & \text{otherwise} \end{cases}$$

$$\xi_{9,156} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,150} & \text{otherwise} \end{cases}$$

$$\xi_{9,157} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,151} & \text{otherwise} \end{cases}$$

$$\xi_{9,158} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,152} & \text{otherwise} \end{cases}$$

$$\xi_{9,159} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,153} & \text{otherwise} \end{cases}$$

$$\xi_{9,160} = \frac{\xi_{9,133} - \xi_{9,147}}{\xi_{9,145} - \xi_{9,147}}$$

$$\xi_{9,161} = \frac{\xi_{9,134}}{\xi_{9,145} - \xi_{9,147}}$$

$$\xi_{9,162} = \frac{\xi_{9,135}}{\xi_{9,145} - \xi_{9,147}}$$

$$\xi_{9,163} = \frac{\xi_{9,136} - \xi_{9,147}}{\xi_{9,145} - \xi_{9,147}}$$

$$\xi_{9,164} = \frac{\xi_{9,137}}{\xi_{9,145} - \xi_{9,147}}$$

$$\xi_{9,165} = \frac{\xi_{9,138} - \xi_{9,147}}{\xi_{9,145} - \xi_{9,147}}$$

$$\xi_{9,166} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,147} \\ \xi_{9,160} & \text{otherwise} \end{cases}$$

$$\xi_{9,167} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,147} \\ \xi_{9,161} & \text{otherwise} \end{cases}$$

$$\xi_{9,168} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,147} \\ \xi_{9,162} & \text{otherwise} \end{cases}$$

$$\xi_{9,169} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,147} \\ \xi_{9,163} & \text{otherwise} \end{cases}$$

$$\xi_{9,170} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,147} \\ \xi_{9,164} & \text{otherwise} \end{cases}$$

$$\xi_{9,171} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,147} \\ \xi_{9,165} & \text{otherwise} \end{cases}$$

$$\xi_{9,172} = \xi_{9,154}\xi_{9,166} + \xi_{9,155}\xi_{9,167} + \xi_{9,156}\xi_{9,168}$$

$$\xi_{9,173} = \xi_{9,154}\xi_{9,167} + \xi_{9,155}\xi_{9,169} + \xi_{9,156}\xi_{9,170}$$

$$\xi_{9,174} = \xi_{9,154}\xi_{9,168} + \xi_{9,155}\xi_{9,170} + \xi_{9,156}\xi_{9,171}$$

$$\xi_{9,175} = \xi_{9,155}\xi_{9,166} + \xi_{9,157}\xi_{9,167} + \xi_{9,158}\xi_{9,168}$$

$$\xi_{9,176} = \xi_{9,155}\xi_{9,167} + \xi_{9,157}\xi_{9,169} + \xi_{9,158}\xi_{9,170}$$

$$\xi_{9,177} = \xi_{9,155}\xi_{9,168} + \xi_{9,157}\xi_{9,170} + \xi_{9,158}\xi_{9,171}$$

$$\xi_{9,178} = \xi_{9,156}\xi_{9,166} + \xi_{9,158}\xi_{9,167} + \xi_{9,159}\xi_{9,168}$$

$$\xi_{9,179} = \xi_{9,156}\xi_{9,167} + \xi_{9,158}\xi_{9,169} + \xi_{9,159}\xi_{9,170}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,180} = \xi_{9,156}\xi_{9,168} + \xi_{9,158}\xi_{9,170} + \xi_{9,159}\xi_{9,171}$$

$$\xi_{9,181} = \sqrt{\xi_{9,172}}$$

$$\xi_{9,182} = \frac{\xi_{9,172}}{\xi_{9,181}}$$

$$\xi_{9,183} = \frac{\xi_{9,175}}{\xi_{9,181}}$$

$$\xi_{9,184} = \frac{\xi_{9,178}}{\xi_{9,181}}$$

$$\xi_{9,185} = \sqrt{\xi_{9,176}}$$

$$\xi_{9,186} = \frac{\xi_{9,173}}{\xi_{9,185}}$$

$$\xi_{9,187} = \frac{\xi_{9,176}}{\xi_{9,185}}$$

$$\xi_{9,188} = \frac{\xi_{9,179}}{\xi_{9,185}}$$

$$\xi_{9,189} = \sqrt{\xi_{9,180}}$$

$$\xi_{9,190} = \frac{\xi_{9,174}}{\xi_{9,189}}$$

$$\xi_{9,191} = \frac{\xi_{9,177}}{\xi_{9,189}}$$

$$\xi_{9,192} = \frac{\xi_{9,180}}{\xi_{9,189}}$$

$$\xi_{9,193} = \begin{cases} \xi_{9,182} & \text{for } \xi_{9,172} \geq \xi_{9,176} \wedge \xi_{9,172} \geq \xi_{9,180} \\ \xi_{9,186} & \text{for } \xi_{9,176} \geq \xi_{9,180} \wedge \xi_{9,172} \leq \xi_{9,176} \\ \xi_{9,190} & \text{otherwise} \end{cases}$$

$$\xi_{9,194} = \begin{cases} \xi_{9,183} & \text{for } \xi_{9,172} \geq \xi_{9,176} \wedge \xi_{9,172} \geq \xi_{9,180} \\ \xi_{9,187} & \text{for } \xi_{9,176} \geq \xi_{9,180} \wedge \xi_{9,172} \leq \xi_{9,176} \\ \xi_{9,191} & \text{otherwise} \end{cases}$$

$$\xi_{9,195} = \begin{cases} \xi_{9,184} & \text{for } \xi_{9,172} \geq \xi_{9,176} \wedge \xi_{9,172} \geq \xi_{9,180} \\ \xi_{9,188} & \text{for } \xi_{9,176} \geq \xi_{9,180} \wedge \xi_{9,172} \leq \xi_{9,176} \\ \xi_{9,192} & \text{otherwise} \end{cases}$$

$$\xi_{9,196} = \frac{\xi_{9,133} - \xi_{9,145}}{-\xi_{9,145} + \xi_{9,146}}$$

$$\xi_{9,197} = \frac{\xi_{9,134}}{-\xi_{9,145} + \xi_{9,146}}$$

$$\xi_{9,198} = \frac{\xi_{9,135}}{-\xi_{9,145} + \xi_{9,146}}$$

$$\xi_{9,199} = \frac{\xi_{9,136} - \xi_{9,145}}{-\xi_{9,145} + \xi_{9,146}}$$

$$\xi_{9,200} = \frac{\xi_{9,137}}{-\xi_{9,145} + \xi_{9,146}}$$

$$\xi_{9,201} = \frac{\xi_{9,138} - \xi_{9,145}}{-\xi_{9,145} + \xi_{9,146}}$$

$$\xi_{9,202} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,196} & \text{otherwise} \end{cases}$$

$$\xi_{9,203} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,197} & \text{otherwise} \end{cases}$$

$$\xi_{9,204} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,198} & \text{otherwise} \end{cases}$$

$$\xi_{9,205} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,199} & \text{otherwise} \end{cases}$$

$$\xi_{9,206} = \begin{cases} 0 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,200} & \text{otherwise} \end{cases}$$

$$\xi_{9,207} = \begin{cases} 1 & \text{for } \xi_{9,145} = \xi_{9,146} \\ \xi_{9,201} & \text{otherwise} \end{cases}$$

$$\xi_{9,208} = \frac{\xi_{9,133} - \xi_{9,147}}{\xi_{9,146} - \xi_{9,147}}$$

$$\xi_{9,209} = \frac{\xi_{9,134}}{\xi_{9,146} - \xi_{9,147}}$$

$$\xi_{9,210} = \frac{\xi_{9,135}}{\xi_{9,146} - \xi_{9,147}}$$

$$\xi_{9,211} = \frac{\xi_{9,136} - \xi_{9,147}}{\xi_{9,146} - \xi_{9,147}}$$

$$\xi_{9,212} = \frac{\xi_{9,137}}{\xi_{9,146} - \xi_{9,147}}$$

$$\xi_{9,213} = \frac{\xi_{9,138} - \xi_{9,147}}{\xi_{9,146} - \xi_{9,147}}$$

$$\xi_{9,214} = \begin{cases} 1 & \text{for } \xi_{9,146} = \xi_{9,147} \\ \xi_{9,208} & \text{otherwise} \end{cases}$$

$$\xi_{9,215} = \begin{cases} 0 & \text{for } \xi_{9,146} = \xi_{9,147} \\ \xi_{9,209} & \text{otherwise} \end{cases}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,216} = \begin{cases} 0 & \text{for } \xi_{9,146} = \xi_{9,147} \\ \xi_{9,210} & \text{otherwise} \end{cases}$$

$$\xi_{9,217} = \begin{cases} 1 & \text{for } \xi_{9,146} = \xi_{9,147} \\ \xi_{9,211} & \text{otherwise} \end{cases}$$

$$\xi_{9,218} = \begin{cases} 0 & \text{for } \xi_{9,146} = \xi_{9,147} \\ \xi_{9,212} & \text{otherwise} \end{cases}$$

$$\xi_{9,219} = \begin{cases} 1 & \text{for } \xi_{9,146} = \xi_{9,147} \\ \xi_{9,213} & \text{otherwise} \end{cases}$$

$$\xi_{9,220} = \xi_{9,202}\xi_{9,214} + \xi_{9,203}\xi_{9,215} + \xi_{9,204}\xi_{9,216}$$

$$\xi_{9,221} = \xi_{9,202}\xi_{9,215} + \xi_{9,203}\xi_{9,217} + \xi_{9,204}\xi_{9,218}$$

$$\xi_{9,222} = \xi_{9,202}\xi_{9,216} + \xi_{9,203}\xi_{9,218} + \xi_{9,204}\xi_{9,219}$$

$$\xi_{9,223} = \xi_{9,203}\xi_{9,214} + \xi_{9,205}\xi_{9,215} + \xi_{9,206}\xi_{9,216}$$

$$\xi_{9,224} = \xi_{9,203}\xi_{9,215} + \xi_{9,205}\xi_{9,217} + \xi_{9,206}\xi_{9,218}$$

$$\xi_{9,225} = \xi_{9,203}\xi_{9,216} + \xi_{9,205}\xi_{9,218} + \xi_{9,206}\xi_{9,219}$$

$$\xi_{9,226} = \xi_{9,204}\xi_{9,214} + \xi_{9,206}\xi_{9,215} + \xi_{9,207}\xi_{9,216}$$

$$\xi_{9,227} = \xi_{9,204}\xi_{9,215} + \xi_{9,206}\xi_{9,217} + \xi_{9,207}\xi_{9,218}$$

$$\xi_{9,228} = \xi_{9,204}\xi_{9,216} + \xi_{9,206}\xi_{9,218} + \xi_{9,207}\xi_{9,219}$$

$$\xi_{9,229} = \begin{cases} \xi_{9,193}^2 & \text{for } \xi_{9,145} = \xi_{9,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,230} = \begin{cases} \xi_{9,193}\xi_{9,194} & \text{for } \xi_{9,145} = \xi_{9,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,231} = \begin{cases} \xi_{9,193}\xi_{9,195} & \text{for } \xi_{9,145} = \xi_{9,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,232} = \begin{cases} \xi_{9,194}^2 & \text{for } \xi_{9,145} = \xi_{9,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,233} = \begin{cases} \xi_{9,194}\xi_{9,195} & \text{for } \xi_{9,145} = \xi_{9,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,234} = \begin{cases} \xi_{9,195}^2 & \text{for } \xi_{9,145} = \xi_{9,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{9,235} = \xi_{9,220} - \xi_{9,229}$$

$$\xi_{9,236} = \xi_{9,221} - \xi_{9,230}$$

$$\xi_{9,237} = \xi_{9,222} - \xi_{9,231}$$

$$\xi_{9,238} = \xi_{9,223} - \xi_{9,230}$$

$$\xi_{9,239} = \xi_{9,224} - \xi_{9,232}$$

$$\xi_{9,240} = \xi_{9,225} - \xi_{9,233}$$

$$\xi_{9,241} = \xi_{9,226} - \xi_{9,231}$$

$$\xi_{9,242} = \xi_{9,227} - \xi_{9,233}$$

$$\xi_{9,243} = \xi_{9,228} - \xi_{9,234}$$

$$\xi_{9,244} = \sqrt{\xi_{9,235}}$$

$$\xi_{9,245} = \frac{\xi_{9,235}}{\xi_{9,244}}$$

$$\xi_{9,246} = \frac{\xi_{9,238}}{\xi_{9,244}}$$

$$\xi_{9,247} = \frac{\xi_{9,241}}{\xi_{9,244}}$$

$$\xi_{9,248} = \sqrt{\xi_{9,239}}$$

$$\xi_{9,249} = \frac{\xi_{9,236}}{\xi_{9,248}}$$

$$\xi_{9,250} = \frac{\xi_{9,239}}{\xi_{9,248}}$$

$$\xi_{9,251} = \frac{\xi_{9,242}}{\xi_{9,248}}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,252} = \sqrt{\xi_{9,243}}$$

$$\xi_{9,253} = \frac{\xi_{9,237}}{\xi_{9,252}}$$

$$\xi_{9,254} = \frac{\xi_{9,240}}{\xi_{9,252}}$$

$$\xi_{9,255} = \frac{\xi_{9,243}}{\xi_{9,252}}$$

$$\xi_{9,256} = \begin{cases} \xi_{9,245} & \text{for } \xi_{9,235} \geq \xi_{9,239} \wedge \xi_{9,235} \geq \xi_{9,243} \\ \xi_{9,249} & \text{for } \xi_{9,239} \geq \xi_{9,243} \wedge \xi_{9,235} \leq \xi_{9,239} \\ \xi_{9,253} & \text{otherwise} \end{cases}$$

$$\xi_{9,257} = \begin{cases} \xi_{9,246} & \text{for } \xi_{9,235} \geq \xi_{9,239} \wedge \xi_{9,235} \geq \xi_{9,243} \\ \xi_{9,250} & \text{for } \xi_{9,239} \geq \xi_{9,243} \wedge \xi_{9,235} \leq \xi_{9,239} \\ \xi_{9,254} & \text{otherwise} \end{cases}$$

$$\xi_{9,258} = \begin{cases} \xi_{9,247} & \text{for } \xi_{9,235} \geq \xi_{9,239} \wedge \xi_{9,235} \geq \xi_{9,243} \\ \xi_{9,251} & \text{for } \xi_{9,239} \geq \xi_{9,243} \wedge \xi_{9,235} \leq \xi_{9,239} \\ \xi_{9,255} & \text{otherwise} \end{cases}$$

$$\xi_{9,259} = \xi_{9,194}\xi_{9,258} - \xi_{9,195}\xi_{9,257}$$

$$\xi_{9,260} = -\xi_{9,193}\xi_{9,258} + \xi_{9,195}\xi_{9,256}$$

$$\xi_{9,261} = \xi_{9,193}\xi_{9,257} - \xi_{9,194}\xi_{9,256}$$

$$\xi_{9,262} = \xi_{9,139}^2 + \xi_{9,142}^2$$

$$\xi_{9,263} = \xi_{9,140}^2 + \xi_{9,143}^2$$

$$\xi_{9,264} = \xi_{9,141}^2 + \xi_{9,144}^2$$

$$\xi_{9,265} = \frac{\xi_{9,126}\xi_{9,142}\xi_{9,193} + \xi_{9,194}(\xi_{9,119}\xi_{9,139} + \xi_{9,127}\xi_{9,142}) + \xi_{9,195}(\xi_{9,120}\xi_{9,139} + \xi_{9,128}\xi_{9,142})}{\xi_{9,262}}$$

$$\xi_{9,266} = \frac{\xi_{9,129}\xi_{9,142}\xi_{9,193} + \xi_{9,194}(\xi_{9,121}\xi_{9,139} + \xi_{9,130}\xi_{9,142}) + \xi_{9,195}(\xi_{9,122}\xi_{9,139} + \xi_{9,131}\xi_{9,142})}{\xi_{9,262}}$$

$$\xi_{9,267} = \frac{\xi_{9,124}\xi_{9,139}\xi_{9,194} + \xi_{9,125}\xi_{9,139}\xi_{9,195} + \xi_{9,193}(\xi_{9,123}\xi_{9,139} + \xi_{9,132}\xi_{9,142})}{\xi_{9,262}}$$

$$\xi_{9,268} = \frac{\xi_{9,126}\xi_{9,143}\xi_{9,256} + \xi_{9,257}(\xi_{9,119}\xi_{9,140} + \xi_{9,127}\xi_{9,143}) + \xi_{9,258}(\xi_{9,120}\xi_{9,140} + \xi_{9,128}\xi_{9,143})}{\xi_{9,263}}$$

$$\xi_{9,269} = \frac{\xi_{9,129}\xi_{9,143}\xi_{9,256} + \xi_{9,257}(\xi_{9,121}\xi_{9,140} + \xi_{9,130}\xi_{9,143}) + \xi_{9,258}(\xi_{9,122}\xi_{9,140} + \xi_{9,131}\xi_{9,143})}{\xi_{9,263}}$$

$$\xi_{9,270} = \frac{\xi_{9,124}\xi_{9,140}\xi_{9,257} + \xi_{9,125}\xi_{9,140}\xi_{9,258} + \xi_{9,256}(\xi_{9,123}\xi_{9,140} + \xi_{9,132}\xi_{9,143})}{\xi_{9,263}}$$

$$\xi_{9,271} = \frac{\xi_{9,126}\xi_{9,144}\xi_{9,259} + \xi_{9,260}(\xi_{9,119}\xi_{9,141} + \xi_{9,127}\xi_{9,144}) + \xi_{9,261}(\xi_{9,120}\xi_{9,141} + \xi_{9,128}\xi_{9,144})}{\xi_{9,264}}$$

$$\xi_{9,272} = \frac{\xi_{9,129}\xi_{9,144}\xi_{9,259} + \xi_{9,260}(\xi_{9,121}\xi_{9,141} + \xi_{9,130}\xi_{9,144}) + \xi_{9,261}(\xi_{9,122}\xi_{9,141} + \xi_{9,131}\xi_{9,144})}{\xi_{9,264}}$$

$$\xi_{9,273} = \frac{\xi_{9,124}\xi_{9,141}\xi_{9,260} + \xi_{9,125}\xi_{9,141}\xi_{9,261} + \xi_{9,259}(\xi_{9,123}\xi_{9,141} + \xi_{9,132}\xi_{9,144})}{\xi_{9,264}}$$

$$\xi_{9,274} = 1 - \xi_{9,265}^2$$

$$\xi_{9,275} = -\xi_{9,265}\xi_{9,266}$$

$$\xi_{9,276} = -\xi_{9,265}\xi_{9,267}$$

$$\xi_{9,277} = 1 - \xi_{9,266}^2$$

$$\xi_{9,278} = -\xi_{9,266}\xi_{9,267}$$

$$\xi_{9,279} = 1 - \xi_{9,267}^2$$

$$\xi_{9,280} = \xi_{9,274}^2 + \xi_{9,275}^2 + \xi_{9,276}^2$$

$$\xi_{9,281} = \xi_{9,275}^2 + \xi_{9,277}^2 + \xi_{9,278}^2$$

$$\xi_{9,282} = \xi_{9,276}^2 + \xi_{9,278}^2 + \xi_{9,279}^2$$

$$\xi_{9,283} = \sqrt{\xi_{9,280}}$$

$$\xi_{9,284} = \frac{\xi_{9,274}}{\xi_{9,283}}$$

$$\xi_{9,285} = \frac{\xi_{9,275}}{\xi_{9,283}}$$

$$\xi_{9,286} = \frac{\xi_{9,276}}{\xi_{9,283}}$$

$$\xi_{9,287} = \sqrt{\xi_{9,281}}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,288} = \frac{\xi_{9,277}}{\xi_{9,287}}$$

$$\xi_{9,289} = \frac{\xi_{9,275}}{\xi_{9,287}}$$

$$\xi_{9,290} = \frac{\xi_{9,278}}{\xi_{9,287}}$$

$$\xi_{9,291} = \sqrt{\xi_{9,282}}$$

$$\xi_{9,292} = \frac{\xi_{9,279}}{\xi_{9,291}}$$

$$\xi_{9,293} = \frac{\xi_{9,276}}{\xi_{9,291}}$$

$$\xi_{9,294} = \frac{\xi_{9,278}}{\xi_{9,291}}$$

$$\xi_{9,295} = \begin{cases} \xi_{9,284} & \text{for } \xi_{9,280} \geq \xi_{9,281} \wedge \xi_{9,280} \geq \xi_{9,282} \\ \xi_{9,289} & \text{for } \xi_{9,281} \geq \xi_{9,282} \wedge \xi_{9,280} \leq \xi_{9,281} \\ \xi_{9,293} & \text{otherwise} \end{cases}$$

$$\xi_{9,296} = \begin{cases} \xi_{9,285} & \text{for } \xi_{9,280} \geq \xi_{9,281} \wedge \xi_{9,280} \geq \xi_{9,282} \\ \xi_{9,288} & \text{for } \xi_{9,281} \geq \xi_{9,282} \wedge \xi_{9,280} \leq \xi_{9,281} \\ \xi_{9,294} & \text{otherwise} \end{cases}$$

$$\xi_{9,297} = \begin{cases} \xi_{9,286} & \text{for } \xi_{9,280} \geq \xi_{9,281} \wedge \xi_{9,280} \geq \xi_{9,282} \\ \xi_{9,290} & \text{for } \xi_{9,281} \geq \xi_{9,282} \wedge \xi_{9,280} \leq \xi_{9,281} \\ \xi_{9,292} & \text{otherwise} \end{cases}$$

$$\xi_{9,298} = \xi_{9,266}\xi_{9,297} - \xi_{9,267}\xi_{9,296}$$

$$\xi_{9,299} = -\xi_{9,265}\xi_{9,297} + \xi_{9,267}\xi_{9,295}$$

$$\xi_{9,300} = \xi_{9,265}\xi_{9,296} - \xi_{9,266}\xi_{9,295}$$

$$\xi_{9,301} = 1 - \xi_{9,268}^2$$

$$\xi_{9,302} = -\xi_{9,268}\xi_{9,269}$$

$$\xi_{9,303} = -\xi_{9,268}\xi_{9,270}$$

$$\xi_{9,304} = 1 - \xi_{9,269}^2$$

$$\xi_{9,305} = -\xi_{9,269}\xi_{9,270}$$

$$\xi_{9,306} = 1 - \xi_{9,270}^2$$

$$\xi_{9,307} = \xi_{9,301}^2 + \xi_{9,302}^2 + \xi_{9,303}^2$$

$$\xi_{9,308} = \xi_{9,302}^2 + \xi_{9,304}^2 + \xi_{9,305}^2$$

$$\xi_{9,309} = \xi_{9,303}^2 + \xi_{9,305}^2 + \xi_{9,306}^2$$

$$\xi_{9,310} = \sqrt{\xi_{9,307}}$$

$$\xi_{9,311} = \frac{\xi_{9,301}}{\xi_{9,310}}$$

$$\xi_{9,312} = \frac{\xi_{9,302}}{\xi_{9,310}}$$

$$\xi_{9,313} = \frac{\xi_{9,303}}{\xi_{9,310}}$$

$$\xi_{9,314} = \sqrt{\xi_{9,308}}$$

$$\xi_{9,315} = \frac{\xi_{9,304}}{\xi_{9,314}}$$

$$\xi_{9,316} = \frac{\xi_{9,302}}{\xi_{9,314}}$$

$$\xi_{9,317} = \frac{\xi_{9,305}}{\xi_{9,314}}$$

$$\xi_{9,318} = \sqrt{\xi_{9,309}}$$

$$\xi_{9,319} = \frac{\xi_{9,306}}{\xi_{9,318}}$$

$$\xi_{9,320} = \frac{\xi_{9,303}}{\xi_{9,318}}$$

$$\xi_{9,321} = \frac{\xi_{9,305}}{\xi_{9,318}}$$

$$\xi_{9,322} = \begin{cases} \xi_{9,311} & \text{for } \xi_{9,307} \geq \xi_{9,308} \wedge \xi_{9,307} \geq \xi_{9,309} \\ \xi_{9,316} & \text{for } \xi_{9,308} \geq \xi_{9,309} \wedge \xi_{9,307} \leq \xi_{9,308} \\ \xi_{9,320} & \text{otherwise} \end{cases}$$

$$\xi_{9,323} = \begin{cases} \xi_{9,312} & \text{for } \xi_{9,307} \geq \xi_{9,308} \wedge \xi_{9,307} \geq \xi_{9,309} \\ \xi_{9,315} & \text{for } \xi_{9,308} \geq \xi_{9,309} \wedge \xi_{9,307} \leq \xi_{9,308} \\ \xi_{9,321} & \text{otherwise} \end{cases}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,324} = \begin{cases} \xi_{9,313} & \text{for } \xi_{9,307} \geq \xi_{9,308} \wedge \xi_{9,307} \geq \xi_{9,309} \\ \xi_{9,317} & \text{for } \xi_{9,308} \geq \xi_{9,309} \wedge \xi_{9,307} \leq \xi_{9,308} \\ \xi_{9,319} & \text{otherwise} \end{cases}$$

$$\xi_{9,325} = \xi_{9,269}\xi_{9,324} - \xi_{9,270}\xi_{9,323}$$

$$\xi_{9,326} = -\xi_{9,268}\xi_{9,324} + \xi_{9,270}\xi_{9,322}$$

$$\xi_{9,327} = \xi_{9,268}\xi_{9,323} - \xi_{9,269}\xi_{9,322}$$

$$\xi_{9,328} = \xi_{9,266}\xi_{9,270} - \xi_{9,267}\xi_{9,269}$$

$$\xi_{9,329} = -\xi_{9,265}\xi_{9,270} + \xi_{9,267}\xi_{9,268}$$

$$\xi_{9,330} = \xi_{9,265}\xi_{9,269} - \xi_{9,266}\xi_{9,268}$$

$$\xi_{9,331} = 1 - \xi_{9,271}^2$$

$$\xi_{9,332} = -\xi_{9,271}\xi_{9,272}$$

$$\xi_{9,333} = -\xi_{9,271}\xi_{9,273}$$

$$\xi_{9,334} = 1 - \xi_{9,272}^2$$

$$\xi_{9,335} = -\xi_{9,272}\xi_{9,273}$$

$$\xi_{9,336} = 1 - \xi_{9,273}^2$$

$$\xi_{9,337} = \xi_{9,331}^2 + \xi_{9,332}^2 + \xi_{9,333}^2$$

$$\xi_{9,338} = \xi_{9,332}^2 + \xi_{9,334}^2 + \xi_{9,335}^2$$

$$\xi_{9,339} = \xi_{9,333}^2 + \xi_{9,335}^2 + \xi_{9,336}^2$$

$$\xi_{9,340} = \sqrt{\xi_{9,337}}$$

$$\xi_{9,341} = \frac{\xi_{9,331}}{\xi_{9,340}}$$

$$\xi_{9,342} = \frac{\xi_{9,332}}{\xi_{9,340}}$$

$$\xi_{9,343} = \frac{\xi_{9,333}}{\xi_{9,340}}$$

$$\xi_{9,344} = \sqrt{\xi_{9,338}}$$

$$\xi_{9,345} = \frac{\xi_{9,334}}{\xi_{9,344}}$$

$$\xi_{9,346} = \frac{\xi_{9,332}}{\xi_{9,344}}$$

$$\xi_{9,347} = \frac{\xi_{9,335}}{\xi_{9,344}}$$

$$\xi_{9,348} = \sqrt{\xi_{9,339}}$$

$$\xi_{9,349} = \frac{\xi_{9,336}}{\xi_{9,348}}$$

$$\xi_{9,350} = \frac{\xi_{9,333}}{\xi_{9,348}}$$

$$\xi_{9,351} = \frac{\xi_{9,335}}{\xi_{9,348}}$$

$$\xi_{9,352} = \begin{cases} \xi_{9,341} & \text{for } \xi_{9,337} \geq \xi_{9,338} \wedge \xi_{9,337} \geq \xi_{9,339} \\ \xi_{9,346} & \text{for } \xi_{9,338} \geq \xi_{9,339} \wedge \xi_{9,337} \leq \xi_{9,338} \\ \xi_{9,350} & \text{otherwise} \end{cases}$$

$$\xi_{9,353} = \begin{cases} \xi_{9,342} & \text{for } \xi_{9,337} \geq \xi_{9,338} \wedge \xi_{9,337} \geq \xi_{9,339} \\ \xi_{9,345} & \text{for } \xi_{9,338} \geq \xi_{9,339} \wedge \xi_{9,337} \leq \xi_{9,338} \\ \xi_{9,351} & \text{otherwise} \end{cases}$$

$$\xi_{9,354} = \begin{cases} \xi_{9,343} & \text{for } \xi_{9,337} \geq \xi_{9,338} \wedge \xi_{9,337} \geq \xi_{9,339} \\ \xi_{9,347} & \text{for } \xi_{9,338} \geq \xi_{9,339} \wedge \xi_{9,337} \leq \xi_{9,338} \\ \xi_{9,349} & \text{otherwise} \end{cases}$$

$$\xi_{9,355} = \xi_{9,272}\xi_{9,354} - \xi_{9,273}\xi_{9,353}$$

$$\xi_{9,356} = -\xi_{9,271}\xi_{9,354} + \xi_{9,273}\xi_{9,352}$$

$$\xi_{9,357} = \xi_{9,271}\xi_{9,353} - \xi_{9,272}\xi_{9,352}$$

$$\xi_{9,358} = -\xi_{9,266}\xi_{9,273} + \xi_{9,267}\xi_{9,272}$$

$$\xi_{9,359} = \xi_{9,265}\xi_{9,273} - \xi_{9,267}\xi_{9,271}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,360} = -\xi_{9,265}\xi_{9,272} + \xi_{9,266}\xi_{9,271}$$

$$\xi_{9,361} = \xi_{9,269}\xi_{9,273} - \xi_{9,270}\xi_{9,272}$$

$$\xi_{9,362} = -\xi_{9,268}\xi_{9,273} + \xi_{9,270}\xi_{9,271}$$

$$\xi_{9,363} = \xi_{9,268}\xi_{9,272} - \xi_{9,269}\xi_{9,271}$$

$$\xi_{9,364} = \begin{cases} 1 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \\ \xi_{9,265} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,325} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,265} & \text{for } \xi_{9,264} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,352} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,265} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,361} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,265} & \text{for } \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,365} = \begin{cases} 0 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \\ \xi_{9,295} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,268} & \text{for } \xi_{9,264} = 0 \wedge (\xi_{9,262} = 0 \vee \xi_{9,264} = 0) \wedge (\xi_{9,262} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,264} = 0 \vee \xi_{9,262} \neq 0) \wedge (\xi_{9,264} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,263} \neq 0) \wedge \xi_{9,263} \neq 0 \\ \xi_{9,355} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,358} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,268} & \text{for } (\xi_{9,262} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} = 0 \vee \xi_{9,264} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,264} \neq 0) \wedge (\xi_{9,263} \neq 0 \vee \xi_{9,264} \neq 0) \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,366} = \begin{cases} 0 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \\ \xi_{9,266} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,326} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,266} & \text{for } \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,353} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,266} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,362} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,266} & \text{for } \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,367} = \begin{cases} 0 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \\ \xi_{9,266} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,326} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,266} & \text{for } \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,353} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,266} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,362} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,266} & \text{for } \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,368} = \begin{cases} 1 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \\ \xi_{9,296} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,269} & \text{for } \xi_{9,264} = 0 \wedge (\xi_{9,262} = 0 \vee \xi_{9,264} = 0) \wedge (\xi_{9,262} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,264} = 0 \vee \xi_{9,262} \neq 0) \wedge (\xi_{9,264} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,263} \neq 0) \wedge \xi_{9,263} \neq 0 \\ \xi_{9,356} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,359} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,269} & \text{for } (\xi_{9,262} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} = 0 \vee \xi_{9,264} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,264} \neq 0) \wedge (\xi_{9,263} \neq 0 \vee \xi_{9,264} \neq 0) \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,369} = \begin{cases} 0 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \\ \xi_{9,267} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,327} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,267} & \text{for } \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,354} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,267} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,363} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,267} & \text{for } \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,370} = \begin{cases} 0 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \\ \xi_{9,267} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,327} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,267} & \text{for } \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,354} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,267} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,363} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,267} & \text{for } \xi_{9,265} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,371} = \begin{cases} 0 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \\ \xi_{9,297} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,270} & \text{for } \xi_{9,264} = 0 \wedge (\xi_{9,262} = 0 \vee \xi_{9,264} = 0) \wedge (\xi_{9,262} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,264} = 0 \vee \xi_{9,262} \neq 0) \wedge (\xi_{9,264} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,263} \neq 0) \wedge \xi_{9,263} \neq 0 \\ \xi_{9,357} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,360} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,270} & \text{for } (\xi_{9,262} = 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} = 0 \vee \xi_{9,264} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,263} \neq 0) \wedge (\xi_{9,262} \neq 0 \vee \xi_{9,264} \neq 0) \wedge (\xi_{9,263} \neq 0 \vee \xi_{9,264} \neq 0) \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,372} = \begin{cases} 0 & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \\ \xi_{9,298} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \wedge \xi_{9,262} \neq 0 \\ \xi_{9,328} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,271} & \text{for } \xi_{9,264} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \\ \xi_{9,355} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,263} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,271} & \text{for } \xi_{9,263} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,364} & \text{for } \xi_{9,262} = 0 \wedge \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \\ \xi_{9,271} & \text{for } \xi_{9,265} = 0 \wedge \xi_{9,266} = 0 \wedge \xi_{9,262} \neq 0 \wedge \xi_{9,263} \neq 0 \wedge \xi_{9,264} \neq 0 \end{cases}$$

$$\xi_{9,373} = \frac{\xi_{9,364}}{4} + \frac{\xi_{9,368}}{4} + \frac{\xi_{9,372}}{4} + \frac{1}{4}$$

$$\xi_{9,374} = \frac{\xi_{9,364}}{4} - \frac{\xi_{9,368}}{4} - \frac{\xi_{9,372}}{4} + \frac{1}{4}$$

$$\xi_{9,375} = -\frac{\xi_{9,364}}{4} + \frac{\xi_{9,368}}{4} - \frac{\xi_{9,372}}{4} + \frac{1}{4}$$

$$\xi_{9,376} = -\frac{\xi_{9,364}}{4} - \frac{\xi_{9,368}}{4} + \frac{\xi_{9,372}}{4} + \frac{1}{4}$$

$$\xi_{9,377} = \sqrt{\xi_{9,373}}$$

$$\xi_{9,378} = \frac{-\xi_{9,369}+\xi_{9,371}}{4\xi_{9,377}}$$

$$\xi_{9,379} = \frac{\xi_{9,366}-\xi_{9,370}}{4\xi_{9,377}}$$

$$\xi_{9,380} = \frac{-\xi_{9,365}+\xi_{9,367}}{4\xi_{9,377}}$$

$$\xi_{9,381} = \sqrt{\xi_{9,374}}$$

$$\xi_{9,382} = \frac{-\xi_{9,369}+\xi_{9,371}}{4\xi_{9,381}}$$

$$\xi_{9,383} = \frac{\xi_{9,365}+\xi_{9,367}}{4\xi_{9,381}}$$

$$\xi_{9,384} = \frac{\xi_{9,366}+\xi_{9,370}}{4\xi_{9,381}}$$

$$\xi_{9,385} = \sqrt{\xi_{9,375}}$$

$$\xi_{9,386} = \frac{\xi_{9,366}-\xi_{9,370}}{4\xi_{9,385}}$$

$$\xi_{9,387} = \frac{\xi_{9,365}+\xi_{9,367}}{4\xi_{9,385}}$$

$$\xi_{9,388} = \frac{\xi_{9,369}+\xi_{9,371}}{4\xi_{9,385}}$$

$$\xi_{9,389} = \sqrt{\xi_{9,376}}$$

$$\xi_{9,390} = \frac{-\xi_{9,365}+\xi_{9,367}}{4\xi_{9,389}}$$

$$\xi_{9,391} = \frac{\xi_{9,366}+\xi_{9,370}}{4\xi_{9,389}}$$

$$\xi_{9,392} = \frac{\xi_{9,369}+\xi_{9,371}}{4\xi_{9,389}}$$

$$\xi_{9,393} = \begin{cases} \xi_{9,377} & \text{for } \xi_{9,373} \geq \xi_{9,374} \wedge \xi_{9,373} \geq \xi_{9,375} \wedge \xi_{9,373} \geq \xi_{9,376} \\ \xi_{9,382} & \text{for } \xi_{9,374} \geq \xi_{9,375} \wedge \xi_{9,374} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,374} \\ \xi_{9,386} & \text{for } \xi_{9,375} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,375} \wedge \xi_{9,374} \leq \xi_{9,375} \\ \xi_{9,390} & \text{otherwise} \end{cases}$$

$$\xi_{9,394} = \begin{cases} \xi_{9,378} & \text{for } \xi_{9,373} \geq \xi_{9,374} \wedge \xi_{9,373} \geq \xi_{9,375} \wedge \xi_{9,373} \geq \xi_{9,376} \\ \xi_{9,381} & \text{for } \xi_{9,374} \geq \xi_{9,375} \wedge \xi_{9,374} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,374} \\ \xi_{9,387} & \text{for } \xi_{9,375} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,375} \wedge \xi_{9,374} \leq \xi_{9,375} \\ \xi_{9,391} & \text{otherwise} \end{cases}$$

$$\xi_{9,395} = \begin{cases} \xi_{9,379} & \text{for } \xi_{9,373} \geq \xi_{9,374} \wedge \xi_{9,373} \geq \xi_{9,375} \wedge \xi_{9,373} \geq \xi_{9,376} \\ \xi_{9,383} & \text{for } \xi_{9,374} \geq \xi_{9,375} \wedge \xi_{9,374} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,374} \\ \xi_{9,385} & \text{for } \xi_{9,375} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,375} \wedge \xi_{9,374} \leq \xi_{9,375} \\ \xi_{9,392} & \text{otherwise} \end{cases}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,396} = \begin{cases} \xi_{9,380} & \text{for } \xi_{9,373} \geq \xi_{9,374} \wedge \xi_{9,373} \geq \xi_{9,375} \wedge \xi_{9,373} \geq \xi_{9,376} \\ \xi_{9,384} & \text{for } \xi_{9,374} \geq \xi_{9,375} \wedge \xi_{9,374} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,374} \\ \xi_{9,388} & \text{for } \xi_{9,375} \geq \xi_{9,376} \wedge \xi_{9,373} \leq \xi_{9,375} \wedge \xi_{9,374} \leq \xi_{9,375} \\ \xi_{9,389} & \text{otherwise} \end{cases}$$

$$\xi_{9,397} = \frac{\xi_{9,193}}{4} + \frac{\xi_{9,257}}{4} + \frac{\xi_{9,261}}{4} + \frac{1}{4}$$

$$\xi_{9,398} = \frac{\xi_{9,193}}{4} - \frac{\xi_{9,257}}{4} - \frac{\xi_{9,261}}{4} + \frac{1}{4}$$

$$\xi_{9,399} = -\frac{\xi_{9,193}}{4} + \frac{\xi_{9,257}}{4} - \frac{\xi_{9,261}}{4} + \frac{1}{4}$$

$$\xi_{9,400} = -\frac{\xi_{9,193}}{4} - \frac{\xi_{9,257}}{4} + \frac{\xi_{9,261}}{4} + \frac{1}{4}$$

$$\xi_{9,401} = \sqrt{\xi_{9,397}}$$

$$\xi_{9,402} = \frac{\xi_{9,258} - \xi_{9,260}}{4\xi_{9,401}}$$

$$\xi_{9,403} = \frac{-\xi_{9,195} + \xi_{9,259}}{4\xi_{9,401}}$$

$$\xi_{9,404} = \frac{\xi_{9,194} - \xi_{9,256}}{4\xi_{9,401}}$$

$$\xi_{9,405} = \sqrt{\xi_{9,398}}$$

$$\xi_{9,406} = \frac{\xi_{9,258} - \xi_{9,260}}{4\xi_{9,405}}$$

$$\xi_{9,407} = \frac{\xi_{9,194} + \xi_{9,256}}{4\xi_{9,405}}$$

$$\xi_{9,408} = \frac{\xi_{9,195} + \xi_{9,259}}{4\xi_{9,405}}$$

$$\xi_{9,409} = \sqrt{\xi_{9,399}}$$

$$\xi_{9,410} = \frac{-\xi_{9,195} + \xi_{9,259}}{4\xi_{9,409}}$$

$$\xi_{9,411} = \frac{\xi_{9,194} + \xi_{9,256}}{4\xi_{9,409}}$$

$$\xi_{9,412} = \frac{\xi_{9,258} + \xi_{9,260}}{4\xi_{9,409}}$$

$$\xi_{9,413} = \sqrt{\xi_{9,400}}$$

$$\xi_{9,414} = \frac{\xi_{9,194} - \xi_{9,256}}{4\xi_{9,413}}$$

$$\xi_{9,415} = \frac{\xi_{9,195} + \xi_{9,259}}{4\xi_{9,413}}$$

$$\xi_{9,416} = \frac{\xi_{9,258} + \xi_{9,260}}{4\xi_{9,413}}$$

$$\xi_{9,417} = \begin{cases} \xi_{9,401} & \text{for } \xi_{9,397} \geq \xi_{9,398} \wedge \xi_{9,397} \geq \xi_{9,399} \wedge \xi_{9,397} \geq \xi_{9,400} \\ \xi_{9,406} & \text{for } \xi_{9,398} \geq \xi_{9,399} \wedge \xi_{9,398} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,398} \\ \xi_{9,410} & \text{for } \xi_{9,399} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,399} \wedge \xi_{9,398} \leq \xi_{9,399} \\ \xi_{9,414} & \text{otherwise} \end{cases}$$

$$\xi_{9,418} = \begin{cases} \xi_{9,402} & \text{for } \xi_{9,397} \geq \xi_{9,398} \wedge \xi_{9,397} \geq \xi_{9,399} \wedge \xi_{9,397} \geq \xi_{9,400} \\ \xi_{9,405} & \text{for } \xi_{9,398} \geq \xi_{9,399} \wedge \xi_{9,398} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,398} \\ \xi_{9,411} & \text{for } \xi_{9,399} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,399} \wedge \xi_{9,398} \leq \xi_{9,399} \\ \xi_{9,415} & \text{otherwise} \end{cases}$$

$$\xi_{9,419} = \begin{cases} \xi_{9,403} & \text{for } \xi_{9,397} \geq \xi_{9,398} \wedge \xi_{9,397} \geq \xi_{9,399} \wedge \xi_{9,397} \geq \xi_{9,400} \\ \xi_{9,407} & \text{for } \xi_{9,398} \geq \xi_{9,399} \wedge \xi_{9,398} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,398} \\ \xi_{9,409} & \text{for } \xi_{9,399} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,399} \wedge \xi_{9,398} \leq \xi_{9,399} \\ \xi_{9,416} & \text{otherwise} \end{cases}$$

$$\xi_{9,420} = \begin{cases} \xi_{9,404} & \text{for } \xi_{9,397} \geq \xi_{9,398} \wedge \xi_{9,397} \geq \xi_{9,399} \wedge \xi_{9,397} \geq \xi_{9,400} \\ \xi_{9,408} & \text{for } \xi_{9,398} \geq \xi_{9,399} \wedge \xi_{9,398} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,398} \\ \xi_{9,412} & \text{for } \xi_{9,399} \geq \xi_{9,400} \wedge \xi_{9,397} \leq \xi_{9,399} \wedge \xi_{9,398} \leq \xi_{9,399} \\ \xi_{9,413} & \text{otherwise} \end{cases}$$

$$\xi_{9,421} = \xi_{9,112}\left(\xi_{9,117} + i\xi_{9,118}\right)$$

$$\xi_{9,422} = -\frac{\sqrt{2}\xi_{9,112}\xi_{9,114}}{2} + \frac{\sqrt{2}i\xi_{9,113}(\xi_{9,117}+i\xi_{9,118})}{2}$$

$$\xi_{9,423} = \frac{\sqrt{2}i\xi_{9,112}\xi_{9,114}}{2} - \frac{\sqrt{2}\xi_{9,113}(\xi_{9,117}+i\xi_{9,118})}{2}$$

$$\xi_{9,424} = -\xi_{9,113}\xi_{9,114}$$

$$\xi_{9,425} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{9,112}(\xi_{9,117}+i\xi_{9,118})}{2} - \frac{\sqrt{2}i\xi_{9,113}\xi_{9,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{9,112}(\xi_{9,117}-i\xi_{9,118})}{2} - \frac{\sqrt{2}\xi_{9,113}\xi_{9,114}}{2}\right)}{2}$$

$$\xi_{9,426} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{9,112}(\xi_{9,117}+i\xi_{9,118})}{2} - \frac{\sqrt{2}i\xi_{9,113}\xi_{9,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{9,112}(\xi_{9,117}-i\xi_{9,118})}{2} - \frac{\sqrt{2}\xi_{9,113}\xi_{9,114}}{2}\right)}{2}$$

$$\xi_{9,427} = \frac{\sqrt{2}i\xi_{9,112}\xi_{9,114}}{2} - \frac{\sqrt{2}\xi_{9,113}(\xi_{9,117}-i\xi_{9,118})}{2}$$

$$\xi_{9,428} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{9,112}(\xi_{9,117}-i\xi_{9,118})}{2} - \frac{\sqrt{2}i\xi_{9,113}\xi_{9,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{9,112}(\xi_{9,117}+i\xi_{9,118})}{2} - \frac{\sqrt{2}\xi_{9,113}\xi_{9,114}}{2}\right)}{2}$$

$$\xi_{9,429} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{9,112}(\xi_{9,117}-i\xi_{9,118})}{2} - \frac{\sqrt{2}i\xi_{9,113}\xi_{9,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{9,112}(\xi_{9,117}+i\xi_{9,118})}{2} - \frac{\sqrt{2}\xi_{9,113}\xi_{9,114}}{2}\right)}{2}$$

$$\xi_{9,430} = -\frac{\sqrt{2}\xi_{9,112}\xi_{9,114}}{2} + \frac{\sqrt{2}i\xi_{9,113}(\xi_{9,117}-i\xi_{9,118})}{2}$$

$$\xi_{9,431} = \xi_{9,112}\left(\xi_{9,117} - i\xi_{9,118}\right)$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,432} = (\xi_{9,393} - i\xi_{9,396})\,(\xi_{9,417} - i\xi_{9,420})$$

$$\xi_{9,433} = (\xi_{9,393} - i\xi_{9,396})\,(-i\xi_{9,418} - \xi_{9,419})$$

$$\xi_{9,434} = (\xi_{9,417} - i\xi_{9,420})\,(-i\xi_{9,394} - \xi_{9,395})$$

$$\xi_{9,435} = (-i\xi_{9,394} - \xi_{9,395})\,(-i\xi_{9,418} - \xi_{9,419})$$

$$\xi_{9,436} = (\xi_{9,393} - i\xi_{9,396})\,(-i\xi_{9,418} + \xi_{9,419})$$

$$\xi_{9,437} = (\xi_{9,393} - i\xi_{9,396})\,(\xi_{9,417} + i\xi_{9,420})$$

$$\xi_{9,438} = (-i\xi_{9,394} - \xi_{9,395})\,(-i\xi_{9,418} + \xi_{9,419})$$

$$\xi_{9,439} = (\xi_{9,417} + i\xi_{9,420})\,(-i\xi_{9,394} - \xi_{9,395})$$

$$\xi_{9,440} = (\xi_{9,417} - i\xi_{9,420})\,(-i\xi_{9,394} + \xi_{9,395})$$

$$\xi_{9,441} = (-i\xi_{9,394} + \xi_{9,395})\,(-i\xi_{9,418} - \xi_{9,419})$$

$$\xi_{9,442} = (\xi_{9,393} + i\xi_{9,396})\,(\xi_{9,417} - i\xi_{9,420})$$

$$\xi_{9,443} = (\xi_{9,393} + i\xi_{9,396})\,(-i\xi_{9,418} - \xi_{9,419})$$

$$\xi_{9,444} = (-i\xi_{9,394} + \xi_{9,395})\,(-i\xi_{9,418} + \xi_{9,419})$$

$$\xi_{9,445} = (\xi_{9,417} + i\xi_{9,420})\,(-i\xi_{9,394} + \xi_{9,395})$$

$$\xi_{9,446} = (\xi_{9,393} + i\xi_{9,396})\,(-i\xi_{9,418} + \xi_{9,419})$$

$$\xi_{9,447} = (\xi_{9,393} + i\xi_{9,396})\,(\xi_{9,417} + i\xi_{9,420})$$

$$\xi_{9,448} = \xi_{9,421}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}}\right) + \xi_{9,422}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}}\right) + \xi_{9,423}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}}\right) + \xi_{9,424}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,444}}\right)$$

$$\xi_{9,449} = \xi_{9,422}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}}\right) + \xi_{9,425}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}}\right) + \xi_{9,427}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,444}}\right) + \xi_{9,428}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}}\right)$$

$$\xi_{9,450} = \xi_{9,423}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}}\right) + \xi_{9,426}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}}\right) + \xi_{9,429}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}}\right) + \xi_{9,430}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,444}}\right)$$

$$\xi_{9,451} = \xi_{9,424}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}}\right) + \xi_{9,427}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}}\right) + \xi_{9,430}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}}\right) + \xi_{9,431}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,444}}\right)$$

$$\xi_{9,452} = \xi_{9,421}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}}\right) + \xi_{9,422}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}}\right) + \xi_{9,423}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}}\right) + \xi_{9,424}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,441}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}}\right)$$

$$\xi_{9,453} = \xi_{9,422}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}}\right) + \xi_{9,425}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}}\right) + \xi_{9,427}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,444}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}}\right) + \xi_{9,428}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}}\right)$$

$$\xi_{9,454} = \xi_{9,423}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}}\right) + \xi_{9,426}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}}\right) + \xi_{9,429}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}}\right) + \xi_{9,430}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,441}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}}\right)$$

$$\xi_{9,455} = \xi_{9,424}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,432}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,435}}\right) + \xi_{9,427}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,436}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,439}}\right) + \xi_{9,430}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,440}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,443}}\right) + \xi_{9,431}\left(-ie^{-i\xi_{9,105}}\sin\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,441}} + e^{-i\xi_{9,105}}\cos\left(\xi_{9,103} - \xi_{9,104}\right)\overline{\xi_{9,447}}\right)$$

$$\xi_{9,456} = \xi_{9,421}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}}\right) + \xi_{9,422}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}}\right) + \xi_{9,423}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}}\right) + \xi_{9,424}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}}\right)$$

$$\xi_{9,457} = \xi_{9,422}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}}\right) + \xi_{9,425}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}}\right) + \xi_{9,427}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}}\right) + \xi_{9,428}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}}\right)$$

$$\xi_{9,458} = \xi_{9,423}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}}\right) + \xi_{9,426}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}}\right) + \xi_{9,429}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}}\right) + \xi_{9,430}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}}\right)$$

$$\xi_{9,459} = \xi_{9,424}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}}\right) + \xi_{9,427}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}}\right) + \xi_{9,430}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}}\right) + \xi_{9,431}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}}\right)$$

$$\xi_{9,460} = \xi_{9,421}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}}\right) + \xi_{9,422}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}}\right) + \xi_{9,423}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}}\right) + \xi_{9,424}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}}\right)$$

$$\xi_{9,461} = \xi_{9,422}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}}\right) + \xi_{9,425}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}}\right) + \xi_{9,427}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}}\right) + \xi_{9,428}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}}\right)$$

$$\xi_{9,462} = \xi_{9,423}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}}\right) + \xi_{9,426}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}}\right) + \xi_{9,429}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}}\right) + \xi_{9,430}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}}\right)$$

$$\xi_{9,463} = \xi_{9,424}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,433}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,434}}\right) + \xi_{9,427}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,437}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,438}}\right) + \xi_{9,430}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,441}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,442}}\right) + \xi_{9,431}\left(-ie^{i\xi_{9,105}}\sin\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,445}} + e^{i\xi_{9,105}}\cos\left(\xi_{9,103} + \xi_{9,104}\right)\overline{\xi_{9,446}}\right)$$

$$\xi_{9,464} = \frac{\operatorname{re}\left(\xi_{9,448}\right)}{4} + \frac{\operatorname{re}\left(\xi_{9,455}\right)}{4} + \frac{\operatorname{re}\left(\xi_{9,457}\right)}{4} + \frac{\operatorname{re}\left(\xi_{9,462}\right)}{4}$$

$$\xi_{9,465} = -\frac{\operatorname{im}\left(\xi_{9,449}\right)}{4} - \frac{\operatorname{im}\left(\xi_{9,454}\right)}{4} - \frac{\operatorname{im}\left(\xi_{9,456}\right)}{4} - \frac{\operatorname{im}\left(\xi_{9,463}\right)}{4}$$

$$\xi_{9,466} = -\frac{\operatorname{re}\left(\xi_{9,449}\right)}{4} + \frac{\operatorname{re}\left(\xi_{9,454}\right)}{4} + \frac{\operatorname{re}\left(\xi_{9,456}\right)}{4} - \frac{\operatorname{re}\left(\xi_{9,463}\right)}{4}$$

$$\xi_{9,467} = -\frac{\operatorname{im}\left(\xi_{9,448}\right)}{4} + \frac{\operatorname{im}\left(\xi_{9,455}\right)}{4} + \frac{\operatorname{im}\left(\xi_{9,457}\right)}{4} - \frac{\operatorname{im}\left(\xi_{9,462}\right)}{4}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,468} = -\frac{\operatorname{im}(\xi_{9,450})}{4} - \frac{\operatorname{im}(\xi_{9,453})}{4} - \frac{\operatorname{im}(\xi_{9,459})}{4} - \frac{\operatorname{im}(\xi_{9,460})}{4}$$

$$\xi_{9,469} = -\frac{\operatorname{re}(\xi_{9,451})}{4} - \frac{\operatorname{re}(\xi_{9,452})}{4} - \frac{\operatorname{re}(\xi_{9,458})}{4} - \frac{\operatorname{re}(\xi_{9,461})}{4}$$

$$\xi_{9,470} = \frac{\operatorname{im}(\xi_{9,451})}{4} - \frac{\operatorname{im}(\xi_{9,452})}{4} - \frac{\operatorname{im}(\xi_{9,458})}{4} + \frac{\operatorname{im}(\xi_{9,461})}{4}$$

$$\xi_{9,471} = -\frac{\operatorname{re}(\xi_{9,450})}{4} + \frac{\operatorname{re}(\xi_{9,453})}{4} + \frac{\operatorname{re}(\xi_{9,459})}{4} - \frac{\operatorname{re}(\xi_{9,460})}{4}$$

$$\xi_{9,472} = -\frac{\operatorname{re}(\xi_{9,450})}{4} + \frac{\operatorname{re}(\xi_{9,453})}{4} - \frac{\operatorname{re}(\xi_{9,459})}{4} + \frac{\operatorname{re}(\xi_{9,460})}{4}$$

$$\xi_{9,473} = \frac{\operatorname{im}(\xi_{9,451})}{4} - \frac{\operatorname{im}(\xi_{9,452})}{4} + \frac{\operatorname{im}(\xi_{9,458})}{4} - \frac{\operatorname{im}(\xi_{9,461})}{4}$$

$$\xi_{9,474} = \frac{\operatorname{re}(\xi_{9,451})}{4} + \frac{\operatorname{re}(\xi_{9,452})}{4} - \frac{\operatorname{re}(\xi_{9,458})}{4} - \frac{\operatorname{re}(\xi_{9,461})}{4}$$

$$\xi_{9,475} = \frac{\operatorname{im}(\xi_{9,450})}{4} + \frac{\operatorname{im}(\xi_{9,453})}{4} - \frac{\operatorname{im}(\xi_{9,459})}{4} - \frac{\operatorname{im}(\xi_{9,460})}{4}$$

$$\xi_{9,476} = -\frac{\operatorname{im}(\xi_{9,448})}{4} + \frac{\operatorname{im}(\xi_{9,455})}{4} - \frac{\operatorname{im}(\xi_{9,457})}{4} + \frac{\operatorname{im}(\xi_{9,462})}{4}$$

$$\xi_{9,477} = -\frac{\operatorname{re}(\xi_{9,449})}{4} + \frac{\operatorname{re}(\xi_{9,454})}{4} - \frac{\operatorname{re}(\xi_{9,456})}{4} + \frac{\operatorname{re}(\xi_{9,463})}{4}$$

$$\xi_{9,478} = \frac{\operatorname{im}(\xi_{9,449})}{4} + \frac{\operatorname{im}(\xi_{9,454})}{4} - \frac{\operatorname{im}(\xi_{9,456})}{4} - \frac{\operatorname{im}(\xi_{9,463})}{4}$$

$$\xi_{9,479} = -\frac{\operatorname{re}(\xi_{9,448})}{4} - \frac{\operatorname{re}(\xi_{9,455})}{4} + \frac{\operatorname{re}(\xi_{9,457})}{4} + \frac{\operatorname{re}(\xi_{9,462})}{4}$$

$$\xi_{9,480} = \xi_{9,464}^2 + \xi_{9,468}^2 + \xi_{9,472}^2 + \xi_{9,476}^2$$

$$\xi_{9,481} = \xi_{9,465}^2 + \xi_{9,469}^2 + \xi_{9,473}^2 + \xi_{9,477}^2$$

$$\xi_{9,482} = \xi_{9,466}^2 + \xi_{9,470}^2 + \xi_{9,474}^2 + \xi_{9,478}^2$$

$$\xi_{9,483} = \xi_{9,467}^2 + \xi_{9,471}^2 + \xi_{9,475}^2 + \xi_{9,479}^2$$

$$\xi_{9,484} = \sqrt{\xi_{9,480}}$$

$$\xi_{9,485} = \frac{\xi_{9,464}}{\xi_{9,484}}$$

$$\xi_{9,486} = \frac{\xi_{9,468}}{\xi_{9,484}}$$

$$\xi_{9,487} = \frac{\xi_{9,472}}{\xi_{9,484}}$$

$$\xi_{9,488} = \frac{\xi_{9,476}}{\xi_{9,484}}$$

$$\xi_{9,489} = \sqrt{\xi_{9,481}}$$

$$\xi_{9,490} = \frac{\xi_{9,465}}{\xi_{9,489}}$$

$$\xi_{9,491} = \frac{\xi_{9,469}}{\xi_{9,489}}$$

$$\xi_{9,492} = \frac{\xi_{9,473}}{\xi_{9,489}}$$

$$\xi_{9,493} = \frac{\xi_{9,477}}{\xi_{9,489}}$$

$$\xi_{9,494} = \sqrt{\xi_{9,482}}$$

$$\xi_{9,495} = \frac{\xi_{9,466}}{\xi_{9,494}}$$

$$\xi_{9,496} = \frac{\xi_{9,470}}{\xi_{9,494}}$$

$$\xi_{9,497} = \frac{\xi_{9,474}}{\xi_{9,494}}$$

$$\xi_{9,498} = \frac{\xi_{9,478}}{\xi_{9,494}}$$

$$\xi_{9,499} = \sqrt{\xi_{9,483}}$$

$$\xi_{9,500} = \frac{\xi_{9,467}}{\xi_{9,499}}$$

$$\xi_{9,501} = \frac{\xi_{9,471}}{\xi_{9,499}}$$

$$\xi_{9,502} = \frac{\xi_{9,475}}{\xi_{9,499}}$$

$$\xi_{9,503} = \frac{\xi_{9,479}}{\xi_{9,499}}$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,504} = \begin{cases} \xi_{9,485} & \text{for } \xi_{9,480} \geq \xi_{9,481} \wedge \xi_{9,480} \geq \xi_{9,482} \wedge \xi_{9,480} \geq \xi_{9,483} \\ \xi_{9,490} & \text{for } \xi_{9,481} \geq \xi_{9,482} \wedge \xi_{9,481} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,481} \\ \xi_{9,495} & \text{for } \xi_{9,482} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,482} \wedge \xi_{9,481} \leq \xi_{9,482} \\ \xi_{9,500} & \text{otherwise} \end{cases}$$

$$\xi_{9,505} = \begin{cases} \xi_{9,486} & \text{for } \xi_{9,480} \geq \xi_{9,481} \wedge \xi_{9,480} \geq \xi_{9,482} \wedge \xi_{9,480} \geq \xi_{9,483} \\ \xi_{9,491} & \text{for } \xi_{9,481} \geq \xi_{9,482} \wedge \xi_{9,481} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,481} \\ \xi_{9,496} & \text{for } \xi_{9,482} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,482} \wedge \xi_{9,481} \leq \xi_{9,482} \\ \xi_{9,501} & \text{otherwise} \end{cases}$$

$$\xi_{9,506} = \begin{cases} \xi_{9,487} & \text{for } \xi_{9,480} \geq \xi_{9,481} \wedge \xi_{9,480} \geq \xi_{9,482} \wedge \xi_{9,480} \geq \xi_{9,483} \\ \xi_{9,492} & \text{for } \xi_{9,481} \geq \xi_{9,482} \wedge \xi_{9,481} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,481} \\ \xi_{9,497} & \text{for } \xi_{9,482} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,482} \wedge \xi_{9,481} \leq \xi_{9,482} \\ \xi_{9,502} & \text{otherwise} \end{cases}$$

$$\xi_{9,507} = \begin{cases} \xi_{9,488} & \text{for } \xi_{9,480} \geq \xi_{9,481} \wedge \xi_{9,480} \geq \xi_{9,482} \wedge \xi_{9,480} \geq \xi_{9,483} \\ \xi_{9,493} & \text{for } \xi_{9,481} \geq \xi_{9,482} \wedge \xi_{9,481} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,481} \\ \xi_{9,498} & \text{for } \xi_{9,482} \geq \xi_{9,483} \wedge \xi_{9,480} \leq \xi_{9,482} \wedge \xi_{9,481} \leq \xi_{9,482} \\ \xi_{9,503} & \text{otherwise} \end{cases}$$

$$\xi_{9,508} = \xi_{9,464}\xi_{9,504} + \xi_{9,468}\xi_{9,505} + \xi_{9,472}\xi_{9,506} + \xi_{9,476}\xi_{9,507}$$

$$\xi_{9,509} = \xi_{9,465}\xi_{9,504} + \xi_{9,469}\xi_{9,505} + \xi_{9,473}\xi_{9,506} + \xi_{9,477}\xi_{9,507}$$

$$\xi_{9,510} = \xi_{9,466}\xi_{9,504} + \xi_{9,470}\xi_{9,505} + \xi_{9,474}\xi_{9,506} + \xi_{9,478}\xi_{9,507}$$

$$\xi_{9,511} = \xi_{9,467}\xi_{9,504} + \xi_{9,471}\xi_{9,505} + \xi_{9,475}\xi_{9,506} + \xi_{9,479}\xi_{9,507}$$

$$\xi_{9,512} = \xi_{9,41}\xi_{9,52} - \xi_{9,42}\xi_{9,55} - \xi_{9,43}\xi_{9,56} - \xi_{9,44}\xi_{9,57}$$

$$\xi_{9,513} = \xi_{9,41}\xi_{9,55} + \xi_{9,42}\xi_{9,52} - \xi_{9,43}\xi_{9,57} + \xi_{9,44}\xi_{9,56}$$

$$\xi_{9,514} = \xi_{9,41}\xi_{9,56} + \xi_{9,42}\xi_{9,57} + \xi_{9,43}\xi_{9,52} - \xi_{9,44}\xi_{9,55}$$

$$\xi_{9,515} = \xi_{9,41}\xi_{9,57} - \xi_{9,42}\xi_{9,56} + \xi_{9,43}\xi_{9,55} + \xi_{9,44}\xi_{9,52}$$

$$\xi_{9,516} = \xi_{9,10}\xi_{9,55} + \xi_{9,11}\xi_{9,55} + \xi_{9,12}\xi_{9,55} + \xi_{9,54}\xi_{9,9}$$

$$\xi_{9,517} = -\xi_{9,10}\xi_{9,54} + \xi_{9,11}\xi_{9,55} - \xi_{9,12}\xi_{9,55} + \xi_{9,55}\xi_{9,9}$$

$$\xi_{9,518} = -\xi_{9,10}\xi_{9,55} - \xi_{9,11}\xi_{9,54} + \xi_{9,12}\xi_{9,55} + \xi_{9,55}\xi_{9,9}$$

$$\xi_{9,519} = \xi_{9,10}\xi_{9,55} - \xi_{9,11}\xi_{9,55} - \xi_{9,12}\xi_{9,54} + \xi_{9,55}\xi_{9,9}$$

$$\xi_{9,520} = \xi_{9,52}\left(\xi_{9,100}\xi_{9,505} + \xi_{9,101}\xi_{9,506} + \xi_{9,102}\xi_{9,507} + \xi_{9,504}\xi_{9,99}\right) - \xi_{9,52}\left(\xi_{9,100}\xi_{9,506} - \xi_{9,101}\xi_{9,505} + \xi_{9,102}\xi_{9,504} - \xi_{9,507}\xi_{9,99}\right) - \xi_{9,53}\left(\xi_{9,100}\xi_{9,504} + \xi_{9,101}\xi_{9,507} - \xi_{9,102}\xi_{9,506} - \xi_{9,505}\xi_{9,99}\right) - \xi_{9,54}\left(-\xi_{9,100}\xi_{9,507} + \xi_{9,101}\xi_{9,504} + \xi_{9,102}\xi_{9,505} - \xi_{9,506}\xi_{9,99}\right)$$

$$\xi_{9,521} = \xi_{9,52}\left(\xi_{9,100}\xi_{9,504} + \xi_{9,101}\xi_{9,507} - \xi_{9,102}\xi_{9,506} - \xi_{9,505}\xi_{9,99}\right) + \xi_{9,52}\left(-\xi_{9,100}\xi_{9,507} + \xi_{9,101}\xi_{9,504} + \xi_{9,102}\xi_{9,505} - \xi_{9,506}\xi_{9,99}\right) + \xi_{9,53}\left(\xi_{9,100}\xi_{9,505} + \xi_{9,101}\xi_{9,506} + \xi_{9,102}\xi_{9,507} + \xi_{9,504}\xi_{9,99}\right) - \xi_{9,54}\left(\xi_{9,100}\xi_{9,506} - \xi_{9,101}\xi_{9,505} + \xi_{9,102}\xi_{9,504} - \xi_{9,507}\xi_{9,99}\right)$$

$$\xi_{9,522} = -\xi_{9,52}\left(\xi_{9,100}\xi_{9,504} + \xi_{9,101}\xi_{9,507} - \xi_{9,102}\xi_{9,506} - \xi_{9,505}\xi_{9,99}\right) + \xi_{9,52}\left(-\xi_{9,100}\xi_{9,507} + \xi_{9,101}\xi_{9,504} + \xi_{9,102}\xi_{9,505} - \xi_{9,506}\xi_{9,99}\right) + \xi_{9,53}\left(\xi_{9,100}\xi_{9,506} - \xi_{9,101}\xi_{9,505} + \xi_{9,102}\xi_{9,501} - \xi_{9,507}\xi_{9,99}\right) + \xi_{9,54}\left(\xi_{9,100}\xi_{9,505} + \xi_{9,101}\xi_{9,506} + \xi_{9,102}\xi_{9,507} + \xi_{9,504}\xi_{9,99}\right)$$

$$\xi_{9,523} = \xi_{9,52}\left(\xi_{9,100}\xi_{9,505} + \xi_{9,101}\xi_{9,506} + \xi_{9,102}\xi_{9,507} + \xi_{9,504}\xi_{9,99}\right) + \xi_{9,52}\left(\xi_{9,100}\xi_{9,506} - \xi_{9,101}\xi_{9,505} + \xi_{9,102}\xi_{9,504} - \xi_{9,507}\xi_{9,99}\right) - \xi_{9,53}\left(-\xi_{9,100}\xi_{9,507} + \xi_{9,101}\xi_{9,504} + \xi_{9,102}\xi_{9,505} - \xi_{9,506}\xi_{9,99}\right) + \xi_{9,54}\left(\xi_{9,100}\xi_{9,504} + \xi_{9,101}\xi_{9,507} - \xi_{9,102}\xi_{9,506} - \xi_{9,505}\xi_{9,99}\right)$$

$$\xi_{9,524} = -\xi_{9,53}\left(-\xi_{9,508}\xi_{9,68} - \xi_{9,509}\xi_{9,67} + \xi_{9,510}\xi_{9,70} - \xi_{9,511}\xi_{9,69}\right) - \xi_{9,53}\left(-\xi_{9,508}\xi_{9,69} - \xi_{9,509}\xi_{9,70} - \xi_{9,510}\xi_{9,67} + \xi_{9,511}\xi_{9,68}\right) - \xi_{9,53}\left(-\xi_{9,508}\xi_{9,70} + \xi_{9,509}\xi_{9,69} - \xi_{9,510}\xi_{9,68} - \xi_{9,511}\xi_{9,67}\right) + \xi_{9,54}\left(\xi_{9,508}\xi_{9,67} - \xi_{9,509}\xi_{9,68} - \xi_{9,510}\xi_{9,69} - \xi_{9,511}\xi_{9,70}\right)$$

$$\xi_{9,525} = \xi_{9,53}\left(\xi_{9,508}\xi_{9,67} - \xi_{9,509}\xi_{9,68} - \xi_{9,510}\xi_{9,69} - \xi_{9,511}\xi_{9,70}\right) + \xi_{9,53}\left(-\xi_{9,508}\xi_{9,69} - \xi_{9,509}\xi_{9,70} - \xi_{9,510}\xi_{9,67} + \xi_{9,511}\xi_{9,68}\right) - \xi_{9,53}\left(-\xi_{9,508}\xi_{9,70} + \xi_{9,509}\xi_{9,69} - \xi_{9,510}\xi_{9,68} - \xi_{9,511}\xi_{9,67}\right) + \xi_{9,54}\left(-\xi_{9,508}\xi_{9,68} - \xi_{9,509}\xi_{9,67} + \xi_{9,510}\xi_{9,70} - \xi_{9,511}\xi_{9,69}\right)$$

$$\xi_{9,526} = \xi_{9,53}\left(\xi_{9,508}\xi_{9,67} - \xi_{9,509}\xi_{9,68} - \xi_{9,510}\xi_{9,69} - \xi_{9,511}\xi_{9,70}\right) - \xi_{9,53}\left(-\xi_{9,508}\xi_{9,68} - \xi_{9,509}\xi_{9,67} + \xi_{9,510}\xi_{9,70} - \xi_{9,511}\xi_{9,69}\right) + \xi_{9,53}\left(-\xi_{9,508}\xi_{9,70} + \xi_{9,509}\xi_{9,69} - \xi_{9,510}\xi_{9,68} - \xi_{9,511}\xi_{9,67}\right) + \xi_{9,54}\left(-\xi_{9,508}\xi_{9,69} - \xi_{9,509}\xi_{9,70} - \xi_{9,510}\xi_{9,67} + \xi_{9,511}\xi_{9,68}\right)$$

$$\xi_{9,527} = \xi_{9,53}\left(\xi_{9,508}\xi_{9,67} - \xi_{9,509}\xi_{9,68} - \xi_{9,510}\xi_{9,69} - \xi_{9,511}\xi_{9,70}\right) + \xi_{9,53}\left(-\xi_{9,508}\xi_{9,68} - \xi_{9,509}\xi_{9,67} + \xi_{9,510}\xi_{9,70} - \xi_{9,511}\xi_{9,69}\right) - \xi_{9,53}\left(-\xi_{9,508}\xi_{9,69} - \xi_{9,509}\xi_{9,70} - \xi_{9,510}\xi_{9,67} + \xi_{9,511}\xi_{9,68}\right) + \xi_{9,54}\left(-\xi_{9,508}\xi_{9,70} + \xi_{9,509}\xi_{9,69} - \xi_{9,510}\xi_{9,68} - \xi_{9,511}\xi_{9,67}\right)$$

$$\xi_{9,528} = \xi_{9,393}\left(\xi_{9,33}\xi_{9,91} - \xi_{9,34}\xi_{9,92} - \xi_{9,35}\xi_{9,93} - \xi_{9,36}\xi_{9,94}\right) + \xi_{9,394}\left(\xi_{9,33}\xi_{9,92} + \xi_{9,34}\xi_{9,91} + \xi_{9,35}\xi_{9,94} - \xi_{9,36}\xi_{9,93}\right) + \xi_{9,395}\left(\xi_{9,33}\xi_{9,93} - \xi_{9,34}\xi_{9,94} + \xi_{9,35}\xi_{9,91} + \xi_{9,36}\xi_{9,92}\right) + \xi_{9,396}\left(\xi_{9,33}\xi_{9,94} + \xi_{9,34}\xi_{9,93} - \xi_{9,35}\xi_{9,92} + \xi_{9,36}\xi_{9,91}\right)$$

$$\xi_{9,529} = \xi_{9,393}\left(\xi_{9,33}\xi_{9,92} + \xi_{9,34}\xi_{9,91} + \xi_{9,35}\xi_{9,94} - \xi_{9,36}\xi_{9,93}\right) - \xi_{9,394}\left(\xi_{9,33}\xi_{9,91} - \xi_{9,34}\xi_{9,92} - \xi_{9,35}\xi_{9,93} - \xi_{9,36}\xi_{9,94}\right) + \xi_{9,395}\left(\xi_{9,33}\xi_{9,94} + \xi_{9,34}\xi_{9,93} - \xi_{9,35}\xi_{9,92} + \xi_{9,36}\xi_{9,91}\right) - \xi_{9,396}\left(\xi_{9,33}\xi_{9,93} - \xi_{9,34}\xi_{9,94} + \xi_{9,35}\xi_{9,91} + \xi_{9,36}\xi_{9,92}\right)$$

$$\xi_{9,530} = \xi_{9,393}\left(\xi_{9,33}\xi_{9,93} - \xi_{9,34}\xi_{9,94} + \xi_{9,35}\xi_{9,91} + \xi_{9,36}\xi_{9,92}\right) - \xi_{9,394}\left(\xi_{9,33}\xi_{9,94} + \xi_{9,34}\xi_{9,93} - \xi_{9,35}\xi_{9,92} + \xi_{9,36}\xi_{9,91}\right) - \xi_{9,395}\left(\xi_{9,33}\xi_{9,91} - \xi_{9,34}\xi_{9,92} - \xi_{9,35}\xi_{9,93} - \xi_{9,36}\xi_{9,94}\right) + \xi_{9,396}\left(\xi_{9,33}\xi_{9,92} + \xi_{9,34}\xi_{9,91} + \xi_{9,35}\xi_{9,94} - \xi_{9,36}\xi_{9,93}\right)$$

$$\xi_{9,531} = \xi_{9,393}\left(\xi_{9,33}\xi_{9,94} + \xi_{9,34}\xi_{9,93} - \xi_{9,35}\xi_{9,92} + \xi_{9,36}\xi_{9,91}\right) + \xi_{9,394}\left(\xi_{9,33}\xi_{9,93} - \xi_{9,34}\xi_{9,94} + \xi_{9,35}\xi_{9,91} + \xi_{9,36}\xi_{9,92}\right) - \xi_{9,395}\left(\xi_{9,33}\xi_{9,92} + \xi_{9,34}\xi_{9,91} + \xi_{9,35}\xi_{9,94} - \xi_{9,36}\xi_{9,93}\right) - \xi_{9,396}\left(\xi_{9,33}\xi_{9,91} - \xi_{9,34}\xi_{9,92} - \xi_{9,35}\xi_{9,93} - \xi_{9,36}\xi_{9,94}\right)$$

$$\xi_{9,532} = \xi_{9,417}\left(\xi_{9,37}\xi_{9,95} - \xi_{9,38}\xi_{9,96} - \xi_{9,39}\xi_{9,97} - \xi_{9,40}\xi_{9,98}\right) + \xi_{9,418}\left(\xi_{9,37}\xi_{9,96} + \xi_{9,38}\xi_{9,95} + \xi_{9,39}\xi_{9,98} - \xi_{9,40}\xi_{9,97}\right) + \xi_{9,419}\left(\xi_{9,37}\xi_{9,97} - \xi_{9,38}\xi_{9,98} + \xi_{9,39}\xi_{9,95} + \xi_{9,40}\xi_{9,96}\right) + \xi_{9,420}\left(\xi_{9,37}\xi_{9,98} + \xi_{9,38}\xi_{9,97} - \xi_{9,39}\xi_{9,96} + \xi_{9,40}\xi_{9,95}\right)$$

$$\xi_{9,533} = \xi_{9,417}\left(\xi_{9,37}\xi_{9,96} + \xi_{9,38}\xi_{9,95} + \xi_{9,39}\xi_{9,98} - \xi_{9,40}\xi_{9,97}\right) - \xi_{9,418}\left(\xi_{9,37}\xi_{9,95} - \xi_{9,38}\xi_{9,96} - \xi_{9,39}\xi_{9,97} - \xi_{9,40}\xi_{9,98}\right) + \xi_{9,419}\left(\xi_{9,37}\xi_{9,98} + \xi_{9,38}\xi_{9,97} - \xi_{9,39}\xi_{9,96} + \xi_{9,40}\xi_{9,95}\right) - \xi_{9,420}\left(\xi_{9,37}\xi_{9,97} - \xi_{9,38}\xi_{9,98} + \xi_{9,39}\xi_{9,95} + \xi_{9,40}\xi_{9,96}\right)$$

$$\xi_{9,534} = \xi_{9,417}\left(\xi_{9,37}\xi_{9,97} - \xi_{9,38}\xi_{9,98} + \xi_{9,39}\xi_{9,95} + \xi_{9,40}\xi_{9,96}\right) - \xi_{9,418}\left(\xi_{9,37}\xi_{9,98} + \xi_{9,38}\xi_{9,97} - \xi_{9,39}\xi_{9,96} + \xi_{9,40}\xi_{9,95}\right) - \xi_{9,419}\left(\xi_{9,37}\xi_{9,95} - \xi_{9,38}\xi_{9,96} - \xi_{9,39}\xi_{9,97} - \xi_{9,40}\xi_{9,98}\right) + \xi_{9,420}\left(\xi_{9,37}\xi_{9,96} + \xi_{9,38}\xi_{9,95} + \xi_{9,39}\xi_{9,98} - \xi_{9,40}\xi_{9,97}\right)$$

$$\xi_{9,535} = \xi_{9,417}\left(\xi_{9,37}\xi_{9,98} + \xi_{9,38}\xi_{9,97} - \xi_{9,39}\xi_{9,96} + \xi_{9,40}\xi_{9,95}\right) + \xi_{9,418}\left(\xi_{9,37}\xi_{9,97} - \xi_{9,38}\xi_{9,98} + \xi_{9,39}\xi_{9,95} + \xi_{9,40}\xi_{9,96}\right) - \xi_{9,419}\left(\xi_{9,37}\xi_{9,96} + \xi_{9,38}\xi_{9,95} + \xi_{9,39}\xi_{9,98} - \xi_{9,40}\xi_{9,97}\right) - \xi_{9,420}\left(\xi_{9,37}\xi_{9,95} - \xi_{9,38}\xi_{9,96} - \xi_{9,39}\xi_{9,97} - \xi_{9,40}\xi_{9,98}\right)$$

$$\xi_{9,536} = \xi_{9,106} + \xi_{9,48}$$

$$\xi_{9,537} = \begin{cases} 0 & \text{for } \xi_{9,512}^2 + \xi_{9,515}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,515}, \xi_{9,512}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,538} = \begin{cases} 0 & \text{for } \xi_{9,513}^2 + \xi_{9,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,513}, \xi_{9,514}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,539} = 2\operatorname{atan}_2\left(\sqrt{\xi_{9,513}^2 + \xi_{9,514}^2}, \sqrt{\xi_{9,512}^2 + \xi_{9,515}^2}\right)$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,540} = -\xi_{9,537} + \xi_{9,538}$$

$$\xi_{9,541} = -\xi_{9,537} - \xi_{9,538}$$

$$\xi_{9,542} = \xi_{9,536} + \xi_{9,537}$$

$$\xi_{9,543} = \begin{cases} 0 & \text{for } \xi_{9,516}^2 + \xi_{9,519}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,519}, \xi_{9,516}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,544} = \begin{cases} 0 & \text{for } \xi_{9,517}^2 + \xi_{9,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,517}, \xi_{9,518}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,545} = 2\operatorname{atan}_2\left(\sqrt{\xi_{9,517}^2 + \xi_{9,518}^2}, \sqrt{\xi_{9,516}^2 + \xi_{9,519}^2}\right)$$

$$\xi_{9,546} = -\xi_{9,543} + \xi_{9,544}$$

$$\xi_{9,547} = -\xi_{9,543} - \xi_{9,544}$$

$$\xi_{9,548} = \xi_{9,542} + \xi_{9,543}$$

$$\xi_{9,549} = \begin{cases} 0 & \text{for } \xi_{9,520}^2 + \xi_{9,523}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,523}, \xi_{9,520}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,550} = \begin{cases} 0 & \text{for } \xi_{9,521}^2 + \xi_{9,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,521}, \xi_{9,522}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,551} = 2\operatorname{atan}_2\left(\sqrt{\xi_{9,521}^2 + \xi_{9,522}^2}, \sqrt{\xi_{9,520}^2 + \xi_{9,523}^2}\right)$$

$$\xi_{9,552} = -\xi_{9,549} + \xi_{9,550}$$

$$\xi_{9,553} = -\xi_{9,549} - \xi_{9,550}$$

$$\xi_{9,554} = \xi_{9,548} + \xi_{9,549}$$

$$\xi_{9,555} = \begin{cases} 0 & \text{for } \xi_{9,524}^2 + \xi_{9,527}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,527}, \xi_{9,524}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,556} = \begin{cases} 0 & \text{for } \xi_{9,525}^2 + \xi_{9,526}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,525}, \xi_{9,526}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,557} = 2\operatorname{atan}_2\left(\sqrt{\xi_{9,525}^2 + \xi_{9,526}^2}, \sqrt{\xi_{9,524}^2 + \xi_{9,527}^2}\right)$$

$$\xi_{9,558} = -\xi_{9,555} + \xi_{9,556}$$

$$\xi_{9,559} = -\xi_{9,555} - \xi_{9,556}$$

$$\xi_{9,560} = \xi_{9,554} + \xi_{9,555}$$

$$\xi_{9,561} = \begin{cases} 0 & \text{for } \xi_{9,117}^2 + \xi_{9,118}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{9,118}, \xi_{9,117}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,562} = \begin{cases} 0 & \text{for } \xi_{9,114}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{9,114}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,563} = 2\operatorname{atan}_2\left(\left|\xi_{9,114}\right|, \sqrt{\xi_{9,117}^2 + \xi_{9,118}^2}\right)$$

$$\xi_{9,564} = -\xi_{9,561} + \xi_{9,562}$$

$$\xi_{9,565} = -\xi_{9,561} - \xi_{9,562}$$

$$\xi_{9,566} = \xi_{9,560} + \xi_{9,561}$$

$$\xi_{9,567} = \begin{cases} 0 & \text{for } \xi_{9,112}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{9,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,568} = \begin{cases} 0 & \text{for } \xi_{9,113}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{9,113}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,569} = 2\operatorname{atan}_2\left(\left|\xi_{9,113}\right|, \left|\xi_{9,112}\right|\right)$$

$$\xi_{9,570} = -\xi_{9,567} + \xi_{9,568}$$

$$\xi_{9,571} = -\xi_{9,567} - \xi_{9,568}$$

$$\xi_{9,572} = \xi_{9,566} + \xi_{9,567}$$

$$\xi_{9,573} = \begin{cases} 0 & \text{for } \xi_{9,528}^2 + \xi_{9,531}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,531}, \xi_{9,528}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,574} = \begin{cases} 0 & \text{for } \xi_{9,529}^2 + \xi_{9,530}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,529}, \xi_{9,530}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,575} = 2\operatorname{atan}_2\left(\sqrt{\xi_{9,529}^2 + \xi_{9,530}^2}, \sqrt{\xi_{9,528}^2 + \xi_{9,531}^2}\right)$$

Exact shared-expression program 9

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{9,576} = -\xi_{9,573} + \xi_{9,574}$$

$$\xi_{9,577} = -\xi_{9,573} - \xi_{9,574}$$

$$\xi_{9,578} = \xi_{9,572} + \xi_{9,573}$$

$$\xi_{9,579} = \begin{cases} 0 & \text{for } \xi_{9,532}^2 + \xi_{9,535}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,535}, \xi_{9,532}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,580} = \begin{cases} 0 & \text{for } \xi_{9,533}^2 + \xi_{9,534}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{9,533}, \xi_{9,534}\right) & \text{otherwise} \end{cases}$$

$$\xi_{9,581} = 2\operatorname{atan}_2\left(\sqrt{\xi_{9,533}^2 + \xi_{9,534}^2}, \sqrt{\xi_{9,532}^2 + \xi_{9,535}^2}\right)$$

$$\xi_{9,582} = -\xi_{9,579} + \xi_{9,580}$$

$$\xi_{9,583} = -\xi_{9,579} - \xi_{9,580}$$

$$\xi_{9,584} = \xi_{9,578} + \xi_{9,579}$$

$$\xi_{9,585} = -\frac{\xi_{9,539}}{2} - \frac{\xi_{9,545}}{2} - \frac{\xi_{9,551}}{2} - \frac{\xi_{9,557}}{2} - \frac{\xi_{9,563}}{2} - \frac{\xi_{9,569}}{2} - \frac{\xi_{9,575}}{2} - \frac{\xi_{9,581}}{2} + \xi_{9,584}$$

Derived parameters for program 9

Exact output expressions used by the native gate labels.

$$t_{9,0} = \frac{2\xi_{9,539}+2\xi_{9,585}+\pi\left(-\frac{\xi_{9,541}}{\pi}+\frac{\xi_{9,540}+\xi_{9,541}}{\pi}+\frac{1}{2}\right)-\frac{3\pi}{2}}{\pi}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,0} = \left\lfloor \frac{1}{2} - a \right\rfloor$$

$$\xi_{10,1} = \left\lfloor \frac{1}{2} - b \right\rfloor$$

$$\xi_{10,2} = \left\lfloor \frac{1}{2} - c \right\rfloor$$

$$\xi_{10,3} = -\frac{\pi \xi_{10,0}}{2} - \frac{\pi a}{2}$$

$$\xi_{10,4} = -\frac{\pi \xi_{10,1}}{2} - \frac{\pi b}{2}$$

$$\xi_{10,5} = -\frac{\pi \xi_{10,2}}{2} - \frac{\pi c}{2}$$

$$\xi_{10,6} = \begin{cases} \xi_{10,3} & \text{for } (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \\ \xi_{10,4} & \text{for } (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,3}| \leq |\xi_{10,4}|) \wedge (|\xi_{10,3}| \leq |\xi_{10,4}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \\ \xi_{10,5} & \text{otherwise} \end{cases}$$

$$\xi_{10,7} = \begin{cases} \xi_{10,4} & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,4}| \geq |\xi_{10,5}| \\ \xi_{10,5} & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge |\xi_{10,4}| \leq |\xi_{10,5}| \\ \xi_{10,3} & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge |\xi_{10,3}| \leq |\xi_{10,4}| \\ \xi_{10,5} & \text{for } |\xi_{10,4}| \geq |\xi_{10,5}| \wedge |\xi_{10,3}| \leq |\xi_{10,5}| \\ \xi_{10,3} & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,3}| \leq |\xi_{10,5}| \\ \xi_{10,4} & \text{otherwise} \end{cases}$$

$$\xi_{10,8} = \begin{cases} \xi_{10,5} & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,4}| \geq |\xi_{10,5}| \\ \xi_{10,4} & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge |\xi_{10,4}| \leq |\xi_{10,5}| \\ \xi_{10,5} & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge |\xi_{10,3}| \leq |\xi_{10,4}| \\ \xi_{10,3} & \text{for } |\xi_{10,4}| \geq |\xi_{10,5}| \wedge |\xi_{10,3}| \leq |\xi_{10,5}| \\ \xi_{10,4} & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,3}| \leq |\xi_{10,5}| \\ \xi_{10,3} & \text{otherwise} \end{cases}$$

$$\xi_{10,9} = \begin{cases} 1 & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,4}| \geq |\xi_{10,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge |\xi_{10,4}| \leq |\xi_{10,5}| \\ 0 & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge |\xi_{10,3}| \leq |\xi_{10,4}| \\ \frac{1}{2} & \text{for } |\xi_{10,3}| \leq |\xi_{10,5}| \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{10,10} = \begin{cases} 0 & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,4}| \geq |\xi_{10,5}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,3}| \leq |\xi_{10,4}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \leq |\xi_{10,4}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \\ \frac{1}{2} & \text{for } |\xi_{10,3}| \leq |\xi_{10,5}| \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,11} = \begin{cases} 0 & \text{for } (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{10,3}| \geq |\xi_{10,5}| \wedge |\xi_{10,3}| \leq |\xi_{10,4}| \\ \frac{1}{2} & \text{for } |\xi_{10,3}| \leq |\xi_{10,5}| \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,3}| \leq |\xi_{10,5}|) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{10,12} = \begin{cases} 0 & \text{for } (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,3}| \leq |\xi_{10,4}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,4}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,3}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \geq |\xi_{10,5}|) \wedge (|\xi_{10,4}| \geq |\xi_{10,5}| \vee |\xi_{10,4}| \leq |\xi_{10,5}|) \\ \frac{1}{2} & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,4}| \leq |\xi_{10,5}| \\ -\frac{1}{2} & \text{for } |\xi_{10,3}| \geq |\xi_{10,4}| \wedge |\xi_{10,4}| \leq |\xi_{10,5}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,13} = \begin{cases} -1 & \text{for } \xi_{10,6} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{10,14} = \begin{cases} -1 & \text{for } \xi_{10,7} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{10,15} = \begin{cases} 1 & \text{for } \xi_{10,13} = 1 \wedge \xi_{10,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,16} = \begin{cases} 0 & \text{for } \xi_{10,13} = 1 \wedge \xi_{10,14} = 1 \\ 1 & \text{for } \xi_{10,13} = 1 \wedge \xi_{10,14} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,17} = \begin{cases} 0 & \text{for } \xi_{10,13} = 1 \wedge (\xi_{10,13} = 1 \vee \xi_{10,14} = -1) \wedge (\xi_{10,13} = 1 \vee \xi_{10,14} = 1) \wedge (\xi_{10,14} = -1 \vee \xi_{10,14} = 1) \\ 1 & \text{for } \xi_{10,13} = -1 \wedge \xi_{10,14} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,18} = \begin{cases} 0 & \text{for } (\xi_{10,13} = -1 \vee \xi_{10,13} = 1) \wedge (\xi_{10,13} = 1 \vee \xi_{10,14} = 1) \wedge (\xi_{10,14} = -1 \vee \xi_{10,14} = 1) \wedge (\xi_{10,13} = -1 \vee \xi_{10,13} = 1 \vee \xi_{10,14} = -1) \wedge (\xi_{10,13} = -1 \vee \xi_{10,13} = 1 \vee \xi_{10,14} = 1) \wedge (\xi_{10,13} = -1 \vee \xi_{10,14} = -1 \vee \xi_{10,14} = 1) \wedge (\xi_{10,13} = 1 \vee \xi_{10,14} = -1 \vee \xi_{10,14} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{10,19} = \cos\left(\frac{\pi \xi_{10,0}}{2}\right)$$

$$\xi_{10,20} = -\sin\left(\frac{\pi \xi_{10,0}}{2}\right)$$

$$\xi_{10,21} = \cos\left(\frac{\pi \xi_{10,1}}{2}\right)$$

$$\xi_{10,22} = -\sin\left(\frac{\pi \xi_{10,1}}{2}\right)$$

$$\xi_{10,23} = \xi_{10,19}\xi_{10,21}$$

$$\xi_{10,24} = \xi_{10,20}\xi_{10,21}$$

$$\xi_{10,25} = \xi_{10,19}\xi_{10,22}$$

$$\xi_{10,26} = -\xi_{10,20}\xi_{10,22}$$

$$\xi_{10,27} = \cos\left(\frac{\pi \xi_{10,2}}{2}\right)$$

$$\xi_{10,28} = -\sin\left(\frac{\pi \xi_{10,2}}{2}\right)$$

$$\xi_{10,29} = \xi_{10,23}\xi_{10,27} - \xi_{10,26}\xi_{10,28}$$

$$\xi_{10,30} = \xi_{10,24}\xi_{10,27} - \xi_{10,25}\xi_{10,28}$$

$$\xi_{10,31} = \xi_{10,24}\xi_{10,28} + \xi_{10,25}\xi_{10,27}$$

$$\xi_{10,32} = \xi_{10,23}\xi_{10,28} + \xi_{10,26}\xi_{10,27}$$

$$\xi_{10,33} = \xi_{10,15}\left(-\xi_{10,10}\xi_{10,30} - \xi_{10,11}\xi_{10,31} - \xi_{10,12}\xi_{10,32} + \xi_{10,29}\xi_{10,9}\right) - \xi_{10,16}\left(\xi_{10,10}\xi_{10,29} - \xi_{10,11}\xi_{10,32} + \xi_{10,12}\xi_{10,31} + \xi_{10,30}\xi_{10,9}\right) - \xi_{10,17}\left(\xi_{10,10}\xi_{10,32} + \xi_{10,11}\xi_{10,29} - \xi_{10,12}\xi_{10,30} + \xi_{10,31}\xi_{10,9}\right) + \xi_{10,18}\left(-\xi_{10,10}\xi_{10,31} + \xi_{10,11}\xi_{10,30} + \xi_{10,12}\xi_{10,29} + \xi_{10,32}\xi_{10,9}\right)$$

$$\xi_{10,34} = \xi_{10,15}\left(\xi_{10,10}\xi_{10,29} - \xi_{10,11}\xi_{10,32} + \xi_{10,12}\xi_{10,31} + \xi_{10,30}\xi_{10,9}\right) + \xi_{10,16}\left(-\xi_{10,10}\xi_{10,30} - \xi_{10,11}\xi_{10,31} - \xi_{10,12}\xi_{10,32} + \xi_{10,29}\xi_{10,9}\right) - \xi_{10,17}\left(-\xi_{10,10}\xi_{10,31} + \xi_{10,11}\xi_{10,30} + \xi_{10,12}\xi_{10,29} + \xi_{10,32}\xi_{10,9}\right) + \xi_{10,18}\left(\xi_{10,10}\xi_{10,32} + \xi_{10,11}\xi_{10,29} - \xi_{10,12}\xi_{10,30} + \xi_{10,31}\xi_{10,9}\right)$$

$$\xi_{10,35} = \xi_{10,15}\left(\xi_{10,10}\xi_{10,32} + \xi_{10,11}\xi_{10,29} - \xi_{10,12}\xi_{10,30} + \xi_{10,31}\xi_{10,9}\right) + \xi_{10,16}\left(-\xi_{10,10}\xi_{10,31} + \xi_{10,11}\xi_{10,30} + \xi_{10,12}\xi_{10,29} + \xi_{10,32}\xi_{10,9}\right) + \xi_{10,17}\left(-\xi_{10,10}\xi_{10,30} - \xi_{10,11}\xi_{10,31} - \xi_{10,12}\xi_{10,32} + \xi_{10,29}\xi_{10,9}\right) - \xi_{10,18}\left(\xi_{10,10}\xi_{10,29} - \xi_{10,11}\xi_{10,32} + \xi_{10,12}\xi_{10,31} + \xi_{10,30}\xi_{10,9}\right)$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,36} = \xi_{10,15}\left(-\xi_{10,10}\xi_{10,31} + \xi_{10,11}\xi_{10,30} + \xi_{10,12}\xi_{10,29} + \xi_{10,32}\xi_{10,9}\right) - \xi_{10,16}\left(\xi_{10,10}\xi_{10,32} + \xi_{10,11}\xi_{10,29} - \xi_{10,12}\xi_{10,30} + \xi_{10,31}\xi_{10,9}\right) + \xi_{10,17}\left(\xi_{10,10}\xi_{10,29} - \xi_{10,11}\xi_{10,32} + \xi_{10,12}\xi_{10,31} + \xi_{10,30}\xi_{10,9}\right) + \xi_{10,18}\left(-\xi_{10,10}\xi_{10,30} - \xi_{10,11}\xi_{10,31} - \xi_{10,12}\xi_{10,32} + \xi_{10,29}\xi_{10,9}\right)$$

$$\xi_{10,37} = -\xi_{10,10}\xi_{10,30} - \xi_{10,11}\xi_{10,31} - \xi_{10,12}\xi_{10,32} + \xi_{10,29}\xi_{10,9}$$

$$\xi_{10,38} = \xi_{10,10}\xi_{10,29} - \xi_{10,11}\xi_{10,32} + \xi_{10,12}\xi_{10,31} + \xi_{10,30}\xi_{10,9}$$

$$\xi_{10,39} = \xi_{10,10}\xi_{10,32} + \xi_{10,11}\xi_{10,29} - \xi_{10,12}\xi_{10,30} + \xi_{10,31}\xi_{10,9}$$

$$\xi_{10,40} = -\xi_{10,10}\xi_{10,31} + \xi_{10,11}\xi_{10,30} + \xi_{10,12}\xi_{10,29} + \xi_{10,32}\xi_{10,9}$$

$$\xi_{10,41} = -\xi_{10,10}\xi_{10,16} - \xi_{10,11}\xi_{10,17} - \xi_{10,12}\xi_{10,18} + \xi_{10,15}\xi_{10,9}$$

$$\xi_{10,42} = -\xi_{10,10}\xi_{10,15} - \xi_{10,11}\xi_{10,18} + \xi_{10,12}\xi_{10,17} - \xi_{10,16}\xi_{10,9}$$

$$\xi_{10,43} = \xi_{10,10}\xi_{10,18} - \xi_{10,11}\xi_{10,15} - \xi_{10,12}\xi_{10,16} - \xi_{10,17}\xi_{10,9}$$

$$\xi_{10,44} = -\xi_{10,10}\xi_{10,17} + \xi_{10,11}\xi_{10,16} - \xi_{10,12}\xi_{10,15} - \xi_{10,18}\xi_{10,9}$$

$$\xi_{10,45} = \xi_{10,13}\xi_{10,6}$$

$$\xi_{10,46} = \xi_{10,14}\xi_{10,7}$$

$$\xi_{10,47} = \xi_{10,13}\xi_{10,14}\xi_{10,8}$$

$$\xi_{10,48} = -\frac{\pi(\xi_{10,0}+\xi_{10,1}+\xi_{10,2})}{2}$$

$$\xi_{10,49} = \begin{cases} 0 & \text{for } \left(\xi_{10,46} > \frac{\pi}{8} \vee \xi_{10,47} < 0\right) \wedge \xi_{10,46} > \frac{\pi}{8} \\ -\frac{\pi}{8} & \text{otherwise} \end{cases}$$

$$\xi_{10,50} = \frac{\pi}{8}$$

$$\xi_{10,51} = \begin{cases} -\frac{\pi}{8} & \text{for } \xi_{10,46} > \frac{\pi}{8} \wedge \xi_{10,47} < 0 \\ \frac{\pi}{8} & \text{for } \xi_{10,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,52} = \begin{cases} -\frac{1}{2} & \text{for } \xi_{10,46} > \frac{\pi}{8} \wedge \xi_{10,47} < 0 \\ \frac{1}{2} & \text{for } \xi_{10,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,53} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{10,46} > \frac{\pi}{8} \vee \xi_{10,47} < 0\right) \wedge \xi_{10,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,54} = \begin{cases} \frac{1}{2} & \text{for } \left(\xi_{10,46} > \frac{\pi}{8} \vee \xi_{10,47} < 0\right) \wedge \xi_{10,46} > \frac{\pi}{8} \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{10,55} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{10,46} > \frac{\pi}{8} \vee \xi_{10,47} < 0\right) \wedge \xi_{10,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,56} = \begin{cases} -\frac{1}{2} & \text{for } \left(\xi_{10,46} > \frac{\pi}{8} \vee \xi_{10,47} < 0\right) \wedge \xi_{10,46} > \frac{\pi}{8} \\ -1 & \text{otherwise} \end{cases}$$

$$\xi_{10,57} = \begin{cases} \frac{1}{2} & \text{for } \xi_{10,46} > \frac{\pi}{8} \wedge \xi_{10,47} < 0 \\ -\frac{1}{2} & \text{for } \xi_{10,46} > \frac{\pi}{8} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,58} = \left\lfloor \frac{2\xi_{10,45} - 2\xi_{10,49}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{10,59} = \left\lfloor \frac{2\xi_{10,46} - 2\xi_{10,50}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{10,60} = \left\lfloor \frac{2\xi_{10,47} - 2\xi_{10,51}}{\pi} + \frac{1}{2} \right\rfloor$$

$$\xi_{10,61} = \xi_{10,45} - \xi_{10,49} - \frac{\pi\xi_{10,58}}{2}$$

$$\xi_{10,62} = \xi_{10,46} - \xi_{10,50} - \frac{\pi\xi_{10,59}}{2}$$

$$\xi_{10,63} = \xi_{10,47} - \xi_{10,51} - \frac{\pi\xi_{10,60}}{2}$$

$$\xi_{10,64} = \begin{cases} \xi_{10,61} & \text{for } \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \\ \xi_{10,62} & \text{for } \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,61}| \leq |\xi_{10,62}|\right) \wedge \left(|\xi_{10,61}| \leq |\xi_{10,62}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \\ \xi_{10,63} & \text{otherwise} \end{cases}$$

$$\xi_{10,65} = \begin{cases} \xi_{10,62} & \text{for } |\xi_{10,61}| \geq |\xi_{10,62}| \wedge |\xi_{10,62}| \geq |\xi_{10,63}| \\ \xi_{10,63} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,62}| \leq |\xi_{10,63}| \\ \xi_{10,61} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,61}| \leq |\xi_{10,62}| \\ \xi_{10,63} & \text{for } |\xi_{10,62}| \geq |\xi_{10,63}| \wedge |\xi_{10,61}| \leq |\xi_{10,63}| \\ \xi_{10,61} & \text{for } |\xi_{10,61}| \geq |\xi_{10,62}| \wedge |\xi_{10,61}| \leq |\xi_{10,63}| \\ \xi_{10,62} & \text{otherwise} \end{cases}$$

$$\xi_{10,66} = \begin{cases} \xi_{10,63} & \text{for } |\xi_{10,61}| \geq |\xi_{10,62}| \wedge |\xi_{10,62}| \geq |\xi_{10,63}| \\ \xi_{10,62} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,62}| \leq |\xi_{10,63}| \\ \xi_{10,63} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,61}| \leq |\xi_{10,62}| \\ \xi_{10,61} & \text{for } |\xi_{10,62}| \geq |\xi_{10,63}| \wedge |\xi_{10,61}| \leq |\xi_{10,63}| \\ \xi_{10,62} & \text{for } |\xi_{10,61}| \geq |\xi_{10,62}| \wedge |\xi_{10,61}| \leq |\xi_{10,63}| \\ \xi_{10,61} & \text{otherwise} \end{cases}$$

$$\xi_{10,67} = \begin{cases} 1 & \text{for } |\xi_{10,61}| \geq |\xi_{10,62}| \wedge |\xi_{10,62}| \geq |\xi_{10,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,62}| \leq |\xi_{10,63}| \\ 0 & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,61}| \leq |\xi_{10,62}| \\ \frac{1}{2} & \text{for } |\xi_{10,61}| \leq |\xi_{10,63}| \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{10,68} = \begin{cases} 0 & \text{for } |\xi_{10,61}| \geq |\xi_{10,62}| \wedge |\xi_{10,62}| \geq |\xi_{10,63}| \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,61}| \leq |\xi_{10,62}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \leq |\xi_{10,62}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \\ \frac{1}{2} & \text{for } |\xi_{10,61}| \leq |\xi_{10,63}| \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,69} = \begin{cases} 0 & \text{for } \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,61}| \leq |\xi_{10,62}| \\ \frac{1}{2} & \text{for } |\xi_{10,61}| \leq |\xi_{10,63}| \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \\ \frac{\sqrt{2}}{2} & \text{otherwise} \end{cases}$$

$$\xi_{10,70} = \begin{cases} 0 & \text{for } \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,61}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,62}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,61}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \geq |\xi_{10,63}|\right) \wedge \left(|\xi_{10,62}| \geq |\xi_{10,63}| \vee |\xi_{10,62}| \leq |\xi_{10,63}|\right) \\ \frac{1}{2} & \text{for } |\xi_{10,61}| \geq |\xi_{10,63}| \wedge |\xi_{10,61}| \leq |\xi_{10,62}| \\ -\frac{1}{2} & \text{for } |\xi_{10,61}| \geq |\xi_{10,62}| \wedge |\xi_{10,61}| \leq |\xi_{10,63}| \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,71} = \begin{cases} -1 & \text{for } \xi_{10,64} < 0 \\ 1 & \text{otherwise} \end{cases}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,72} = \begin{cases} -1 & \text{for } \xi_{10,65} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{10,73} = \begin{cases} 1 & \text{for } \xi_{10,71} = 1 \wedge \xi_{10,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,74} = \begin{cases} 0 & \text{for } \xi_{10,71} = 1 \wedge \xi_{10,72} = 1 \\ 1 & \text{for } \xi_{10,71} = 1 \wedge \xi_{10,72} = -1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,75} = \begin{cases} 0 & \text{for } \xi_{10,71} = 1 \wedge (\xi_{10,71} = 1 \vee \xi_{10,72} = -1) \wedge (\xi_{10,71} = 1 \vee \xi_{10,72} = 1) \wedge (\xi_{10,72} = -1 \vee \xi_{10,72} = 1) \\ 1 & \text{for } \xi_{10,71} = -1 \wedge \xi_{10,72} = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,76} = \begin{cases} 0 & \text{for } (\xi_{10,71} = -1 \vee \xi_{10,71} = 1) \wedge (\xi_{10,71} = 1 \vee \xi_{10,72} = 1) \wedge (\xi_{10,72} = -1 \vee \xi_{10,72} = 1) \wedge (\xi_{10,71} = -1 \vee \xi_{10,71} = 1 \vee \xi_{10,72} = -1) \wedge (\xi_{10,71} = -1 \vee \xi_{10,71} = 1 \vee \xi_{10,72} = 1) \wedge (\xi_{10,71} = -1 \vee \xi_{10,72} = -1 \vee \xi_{10,72} = 1) \wedge (\xi_{10,71} = 1 \vee \xi_{10,72} = -1 \vee \xi_{10,72} = 1) \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{10,77} = \cos\left(\frac{\pi \xi_{10,58}}{2}\right)$$

$$\xi_{10,78} = -\sin\left(\frac{\pi \xi_{10,58}}{2}\right)$$

$$\xi_{10,79} = \cos\left(\frac{\pi \xi_{10,59}}{2}\right)$$

$$\xi_{10,80} = -\sin\left(\frac{\pi \xi_{10,59}}{2}\right)$$

$$\xi_{10,81} = \xi_{10,77}\xi_{10,79}$$

$$\xi_{10,82} = \xi_{10,78}\xi_{10,79}$$

$$\xi_{10,83} = \xi_{10,77}\xi_{10,80}$$

$$\xi_{10,84} = -\xi_{10,78}\xi_{10,80}$$

$$\xi_{10,85} = \cos\left(\frac{\pi \xi_{10,60}}{2}\right)$$

$$\xi_{10,86} = -\sin\left(\frac{\pi \xi_{10,60}}{2}\right)$$

$$\xi_{10,87} = \xi_{10,81}\xi_{10,85} - \xi_{10,84}\xi_{10,86}$$

$$\xi_{10,88} = \xi_{10,82}\xi_{10,85} - \xi_{10,83}\xi_{10,86}$$

$$\xi_{10,89} = \xi_{10,82}\xi_{10,86} + \xi_{10,83}\xi_{10,85}$$

$$\xi_{10,90} = \xi_{10,81}\xi_{10,86} + \xi_{10,84}\xi_{10,85}$$

$$\xi_{10,91} = \xi_{10,73}\left(\xi_{10,67}\xi_{10,87} - \xi_{10,68}\xi_{10,88} - \xi_{10,69}\xi_{10,89} - \xi_{10,70}\xi_{10,90}\right) - \xi_{10,74}\left(\xi_{10,67}\xi_{10,88} + \xi_{10,68}\xi_{10,87} - \xi_{10,69}\xi_{10,90} + \xi_{10,70}\xi_{10,89}\right) - \xi_{10,75}\left(\xi_{10,67}\xi_{10,89} + \xi_{10,68}\xi_{10,90} + \xi_{10,69}\xi_{10,87} - \xi_{10,70}\xi_{10,88}\right) - \xi_{10,76}\left(\xi_{10,67}\xi_{10,90} - \xi_{10,68}\xi_{10,89} + \xi_{10,69}\xi_{10,88} + \xi_{10,70}\xi_{10,87}\right)$$

$$\xi_{10,92} = \xi_{10,73}\left(\xi_{10,67}\xi_{10,88} + \xi_{10,68}\xi_{10,87} - \xi_{10,69}\xi_{10,90} + \xi_{10,70}\xi_{10,89}\right) + \xi_{10,74}\left(\xi_{10,67}\xi_{10,87} - \xi_{10,68}\xi_{10,88} - \xi_{10,69}\xi_{10,89} - \xi_{10,70}\xi_{10,90}\right) - \xi_{10,75}\left(\xi_{10,67}\xi_{10,90} - \xi_{10,68}\xi_{10,89} + \xi_{10,69}\xi_{10,88} + \xi_{10,70}\xi_{10,87}\right) + \xi_{10,76}\left(\xi_{10,67}\xi_{10,89} + \xi_{10,68}\xi_{10,90} + \xi_{10,69}\xi_{10,87} - \xi_{10,70}\xi_{10,88}\right)$$

$$\xi_{10,93} = \xi_{10,73}\left(\xi_{10,67}\xi_{10,89} + \xi_{10,68}\xi_{10,90} + \xi_{10,69}\xi_{10,87} - \xi_{10,70}\xi_{10,88}\right) + \xi_{10,74}\left(\xi_{10,67}\xi_{10,90} - \xi_{10,68}\xi_{10,89} + \xi_{10,69}\xi_{10,88} + \xi_{10,70}\xi_{10,87}\right) + \xi_{10,75}\left(\xi_{10,67}\xi_{10,87} - \xi_{10,68}\xi_{10,88} - \xi_{10,69}\xi_{10,89} - \xi_{10,70}\xi_{10,90}\right) - \xi_{10,76}\left(\xi_{10,67}\xi_{10,88} + \xi_{10,68}\xi_{10,87} - \xi_{10,69}\xi_{10,90} + \xi_{10,70}\xi_{10,89}\right)$$

$$\xi_{10,94} = \xi_{10,73}\left(\xi_{10,67}\xi_{10,90} - \xi_{10,68}\xi_{10,89} + \xi_{10,69}\xi_{10,88} + \xi_{10,70}\xi_{10,87}\right) - \xi_{10,74}\left(\xi_{10,67}\xi_{10,89} + \xi_{10,68}\xi_{10,90} + \xi_{10,69}\xi_{10,87} - \xi_{10,70}\xi_{10,88}\right) + \xi_{10,75}\left(\xi_{10,67}\xi_{10,88} + \xi_{10,68}\xi_{10,87} - \xi_{10,69}\xi_{10,90} + \xi_{10,70}\xi_{10,89}\right) + \xi_{10,76}\left(\xi_{10,67}\xi_{10,87} - \xi_{10,68}\xi_{10,88} - \xi_{10,69}\xi_{10,89} - \xi_{10,70}\xi_{10,90}\right)$$

$$\xi_{10,95} = \xi_{10,67}\xi_{10,87} - \xi_{10,68}\xi_{10,88} - \xi_{10,69}\xi_{10,89} - \xi_{10,70}\xi_{10,90}$$

$$\xi_{10,96} = \xi_{10,67}\xi_{10,88} + \xi_{10,68}\xi_{10,87} - \xi_{10,69}\xi_{10,90} + \xi_{10,70}\xi_{10,89}$$

$$\xi_{10,97} = \xi_{10,67}\xi_{10,89} + \xi_{10,68}\xi_{10,90} + \xi_{10,69}\xi_{10,87} - \xi_{10,70}\xi_{10,88}$$

$$\xi_{10,98} = \xi_{10,67}\xi_{10,90} - \xi_{10,68}\xi_{10,89} + \xi_{10,69}\xi_{10,88} + \xi_{10,70}\xi_{10,87}$$

$$\xi_{10,99} = \xi_{10,67}\xi_{10,73} - \xi_{10,68}\xi_{10,74} - \xi_{10,69}\xi_{10,75} - \xi_{10,70}\xi_{10,76}$$

$$\xi_{10,100} = -\xi_{10,67}\xi_{10,74} - \xi_{10,68}\xi_{10,73} - \xi_{10,69}\xi_{10,76} + \xi_{10,70}\xi_{10,75}$$

$$\xi_{10,101} = -\xi_{10,67}\xi_{10,75} + \xi_{10,68}\xi_{10,76} - \xi_{10,69}\xi_{10,73} - \xi_{10,70}\xi_{10,74}$$

$$\xi_{10,102} = -\xi_{10,67}\xi_{10,76} - \xi_{10,68}\xi_{10,75} + \xi_{10,69}\xi_{10,74} - \xi_{10,70}\xi_{10,73}$$

$$\xi_{10,103} = \xi_{10,64}\xi_{10,71}$$

$$\xi_{10,104} = \xi_{10,65}\xi_{10,72}$$

$$\xi_{10,105} = \xi_{10,66}\xi_{10,71}\xi_{10,72}$$

$$\xi_{10,106} = -\frac{\pi\left(\xi_{10,58} + \xi_{10,59} + \xi_{10,60}\right)}{2}$$

$$\xi_{10,107} = \cos\left(2\xi_{10,103}\right)$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,108} = \cos\left(2\xi_{10,104}\right)$$

$$\xi_{10,109} = \cos\left(2\xi_{10,105}\right)$$

$$\xi_{10,110} = \sqrt{\xi_{10,107}^2 - 2\xi_{10,107}\xi_{10,108}\xi_{10,109} + \xi_{10,108}^2 + \xi_{10,109}^2 - 1}$$

$$\xi_{10,111} = \xi_{10,107} - \xi_{10,108} + \xi_{10,109}$$

$$\xi_{10,112} = \sqrt{-\frac{\xi_{10,110}}{2} + \frac{\xi_{10,111}}{2} + \frac{1}{2}}$$

$$\xi_{10,113} = \sqrt{\frac{\xi_{10,110}}{2} - \frac{\xi_{10,111}}{2} + \frac{1}{2}}$$

$$\xi_{10,114} = \sqrt{-\frac{\xi_{10,110}}{2} - \frac{\xi_{10,111}}{2} + \frac{1}{2}}$$

$$\xi_{10,115} = \frac{(1-\xi_{10,108})(\xi_{10,107}+1)(\xi_{10,109}+1)}{2}$$

$$\xi_{10,116} = \begin{cases} -1 & \text{for } \xi_{10,105} < 0 \\ 1 & \text{otherwise} \end{cases}$$

$$\xi_{10,117} = \begin{cases} 0 & \text{for } \xi_{10,112} = 0 \\ \frac{\sqrt{\xi_{10,115}}\xi_{10,116}}{\xi_{10,112}} & \text{otherwise} \end{cases}$$

$$\xi_{10,118} = \begin{cases} 1 & \text{for } \xi_{10,112} = 0 \\ \frac{\sqrt{\xi_{10,107}\xi_{10,108}\xi_{10,109}}}{\xi_{10,112}} & \text{otherwise} \end{cases}$$

$$\xi_{10,119} = 2\xi_{10,112}\xi_{10,113}\xi_{10,114}\xi_{10,118}$$

$$\xi_{10,120} = \sqrt{2}\xi_{10,118}\left(\xi_{10,112}^2\xi_{10,114} - \xi_{10,114}\right)$$

$$\xi_{10,121} = 2\xi_{10,112}\xi_{10,113}\xi_{10,114}\xi_{10,117}$$

$$\xi_{10,122} = \sqrt{2}\left(2\xi_{10,112}^2\xi_{10,114}\xi_{10,117} - \xi_{10,114}\xi_{10,117}\right)$$

$$\xi_{10,123} = \sqrt{2}\xi_{10,112}\xi_{10,113}\xi_{10,117}\xi_{10,118}$$

$$\xi_{10,124} = \sqrt{2}\xi_{10,112}\xi_{10,113}\left(-\xi_{10,114}^2 + \xi_{10,117}^2\right)$$

$$\xi_{10,125} = -\xi_{10,112}^2\xi_{10,114}^2 + \xi_{10,112}^2\xi_{10,117}^2 + \xi_{10,114}^2$$

$$\xi_{10,126} = \xi_{10,112}^2\xi_{10,114}^2 - \xi_{10,112}^2\xi_{10,117}^2 + \xi_{10,117}^2$$

$$\xi_{10,127} = 2\xi_{10,118}\left(\xi_{10,112}^2\xi_{10,117} - \frac{\xi_{10,117}}{2}\right)$$

$$\xi_{10,128} = -\sqrt{2}\xi_{10,112}\xi_{10,113}\xi_{10,117}\xi_{10,118}$$

$$\xi_{10,129} = -\xi_{10,117}\xi_{10,118}$$

$$\xi_{10,130} = \xi_{10,112}^2\xi_{10,114}^2 + \xi_{10,112}^2\xi_{10,117}^2 - \xi_{10,112}^2 - \xi_{10,114}^2 - \xi_{10,117}^2 + 1$$

$$\xi_{10,131} = -\sqrt{2}\left(\xi_{10,112}\xi_{10,113}\xi_{10,114}^2 + \xi_{10,112}\xi_{10,113}\xi_{10,117}^2 - \xi_{10,112}\xi_{10,113}\right)$$

$$\xi_{10,132} = \sqrt{2}\xi_{10,112}^2\xi_{10,114}\xi_{10,118}$$

$$\xi_{10,133} = \xi_{10,123}^2 - \xi_{10,126}^2 - \xi_{10,129}^2 - \xi_{10,132}^2$$

$$\xi_{10,134} = \xi_{10,123}\xi_{10,124} - \xi_{10,126}\xi_{10,127} - \xi_{10,129}\xi_{10,130}$$

$$\xi_{10,135} = \xi_{10,123}\xi_{10,125} - \xi_{10,126}\xi_{10,128} - \xi_{10,129}\xi_{10,131}$$

$$\xi_{10,136} = \xi_{10,119}^2 + \xi_{10,121}^2 + \xi_{10,124}^2 - \xi_{10,127}^2 - \xi_{10,130}^2$$

$$\xi_{10,137} = \xi_{10,119}\xi_{10,120} + \xi_{10,121}\xi_{10,122} + \xi_{10,124}\xi_{10,125} - \xi_{10,127}\xi_{10,128} - \xi_{10,130}\xi_{10,131}$$

$$\xi_{10,138} = \xi_{10,120}^2 + \xi_{10,122}^2 + \xi_{10,125}^2 - \xi_{10,128}^2 - \xi_{10,131}^2$$

$$\xi_{10,139} = \sin\left(2\xi_{10,104}\right)\sin\left(2\xi_{10,105}\right)\cos\left(2\xi_{10,103}\right)$$

$$\xi_{10,140} = \sin\left(2\xi_{10,103}\right)\sin\left(2\xi_{10,105}\right)\cos\left(2\xi_{10,104}\right)$$

$$\xi_{10,141} = \sin\left(2\xi_{10,103}\right)\sin\left(2\xi_{10,104}\right)\cos\left(2\xi_{10,105}\right)$$

$$\xi_{10,142} = \sin\left(2\xi_{10,103}\right)\cos\left(2\xi_{10,104}\right)\cos\left(2\xi_{10,105}\right)$$

$$\xi_{10,143} = \sin\left(2\xi_{10,104}\right)\cos\left(2\xi_{10,103}\right)\cos\left(2\xi_{10,105}\right)$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{10,144} = \sin(2\xi_{10,105})\cos(2\xi_{10,103})\cos(2\xi_{10,104})$$

$$\xi_{10,145} = \xi_{10,139}^2 - \xi_{10,142}^2$$

$$\xi_{10,146} = \xi_{10,140}^2 - \xi_{10,143}^2$$

$$\xi_{10,147} = \xi_{10,141}^2 - \xi_{10,144}^2$$

$$\xi_{10,148} = \frac{\xi_{10,133}-\xi_{10,146}}{\xi_{10,145}-\xi_{10,146}}$$

$$\xi_{10,149} = \frac{\xi_{10,134}}{\xi_{10,145}-\xi_{10,146}}$$

$$\xi_{10,150} = \frac{\xi_{10,135}}{\xi_{10,145}-\xi_{10,146}}$$

$$\xi_{10,151} = \frac{\xi_{10,136}-\xi_{10,146}}{\xi_{10,145}-\xi_{10,146}}$$

$$\xi_{10,152} = \frac{\xi_{10,137}}{\xi_{10,145}-\xi_{10,146}}$$

$$\xi_{10,153} = \frac{\xi_{10,138}-\xi_{10,146}}{\xi_{10,145}-\xi_{10,146}}$$

$$\xi_{10,154} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,148} & \text{otherwise} \end{cases}$$

$$\xi_{10,155} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,149} & \text{otherwise} \end{cases}$$

$$\xi_{10,156} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,150} & \text{otherwise} \end{cases}$$

$$\xi_{10,157} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,151} & \text{otherwise} \end{cases}$$

$$\xi_{10,158} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,152} & \text{otherwise} \end{cases}$$

$$\xi_{10,159} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,153} & \text{otherwise} \end{cases}$$

$$\xi_{10,160} = \frac{\xi_{10,133}-\xi_{10,147}}{\xi_{10,145}-\xi_{10,147}}$$

$$\xi_{10,161} = \frac{\xi_{10,134}}{\xi_{10,145}-\xi_{10,147}}$$

$$\xi_{10,162} = \frac{\xi_{10,135}}{\xi_{10,145}-\xi_{10,147}}$$

$$\xi_{10,163} = \frac{\xi_{10,136}-\xi_{10,147}}{\xi_{10,145}-\xi_{10,147}}$$

$$\xi_{10,164} = \frac{\xi_{10,137}}{\xi_{10,145}-\xi_{10,147}}$$

$$\xi_{10,165} = \frac{\xi_{10,138}-\xi_{10,147}}{\xi_{10,145}-\xi_{10,147}}$$

$$\xi_{10,166} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,147} \\ \xi_{10,160} & \text{otherwise} \end{cases}$$

$$\xi_{10,167} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,147} \\ \xi_{10,161} & \text{otherwise} \end{cases}$$

$$\xi_{10,168} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,147} \\ \xi_{10,162} & \text{otherwise} \end{cases}$$

$$\xi_{10,169} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,147} \\ \xi_{10,163} & \text{otherwise} \end{cases}$$

$$\xi_{10,170} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,147} \\ \xi_{10,164} & \text{otherwise} \end{cases}$$

$$\xi_{10,171} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,147} \\ \xi_{10,165} & \text{otherwise} \end{cases}$$

$$\xi_{10,172} = \xi_{10,154}\xi_{10,166} + \xi_{10,155}\xi_{10,167} + \xi_{10,156}\xi_{10,168}$$

$$\xi_{10,173} = \xi_{10,154}\xi_{10,167} + \xi_{10,155}\xi_{10,169} + \xi_{10,156}\xi_{10,170}$$

$$\xi_{10,174} = \xi_{10,154}\xi_{10,168} + \xi_{10,155}\xi_{10,170} + \xi_{10,156}\xi_{10,171}$$

$$\xi_{10,175} = \xi_{10,155}\xi_{10,166} + \xi_{10,157}\xi_{10,167} + \xi_{10,158}\xi_{10,168}$$

$$\xi_{10,176} = \xi_{10,155}\xi_{10,167} + \xi_{10,157}\xi_{10,169} + \xi_{10,158}\xi_{10,170}$$

$$\xi_{10,177} = \xi_{10,155}\xi_{10,168} + \xi_{10,157}\xi_{10,170} + \xi_{10,158}\xi_{10,171}$$

$$\xi_{10,178} = \xi_{10,156}\xi_{10,166} + \xi_{10,158}\xi_{10,167} + \xi_{10,159}\xi_{10,168}$$

$$\xi_{10,179} = \xi_{10,156}\xi_{10,167} + \xi_{10,158}\xi_{10,169} + \xi_{10,159}\xi_{10,170}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{10,180} = \xi_{10,156}\xi_{10,168} + \xi_{10,158}\xi_{10,170} + \xi_{10,159}\xi_{10,171}$$

$$\xi_{10,181} = \sqrt{\xi_{10,172}}$$

$$\xi_{10,182} = \frac{\xi_{10,172}}{\xi_{10,181}}$$

$$\xi_{10,183} = \frac{\xi_{10,175}}{\xi_{10,181}}$$

$$\xi_{10,184} = \frac{\xi_{10,178}}{\xi_{10,181}}$$

$$\xi_{10,185} = \sqrt{\xi_{10,176}}$$

$$\xi_{10,186} = \frac{\xi_{10,173}}{\xi_{10,185}}$$

$$\xi_{10,187} = \frac{\xi_{10,176}}{\xi_{10,185}}$$

$$\xi_{10,188} = \frac{\xi_{10,179}}{\xi_{10,185}}$$

$$\xi_{10,189} = \sqrt{\xi_{10,180}}$$

$$\xi_{10,190} = \frac{\xi_{10,174}}{\xi_{10,189}}$$

$$\xi_{10,191} = \frac{\xi_{10,177}}{\xi_{10,189}}$$

$$\xi_{10,192} = \frac{\xi_{10,180}}{\xi_{10,189}}$$

$$\xi_{10,193} = \begin{cases} \xi_{10,182} & \text{for } \xi_{10,172} \geq \xi_{10,176} \wedge \xi_{10,172} \geq \xi_{10,180} \\ \xi_{10,186} & \text{for } \xi_{10,176} \geq \xi_{10,180} \wedge \xi_{10,172} \leq \xi_{10,176} \\ \xi_{10,190} & \text{otherwise} \end{cases}$$

$$\xi_{10,194} = \begin{cases} \xi_{10,183} & \text{for } \xi_{10,172} \geq \xi_{10,176} \wedge \xi_{10,172} \geq \xi_{10,180} \\ \xi_{10,187} & \text{for } \xi_{10,176} \geq \xi_{10,180} \wedge \xi_{10,172} \leq \xi_{10,176} \\ \xi_{10,191} & \text{otherwise} \end{cases}$$

$$\xi_{10,195} = \begin{cases} \xi_{10,184} & \text{for } \xi_{10,172} \geq \xi_{10,176} \wedge \xi_{10,172} \geq \xi_{10,180} \\ \xi_{10,188} & \text{for } \xi_{10,176} \geq \xi_{10,180} \wedge \xi_{10,172} \leq \xi_{10,176} \\ \xi_{10,192} & \text{otherwise} \end{cases}$$

$$\xi_{10,196} = \frac{\xi_{10,133} - \xi_{10,145}}{-\xi_{10,145} + \xi_{10,146}}$$

$$\xi_{10,197} = \frac{\xi_{10,134}}{-\xi_{10,145} + \xi_{10,146}}$$

$$\xi_{10,198} = \frac{\xi_{10,135}}{-\xi_{10,145} + \xi_{10,146}}$$

$$\xi_{10,199} = \frac{\xi_{10,136} - \xi_{10,145}}{-\xi_{10,145} + \xi_{10,146}}$$

$$\xi_{10,200} = \frac{\xi_{10,137}}{-\xi_{10,145} + \xi_{10,146}}$$

$$\xi_{10,201} = \frac{\xi_{10,138} - \xi_{10,145}}{-\xi_{10,145} + \xi_{10,146}}$$

$$\xi_{10,202} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,196} & \text{otherwise} \end{cases}$$

$$\xi_{10,203} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,197} & \text{otherwise} \end{cases}$$

$$\xi_{10,204} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,198} & \text{otherwise} \end{cases}$$

$$\xi_{10,205} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,199} & \text{otherwise} \end{cases}$$

$$\xi_{10,206} = \begin{cases} 0 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,200} & \text{otherwise} \end{cases}$$

$$\xi_{10,207} = \begin{cases} 1 & \text{for } \xi_{10,145} = \xi_{10,146} \\ \xi_{10,201} & \text{otherwise} \end{cases}$$

$$\xi_{10,208} = \frac{\xi_{10,133} - \xi_{10,147}}{\xi_{10,146} - \xi_{10,147}}$$

$$\xi_{10,209} = \frac{\xi_{10,134}}{\xi_{10,146} - \xi_{10,147}}$$

$$\xi_{10,210} = \frac{\xi_{10,135}}{\xi_{10,146} - \xi_{10,147}}$$

$$\xi_{10,211} = \frac{\xi_{10,136} - \xi_{10,147}}{\xi_{10,146} - \xi_{10,147}}$$

$$\xi_{10,212} = \frac{\xi_{10,137}}{\xi_{10,146} - \xi_{10,147}}$$

$$\xi_{10,213} = \frac{\xi_{10,138} - \xi_{10,147}}{\xi_{10,146} - \xi_{10,147}}$$

$$\xi_{10,214} = \begin{cases} 1 & \text{for } \xi_{10,146} = \xi_{10,147} \\ \xi_{10,208} & \text{otherwise} \end{cases}$$

$$\xi_{10,215} = \begin{cases} 0 & \text{for } \xi_{10,146} = \xi_{10,147} \\ \xi_{10,209} & \text{otherwise} \end{cases}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{10,216} = \begin{cases} 0 & \text{for } \xi_{10,146} = \xi_{10,147} \\ \xi_{10,210} & \text{otherwise} \end{cases}$$

$$\xi_{10,217} = \begin{cases} 1 & \text{for } \xi_{10,146} = \xi_{10,147} \\ \xi_{10,211} & \text{otherwise} \end{cases}$$

$$\xi_{10,218} = \begin{cases} 0 & \text{for } \xi_{10,146} = \xi_{10,147} \\ \xi_{10,212} & \text{otherwise} \end{cases}$$

$$\xi_{10,219} = \begin{cases} 1 & \text{for } \xi_{10,146} = \xi_{10,147} \\ \xi_{10,213} & \text{otherwise} \end{cases}$$

$$\xi_{10,220} = \xi_{10,202}\xi_{10,214} + \xi_{10,203}\xi_{10,215} + \xi_{10,204}\xi_{10,216}$$

$$\xi_{10,221} = \xi_{10,202}\xi_{10,215} + \xi_{10,203}\xi_{10,217} + \xi_{10,204}\xi_{10,218}$$

$$\xi_{10,222} = \xi_{10,202}\xi_{10,216} + \xi_{10,203}\xi_{10,218} + \xi_{10,204}\xi_{10,219}$$

$$\xi_{10,223} = \xi_{10,203}\xi_{10,214} + \xi_{10,205}\xi_{10,215} + \xi_{10,206}\xi_{10,216}$$

$$\xi_{10,224} = \xi_{10,203}\xi_{10,215} + \xi_{10,205}\xi_{10,217} + \xi_{10,206}\xi_{10,218}$$

$$\xi_{10,225} = \xi_{10,203}\xi_{10,216} + \xi_{10,205}\xi_{10,218} + \xi_{10,206}\xi_{10,219}$$

$$\xi_{10,226} = \xi_{10,204}\xi_{10,214} + \xi_{10,206}\xi_{10,215} + \xi_{10,207}\xi_{10,216}$$

$$\xi_{10,227} = \xi_{10,204}\xi_{10,215} + \xi_{10,206}\xi_{10,217} + \xi_{10,207}\xi_{10,218}$$

$$\xi_{10,228} = \xi_{10,204}\xi_{10,216} + \xi_{10,206}\xi_{10,218} + \xi_{10,207}\xi_{10,219}$$

$$\xi_{10,229} = \begin{cases} \xi_{10,193}^2 & \text{for } \xi_{10,145} = \xi_{10,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,230} = \begin{cases} \xi_{10,193}\xi_{10,194} & \text{for } \xi_{10,145} = \xi_{10,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,231} = \begin{cases} \xi_{10,193}\xi_{10,195} & \text{for } \xi_{10,145} = \xi_{10,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,232} = \begin{cases} \xi_{10,194}^2 & \text{for } \xi_{10,145} = \xi_{10,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,233} = \begin{cases} \xi_{10,194}\xi_{10,195} & \text{for } \xi_{10,145} = \xi_{10,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,234} = \begin{cases} \xi_{10,195}^2 & \text{for } \xi_{10,145} = \xi_{10,146} \\ 0 & \text{otherwise} \end{cases}$$

$$\xi_{10,235} = \xi_{10,220} - \xi_{10,229}$$

$$\xi_{10,236} = \xi_{10,221} - \xi_{10,230}$$

$$\xi_{10,237} = \xi_{10,222} - \xi_{10,231}$$

$$\xi_{10,238} = \xi_{10,223} - \xi_{10,230}$$

$$\xi_{10,239} = \xi_{10,224} - \xi_{10,232}$$

$$\xi_{10,240} = \xi_{10,225} - \xi_{10,233}$$

$$\xi_{10,241} = \xi_{10,226} - \xi_{10,231}$$

$$\xi_{10,242} = \xi_{10,227} - \xi_{10,233}$$

$$\xi_{10,243} = \xi_{10,228} - \xi_{10,234}$$

$$\xi_{10,244} = \sqrt{\xi_{10,235}}$$

$$\xi_{10,245} = \frac{\xi_{10,235}}{\xi_{10,244}}$$

$$\xi_{10,246} = \frac{\xi_{10,238}}{\xi_{10,244}}$$

$$\xi_{10,247} = \frac{\xi_{10,241}}{\xi_{10,244}}$$

$$\xi_{10,248} = \sqrt{\xi_{10,239}}$$

$$\xi_{10,249} = \frac{\xi_{10,236}}{\xi_{10,248}}$$

$$\xi_{10,250} = \frac{\xi_{10,239}}{\xi_{10,248}}$$

$$\xi_{10,251} = \frac{\xi_{10,242}}{\xi_{10,248}}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

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$$\xi_{10,252} = \sqrt{\xi_{10,243}}$$

$$\xi_{10,253} = \frac{\xi_{10,237}}{\xi_{10,252}}$$

$$\xi_{10,254} = \frac{\xi_{10,240}}{\xi_{10,252}}$$

$$\xi_{10,255} = \frac{\xi_{10,243}}{\xi_{10,252}}$$

$$\xi_{10,256} = \begin{cases} \xi_{10,245} & \text{for } \xi_{10,235} \geq \xi_{10,239} \wedge \xi_{10,235} \geq \xi_{10,243} \\ \xi_{10,249} & \text{for } \xi_{10,239} \geq \xi_{10,243} \wedge \xi_{10,235} \leq \xi_{10,239} \\ \xi_{10,253} & \text{otherwise} \end{cases}$$

$$\xi_{10,257} = \begin{cases} \xi_{10,246} & \text{for } \xi_{10,235} \geq \xi_{10,239} \wedge \xi_{10,235} \geq \xi_{10,243} \\ \xi_{10,250} & \text{for } \xi_{10,239} \geq \xi_{10,243} \wedge \xi_{10,235} \leq \xi_{10,239} \\ \xi_{10,254} & \text{otherwise} \end{cases}$$

$$\xi_{10,258} = \begin{cases} \xi_{10,247} & \text{for } \xi_{10,235} \geq \xi_{10,239} \wedge \xi_{10,235} \geq \xi_{10,243} \\ \xi_{10,251} & \text{for } \xi_{10,239} \geq \xi_{10,243} \wedge \xi_{10,235} \leq \xi_{10,239} \\ \xi_{10,255} & \text{otherwise} \end{cases}$$

$$\xi_{10,259} = \xi_{10,194}\xi_{10,258} - \xi_{10,195}\xi_{10,257}$$

$$\xi_{10,260} = -\xi_{10,193}\xi_{10,258} + \xi_{10,195}\xi_{10,256}$$

$$\xi_{10,261} = \xi_{10,193}\xi_{10,257} - \xi_{10,194}\xi_{10,256}$$

$$\xi_{10,262} = \xi_{10,139}^2 + \xi_{10,142}^2$$

$$\xi_{10,263} = \xi_{10,140}^2 + \xi_{10,143}^2$$

$$\xi_{10,264} = \xi_{10,141}^2 + \xi_{10,144}^2$$

$$\xi_{10,265} = \frac{\xi_{10,126}\xi_{10,142}\xi_{10,193}+\xi_{10,194}(\xi_{10,119}\xi_{10,139}+\xi_{10,127}\xi_{10,142})+\xi_{10,195}(\xi_{10,120}\xi_{10,139}+\xi_{10,128}\xi_{10,142})}{\xi_{10,262}}$$

$$\xi_{10,266} = \frac{\xi_{10,129}\xi_{10,142}\xi_{10,193}+\xi_{10,194}(\xi_{10,121}\xi_{10,139}+\xi_{10,130}\xi_{10,142})+\xi_{10,195}(\xi_{10,122}\xi_{10,139}+\xi_{10,131}\xi_{10,142})}{\xi_{10,262}}$$

$$\xi_{10,267} = \frac{\xi_{10,124}\xi_{10,139}\xi_{10,194}+\xi_{10,125}\xi_{10,139}\xi_{10,195}+\xi_{10,193}(\xi_{10,123}\xi_{10,139}+\xi_{10,132}\xi_{10,142})}{\xi_{10,262}}$$

$$\xi_{10,268} = \frac{\xi_{10,126}\xi_{10,143}\xi_{10,256}+\xi_{10,257}(\xi_{10,119}\xi_{10,140}+\xi_{10,127}\xi_{10,143})+\xi_{10,258}(\xi_{10,120}\xi_{10,140}+\xi_{10,128}\xi_{10,143})}{\xi_{10,263}}$$

$$\xi_{10,269} = \frac{\xi_{10,129}\xi_{10,143}\xi_{10,256}+\xi_{10,257}(\xi_{10,121}\xi_{10,140}+\xi_{10,130}\xi_{10,143})+\xi_{10,258}(\xi_{10,122}\xi_{10,140}+\xi_{10,131}\xi_{10,143})}{\xi_{10,263}}$$

$$\xi_{10,270} = \frac{\xi_{10,124}\xi_{10,140}\xi_{10,257}+\xi_{10,125}\xi_{10,140}\xi_{10,258}+\xi_{10,256}(\xi_{10,123}\xi_{10,140}+\xi_{10,132}\xi_{10,143})}{\xi_{10,263}}$$

$$\xi_{10,271} = \frac{\xi_{10,126}\xi_{10,144}\xi_{10,259}+\xi_{10,260}(\xi_{10,119}\xi_{10,141}+\xi_{10,127}\xi_{10,144})+\xi_{10,261}(\xi_{10,120}\xi_{10,141}+\xi_{10,128}\xi_{10,144})}{\xi_{10,264}}$$

$$\xi_{10,272} = \frac{\xi_{10,129}\xi_{10,144}\xi_{10,259}+\xi_{10,260}(\xi_{10,121}\xi_{10,141}+\xi_{10,130}\xi_{10,144})+\xi_{10,261}(\xi_{10,122}\xi_{10,141}+\xi_{10,131}\xi_{10,144})}{\xi_{10,264}}$$

$$\xi_{10,273} = \frac{\xi_{10,124}\xi_{10,141}\xi_{10,260}+\xi_{10,125}\xi_{10,141}\xi_{10,261}+\xi_{10,259}(\xi_{10,123}\xi_{10,141}+\xi_{10,132}\xi_{10,144})}{\xi_{10,264}}$$

$$\xi_{10,274} = 1 - \xi_{10,265}^2$$

$$\xi_{10,275} = -\xi_{10,265}\xi_{10,266}$$

$$\xi_{10,276} = -\xi_{10,265}\xi_{10,267}$$

$$\xi_{10,277} = 1 - \xi_{10,266}^2$$

$$\xi_{10,278} = -\xi_{10,266}\xi_{10,267}$$

$$\xi_{10,279} = 1 - \xi_{10,267}^2$$

$$\xi_{10,280} = \xi_{10,274}^2 + \xi_{10,275}^2 + \xi_{10,276}^2$$

$$\xi_{10,281} = \xi_{10,275}^2 + \xi_{10,277}^2 + \xi_{10,278}^2$$

$$\xi_{10,282} = \xi_{10,276}^2 + \xi_{10,278}^2 + \xi_{10,279}^2$$

$$\xi_{10,283} = \sqrt{\xi_{10,280}}$$

$$\xi_{10,284} = \frac{\xi_{10,274}}{\xi_{10,283}}$$

$$\xi_{10,285} = \frac{\xi_{10,275}}{\xi_{10,283}}$$

$$\xi_{10,286} = \frac{\xi_{10,276}}{\xi_{10,283}}$$

$$\xi_{10,287} = \sqrt{\xi_{10,281}}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,288} = \frac{\xi_{10,277}}{\xi_{10,287}}$$

$$\xi_{10,289} = \frac{\xi_{10,275}}{\xi_{10,287}}$$

$$\xi_{10,290} = \frac{\xi_{10,278}}{\xi_{10,287}}$$

$$\xi_{10,291} = \sqrt{\xi_{10,282}}$$

$$\xi_{10,292} = \frac{\xi_{10,279}}{\xi_{10,291}}$$

$$\xi_{10,293} = \frac{\xi_{10,276}}{\xi_{10,291}}$$

$$\xi_{10,294} = \frac{\xi_{10,278}}{\xi_{10,291}}$$

$$\xi_{10,295} = \begin{cases} \xi_{10,284} & \text{for } \xi_{10,280} \geq \xi_{10,281} \wedge \xi_{10,280} \geq \xi_{10,282} \\ \xi_{10,289} & \text{for } \xi_{10,281} \geq \xi_{10,282} \wedge \xi_{10,280} \leq \xi_{10,281} \\ \xi_{10,293} & \text{otherwise} \end{cases}$$

$$\xi_{10,296} = \begin{cases} \xi_{10,285} & \text{for } \xi_{10,280} \geq \xi_{10,281} \wedge \xi_{10,280} \geq \xi_{10,282} \\ \xi_{10,288} & \text{for } \xi_{10,281} \geq \xi_{10,282} \wedge \xi_{10,280} \leq \xi_{10,281} \\ \xi_{10,294} & \text{otherwise} \end{cases}$$

$$\xi_{10,297} = \begin{cases} \xi_{10,286} & \text{for } \xi_{10,280} \geq \xi_{10,281} \wedge \xi_{10,280} \geq \xi_{10,282} \\ \xi_{10,290} & \text{for } \xi_{10,281} \geq \xi_{10,282} \wedge \xi_{10,280} \leq \xi_{10,281} \\ \xi_{10,292} & \text{otherwise} \end{cases}$$

$$\xi_{10,298} = \xi_{10,266}\xi_{10,297} - \xi_{10,267}\xi_{10,296}$$

$$\xi_{10,299} = -\xi_{10,265}\xi_{10,297} + \xi_{10,267}\xi_{10,295}$$

$$\xi_{10,300} = \xi_{10,265}\xi_{10,296} - \xi_{10,266}\xi_{10,295}$$

$$\xi_{10,301} = 1 - \xi_{10,268}^2$$

$$\xi_{10,302} = -\xi_{10,268}\xi_{10,269}$$

$$\xi_{10,303} = -\xi_{10,268}\xi_{10,270}$$

$$\xi_{10,304} = 1 - \xi_{10,269}^2$$

$$\xi_{10,305} = -\xi_{10,269}\xi_{10,270}$$

$$\xi_{10,306} = 1 - \xi_{10,270}^2$$

$$\xi_{10,307} = \xi_{10,301}^2 + \xi_{10,302}^2 + \xi_{10,303}^2$$

$$\xi_{10,308} = \xi_{10,302}^2 + \xi_{10,304}^2 + \xi_{10,305}^2$$

$$\xi_{10,309} = \xi_{10,303}^2 + \xi_{10,305}^2 + \xi_{10,306}^2$$

$$\xi_{10,310} = \sqrt{\xi_{10,307}}$$

$$\xi_{10,311} = \frac{\xi_{10,301}}{\xi_{10,310}}$$

$$\xi_{10,312} = \frac{\xi_{10,302}}{\xi_{10,310}}$$

$$\xi_{10,313} = \frac{\xi_{10,303}}{\xi_{10,310}}$$

$$\xi_{10,314} = \sqrt{\xi_{10,308}}$$

$$\xi_{10,315} = \frac{\xi_{10,304}}{\xi_{10,314}}$$

$$\xi_{10,316} = \frac{\xi_{10,302}}{\xi_{10,314}}$$

$$\xi_{10,317} = \frac{\xi_{10,305}}{\xi_{10,314}}$$

$$\xi_{10,318} = \sqrt{\xi_{10,309}}$$

$$\xi_{10,319} = \frac{\xi_{10,306}}{\xi_{10,318}}$$

$$\xi_{10,320} = \frac{\xi_{10,303}}{\xi_{10,318}}$$

$$\xi_{10,321} = \frac{\xi_{10,305}}{\xi_{10,318}}$$

$$\xi_{10,322} = \begin{cases} \xi_{10,311} & \text{for } \xi_{10,307} \geq \xi_{10,308} \wedge \xi_{10,307} \geq \xi_{10,309} \\ \xi_{10,316} & \text{for } \xi_{10,308} \geq \xi_{10,309} \wedge \xi_{10,307} \leq \xi_{10,308} \\ \xi_{10,320} & \text{otherwise} \end{cases}$$

$$\xi_{10,323} = \begin{cases} \xi_{10,312} & \text{for } \xi_{10,307} \geq \xi_{10,308} \wedge \xi_{10,307} \geq \xi_{10,309} \\ \xi_{10,315} & \text{for } \xi_{10,308} \geq \xi_{10,309} \wedge \xi_{10,307} \leq \xi_{10,308} \\ \xi_{10,321} & \text{otherwise} \end{cases}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,324} = \begin{cases} \xi_{10,313} & \text{for } \xi_{10,307} \geq \xi_{10,308} \wedge \xi_{10,307} \geq \xi_{10,309} \\ \xi_{10,317} & \text{for } \xi_{10,308} \geq \xi_{10,309} \wedge \xi_{10,307} \leq \xi_{10,308} \\ \xi_{10,319} & \text{otherwise} \end{cases}$$

$$\xi_{10,325} = \xi_{10,269}\xi_{10,324} - \xi_{10,270}\xi_{10,323}$$

$$\xi_{10,326} = -\xi_{10,268}\xi_{10,324} + \xi_{10,270}\xi_{10,322}$$

$$\xi_{10,327} = \xi_{10,268}\xi_{10,323} - \xi_{10,269}\xi_{10,322}$$

$$\xi_{10,328} = \xi_{10,266}\xi_{10,270} - \xi_{10,267}\xi_{10,269}$$

$$\xi_{10,329} = -\xi_{10,265}\xi_{10,270} + \xi_{10,267}\xi_{10,268}$$

$$\xi_{10,330} = \xi_{10,265}\xi_{10,269} - \xi_{10,266}\xi_{10,268}$$

$$\xi_{10,331} = 1 - \xi_{10,271}^2$$

$$\xi_{10,332} = -\xi_{10,271}\xi_{10,272}$$

$$\xi_{10,333} = -\xi_{10,271}\xi_{10,273}$$

$$\xi_{10,334} = 1 - \xi_{10,272}^2$$

$$\xi_{10,335} = -\xi_{10,272}\xi_{10,273}$$

$$\xi_{10,336} = 1 - \xi_{10,273}^2$$

$$\xi_{10,337} = \xi_{10,331}^2 + \xi_{10,332}^2 + \xi_{10,333}^2$$

$$\xi_{10,338} = \xi_{10,332}^2 + \xi_{10,334}^2 + \xi_{10,335}^2$$

$$\xi_{10,339} = \xi_{10,333}^2 + \xi_{10,335}^2 + \xi_{10,336}^2$$

$$\xi_{10,340} = \sqrt{\xi_{10,337}}$$

$$\xi_{10,341} = \frac{\xi_{10,331}}{\xi_{10,340}}$$

$$\xi_{10,342} = \frac{\xi_{10,332}}{\xi_{10,340}}$$

$$\xi_{10,343} = \frac{\xi_{10,333}}{\xi_{10,340}}$$

$$\xi_{10,344} = \sqrt{\xi_{10,338}}$$

$$\xi_{10,345} = \frac{\xi_{10,334}}{\xi_{10,344}}$$

$$\xi_{10,346} = \frac{\xi_{10,332}}{\xi_{10,344}}$$

$$\xi_{10,347} = \frac{\xi_{10,335}}{\xi_{10,344}}$$

$$\xi_{10,348} = \sqrt{\xi_{10,339}}$$

$$\xi_{10,349} = \frac{\xi_{10,336}}{\xi_{10,348}}$$

$$\xi_{10,350} = \frac{\xi_{10,333}}{\xi_{10,348}}$$

$$\xi_{10,351} = \frac{\xi_{10,335}}{\xi_{10,348}}$$

$$\xi_{10,352} = \begin{cases} \xi_{10,341} & \text{for } \xi_{10,337} \geq \xi_{10,338} \wedge \xi_{10,337} \geq \xi_{10,339} \\ \xi_{10,346} & \text{for } \xi_{10,338} \geq \xi_{10,339} \wedge \xi_{10,337} \leq \xi_{10,338} \\ \xi_{10,350} & \text{otherwise} \end{cases}$$

$$\xi_{10,353} = \begin{cases} \xi_{10,342} & \text{for } \xi_{10,337} \geq \xi_{10,338} \wedge \xi_{10,337} \geq \xi_{10,339} \\ \xi_{10,345} & \text{for } \xi_{10,338} \geq \xi_{10,339} \wedge \xi_{10,337} \leq \xi_{10,338} \\ \xi_{10,351} & \text{otherwise} \end{cases}$$

$$\xi_{10,354} = \begin{cases} \xi_{10,343} & \text{for } \xi_{10,337} \geq \xi_{10,338} \wedge \xi_{10,337} \geq \xi_{10,339} \\ \xi_{10,347} & \text{for } \xi_{10,338} \geq \xi_{10,339} \wedge \xi_{10,337} \leq \xi_{10,338} \\ \xi_{10,349} & \text{otherwise} \end{cases}$$

$$\xi_{10,355} = \xi_{10,272}\xi_{10,354} - \xi_{10,273}\xi_{10,353}$$

$$\xi_{10,356} = -\xi_{10,271}\xi_{10,354} + \xi_{10,273}\xi_{10,352}$$

$$\xi_{10,357} = \xi_{10,271}\xi_{10,353} - \xi_{10,272}\xi_{10,352}$$

$$\xi_{10,358} = -\xi_{10,266}\xi_{10,273} + \xi_{10,267}\xi_{10,272}$$

$$\xi_{10,359} = \xi_{10,265}\xi_{10,273} - \xi_{10,267}\xi_{10,271}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,360} = -\xi_{10,265}\xi_{10,272} + \xi_{10,266}\xi_{10,271}$$

$$\xi_{10,361} = \xi_{10,269}\xi_{10,273} - \xi_{10,270}\xi_{10,272}$$

$$\xi_{10,362} = -\xi_{10,268}\xi_{10,273} + \xi_{10,270}\xi_{10,271}$$

$$\xi_{10,363} = \xi_{10,268}\xi_{10,272} - \xi_{10,269}\xi_{10,271}$$

$$\xi_{10,364} = \begin{cases} 1 & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} = 0 \\ \xi_{10,265} & \text{for } \xi_{10,263} = 0 \land \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \\ \xi_{10,325} & \text{for } \xi_{10,262} = 0 \land \xi_{10,264} = 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,265} & \text{for } \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,352} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,265} & \text{for } \xi_{10,263} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,361} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,265} & \text{for } \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \end{cases}$$

$$\xi_{10,365} = \begin{cases} 0 & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} = 0 \\ \xi_{10,295} & \text{for } \xi_{10,263} = 0 \land \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \\ \xi_{10,268} & \text{for } \xi_{10,264} = 0 \land (\xi_{10,262} = 0 \lor \xi_{10,264} = 0) \land (\xi_{10,262} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,264} = 0 \lor \xi_{10,262} \neq 0) \land (\xi_{10,264} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,263} \neq 0) \land \xi_{10,263} \neq 0 \\ \xi_{10,355} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,358} & \text{for } \xi_{10,263} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,268} & \text{for } (\xi_{10,262} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} = 0 \lor \xi_{10,264} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,264} \neq 0) \land (\xi_{10,263} \neq 0 \lor \xi_{10,264} \neq 0) \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \end{cases}$$

$$\xi_{10,366} = \begin{cases} 0 & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} = 0 \\ \xi_{10,266} & \text{for } \xi_{10,263} = 0 \land \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \\ \xi_{10,326} & \text{for } \xi_{10,262} = 0 \land \xi_{10,264} = 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,266} & \text{for } \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,353} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,266} & \text{for } \xi_{10,263} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,362} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,266} & \text{for } \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \end{cases}$$

$$\xi_{10,367} = \begin{cases} 0 & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} = 0 \\ \xi_{10,266} & \text{for } \xi_{10,263} = 0 \land \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \\ \xi_{10,326} & \text{for } \xi_{10,262} = 0 \land \xi_{10,264} = 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,266} & \text{for } \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,353} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,266} & \text{for } \xi_{10,263} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,362} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,266} & \text{for } \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \end{cases}$$

$$\xi_{10,368} = \begin{cases} 1 & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} = 0 \\ \xi_{10,296} & \text{for } \xi_{10,263} = 0 \land \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \\ \xi_{10,269} & \text{for } \xi_{10,264} = 0 \land (\xi_{10,262} = 0 \lor \xi_{10,264} = 0) \land (\xi_{10,262} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,264} = 0 \lor \xi_{10,262} \neq 0) \land (\xi_{10,264} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,263} \neq 0) \land \xi_{10,263} \neq 0 \\ \xi_{10,356} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,359} & \text{for } \xi_{10,263} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,269} & \text{for } (\xi_{10,262} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} = 0 \lor \xi_{10,264} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,264} \neq 0) \land (\xi_{10,263} \neq 0 \lor \xi_{10,264} \neq 0) \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \end{cases}$$

$$\xi_{10,370} = \begin{cases} 0 & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} = 0 \\ \xi_{10,267} & \text{for } \xi_{10,263} = 0 \land \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \\ \xi_{10,327} & \text{for } \xi_{10,262} = 0 \land \xi_{10,264} = 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,267} & \text{for } \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \\ \xi_{10,354} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,267} & \text{for } \xi_{10,263} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,363} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,267} & \text{for } \xi_{10,262} \neq 0 \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \end{cases}$$

$$\xi_{10,371} = \begin{cases} 0 & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} = 0 \\ \xi_{10,297} & \text{for } \xi_{10,263} = 0 \land \xi_{10,264} = 0 \land \xi_{10,262} \neq 0 \\ \xi_{10,270} & \text{for } \xi_{10,264} = 0 \land (\xi_{10,262} = 0 \lor \xi_{10,264} = 0) \land (\xi_{10,262} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,264} = 0 \lor \xi_{10,262} \neq 0) \land (\xi_{10,264} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,263} \neq 0) \land \xi_{10,263} \neq 0 \\ \xi_{10,357} & \text{for } \xi_{10,262} = 0 \land \xi_{10,263} = 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,360} & \text{for } \xi_{10,263} = 0 \land \xi_{10,262} \neq 0 \land \xi_{10,264} \neq 0 \\ \xi_{10,270} & \text{for } (\xi_{10,262} = 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} = 0 \lor \xi_{10,264} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,263} \neq 0) \land (\xi_{10,262} \neq 0 \lor \xi_{10,264} \neq 0) \land (\xi_{10,263} \neq 0 \lor \xi_{10,264} \neq 0) \land \xi_{10,263} \neq 0 \land \xi_{10,264} \neq 0 \end{cases}$$

$$\xi_{10,373} = \frac{\xi_{10,364}}{4} + \frac{\xi_{10,368}}{4} + \frac{\xi_{10,372}}{4} + \frac{1}{4}$$

$$\xi_{10,374} = \frac{\xi_{10,364}}{4} - \frac{\xi_{10,368}}{4} - \frac{\xi_{10,372}}{4} + \frac{1}{4}$$

$$\xi_{10,375} = -\frac{\xi_{10,364}}{4} + \frac{\xi_{10,368}}{4} - \frac{\xi_{10,372}}{4} + \frac{1}{4}$$

$$\xi_{10,376} = -\frac{\xi_{10,364}}{4} - \frac{\xi_{10,368}}{4} + \frac{\xi_{10,372}}{4} + \frac{1}{4}$$

$$\xi_{10,377} = \sqrt{\xi_{10,373}}$$

$$\xi_{10,378} = \frac{-\xi_{10,369} + \xi_{10,371}}{4\xi_{10,377}}$$

$$\xi_{10,379} = \frac{\xi_{10,366} - \xi_{10,370}}{4\xi_{10,377}}$$

$$\xi_{10,380} = \frac{-\xi_{10,365} + \xi_{10,367}}{4\xi_{10,377}}$$

$$\xi_{10,381} = \sqrt{\xi_{10,374}}$$

$$\xi_{10,382} = \frac{-\xi_{10,369} + \xi_{10,371}}{4\xi_{10,381}}$$

$$\xi_{10,383} = \frac{\xi_{10,365} + \xi_{10,367}}{4\xi_{10,381}}$$

$$\xi_{10,384} = \frac{\xi_{10,366} + \xi_{10,370}}{4\xi_{10,381}}$$

$$\xi_{10,385} = \sqrt{\xi_{10,375}}$$

$$\xi_{10,386} = \frac{\xi_{10,366} - \xi_{10,370}}{4\xi_{10,385}}$$

$$\xi_{10,387} = \frac{\xi_{10,365} + \xi_{10,367}}{4\xi_{10,385}}$$

$$\xi_{10,388} = \frac{\xi_{10,369} + \xi_{10,371}}{4\xi_{10,385}}$$

$$\xi_{10,389} = \sqrt{\xi_{10,376}}$$

$$\xi_{10,390} = \frac{-\xi_{10,365} + \xi_{10,367}}{4\xi_{10,389}}$$

$$\xi_{10,391} = \frac{\xi_{10,366} + \xi_{10,370}}{4\xi_{10,389}}$$

$$\xi_{10,392} = \frac{\xi_{10,369} + \xi_{10,371}}{4\xi_{10,389}}$$

$$\xi_{10,393} = \begin{cases} \xi_{10,377} & \text{for } \xi_{10,373} \geq \xi_{10,374} \land \xi_{10,373} \geq \xi_{10,375} \land \xi_{10,373} \geq \xi_{10,376} \\ \xi_{10,382} & \text{for } \xi_{10,374} \geq \xi_{10,375} \land \xi_{10,374} \geq \xi_{10,376} \land \xi_{10,373} \leq \xi_{10,374} \\ \xi_{10,386} & \text{for } \xi_{10,375} \geq \xi_{10,376} \land \xi_{10,373} \leq \xi_{10,375} \land \xi_{10,374} \leq \xi_{10,375} \\ \xi_{10,390} & \text{otherwise} \end{cases}$$

$$\xi_{10,394} = \begin{cases} \xi_{10,378} & \text{for } \xi_{10,373} \geq \xi_{10,374} \land \xi_{10,373} \geq \xi_{10,375} \land \xi_{10,373} \geq \xi_{10,376} \\ \xi_{10,381} & \text{for } \xi_{10,374} \geq \xi_{10,375} \land \xi_{10,374} \geq \xi_{10,376} \land \xi_{10,373} \leq \xi_{10,374} \\ \xi_{10,387} & \text{for } \xi_{10,375} \geq \xi_{10,376} \land \xi_{10,373} \leq \xi_{10,375} \land \xi_{10,374} \leq \xi_{10,375} \\ \xi_{10,391} & \text{otherwise} \end{cases}$$

$$\xi_{10,395} = \begin{cases} \xi_{10,379} & \text{for } \xi_{10,373} \geq \xi_{10,374} \land \xi_{10,373} \geq \xi_{10,375} \land \xi_{10,373} \geq \xi_{10,376} \\ \xi_{10,383} & \text{for } \xi_{10,374} \geq \xi_{10,375} \land \xi_{10,374} \geq \xi_{10,376} \land \xi_{10,373} \leq \xi_{10,374} \\ \xi_{10,385} & \text{for } \xi_{10,375} \geq \xi_{10,376} \land \xi_{10,373} \leq \xi_{10,375} \land \xi_{10,374} \leq \xi_{10,375} \\ \xi_{10,392} & \text{otherwise} \end{cases}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,396} = \begin{cases} \xi_{10,380} & \text{for } \xi_{10,373} \geq \xi_{10,374} \wedge \xi_{10,373} \geq \xi_{10,375} \wedge \xi_{10,373} \geq \xi_{10,376} \\ \xi_{10,384} & \text{for } \xi_{10,374} \geq \xi_{10,375} \wedge \xi_{10,374} \geq \xi_{10,376} \wedge \xi_{10,373} \leq \xi_{10,374} \\ \xi_{10,388} & \text{for } \xi_{10,375} \geq \xi_{10,376} \wedge \xi_{10,373} \leq \xi_{10,375} \wedge \xi_{10,374} \leq \xi_{10,375} \\ \xi_{10,389} & \text{otherwise} \end{cases}$$

$$\xi_{10,397} = \frac{\xi_{10,193}}{4} + \frac{\xi_{10,257}}{4} + \frac{\xi_{10,261}}{4} + \frac{1}{4}$$

$$\xi_{10,398} = \frac{\xi_{10,193}}{4} - \frac{\xi_{10,257}}{4} - \frac{\xi_{10,261}}{4} + \frac{1}{4}$$

$$\xi_{10,399} = -\frac{\xi_{10,193}}{4} + \frac{\xi_{10,257}}{4} - \frac{\xi_{10,261}}{4} + \frac{1}{4}$$

$$\xi_{10,400} = -\frac{\xi_{10,193}}{4} - \frac{\xi_{10,257}}{4} + \frac{\xi_{10,261}}{4} + \frac{1}{4}$$

$$\xi_{10,401} = \sqrt{\xi_{10,397}}$$

$$\xi_{10,402} = \frac{\xi_{10,258} - \xi_{10,260}}{4\xi_{10,401}}$$

$$\xi_{10,403} = \frac{-\xi_{10,195} + \xi_{10,259}}{4\xi_{10,401}}$$

$$\xi_{10,404} = \frac{\xi_{10,194} - \xi_{10,256}}{4\xi_{10,401}}$$

$$\xi_{10,405} = \sqrt{\xi_{10,398}}$$

$$\xi_{10,406} = \frac{\xi_{10,258} - \xi_{10,260}}{4\xi_{10,405}}$$

$$\xi_{10,407} = \frac{\xi_{10,194} + \xi_{10,256}}{4\xi_{10,405}}$$

$$\xi_{10,408} = \frac{\xi_{10,195} + \xi_{10,259}}{4\xi_{10,405}}$$

$$\xi_{10,409} = \sqrt{\xi_{10,399}}$$

$$\xi_{10,410} = \frac{-\xi_{10,195} + \xi_{10,259}}{4\xi_{10,409}}$$

$$\xi_{10,411} = \frac{\xi_{10,194} + \xi_{10,256}}{4\xi_{10,409}}$$

$$\xi_{10,412} = \frac{\xi_{10,258} + \xi_{10,260}}{4\xi_{10,409}}$$

$$\xi_{10,413} = \sqrt{\xi_{10,400}}$$

$$\xi_{10,414} = \frac{\xi_{10,194} - \xi_{10,256}}{4\xi_{10,413}}$$

$$\xi_{10,415} = \frac{\xi_{10,195} + \xi_{10,259}}{4\xi_{10,413}}$$

$$\xi_{10,416} = \frac{\xi_{10,258} + \xi_{10,260}}{4\xi_{10,413}}$$

$$\xi_{10,417} = \begin{cases} \xi_{10,401} & \text{for } \xi_{10,397} \geq \xi_{10,398} \wedge \xi_{10,397} \geq \xi_{10,399} \wedge \xi_{10,397} \geq \xi_{10,400} \\ \xi_{10,406} & \text{for } \xi_{10,398} \geq \xi_{10,399} \wedge \xi_{10,398} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,398} \\ \xi_{10,410} & \text{for } \xi_{10,399} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,399} \wedge \xi_{10,398} \leq \xi_{10,399} \\ \xi_{10,414} & \text{otherwise} \end{cases}$$

$$\xi_{10,418} = \begin{cases} \xi_{10,402} & \text{for } \xi_{10,397} \geq \xi_{10,398} \wedge \xi_{10,397} \geq \xi_{10,399} \wedge \xi_{10,397} \geq \xi_{10,400} \\ \xi_{10,405} & \text{for } \xi_{10,398} \geq \xi_{10,399} \wedge \xi_{10,398} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,398} \\ \xi_{10,411} & \text{for } \xi_{10,399} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,399} \wedge \xi_{10,398} \leq \xi_{10,399} \\ \xi_{10,415} & \text{otherwise} \end{cases}$$

$$\xi_{10,419} = \begin{cases} \xi_{10,403} & \text{for } \xi_{10,397} \geq \xi_{10,398} \wedge \xi_{10,397} \geq \xi_{10,399} \wedge \xi_{10,397} \geq \xi_{10,400} \\ \xi_{10,407} & \text{for } \xi_{10,398} \geq \xi_{10,399} \wedge \xi_{10,398} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,398} \\ \xi_{10,409} & \text{for } \xi_{10,399} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,399} \wedge \xi_{10,398} \leq \xi_{10,399} \\ \xi_{10,416} & \text{otherwise} \end{cases}$$

$$\xi_{10,420} = \begin{cases} \xi_{10,404} & \text{for } \xi_{10,397} \geq \xi_{10,398} \wedge \xi_{10,397} \geq \xi_{10,399} \wedge \xi_{10,397} \geq \xi_{10,400} \\ \xi_{10,408} & \text{for } \xi_{10,398} \geq \xi_{10,399} \wedge \xi_{10,398} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,398} \\ \xi_{10,412} & \text{for } \xi_{10,399} \geq \xi_{10,400} \wedge \xi_{10,397} \leq \xi_{10,399} \wedge \xi_{10,398} \leq \xi_{10,399} \\ \xi_{10,413} & \text{otherwise} \end{cases}$$

$$\xi_{10,421} = \xi_{10,112}\left(\xi_{10,117} + i\xi_{10,118}\right)$$

$$\xi_{10,422} = -\frac{\sqrt{2}\xi_{10,112}\xi_{10,114}}{2} + \frac{\sqrt{2}i\xi_{10,113}(\xi_{10,117} + i\xi_{10,118})}{2}$$

$$\xi_{10,423} = \frac{\sqrt{2}i\xi_{10,112}\xi_{10,114}}{2} - \frac{\sqrt{2}\xi_{10,113}(\xi_{10,117} + i\xi_{10,118})}{2}$$

$$\xi_{10,424} = -\xi_{10,113}\xi_{10,114}$$

$$\xi_{10,425} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{10,112}(\xi_{10,117} + i\xi_{10,118})}{2} - \frac{\sqrt{2}i\xi_{10,113}\xi_{10,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{10,112}(\xi_{10,117} - i\xi_{10,118})}{2} - \frac{\sqrt{2}\xi_{10,113}\xi_{10,114}}{2}\right)}{2}$$

$$\xi_{10,426} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{10,112}(\xi_{10,117} + i\xi_{10,118})}{2} - \frac{\sqrt{2}i\xi_{10,113}\xi_{10,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{10,112}(\xi_{10,117} - i\xi_{10,118})}{2} - \frac{\sqrt{2}\xi_{10,113}\xi_{10,114}}{2}\right)}{2}$$

$$\xi_{10,427} = \frac{\sqrt{2}i\xi_{10,112}\xi_{10,114}}{2} - \frac{\sqrt{2}\xi_{10,113}(\xi_{10,117} - i\xi_{10,118})}{2}$$

$$\xi_{10,428} = \frac{\sqrt{2}i\left(\frac{\sqrt{2}\xi_{10,112}(\xi_{10,117} - i\xi_{10,118})}{2} - \frac{\sqrt{2}i\xi_{10,113}\xi_{10,114}}{2}\right)}{2} + \frac{\sqrt{2}\left(\frac{\sqrt{2}i\xi_{10,112}(\xi_{10,117} + i\xi_{10,118})}{2} - \frac{\sqrt{2}\xi_{10,113}\xi_{10,114}}{2}\right)}{2}$$

$$\xi_{10,429} = \frac{\sqrt{2}\left(\frac{\sqrt{2}\xi_{10,112}(\xi_{10,117} - i\xi_{10,118})}{2} - \frac{\sqrt{2}i\xi_{10,113}\xi_{10,114}}{2}\right)}{2} + \frac{\sqrt{2}i\left(\frac{\sqrt{2}i\xi_{10,112}(\xi_{10,117} + i\xi_{10,118})}{2} - \frac{\sqrt{2}\xi_{10,113}\xi_{10,114}}{2}\right)}{2}$$

$$\xi_{10,430} = -\frac{\sqrt{2}\xi_{10,112}\xi_{10,114}}{2} + \frac{\sqrt{2}i\xi_{10,113}(\xi_{10,117} - i\xi_{10,118})}{2}$$

$$\xi_{10,431} = \xi_{10,112}\left(\xi_{10,117} - i\xi_{10,118}\right)$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,432} = (\xi_{10,393} - i\xi_{10,396})\,(\xi_{10,417} - i\xi_{10,420})$$

$$\xi_{10,433} = (\xi_{10,393} - i\xi_{10,396})\,(-i\xi_{10,418} - \xi_{10,419})$$

$$\xi_{10,434} = (\xi_{10,417} - i\xi_{10,420})\,(-i\xi_{10,394} - \xi_{10,395})$$

$$\xi_{10,435} = (-i\xi_{10,394} - \xi_{10,395})\,(-i\xi_{10,418} - \xi_{10,419})$$

$$\xi_{10,436} = (\xi_{10,393} - i\xi_{10,396})\,(-i\xi_{10,418} + \xi_{10,419})$$

$$\xi_{10,437} = (\xi_{10,393} - i\xi_{10,396})\,(\xi_{10,417} + i\xi_{10,420})$$

$$\xi_{10,438} = (-i\xi_{10,394} - \xi_{10,395})\,(-i\xi_{10,418} + \xi_{10,419})$$

$$\xi_{10,439} = (\xi_{10,417} + i\xi_{10,420})\,(-i\xi_{10,394} - \xi_{10,395})$$

$$\xi_{10,440} = (\xi_{10,417} - i\xi_{10,420})\,(-i\xi_{10,394} + \xi_{10,395})$$

$$\xi_{10,441} = (-i\xi_{10,394} + \xi_{10,395})\,(-i\xi_{10,418} - \xi_{10,419})$$

$$\xi_{10,442} = (\xi_{10,393} + i\xi_{10,396})\,(\xi_{10,417} - i\xi_{10,420})$$

$$\xi_{10,443} = (\xi_{10,393} + i\xi_{10,396})\,(-i\xi_{10,418} - \xi_{10,419})$$

$$\xi_{10,444} = (-i\xi_{10,394} + \xi_{10,395})\,(-i\xi_{10,418} + \xi_{10,419})$$

$$\xi_{10,445} = (\xi_{10,417} + i\xi_{10,420})\,(-i\xi_{10,394} + \xi_{10,395})$$

$$\xi_{10,446} = (\xi_{10,393} + i\xi_{10,396})\,(-i\xi_{10,418} + \xi_{10,419})$$

$$\xi_{10,447} = (\xi_{10,393} + i\xi_{10,396})\,(\xi_{10,417} + i\xi_{10,420})$$

$$\xi_{10,448} = \xi_{10,421}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}}\right) + \xi_{10,422}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}}\right) + \xi_{10,423}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}}\right) + \xi_{10,424}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,447}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,444}}\right)$$

$$\xi_{10,449} = \xi_{10,422}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}}\right) + \xi_{10,425}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}}\right) + \xi_{10,427}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,447}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,444}}\right) + \xi_{10,428}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}}\right)$$

$$\xi_{10,450} = \xi_{10,423}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}}\right) + \xi_{10,426}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}}\right) + \xi_{10,429}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}}\right) + \xi_{10,430}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,447}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,444}}\right)$$

$$\xi_{10,451} = \xi_{10,424}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}}\right) + \xi_{10,427}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}}\right) + \xi_{10,430}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}}\right) + \xi_{10,431}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,447}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,444}}\right)$$

$$\xi_{10,452} = \xi_{10,421}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}}\right) + \xi_{10,422}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}}\right) + \xi_{10,423}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}}\right) + \xi_{10,424}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,441}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,447}}\right)$$

$$\xi_{10,453} = \xi_{10,422}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}}\right) + \xi_{10,425}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}}\right) + \xi_{10,427}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,444}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}}\right) + \xi_{10,428}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}}\right)$$

$$\xi_{10,454} = \xi_{10,423}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}}\right) + \xi_{10,426}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}}\right) + \xi_{10,429}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}}\right) + \xi_{10,430}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,441}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,447}}\right)$$

$$\xi_{10,455} = \xi_{10,424}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,432}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,435}}\right) + \xi_{10,427}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,436}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,439}}\right) + \xi_{10,430}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,440}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,443}}\right) + \xi_{10,431}\left(-ie^{-i\xi_{10,105}}\sin\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,444}} + e^{-i\xi_{10,105}}\cos\left(\xi_{10,103} - \xi_{10,104}\right)\overline{\xi_{10,447}}\right)$$

$$\xi_{10,456} = \xi_{10,421}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}}\right) + \xi_{10,422}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}}\right) + \xi_{10,423}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}}\right) + \xi_{10,424}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}}\right)$$

$$\xi_{10,457} = \xi_{10,422}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}}\right) + \xi_{10,425}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}}\right) + \xi_{10,427}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}}\right) + \xi_{10,428}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}}\right)$$

$$\xi_{10,458} = \xi_{10,423}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}}\right) + \xi_{10,426}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}}\right) + \xi_{10,429}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}}\right) + \xi_{10,430}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}}\right)$$

$$\xi_{10,459} = \xi_{10,424}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}}\right) + \xi_{10,427}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}}\right) + \xi_{10,430}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}}\right) + \xi_{10,431}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}}\right)$$

$$\xi_{10,460} = \xi_{10,421}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}}\right) + \xi_{10,422}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}}\right) + \xi_{10,423}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}}\right) + \xi_{10,424}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}}\right)$$

$$\xi_{10,461} = \xi_{10,422}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}}\right) + \xi_{10,425}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}}\right) + \xi_{10,427}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}}\right) + \xi_{10,428}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}}\right)$$

$$\xi_{10,462} = \xi_{10,423}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}}\right) + \xi_{10,426}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}}\right) + \xi_{10,429}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}}\right) + \xi_{10,430}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}}\right)$$

$$\xi_{10,463} = \xi_{10,424}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,433}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,434}}\right) + \xi_{10,427}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,437}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,438}}\right) + \xi_{10,430}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,441}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,442}}\right) + \xi_{10,431}\left(-ie^{i\xi_{10,105}}\sin\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,445}} + e^{i\xi_{10,105}}\cos\left(\xi_{10,103} + \xi_{10,104}\right)\overline{\xi_{10,446}}\right)$$

$$\xi_{10,464} = \frac{\operatorname{re}\left(\xi_{10,448}\right)}{4} + \frac{\operatorname{re}\left(\xi_{10,455}\right)}{4} + \frac{\operatorname{re}\left(\xi_{10,457}\right)}{4} + \frac{\operatorname{re}\left(\xi_{10,462}\right)}{4}$$

$$\xi_{10,465} = -\frac{\operatorname{im}\left(\xi_{10,449}\right)}{4} - \frac{\operatorname{im}\left(\xi_{10,454}\right)}{4} - \frac{\operatorname{im}\left(\xi_{10,456}\right)}{4} - \frac{\operatorname{im}\left(\xi_{10,463}\right)}{4}$$

$$\xi_{10,466} = -\frac{\operatorname{re}\left(\xi_{10,449}\right)}{4} + \frac{\operatorname{re}\left(\xi_{10,454}\right)}{4} + \frac{\operatorname{re}\left(\xi_{10,456}\right)}{4} - \frac{\operatorname{re}\left(\xi_{10,463}\right)}{4}$$

$$\xi_{10,467} = -\frac{\operatorname{im}\left(\xi_{10,448}\right)}{4} + \frac{\operatorname{im}\left(\xi_{10,455}\right)}{4} + \frac{\operatorname{im}\left(\xi_{10,457}\right)}{4} - \frac{\operatorname{im}\left(\xi_{10,462}\right)}{4}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,468} = -\frac{\operatorname{im}(\xi_{10,450})}{4} - \frac{\operatorname{im}(\xi_{10,453})}{4} - \frac{\operatorname{im}(\xi_{10,459})}{4} - \frac{\operatorname{im}(\xi_{10,460})}{4}$$

$$\xi_{10,469} = -\frac{\operatorname{re}(\xi_{10,451})}{4} - \frac{\operatorname{re}(\xi_{10,452})}{4} - \frac{\operatorname{re}(\xi_{10,458})}{4} - \frac{\operatorname{re}(\xi_{10,461})}{4}$$

$$\xi_{10,470} = \frac{\operatorname{im}(\xi_{10,451})}{4} - \frac{\operatorname{im}(\xi_{10,452})}{4} - \frac{\operatorname{im}(\xi_{10,458})}{4} + \frac{\operatorname{im}(\xi_{10,461})}{4}$$

$$\xi_{10,471} = -\frac{\operatorname{re}(\xi_{10,450})}{4} + \frac{\operatorname{re}(\xi_{10,453})}{4} + \frac{\operatorname{re}(\xi_{10,459})}{4} - \frac{\operatorname{re}(\xi_{10,460})}{4}$$

$$\xi_{10,472} = -\frac{\operatorname{re}(\xi_{10,450})}{4} + \frac{\operatorname{re}(\xi_{10,453})}{4} - \frac{\operatorname{re}(\xi_{10,459})}{4} + \frac{\operatorname{re}(\xi_{10,460})}{4}$$

$$\xi_{10,473} = \frac{\operatorname{im}(\xi_{10,451})}{4} - \frac{\operatorname{im}(\xi_{10,452})}{4} + \frac{\operatorname{im}(\xi_{10,458})}{4} - \frac{\operatorname{im}(\xi_{10,461})}{4}$$

$$\xi_{10,474} = \frac{\operatorname{re}(\xi_{10,451})}{4} + \frac{\operatorname{re}(\xi_{10,452})}{4} - \frac{\operatorname{re}(\xi_{10,458})}{4} - \frac{\operatorname{re}(\xi_{10,461})}{4}$$

$$\xi_{10,475} = \frac{\operatorname{im}(\xi_{10,450})}{4} + \frac{\operatorname{im}(\xi_{10,453})}{4} - \frac{\operatorname{im}(\xi_{10,459})}{4} - \frac{\operatorname{im}(\xi_{10,460})}{4}$$

$$\xi_{10,476} = -\frac{\operatorname{im}(\xi_{10,448})}{4} + \frac{\operatorname{im}(\xi_{10,455})}{4} - \frac{\operatorname{im}(\xi_{10,457})}{4} + \frac{\operatorname{im}(\xi_{10,462})}{4}$$

$$\xi_{10,477} = -\frac{\operatorname{re}(\xi_{10,449})}{4} + \frac{\operatorname{re}(\xi_{10,454})}{4} - \frac{\operatorname{re}(\xi_{10,456})}{4} + \frac{\operatorname{re}(\xi_{10,463})}{4}$$

$$\xi_{10,478} = \frac{\operatorname{im}(\xi_{10,449})}{4} + \frac{\operatorname{im}(\xi_{10,454})}{4} - \frac{\operatorname{im}(\xi_{10,456})}{4} - \frac{\operatorname{im}(\xi_{10,463})}{4}$$

$$\xi_{10,479} = -\frac{\operatorname{re}(\xi_{10,448})}{4} - \frac{\operatorname{re}(\xi_{10,455})}{4} + \frac{\operatorname{re}(\xi_{10,457})}{4} + \frac{\operatorname{re}(\xi_{10,462})}{4}$$

$$\xi_{10,480} = \xi_{10,464}^2 + \xi_{10,468}^2 + \xi_{10,472}^2 + \xi_{10,476}^2$$

$$\xi_{10,481} = \xi_{10,465}^2 + \xi_{10,469}^2 + \xi_{10,473}^2 + \xi_{10,477}^2$$

$$\xi_{10,482} = \xi_{10,466}^2 + \xi_{10,470}^2 + \xi_{10,474}^2 + \xi_{10,478}^2$$

$$\xi_{10,483} = \xi_{10,467}^2 + \xi_{10,471}^2 + \xi_{10,475}^2 + \xi_{10,479}^2$$

$$\xi_{10,484} = \sqrt{\xi_{10,480}}$$

$$\xi_{10,485} = \frac{\xi_{10,464}}{\xi_{10,484}}$$

$$\xi_{10,486} = \frac{\xi_{10,468}}{\xi_{10,484}}$$

$$\xi_{10,487} = \frac{\xi_{10,472}}{\xi_{10,484}}$$

$$\xi_{10,488} = \frac{\xi_{10,476}}{\xi_{10,484}}$$

$$\xi_{10,489} = \sqrt{\xi_{10,481}}$$

$$\xi_{10,490} = \frac{\xi_{10,465}}{\xi_{10,489}}$$

$$\xi_{10,491} = \frac{\xi_{10,469}}{\xi_{10,489}}$$

$$\xi_{10,492} = \frac{\xi_{10,473}}{\xi_{10,489}}$$

$$\xi_{10,493} = \frac{\xi_{10,477}}{\xi_{10,489}}$$

$$\xi_{10,494} = \sqrt{\xi_{10,482}}$$

$$\xi_{10,495} = \frac{\xi_{10,466}}{\xi_{10,494}}$$

$$\xi_{10,496} = \frac{\xi_{10,470}}{\xi_{10,494}}$$

$$\xi_{10,497} = \frac{\xi_{10,474}}{\xi_{10,494}}$$

$$\xi_{10,498} = \frac{\xi_{10,478}}{\xi_{10,494}}$$

$$\xi_{10,499} = \sqrt{\xi_{10,483}}$$

$$\xi_{10,500} = \frac{\xi_{10,467}}{\xi_{10,499}}$$

$$\xi_{10,501} = \frac{\xi_{10,471}}{\xi_{10,499}}$$

$$\xi_{10,502} = \frac{\xi_{10,475}}{\xi_{10,499}}$$

$$\xi_{10,503} = \frac{\xi_{10,479}}{\xi_{10,499}}$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,504} = \begin{cases} \xi_{10,485} & \text{for } \xi_{10,480} \geq \xi_{10,481} \wedge \xi_{10,480} \geq \xi_{10,482} \wedge \xi_{10,480} \geq \xi_{10,483} \\ \xi_{10,490} & \text{for } \xi_{10,481} \geq \xi_{10,482} \wedge \xi_{10,481} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,481} \\ \xi_{10,495} & \text{for } \xi_{10,482} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,482} \wedge \xi_{10,481} \leq \xi_{10,482} \\ \xi_{10,500} & \text{otherwise} \end{cases}$$

$$\xi_{10,505} = \begin{cases} \xi_{10,486} & \text{for } \xi_{10,480} \geq \xi_{10,481} \wedge \xi_{10,480} \geq \xi_{10,482} \wedge \xi_{10,480} \geq \xi_{10,483} \\ \xi_{10,491} & \text{for } \xi_{10,481} \geq \xi_{10,482} \wedge \xi_{10,481} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,481} \\ \xi_{10,496} & \text{for } \xi_{10,482} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,482} \wedge \xi_{10,481} \leq \xi_{10,482} \\ \xi_{10,501} & \text{otherwise} \end{cases}$$

$$\xi_{10,506} = \begin{cases} \xi_{10,487} & \text{for } \xi_{10,480} \geq \xi_{10,481} \wedge \xi_{10,480} \geq \xi_{10,482} \wedge \xi_{10,480} \geq \xi_{10,483} \\ \xi_{10,492} & \text{for } \xi_{10,481} \geq \xi_{10,482} \wedge \xi_{10,481} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,481} \\ \xi_{10,497} & \text{for } \xi_{10,482} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,482} \wedge \xi_{10,481} \leq \xi_{10,482} \\ \xi_{10,502} & \text{otherwise} \end{cases}$$

$$\xi_{10,507} = \begin{cases} \xi_{10,488} & \text{for } \xi_{10,480} \geq \xi_{10,481} \wedge \xi_{10,480} \geq \xi_{10,482} \wedge \xi_{10,480} \geq \xi_{10,483} \\ \xi_{10,493} & \text{for } \xi_{10,481} \geq \xi_{10,482} \wedge \xi_{10,481} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,481} \\ \xi_{10,498} & \text{for } \xi_{10,482} \geq \xi_{10,483} \wedge \xi_{10,480} \leq \xi_{10,482} \wedge \xi_{10,481} \leq \xi_{10,482} \\ \xi_{10,503} & \text{otherwise} \end{cases}$$

$$\xi_{10,508} = \xi_{10,464}\xi_{10,504} + \xi_{10,468}\xi_{10,505} + \xi_{10,472}\xi_{10,506} + \xi_{10,476}\xi_{10,507}$$

$$\xi_{10,509} = \xi_{10,465}\xi_{10,504} + \xi_{10,469}\xi_{10,505} + \xi_{10,473}\xi_{10,506} + \xi_{10,477}\xi_{10,507}$$

$$\xi_{10,510} = \xi_{10,466}\xi_{10,504} + \xi_{10,470}\xi_{10,505} + \xi_{10,474}\xi_{10,506} + \xi_{10,478}\xi_{10,507}$$

$$\xi_{10,511} = \xi_{10,467}\xi_{10,504} + \xi_{10,471}\xi_{10,505} + \xi_{10,475}\xi_{10,506} + \xi_{10,479}\xi_{10,507}$$

$$\xi_{10,512} = \xi_{10,41}\xi_{10,52} - \xi_{10,42}\xi_{10,55} - \xi_{10,43}\xi_{10,56} - \xi_{10,44}\xi_{10,57}$$

$$\xi_{10,513} = \xi_{10,41}\xi_{10,55} + \xi_{10,42}\xi_{10,52} - \xi_{10,43}\xi_{10,57} + \xi_{10,44}\xi_{10,56}$$

$$\xi_{10,514} = \xi_{10,41}\xi_{10,56} + \xi_{10,42}\xi_{10,57} + \xi_{10,43}\xi_{10,52} - \xi_{10,44}\xi_{10,55}$$

$$\xi_{10,515} = \xi_{10,41}\xi_{10,57} - \xi_{10,42}\xi_{10,56} + \xi_{10,43}\xi_{10,55} + \xi_{10,44}\xi_{10,52}$$

$$\xi_{10,516} = \xi_{10,10}\xi_{10,55} + \xi_{10,11}\xi_{10,55} + \xi_{10,12}\xi_{10,55} + \xi_{10,54}\xi_{10,9}$$

$$\xi_{10,517} = -\xi_{10,10}\xi_{10,54} + \xi_{10,11}\xi_{10,55} - \xi_{10,12}\xi_{10,55} + \xi_{10,55}\xi_{10,9}$$

$$\xi_{10,518} = -\xi_{10,10}\xi_{10,55} - \xi_{10,11}\xi_{10,54} + \xi_{10,12}\xi_{10,55} + \xi_{10,55}\xi_{10,9}$$

$$\xi_{10,519} = \xi_{10,10}\xi_{10,55} - \xi_{10,11}\xi_{10,55} - \xi_{10,12}\xi_{10,54} + \xi_{10,55}\xi_{10,9}$$

$$\xi_{10,520} = \xi_{10,52}\left(\xi_{10,100}\xi_{10,505} + \xi_{10,101}\xi_{10,506} + \xi_{10,102}\xi_{10,507} + \xi_{10,504}\xi_{10,99}\right) - \xi_{10,52}\left(\xi_{10,100}\xi_{10,506} - \xi_{10,101}\xi_{10,505} + \xi_{10,102}\xi_{10,504} - \xi_{10,507}\xi_{10,99}\right) - \xi_{10,53}\left(\xi_{10,100}\xi_{10,504} + \xi_{10,101}\xi_{10,507} - \xi_{10,102}\xi_{10,506} - \xi_{10,505}\xi_{10,99}\right) - \xi_{10,54}\left(-\xi_{10,100}\xi_{10,507} + \xi_{10,101}\xi_{10,504} + \xi_{10,102}\xi_{10,505} - \xi_{10,506}\xi_{10,99}\right)$$

$$\xi_{10,521} = \xi_{10,52}\left(\xi_{10,100}\xi_{10,504} + \xi_{10,101}\xi_{10,507} - \xi_{10,102}\xi_{10,506} - \xi_{10,505}\xi_{10,99}\right) + \xi_{10,52}\left(-\xi_{10,100}\xi_{10,507} + \xi_{10,101}\xi_{10,504} + \xi_{10,102}\xi_{10,505} - \xi_{10,506}\xi_{10,99}\right) + \xi_{10,53}\left(\xi_{10,100}\xi_{10,505} + \xi_{10,101}\xi_{10,506} + \xi_{10,102}\xi_{10,507} + \xi_{10,504}\xi_{10,99}\right) - \xi_{10,54}\left(\xi_{10,100}\xi_{10,506} - \xi_{10,101}\xi_{10,505} + \xi_{10,102}\xi_{10,504} - \xi_{10,507}\xi_{10,99}\right)$$

$$\xi_{10,522} = -\xi_{10,52}\left(\xi_{10,100}\xi_{10,504} + \xi_{10,101}\xi_{10,507} - \xi_{10,102}\xi_{10,506} - \xi_{10,505}\xi_{10,99}\right) + \xi_{10,52}\left(-\xi_{10,100}\xi_{10,507} + \xi_{10,101}\xi_{10,504} + \xi_{10,102}\xi_{10,505} - \xi_{10,506}\xi_{10,99}\right) + \xi_{10,53}\left(\xi_{10,100}\xi_{10,506} - \xi_{10,101}\xi_{10,505} + \xi_{10,102}\xi_{10,504} - \xi_{10,507}\xi_{10,99}\right) + \xi_{10,54}\left(\xi_{10,100}\xi_{10,505} + \xi_{10,101}\xi_{10,506} + \xi_{10,102}\xi_{10,507} + \xi_{10,504}\xi_{10,99}\right)$$

$$\xi_{10,523} = \xi_{10,52}\left(\xi_{10,100}\xi_{10,505} + \xi_{10,101}\xi_{10,506} + \xi_{10,102}\xi_{10,507} + \xi_{10,504}\xi_{10,99}\right) + \xi_{10,52}\left(\xi_{10,100}\xi_{10,506} - \xi_{10,101}\xi_{10,505} + \xi_{10,102}\xi_{10,504} - \xi_{10,507}\xi_{10,99}\right) - \xi_{10,53}\left(-\xi_{10,100}\xi_{10,507} + \xi_{10,101}\xi_{10,504} + \xi_{10,102}\xi_{10,505} - \xi_{10,506}\xi_{10,99}\right) + \xi_{10,54}\left(\xi_{10,100}\xi_{10,504} + \xi_{10,101}\xi_{10,507} - \xi_{10,102}\xi_{10,506} - \xi_{10,505}\xi_{10,99}\right)$$

$$\xi_{10,524} = -\xi_{10,53}\left(-\xi_{10,508}\xi_{10,68} - \xi_{10,509}\xi_{10,67} + \xi_{10,510}\xi_{10,70} - \xi_{10,511}\xi_{10,69}\right) - \xi_{10,53}\left(-\xi_{10,508}\xi_{10,69} - \xi_{10,509}\xi_{10,70} - \xi_{10,510}\xi_{10,67} + \xi_{10,511}\xi_{10,68}\right) - \xi_{10,53}\left(-\xi_{10,508}\xi_{10,70} + \xi_{10,509}\xi_{10,69} - \xi_{10,510}\xi_{10,68} - \xi_{10,511}\xi_{10,67}\right) + \xi_{10,54}\left(\xi_{10,508}\xi_{10,67} - \xi_{10,509}\xi_{10,68} - \xi_{10,510}\xi_{10,69} - \xi_{10,511}\xi_{10,70}\right)$$

$$\xi_{10,525} = \xi_{10,53}\left(\xi_{10,508}\xi_{10,67} - \xi_{10,509}\xi_{10,68} - \xi_{10,510}\xi_{10,69} - \xi_{10,511}\xi_{10,70}\right) + \xi_{10,53}\left(-\xi_{10,508}\xi_{10,69} - \xi_{10,509}\xi_{10,70} - \xi_{10,510}\xi_{10,67} + \xi_{10,511}\xi_{10,68}\right) - \xi_{10,53}\left(-\xi_{10,508}\xi_{10,70} + \xi_{10,509}\xi_{10,69} - \xi_{10,510}\xi_{10,68} - \xi_{10,511}\xi_{10,67}\right) + \xi_{10,54}\left(-\xi_{10,508}\xi_{10,68} - \xi_{10,509}\xi_{10,67} + \xi_{10,510}\xi_{10,70} - \xi_{10,511}\xi_{10,69}\right)$$

$$\xi_{10,526} = \xi_{10,53}\left(\xi_{10,508}\xi_{10,67} - \xi_{10,509}\xi_{10,68} - \xi_{10,510}\xi_{10,69} - \xi_{10,511}\xi_{10,70}\right) - \xi_{10,53}\left(-\xi_{10,508}\xi_{10,68} - \xi_{10,509}\xi_{10,67} + \xi_{10,510}\xi_{10,70} - \xi_{10,511}\xi_{10,69}\right) + \xi_{10,53}\left(-\xi_{10,508}\xi_{10,70} + \xi_{10,509}\xi_{10,69} - \xi_{10,510}\xi_{10,68} - \xi_{10,511}\xi_{10,67}\right) + \xi_{10,54}\left(-\xi_{10,508}\xi_{10,69} - \xi_{10,509}\xi_{10,70} - \xi_{10,510}\xi_{10,67} + \xi_{10,511}\xi_{10,68}\right)$$

$$\xi_{10,527} = \xi_{10,53}\left(\xi_{10,508}\xi_{10,67} - \xi_{10,509}\xi_{10,68} - \xi_{10,510}\xi_{10,69} - \xi_{10,511}\xi_{10,70}\right) + \xi_{10,53}\left(-\xi_{10,508}\xi_{10,68} - \xi_{10,509}\xi_{10,67} + \xi_{10,510}\xi_{10,70} - \xi_{10,511}\xi_{10,69}\right) - \xi_{10,53}\left(-\xi_{10,508}\xi_{10,69} - \xi_{10,509}\xi_{10,70} - \xi_{10,510}\xi_{10,67} + \xi_{10,511}\xi_{10,68}\right) + \xi_{10,54}\left(-\xi_{10,508}\xi_{10,70} + \xi_{10,509}\xi_{10,69} - \xi_{10,510}\xi_{10,68} - \xi_{10,511}\xi_{10,67}\right)$$

$$\xi_{10,528} = \xi_{10,393}\left(\xi_{10,33}\xi_{10,91} - \xi_{10,34}\xi_{10,92} - \xi_{10,35}\xi_{10,93} - \xi_{10,36}\xi_{10,94}\right) + \xi_{10,394}\left(\xi_{10,33}\xi_{10,92} + \xi_{10,34}\xi_{10,91} + \xi_{10,35}\xi_{10,94} - \xi_{10,36}\xi_{10,93}\right) + \xi_{10,395}\left(\xi_{10,33}\xi_{10,93} - \xi_{10,34}\xi_{10,94} + \xi_{10,35}\xi_{10,91} + \xi_{10,36}\xi_{10,92}\right) + \xi_{10,396}\left(\xi_{10,33}\xi_{10,94} + \xi_{10,34}\xi_{10,93} - \xi_{10,35}\xi_{10,92} + \xi_{10,36}\xi_{10,91}\right)$$

$$\xi_{10,529} = \xi_{10,393}\left(\xi_{10,33}\xi_{10,92} + \xi_{10,34}\xi_{10,91} + \xi_{10,35}\xi_{10,94} - \xi_{10,36}\xi_{10,93}\right) - \xi_{10,394}\left(\xi_{10,33}\xi_{10,91} - \xi_{10,34}\xi_{10,92} - \xi_{10,35}\xi_{10,93} - \xi_{10,36}\xi_{10,94}\right) + \xi_{10,395}\left(\xi_{10,33}\xi_{10,94} + \xi_{10,34}\xi_{10,93} - \xi_{10,35}\xi_{10,92} + \xi_{10,36}\xi_{10,91}\right) - \xi_{10,396}\left(\xi_{10,33}\xi_{10,93} - \xi_{10,34}\xi_{10,94} + \xi_{10,35}\xi_{10,91} + \xi_{10,36}\xi_{10,92}\right)$$

$$\xi_{10,530} = \xi_{10,393}\left(\xi_{10,33}\xi_{10,93} - \xi_{10,34}\xi_{10,94} + \xi_{10,35}\xi_{10,91} + \xi_{10,36}\xi_{10,92}\right) - \xi_{10,394}\left(\xi_{10,33}\xi_{10,94} + \xi_{10,34}\xi_{10,93} - \xi_{10,35}\xi_{10,92} + \xi_{10,36}\xi_{10,91}\right) - \xi_{10,395}\left(\xi_{10,33}\xi_{10,91} - \xi_{10,34}\xi_{10,92} - \xi_{10,35}\xi_{10,93} - \xi_{10,36}\xi_{10,94}\right) + \xi_{10,396}\left(\xi_{10,33}\xi_{10,92} + \xi_{10,34}\xi_{10,91} + \xi_{10,35}\xi_{10,94} - \xi_{10,36}\xi_{10,93}\right)$$

$$\xi_{10,531} = \xi_{10,393}\left(\xi_{10,33}\xi_{10,94} + \xi_{10,34}\xi_{10,93} - \xi_{10,35}\xi_{10,92} + \xi_{10,36}\xi_{10,91}\right) + \xi_{10,394}\left(\xi_{10,33}\xi_{10,93} - \xi_{10,34}\xi_{10,94} + \xi_{10,35}\xi_{10,91} + \xi_{10,36}\xi_{10,92}\right) - \xi_{10,395}\left(\xi_{10,33}\xi_{10,92} + \xi_{10,34}\xi_{10,91} + \xi_{10,35}\xi_{10,94} - \xi_{10,36}\xi_{10,93}\right) - \xi_{10,396}\left(\xi_{10,33}\xi_{10,91} - \xi_{10,34}\xi_{10,92} - \xi_{10,35}\xi_{10,93} - \xi_{10,36}\xi_{10,94}\right)$$

$$\xi_{10,532} = \xi_{10,417}\left(\xi_{10,37}\xi_{10,95} - \xi_{10,38}\xi_{10,96} - \xi_{10,39}\xi_{10,97} - \xi_{10,40}\xi_{10,98}\right) + \xi_{10,418}\left(\xi_{10,37}\xi_{10,96} + \xi_{10,38}\xi_{10,95} + \xi_{10,39}\xi_{10,98} - \xi_{10,40}\xi_{10,97}\right) + \xi_{10,419}\left(\xi_{10,37}\xi_{10,97} - \xi_{10,38}\xi_{10,98} + \xi_{10,39}\xi_{10,95} + \xi_{10,40}\xi_{10,96}\right) + \xi_{10,420}\left(\xi_{10,37}\xi_{10,98} + \xi_{10,38}\xi_{10,97} - \xi_{10,39}\xi_{10,96} + \xi_{10,40}\xi_{10,95}\right)$$

$$\xi_{10,533} = \xi_{10,417}\left(\xi_{10,37}\xi_{10,96} + \xi_{10,38}\xi_{10,95} + \xi_{10,39}\xi_{10,98} - \xi_{10,40}\xi_{10,97}\right) - \xi_{10,418}\left(\xi_{10,37}\xi_{10,95} - \xi_{10,38}\xi_{10,96} - \xi_{10,39}\xi_{10,97} - \xi_{10,40}\xi_{10,98}\right) + \xi_{10,419}\left(\xi_{10,37}\xi_{10,98} + \xi_{10,38}\xi_{10,97} - \xi_{10,39}\xi_{10,96} + \xi_{10,40}\xi_{10,95}\right) - \xi_{10,420}\left(\xi_{10,37}\xi_{10,97} - \xi_{10,38}\xi_{10,98} + \xi_{10,39}\xi_{10,95} + \xi_{10,40}\xi_{10,96}\right)$$

$$\xi_{10,534} = \xi_{10,417}\left(\xi_{10,37}\xi_{10,97} - \xi_{10,38}\xi_{10,98} + \xi_{10,39}\xi_{10,95} + \xi_{10,40}\xi_{10,96}\right) - \xi_{10,418}\left(\xi_{10,37}\xi_{10,98} + \xi_{10,38}\xi_{10,97} - \xi_{10,39}\xi_{10,96} + \xi_{10,40}\xi_{10,95}\right) - \xi_{10,419}\left(\xi_{10,37}\xi_{10,95} - \xi_{10,38}\xi_{10,96} - \xi_{10,39}\xi_{10,97} - \xi_{10,40}\xi_{10,98}\right) + \xi_{10,420}\left(\xi_{10,37}\xi_{10,96} + \xi_{10,38}\xi_{10,95} + \xi_{10,39}\xi_{10,98} - \xi_{10,40}\xi_{10,97}\right)$$

$$\xi_{10,535} = \xi_{10,417}\left(\xi_{10,37}\xi_{10,98} + \xi_{10,38}\xi_{10,97} - \xi_{10,39}\xi_{10,96} + \xi_{10,40}\xi_{10,95}\right) + \xi_{10,418}\left(\xi_{10,37}\xi_{10,97} - \xi_{10,38}\xi_{10,98} + \xi_{10,39}\xi_{10,95} + \xi_{10,40}\xi_{10,96}\right) - \xi_{10,419}\left(\xi_{10,37}\xi_{10,96} + \xi_{10,38}\xi_{10,95} + \xi_{10,39}\xi_{10,98} - \xi_{10,40}\xi_{10,97}\right) - \xi_{10,420}\left(\xi_{10,37}\xi_{10,95} - \xi_{10,38}\xi_{10,96} - \xi_{10,39}\xi_{10,97} - \xi_{10,40}\xi_{10,98}\right)$$

$$\xi_{10,536} = \xi_{10,106} + \xi_{10,48}$$

$$\xi_{10,537} = \begin{cases} 0 & \text{for } \xi_{10,512}^2 + \xi_{10,515}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,515}, \xi_{10,512}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,538} = \begin{cases} 0 & \text{for } \xi_{10,513}^2 + \xi_{10,514}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,513}, \xi_{10,514}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,539} = 2\operatorname{atan}_2\left(\sqrt{\xi_{10,513}^2 + \xi_{10,514}^2}, \sqrt{\xi_{10,512}^2 + \xi_{10,515}^2}\right)$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,540} = -\xi_{10,537} + \xi_{10,538}$$

$$\xi_{10,541} = -\xi_{10,537} - \xi_{10,538}$$

$$\xi_{10,542} = \xi_{10,536} + \xi_{10,537}$$

$$\xi_{10,543} = \begin{cases} 0 & \text{for } \xi_{10,516}^2 + \xi_{10,519}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,519}, \xi_{10,516}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,544} = \begin{cases} 0 & \text{for } \xi_{10,517}^2 + \xi_{10,518}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,517}, \xi_{10,518}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,545} = 2\operatorname{atan}_2\left(\sqrt{\xi_{10,517}^2 + \xi_{10,518}^2}, \sqrt{\xi_{10,516}^2 + \xi_{10,519}^2}\right)$$

$$\xi_{10,546} = -\xi_{10,543} + \xi_{10,544}$$

$$\xi_{10,547} = -\xi_{10,543} - \xi_{10,544}$$

$$\xi_{10,548} = \xi_{10,542} + \xi_{10,543}$$

$$\xi_{10,549} = \begin{cases} 0 & \text{for } \xi_{10,520}^2 + \xi_{10,523}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,523}, \xi_{10,520}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,550} = \begin{cases} 0 & \text{for } \xi_{10,521}^2 + \xi_{10,522}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,521}, \xi_{10,522}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,551} = 2\operatorname{atan}_2\left(\sqrt{\xi_{10,521}^2 + \xi_{10,522}^2}, \sqrt{\xi_{10,520}^2 + \xi_{10,523}^2}\right)$$

$$\xi_{10,552} = -\xi_{10,549} + \xi_{10,550}$$

$$\xi_{10,553} = -\xi_{10,549} - \xi_{10,550}$$

$$\xi_{10,554} = \xi_{10,548} + \xi_{10,549}$$

$$\xi_{10,555} = \begin{cases} 0 & \text{for } \xi_{10,524}^2 + \xi_{10,527}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,527}, \xi_{10,524}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,556} = \begin{cases} 0 & \text{for } \xi_{10,525}^2 + \xi_{10,526}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,525}, \xi_{10,526}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,557} = 2\operatorname{atan}_2\left(\sqrt{\xi_{10,525}^2 + \xi_{10,526}^2}, \sqrt{\xi_{10,524}^2 + \xi_{10,527}^2}\right)$$

$$\xi_{10,558} = -\xi_{10,555} + \xi_{10,556}$$

$$\xi_{10,559} = -\xi_{10,555} - \xi_{10,556}$$

$$\xi_{10,560} = \xi_{10,554} + \xi_{10,555}$$

$$\xi_{10,561} = \begin{cases} 0 & \text{for } \xi_{10,117}^2 + \xi_{10,118}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{10,118}, \xi_{10,117}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,562} = \begin{cases} 0 & \text{for } \xi_{10,114}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{10,114}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,563} = 2\operatorname{atan}_2\left(\left|\xi_{10,114}\right|, \sqrt{\xi_{10,117}^2 + \xi_{10,118}^2}\right)$$

$$\xi_{10,564} = -\xi_{10,561} + \xi_{10,562}$$

$$\xi_{10,565} = -\xi_{10,561} - \xi_{10,562}$$

$$\xi_{10,566} = \xi_{10,560} + \xi_{10,561}$$

$$\xi_{10,567} = \begin{cases} 0 & \text{for } \xi_{10,112}^2 = 0 \\ \operatorname{atan}_2\left(0, \xi_{10,112}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,568} = \begin{cases} 0 & \text{for } \xi_{10,113}^2 = 0 \\ \operatorname{atan}_2\left(\xi_{10,113}, 0\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,569} = 2\operatorname{atan}_2\left(\left|\xi_{10,113}\right|, \left|\xi_{10,112}\right|\right)$$

$$\xi_{10,570} = -\xi_{10,567} + \xi_{10,568}$$

$$\xi_{10,571} = -\xi_{10,567} - \xi_{10,568}$$

$$\xi_{10,572} = \xi_{10,566} + \xi_{10,567}$$

$$\xi_{10,573} = \begin{cases} 0 & \text{for } \xi_{10,528}^2 + \xi_{10,531}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,531}, \xi_{10,528}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,574} = \begin{cases} 0 & \text{for } \xi_{10,529}^2 + \xi_{10,530}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,529}, \xi_{10,530}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,575} = 2\operatorname{atan}_2\left(\sqrt{\xi_{10,529}^2 + \xi_{10,530}^2}, \sqrt{\xi_{10,528}^2 + \xi_{10,531}^2}\right)$$

Exact shared-expression program 10

Original parameters: a, b, c. Bindings are definitions, not additional free angles.

Evaluate definitions in order. Piecewise expressions select the first satisfied case; unused singular branches are not evaluated. The source bundle also includes the machine-readable program.

$$\xi_{10,576} = -\xi_{10,573} + \xi_{10,574}$$

$$\xi_{10,577} = -\xi_{10,573} - \xi_{10,574}$$

$$\xi_{10,578} = \xi_{10,572} + \xi_{10,573}$$

$$\xi_{10,579} = \begin{cases} 0 & \text{for } \xi_{10,532}^2 + \xi_{10,535}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,535}, \xi_{10,532}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,580} = \begin{cases} 0 & \text{for } \xi_{10,533}^2 + \xi_{10,534}^2 = 0 \\ \operatorname{atan}_2\left(-\xi_{10,533}, \xi_{10,534}\right) & \text{otherwise} \end{cases}$$

$$\xi_{10,581} = 2\operatorname{atan}_2\left(\sqrt{\xi_{10,533}^2 + \xi_{10,534}^2}, \sqrt{\xi_{10,532}^2 + \xi_{10,535}^2}\right)$$

$$\xi_{10,582} = -\xi_{10,579} + \xi_{10,580}$$

$$\xi_{10,583} = -\xi_{10,579} - \xi_{10,580}$$

$$\xi_{10,584} = \xi_{10,578} + \xi_{10,579}$$